

Connectivity Bounds in Graphs and Beyond

Erdős Problem 915, from Proof to Search

Louis Vandenbruwaene

Supervisor: Prof. Stijn Cambie
KU Leuven

Thesis presented in
fulfillment of the requirements
for the degree of Master of Science
in Mathematics

Academic year 2026–2027

Open Access Disclaimer

This master's thesis is a student work submitted as part of an academic programme at KU Leuven. The work is made available for examination and archiving in accordance with the applicable KU Leuven regulations and open-access policy for master's theses.

© Copyright by KU Leuven

Without written permission from the supervisors and the author, it is forbidden to reproduce or adapt in any form or by any means any part of this publication. Requests for obtaining the right to reproduce or utilize parts of this publication should be addressed to KU Leuven, Faculty of Science, Celestijnenlaan 200H - box 2100, 3001 Leuven (Heverlee), Telephone +32 16 32 14 01.

Written permission from the supervisor is also required to use the methods, products, schematics, and programs described in this work for industrial or commercial use, and for submitting this publication in scientific contests.

Foreword

This thesis studies how many edges or arcs a network can contain when no pair of vertices is allowed to have too many independent routes between them. The starting point is Erdős Problem 915, an extremal question about edge-disjoint paths in simple graphs. The work is told as a journey. It begins with the variants whose answers can be proved cleanly by hand, watches those proofs grow longer and more delicate as the model changes, and at the point where a single argument no longer suffices it turns to a unified program. That program holds all twelve variants behind one common interface, so a single piece of code serves them all. It measures connectivity exactly with a maximum-flow routine, the standard way to count the independent routes between two points. It runs a prover that sweeps every graph of a small fixed size and, when none beats a target, reports that sweep as a genuine upper-bound proof. And it runs a search that, like hot metal cooling slowly into shape, settles random graphs toward the densest ones the rules allow.

I am grateful to my supervisor, Prof. Stijn Cambie, and my daily advisor for their guidance, corrections, and mathematical feedback throughout the project. One review in particular changed the shape of this thesis: Prof. Cambie put a late draft through a language model different from the one I had been working with, and it came back with a misapplied inequality I had read past many times, together with the counting idea that grew into [Proposition A.22](#) and [Theorem A.24](#). I thank my promotor, Prof. Jan Goedgebeur, whose suggestion of the `nauty` generation pipeline sharply accelerated the directed enumeration. I also thank the people who discussed early versions of the arguments, checked examples, or helped clarify the presentation. Above all I thank my girlfriend Sara, whose patience and encouragement carried me through the long stretches of this work. Any remaining mistakes are my own.

Contribution Statement

This thesis is my own work. I carried out the literature study, recast the twelve variants of the problem in one common language, and designed and wrote the single program that runs through all of them. I built every example, by hand or by computer search, checked each one with that program, wrote the proofs that appear in the thesis, and typeset the whole document. Where a result comes from the literature I say so and cite it. The theorems I lean on but do not claim as mine are those of Mader, Leonard, Sørensen and Thomassen, Menger, Gomory and Hu, and Baranyai, the hypergraph connectivity results of Bérczi, Chandrasekaran, Király and Kulkarni, and the standard probabilistic tools used for the random graphs.

The program has three jobs, and it keeps them separate. It *measures*: given any graph, it computes exactly, through maximum flow, how many independent routes the busiest pair of vertices has. It *proves*: for a fixed number of vertices it sweeps a finite space of graphs, and when nothing in that space beats a target, that sweep is a genuine upper-bound proof rather than a guess. And it *discovers*: a search that cools like a heated metal settling into shape wanders the space of graphs and finds dense ones, which give lower bounds and suggest conjectures but never prove an upper bound by themselves. Every general statement in the thesis rests on a proof by hand or a cited theorem. The computer searches are evidence that points the way, never a replacement for the argument. I used AI language models, including Claude Code, ChatGPT, and Gemini, as tools for literature search, L^AT_EX editing, proofreading, and trying out proof drafts. I checked every mathematical statement and proof in the thesis myself and take full responsibility for the content.

One thing about that use is worth recording, because it is a methodological point and not just a disclosure. A model that has helped write an argument is a poor reviewer of it: it reads the text the way its author does, follows the same intended reading of each sentence, and slides over the same gaps. Handing the finished draft to a *different* model, cold, behaved much more like handing it to a different person. It read each sentence as a standalone claim rather than as a step in an argument it already believed, and that is exactly how it caught a use of Whitney's inequality that runs the wrong way, in a sentence that had survived several careful readings by me and by the model that helped write it. The same pass also proposed a counting argument I had not tried. It was wrong about a good deal else, and every item it raised had to be checked against the source before anything was changed, several of them turning out to be objections to conventions the thesis states rather than to errors. But the pattern is the familiar one from working with people: the value came from the disagreement, not from the agreement, and from the fact that the second reader had no stake in the first reader's phrasing.

The genuinely new results are the following. For hyperedge networks I settle the vertex-disjoint problem completely at the two strictest route limits, $m = 2$ and $m = 3$, for every hyper-

edge size r and every number of vertices n (Theorems A.50 and A.53): the vertex-disjoint and edge-disjoint answers turn out to be equal, both $\lfloor (m-1)(n-1)/(r-1) \rfloor$. The $m=3$ case rests on a bound for incidence graphs that I prove using Tutte's decomposition of a graph into its three-connected pieces (Lemma A.52). For one-way multigraphs I prove the exact value for every n and every m (Theorem A.30), closing what had stood as a conjecture verified only for $n \leq 6$. The route is a reachability-preserving spanning subgraph, always sparse, whose deletion is guaranteed to lower every pairwise connectivity by exactly one, which sidesteps the fractional obstruction that stopped a direct vertex-deletion argument at odd n and needs no minimum-degree conjecture and no restriction on m . As a three-line corollary this reproves, by hand, the one finite fact ($L_3^{\text{dir}}(7) = 24$) the earlier route had reduced the whole $m=3$ problem to and left to a computer. That argument is deliberately blind to which multigraph it takes apart, which is why it had seemed to say nothing about *uniqueness*. Run backwards from the assumption that its bound is met exactly, it turns out to be rigid, and I use that to prove the extremiser unique on the quadratic branch whenever $n \geq \max(8, 2m)$ (Theorem A.44): it is the balanced one-way complete bipartite digraph at multiplicity $m-1$. Equality pins the number of strongly connected components, forcing the extremiser to be acyclic, and the triangle-freeness that had only been used to keep the sparse subgraph small becomes the equality case of Mantel's theorem, which pins its shape. For one-way simple graphs I prove an unconditional quadratic bound (Proposition A.22 and Theorem A.24): under any route limit m , such a graph has at most $\lfloor n^2/4 \rfloor + 4(m-1)(n-1)$ arcs, so its size is within a linear amount of the one-way bipartite wall, and separately it can always be turned into such a wall by deleting at most $\sqrt{m} n^{3/2}$ arcs. This fixes both the leading constant and the order of the second-order term of the standing conjecture (Conjecture 1.14) without any assumption about the shape of the extremal graph, which is what every previous route to that bound needed, and leaves only a constant factor of about eight in one coefficient. The counting idea behind it came out of the external review recorded in the foreword, and I verified it and worked out the sharpening myself. For multigraphs under the alternative counting convention I settle the vertex problem at $m=2$ and disprove the natural guess, suggested by the data through $m=4$, that a thickened tree stays optimal for all larger n : a bouquet of thickened cliques sharing a single vertex (Theorem A.63) beats it by a margin growing linearly in n , at a rate that is quadratic rather than linear in m (Section A.9). A matching upper bound (Proposition A.64) shows that construction has the right order in n and in m both, by way of the observation that the underlying simple graph of a feasible multigraph is itself feasible for the simple vertex problem, which ties this variant to the oldest one in the family. I then show that the value is exactly a *block* problem (Theorem A.66): the objective is a sum of local terms over the edges of the underlying simple graph, no route between adjacent vertices ever leaves their common block, so the total is additive over blocks and the whole question reduces to a knapsack over the best two-connected block of each size. That reduction settles the value exactly for every $n \leq 8$ and $m \leq 8$, and it shows the bouquet is not the shape of the answer: from $m=5$ on, a single two-connected block beats every way of splitting the vertices. The blocks that win are bipartite rather than complete, and thickening them instead (Theorem A.67) improves the constant by a factor $8(3-2\sqrt{2}) \approx 1.373$, closing part of the gap to the upper bound.

Two further results come from isolating what the quadratic bound above actually uses, which is one cap on one family of routes and after that only degree counting (Lemma A.27). Stated that way it applies well beyond the case it was proved for. Since the counting it rests on is the reviewer's idea, credited above, these two inherit that debt as well, and what I claim in them is the observation that the hypothesis can be weakened, and the two arguments that then supply it. The multigraph upper bound just mentioned is not one of them, and reaches its conclusion by an unrelated route through Mader's density theorem. For one-way simple graphs under the *vertex* separation, a problem this thesis is otherwise careful to keep separate from the arc one because Whitney's inequality runs the wrong way, it gives $k_m^{\text{dir}}(n) = n^2/4 + \Theta_m(n)$ unconditionally (Theorem A.28), so the leading constant is settled there too and the two directed problems agree to first order. What transfers is the route family rather than the bound, and that is why it is allowed. For directed hyperedge networks it settles the leading constant outright (Theorem A.59): the maximum is $(1 + o(1))(m - 1)n^2/(4(r - 1))$ for every fixed r and m , under both separations at once, so the bipartite construction is asymptotically optimal and the factor of four that had separated the known bounds is closed. The obstacle there was that a hyperedge carries $r - 1$ heads at once, so the routes being counted are not disjoint. The argument does not repair that. It pays the factor $r - 1$ and then shows the payment lands in the linear error term rather than in the leading one, while a second factor of $r - 1$, of different origin, divides the leading term. That proof reads a directed hyperedge as one tail fanning to many heads, and I show separately that the same leading constant holds for *every* way of orienting a hyperedge, any split into tails and heads (Theorem A.60). The single tail had been used twice, once for the routes entering a hyperedge and once for those leaving it, and with several tails the two sides interfere. The repair is to stop constructing a large family of routes and instead take a maximum one and read off what stopped it, since every route it leaves out is blocked by a hyperedge it has already used, and a hyperedge has only r vertices to block with. This closes the orientation axis to first order: forward, backward and general all have the same leading term.

Short Summary

Erdős Problem 915, posed by Bollobás and Erdős [1], asks for the maximum number of edges in a graph on n vertices such that no two vertices are connected by m edge-disjoint paths. In modern notation this asks for

$$\ell_m(n) = \max\{|E(G)| : |V(G)| = n, \lambda_G^{\max} \leq m - 1\}.$$

Mader proved that, for simple graphs and $n \geq m \geq 2$,

$$\ell_m(n) = \left\lfloor \frac{m(n-1)}{2} \right\rfloor.$$

This thesis revisits this theorem and studies the corresponding extremal problem under three independent changes: the graph model (simple graphs, multigraphs, and hypergraphs), the path-separation notion (edge-disjoint or internally vertex-disjoint), and the presence or absence of directions. Sampled random graphs $G(n, p)$ run alongside throughout as a lens on the typical case rather than as a separate model.

The thesis is organised as a journey from hand proofs to computation. The early variants are settled rigorously: the multigraph value $L_m(n) = (m-1)(n-1)$, the random-graph threshold $p^* = m/n$, the equality of the edge and vertex problems for $m \leq 4$, and the first divergence at $m = 5$, where Sørensen and Thomassen prove $k_5(n) = \lfloor 8n/3 \rfloor - 4$, which pulls above the edge value from $n = 14$ onwards. The directed case forces the turn: the value $\ell_2^{\text{dir}}(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$ is proved by a long induction, and for $m \geq 3$ the hand proof gives way to a conjecture, after an explicit ten-vertex graph rules out the naive guess. The centrepiece is then a unified program. One multiplicity-matrix representation models almost all twelve variants, an exact maximum-flow checker measures local connectivity, a cut-counting optimisation certifies upper bounds for small sizes, and a search guided by edge sensitivity, run as both simulated annealing and tabu search, discovers extremal graphs. The program rediscovers, each in its own variant, the double star (undirected multigraphs at $m = 2$), the directed hub, the one-directional bipartite digraph, and the augmented-bipartite construction (the directed cases), certifies the directed multigraph values $L_m^{\text{dir}}(n) = 2(n-1)(m-1)$ at every $n \leq 6$, which was as far as any proof then reached, and frames the remaining open problems on the directed frontier.

The interplay then pays back in the other direction. The hypergraph vertex problem is settled completely at $m = 2$ and $m = 3$: $k_m^{(r)}(n) = \lfloor (m-1)(n-1)/(r-1) \rfloor$ for every uniformity r , the $m = 3$ case through a new rank bound on incidence graphs proved by way of Tutte's triconnected decomposition. On the directed side the multigraph problem closes completely, $L_m^{\text{dir}}(n) = (m-1) \max(2(n-1), \lfloor n^2/4 \rfloor)$ for every n and every m , by deleting a sparse reachability-preserving subgraph at a time rather than a vertex at a time, and the simple digraph problem, arc and

vertex alike, is pinned at $n^2/4 + \Theta_m(n)$ with no assumption on the shape of the extremiser. The directed hypergraph, which had only bounds a factor of four apart, has its leading term settled at $(m-1)n^2/(4(r-1))$, so its bipartite construction is asymptotically optimal.

Abbreviations and Symbols

GH	Gomory-Hu
MILP	mixed-integer linear program
SA	simulated annealing
SPQR	the triconnected decomposition tree (series, parallel, rigid pieces)
$G = (V, E)$	graph with vertex set V and edge set E
$D = (V, A)$	directed graph with vertex set V and arc set A
n	number of vertices
m	forbidden-connectivity parameter, with all local connectivities at most $m - 1$
r	hyperedge size (every hyperedge spans r vertices)
$\lambda_G(u, v)$	maximum number of edge-disjoint u - v paths
$\lambda_G^{\max}$	maximum local edge-connectivity over all vertex pairs
$\kappa_G(u, v)$	maximum number of internally vertex-disjoint u - v paths
$\kappa_G^{\max}$	maximum local vertex-connectivity over all vertex pairs
$\ell_m(n)$	simple undirected edge-disjoint extremal function
$L_m(n)$	undirected multigraph edge-disjoint extremal function
$k_m(n)$	simple undirected vertex-disjoint extremal function
$\ell_m^{\text{dir}}(n)$	simple directed arc-disjoint extremal function
$L_m^{\text{dir}}(n)$	directed multigraph arc-disjoint extremal function
$k_m^{\text{dir}}(n)$	simple directed vertex-disjoint extremal function
$\ell_m^{(r)}(n), k_m^{(r)}(n)$	r -uniform hypergraph edge- and vertex-disjoint extremal functions
$K_m^{\text{multi}}(n)$	multigraph vertex-disjoint extremal function with parallel copies counted as distinct routes (Section A.9)
$M(n)$	$\max(2(n - 1), \lfloor n^2/4 \rfloor)$, the directed multigraph value per unit of $m - 1$
$M^*(n)$	the m -free directed multigraph optimum, $L_m^{\text{dir}}(n) = (m - 1) M^*(n)$, equal to $M(n)$ by Corollary A.37
$B_{p,q}$	one-directional complete bipartite digraph, all arcs from a p -part to a q -part
$\mu(u, v)$	multiplicity of an edge or arc between u and v
$\sigma(e)$	edge sensitivity $\lambda^{\max}(G) - \lambda^{\max}(G - e)$
$G(n, p)$	binomial random graph

Contents

Foreword	II
Contribution Statement	III
Short Summary	VI
Abbreviations and Symbols	VIII
1 The Problem and Its Base Cases	1
1.1 Graphs and routes	1
1.2 Erdős Problem 915	2
1.3 Three axes of variation	4
1.4 Cases with short proofs	6
1.5 The first divergence: vertex-disjoint paths	8
1.6 The directed case and the quadratic phenomenon	10
1.7 The failure of direct proof for $m \geq 3$	14
1.8 The hypergraph variant	16
1.9 Random graphs as intuition	18
1.10 Where this work sits	21
2 Certifying Bounds by Machine	23
2.1 The solver	24
2.2 A single representation for all variants	24
2.3 The connectivity checker built on maximum flow	26
2.3.1 The directed hypergraph checker	32
2.4 Checking every graph of a fixed size	33
2.5 Reaching further: a pruned search for the directed base cases	35
2.6 Generating the directed cases faster	36
2.7 Proving every m at once	37

2.8	What the solver may claim	43
2.9	Reproducibility	43
2.9.1	What a machine-checked claim in this thesis means	44
3	Discovering Bounds by Search	46
3.1	The wall: how enumeration explodes	46
3.2	The temperature search	47
3.3	Temperature and edge sensitivity	49
3.4	Keeping the checker cheap inside the search	51
3.5	Rediscovering the known extremisers	53
3.5.1	Simulated annealing versus tabu search	54
3.6	The trichotomy: proved, certified, conjectured	56
4	Synthesis, Results, and Open Problems	63
4.1	The directed frontier and the missing decomposition	63
4.2	Two principal phenomena	67
4.3	Summary of the twelve variants	69
4.3.1	Orientation models for the directed hypergraph	74
4.4	Contributions and limitations of the program	77
4.5	Open problems	77
A	Full Proofs and Computational Transcripts	81
A.1	Menger, Gomory–Hu, and the degree bound	81
A.2	Mader’s theorem in full	83
A.3	The directed arc value at $m = 2$	87
A.4	Structural propositions on the directed frontier	91
A.4.1	What that proof actually used	97
A.5	The directed multigraph problem, solved in full	98
A.5.1	The lower bound: thickening	99
A.5.2	The upper bound: a reachability skeleton	100
A.5.3	Uniqueness on the quadratic branch	106
A.6	The hypergraph bounds	109

A.7	The hypergraph vertex problem	112
A.8	The directed hypergraph	119
A.9	The multigraph vertex problem under the other convention	125
A.9.1	The value is a block problem	130
A.10	The random threshold	134
A.11	Computational transcripts	136
A.12	How the program checks itself	137
B	Supplementary Figures	139
B.1	A gallery of extremal graphs	139
B.2	Search traces across the variants	139
B.3	The variant landscape at $m = 6$ and in three dimensions	141
C	The Complete Source Code	145
C.1	The main program	145
C.1.1	How the file opens	145
C.1.2	The objects: graphs, hypergraphs, and the values we already know	149
C.1.3	Measuring, and proving	160
C.1.4	Searching	169
C.1.5	The random model	189
C.1.6	Drawing the figures	193
C.1.7	Walking the whole space	210
C.1.8	The open variants	226
C.1.9	The gallery	236
C.1.10	The self-check	246
C.2	The figure script	252
C.3	The C accelerator	263
C.4	The test suite	266
C.4.1	Shared setup	266
C.4.2	The graph model	267
C.4.3	The closed-form bounds	269
C.4.4	The named constructions	271

C.4.5 The connectivity checker	272
C.4.6 Hypergraphs	275
C.4.7 Edge sensitivity	277
C.4.8 The search	279
C.4.9 The prover	281
C.4.10 The random model	283
C.4.11 The solver	284

Chapter 1

The Problem and Its Base Cases

A network is useful not only because it connects, but because it connects in more than one way. A single road between two towns is a liability, and a second independent road is insurance. The mathematical content of that intuition is the number of routes between two points that share no segment, and the question this thesis circles is how many connections a network can hold before some pair of points is forced to have too many such routes. This chapter builds the question from nothing, settles the variants whose proofs are short enough to read in an afternoon, and then follows the same question along three axes until the proofs stretch past breaking. By the end the case for a machine has made itself.

1.1 Graphs and routes

To state the question precisely we need only a few ideas. A *graph* G is a finite set of points, called *vertices*, together with a set of links, called *edges*, each joining two of the vertices. We write $V(G)$ for the set of vertices and $E(G)$ for the set of edges, so that $|E(G)|$ is the number of edges, the very quantity this thesis tries to make as large as possible. One pictures the vertices as dots and the edges as lines between them.

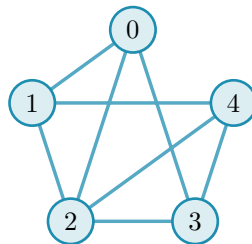


Figure 1.1. A simple graph on five vertices. It is K_5 , the *complete* graph in which every one of the $\binom{5}{2} = 10$ pairs is joined by an edge (the binomial coefficient $\binom{5}{2}$ counts the ways to choose two of the five vertices), with the two edges $\{0, 4\}$ and $\{1, 3\}$ removed, so its edge set $E(G)$ holds eight edges. The five dots are the vertex set $V(G)$ and the lines are the edges. Because at most one line joins any given pair and no edge loops back to its own endpoint, this is a simple graph. The same graph returns in [Figure 1.2](#), once we can count the independent routes between a pair.

A graph is the bare skeleton of a network: the vertices are the towns or the devices, and the edges are the roads or the links between them. We write n for the number of vertices, and the *degree* of a vertex is the number of edges meeting it. Unless we say otherwise our graphs are *simple*, meaning that at most one edge joins any given pair of vertices and its endpoints are distinct.

A *path* from a vertex u to a vertex v is a sequence of distinct vertices that starts at u , ends at v , and joins each consecutive pair by an edge. It is the formal version of a route through the network. Two paths between the same endpoints are *edge-disjoint* if no edge belongs to both, so that the loss of a single edge can break at most one of them. The largest number of pairwise edge-disjoint paths between u and v is their *local edge-connectivity* $\lambda_G(u, v)$, the count of genuinely independent routes the network offers to that pair. The pair with the most independent routes sets the ceiling for the whole graph:

$$\lambda_G^{\max} = \max_{u \neq v} \lambda_G(u, v),$$

the largest local edge-connectivity anywhere in G . A theorem of Menger [2], recalled in [Appendix A](#), gives this quantity a second face: $\lambda_G(u, v)$ also equals the least number of edges one must remove to cut every route from u to v ([Figure 1.2](#)). Many independent routes between a pair is exactly the same thing as a costly cut between them.

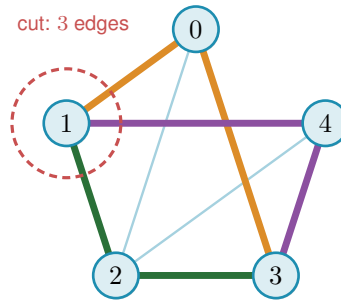


Figure 1.2. Menger’s theorem on the graph of [Figure 1.1](#). The pair 1, 3 carries no edge of its own, yet three routes still join them with no shared edge, 1-0-3 (orange), 1-2-3 (green), and 1-4-3 (purple), each through a different middle vertex, so $\lambda(1, 3) = 3$. The three edges meeting vertex 1 (inside the red dashed ring) are a smallest set whose removal cuts every 1-3 route, and no set of just two edges can cut them all, because the three routes share no edge, so each removed edge breaks at most one route and two removals leave one route intact. The cheapest cut therefore also has size three. The most independent routes a pair can muster always equals the cheapest cut between them, the fact every bound in this thesis leans on.

1.2 Erdős Problem 915

The constraint at the heart of this thesis is a ceiling on connectivity. Requiring $\lambda_G^{\max} \leq m - 1$ says that no pair of vertices is joined by as many as m edge-disjoint paths, or equivalently that every pair can be severed by removing fewer than m edges. The question is how many edges a graph can hold while staying under that ceiling.

Question 1.1 (Erdős Problem 915 [3]). How many edges can a graph on n vertices have if no two of its vertices are joined by m edge-disjoint paths?

Writing the answer as an extremal function,

$$\ell_m(n) = \max\{|E(G)| : |V(G)| = n, \lambda_G^{\max} \leq m - 1\},$$

the problem asks for $\ell_m(n)$, the most edges possible under the ceiling. Mader settled the simple-graph case completely, proving both halves at once: no graph under the ceiling has more than $\lfloor m(n-1)/2 \rfloor$ edges (the upper bound), and some graph reaches that count (the lower bound). The brackets $\lfloor \cdot \rfloor$ round their content down to the nearest whole number, and their ceiling counterpart $\lceil \cdot \rceil$ rounds up. Both recur throughout, because an edge count is always an integer while the formulas that bound it need not be.

Theorem 1.2 (Mader [4]). *For all $n \geq m \geq 2$,*

$$\ell_m(n) = \left\lfloor \frac{m(n-1)}{2} \right\rfloor.$$

Mader's own proof is a direct structural induction. The proof given in full in [Appendix A](#) takes a different road, by way of the Gomory–Hu tree, found independently of Mader, together with a characterisation of the graphs that attain equality ([Theorem A.11](#)). This is the road the thesis adopts throughout, because the very same tree returns for the multigraph and hypergraph variants ([Theorem 1.3](#), [Proposition 1.15](#)), so a single argument serves all of them rather than each needing its own. Every graph has such a tree ([Theorem A.2](#)). A Gomory–Hu tree of G is a weighted tree on the same vertex set in which $\lambda_G(u, v)$ equals the lightest edge on the tree path from u to v . We single out this one identity because the same idea reappears when the program of [Chapter 2](#) stores connectivities as a tree (a graph without cycles). All $\binom{n}{2}$ local connectivities are therefore encoded by $n-1$ numbers, and $\lambda_G^{\max}$ is simply the heaviest edge of the tree: each pairwise value is the lightest edge on some path through the tree, so no pair can exceed the single heaviest edge anywhere in it, and the two endpoints of that heaviest edge attain it directly, since their own tree path is that one edge.

The whole argument ([Appendix A](#)) is one act of double counting on that tree. The word *charged* below is just accounting: to charge an edge to a tree edge is to assign it there for the count, and the proof works by charging every edge of G to tree edges and then adding the charges up. Each of the $n-1$ tree edges stands for a minimum cut of G , and an original edge of G is charged once for every tree edge lying between its endpoints, so summing the tree-edge weights counts every edge of G at least once and, crucially, counts most of them at least twice. The only edges charged a single time are those joining tree-adjacent vertices, and a *simple* graph has at most $n-1$ of those, one per tree edge. Every other edge is charged twice. Since each tree weight is a local connectivity and so at most $m-1$, the two counts meet at $2|E(G)| - (n-1) \leq (m-1)(n-1)$, which rearranges to the floor. The factor of two, and with it the gap between Mader's $\lfloor m(n-1)/2 \rfloor$ and the larger value $(m-1)(n-1)$ that parallel edges will buy in [Theorem 1.3](#), is exactly the dividend of simplicity, the one place the argument uses that no pair carries a parallel edge.

One example fixes the scale. The complete graph K_n joins every pair, so every pair has $n-1$ routes that share no edge, $\lambda(u, v) = n-1$, while it carries all $\binom{n}{2}$ edges. It therefore meets the ceiling $\lambda^{\max} \leq m-1$ exactly when $m \geq n$, and from that point on the question is settled trivially:

the complete graph is feasible, no simple graph on n vertices is denser, so $\ell_m(n) = \binom{n}{2}$ for every $m \geq n$ and raising m further changes nothing. For smaller m the extremal graph must be strictly sparser than complete, and pinning down exactly how much sparser is the whole problem. That is why the theorem, and the thesis, assume $m \leq n$.

1.3 Three axes of variation

Mader's theorem is the floor we build on, and everything afterwards is a variation on it. There are three independent ways to change the rules, and they generate the landscape this thesis traverses.

The first axis is the *model*, drawn in Figure 1.3. Two independent choices generate it. The first is *arity*: an edge may join exactly two vertices, or a *hyperedge* may join any $r \geq 2$ of them, a route then passing through it between any two of its members. At $r = 2$ a hyperedge is an ordinary edge, so graphs are the $r = 2$ special case and the genuinely new setting begins at $r \geq 3$. The second is *multiplicity*: a *simple* object carries at most one copy of each edge, while a *multi* object allows parallel copies, recorded by a multiplicity $\mu(u, v)$, so a single pair can already carry several independent routes. Because the two choices are independent they form a small lattice. At the top sits the most general object, the *multi-hypergraph*, and lowering one toggle at a time restricts it downward: forbidding repeats gives the simple hypergraph, dropping the arity to two gives the multigraph, and doing both gives the simple graph at the bottom. Every theorem in this thesis is a statement about one node of this lattice, and the three rows of every comparison figure are its simple-graph, multigraph, and hypergraph levels.

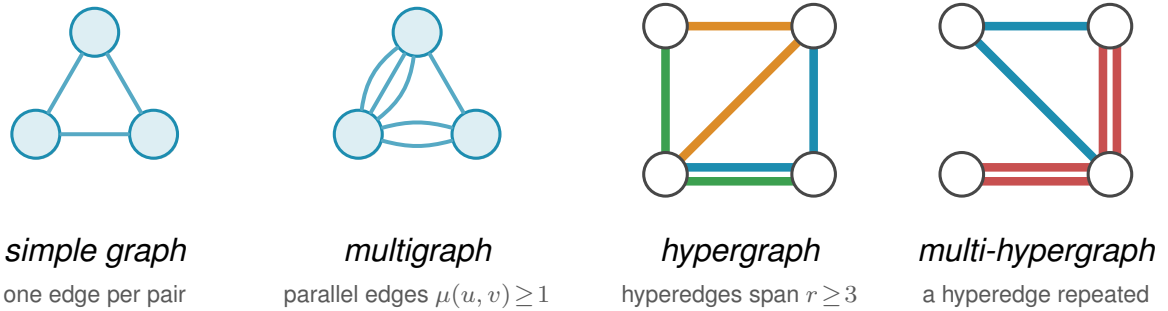


Figure 1.3. The first axis of variation, the *model*, shown for the four object classes this thesis uses. In a *simple graph* each pair of vertices is joined by at most one edge. A *multigraph* permits parallel edges, here a triple and a double, so the multiplicity $\mu(u, v)$ counts how many copies join a pair. A *hypergraph* replaces pairwise edges by hyperedges, each a set of $r \geq 2$ vertices ($r = 2$ gives back ordinary edges, and the depicted examples use $r = 3$), drawn the way a metro map draws a line, threading all of its stations, one colour per hyperedge. Where two hyperedges share a stretch between stations the two lines run side by side, one colour each, exactly as a metro map separates lines along a shared stretch of track. A *multi-hypergraph* lets a hyperedge be repeated, drawn as two parallel lines through the same stations, the hypergraph counterpart of parallel edges. When all hyperedges have the same size r the hypergraph is r -uniform. The same connectivity question, how many independent routes a pair can be forced to share, is asked of all four.

The second axis is the *direction*, a third toggle independent of the other two. Setting it at the top of the model lattice gives the single most general object in the thesis, the *directed multi-*

hypergraph, of which every other class is a restriction. Figure 1.4 redraws the four models of Figure 1.3 with arcs in place of edges. An undirected object is the symmetric special case of a directed one, each edge standing for a matched pair of opposite arcs $u \rightarrow v$ and $v \rightarrow u$, so direction strictly enlarges the family of objects. What it does *not* do is let the directed and undirected problems merge into one. The two opposite arcs offer a round trip the single edge uv does not. This is why the extremal digraphs (directed graphs) are deliberately *lopsided*, with out-degree and in-degree pulled apart in a way no symmetric object can imitate, and why the Gomory-Hu tree that settles every undirected case has no directed analogue (Appendix A). The one-way variants therefore stay distinct in both difficulty and method.

A digraph carries three degree counts at each vertex. The *out-degree* $d^+(v)$ counts the arcs leaving v , and the vertices they reach are its *out-neighbours*. The *in-degree* $d^-(v)$ counts the arcs entering v , and the vertices they come from are its *in-neighbours*. The *total degree* $d(v) = d^+(v) + d^-(v)$ adds the two. A vertex with high out-degree fans arcs outward, one with high in-degree gathers them, and keeping the two balanced is the structural idea behind several of the directed constructions to come. The directed measure itself is easiest to meet on one concrete digraph. Figure 1.5 shows a digraph carrying three arc-disjoint routes between one chosen pair of vertices. The directed form of Menger’s theorem says that route count equals the size of the smallest arc-cut separating the pair, and the figure also marks the out-degree and in-degree that bound both quantities from above, which is exactly how Menger reads in a one-way network.

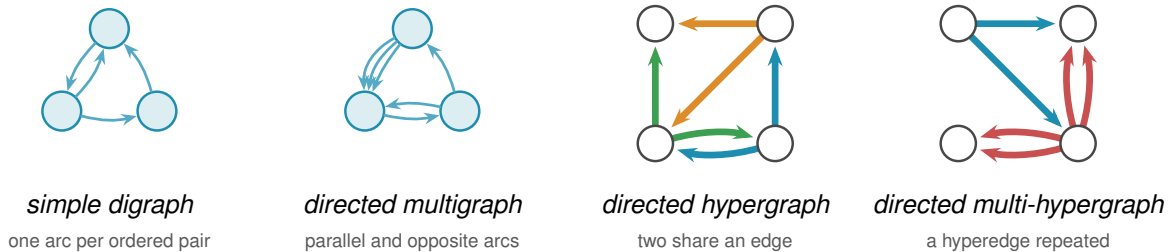


Figure 1.4. The four models of Figure 1.3 redrawn *directed*, the second axis applied to the first. Every edge becomes an arc, and a pair may now carry arcs both ways, the round trip an undirected edge cannot offer. The *simple digraph* still allows at most one arc per ordered pair, so the two-cycle on the top pair is simple. The *directed multigraph* adds parallel arcs, here a triple one way and a bidirected pair. The *directed hypergraph* draws the same three hyperedges as Figure 1.3, in the same colours, each now a metro-styled star, the tail sending one thick coloured arc to each of its heads, the star convention of Figure A.7. Taking each star’s tail to be the middle station of the undirected line keeps every arc on the same edge as before. The blue and green hyperedges still share the bottom edge, now one arc running each way, the directed counterpart of two metro lines sharing a stretch of track. The *directed multi-hypergraph* repeats the red hyperedge, so each of its arcs becomes a parallel pair. This last panel is the directed multi-hypergraph at the top of the model lattice, the most general object in the thesis.

The third axis is the *separation*: routes may be required to avoid sharing an edge, as above, or the stricter *internally vertex-disjoint*, sharing no intermediate vertex, which gives the local vertex-connectivity $\kappa_G(u, v)$ and its maximum $\kappa_G^{\max}$. Whitney’s inequality $\kappa_G(u, v) \leq \lambda_G(u, v)$ [5] records that avoiding shared vertices is at least as demanding as avoiding shared edges, and

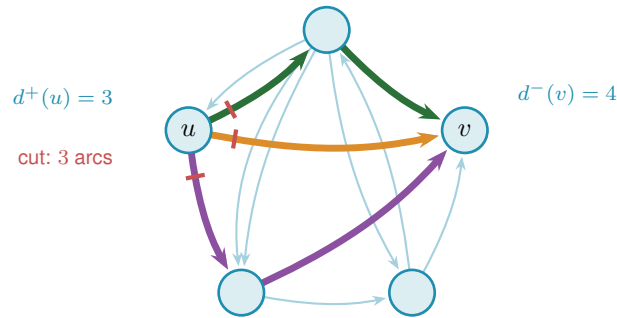


Figure 1.5. A directed multigraph on the same five vertices as Figure 1.1, drawn in the same pentagon. It is not that graph with arrows added: the pair $\{0, 4\}$ is joined here and the pair $\{0, 2\}$ carries two parallel arcs, so this is a directed multigraph rather than a simple graph. Three arc-disjoint directed routes connect u to v (orange, green, purple). The three red marks cut the arcs leaving u , the first arc of each route, so $\lambda(u, v) = 3$ by Menger's theorem. A directed cut counts only the arcs leaving u 's side, so the lone arc entering u from the top is not part of it. The out-degree $d^+(u) = 3$ and in-degree $d^-(v) = 4$ bound the value from above, and $\lambda(u, v) = \min(3, 4) = 3$ meets the smaller one, at u .

Figure 1.6 shows the gap on a graph where two routes are edge-disjoint yet forced through one common vertex.

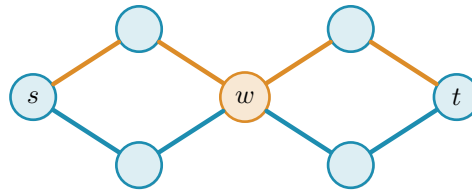


Figure 1.6. The third axis, the *separation*. The two s - t routes drawn here, one orange and one blue, share no edge, so they count as two edge-disjoint paths and $\lambda(s, t) = 2$. Both pass through the single vertex w , however, so they are *not* internally vertex-disjoint, and since every s - t route must cross w we have $\kappa(s, t) = 1$. This is Whitney's inequality $\kappa \leq \lambda$ made concrete: the gap between the two separations is exactly what the vertex-disjoint problem measures, and the rest of the chapter shows it stays shut until $m = 5$.

Three models, two directions, two separations: twelve versions of one question. The thesis is the attempt to answer as many of them as honest mathematics allows, and to build a machine for the rest. Some answers are short and we prove them here. Some answers are long, and their length is the reason Chapter 2 exists. The rest of this chapter collects the variants whose proofs are short enough to belong in the body, so that the reader meets the easy slope before the climb. Figure 1.7 draws the whole space as one tree, so that the reader has a single map to return to as the chapters visit its branches one at a time.

1.4 Cases with short proofs

Allowing parallel edges removes the parity subtlety, the dependence on whether $m(n-1)$ is even or odd, that produces the floor in Mader's theorem, and the answer becomes exactly linear. Write $L_m(n)$ for the multigraph analogue of $\ell_m(n)$.

Theorem 1.3. *For undirected multigraphs on n vertices, $L_m(n) = (m-1)(n-1)$.*

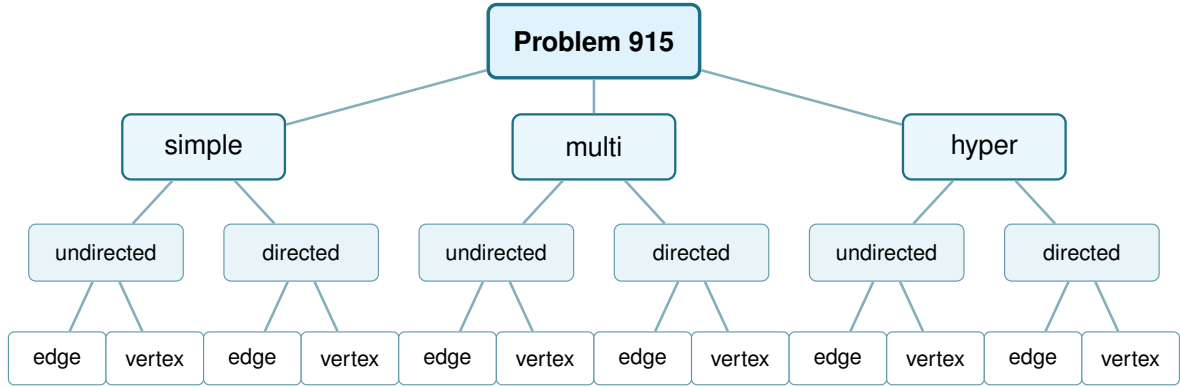


Figure 1.7. The whole space this thesis studies, drawn as one tree. Three model choices (simple graph, multigraph, r -uniform hypergraph), each splitting into undirected or directed, each in turn splitting into edge-disjoint or internally vertex-disjoint routes: twelve leaves, twelve variants. [Figure 4.7](#) redraws this exact tree once every leaf's status is known.

Proof. For the lower bound, take a spanning tree of K_n , that is, a connected cycle-free subgraph that touches all n vertices and so has exactly $n - 1$ edges, and give every tree edge multiplicity $m - 1$, meaning $m - 1$ parallel copies. The result has $(m - 1)(n - 1)$ edges. Any two vertices are joined by a unique path in the tree, of some length $k \geq 1$. To separate them one must cut some edge of that path, and the cheapest choice is the lightest tree edge on it, which still carries $m - 1$ copies, so $\lambda(u, v) = m - 1$ for every pair and $\lambda^{\max} = m - 1$. The graph sits right at the ceiling.

For the upper bound, suppose $\lambda_G^{\max} \leq m - 1$ and build a Gomory–Hu tree T of G ([Theorem A.2](#)), the same $(n - 1)$ -edge summary of all connectivities used for Mader's theorem. Each tree edge e stands for a minimum cut of G , of capacity $c(e) = \lambda_G(u, v) \leq m - 1$ for the pair (u, v) it represents. Now charge each edge of G to a tree edge: an edge $\{x, y\}$ crosses the cut of the lightest tree edge on the tree path from x to y , so it is counted inside that tree edge's capacity. Summing the $n - 1$ capacities therefore counts every edge of G at least once, which already gives $|E(G)| \leq \sum_{e \in T} c(e) \leq (m - 1)(n - 1)$. Unlike Mader's simple-graph count there is no second charge to halve, because parallel edges are now allowed and nothing forces an edge to be counted twice. The tree-of-multiplicity- $(m - 1)$ construction above meets this bound exactly, which is why the answer is the clean product $(m - 1)(n - 1)$ and not a floor. $\square$

The vertex-disjoint question is just as quick at the first non-trivial value. Forbidding two internally vertex-disjoint paths is a strong constraint: it forbids cycles.

Proposition 1.4. *For simple graphs, $k_2(n) = n - 1$, attained exactly by trees.*

Proof. If G contains a cycle, a closed loop of distinct vertices, then any two vertices on that cycle are joined by the two paths of the cycle, which are internally vertex-disjoint, so $\kappa_G^{\max} \geq 2$. Hence $\kappa_G^{\max} \leq 1$ forces G to be acyclic, free of cycles, and an acyclic graph on n vertices has at most $n - 1$ edges, with equality for trees. A tree indeed has $\kappa^{\max} = 1$, since removing the single intermediate vertex on any path disconnects its endpoints. $\square$

Here $k_m(n)$ denotes the vertex-disjoint extremal function, the counterpart of $\ell_m(n)$ for $\kappa^{\max}$. We have just seen $k_2(n) = n - 1 = \ell_2(n)$. The agreement of the edge and vertex problems is no accident at small m , and the moment it breaks is the first place the proofs begin to strain.

1.5 The first divergence: vertex-disjoint paths

Replacing edge-disjoint by internally vertex-disjoint routes changes nothing at first. Leonard proved that the edge and vertex problems give the same answer while m stays small.

Theorem 1.5 (Leonard [6]). *For simple graphs, all $n \geq m$ and $m \leq 4$,*

$$k_m(n) = \ell_m(n) = \left\lfloor \frac{m(n-1)}{2} \right\rfloor.$$

The hypothesis $n \geq m$ is the same one Mader's theorem carries, and it is not decoration. Below it the complete graph K_n is itself feasible, since $\kappa_{K_n}^{\max} = \lambda_{K_n}^{\max} = n - 1 \leq m - 1$, so the answer is the trivial $\binom{n}{2}$ and the formula overshoots it. At $n = 3$, $m = 4$ the formula gives 4 where a graph on three vertices has only 3 edges. Every closed form in this thesis carries the same caveat, stated or inherited from its constructions, and the figures cap each curve at the trivial maximum for exactly this reason.

For $m \leq 4$ the extremal graphs for the two problems coincide: Whitney's inequality $\kappa \leq \lambda$ is tight where it matters and the harder vertex constraint costs nothing (Figure 1.9, the three overlapping blue curves). One is tempted to expect this forever. It fails at the very next value, and the failure is sharp.

Theorem 1.6 (Sørensen-Thomassen [7]). *For simple graphs and all $n \geq 6$ with $n \neq 7$ and $n \neq 12$,*

$$k_5(n) = \left\lfloor \frac{8n}{3} \right\rfloor - 4,$$

while $k_5(7) = 15$ and $k_5(12) = 27$. The formula first exceeds the edge value at $n = 14$: for $6 \leq n \leq 13$ one has $k_5(n) = \left\lfloor \frac{5(n-1)}{2} \right\rfloor = \ell_5(n)$, and for every $n \geq 14$,

$$k_5(n) > \left\lfloor \frac{5(n-1)}{2} \right\rfloor = \ell_5(n).$$

Two points about how this is stated. The constant is -4 rather than the -3 printed in [7], because that paper counts the other way round, and Remark 1.7 does the conversion. The two exceptional sizes are genuine holes in the closed form rather than an artefact of the proof: at $n = 7$ it predicts 14 against the true 15, and at $n = 12$ it predicts 28 against the true 27, which is why Sørensen and Thomassen exclude both and give those two values separately. The divergence point matters for Chapter 4, which reads this theorem as the first place the edge and vertex problems part company. They agree throughout $6 \leq n \leq 13$, where the paper proves $k_5(n) = \lfloor 5(n-1)/2 \rfloor$ outright, and they separate from $n = 14$ onwards, where $k_5(14) = 33$

against $\ell_5(14) = 32$.

Remark 1.7 (A counting convention that shifts every quoted constant by one). The additive constant above needs a word of explanation, because two conventions are in circulation and they differ by exactly one. This thesis defines $\ell_m(n)$ and $k_m(n)$ as the *largest* number of edges a graph can carry while *no* pair is joined by m independent routes. Sørensen and Thomassen, and the Erdős Problems database [3] after them, state the same results the other way round, as the *smallest* number of edges that *forces* such a pair. Those two quantities are adjacent integers by construction: if k is the largest feasible edge count then no graph on n vertices with $k + 1$ edges is feasible, so $k + 1$ is exactly the forcing threshold.

The primary source settles which is which. [7] defines $f_k(n)$ as “the least integer r so that every graph with n vertices and r or more edges contains a k -rail”, which is the forcing convention, and its Theorem 4 reads $f_5(n) = \lfloor 8n/3 \rfloor - 3$. Subtracting one gives the $\lfloor 8n/3 \rfloor - 4$ of Theorem 1.6. A k -rail there is a union of k paths pairwise meeting only at their two common endvertices, so it is precisely a pair at vertex-connectivity k , and the two problems are the same problem.

The shift is visible in every entry the database and this thesis both state. It gives $k_2(n) = n$ where a tree has $n - 1$ edges, $k_3(2n) = 3n - 1$ against $\lfloor 3(2n - 1)/2 \rfloor = 3n - 2$ here, $k_4(n) = 2n - 1$ against $2n - 2$, $\ell_5(2n) = 5n - 2$ against $5n - 3$, $\ell_6(n) = 3n - 2$ against $3n - 3$, and it writes the Bollobás-Erdős conjecture as $1 + \binom{m}{2}n$ where the convention here gives $\binom{m}{2}n$. Every one is exactly one apart, and the values in this thesis are the ones an exhaustive search returns: $k_3(4) = 4$, $k_3(5) = 6$ and $k_4(5) = 8$ were each confirmed by walking every graph of that size.

One trap is worth naming, because an earlier draft of this thesis fell into it. The comparison $k_5 > \ell_5$ is invariant under the shift, since *both* functions move by one, so no internal check can detect a half-converted statement. What does detect it is evaluating the two sides in different conventions, which manufactures a spurious divergence one size early. Reading $\lfloor 8 \cdot 13/3 \rfloor - 3 = 31$ against this thesis’s $\ell_5(13) = 30$ appears to separate the two problems at $n = 13$, but the 31 is a forcing count and the 30 is an avoiding count. Done consistently, $k_5(13) = 30 = \ell_5(13)$ and the separation begins at $n = 14$, which is also what [7] proves directly by showing $f_5(n) = \lfloor 5(n - 1)/2 \rfloor + 1$ throughout $6 \leq n \leq 13$.

The range likewise comes from the paper rather than from the database, which records only the clean tail $n \geq 13$. Theorem 4 there covers every $n \geq 6$ apart from $n = 7$ and $n = 12$, and supplies those two values separately, which is where the $k_5(7) = 15$ and $k_5(12) = 27$ of Theorem 1.6 come from.

At $m = 5$ the vertex problem genuinely diverges from the edge problem, visible in Figure 1.9 as the shaded gap that opens between the dashed blue and solid orange curves. Leonard had already found the first crack, an explicit 57-vertex, 141-edge counterexample to the general conjecture at $m = 5$ [8], before Sørensen and Thomassen pinned the exact value down. The reason the crack opens at all is structural. Leonard’s earlier method, the one that closes $m \leq 4$, leans on the extremal graphs being assembled from cliques, vertex sets joined in all pairs, glued

along shared cliques. A k -tree is built up one vertex at a time: start from a complete graph on $k + 1$ vertices, then again and again add a single new vertex and join it to all k vertices of some clique of size k that is already there. Figure 1.8 grows a 2-tree, hanging each new vertex on an edge already present. These clique scaffolds are exactly the building blocks the method uses. Through $m = 4$ those scaffolds are forced, and counting their edges gives the bound.

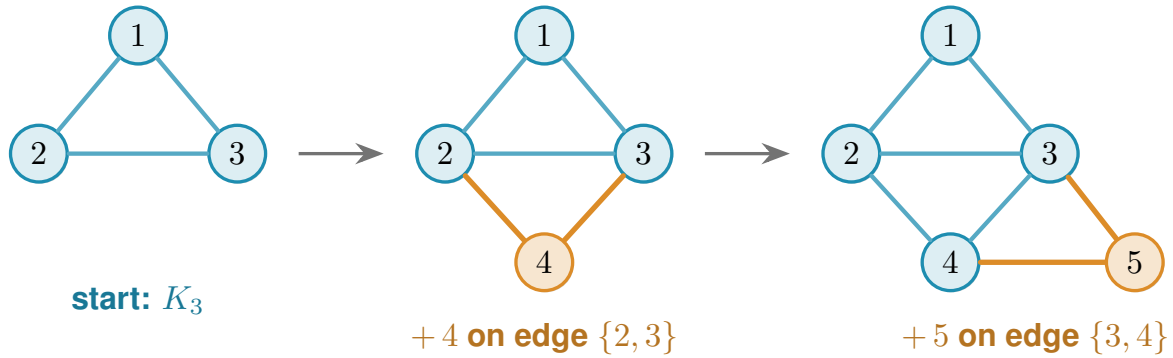


Figure 1.8. A 2-tree grown one vertex at a time, read left to right. The blue triangle on $\{1, 2, 3\}$ is the starting K_3 . Vertex 4 is then joined to the two ends of the edge $\{2, 3\}$, and vertex 5 to the two ends of the edge $\{3, 4\}$ (the new vertex and its two orange edges at each step), each new vertex hung on an edge already present. A general k -tree follows the same recipe with K_{k+1} in place of the triangle and a k -clique in place of the edge. These are the clique scaffolds Leonard’s count rests on.

At $m = 5$ they are no longer the unique extremal shape. Additional configurations appear, and the clean count breaks. Sørensen and Thomassen recover the exact value by a delicate structural induction, which we cite rather than reproduce. For $m \geq 6$ even that much is unknown: the vertex-disjoint problem has stood open since 1974, and the extremal structure is genuinely not understood. It is the one direction this thesis does not attempt to force, and Chapter 4 explains why.

1.6 The directed case and the quadratic phenomenon

Direction breaks a symmetry that undirected graphs never had. In an undirected graph the number of edges is tied to the connectivity through every vertex’s degree. In a digraph a vertex can have many arcs in and none out, and a whole graph can lean entirely one way. Send every arc from one half of the vertices to the other and nothing comes back: between any two vertices there is at most one route, yet the number of arcs is quadratic.

Construction 1.8 (One-directional bipartite digraph). Split V into parts A and B with $|A| = \lfloor n/2 \rfloor$ and $|B| = \lceil n/2 \rceil$, and include every arc from A to B . All arcs run from A to B , and none within A , within B , or from B back to A . Since B has no outgoing arcs, a vertex in A cannot reach any other vertex of A , and two vertices of B cannot communicate at all. For any $a \in A$ and $b \in B$ the only directed path from a to b is the single arc $a \rightarrow b$, so $\lambda(a, b) = 1$ and $\lambda^{\max} = 1$. The arc count is $|A| \cdot |B| = \lfloor n^2/4 \rfloor$.

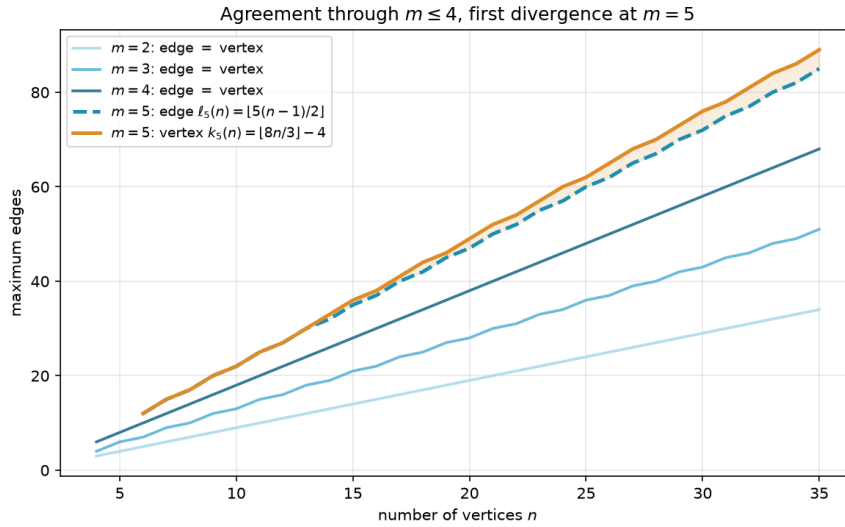


Figure 1.9. The edge and vertex extremal functions for $m = 2, 3, 4$ (blue shades, single curve per m since they agree) and $m = 5$ (dashed blue for edge, solid orange for vertex). Through $m = 4$ the two problems give identical answers, and at $m = 5$ the vertex formula $k_5(n) = \lfloor 8n/3 \rfloor - 4$ separates above the edge formula $\ell_5(n) = \lfloor 5(n-1)/2 \rfloor$, with the gap shaded. The two curves still touch at every $n \leq 13$ and part for good at $n = 14$, so the shading begins there.

Figure 1.10 draws the construction. Every arrowhead lands in B and none leaves it, so the picture is a wall of one-way traffic: from any a on the left there is exactly one way to reach any b on the right, the single arc between them, and there is no way at all to travel between two vertices on the same side. The arc count $|A| \cdot |B|$ is quadratic, yet no pair carries more than one route. This is the lopsidedness direction buys, and it has no undirected analogue. It competes with a second construction built around a single centre.

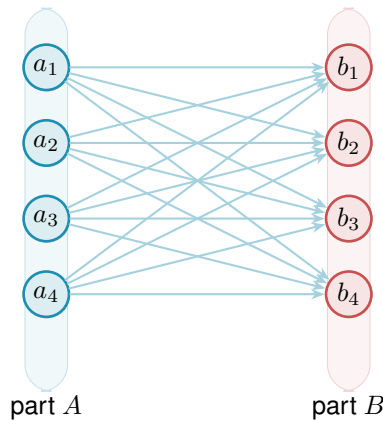


Figure 1.10. The one-directional complete bipartite digraph $B_{k,k}$ of Construction 1.8, drawn here at $n = 8$ with $|A| = |B| = 4$. Every one of the $|A| \cdot |B| = \lfloor n^2/4 \rfloor$ arcs runs left to right, from A to B . There are no arcs inside A , inside B , or back from B to A . Because B has no outgoing arcs and A no incoming ones, the only route from any a_i to any b_j is the single arc $a_i \rightarrow b_j$, so every pair has local edge-connectivity 1 while the arc count grows quadratically.

That second construction is the directed hub. It keeps each vertex's in-degree and out-degree balanced, the directed measure already met in Figure 1.5, and spends its arcs on a central vertex

rather than a one-way wall.

Construction 1.9 (Directed hub). For $n \geq m \geq 2$, take a hub vertex h and label the remaining $N = n - 1$ vertices $0, 1, \dots, N - 1$. Add the $2N$ hub arcs $h \rightarrow i$ and $i \rightarrow h$ for every i , and on the non-hub vertices add the circulant arcs $i \rightarrow i + j \pmod{N}$ for $j = 1, \dots, m - 2$, the target label wrapping around past N back to 0 (that ring pattern is what *circulant* means, and “mod N ” is the wrap-around). Counting these gives $2N$ hub arcs (each of the N spokes joined to h in both directions) plus $(m - 2)N$ circulant arcs ($m - 2$ layers, one arc per spoke), so the total is $2N + (m - 2)N = mN = m(n - 1)$ arcs. Every non-hub vertex has $d^+ = d^- = 1 + (m - 2) = m - 1$. Every ordered pair (u, v) has at least one non-hub member, and that member’s relevant degree is exactly $m - 1$, smaller than the hub’s own degree $n - 1$ since $n \geq m$, so it is always the binding one regardless of which of u, v is the hub. The directed degree bound (Lemma A.3, which says $\lambda(u, v) \leq \min(d^+(u), d^-(v))$, since a route out of u must use one of u ’s out-arcs and one of v ’s in-arcs) therefore gives $\lambda^{\max} \leq m - 1$.

At $m = 2$ this construction is easiest to picture: the circulant layer is empty and what remains is the *bidirected star* of Figure 1.11, a hub joined to each spoke by a matched pair of opposite arcs. Every spoke can reach every other only by going in to the hub and back out, two hops, so no pair is ever joined by two independent routes, and the arc total is exactly $2(n - 1)$. For $m \geq 3$ not all $m(n - 1)$ arcs can touch the hub, since a simple digraph has at most $2(n - 1)$ arcs incident to one vertex: the surplus lives in the circulant layers among the spokes, tuned so that no degree exceeds the budget. The two constructions compete on different scales, linear against quadratic, and which one wins depends on n . At $m = 2$ the competition resolves exactly.

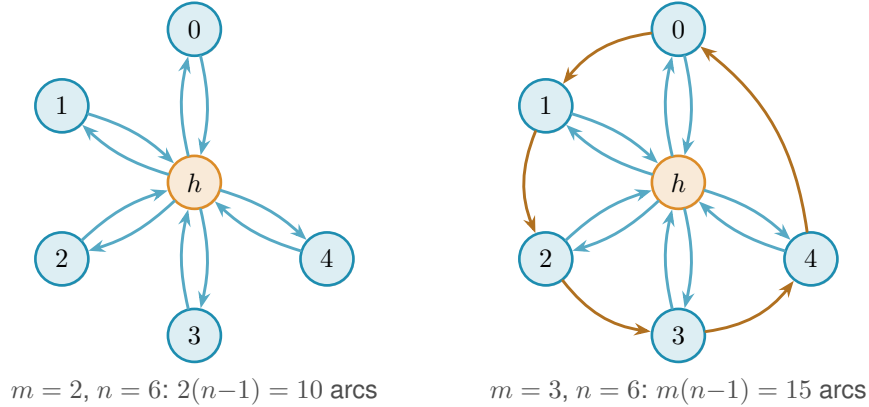


Figure 1.11. The directed hub at $n = 6$ for two values of m . *Left* ($m = 2$): the bidirected star. The hub h is joined to each spoke by one arc in each direction, giving $2(n - 1) = 10$ arcs. No circulant layer is added, so all spoke communication routes through h , so $\lambda^{\max} = 1$. *Right* ($m = 3$): a single circulant layer is added among the five spokes. The blue arcs are the same hub-spoke pairs as on the left. The orange arcs are that one circulant layer, the cycle $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 0$, so each spoke has $d^+ = d^- = m - 1 = 2$. The total is $m(n - 1) = 15$ arcs and $\lambda^{\max} = 2$. For $m \geq 3$ the spokes are no longer isolated from one another, and tracing all the routes becomes the argument that fills the rest of the chapter.

Theorem 1.10 (Directed arc value at $m = 2$). *For simple digraphs,*

$$\ell_2^{\text{dir}}(n) = \max\left(2(n-1), \left\lfloor \frac{n^2}{4} \right\rfloor\right).$$

The hub construction leads for $n < 7$, the one-directional bipartite construction leads for $n > 7$, and the two tie at $n = 7$, where both reach 12 arcs.

The lower bound is the better of the two constructions. The upper bound is where the trouble starts.

Idea of the upper bound at $m = 2$. The value $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$ is a contest between a linear term and a quadratic one, and which leads is pure arithmetic: $2(n-1) \geq \lfloor n^2/4 \rfloor$ precisely for $n \leq 7$, with equality at $n = 7$, where both give 12. So M follows the bidirected star up to the crossover and the bipartite digraph past it, and $n = 7$ is the single value where the two constructions tie. The proof that no digraph beats $M(n)$ is an induction on n (Appendix A). Write $a = |A(D)|$ for the number of arcs and delete a vertex v of *minimum* total degree. Two facts about that deletion meet. The degrees sum to twice the arc count, so the smallest of them is at most the average,

$$\deg(v) \leq \frac{2a}{n},$$

and the remaining digraph on $n-1$ vertices still respects the ceiling, so once $n-1 \geq 7$ the inductive bound applies to it,

$$|A(D-v)| \leq \left\lfloor \frac{(n-1)^2}{4} \right\rfloor.$$

The arc count of D itself splits as $a = |A(D-v)| + \deg(v)$, the arcs away from the deleted vertex plus the arcs at it. Feeding both facts into that split leaves a copy of a on each side, and gathering them before dividing by $1 - 2/n$ finishes the step:

$$\begin{aligned} a &\leq \left\lfloor \frac{(n-1)^2}{4} \right\rfloor + \frac{2a}{n}, \\ a\left(1 - \frac{2}{n}\right) &\leq \left\lfloor \frac{(n-1)^2}{4} \right\rfloor, \\ a &\leq \frac{n}{n-2} \left\lfloor \frac{(n-1)^2}{4} \right\rfloor. \end{aligned}$$

A short parity check then shows the right-hand side never exceeds $\lfloor n^2/4 \rfloor$ once $n \geq 8$: for even n it lands exactly on the bipartite value, and for odd n a sub-unit slack is swallowed by the fact that a is an integer. The averaging closes the step with no separate hub case, which is why the difficulty is not depth but bulk. Nothing in the argument is deep. It is a careful walk through two parity regimes, with the base cases $n \leq 7$ settled by exhaustive computer search rather than by hand. That experience of a correct but long and almost entirely mechanical case-check is exactly what motivates the chapters that follow. One genuinely wants to hand the case-checking to a machine.

The vertex-disjoint version of the same question has the same answer at $m = 2$, but it is worth being careful about why, because Whitney's inequality is easy to read backwards here. Since $\kappa \leq \lambda$ pairwise, a digraph that respects the arc ceiling automatically respects the vertex one, so the vertex-feasible family *contains* the arc-feasible one and is in general strictly larger. The free consequence therefore runs one way only: the two constructions above keep $\kappa^{\max} = 1$ as readily as $\lambda^{\max} = 1$, which makes them lower bounds for the vertex problem too. No upper bound comes with them. What settles the vertex value is that the same induction goes through unchanged, because deleting a vertex lowers $\kappa^{\max}$ for exactly the reason it lowers $\lambda^{\max}$, and because the base cases $n \leq 7$ were re-run under the stricter separation and returned the same six numbers. [Appendix A](#) gives both halves, and [Remark A.16](#) there spells out the direction trap in full.

Theorem 1.11 (Directed vertex value at $m = 2$). *For simple digraphs,*

$$k_2^{\text{dir}}(n) = \max\left(2(n-1), \left\lfloor \frac{n^2}{4} \right\rfloor\right) = \ell_2^{\text{dir}}(n).$$

1.7 The failure of direct proof for $m \geq 3$

For $m \geq 3$ the induction no longer closes, and the natural guess is wrong. The natural initial conjecture was that the two constructions above continued to be the whole story, giving $\max(m(n-1), \lfloor n^2/4 \rfloor)$. A single graph on ten vertices refutes it.

Remark 1.12 (A thirty-arc counterexample). Take $A = \{a_1, \dots, a_5\}$ and $B = \{b_1, \dots, b_5\}$ with all 25 arcs from A to B , and add a directed five-cycle $b_1 \rightarrow b_2 \rightarrow b_3 \rightarrow b_4 \rightarrow b_5 \rightarrow b_1$ inside B . This digraph has 30 arcs. Every vertex of B has exactly one extra in-arc from inside B , so two arc-disjoint routes from a_i to b_j exist: the direct arc $a_i \rightarrow b_j$, and the detour $a_i \rightarrow b_{j-1} \rightarrow b_j$ using the 5-cycle predecessor of b_j (indices mod 5). For the matching upper bound, the only in-arcs of b_j reachable on a path from a_i are the direct arc $a_i \rightarrow b_j$ and the cycle arc $b_{j-1} \rightarrow b_j$. Removing both disconnects a_i from b_j , while neither alone does, so the minimum cut has size 2. Hence $\lambda(a_i, b_j) = 2$. Routes between two vertices of B never leave B and give $\lambda(b_i, b_j) = 1$. Hence $\lambda^{\max} = 2$, so the graph is feasible at $m = 3$, yet $30 > \max(3 \cdot 9, 25) = 27$.

[Figure 1.12](#) draws this thirty-arc digraph and traces the mechanism. The twenty-five faint arcs are the complete one-directional bipartite layer from A to B , the construction we already understand. On top of them the orange five-cycle gives each vertex of B a single extra in-arc from inside B . That one extra arc is exactly what doubles the connectivity. For the marked pair a_1, b_2 the two heavy routes show how: the blue arc is the direct hop $a_1 \rightarrow b_2$, and the green path $a_1 \rightarrow b_1 \rightarrow b_2$ borrows the cycle arc $b_1 \rightarrow b_2$ for its second step. The two share no arc, so $\lambda(a_1, b_2) = 2$, and the same pair of routes exists for every a_i, b_j . Five extra arcs lift the whole digraph from one independent route per pair to two, and the thirty arcs overshoot the old prediction of twenty-seven.

The counterexample generalises: the five-cycle gave every vertex of B exactly one extra in-

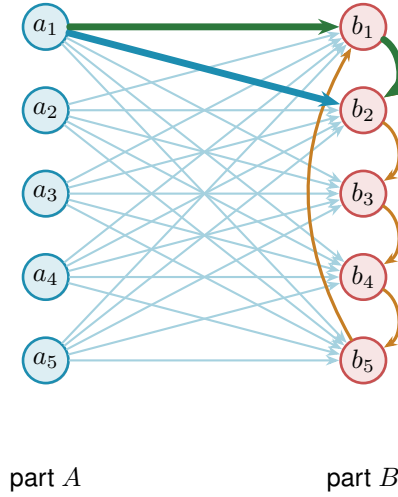


Figure 1.12. The thirty-arc counterexample of Remark 1.12: the augmented bipartite digraph (Construction 1.13) at $m = 3$, $n = 10$, $|A| = |B| = 5$. The 25 faint arcs are the complete one-directional layer from A to B , and the 5 orange arcs form the directed cycle $b_1 \rightarrow b_2 \rightarrow b_3 \rightarrow b_4 \rightarrow b_5 \rightarrow b_1$ inside B , one extra in-arc per vertex of B . The heavy blue arc and the heavy green detour show the pair a_1, b_2 carrying 2 arc-disjoint routes: the direct arc $a_1 \rightarrow b_2$ (blue) and the two-step detour $a_1 \rightarrow b_1 \rightarrow b_2$ (green) through the cycle predecessor b_1 . Hence $\lambda^{\max} = 2$, feasible at $m = 3$, while $30 > \max(3 \cdot 9, 25) = 27$.

arc from inside B , and the budget for such arcs is $m - 2$ per vertex, not one. Figure 1.12 is the instance $m = 3$, where that budget is one and a single cycle suffices. For larger m each b_j collects $m - 2$ cyclic predecessors and the same picture acquires more concentric chords inside B .

Construction 1.13 (Augmented bipartite digraph). For $m \geq 2$ and $n \geq m$, set $|B| = \lceil (n + m - 2)/2 \rceil$ and $|A| = n - |B| = \lfloor (n - m + 2)/2 \rfloor$, include every arc from A to B , and inside B give each vertex b_j the $m - 2$ in-arcs from its cyclic predecessors $b_{j-1}, \dots, b_{j-(m-2)}$ (indices mod $|B|$). The total is $\lfloor (n + m - 2)^2/4 \rfloor$ arcs.

For the connectivity, fix $a \in A$ and $b \in B$: the direct arc and the $m - 2$ two-step detours $a \rightarrow b' \rightarrow b$ through the in-neighbours b' of b inside B are $m - 1$ arc-disjoint routes, and they are all there are, because the only in-arcs of b reachable from a are exactly those $m - 1$ arcs, which therefore form a cut. Pairs inside B are limited by the in-degree $m - 2$, pairs inside A and pairs from B to A have no routes at all, so $\lambda^{\max} = m - 1$.

At $m = 2$ the inside of B is empty and the construction reduces to Construction 1.8, with the balanced partition $(\lfloor n/2 \rfloor, \lceil n/2 \rceil)$. At $m = 3$ the formula gives $|B| = \lceil (n + 1)/2 \rceil$, which coincides with $\lceil n/2 \rceil$ at odd n but gives $|B| = n/2 + 1$ at even n . Both the balanced and the shifted partition achieve the same arc count $\lfloor n^2/4 \rfloor + \lceil n/2 \rceil$ in this case, so the two are interchangeable at $m = 3$. The shift to a strictly larger B is structurally significant only at $m \geq 4$, where the balanced partition is suboptimal. At $m = 3$, $n = 10$ it is the thirty-arc counterexample of Remark 1.12, drawn in Figure 1.12 (using the equivalent balanced partition $|A| = |B| = 5$).

The two rival shapes invite a political picture. The hub is a single king through whom all

traffic must pass, cheap to build but limited by the one throne. The bipartite family is a flat cabinet of ministers, each handling its own dossier with no central seat, and it scales quadratically. Which arrangement packs more arcs under the ceiling depends on n , and for $m \geq 3$ the cabinet overtakes the king once n is large. This is the corrected shape, and it leads to the standing conjecture for the directed arc problem.

Conjecture 1.14. *For simple digraphs and all $n \geq m \geq 2$,*

$$\ell_m^{\text{dir}}(n) = \max\left(m(n-1), \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor\right).$$

The hypothesis $n \geq m$ is the one both constructions already carry, and it is needed for the same reason as in [Theorem 1.5](#): below it the complete digraph is feasible and the answer is the trivial $n(n-1)$, which the formula exceeds. At $n = 3, m = 4$ the formula gives 8 where three vertices carry only 6 arcs. The exhaustive search confirms the value is 6 there.

For $m = 2$ this is [Theorem 1.10](#). For $m = 3$ the formula $\lfloor (n+1)^2/4 \rfloor$ equals $\lfloor n^2/4 \rfloor + \lceil n/2 \rceil$, so the conjecture reduces to the previous formula for $m = 2$ and $m = 3$. For $m = 3, n = 10$ the second branch gives $\lfloor 121/4 \rfloor = 30$, matching [Remark 1.12](#). For $m \geq 4$ the new formula and the old balanced-partition count $\lfloor n^2/4 \rfloor + (m-2) \lceil n/2 \rceil$ no longer agree: the quadratic branches already differ (for $m = 4$ the new one is larger by exactly one at every even n), but the hub term $m(n-1)$ dominates both at small n , so the full conjectured value first overtakes the old one at $n = 12$ for $m = 4$. The construction of [Construction 1.13](#) with the shifted partition witnesses the difference. The lower bound holds for all m , witnessed by [Construction 1.9](#) on the first branch and [Construction 1.13](#) on the second. The matching upper bound is open, though not entirely: [Theorem A.24](#) proves unconditionally that the value is $n^2/4 + \Theta_m(n)$, so what the conjecture still claims beyond what is known is the coefficient of that linear term rather than the shape of the answer. [Chapter 4](#) returns to it with what structural progress the machine and the hand can jointly supply.

1.8 The hypergraph variant

The third model deserves its own short treatment, not because its base case is hard but because it is the one variant whose objects are stored and manipulated differently from all the rest, and it is worth meeting that difference now rather than at the moment of computation. A hypergraph keeps no multiplicity matrix, only the list of its hyperedges, each a set of r vertices. We fix the edge size r , following the extremal literature, for a structural reason: under a connectivity ceiling a small edge is always at least as efficient per route as a large one, so a hypergraph free to mix sizes would maximise its edge count by using pairs only, and the question would collapse back to the graph case. Fixing r is what makes the hypergraph question a new question.

The route between two vertices is a *Berge path*, which alternates vertices and hyperedges so that each consecutive pair of vertices lies in the hyperedge listed between them, and two such

routes are edge-disjoint when they share no hyperedge. Concretely, in a three-uniform hypergraph a route from u to w might be the single step $u, \{u, x, w\}, w$ that crosses one hyperedge, or a longer alternation $u, \{u, x, y\}, y, \{y, z, w\}, w$ that switches hyperedge at the intermediate vertex y .

With routes redefined this way the same Gomory–Hu machinery that proved the multigraph value carries over, and the base case has the same linear shape. The picture to keep in mind is a metro map: each hyperedge is a line, and a route from one vertex to another rides a line and changes to another line at a station the two share.

A single hyperedge may serve only one route, so two Berge routes that share no hyperedge are edge-disjoint, the hypergraph counterpart of two edge-disjoint paths in a graph. Reading routes off a metro map is a matter of seeing which lines share track: two hyperedges that share several vertices run side by side along the stretch between them, as in [Figure 1.3](#), and two routes are hyperedge-disjoint exactly when the lines they ride share no track at all.

The separation axis needs its own definition here, and it is worth fixing now because the natural guess is not the one this thesis uses. Two Berge routes are called *internally disjoint* when they share neither an intermediate vertex nor a hyperedge. Both halves are required. Demanding only that the routes avoid each other’s intermediate stations would let two routes ride the same line, and a line that can run only one train at a time cannot serve two routes at once, so the resulting count would not correspond to anything the metro map can carry. Requiring both is also what makes the vertex measure the exact analogue of the graph one under the translation of [Appendix A](#), where a hypergraph becomes its incidence graph, each hyperedge becomes a node, and internally disjoint routes become internally disjoint paths in the ordinary sense. It has the pleasant side effect that Whitney’s inequality survives the move to hypergraphs by definition, since an internally disjoint family is in particular hyperedge-disjoint. Write $\kappa_{\mathcal{H}}(u, v)$ for the largest such family and $\kappa_{\mathcal{H}}^{\max}$ for its maximum over pairs.

Proposition 1.15 (Hypergraph edge bound). *An r -uniform hypergraph on n vertices with Berge edge-connectivity at most $m - 1$ has at most $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ hyperedges. A simple hypergraph attains the bound whenever $m - 1 \leq \binom{n-2}{r-2}$, and a multihypergraph attains it whenever $(r - 1) \mid n - 1$.*

The proof, given in [Appendix A](#), runs the multigraph argument through the hypergraph Gomory–Hu theorem ([Theorem A.46](#), the same tree construction of [Theorem A.2](#) applied to the hyperedge-cut function in place of the edge-cut function), charging each hyperedge to the at least $r - 1$ tree cuts it crosses. The star hypertree ([Figure 1.13](#)) attains the $m = 2$ case, a hub joined to disjoint blocks of $r - 1$ vertices, with Berge connectivity one everywhere and $\lfloor (n - 1)/(r - 1) \rfloor$ hyperedges, exactly as a tree attained the multigraph bound. One genuine surprise is worth recording here: for $r \geq 3$ the bound is attained by *simple* hypergraphs, with no repeated hyperedge ([Theorem A.47](#)). In the graph case simplicity is exactly what separates Mader’s $\lfloor m(n - 1)/2 \rfloor$ from the multigraph value $(m - 1)(n - 1)$. One edge size higher, it costs

nothing.

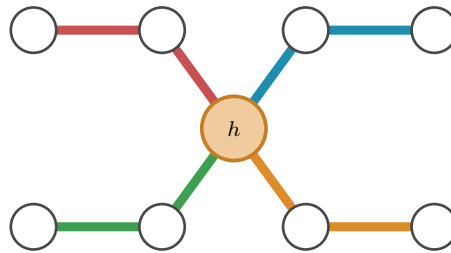


Figure 1.13. The star hypertree that attains the $m = 2$ hypergraph bound, here $r = 3$ and $n = 9$, drawn as a metro map with the hub h centred between two rows of stations. The single hub station lies on every line, and each of the $\lfloor (n-1)/(r-1) \rfloor = 4$ hyperedges is one metro line of $r = 3$ stations: it leaves the hub diagonally to an inner station and then runs horizontally to its outer station, hanging a fresh block of $r-1 = 2$ vertices off the hub. Because every line meets every other only at h , no pair has two hyperedge-disjoint routes, so the Berge connectivity is one everywhere, exactly as a spanning tree attained the multigraph bound.

The proposition counts hyperedges, but it does not say how to *measure* the Berge connectivity of a hypergraph we are handed, and that measurement is the genuinely new part. The metro picture shows why. An ordinary edge is a one-stop link between a single pair, so policing edge-disjointness means watching each pair on its own. A hyperedge is a whole line: it stops at all r of its vertices and so touches $\binom{r}{2}$ pairs at once, yet like a line that can run only one train at a time it may serve only one route. When several routes want to board the same line at different stations, the checker must wave exactly one through and turn the rest away, and it must do this for every line simultaneously. That bookkeeping is the one new piece of machinery the next chapter builds, and it is why the hypergraph rejoins the family only once the checker is in place.

1.9 Random graphs as intuition

The short proofs above hinted that the edge and vertex problems travel together for a while, agreeing through $m = 4$ before the split at $m = 5$. That separation is a statement about extremal graphs, the densest ones the ceiling allows, and it is worth asking whether the same split can be seen in ordinary graphs drawn at random, with no extremal tuning at all. The random model is the natural place to build intuition, because it shows the qualitative behaviour of the connectivity notions before any construction is pushed to its limit, and we will return to it whenever a picture of the typical graph clarifies a result. It is not a detour from the program to come, either: a sampled $G(n, p)$ is an ordinary graph, measured by the very same connectivity checker the next chapter builds for every other variant, so the random model is simply the one pipeline turned to *observe* the typical case rather than to prove or to discover the extremal one.

Each of the $\binom{n}{2}$ possible edges of $G(n, p)$ is included independently with probability p , and on the resulting graph we read off both binding connectivities, the edge version $\lambda_G^{\max}$ and the vertex version $\kappa_G^{\max}$, from the very same sample. Whitney's inequality $\kappa_G^{\max} \leq \lambda_G^{\max}$ guarantees the vertex value never exceeds the edge one, but it is silent on how often the two actually come apart.

Figure 1.14 looks directly at the distributions, fixing $n = 16$ and histogramming both connectivities over many samples at three densities. Two things are visible at once. As the density rises the whole distribution marches rightward, from a binding connectivity around five at $p = \frac{1}{4}$ to around thirteen at $p = \frac{3}{4}$, so by $n = 16$ the typical graph already sits well past the $m = 4$ at which the proofs agreed. And comparing the edge row to the vertex row, the vertex distribution sits at or just to the left of the edge one, never to its right, which is Whitney's inequality seen across a whole distribution rather than a single graph.

The two pull apart on a fraction of samples that shrinks as the density grows, from about a fifth of the samples at $p = \frac{1}{4}$ to only a handful by $p = \frac{3}{4}$. The reason is structural. For $\kappa^{\max}$ to fall below $\lambda^{\max}$ some pair must carry more edge-disjoint routes than internally vertex-disjoint ones, which happens only when several of those routes are forced through one shared intermediate vertex. In a sparse graph routes are scarce and a single low-degree vertex can be exactly such a bottleneck, but as p grows every vertex gains degree and the graph offers many alternative intermediate vertices, so independent routes almost never need to reuse one. In the dense limit the graph approaches the complete graph, where $\kappa^{\max} = \lambda^{\max} = n - 1$ for every pair and the gap closes entirely. The divergence the thesis records at $m = 5$ is therefore not an artefact of the extremal construction. It is already there, in miniature, in the random model.

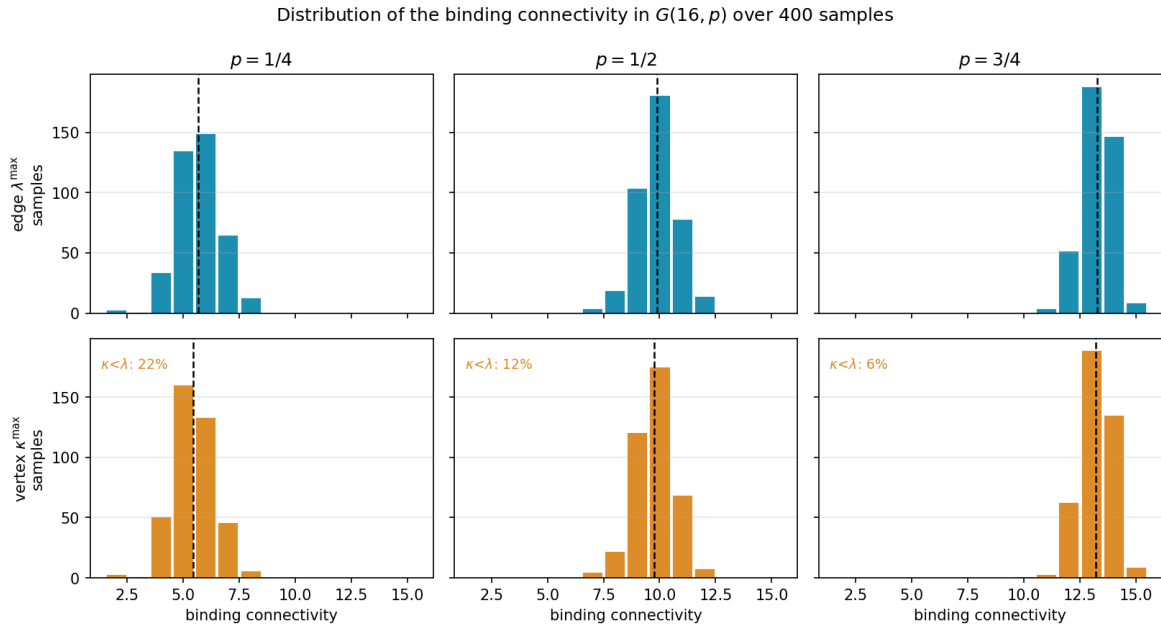


Figure 1.14. Distributions of the binding connectivity in $G(16, p)$ over 400 samples, at densities $p = \frac{1}{4}, \frac{1}{2}, \frac{3}{4}$. There are $2^{\binom{16}{2}} = 2^{120}$ labelled graphs on 16 vertices, far too many to enumerate, so these are Monte Carlo samples rather than the full population, and 400 is plenty to show the shape of each distribution. The top row histograms the edge-connectivity $\lambda^{\max}$, the bottom row the vertex-connectivity $\kappa^{\max}$, measured on the same samples, with the mean marked by a dashed line and the fraction of samples on which $\kappa^{\max} < \lambda^{\max}$ printed on each vertex panel. Reading a column top to bottom, the vertex distribution sits at or to the left of the edge one, which is Whitney's inequality, and the gap is largest at the sparsest density (22% of samples) and closes as p grows (6%). All six panels share one axis range.

The random model also supplies a landmark of a different flavour. Below a sharp density the forbidden configuration is almost surely absent, and above it almost surely present, and the location of that threshold is the same for every one of the connectivity notions above.

Theorem 1.16 (Appearance threshold). *Let $m = m(n) \geq 2$ grow with n so that $m/\ln n \rightarrow \infty$ and $m = o(n)$, and let $c > 0$ be a constant. In the binomial random graph $G(n, p)$ with $p = c \cdot m/n$,*

$$\lim_{n \rightarrow \infty} \Pr[\lambda^{\max} \geq m] = \begin{cases} 1 & \text{if } c > 1, \\ 0 & \text{if } c < 1, \end{cases}$$

so the threshold for the appearance of a pair with local edge-connectivity at least m is $p^ = m/n$. The same threshold governs local vertex-connectivity and the directed analogues ([Remark A.70](#)). It does not govern the hypergraph models, which live on a different density scale ([Remark A.71](#)).*

The proof is a concentration argument with two halves ([Appendix A](#)). Above the threshold the edge count concentrates past Mader's bound, so [Theorem 1.2](#) itself forces a high pair into existence. Below it the maximum degree stays under m , and the degree bound caps every local connectivity. The two hypotheses earn their keep. The first, $m/\ln n \rightarrow \infty$, says m must outgrow the natural logarithm of n , and that is what lets a union bound over the n vertices close, the union bound being nothing more than adding up the n individual failure chances and needing even their sum to vanish. The second, $m = o(n)$, says m stays negligible next to n , which keeps the density $p = cm/n$ below one and in the sparse regime where the threshold lives. Two caveats on scope, and the second is the one that limits how far the result may be quoted. The upper half quoted here uses Mader's bound, which is the simple undirected edge case, so it is that variant the proof settles directly. The vertex-disjoint and directed analogues sit at the same threshold $p^* = m/n$ but reach it through their own extremal bounds rather than Mader's, possibly with a different leading constant, as [Remark A.70](#) records. The hypergraph models are a different matter and are *not* covered. A threshold of this kind sits where a vertex's expected degree reaches m , and in a binomial r -uniform hypergraph a vertex lies in an expected $p \binom{n-1}{r-1}$ hyperedges, so the balance point is $m / \binom{n-1}{r-1}$, smaller than m/n by a factor of order n^{r-2} once $r \geq 3$. [Remark A.71](#) works this out and [Figure B.8](#) plots each model against its own scale. So the collapse of complexity is real but bounded: within a fixed edge size the distinctions between edges and vertices and between the two directions wash out and one density decides the matter, while changing the edge size changes the density at which everything happens.

One limitation must be stated plainly, lest the threshold be read as evidence about the problem this thesis actually studies. The growth hypothesis $m/\ln n \rightarrow \infty$ forces m to climb with n , so the theorem says nothing whatever about a fixed small m , the regime of every exact value above, where m is 2, 3, or 4 and n runs to infinity around it. There the union bound is vacuous and the appearance of a dense pair is governed by rare low-degree configurations that need finer, Poisson-type tools. The random model therefore enters as contrast and not as evidence: it shows how cleanly the question collapses in one limit, which throws into relief how stubbornly it

resists in the fixed- m limit the rest of the thesis must attack by hand and by machine.

1.10 Where this work sits

It is worth pausing, before the machine, to place the question and the method against the two traditions they descend from: the extremal graph theory that posed the problem and the computational mathematics that will help settle it.

The mathematical lineage is the older of the two. The problem belongs to the family of extremal connectivity questions opened by Bollobás and Erdős, who asked how dense a graph can be while every pair of vertices stays below a fixed connectivity [1], logged today as Erdős Problem 915 [3]. Mader’s theorem [4] closed the simple undirected edge case with the clean floor $\lfloor m(n-1)/2 \rfloor$ recalled in [Theorem 1.2](#), and it is the floor every variant in this thesis is measured against. The vertex-disjoint branch has a longer and rougher history: Leonard [6] showed the edge and vertex problems coincide through $m \leq 4$, and Sørensen and Thomassen [7] found the first divergence at $m = 5$ with $k_5(n) = \lfloor 8n/3 \rfloor - 4$, after which the exact values $k_m(n)$ remain unknown. That gap between a settled edge case and a stalled vertex case is exactly the terrain [Chapter 1](#) has been mapping, and the directed and hypergraph axes extend it further. The thesis adds to this lineage not a single new closed form but a uniform way of pushing each axis as far as exact computation allows.

The computational tradition this thesis’s pipeline belongs to is the practice of settling combinatorial questions by machine in a way a human can audit, and it splits into a few schools worth contrasting with the certify-then-prove loop built next. The oldest is *exhaustive generation*: enumerate every object of a given size up to isomorphism and check each one. McKay and Piperno’s *nauty/geng* toolkit [9] is the standard engine for this, and its flagship results include exact small Ramsey numbers such as $R(4, 5) = 25$ [10]. The enumeration of [Chapter 2](#) is the same idea, run on the twelve variants of Problem 915. The second school is *SAT-assisted proof*, in which a question is encoded in propositional logic and a solver emits a machine-checkable certificate: Heule, Kullmann and Marek’s resolution of the Boolean Pythagorean triples problem [11], whose verified proof runs to some two hundred terabytes, and Heule’s determination of Schur number five [12] are the canonical examples of certified solver output at scale. The third is *flag algebras*, Razborov’s framework [13] that converts asymptotic extremal bounds into semidefinite-programming certificates, the dominant tool for density problems in the large- n limit. Closest of all to the method here is the fourth, *certification by linear and integer programming* (LP/ILP), where a feasibility or optimality argument over the rationals or integers is written as a linear program, an optimisation with linear constraints, and the proof is the certificate of optimality the solved program hands back, a short list of numbers anyone can verify by direct arithmetic. This is precisely the form the cut-counting certifier of [Chapter 2](#) takes.

Against that backdrop, what is distinctive here is consolidation rather than a new solver. A single checker measures connectivity exactly across all twelve problem variants on one data model, so search and certification share one codebase rather than living in separate tools. The

certificates it emits are exact rational or integral objects, not floating-point estimates, so an upper bound it reports is a proof and not evidence. And because the same program both hunts for dense graphs and certifies the bounds they suggest, the discover step and the prove step are two modes of one loop, which is the arrangement the next chapter builds. What makes the consolidation honest is that it never blurs three kinds of knowledge: a value can be *proved* (by hand, or by exhausting every graph of a size), *certified* (by the same optimisation, reaching a vertex or two further), or only *conjectured* (witnessed by a dense graph the search exhibits, with no matching upper bound). This trichotomy, named in [Section 3.6](#) and obeyed by the summary of [Chapter 4](#), is the through-line of everything that follows.

Computer note

We are now stuck in three different ways: a fifty-year-old open problem at $m \geq 6$ for vertices, a long proof we would rather not check by hand at $m = 2$ for arcs, and at $m \geq 3$ a conjecture of which only the leading behaviour can be proved. The base cases of the long induction were verified by exhaustive search, and the thirty-arc counterexample was found the same way. The next chapter builds a single program that models all of these cases at once, measures their connectivity exactly, and proves the small-case upper bounds outright.

Chapter 2

Certifying Bounds by Machine

The last chapter ended in three different kinds of stuckness, and all three share a shape. The objects are graphs of one flavour or another, the constraint is always a ceiling on connectivity, and the work is always either measuring a given graph or checking many small ones. That shared shape is what a single program can exploit. This chapter builds it, and then uses it for one purpose only: to *prove*. It gives every variant one representation, measures connectivity exactly through a max-flow checker, checks small cases by honest enumeration, and turns the finite checks the proofs of [Chapter 1](#) relied on into a single auditable certifier. The hunt for dense examples, which proves nothing on its own, waits until [Chapter 3](#).

A word on what the program is for. It is a microscope, not a source of final answers. It measures connectivity exactly, so what it reports about a given graph is a fact. It enumerates graphs of a fixed size, so for that size it can prove a theorem. It will later discover dense graphs, which are only lower bounds, but not here. Each of these has a different epistemic weight, and the program is built so that the three never get confused. [Figure 2.1](#) draws this spine: a single *model* holds every variant, one exact *measure* reads connectivity off it, and from that measurement a *prove* step bounds a fixed size from above, with the *discover* step of the next chapter and a final *observe* step for the random model added later. Every routine introduced from here on is shown as a small card carrying its name, its inputs and output, a one-line account of how it works, and a coloured badge naming its place in the spine. The function bodies are not reproduced, because the point is the interface, not the code.

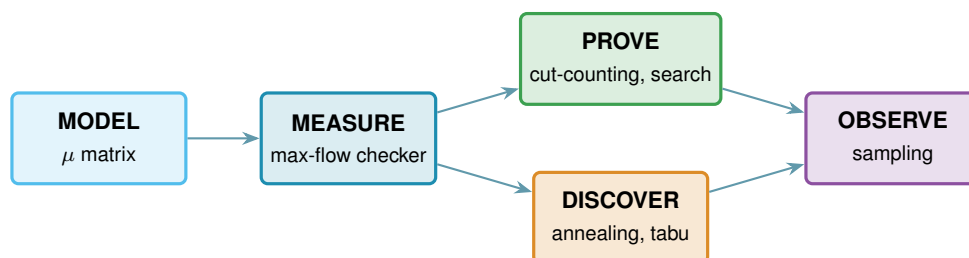


Figure 2.1. The architecture as one pipeline. A single data model, the multiplicity matrix μ , feeds an exact measurement, the max-flow checker. That measurement feeds the two bound-producing roles: a prover for upper bounds (the cut-counting optimisation and the exhaustive search), built in this chapter, and a discoverer for lower bounds (simulated annealing, or tabu search), built in [Chapter 3](#). A final sampling role studies the random model. The badge colours on the code cards below match the boxes here.

2.1 The solver

Everything in this chapter is a single function, `solve`, and it helps to hold its most basic version in mind from the start, because every later idea is a refinement of it and nothing is ever a separate program for a separate case. Given n and m , the basic strategy is as plain as it sounds. Walk through every graph on n vertices, measure the largest local connectivity $\lambda^{\max}$ of each with a max-flow computation, and keep the densest graph that stays at or below the ceiling $m - 1$. Because it has looked at every graph, the count it returns is the exact answer, not an estimate.

That one loop already settles the smallest cases, and the rest of the chapter is built on top of it without ever leaving `solve`. We first make the measurement step trustworthy and quick, via the connectivity checker and a Gomory–Hu shortcut for undirected graphs. We then make the enumeration step skip whole families of hopeless graphs, via a pruned search. Finally, for the directed multigraph case, we replace enumeration altogether with one small optimisation that proves every m at once, which is the certifier. Each of these is the same `solve` doing less work for the same guarantee, and the choice among them is made automatically from the booleans it receives.

```
solve(n, m, directed, simple, separation, exhaustive, max_seconds)
```

DRIVER

in the variant booleans `directed` and `simple` (or hypergraph with uniformity r), the `separation` (edge or vertex), n , m , the method, and a time budget `max_seconds`

out a `SolveResult` carrying a value with an honest label: `exact`, `lower bound`, or `upper bound`

how picks the method the case allows: brute-force enumeration, the pruned search, or the cut-counting certifier when proving, or the temperature search or tabu search when discovering (Chapter 3). It always respects the budget, and it is the one function the rest of the thesis ever calls.

2.2 A single representation for all variants

Two of the three axes from Chapter 1, the model and the direction, are properties of the object, and a single data structure holds almost all of them. Of the three structural models, simple graphs and multigraphs fit the matrix directly, while the hypergraph keeps the separate storage promised in Chapter 1 and rejoins at the end of this chapter. The hypergraph could in principle be pressed into the same shape as a higher-order tensor, an r -dimensional array marking which r -sets are present, but that array is almost entirely zero and costs n^r entries, so the program keeps the compact list of hyperedges instead and pays nothing for the empty cells. Sampled $G(n, p)$ graphs are ordinary simple graphs and therefore use this same representation without any change. A graph or digraph on n vertices is stored as an integer matrix μ , where $\mu(u, v)$ is the multiplicity of the edge or arc from u to v . A simple graph is the case $\mu \in \{0, 1\}$, an undirected graph satisfies $\mu = \mu^\top$, and a multigraph allows larger entries. The third axis, edge

against vertex separation, is a property of the question rather than the object, so it lives in how we measure μ , not in μ itself. Figure 2.2 illustrates the encoding.

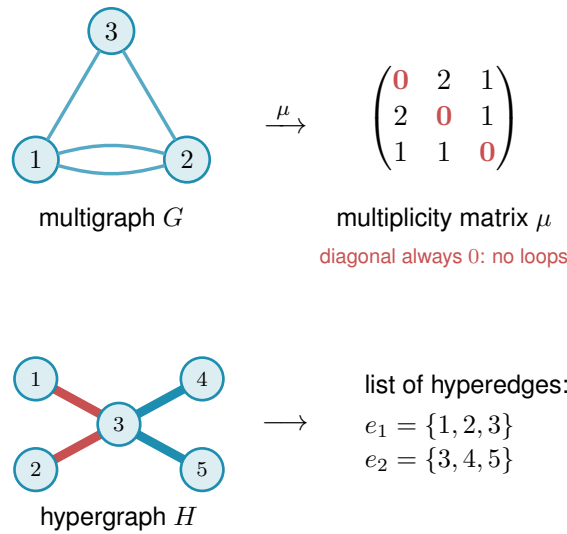


Figure 2.2. How the program stores a graph. Top: a graph or digraph on n vertices is one $n \times n$ integer matrix μ , with $\mu(u, v)$ the number of parallel edges or arcs from u to v . The double edge between 1 and 2 gives $\mu(1, 2) = 2$. The diagonal is always zero (red), since no vertex carries an edge to itself, an undirected graph keeps $\mu = \mu^\top$, a directed graph allows $\mu(u, v) \neq \mu(v, u)$, and a simple graph caps every entry at one. Bottom: the hypergraph is the one model that does not use the matrix at all, since a matrix has no room for an edge on three vertices, so it is kept as a plain list of its hyperedges, here two metro lines that share vertex 3.

```
Graph(n, Variant(directed, simple))
```

MODEL

in the vertex count n and a `Variant`, the small record holding the two booleans
out a graph whose whole state is the matrix μ , stored as the numpy array `self.mu`
how one $n \times n$ integer numpy array is the object, an undirected graph keeps $\mu = \mu^\top$, and a simple graph caps every entry at one.

Keeping a single representation is what lets one measurement routine, one certifier, and one search serve every case. Nothing downstream asks which of the twelve problems it is solving. It asks only the two booleans (directed or undirected, simple or multi) and the matrix. Everything that builds or edits a graph is a thin wrapper around μ that respects those booleans, and these operations are so routine that they need no individual exposition, and Table 2.1 lists them together.

The two booleans are all the mirroring above needs, and the whole trick fits in one short routine. Every write to μ passes through it, so this is the single place the undirected symmetry is kept, not an invariant a caller has to remember to maintain.

```
1 def _assign(self, u, v, value):
2     self.mu[u, v] = value
3     if not self.variant.directed:
```

Operation	Meaning
<code>addEdge(G, u, v, k)</code>	add k parallel edges or arcs $u \rightarrow v$ (a simple graph saturates at one)
<code>removeEdge(G, u, v, k)</code>	remove k copies, never below zero
<code>setMultiplicity(G, u, v, c)</code>	set $\mu(u, v) = c$ directly
<code>multiplicity(G, u, v)</code>	read $\mu(u, v)$
<code>hasEdge(G, u, v)</code>	whether $\mu(u, v) > 0$
<code>degree(G, v)</code>	total degree of v (in- plus out-degree if directed)
<code>edgeCount(G)</code>	number of edges or arcs, counted with multiplicity
<code>edges(G), vertices(G)</code>	iterate the present edges, and the vertex set
<code>copy(G)</code>	an independent duplicate

Table 2.1. The routine operations on the graph model. Each is a one-line wrapper that reads or writes the matrix μ while keeping an undirected graph symmetric and a simple graph binary. They are bookkeeping, and the chapter’s attention belongs to the routines that follow.

```
4 | self.mu[v, u] = value    # keep undirected graphs symmetric
```

Listing 2.1. How every write keeps an undirected graph symmetric. A directed graph simply skips the mirrored write.

2.3 The connectivity checker built on maximum flow

The whole question reduces to a flow computation. A *maximum flow* asks how much can be pushed from one point to another through a network whose links each carry a limited capacity. Menger’s theorem says the most edge-disjoint u – v paths a graph offers equals its smallest u – v cut, the fewest links whose removal severs every route between the two. The max-flow / min-cut theorem of Ford and Fulkerson [14] says that smallest cut is exactly the value of a maximum flow, once each multiplicity $\mu(u, v)$ is read as the capacity of the link $u \rightarrow v$. So the route count we are after is just a maximum-flow value: measure the flow and you have measured the connectivity. The routine that runs this flow and reports the route count is the connectivity *checker*. It is the single most important routine in the program: it is the only place graph theory meets computation, and every later claim about connectivity passes through it. Because the search of Chapter 3 calls it by the million on tiny graphs, the program ships an optional compiled C helper that runs this same flow (named later in this chapter’s reproducibility section). It is a speed-up only: it returns the identical value the pure-Python checker does, verified against it, so no number in the thesis depends on whether the helper is built.

```
local_edge_connectivity(G, s, t) → ℕ
```

MEASURE

in a graph G and an ordered pair (s, t)

out $\lambda_G(s, t)$, the maximum number of edge-disjoint s – t paths

how a single `scipy.sparse.csgraph.maximum_flow` on the network whose arc capacities are the multiplicities μ , so by Menger the flow value is the path count.

`max_edge_connectivity(G) → ℕ`

MEASURE

in a graph G

out $\lambda^{\max}(G)$, the largest local edge-connectivity over all pairs

how one max-flow per pair (ordered if directed), take the largest. This is the quantity the edge problem caps at $m - 1$.

The vertex-disjoint version follows the same idea after one construction: split each vertex v into an in-copy v^{in} and an out-copy v^{out} joined by a single unit-capacity arc (Figure 2.3). Every arc $u \rightarrow v$ of the original graph becomes $u^{\text{out}} \rightarrow v^{\text{in}}$ in the split network. The construction is not special to digraphs: for an undirected graph each edge $\{u, v\}$ is first read as the two opposite arcs $u \rightarrow v$ and $v \rightarrow u$, and then the same split applies, so one routine serves the undirected and directed vertex problems alike. Routing through v now consumes the one unit of capacity on its internal arc, so two paths sharing v as an intermediate vertex compete for that unit and cannot coexist in any maximum flow. A max-flow from u^{out} to v^{in} in the split network therefore equals the number of internally vertex-disjoint u – v paths in the original graph.

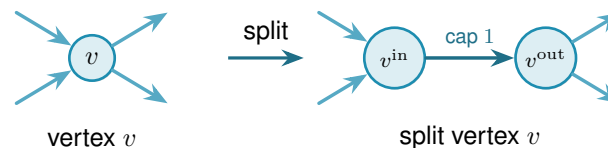


Figure 2.3. Vertex splitting for the vertex-disjoint checker. Incoming arcs attach to v^{in} and outgoing arcs leave from v^{out} , connected by a single unit-capacity arc. Any directed path that uses v as an intermediate vertex passes through this bottleneck, and two such paths would need two units of capacity and are therefore separated.

Figure 2.4 applies the split to a whole digraph. Three arc-disjoint routes are forced through one shared vertex w , so on the split network all three must cross the single unit-capacity arc inside w , which caps the number of internally vertex-disjoint routes below the edge-disjoint count.

`max_vertex_connectivity(G) → ℕ`

MEASURE

in a graph G

out $\kappa^{\max}(G)$, the largest local vertex-connectivity over all pairs

how the same sweep as the edge checker, but each flow runs on the split capacity matrix `_split_capacity_matrix(G)` of Figure 2.3, so the unit internal arcs count internally vertex-disjoint paths. Parallel edges are given capacity one here, so they never raise κ .

One convention in that codecard deserves to be said in prose, because it does not merely change how the checker computes, it changes what the multigraph vertex question asks. In vertex mode every adjacency carries capacity one, so a bundle of parallel edges counts as a single direct route. The rationale is that vertex-disjointness is about intermediate vertices, and parallel copies offer no new intermediate vertices to be disjoint over.

Remark 2.1 (What parallel edges mean in vertex mode). That choice has a consequence which

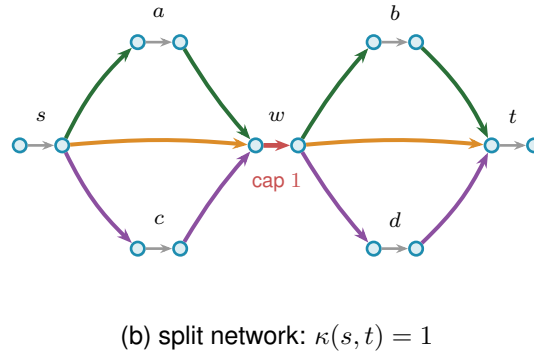
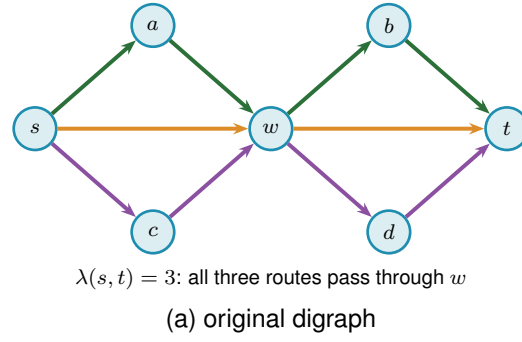


Figure 2.4. The vertex-split checker on a larger example. **(a)** A digraph in which three arc-disjoint s - t routes, $s \rightarrow w \rightarrow t$ (orange), $s \rightarrow a \rightarrow w \rightarrow b \rightarrow t$ (green), and $s \rightarrow c \rightarrow w \rightarrow d \rightarrow t$ (purple), all pass through the single vertex w , so $\lambda(s, t) = 3$. **(b)** The split network replaces every vertex V by an arc $V^{\text{in}} \rightarrow V^{\text{out}}$ of capacity one. All three routes must now traverse the single unit arc inside w (red), which carries only one unit, so at most one internally vertex-disjoint route gets through and $\kappa(s, t) = 1$. Splitting every vertex this way turns the vertex-disjoint count into an ordinary maximum flow.

has to be stated rather than left implicit. If parallel copies never raise κ , they never break feasibility either, so one could pile arbitrarily many copies onto any feasible multigraph and stay under the ceiling forever. Counted with multiplicity, the maximum number of edges would simply be infinite, and the extremal question would be empty. The two multigraph vertex variants are therefore posed here with the objective counting *adjacencies*, that is, pairs joined by at least one edge, rather than edges with multiplicity. Under that reading the question is exactly the simple question on the underlying graph, which is why those two rows of [Table 4.1](#) reduce to their simple counterparts and why every figure in this thesis reports the reduced problem for them.

The other convention is perfectly available and asks something genuinely different. A single edge is a path with no interior, so q parallel copies between the same pair are q internally vertex-disjoint paths, and feasibility caps every multiplicity at $m - 1$ by itself. Nothing runs away, and the multigraph vertex problem separates from the simple one immediately: at $n = 2$ and $m = 3$ two parallel edges are feasible where the simple graph allows one. That is a new extremal problem rather than a variant of a solved one, and [Chapter 4](#) lists it among the open problems. The appendix uses precisely this second reading whenever it argues about a general multigraph, most visibly in [Lemma A.52](#), whose hypothesis (iv) caps a multiplicity at two for exactly this

reason. The two readings never meet, because that lemma is applied only to incidence graphs, which carry no parallel edges at all, but a reader moving between the chapters and the appendix should know that the word “multigraph” is doing different work in the two places.

The two checkers are the entire interface between graph theory and computation in this thesis. As a sanity check, applied to the bidirected star of [Figure 1.11](#) on n vertices, the edge checker reports $\lambda^{\max} = 1$, and the arc count $2(n - 1)$ matches the proved values $\ell_2^{\text{dir}}(n) = 2, 4, 6, 8$ for $n = 2, 3, 4, 5$.

The same checker answers two different questions. Throughout this chapter it runs in its exact-value mode, returning $\lambda^{\max}$ or $\kappa^{\max}$ on the nose, and every proved or reported number rests on that exact reading. The search of [Chapter 3](#) needs only the cheaper yes-or-no question “is the binding connectivity above the bound?”, so there the same flow runs in a capped mode that stops as soon as it has seen one route too many. The capped mode never changes a reported value: it is a faster way to ask for a single bit, and [Chapter 3](#) shows that the search built on it reproduces the exact search step for step.

The Gomory–Hu tree from [Chapter 1](#) reappears here as an efficiency, and it is the honest kind. For undirected graphs all $\binom{n}{2}$ edge-connectivities are read off a single weighted tree, so $\lambda^{\max}$ is the heaviest tree edge rather than the maximum over $\binom{n}{2}$ separate flows. The speed-up is real, but it is bought with a theorem, not a heuristic.

`max_edge_connectivity_via_tree(G) → ℕ`

MEASURE

in an undirected graph G

out $\lambda^{\max}(G)$, identical to the pairwise sweep

how build one Gomory–Hu tree with `networkx.gomory_hu_tree`, then return its heaviest edge, which costs $n - 1$ flows in place of $\binom{n}{2}$ and cross-checks the checker.

The hypergraph is the one model that never used the multiplicity matrix, since it is stored as a plain list of hyperedges, each one a set of r vertices. The checker still has to read it, and the trick is to run the same flow computation as before on a slightly larger network that we build on the spot. The difficulty is that one hyperedge ties several vertices together at once. By the edge-disjointness convention of [Chapter 1](#), only one route may be counted through a hyperedge at a time, so we cannot just treat it as a bundle of ordinary links. Instead, for every hyperedge we add one extra helper point and connect each of that hyperedge’s member vertices to it ([Figure 2.5](#)). A route that wants to use the hyperedge now enters through one member vertex, passes through the helper point, and leaves through another.

The key is to make the helper point carry a single unit of capacity, built by the very same device as the vertex bottleneck of [Figure 2.3](#): the helper point is split into an in-copy and an out-copy joined by one unit-capacity arc, and every route boarding the hyperedge must squeeze through that one arc. A second route would need a second unit of capacity there and so is shut out. That one unit is what captures the rule that one hyperedge serves one route.

Counting routes between two vertices is then the very same maximum flow as before, run on this enlarged network, and when a hyperedge happens to hold just two vertices the helper point collapses back to an ordinary link, so the answer agrees with the graph case. That the flow on the enlarged network counts exactly the hyperedge-disjoint Berge routes is the hypergraph version of Menger’s theorem, proved in [Appendix A \(Theorem A.45\)](#), so the checker rests on a theorem rather than an analogy.

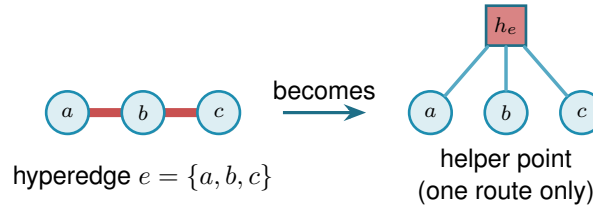


Figure 2.5. Measuring Berge connectivity by maximum flow. A hyperedge e , drawn as a metro line through its members (left), is replaced by a helper point h_e joined to each member (right). The helper point holds a single unit of capacity inside it, so at most one route may pass through h_e and the line runs one train at a time, which is what makes the hyperedge serve only one route. Counting independent routes between two vertices is then an ordinary maximum flow on this enlarged network, and a two-vertex hyperedge collapses to a single ordinary edge.

[Figure 2.5](#) shows one hyperedge in isolation. [Figure 2.6](#) carries the idea to a whole hypergraph in two panels. Panel (a) is the hypergraph drawn as a metro map: each of the eight hyperedges is one coloured line threading its three member stations, vertex to vertex, and a station on several hyperedges shows several coloured lines running through it. Panel (b) reads the same map as the flow network. For one pair of stations, u and v , it adds the helper point of every hyperedge a route boards, drawn as a small gate h_e in that line’s colour, and traces the edge-disjoint Berge routes between u and v as thin black lines passing through those gates, with every hyperedge rail faded to the background. The checker reads $\lambda(u, v) = 4$: two routes run straight through the two hyperedges that contain both u and v , and two longer ones change lines at an intermediate station, four in all, while a fifth would have to reuse a gate.

One way to picture the helper point is the metro map of [Chapter 1](#): each hyperedge is a metro line, its member vertices are the stations on it, and the helper is the line itself. Boarding at any station counts as taking that line. Because only one route may pass through the helper, only one train runs on the line at a time, so a second route must reach its destination without ever boarding it. The same flow computation that counts edge-disjoint paths through ordinary links now counts routes that share no line, which is exactly what Berge edge-disjointness means.

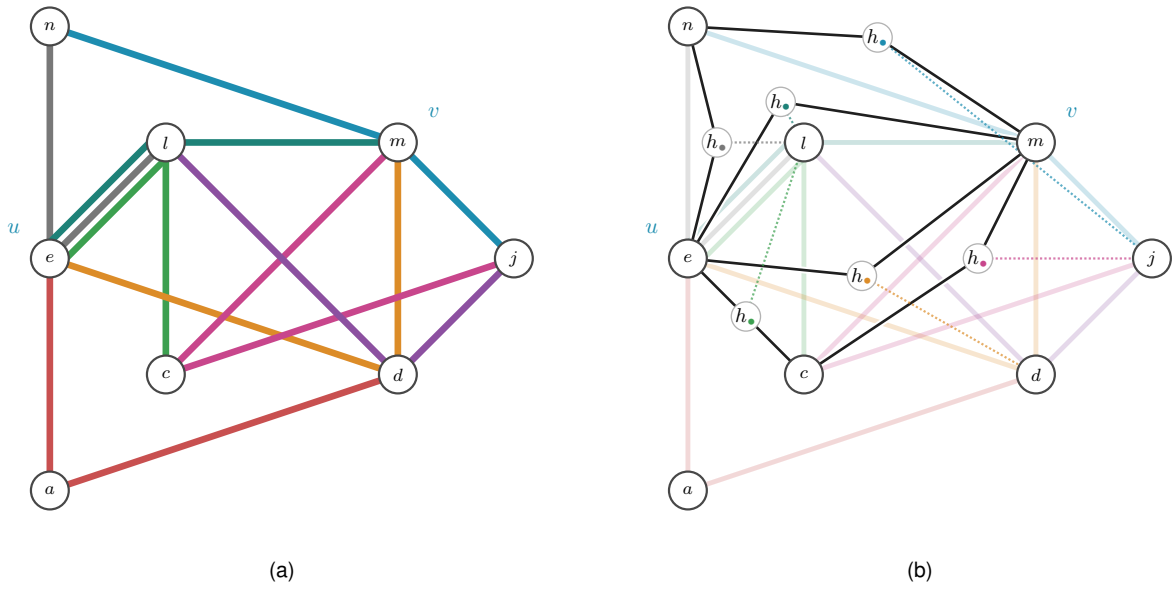


Figure 2.6. A 3-uniform hypergraph on eight stations, with the pair u, v singled out. Panel (a) draws it as a metro map: every hyperedge is one coloured line threading its three members, vertex to vertex, the metro reading of [Chapter 1](#). Panel (b) reads the same picture as the helper-point flow network. Each hyperedge a route boards becomes its one-route gate $h_\bullet$ of [Figure 2.5](#), shown in that line's colour and placed on the route, and the thin black lines are the edge-disjoint Berge routes from u to v threading those gates. A dotted leg ties each gate to the hyperedge's third member, the station that route does not use, and every hyperedge rail is faded to the background so the black routes read clearly. Two routes pass straight through the two hyperedges that hold both u and v , and two longer ones transfer at an intermediate station, so the checker reads $\lambda(u, v) = 4$. A fifth route would have to reuse a gate.

`max_hyperedge_connectivity(H) → ℕ`

MEASURE

in an r -uniform hypergraph H

out $\lambda^{\max}(H)$, the largest Berge edge-connectivity over all pairs

how build the enlarged capacity matrix `_hyper_capacity_matrix(H)` in which every hyper-edge becomes a helper point that only one route may pass through, then take the same maximum flow as the graph checker. On two-vertex hyperedges it matches ordinary edge-connectivity.

On the complete graph K_n the hypergraph checker returns $n - 1$, and on the complete three-uniform hypergraph $K_4^{(3)}$ it returns 3, larger than the two hyperedges through each pair because longer Berge routes also carry flow. The self-test confirms both of these values, so the construction is checked rather than merely asserted. The same helper-point network also answers the vertex-disjoint hypergraph question without alteration: splitting each ordinary vertex as in [Figure 2.3](#) counts internally vertex-disjoint Berge routes. The one remaining direction, the *directed* hypergraph, needs the helper point to learn which way is which, and that is the subject of the next subsection.

2.3.1 The directed hypergraph checker

A *directed* hyperedge carries an orientation: it is a single *tail* vertex together with $r - 1$ *head* vertices, drawn as the star of [Figure A.7](#), and a directed Berge route may only enter it at the tail and leave it at one of its heads. The undirected helper point treated a hyperedge as a line any route could board at any station. The directed one must instead be a one-way turnstile: a route may step onto the hyperedge only at its tail and step off only at a head. The gadget that enforces this is the helper point made directional. For a hyperedge $e = (a; b, c, \dots)$ we add a gate node g_e , send one capacity-one arc from the tail into it, $a \rightarrow g_e$, and send an arc out of it to each head, $g_e \rightarrow b$, $g_e \rightarrow c$, and so on ([Figure 2.7](#)). A route reaches g_e only through the tail a , and the single unit of capacity on $a \rightarrow g_e$ lets just one route use the hyperedge, exactly as the undirected turnstile did, but now it can leave only towards a head. The gate is built once per hyperedge, directed or not, by the same loop:

```
1 gate_in, gate_out = base + 2 * index, base + 2 * index + 1
2 cap[gate_in, gate_out] = 1 # one route through this hyperedge
3 if hypergraph.directed:
4     tails, heads = _dir_tails_heads(edge)
5     for tail in tails:
6         cap[leave(tail), gate_in] = _UNBOUNDED
7     for head in heads:
8         cap[gate_out, enter(head)] = _UNBOUNDED
```

Listing 2.2. The gate loop behind [Figure 2.5](#) and [Figure 2.7](#): one capacity-one gate per hyperedge, wired one-way for a directed hyperedge and both ways for an undirected one.

[Figure 2.8](#) reads the same hypergraph directed, again in two panels. Panel (a) gives the whole

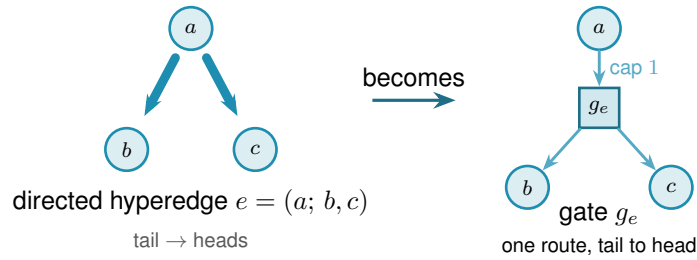


Figure 2.7. The directed helper point. A directed hyperedge $e = (a; b, c)$, the tail a fanning to its heads b, c (left), becomes a gate node g_e fed by one capacity-one arc from the tail and feeding an arc to each head (right). A route may reach g_e only through the tail and may leave only towards a head, so the gate is a one-way version of the turnstile of Figure 2.5, and the single unit of capacity still lets only one route use the hyperedge.

hypergraph oriented, every hyperedge a one-way line whose arrows point away from its tail, so the middle-tailed lines fan out from their centre and the end-tailed ones run one way along their length. Panel (b) traces the routes from v to u , the gate rule of Figure 2.7, fading every hyperedge rail to the background and threading a black arrow from the route's entry station through the one-route gate $h_\bullet$, set in a clear gap off the rails, to its exit, with a dotted leg to the hyperedge's third member. Orientation costs connectivity. Of the four undirected routes between u and v only two survive the arrows, $\lambda(v, u) = 2$, because v can set out only through the two hyperedges of which it is the tail, and the two survivors cross at the interchange l so they share no hyperedge. The same vertex-splitting trick of Figure 2.3 laid on top then counts internally vertex-disjoint directed routes, so this one construction measures all four hypergraph variants and `solve` reaches every one of the twelve problems through a single checker.

The gate is indifferent to how a hyperedge's vertices are split between tails and heads. It draws one capacity-one arc in from every tail and one out to every head, so the single forward tail it was built for is only the simplest case: give it a hyperedge with several tails and one head, or several of each, and it still counts the routes that enter at a tail and leave at a head. The same checker therefore measures every *orientation model* of a directed Berge edge, not just the one-tail one, and Section 4.3.1 takes up which of those models are worth distinguishing and what the program finds for them.

With the checker in place the proposed value of Proposition 1.15 can be tested on any small hypergraph the way every other variant is, which is the subject of the next section.

2.4 Checking every graph of a fixed size

The checker measures one graph. To bound every graph of a given size, the most direct method is the one the base cases of Chapter 1 already needed: enumerate them all. A simple graph on n vertices is a choice of $\mu(u, v) \in \{0, 1\}$ on each off-diagonal cell, a multigraph a choice in $\{0, \dots, m-1\}$, and a digraph varies every ordered pair rather than only the upper triangle. Walking that product, measuring each candidate with the checker, and keeping the densest feasible one settles the value exactly, and because it has looked at every graph it is a genuine upper

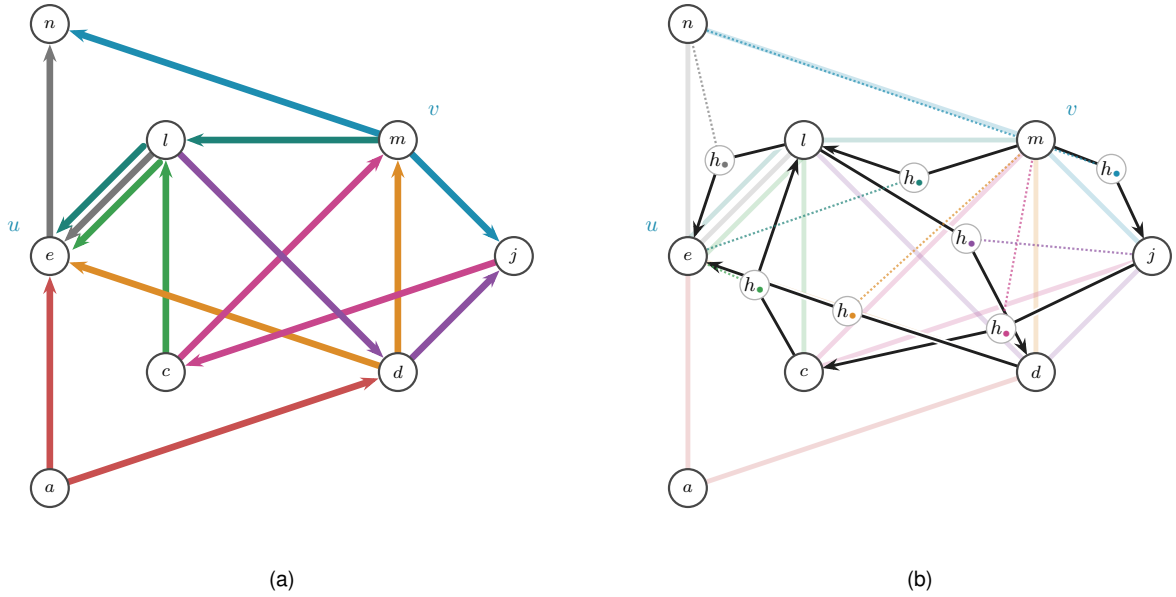


Figure 2.8. The hypergraph of Figure 2.6 read directed. Panel (a) gives the whole directed hypergraph, every hyperedge a one-way line whose arrows all run away from its tail station, so a route may join a line at its tail and ride it to a head. Because the map keeps each hyperedge on the track it already occupied, a line whose tail is an end station is oriented along its own length rather than fanned out as the star of Figure A.7, so both renderings appear here. One rule reads either of them off the picture: the tail of a coloured line is the single station of that line into which none of its arrows points, and the other $r - 1$ stations are its heads. Applied here it gives the eight tail-and-heads splits $a \rightarrow ed$, $m \rightarrow nj$, $c \rightarrow le$, $d \rightarrow em$, $l \rightarrow en$, $m \rightarrow le$, $j \rightarrow cm$ and $l \rightarrow dj$. Panel (b) traces the routes from v to u . Every hyperedge rail is faded to the background, and a black arrow runs from the route's entry station through the hyperedge's one-route gate $h_\bullet$ to its exit, the directed turnstile of Figure 2.7, with the gate set in a clear gap off the rails. A dotted leg in the line's colour ties each gate to the hyperedge's third member, the station the route does not visit. Two edge-disjoint directed Berge routes from v to u survive. The long route $v \rightarrow j \rightarrow c \rightarrow l \rightarrow u$ rides nmj , mcj , elc and nel in turn, and the short route $v \rightarrow l \rightarrow d \rightarrow u$ rides elm , ldj and edm . They cross only at the interchange l , so they share no hyperedge, and the checker reads $\lambda(v, u) = 2$, since v can set out only through the two hyperedges of which it is the tail. Against the undirected four of Figure 2.6, the arrows have halved the count.

bound, not evidence.

This is the first thing `solve` does when it is asked to prove a small case. With its exhaustive flag set it walks that product, measures each candidate with the checker, and keeps the densest feasible one. The value it returns is exact and exhausted, a true upper bound for that n , but only while the count of graphs stays small. No new routine is involved, only `solve` choosing to enumerate.

Run on the small cases this confirms the proved values directly, with no induction and no cleverness. [Table 2.2](#) collects a spread of variants where the enumeration finishes: in every row the machine returns exactly the value [Chapter 1](#) proved by hand. This is the first payoff of the uniform representation. One driver, `solve`, pointed at six different problems, settles all six.

Variant	m	n	<code>solve</code>	Theory
simple undirected, edge	2	5	4	$\lfloor m(n-1)/2 \rfloor = 4$
simple undirected, edge	3	5	6	$\lfloor m(n-1)/2 \rfloor = 6$
multigraph undirected, edge	3	5	8	$(m-1)(n-1) = 8$
simple undirected, vertex	2	6	5	$k_2(n) = n-1 = 5$
simple directed, arc	2	5	8	$\max(2(n-1), \lfloor n^2/4 \rfloor) = 8$
simple directed, arc	2	6	10	$\max(2(n-1), \lfloor n^2/4 \rfloor) = 10$

Table 2.2. Variants that converge under plain enumeration. For each small case `solve`, in its exhaustive mode, walks every graph of the variant, measures it with the checker, and returns the densest feasible one. Every value matches the hand proof of [Chapter 1](#) exactly.

The honesty of the method is also its limit. The number of graphs it must visit grows as $2^{\binom{n}{2}}$ for simple undirected graphs and $2^{n(n-1)}$ for digraphs, and capping multiplicities at $m-1$, so that each cell ranges over the m values $\{0, \dots, m-1\}$, raises the base from two to m , giving $m^{\binom{n}{2}}$ undirected multigraphs and $m^{n(n-1)}$ directed ones. The enumeration is exact wherever it finishes, and it finishes only for the smallest sizes. [Chapter 3](#) opens with exactly how fast this wall arrives. For now the point is that the wall arrives precisely where [Chapter 1](#) needed help, at the directed base cases $n = 6, 7$, so the rest of this chapter is about reaching a little further than blind enumeration can.

2.5 Reaching further: a pruned search for the directed base cases

The base cases $n \leq 7$ of the directed $m = 2$ theorem ([Theorem 1.10](#)) are the finite checks the induction of [Appendix A](#) depends on, and they sit just past where blind enumeration of digraphs is tractable. They can be reached by enumerating more cleverly. Feasibility is monotone: adding an arc can never lower $\lambda^{\max}$, so once a partial digraph is infeasible every completion of it is infeasible too and the whole branch is abandoned at once. A counting bound prunes further. At any partial digraph, count the arcs already placed and add the most the still-undecided cells could ever contribute. That sum is a ceiling on every completion of the branch, so if even the ceiling cannot beat the best feasible digraph found so far, the branch is dropped unexplored ([Figure 2.9](#)). What remains is a branch and bound over simple digraphs with the connectivity checker as the feasibility test, exact and complete at these sizes where the blind walk is not.

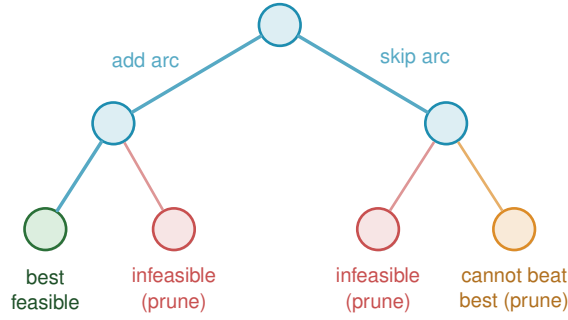


Figure 2.9. Branch and bound over digraphs. Each node fixes one more cell, arc present or absent, so the leaves are whole digraphs. Two rules cut entire subtrees before they are walked. A branch whose partial digraph already breaks the connectivity ceiling is infeasible, and so is every completion of it (red). A branch whose best possible completion still cannot beat the densest feasible digraph found so far is dropped on the counting bound (orange). Only the surviving branches are explored, which is what lets the search reach the base cases $n \leq 7$ the blind walk cannot.

Here `solve` changes its implementation without changing its interface. For a simple digraph it runs this branch and bound in place of the blind walk, still with the connectivity checker as the feasibility test, and so reaches the base cases $n \leq 7$ that blind enumeration cannot. The booleans alone decide which of the two it uses, and the caller never asks.

Run on the directed $m = 2$ problem `solve` returns the complete sequence 2, 4, 6, 8, 10, 12 for $n = 2, \dots, 7$, each proved complete at its size. The densest digraph it finds at each n is exactly the extremal shape [Chapter 1](#) named, the bidirected star (the hub) at every $n < 7$ and, at the crossover $n = 7$ itself, a graph of the tying value 12 that both the hub and the one-directional bipartite digraph attain. The sweep stops there, so the bipartite side of the crossover is witnessed by the construction and the checker rather than by this search ([Chapter 3](#) records that the cooled walk in fact stalls on the hub at $n = 8$). Within its range the search therefore confirms not just the values but the winning construction. These are exactly the base cases the induction requires, and with them the long proof of [Theorem 1.10](#) rests on a single auditable search rather than on anything done by hand. The exhaustive-search transcript is in [Appendix A](#).

2.6 Generating the directed cases faster

[Theorem A.30](#) ([Appendix A](#)) now proves the directed multigraph value $L_m^{\text{dir}}(n) = (m - 1)M(n)$ by hand, for every n and m at once, so nothing in this section is load-bearing for that theorem any more. It remains useful for a narrower and genuinely separate question: not how many arcs an extremal multigraph on $n = 7$ vertices carries, but exactly which multigraphs carry them ([Remark A.41](#)). Answering that means listing every such multigraph up to a renaming of its vertices, and the program does this in two independent ways, on purpose: when two different methods return the same list, that agreement is itself a check.

The first way is a search written from scratch. It is a depth-first walk over multiplicity matrices that abandons a branch the moment it turns infeasible or can no longer beat the best graph found, and it keeps just one representative of each isomorphism class by reducing every finished

graph to a canonical relabelling, the smallest matrix in dictionary order over all relabellings. The canonical form keeps the memory small, but the time is spent walking labelled partial graphs, and time is what runs out: the full classification at $n = 5$ costs this in-house search about 456 seconds, and it climbs steeply from there.

The second way (Table 4.2 names it) does not search at all, it generates, and it was suggested by Prof. Jan Goedgebeur. The idea is to hand the hard part, listing one graph per isomorphism class, to a tool that already does it supremely well: the `geng` program from the `nauty` toolkit of McKay and Piperno [9], invoked through a subprocess and decoded from its `graph6` output (a compact one-line text encoding of a graph). `geng` lists the non-isomorphic *undirected* graphs on n vertices, and the program then decorates each one with the directed multiplicities the problem needs, so the isomorphism reduction is inherited for free from the support graph. The decoration itself is plain to state. For one support graph, walk its edges in a fixed order and assign each edge $\{u, v\}$ an ordered pair of multiplicities $\mu(u, v), \mu(v, u)$, each between 0 and $m-1$ but not both zero, since an edge of the support must carry at least one arc. Along the way the checker measures the partly decorated multigraph, and the moment some pair already exceeds the connectivity ceiling the whole branch of assignments is abandoned, since adding further arcs can never lower a connectivity, the same monotonicity prune as Figure 2.9. A decoration that survives to the end with the target arc count is kept, reduced to its canonical form so that each isomorphism class is reported once. Different supports share nothing, so they are processed independently across the processor cores. The companion orientation tools `directg` and `watercluster2` are not used, because each gives only one orientation per edge and so cannot express the bidirected arcs and repeated multiplicities the extremal graphs are built from. Layering the multiplicities in-house on top of `geng` is what fits the problem. To be precise about credit: Prof. Goedgebeur is not an author of `nauty`, which is the work of McKay and Piperno cited above. He suggested the generation route, and it is his suggestion that the speed-up rests on.

The two ways agree to the last graph. The generated pipeline reproduces the in-house classification exactly at every settled size, two non-isomorphic extremal families at $n = 4$ and three at $n = 5$, and it runs substantially faster, since the isomorphism work it never repeats is precisely what `geng` has already done. One difference decides which to trust further out: the in-house search leans on a conjectured counting bound once $n \geq 7$, while the generated enumeration prunes only with proved bounds, so it is a sound and complete search at every n . That soundness has now paid off. Pointed at the degree-bounded classification at $n = 7$, the generation enumerator completed the sweep in about seven hours across ten processor cores and returned exactly three extremal families, $2 \cdot B_{3,4}$, $2 \cdot B_{4,3}$, and the doubled bidirected path P_7 (Remark A.41).

2.7 Proving every m at once

The pruned search proves one (n, m) at a time. For the directed multigraph arc problem `solve` has a sharper implementation, one that proves every m at a fixed n in a single pass, and it is worth the extra machinery because it closes the directed multigraph case outright for the sizes it

reaches.

The key is that m can be scaled out of the problem completely, so that it never appears in the computation at all. Start with any directed multigraph that is feasible at level m , so $\lambda^{\max} \leq m - 1$, and divide every multiplicity by $m - 1$. Two things happen at once. The multiplicities become weights $w(u, v)$ between 0 and 1, and since dividing every capacity by $m - 1$ divides every flow by the same factor, the maximum flow between each pair drops to at most one. The arc count, meanwhile, turns into the total weight $\sum_{u \neq v} w(u, v) = |A|/(m - 1)$. So a dense feasible multigraph becomes a heavy feasible weight matrix, and the other way round. Figure 2.10 carries one small multigraph through the division. Write

$$M^*(n) = \max \left\{ \sum_{u \neq v} w(u, v) : w(u, v) \in [0, 1], \text{ maxflow}(s, t; w) \leq 1 \text{ for all } s, t \right\}$$

for the largest total weight any such matrix can carry. It is one number for each n , with no m inside it, and scaling a feasible multigraph down gives

$$L_m^{\text{dir}}(n) \leq (m - 1) M^*(n) \quad \text{for every } m \geq 2.$$

That inequality holds always, and it is the direction that does the proving. The reverse does not come for free, and it is worth being exact about why, because the two are easy to run together. Scaling *up* needs an optimal weight matrix whose entries are whole multiples of $1/(m - 1)$, and a real optimum has no reason to oblige. What closes the gap is not an argument but an observation about the answer the solve returns: the heaviest matrix it finds is the *double star*, the bidirected star carrying weight one on both arcs $h \rightarrow v$ and $v \rightarrow h$ between the hub and every other vertex. Its weights are all 0 or 1, so multiplying them by $m - 1$ gives an honest integer multigraph, not a fractional one, with $2(n - 1)(m - 1)$ arcs and $\lambda^{\max} = m - 1$, for every m at once. Whenever the optimum is attained by such a $\{0, 1\}$ matrix, the two bounds meet and

$$L_m^{\text{dir}}(n) = (m - 1) M^*(n) \quad \text{for every } m \geq 2,$$

so that one solve settles infinitely many values. This is exactly what happens at every $n \leq 6$, which is what Theorem 2.2 rests on. At the time these solves were run it was a verified property of those particular optima and not a theorem about all n , which is why `prove_integral_arc_bound` exists at all: it answers an integral question, such as $L_3^{\text{dir}}(7) = 24$, directly, without leaning on an unverified property of the fractional optimum. That property is now a theorem for every n (Corollary A.37), so the identity above holds unconditionally, but the route to it runs the other way round from what this section expected. It is not that a closer look at the fractional problem settled the integral one. It is that the integral problem was settled outright, at every m at once, and the fractional statement then followed by scaling a rational optimum up until its entries become whole. Theorem A.30 (Appendix A) settles the integral question for every n by an entirely different, elementary argument, so this MILP route is no longer the only way to reach it, and in fact never reached it: the $n = 7$ call was abandoned at its time limit and never rerun on a stronger solver

(Corollary A.36). Where the two do meet, at every $n \leq 6$, they agree.

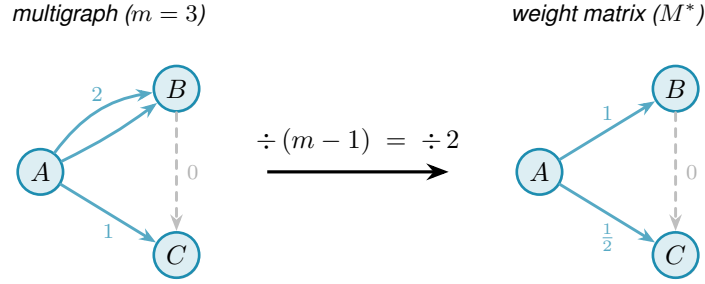


Figure 2.10. The scaling step for $m = 3$. Dividing each multiplicity by $m - 1 = 2$ maps the integer multigraph (left) to a weight matrix with all values in $[0, 1]$ and max-flow at most one between every pair (right). The double arc $A \rightarrow B$ (multiplicity 2) becomes weight 1, the single arc $A \rightarrow C$ (multiplicity 1) becomes weight $\frac{1}{2}$, and the absent arc $B \rightarrow C$ (multiplicity 0, shown dashed) stays absent. Crucially, m drops out on the right, so a single optimisation over all weight matrices bounds $L_m^{\text{dir}}(n) \leq (m - 1) M^*(n)$ for every m simultaneously. Along this route the reverse inequality is not automatic and needs the optimum to be attained by a $\{0, 1\}$ matrix, which is a verified property of the optima at $n \leq 6$ rather than a theorem about all n , as the text above sets out. [Appendix A](#) proves the reverse inequality unconditionally by a different route, so this scaling step is a useful independent lens on the value rather than a gap in it.

What remains is to compute $M^*(n)$, and the difficulty is that the constraint “maximum flow at most one” is itself the answer to a flow problem rather than a plain inequality on w . The max-flow / min-cut theorem removes it. The maximum s – t flow equals the smallest capacity of any *cut* separating s from t . A cut just splits the vertices into a source side holding s and a sink side holding t , and its capacity is the total weight of the arcs that cross from the source side to the sink side. So $\text{maxflow}(s, t) \leq 1$ holds exactly when some such cut has capacity at most one. That lets us drop the flow entirely: instead, for every ordered pair (s, t) , the optimisation picks one cut and we require its capacity to stay at or below one.

A cut is described by a yes/no label x_u^{st} on each vertex, equal to one when u is on the source side, with s fixed on the source side and t on the sink side. An arc $u \rightarrow v$ crosses the cut exactly when u is on the source side and v is not, and a helper variable p_{uv}^{st} then picks up its weight $w(u, v)$. Forcing the crossing weights to sum to at most one for every pair forces every pair to admit a cut of capacity one, hence maximum flow at most one. This is the exact feasible region, not a relaxation of it, a relaxation being a loosened version that would also admit weight matrices no multigraph realises, and that exactness is what turns the result into a proof. The optimisation reports both the best total weight it has found and a value it has shown no matrix can exceed, and when those two meet the gap is zero and $M^*(n)$ is certified.

For the directed multigraph arc problem, then, `solve` runs this cut-counting optimisation in place of any enumeration: it chooses the minimum cut for every ordered pair, and a zero gap makes the reported $M^*(n)$ a proved upper bound, from which $L_m^{\text{dir}}(n) \leq (m - 1) M^*(n)$ delivers a ceiling for every m at once, met from below by the double star whenever the reported optimum is integral.

`prove_directed_multigraph(n) → ProofResult`

PROVE

in the vertex count n (and tightening flags such as `use_deletion_cuts`)

out $M^*(n)$ with a proof status, giving $L_m^{\text{dir}}(n) \leq (m-1)M^*(n)$ for every m , with equality when the optimum found is integral

how builds the cut-counting optimisation, the mixed-integer linear program written out below, once with `_cut_counting_model` as a pulp model, then `_pick_solver` runs it on CBC, the free solver bundled with pulp, or on the commercial Gurobi solver by a one-line switch. A zero optimality gap is a proof. The integer-weight companion `prove_integral_arc_bound(n, m, target)` answers a single feasibility question directly, an independent way to confirm any one value of [Theorem A.30 \(Appendix A\)](#), such as $L_3^{\text{dir}}(7) = 24$.

`solve` assembles this optimisation from n alone, and a reader content to take it on the strength of the cut argument just given can skip to [Theorem 2.2](#) without loss. Written out it reads more heavily than it behaves, so we first name every symbol in plain words, then show the one inequality that looks cryptic is a simple yes-or-no switch.

The optimisation is a *Mixed-Integer Linear Program*, or MILP for short. “Linear” means every constraint and the objective are linear functions of the variables: no products, no squares, no exponentials. “Mixed-integer” means the variables split into two kinds: most are ordinary real numbers that may take any value in a continuous range, but a few are forced to be whole numbers, either 0 or 1. The reason for the split is conceptual. Arc weights w are naturally continuous: they are scaled multiplicities and can sit anywhere in $[0, 1]$. But the cut assignment for a vertex is a hard binary choice: a vertex is either on the source side of a given cut or on the sink side, with no meaningful notion of being “half on each side”. Allowing the cut labels to be fractional would describe a different, weaker condition and could report an optimistic value that no integer-weight multigraph can actually achieve. Keeping those labels in $\{0, 1\}$ is what keeps the optimal gap honest, and it is what makes the program’s result a proof rather than an estimate.

Three quantities appear, and only three. The first is already familiar.

- ★ $w(u, v) \in [0, 1]$ is the *scaled multiplicity* of the arc $u \rightarrow v$, the amount of arc weight the matrix places on $u \rightarrow v$ after the division by $m-1$ of the previous section. It is what we are maximising the total of. This is a continuous variable: it may be any real number in $[0, 1]$.
- ★ $x_u^{st} \in \{0, 1\}$ is the *side label* for vertex u with respect to the ordered pair (s, t) . The optimisation chooses one cut for each pair, and x_u^{st} records whether u is on the source side (1) or the sink side (0). The two endpoints are pinned: $x_s^{st} = 1$ and $x_t^{st} = 0$. This is the integer variable: a vertex cannot straddle the cut.
- ★ $p_{uv}^{st} \geq 0$ is the *crossing weight*, the part of $w(u, v)$ that the pair (s, t) ’s chosen cut is forced to count. It equals $w(u, v)$ when the arc $u \rightarrow v$ crosses from source to sink, and is otherwise left at zero. This is a continuous variable.

With those three named, the program is short:

$$\begin{aligned}
& \text{maximise} && \sum_{u \neq v} w(u, v) \\
& \text{subject to} && p_{uv}^{st} \geq w(u, v) + x_u^{st} - x_v^{st} - 1 && \text{for all } (s, t), (u, v), \\
& && \sum_{u \neq v} p_{uv}^{st} \leq 1 && \text{for all } (s, t), \\
& && w(s, t) + \sum_x \min(w(s, x), w(x, t)) \leq 1 && \text{for all } (s, t), \\
& && d(v_0) \leq d(v_1) \leq \dots \leq d(v_{n-1}), && \text{with } d(v) = \sum_{u \neq v} (w(u, v) + w(v, u)), \\
& && w \in [0, 1], \quad x \in \{0, 1\}.
\end{aligned}$$

The first line is the only one that looks cryptic, and it is just an indicator written as an inequality. Its right-hand side $w(u, v) + x_u^{st} - x_v^{st} - 1$ takes one of four shapes according to which side the cut puts the two endpoints u and v on, and [Table 2.3](#) works all four out. Because $w(u, v) \leq 1$, the right-hand side is positive in one case only, the crossing case, where it equals $w(u, v)$ and so forces $p_{uv}^{st} \geq w(u, v)$. In the other three cases it is zero or negative, so the constraint reads $p_{uv}^{st} \geq (\text{something} \leq 0)$ and leaves p_{uv}^{st} free to stay at zero, which the maximisation is happy to let it do.

x_u^{st}	x_v^{st}	meaning	$w + x_u^{st} - x_v^{st} - 1$	forces
1	0	u source side, v sink side: arc <i>crosses</i>	$w(u, v)$	$p_{uv}^{st} \geq w(u, v)$
1	1	both on the source side	$w(u, v) - 1 \leq 0$	$p_{uv}^{st} \geq 0$
0	0	both on the sink side	$w(u, v) - 1 \leq 0$	$p_{uv}^{st} \geq 0$
0	1	u sink side, v source side	$w(u, v) - 2 \leq 0$	$p_{uv}^{st} \geq 0$

Table 2.3. The first constraint, case by case. This is where the bound $w \leq 1$ earns its keep: it makes the three non-crossing rows land at or below zero, so the helper p_{uv}^{st} is pushed up to the arc weight in exactly one case, the crossing one. The constraint is therefore an exact indicator, “count $w(u, v)$ when the arc crosses this cut, and nothing otherwise”, with no slack and no fudge.

[Figure 2.11](#) shows one chosen cut and the weights it counts. The second line of the program now reads plainly. It says the total crossing weight of this pair’s cut, $\sum_{u \neq v} p_{uv}^{st}$, is at most one. By max-flow / min-cut a cut of capacity at most one exists for (s, t) exactly when $\text{maxflow}(s, t) \leq 1$, so the two lines together say “every ordered pair admits a cut of capacity at most one”, which is precisely the feasibility we want. This is the true feasible region, not a relaxation of it, so an optimal solution reported with zero gap is a proof and not merely evidence.

The last two lines change no answer. They only help the solver finish in time. The third is a redundant shortcut. The direct arc $s \rightarrow t$, together with one two-step detour $s \rightarrow x \rightarrow t$ through each intermediate vertex x , already forms that many routes between s and t . So $w(s, t) + \sum_x \min(w(s, x), w(x, t)) \leq 1$ holds of any feasible matrix anyway, and stating it outright just sharpens the continuous picture the solver reasons from.

The $\min$ in that line is not a linear function, so it cannot appear in the MILP as written. To keep

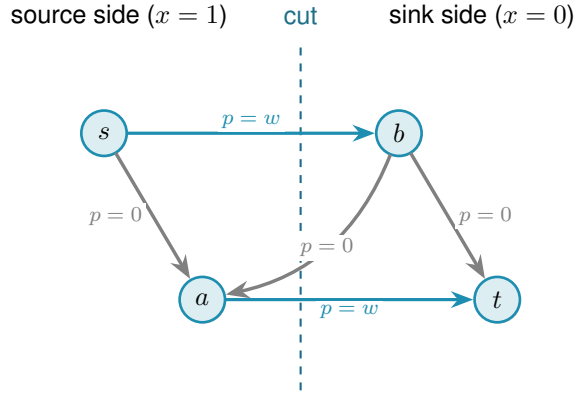


Figure 2.11. One ordered pair's chosen cut. The dashed line splits the vertices into the source side, where $x_u^{st} = 1$, and the sink side, where $x_u^{st} = 0$. Only arcs running from source side to sink side cross the cut, and the first constraint forces their helper p_{uv}^{st} up to the arc weight w (the two thick arcs). Arcs that stay on one side, and arcs that run the other way across the line, contribute nothing. The second constraint then caps the total counted weight, the thick arcs, at one.

the program linear, each $\min(w(s, x), w(x, t))$ is replaced by a fresh helper variable z_x^{st} together with a binary selector $b_x^{st} \in \{0, 1\}$. The selector records which of the two arguments is smaller: $b_x^{st} = 1$ means $w(s, x)$ is no larger than $w(x, t)$. Four linear inequalities then pin z_x^{st} exactly to the minimum. Two upper bounds, $z_x^{st} \leq w(s, x)$ and $z_x^{st} \leq w(x, t)$, prevent z from exceeding either argument. Two lower bounds, one of the form $z_x^{st} \geq w(s, x) - (1 - b_x^{st})$ and the other $z_x^{st} \geq w(x, t) - b_x^{st}$, each become active only when the selector chooses that argument: at any integer value of b exactly one lower bound bites and forces z up to the chosen weight. Together with the upper bounds this pins $z_x^{st} = \min(w(s, x), w(x, t))$ exactly. This technique is called the McCormick linearisation of a minimum. It adds one continuous variable z and one binary variable b per intermediate vertex per ordered pair, all of them appearing only in the tightening constraint and nowhere else, so the proof structure of the program is unchanged.

The fourth line orders the vertices by weighted degree. Renaming vertices cannot change any count, so insisting on this order rules out no graph, and it spares the solver the many relabelled copies of a graph that differ only in vertex names.

Theorem 2.2 (Directed multigraph value, small n). *For $n \leq 6$, $M^*(n) = 2(n - 1)$, and the optimum is attained by a $\{0, 1\}$ weight matrix. Hence for all $m \geq 2$, $L_m^{\text{dir}}(n) = 2(n - 1)(m - 1)$, attained by the double star.*

It is worth being clear about what “for all m ” rests on, because the theorem proves infinitely many values from finitely many solves. It is not that each m was tested. In this mode `solve` never sees m at all. It proves the single m -free number $M^*(n)$, and because that optimum turns out to be attained by a $\{0, 1\}$ matrix, $L_m^{\text{dir}}(n) = (m - 1)M^*(n)$ then carries that one fact to every m at once. The only quantity that is checked case by case, and the only genuine limit on how far the result reaches, is n .

`solve` proves [Theorem 2.2](#) by returning $M^*(3) = 4$, $M^*(4) = 6$, $M^*(5) = 8$, and $M^*(6) =$

10, each optimal with zero gap, so that $M^*(n) = 2(n - 1)$ over this range. The first three finish quickly, $M^*(3)$ and $M^*(4)$ in well under a second and $M^*(5)$ in a few seconds, while $n = 6$ takes about twenty minutes. At $n = 7$ the same fractional encoding does not close within a practical budget, a finite obstacle of scale that once marked the edge of what this method could reach for this variant. It no longer marks the edge of what is *known*: [Appendix A](#) proves $L_m^{\text{dir}}(n)$ for every n and m by an unrelated, elementary argument, so the boundary above is a statement about this one MILP encoding, not about the directed multigraph problem itself. The solver transcript is in [Appendix A](#).

2.8 What the solver may claim

Now that the pieces inside `solve` are built, it is worth stating plainly, because the rest of the thesis depends on it, what each is allowed to claim. The checker *measures*: its connectivity values are exact. Its three proving implementations, the exhaustive enumeration, the pruned search, and the cut-counting, *prove*: an exhausted enumeration or an optimal, zero-gap solve is an upper bound for that fixed n . The search of [Chapter 3](#) will *discover*: it produces concrete graphs, hence lower bounds, and never an upper bound. These three claims are exactly what `solve` labels its answer with, so that an exact value, a lower bound, and an upper bound can never be quietly confused with one another.

With the microscope built and aimed at proof, the variants with small answers are settled and the directed base cases the hand proof needed are in hand. What proof by enumeration cannot do is reach the sizes where the answers are only conjectured, because the number of graphs explodes long before the interesting cases arrive. The next chapter measures that explosion exactly, and then turns the same program loose to *discover* the dense graphs that blind enumeration can never reach.

2.9 Reproducibility

Every claim in this thesis that rests on computation can be re-run from a single Python driver. All of the program's logic lives in `program/erdos915_unified.py`, one self-contained file. Its core needs only `numpy` and `scipy`, because every connectivity value it reports is a single `scipy` integer maximum flow on a capacity matrix. The model, the checker, the search, the enumeration, and the self-test all run on those two libraries alone.

A few more libraries sit on top, each for one job. The certifier uses `pulp`, a Python library for writing an optimisation once and handing it to any solver, and runs it on CBC, the free solver `pulp` bundles, or on the commercial Gurobi solver by a one-line switch. Two others are optional and only for presentation: `matplotlib` draws the figures, and `networkx` supplies the Gomory–Hu tree behind one of them. An optional compiled helper, `_erdos_fast.so` (built from `_erdos_fast.c` by `build_fast.sh`), speeds up the hot-path `_tiny_maxflow` and `_canonical_form` calls, but the pure-Python path returns the same answers.

A handful of named entry points reach the whole pipeline. `solve` drives every variant. The measures are `max_edge_connectivity` and `max_vertex_connectivity`, with the hypergraph and Gomory–Hu views beside them. The provers are `prove_directed_multigraph` and `enumerate_extremal_directed_multigraphs`, and `search_for_dense_graph` is the discoverer. Running `python erdos915_unified.py` executes the built-in self-test, the suite of invariant checks that confirms the connectivity values quoted throughout this chapter, including the sanity checks on the bidirected star and on K_n and $K_4^{(3)}$.

The figures and proofs are just as reproducible. Every figure is regenerated by the companion driver `program/make_figures.py`, which imports the same file and fixes every random seed, so the pictures are reproduced exactly rather than merely resembled, and the table in `program/README.md` records which routine produces which figure. The solver transcripts behind this chapter’s certified values, the pruned-search log for [Theorem 1.10](#) and the cut-counting certificate for [Theorem 2.2](#), are reproduced verbatim in [Appendix A](#), so a reader may check the proofs without running anything at all.

2.9.1 What a machine-checked claim in this thesis means

Reproducibility is not the same as certification, and it is worth stating plainly what standard the computed results here meet, since a solver reporting `OPTIMAL` is not by itself a mathematical certificate. Four practices carry the weight.

First, the exhaustive searches are self-certifying in a way the optimisations are not. A finished include-or-exclude sweep has visited a superset of the objects it needed to visit, and the two prunes are proved sound in [Chapter 3](#), so its verdict depends on no solver and no floating-point arithmetic, only on the checker. Where a value in this thesis is called *proved*, that is usually the reason.

Second, every reported connectivity is exact and integral. The checker is a single integer maximum flow, so no extremal value in the thesis is a rounded floating-point quantity. Real numbers enter in exactly one place that carries a reported extremal value, the fractional optimisation of [Theorem 2.2](#), where the reported optimum is a $\{0, 1\}$ matrix and is checked exactly.

Third, the load-bearing computations are run twice, by implementations that share no code. The base cases of [Theorem 1.10](#) and [Theorem 1.11](#) were each confirmed by the program and by separately written checkers, agreeing on every size. The values of [Section A.9](#) were computed by the program’s routine and by an independent script, agreeing on every cell. The directed multigraph classification at $n = 7$ ([Remark A.41](#)) was produced by the generation enumerator and re-verified graph by graph with the exact checker. The integral MILP of this chapter re-proves $L_3^{\text{dir}}(n)$ independently of the fractional route at every $n \leq 6$, which is as far as it ever got: at $n = 7$ it timed out, and the value $L_3^{\text{dir}}(7) = 24$ rests on the hand proof of [Appendix A](#) alone. The route-counting steps behind [Theorem A.28](#) and [Theorem A.59](#) were checked the same way before they were written down, over several thousand random digraphs and hypergraphs and every ordered pair in each, by a script that shares no code with this program. Those two are hand proofs and

do not need the check to stand, but the check is what caught that both inequalities are attained with equality in real instances, which is how one knows the proof is not throwing anything away. Two implementations agreeing is not a proof, but it is the practical guard against the failure mode a certificate would catch, namely a correct method with a wrong program.

Fourth, the self-test at the foot of the file is a standing regression check on every invariant the text quotes, and it is run after every change to the program.

What this does not amount to is a formally checkable certificate of the kind a SAT solver can emit, such as a resolution proof a small independent verifier could replay. That standard would have been the right one to aim at for $L_3^{\text{dir}}(7) = 24$ had it stayed a solver question, since an INFEASIBLE verdict is exactly the shape of claim a resolution proof records. It did not stay one: [Appendix A](#) proves the same statement, and its general case for every n and m , by hand, so nothing in this thesis currently rests on an infeasibility verdict that lacks such a proof behind it.

Chapter 3

Discovering Bounds by Search

The previous chapter proved what could be proved by looking at every graph of a fixed size. That method is honest and exact, and it stops almost at once. This chapter begins by measuring exactly how fast it stops, because the speed of that collapse is the whole reason a second kind of tool is needed. Then it builds that tool. Where the certifier walks all graphs, the search walks a few of them well, cooling like a physical system toward the densest feasible shape. What it returns is never a proof. It is a concrete graph, hence a lower bound and a conjecture, a stepping stone toward an upper bound that a later proof or certificate must supply. The chapter ends by asking the question any such tool must face: how good is it, and how far from the truth might its answers be.

3.1 The wall: how enumeration explodes

The brute force of [Chapter 2](#) visits every graph of a given size. The trouble is how many that is. A simple undirected graph on n vertices is a yes-or-no choice on each of the $\binom{n}{2}$ possible edges, so there are $2^{\binom{n}{2}}$ of them. A digraph chooses on each of the $n(n-1)$ ordered pairs, giving $2^{n(n-1)}$, and allowing multiplicities up to a cap c raises the base from two to $c+1$. These are not numbers that merely grow. They detonate. [Figure 3.1](#) draws the growth on a logarithmic scale, where each step up the axis multiplies the count tenfold, and the message is the same in every panel: the count passes any reachable budget while n is still in the single digits.

Three readings of the figure matter for what follows. Adding vertices is the sharpest cost: each new vertex multiplies the count by an exponential in n , so a single extra vertex can move a problem from seconds to centuries. Raising the connectivity budget is the next cost, lifting the base of the exponent rather than its height, which is why the multigraph curves sit above the simple one in every panel. Direction is the third: a digraph has twice as many cells to fill as the undirected graph on the same vertices (and a directed 3-uniform hyperedge, a tail with two heads, has three times as many), so the directed panel climbs fastest of all, and those are precisely the variants whose answers [Chapter 1](#) could not prove. The separation, by contrast, costs nothing here, because edge and vertex disjointness search the same graphs and differ only in what the checker measures, which is why a single pair of panels suffices for all of them. The certifier of [Chapter 2](#) reaches a few vertices further than blind enumeration, because its cut-counting optimisation is polynomial in the size of the encoding rather than in the number of graphs, but solving that encoding is itself hard and the gain is measured in single vertices, not orders of magnitude. Past that, no exhaustive method reaches the interesting sizes. Something

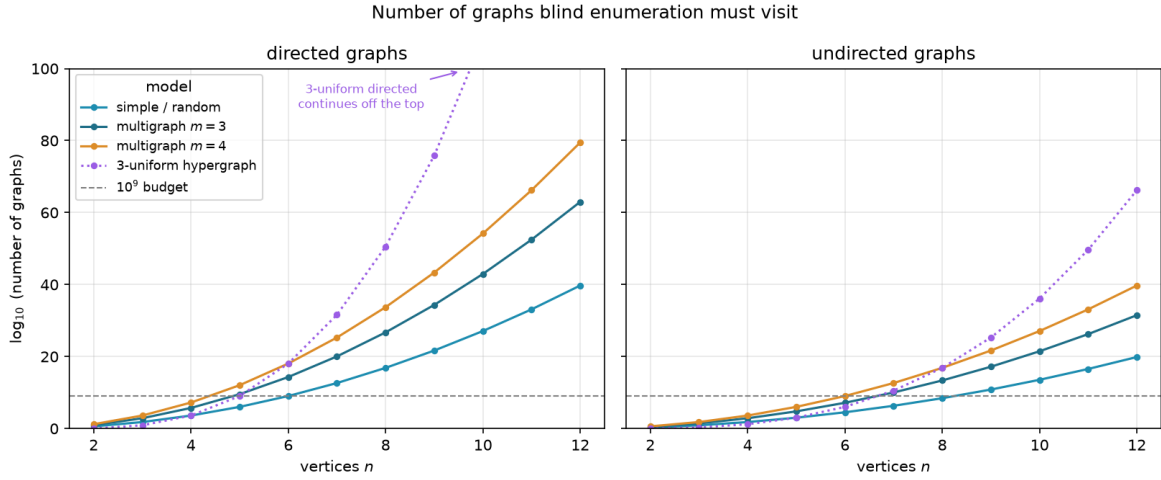


Figure 3.1. The number of graphs blind enumeration must visit, on a logarithmic vertical axis, for directed graphs (left) and undirected graphs (right). Each panel shows the simple model, two multigraph connectivity caps, and the three-uniform hypergraph (dotted), with a dashed line at a generous budget of 10^9 candidates, and every curve crosses that budget while n is still in the single digits. Writing E for the number of off-diagonal cells a model may fill, a simple graph offers 2^E candidates, a multigraph with multiplicity cap $m - 1$ (the connectivity ceiling) offers m^E (one choice per cell from $\{0, \dots, m - 1\}$), and the 3-uniform hypergraph offers $2^{\binom{n}{3}}$ undirected or $2^{n \binom{n-1}{2}}$ directed, since a directed hyperedge is a tail together with two heads. The count depends only on the model and the direction, never on whether one later asks for edge- or vertex-disjoint routes, so a single pair of panels covers all of those variants at once. Direction multiplies the number of cells to fill, by two for graphs (ordered versus unordered pairs) but by three for the 3-uniform hypergraph, so the directed panel sits far above the undirected one, and the directed cases, exactly the ones [Chapter 1](#) left open, are the first to go.

that does not look at every graph is required.

3.2 The temperature search

We want, for fixed n and m , a graph with as many edges as possible subject to $\lambda^{\max} \leq m - 1$ (or $\kappa^{\max} \leq m - 1$ for the vertex variants). The space of graphs on n vertices is enormous and its feasible region has no helpful shape: adding an edge can be harmless until, together with edges added long before, it suddenly creates a forbidden pair. A greedy walk that only ever accepts an improving move stalls at the first such collision, in a graph that is locally maximal but globally far from extremal.

The classical method for this kind of landscape is *simulated annealing* (SA), named after the metallurgical process of cooling a heated material slowly so that its atoms settle into a low-energy crystal rather than a disordered glass. Applied to graphs, annealing escapes local optima by sometimes accepting a worsening move, and by accepting fewer of them as the run proceeds. Each graph G is a state with energy

$$E(G) = -|E(G)| + \Lambda \cdot \max(0, \lambda^{\max}(G) - (m - 1)),$$

Algorithm 1: Temperature search for a dense feasible graph

Input: vertices n , value m , start T_0 , cooling α , penalty Λ , steps S
Output: the densest feasible graph encountered

```
1  $G \leftarrow$  empty graph on  $n$  vertices;  
2  $T \leftarrow T_0$ ;  
3 for  $S$  steps do  
4   propose  $G'$  by toggling one adjacency, biasing removals toward low-sensitivity edges;  
5    $\Delta \leftarrow E(G') - E(G)$ ;  
6   if  $\Delta \leq 0$  or  $\text{rand}() < e^{-\Delta/T}$  then  
7      $G \leftarrow G'$ ;  
8   if  $G$  is feasible and denser than the best so far then  
9     record  $G$  as the best;  
10   $T \leftarrow \alpha T$ ;  
11 return the best recorded graph;
```

which rewards edges and penalises any excess connectivity. The penalty is a soft constraint rather than a hard one, so the walk may pass briefly through infeasible graphs rather than being walled inside the feasible region, and we return below to why that freedom is worth having. A move toggles one adjacency. The Metropolis rule accepts an improving move always and a worsening move only with probability $e^{-\Delta E/T}$, a chance that shrinks both as the move gets worse and as the temperature drops, and the temperature T falls geometrically, multiplied by a fixed factor just below one at every step. Algorithm 1 is the whole loop, the one place in this chapter where a body is worth showing, because the loop *is* the method.

This loop is what `solve` runs in its discovery mode. With the exhaustive flag off it anneals instead of enumerating, following the Metropolis walk of Algorithm 1 under geometric cooling, removals biased toward low-sensitivity edges, and returns the densest feasible graph it encountered together with the full temperature trace. What it returns is a witness, hence a lower bound, never a proof.

`search_for_dense_graph(n, m, variant, separation) → SearchResult`

DISCOVER

in the size n , the ceiling m , the `Variant`, the separation, and a step budget

out a `SearchResult`: the densest feasible graph found, with the full step-by-step trace

how one Metropolis cooling run of Algorithm 1. The driver `best_of_searches` dispatches several independent runs across processor cores and keeps the densest, so a single local optimum never decides the answer.

It is fair to ask why the walk is allowed above the ceiling at all. Reaching every feasible graph does not require it. Removing an edge can only lower $\lambda^{\max}$ and $\kappa^{\max}$, never raise them, so every subgraph of a feasible graph is again feasible, and the empty graph is feasible trivially. From any feasible target we can therefore delete edges one at a time down to the empty graph without ever crossing the ceiling, and reading that path backward builds the target up the same way. The

feasible region is connected through the empty graph by single-edge moves that stay inside it, so a walk that never went infeasible could in principle visit every feasible graph there is.

We let it go infeasible anyway, because reachability and efficiency are different things. A feasible graph can be locally maximal, meaning every edge we might add creates a forbidden pair, and yet still sit far below the densest feasible graph. To improve such a graph the walk has to add an edge and then remove a different one, a swap whose first half is infeasible. A walk confined to feasible graphs could only find that better graph by retreating most of the way back to the empty graph and climbing a different slope, a detour so long that the cooling schedule usually ends first. Allowing a brief excursion above the ceiling lets the walk tunnel straight across instead, trading one load-bearing edge for a better one in two moves rather than dozens. The soft penalty also turns the search into a single smooth landscape with no hard walls for the Metropolis rule to bounce off, and the same machinery then carries over unchanged to variants whose feasible region is not so conveniently connected. Going above the bound is never the goal, then. It is the shortcut the annealer takes on its way back down to it.

3.3 Temperature and edge sensitivity

The phrase “biasing removals toward low-sensitivity edges” in [Algorithm 1](#) is where the temperature acquires its meaning, and it is the heart of the method. For an edge e define its *sensitivity*

$$\sigma(e) = \lambda^{\max}(G) - \lambda^{\max}(G - e),$$

the amount by which deleting e relaxes the binding connectivity. A load-bearing edge sits on a tightest cut and has $\sigma(e) > 0$. Removing it would drop $\lambda^{\max}$ and so loosen the very constraint that limits how many edges the graph may hold. A free edge has $\sigma(e) = 0$: it can be removed without easing the binding pair, which usually means it can be put to better use elsewhere.

`edge_sensitivity(G, u, v) → ℕ`

MEASURE

in a graph G and one of its edges $e = (u, v)$

out $\sigma(e) = \lambda^{\max}(G) - \lambda^{\max}(G - e)$

how delete e and let the checker remeasure, where a positive value marks a load-bearing edge. The whole-graph version `sensitivity_map(G)` reuses one shared baseline measurement. This single number is the dial the cooling turns.

When the search proposes a removal it draws the edge with probability proportional to $e^{-\sigma(e)/T}$. The dial is now explicit. At high temperature these weights flatten and the walk toggles edges regardless of how load-bearing they are, exploring widely and willing to dismantle essential structure. As the temperature falls the weights concentrate on $\sigma = 0$, so the walk almost never disturbs a load-bearing edge and instead rearranges only the free ones, until the graph crystallises onto an extremal shape. This is exactly the behaviour one wants, and it is the behaviour the formulation asks for: the temperature tracks how strongly removing an edge

changes the bound. Figure 3.2 shows one such run settling onto a certified optimum. Section B.2 collects five further traces, covering both matrix models in both directions and one run under the vertex separation, and the behaviour is the same in each.

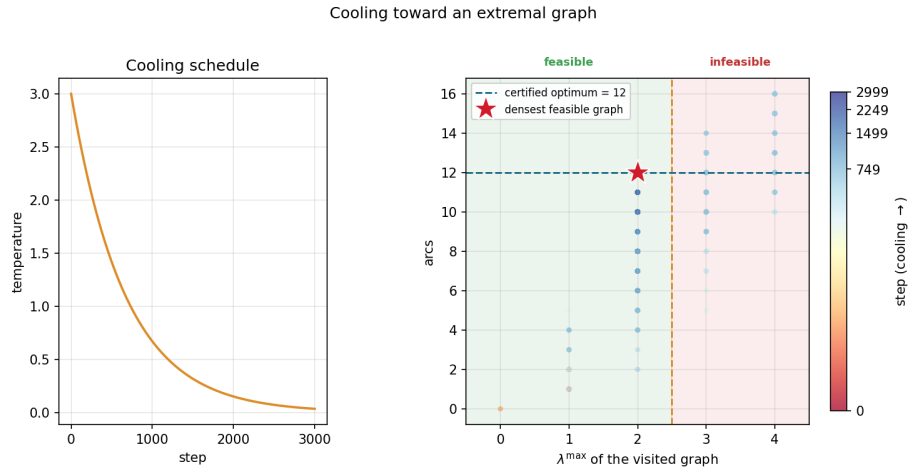


Figure 3.2. A single cooling run on the directed multigraph problem at $n = 4$, $m = 3$. *Left:* the temperature schedule, cooling geometrically from its starting value. *Right:* every visited graph plotted by its binding connectivity $\lambda^{\max}$ (horizontal) against its arc count (vertical), coloured by the step at which it was visited. Annealing uses a soft penalty rather than a hard wall, so while hot (dark points) the walk strays into the infeasible region $\lambda^{\max} > m - 1 = 2$, picking up extra arcs it is not allowed to keep. As the temperature falls (bright points) the penalty drives the walk back across the feasibility boundary and onto the densest graph that stays feasible, the certified optimum of 12 arcs at $\lambda^{\max} = 2$ (star).

The same sensitivities also explain a finished graph. Colouring a graph's arcs by σ shows at a glance which arcs are structural and which are slack. Figure 3.3 illustrates all four sensitivity levels on one directed multigraph: the darkest arcs carry the most flow and lower $\lambda^{\max}$ the most when removed, the cool arc contributes nothing. The picture is not decoration: it is the certificate of tightness made visible, and it tells the search exactly which arcs to leave alone as the temperature falls.

A single cooling schedule can still settle into a local optimum, so in discovery mode `solve` runs several independent schedules from different starting seeds and keeps the densest result across all of them. Because the restarts share no state and none depends on the outcome of another, they run in parallel. The program dispatches them across the available processor cores, so the wall-clock cost of k restarts is a small multiple of a single run rather than k times it, bounded below by the slowest restart and by the cost of starting the worker processes. A handful of parallel restarts is enough to reach the known optima at the sizes we examine. This is a reliability measure, not a change of method, and the parallelism is invisible to every number the thesis reports: the same best graph would be found by running the schedules sequentially, just more slowly.

Parallel arcs drawn explicitly, coloured by sensitivity σ :
cool ($\sigma = 0$, free) $\rightarrow$ warm (load-bearing)

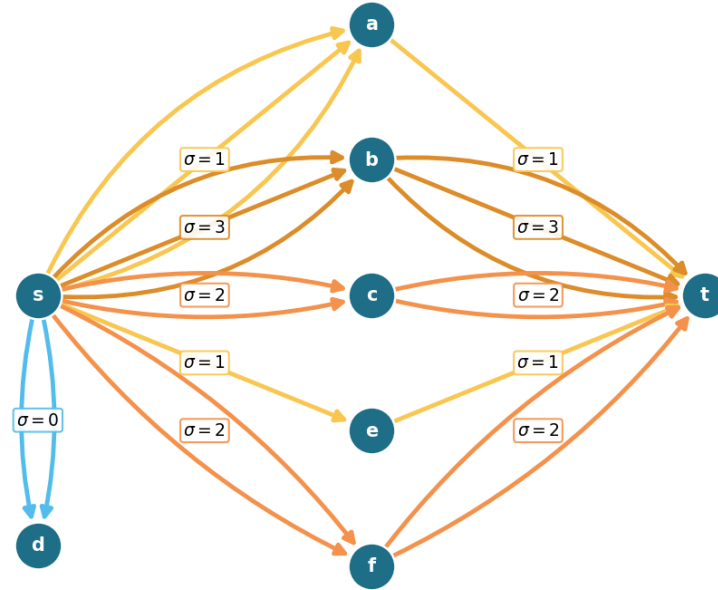


Figure 3.3. A directed multigraph on eight vertices with $\lambda^{\max} = 9$, every parallel arc drawn explicitly and coloured by its sensitivity σ , so multiplicity is read by counting arcs rather than from a label. The point is the contrast between $s \rightarrow a$ and $s \rightarrow b$: both carry three parallel arcs, yet $s \rightarrow a$ is cool ($\sigma = 1$) because the single arc $a \rightarrow t$ downstream throttles that route to one unit, while $s \rightarrow b$ is hot ($\sigma = 3$) because all three of its copies carry flow. The arcs $s \rightarrow d$ are coolest of all ($\sigma = 0$): d is a dead end, so deleting them changes nothing. Removing a hot adjacency drops $\lambda^{\max}$ by its σ while removing a cool one does not, which is exactly what tells the search which arcs to leave alone as the temperature falls.

3.4 Keeping the checker cheap inside the search

The energy of [Algorithm 1](#) reads the binding connectivity $\lambda^{\max}(G)$, and a literal reading of the loop would call the checker afresh on the whole graph at every step. That checker is the most expensive routine in the program, one maximum flow per ordered pair, and the walk runs for thousands of steps with several restarts, so such a search would spend nearly all of its time remeasuring graphs that barely changed from the step before. Two elementary facts remove almost all of that work without altering a single number the search reports.

Proposition 3.1 (Monotonicity of the binding connectivity). *Let G be a graph or directed multigraph, and let G' be obtained from G by changing one adjacency by a single unit of multiplicity. Then*

- (1) (monotonicity) *removing the unit cannot raise $\lambda^{\max}$, and adding it cannot lower $\lambda^{\max}$, and*
- (2) (unit step) *adding the unit raises $\lambda^{\max}$ by at most one.*

Both statements hold verbatim for the vertex measure $\kappa^{\max}$.

Proof. Fix an ordered pair (s, t) . By the max-flow / min-cut theorem $\lambda_G(s, t)$ is the smallest capacity of any cut separating s from t , where the capacity of a cut is the total multiplicity of the arcs that cross it from the source side to the sink side.

For part (1), reducing one adjacency by a unit lowers the capacity of every cut that the changed arc crosses and leaves all other cut capacities untouched, so no cut capacity rises. The smallest cut capacity therefore cannot rise either, which gives $\lambda_{G'}(s, t) \leq \lambda_G(s, t)$. Taking the maximum over all pairs yields $\lambda^{\max}(G') \leq \lambda^{\max}(G)$. Adding a unit is the same statement read backwards.

For part (2), write G'' for G with one unit added to a single adjacency. That extra unit of capacity crosses any fixed cut at most once, so every cut has capacity at most one greater in G'' than in G . The smallest cut capacity thus grows by at most one, giving $\lambda_{G''}(s, t) \leq \lambda_G(s, t) + 1$, and together with part (1) we have $\lambda_G(s, t) \leq \lambda_{G''}(s, t) \leq \lambda_G(s, t) + 1$. The maximum over pairs gives $\lambda^{\max}(G'') \leq \lambda^{\max}(G) + 1$.

For $\kappa^{\max}$ the argument is identical on the vertex-split network of [Figure 2.3](#), where changing an adjacency changes the capacity of its single arc $u^{\text{out}} \rightarrow v^{\text{in}}$ in the same way. Parallel copies leave that capacity at one and so change $\kappa^{\max}$ not at all, which only strengthens both parts. $\square$

These two facts settle almost every step without a flow. The energy depends on $\lambda^{\max}$ only through the excess $\max(0, \lambda^{\max} - (m - 1))$, which is zero exactly when the graph is feasible. So a removal applied to a feasible graph leaves it feasible by part (1), with excess still zero, and needs no measurement at all. An addition made while the graph sits strictly below the bound, at $\lambda^{\max} \leq m - 2$, stays feasible by part (2), which caps the rise from a single added unit at one, so the graph can move at most to $\lambda^{\max} = m - 1$, still inside the feasible region, again with excess zero and no measurement. Only an addition made right at the boundary $\lambda^{\max} = m - 1$ can tip the graph over, and only there does the search run a single flow, capped so that it stops the instant it finds one route too many. The exact connectivity value is recomputed only on the rare step whose proposal is genuinely infeasible, where the penalty term needs it. The three cases of [Proposition 3.1](#) read directly off the code:

```

1 if (is_add and lam_ub <= m - 2) or (not is_add and feasible_current):
2     excess = 0                                # part (2) or part (1): provably feasible
3 elif not exceeds_bound(proposal, m - 1, separation=separation):
4     excess = 0                                # at the boundary, but the flow clears it
5 else:
6     excess = measure_full(proposal) - (m - 1) # infeasible: exact penalty

```

Listing 3.1. The three cases behind [Proposition 3.1](#), in the order the proof gives them: removal from a feasible graph (part 1), addition below the bound (part 2), and only otherwise a single capped flow.

To know which case applies, the search carries a running upper bound on $\lambda^{\max}$, raised by one

Construction	n	m	Search	Theory
spanning tree (simple undirected, $m=2$)	6	2	5	5
multiplicity- $(m-1)$ star (multi undirected, $m=3$)	6	3	10	10
bidirected star (simple directed, $m=2$)	4	2	6	6
bidirected star (simple directed, $m=2$)	6	2	10	10
hub-bipartite tie (simple directed, $m=2$)	7	2	12	12
double star (multi directed, $m=3$)	4	3	12	12
double star (multi directed, $m=3$)	5	3	16	16

Table 3.1. Validation by rediscovery. For each variant the temperature search, run from scratch with a handful of restarts, recovers exactly the extremal value proved or certified in the earlier chapters. The “Search” column is an honest lower bound found by annealing, and the “Theory” column is the proved or certified value.

whenever it accepts an addition (part (2)) and left untouched when it accepts a removal (part (1)), and resynchronised to the exact value at the cadence where it already pays for flows to refresh the sensitivity map. The bound can only ever sit at or above the truth, so any step it clears is certainly feasible.

The key point is that the energy returned on every step is, by [Proposition 3.1](#), exactly the value the naive implementation would have computed. Every accept-or-reject decision, every random draw, and the graph the walk finally returns are therefore unchanged, and the speed-up is invisible to every number the thesis reports. This is verified rather than asserted. A regression test runs both implementations, the fast one and a reference that measures exactly at every step, across all four graph variants and both separations, and checks that the two agree step for step in energy, in every accept-or-reject decision, and in the final graph, and that the returned graph is feasible under the exact checker. Nothing about the optimisation hides inside the bound: the witness the search reports is always certified by an exact measurement, exactly as in the slower search.

It is worth noting where this method earns its keep. For undirected graphs one could resynchronise the exact value cheaply from the Gomory–Hu tree of [Chapter 2](#), which encodes every pairwise edge-connectivity in $n - 1$ flows. The directed and vertex variants admit no such tree, so there the monotone bound and the capped predicate are the only inexpensive handle on feasibility, and they are what let one annealer serve every variant at speed.

3.5 Rediscovering the known extremisers

A search method is only worth trusting if it recovers what is already known before it is pointed at what is not. Run from the empty graph on the cases whose answers [Chapter 1](#) and [Chapter 2](#) settled, and told only to pack in edges without breaching the connectivity ceiling, the cooled search arrives unprompted at the named constructions. [Table 3.1](#) reports the densest graph it found against the value the theory predicts, and in every case the two agree.

What the table hides is that the graphs themselves are the named constructions, not merely

graphs of the right size. The simple undirected run at $m = 2$ settles on a spanning tree, the only shape with $n - 1$ edges and no cycle. The undirected multigraph run settles on a star with every edge at multiplicity $m - 1$, an instance of the multi-tree extremiser behind [Theorem 1.3](#). The directed multigraph runs settle on the double star of [Chapter 2](#), both directions of every spoke at full multiplicity. The simple directed runs find the bidirected star at $n = 4$ and, at $n = 7$, land on twelve arcs, the value at which the hub and the one-directional wall tie, exactly the crossover [Theorem 1.10](#) predicts. The cooled search does not know these names. It rediscovers the structures because, under the connectivity ceiling, there is nowhere denser to go.

Some extremisers are past the size at which a quick search lands reliably, but the exact checker of [Chapter 2](#) settles them immediately by measuring their connectivity. The one-directional bipartite digraph on eight vertices has 16 arcs at $\lambda^{\max} = 1$, and 16 already exceeds the linear hub value $2(n - 1) = 14$: the smallest size at which the quadratic phenomenon strictly wins, made concrete. It is also a measured illustration of the search's limit: repeated cooling runs at $n = 8$, $m = 2$ stall at the hub value of 14, because dismantling a finished hub and rebuilding a one-directional wall is far more than the single-arc moves of the walk can tunnel through. The known answer there comes from the construction and the checker, not from the search, which is exactly the division of labour this chapter argues for. The augmented bipartite digraph on ten vertices at $m = 3$ has 30 arcs at $\lambda^{\max} = 2$, the thirty-arc counterexample of [Remark 1.12](#), confirmed pair by pair against the naive bound of 27 it refutes. The directed double star on seven vertices at $m = 4$ has 36 arcs at $\lambda^{\max} = 3$, matching $2(n - 1)(m - 1)$. None of these is a proof of optimality. Each is an exactly measured feasible graph, a lower bound the checker stamps as genuine.

3.5.1 Simulated annealing versus tabu search

The annealer is not the only way to walk this landscape. A natural alternative, suggested by Jan Goedgebeur, is *tabu search* ([Table 4.2](#) names the implementation), which keeps the energy and the neighbourhood of [Algorithm 1](#) but replaces the cooling schedule with a deterministic best-improving step. At each step it evaluates every single-edge move from the current graph and takes the one that lowers the energy most, then forbids the pair it just changed for a fixed number of steps, the *tabu tenure*, so the walk cannot immediately undo its last move and stall in a two-step cycle. When no improving move is left and the search stalls, it perturbs the incumbent with a few random moves and carries on. There is no temperature and no accept-worse coin: apart from the perturbation kicks the walk is deterministic. Where annealing explores by occasionally stepping uphill and trusts the cooling to settle it, tabu climbs greedily and trusts the tabu list to keep it from retracing its path.

The two engines rediscover the same values, but at very different speeds, and on the harder directed cases the difference is decisive. [Table 3.2](#) runs them head to head at an equal budget of six seconds each, both restarted within that budget. On the simple undirected problem both reach the optimum at once. On the two directed problems past the certified range tabu reaches the optimum within the budget while the annealer plateaus below it, assembling the eighteen-arc simple-directed extremiser and the full twenty-four-arc double star at $n = 7$ where the annealer

stalls at sixteen and twenty. The cause is the one already met at the $n = 8$, $m = 2$ hub stall above. The annealer cools onto whatever dense shape it first reaches and then cannot dismantle it within the budget, whereas tabu’s best-improving step follows the steepest density gradient and the tabu list stops it sliding back, so it assembles the structured extremiser the cooled walk struggles to find.

Variant	n	m	optimum	best in 6 s		time to optimum	
				SA	tabu	SA	tabu
simple undirected	7	3	9	9	9	1.3 s	0.1 s
simple directed	7	3	18	16	18	—	2.6 s
directed multigraph	7	3	24	20	24	—	2.5 s

Table 3.2. Simulated annealing (SA) against tabu search at an equal six-second budget per method, all at $m = 3$, the annealer restarted from fresh seeds and tabu restarted on stalls. Only the simple undirected value is a proved optimum (Mader). The two directed values are the best known rather than proved, found by search and construction past the certified range that stops below $n = 7$ for those variants (Section 3.6), so here the “optimum” column is the target each engine is trying to reach. Both reach the simple undirected optimum. On the two directed problems tabu reaches the best known value while the annealer plateaus below it. The last two columns give the wall-clock time each method first reached that value on the reference machine, with a dash where it did not reach it in the budget.

Figure 3.4 shows the divergence directly, plotting the densest feasible graph each engine has found against wall-clock time on the two directed multigraph cases. Tabu climbs in a few decisive steps and holds at the optimum, while a single annealing schedule rises quickly and then flattens a few arcs below it. The annealer’s independent restarts recover the optimum on the smaller case, as Table 3.1 records, so a single plateau is not the whole story there. But at $n = 7$ the plateau persists within the budget, and one tabu schedule already reaches what the restarted annealer does not.

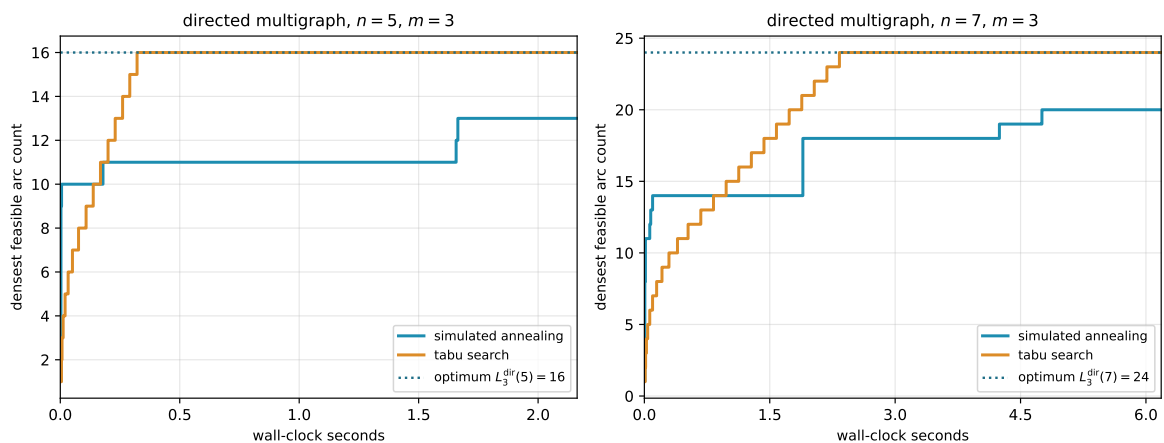


Figure 3.4. Densest feasible graph found against wall-clock time, for one simulated-annealing schedule and one tabu schedule on the directed multigraph at $n = 5$ and $n = 7$, $m = 3$. Tabu climbs in a few best-improving steps to the optimum (dotted) and holds, while a single annealing schedule rises quickly and then flattens below it. The time axis is wall-clock on the reference machine, so unlike the seed-reproducible figures elsewhere this one is a representative timed run.

This advantage comes at a cost the equal-time comparison already absorbs but is worth naming: each tabu step scans the whole neighbourhood, so it is far more expensive than the annealer’s single random proposal, and tabu takes many fewer but better-aimed steps in the same wall-clock. Tabu search is therefore the program’s default for solving the problems, reaching every value this thesis reports, the rediscovery of [Table 3.1](#) and the conjectural lower bounds alike, and reaching the harder directed optima past the certified range where the annealer plateaus. Simulated annealing is kept for the self-check and verification, where the small values fall at once, the lighter per-step cost is all that is needed, and its independent restarts parallelise trivially across cores.

3.6 The trichotomy: proved, certified, conjectured

It is worth stopping to lay out, for every variant, the three kinds of knowledge the computation can hold about it, because the whole value of the pipeline is that they are kept apart and never quietly merged. About any given value the program can do one of three things. It can *prove* the value, closing a fixed size from above by exhausting every graph of that size, an upper bound with the standing of a hand proof. It can *certify* it, the same kind of upper bound but obtained by the cut-counting optimisation rather than blind exhaustion, reaching a vertex or two further. Or, where proof and certificate both run out, it can only *conjecture* the value, exhibiting a concrete dense graph that pins down a lower bound and leaving the matching upper bound to a later argument.

Proof and certificate bound the answer from above. The search bounds it only from below. Where a closed *formula* exists it can tie the two together, threading through the certified small cases and the conjectured large ones as a single curve. This is the discipline the whole thesis runs on, and it is worth stating as one: the search proposes a lower bound, a certificate or a hand proof disposes the upper bound, and only when both meet is a value called settled.

We begin with the single variant where all three kinds of knowledge are non-trivial and visibly distinct, the directed simple arc problem at a fixed $m \geq 3$, and then sweep the whole family. For that variant the knowledge splits cleanly by the size of the graph.

For small n the answer is not conjectured at all. The pruned exhaustion inside `solve` walks every simple digraph on n vertices and returns the exact maximum, complete and certain, an upper bound with the standing of a hand proof. At $m = 3$ it certifies 6, 9, 12 for $n = 3, 4, 5$, and there the exhaustion stops within a reasonable budget, the pruned walk no longer closing $n = 6$ in tractable time. That is the certified region, finite and closed.

Beyond it the ceiling is gone and only lower bounds remain, each witnessed by a concrete feasible graph. The search finds the extremal 15 arcs at $n = 6$ on its own. Past that size it falls behind: across six restarts it reaches 17, 17, 20, 23 for $n = 7, \dots, 10$, while the named constructions of [Chapter 1](#), measured exactly by the checker, supply 18, 21, 25, 30, the last being the thirty-arc graph of [Remark 1.12](#). The reported lower bound at each size is the better of the two, exactly as the rediscovery section combined search and checker by hand. These are conjectures with an honest floor and no matching ceiling. That is the searched region, unbounded and forever

one-sided.

The two regions are tied together by a single closed form. The conjectured value

$$\ell_m^{\text{dir}}(n) = \max\left(m(n-1), \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor\right) \quad (n \geq m)$$

of [Conjecture 1.14](#) is not a fresh guess bolted on past the certified range. It is one formula that already passes through every certified value at small n and then continues, without a seam, through every searched value at large n . The directed-arc panels of [Figure 3.5](#) show this directly: the certified squares, proved exactly at the small sizes the certifier reaches, sit on the one red conjecture curve there, and the open lower-bound circles, the best feasible graph the search finds at each larger size, sit on that same curve past that point, matching it as far out as the search goes without ever crossing above it. The formula is the through-line that makes the small certified cases and the large conjectured ones a single statement rather than two.

Two features of those directed-arc panels are worth naming. The crossover from the linear hub term to the quadratic bipartite term happens at $n = 9$, where the second argument of the maximum first wins, and that is past the last certified size. The shape of the answer therefore changes in exactly the region where only the search can see it, which is precisely why a discovery tool was needed and a certifier alone would not do. And at $n = 10$ the formula reads 30, the count of the very counterexample that overturned the natural initial conjecture, so the closed form is not a smooth fit to easy data but one that survives the case that broke the previous guess.

Asking the same three questions of every variant, the answers sort into three regimes, and [Figure 3.5](#) shows all twelve at once, one panel per variant, in exactly this proved, conjectured, and guessed language.

In the first regime the closed form is already a theorem for every n , proved by hand in [Chapter 1](#) or cited, and the computation only confirms it. The undirected edge families live here for every m , together with the directed problems at $m = 2$, the directed multigraph arc problem at every m ([Theorem A.30](#)) and, up to a threshold in m that differs by model, the undirected vertex families as well: through $m \leq 4$ for simple graphs and multigraphs ([Theorem 1.5](#)) and through $m \leq 3$ for hypergraphs ([Theorems A.50](#) and [A.53](#)). Past those thresholds the same rows drop into the third regime below, which is why the leaf colours of [Figure 4.7](#) are stated at general m and [Table 4.1](#) carries the exact ranges. For these the certified small cases and the rediscovered constructions of [Table 3.1](#) are validation, not new knowledge: the formula was never in doubt, and the program's role is to check that the pipeline reproduces it. That the two multigraph vertex rows measure exactly as their underlying simple graphs is a matter of the counting convention of [Remark 2.1](#) rather than a theorem, so those rows need no computation of their own at all. The directed vertex rows do need their own computation. Whitney's inequality $\kappa^{\max} \leq \lambda^{\max}$ means every graph feasible for the arc problem is automatically feasible for the vertex problem too, since a small $\lambda^{\max}$ forces an even smaller $\kappa^{\max}$. So the arc problem's own dense construction doubles as a valid witness, hence a lower bound, for the vertex problem. But it supplies no upper bound: a

bound proved only for the smaller arc-feasible family says nothing about the larger vertex-feasible family, which may contain denser graphs the arc argument never had to rule out. So they get their own: every exact point on those panels comes from the search and the exhaustion run under the vertex test.

In the second regime the three columns genuinely differ and the computation is load-bearing. This is the regime of the worked example above. The directed simple arc and vertex problems at $m \geq 3$ are certified only at the small n the exhaustion reaches and conjectured by the search beyond, with the formula of [Conjecture 1.14](#) the through-line.

The directed multigraph arc problem used to sit here and no longer does. It has moved up into the first regime, since [Theorem A.30](#) proves $L_m^{\text{dir}}(n) = (m - 1) \max(2(n - 1), \lfloor n^2/4 \rfloor)$ for every n and every m , with no parity split and no auxiliary conjecture. Its panel is worth watching all the same, because it is the one place in the grid where the computation and the proof were developed against each other rather than in sequence: in its cut-counting mode `solve` proves the m -free number $M^*(n) = 2(n - 1)$ outright for $n \leq 6$, and the scaling identity $L_m^{\text{dir}}(n) = (m - 1) M^*(n)$ of [Chapter 2](#) carries that one certified number to every m at once. Those certificates were the evidence that the closed form was right long before the proof of it existed, and they now serve as an independent check on a theorem rather than as the reason to believe a conjecture.

In every row of this regime the formula agrees with proof where proof reaches and with the search where it does not, exactly the linkage the worked example displayed.

In the third regime there is no formula at all, and the honesty is in drawing the guess as a guess. The undirected vertex problem at $m \geq 6$ has been open since 1974 with its extremal structure genuinely unknown. The undirected hypergraph vertex problem joins it, but only from $m \geq 4$: at $m \leq 3$ its value is settled for every uniformity and every n ([Theorems A.50](#) and [A.53](#)), which is why that panel of [Figure 3.5](#) carries a proved blue curve while its counterpart in [Figure 3.6](#) carries a yellow guess. The two *directed* hypergraph variants are in the third regime at every m , and they need a one-tail, many-head reading of a Berge edge that this thesis defines itself only so that `solve` can measure it at all. Here the program still computes exact values at the few sizes that finish, and its search still exhibits feasible graphs at a handful more, but no closed form is in sight. What is known there is asymptotic rather than exact: [Theorem A.59](#) pins the leading term at $(m - 1)n^2/(4(r - 1))$ for both separations, which is a statement about where these curves are heading and not one a panel at $n \leq 10$ can display. The panels of [Figure 3.5](#) carry, instead of a red conjecture curve, a yellow line that simply interpolates the machine points in straight segments, trusted no further than the eye. We draw it *through* the points rather than fitting a smooth curve above them: the points are honest lower bounds, the truth lies on or above them, and a line riding above the points would claim more than the search ever found.

Two cautions govern how to read the dotted curves. At small n the complete graph or hypergraph is itself feasible, its binding connectivity being only $n - 1$, so the earliest points merely trace the trivial maximum and reveal nothing of the open behaviour: the undirected-vertex points run up $\binom{n}{2}$ until n passes m , and only then bend toward the linear growth the proved edge value

predicts.

Past that bend the dotted line is as strong as the best lower bound behind it, which at each size is the better of the quick search and a named construction the checker verifies. The search alone stalls, exactly as it stalled on the quadratic wall in the rediscovery above: where it cannot build a denser construction its own reports climb only linearly. The clearest case is the directed hypergraph, whose bipartite family is quadratic, reaching $\sim (m-1)n^2/(4(r-1))$ hyperarcs by [Proposition A.55](#): there the walk slips below that construction at larger n , so it is the construction that carries the circles, with the search free to beat it wherever it can. We therefore read the dotted trends as honest lower bounds and not as fitted laws, and the one closed form we record on those panels, the quadratic for the directed hypergraph, comes from the construction and not from the shape of the dots. That it is also the right leading term is a theorem ([Theorem A.59](#)), which is a fact about the construction rather than about the dots.

So to the question of whether one formula links what is proved to what is conjectured, the answer is three-valued and depends on the variant, and the grid of [Figure 3.5](#) shows all three at a glance. Where the problem is already solved the curve is blue, a theorem the computation merely agrees with. Where the problem is open but tractable the curve is red, a conjecture the certified squares confirm exactly at the small sizes the certifier reaches and the searched circles confirm, without proving, at every larger size the search reaches, and that is the case the program was built for. Where the problem is genuinely hard the curve is yellow, a guess interpolated through the search points and an open question of what, if anything, the dots truly trace.

One feature stands out in the bottom row: the directed hypergraph panels, both arc and vertex, climb steeply at $m = 3$, the yellow guess curving upward rather than running straight. This is genuine and not search instability: the directed Berge edge packs a tail and $r - 1$ heads into a single object, and the bipartite construction of [Proposition A.55](#) already reaches roughly $(m-1)n^2/(4(r-1))$ hyperarcs, so the value grows with n^2 . The same quadratic shape sits one column over, in the directed arc and vertex panels, whose conjectured red curves carry the $\left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor$ term and rise just as fast. Super-linear growth is the signature of *direction* across the grid, not of the hypergraph alone: every directed column bends upward while the two undirected columns stay linear. [Figure 3.6](#) redraws the same grid at $m = 6$: the proved curves climb linearly with m exactly as their formulas predict, while the open variants' search points and their yellow interpolation climb higher without a proved curve to meet, so raising the threshold does not change which regime a variant lives in, only how far the red and yellow curves have travelled.

The same grid answers the harder question the trichotomy raises, which is what the search is worth where no *matching* upper bound exists to check it against. It says two things at once. First, wherever an upper bound exists, proved or certified, the search lands exactly on it and never above it at every plotted size, which is strong evidence that the search is not leaving edges on the table, with the one instructive exception recorded above: the directed $m = 2$ run at $n = 8$, where it stalls on the hub and the bipartite optimum must be supplied by the construction instead.

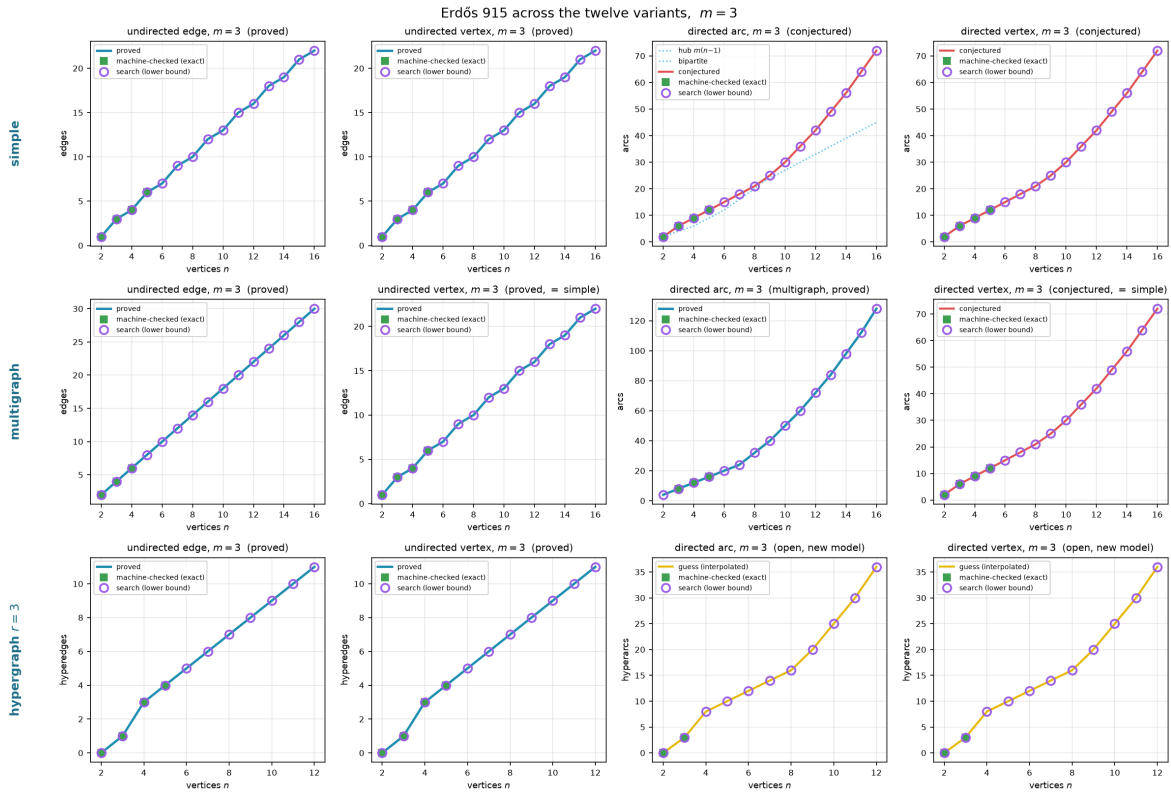


Figure 3.5. All twelve variants at $m = 3$, edges against n , laid out by model (rows: simple, multigraph, 3-uniform hypergraph) and by separation and direction (columns: undirected edge, undirected vertex, directed arc, directed vertex). A *blue* curve is a value proved for all n , a *red* curve a conjectured formula, and a *yellow* curve a bare interpolation of the machine points where no formula is known, the three told apart by colour rather than by dash pattern. Filled squares are the sizes `solve` proved or certified exactly. Open circles are lower bounds at larger n , the better at each size of the search's densest find and a named construction measured by the checker, reported as a running maximum so a feasible graph at one size is never lost at the next. On the open variants the yellow curve is the piecewise-linear interpolation *through* those circles, never above them, and the axes are scaled to the data rather than to a loose certain interval. The directed hypergraph panels in the bottom row rise steeply because their value is quadratic in n , not linear.

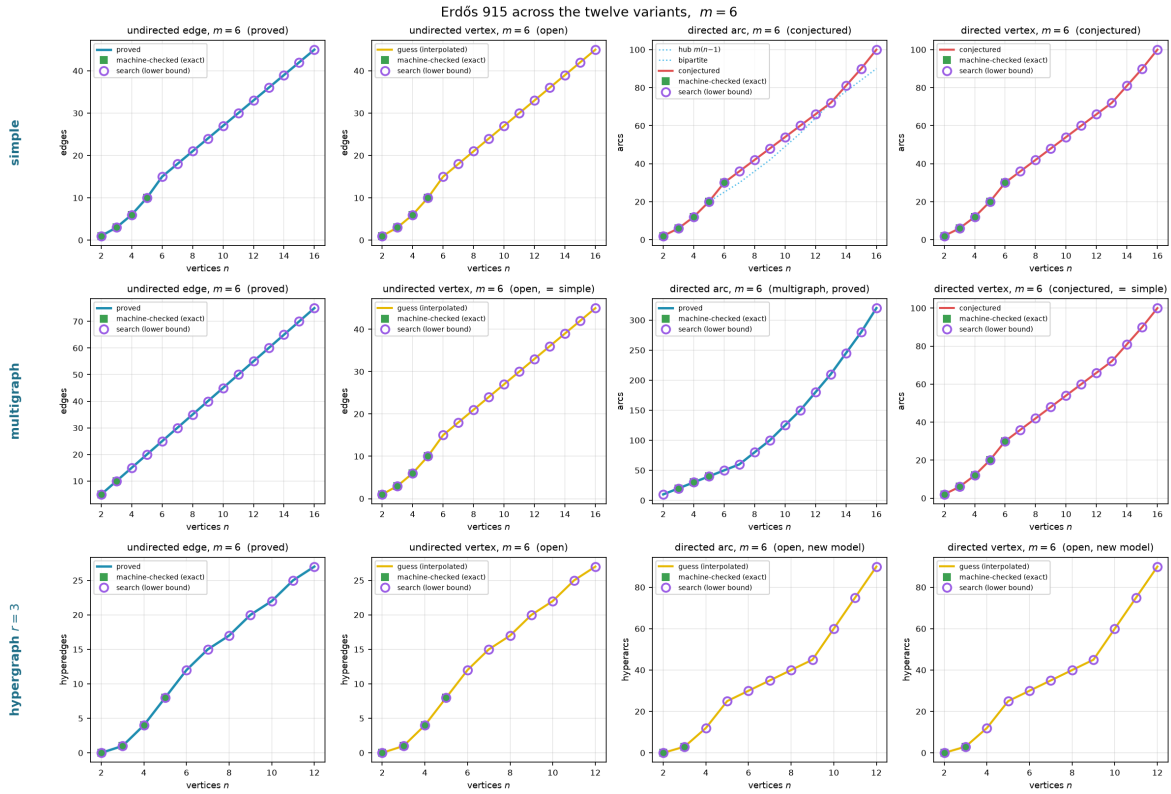


Figure 3.6. The same twelve variants at $m = 6$. The undirected edge and multigraph edge cases remain proved (Mader’s theorem holds for all m), as does the directed multigraph arc case (Theorem A.30 holds for all m too), while the undirected vertex separation is the famous open problem at $m \geq 6$ and the directed *simple* cases carry the same conjectured formula scaled to $m = 6$. On the open vertex panels the circles never sit below the proved edge value at the same size, because every graph feasible for the edge problem is feasible for the vertex problem (Whitney), so the edge extremiser is an honest lower bound there, and the search is free to climb past that floor where the vertex problem is genuinely richer. Comparing this figure with the $m = 3$ version above shows that the proved cases grow linearly with m as the formulas predict, while on the open cases the circles and their yellow interpolation climb steadily without a proved curve to meet.

Second, on the directed arc problem at $m \geq 3$, where the only upper bound past the certified range is the asymptotic one of [Theorem A.24](#), far too loose at these sizes to check anything, the plotted lower bounds land exactly on the conjectured value $\max(m(n-1), \lfloor (n+m-2)^2/4 \rfloor)$ at every reachable n , and the certified points agree with it as far as they go, but past the first open size it is the constructions, not the cooled walk, that carry the bounds there. Run from scratch the search confirms the conjecture at $n = 6$ and then plateaus a few arcs short (17, 17, 20, 23 against 18, 21, 25, 30 at $n = 7, \dots, 10$, across six restarts), for the same reason as the $m = 2$ stall recorded earlier: the extremal shapes past the crossover are bipartite walls, and a walk that has crystallised onto a hub cannot rebuild itself into a wall by single-arc moves within the cooling budget.

So to the question “do we believe the reported lower bounds are best possible,” the grid answers: wherever a proof or a certificate can check them, yes, and on the open sizes they sit on the only value anyone has conjectured, with the search confirming the first open size and the constructions carrying the rest. The gap between those lower bounds and what is proved exactly is the gap [Chapter 4](#) names precisely, and which has since narrowed from the whole answer to the coefficient of a linear term. The search has not closed that gap, and it cannot. What the pipeline supplies is reproducible evidence for which side of the gap the truth lies on.

Honesty

A number found by the search is reported as a lower bound, and is marked as certified or proved only when a separate cut-counting optimisation or a hand proof supplies the matching upper bound. No extremal value is ever asserted on the strength of the search alone. The figures in this chapter use fixed random seeds, so every search point is reproducible.

Chapter 4

Synthesis, Results, and Open Problems

The journey set out from one clean theorem and followed it across twelve variants, proving what proved cleanly, building a machine where the proofs ran out, certifying the small cases the machine could close, and using the search to discover the dense graphs the proofs had singled out and to point at the ones still open. This closing chapter draws the threads together. It maps the directed frontier, where proof and certificate both still fall short of the full answer, names the two genuine surprises the landscape contains, gathers the twelve answers into one picture, and says plainly what the program changed and where it stops.

4.1 The directed frontier and the missing decomposition

Two of the open directions are directed, and they share a structure worth stating exactly rather than hiding. The directed arc problem is a race between two constructions on different scales. The directed hub ([Construction 1.9](#)) reaches $m(n-1)$ arcs and grows linearly. The augmented bipartite construction ([Construction 1.13](#)) packs $\lfloor (n+m-2)^2/4 \rfloor$ arcs and grows quadratically. Which leads depends on n , and the two cross at an order linear in m , near $n = 9$ at the $m = 3$ of [Figure 4.1](#). [Conjecture 1.14](#) asserts that together they are the whole story.

[Conjecture 1.14](#) is proved as a lower bound for all m , and it is a theorem at $m = 2$ by [Theorem 1.10](#). Past $m = 2$ it sits squarely in the second regime of the trichotomy of [Section 3.6](#): certified by exhaustion at the small sizes `solve` can close for $m = 3$, conjectured by the temperature search beyond, and never beaten anywhere the search reaches. What is missing is a general upper bound for $m \geq 3$, and the obstacle is structural rather than computational.

The route to the upper bound is to show that an extremal digraph, once it is past the hub regime, must look like the augmented bipartite construction: its vertices split into a part A and a part B with every arc running $A \rightarrow B$, and B thickened only mildly. Three structural facts, each proved in [Appendix A](#), push in this direction. At $m = 2$ out-neighbourhoods are mutually unreachable and second out-neighbourhoods are disjoint, and for every m a complete $A \rightarrow B$ partition forces each vertex of B to absorb at most $m - 2$ arcs from inside B . That cap of $m - 2$ comes from the completeness of the $A \rightarrow B$ layer itself: every vertex of A already reaches a given $b \in B$ directly, so each further arc into b from another vertex of B supplies one more route from any such vertex of A to b , and the connectivity ceiling allows only $m - 2$

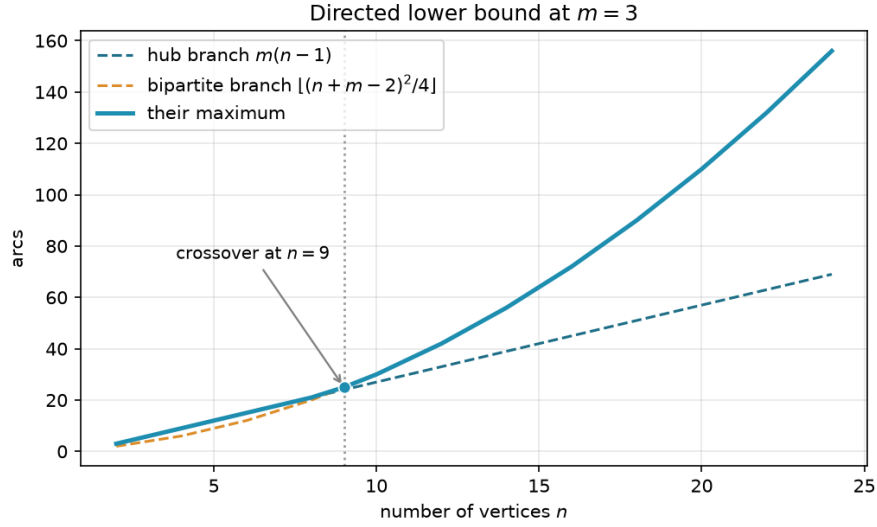


Figure 4.1. The two directed lower bounds at $m = 3$. The linear hub branch leads for small n . The quadratic augmented-bipartite branch overtakes it near $n = 9$ and leads thereafter. [Conjecture 1.14](#) asserts their maximum is the exact value.

such extra routes on top of the direct one. With the partition free to vary, the total arc count $|B|(|A| + m - 2) = |B|(n - |B| + m - 2)$ is maximised at $|B| = \lceil (n + m - 2)/2 \rceil$, which gives $\lfloor (n + m - 2)^2/4 \rfloor$, precisely the conjectured upper bound.

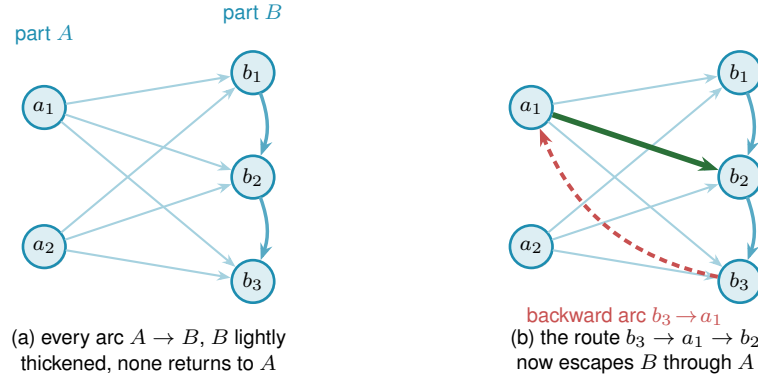


Figure 4.2. The directed frontier reduction and the single obstruction. **(a)** The structure the upper bound wants. Every arc runs from A to B (faint), B carries only a light internal thickening (the $m - 2$ budget, dark blue), and no arc returns to A , so a route between two vertices of B never leaves B and the count $|A||B| + (m - 2)|B|$ is complete. **(b)** A single backward arc $b_3 \rightarrow a_1$ (red) breaks this. The route $b_3 \rightarrow a_1 \rightarrow b_2$ (red leg then green leg) now escapes B through A , an extra route the budget never counted. Ruling out backward arcs in an extremal digraph is the one missing lemma behind the $m \geq 3$ upper bound, and the checker refuted a proposed counterexample by finding exactly such a hidden route.

The reduction is clean except at a single point, and it is worth naming the point exactly rather than hiding it. The count goes through provided the extremal digraph decomposes the way [Figure 4.2\(a\)](#) draws it, and that shape is four conditions rather than one: a partition $V = A \cup B$ exists, every arc from A to B is present, no arc lies inside A , and no arc runs from B back to A . The last condition is the one with a name and a picture. Under it a route between two vertices of

B can never escape B , so the connectivity inside B is governed entirely by B 's own arcs and the two-hop budget applies, and [Figure 4.2\(b\)](#) shows how a single backward arc destroys exactly that. It is also the condition where the natural delete-and-compensate exchange fails to be monotone. The standard move in an argument of this shape is to delete an offending backward arc and add a compensating arc elsewhere so the digraph stays exactly as dense while moving closer to the target shape, then repeat until the shape is reached. Here that move is not reliable: deleting the backward arc does not simply free up room for a compensating arc, since restoring feasibility can force other arcs to be removed too, so the exchange need not preserve or increase the arc count. That is why it is the one usually named as the obstacle.

Naming it alone would nonetheless overstate the position, so the appendix states the whole hypothesis as [Conjecture A.21](#): an extremal non-hub digraph admits such a partition, complete forwards, empty inside A , and with nothing coming back. Nothing proved here forces an arbitrary extremal digraph into that shape, and the small-case evidence for it is exactly the kind of thing the search of [Chapter 3](#) can probe. So the honest summary is that [Conjecture 1.14](#) for $m \geq 3$ awaits one structural decomposition theorem, of which the backward-arc clause is the hardest quarter.

What does not wait is the leading constant, and neither, it turns out, does the order of the term after it. [Proposition A.22](#) proves, with no hypothesis at all beyond feasibility, that a digraph with $\lambda^{\max} \leq m - 1$ has at most $\lfloor n^2/4 \rfloor + \sqrt{m} n^{3/2}$ arcs, and that deleting at most $\sqrt{m} n^{3/2}$ of them leaves a one-directional bipartite digraph. Its engine is a single observation the rest of the chapter has been circling: for any ordered pair, the direct arc and the two-step detours through distinct midpoints are automatically arc-disjoint, so feasibility caps their number. Summed over every ordered pair, the two-step detours through a fixed vertex x are counted once for every in-arc into x paired with every out-arc out of x , that is $d^-(x) d^+(x)$ of them, so the same per-pair cap, added up over every pair, bounds the total $\sum_x d^+(x) d^-(x)$. A vertex with both a large in-degree and a large out-degree is therefore expensive, every vertex is nearly a pure source or a pure sink, and throwing away each vertex's smaller side leaves the bipartite wall standing.

That error term is then sharpened all the way to the right order. The $\sqrt{n}$ is wasted precisely when every vertex carries a middling degree, which a digraph with $\Theta(n^2)$ arcs cannot do, so a dichotomy on the minimum degree removes it: when every vertex already has degree at least $n/2$, the same counting sharpens directly to the linear order, and otherwise some vertex has degree below $n/2$, and deleting it and inducting on the smaller digraph closes the gap. [Theorem A.24](#) proves, still with no structural hypothesis, that

$$|A(D)| \leq \left\lfloor \frac{n^2}{4} \right\rfloor + 4(m-1)(n-1), \quad \text{hence} \quad \ell_m^{\text{dir}}(n) = \frac{n^2}{4} + \Theta_m(n).$$

So both the leading constant $\frac{1}{4}$ of [Conjecture 1.14](#) and the *order* of its second-order term are unconditional theorems. What the conjecture still claims beyond them is one constant: it says the second-order coefficient is exactly $(m-2)/2$, and the theorem pins it between that and $4(m-1)$, a factor of roughly eight. That constant, and not the shape of the answer, is what [Conjecture A.21](#) is now needed for, and [Remark A.25](#) shows that the obvious way to close it, reading the same

two inequalities more carefully, provably cannot work.

The machine has been useful here in the negative direction too: a proposed counterexample to the conjecture, a backward arc bolted onto the thirty-arc graph, was refuted by the checker, which found the third route the construction's author had missed.

The directed vertex-disjoint problem shares the constructions but not the proofs, and the distinction matters more than it looks. Whitney's inequality $\kappa \leq \lambda$ says a digraph obeying the arc ceiling obeys the vertex one, so the arc constructions are vertex-feasible and every lower bound carries across for free. It says nothing in the other direction, and the vertex-feasible family is the larger of the two, so no arc upper bound restricts it. At $m = 2$ the vertex value is nonetheless the same, and [Theorem 1.11](#) proves it by running the induction a second time under the stricter separation, with its own exhaustive base cases through $n = 7$. For $m \geq 3$ the vertex value is conjectured equal to the arc value, on the strength of the same constructions and of exhaustive small-case checks run under the vertex test rather than inherited from the arc one. What it will not do is follow from the arc case, even once [Conjecture A.21](#) is supplied, because an upper bound proved over the smaller family says nothing over the larger one. The vertex problem needs its own upper bound at $m \geq 3$, and [Table 4.1](#) lists it as a separate conjecture rather than a corollary. It has one now, to first order. [Theorem A.28](#) proves $k_m^{\text{dir}}(n) = n^2/4 + \Theta_m(n)$ with no hypothesis beyond feasibility, by the route-counting of [Proposition A.22](#) rather than by its conclusion. The distinction matters and is the whole of why this is legitimate: the counting exhibits a family of routes that turns out to be internally vertex-disjoint and not merely arc-disjoint, so it may be measured against κ directly, over the larger family, without ever invoking the arc bound. The directed multigraph problem has its own two-branch story, parallel to the simple one, but unlike the simple one it is now a closed theorem rather than a conjecture ([Theorem A.30](#), [Appendix A](#)):

$$L_m^{\text{dir}}(n) = (m-1) \max(2(n-1), \left\lfloor \frac{n^2}{4} \right\rfloor),$$

where the double star realises the linear branch and the one-directional bipartite multigraph, every arc at multiplicity $m-1$, realises the quadratic one.

The quadratic branch here uses the balanced partition $|A| = \lfloor n/2 \rfloor$, $|B| = \lceil n/2 \rceil$, giving $(m-1) \lfloor n^2/4 \rfloor$ arcs. The multigraph and the simple digraph disagree about the best partition, and the reason is worth spelling out. In a multigraph the $m-1$ routes per pair are supplied directly by $m-1$ parallel copies of every $A \rightarrow B$ arc, so nothing needs to be added inside B , and the arc count $(m-1)|A||B|$ is largest at the balanced split. In a simple digraph each ordered pair carries at most one arc, so the extra routes must instead come from arcs inside B , and feeding B those in-arcs is what pushes its optimal size up to $|B| = \lceil (n+m-2)/2 \rceil$ once $m \geq 4$ (at $m \leq 3$ the balanced and shifted splits happen to give the same count, as [Construction 1.13](#) records). One consequence is that the multigraph's two branches cross at $n = 7$ for every m , because the factor $m-1$ multiplies both branches and cancels, whereas the simple crossing point grows with m and already sits near $n = 9$ at $m = 3$.

The route to the theorem is worth a sentence, because it is not the route the rest of this thesis's directed work uses. Deleting one vertex at a time, the tool behind [Theorem 1.10](#) and [Theorem 1.2](#), closes the linear branch for every $n \leq 6$ ([Theorem 2.2](#)) and the quadratic branch at even n , but leaves a genuinely fractional remainder at odd n that no minimum-degree statement was ever found to close. [Appendix A](#) sidesteps that obstruction rather than solving it: instead of deleting one vertex, it deletes one whole *reachability-preserving subgraph* at a time, a sparse piece of the digraph that alone reproduces every reachability relation the full digraph has. Such a subgraph always has at most $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$ arcs (a short argument built from strongly connected components and Mantel's theorem), and removing it is guaranteed to lower every pairwise connectivity by exactly one, so peeling off $m-1$ of them in turn proves the bound with no parity split and no restriction on m . In particular $L_3^{\text{dir}}(7) = 24$, the one finite fact this problem used to need a computer for, now follows in three lines.

A separate, narrower question survives this closure: not the arc count of an extremal multigraph on $n = 7$ vertices, but exactly *which* multigraphs attain it. That was settled by machine before the value had a hand proof, and it stands on its own regardless: the sound generation enumerator of [Chapter 2](#), run on the degree-bounded classification at $n = 7$, returns exactly three extremal families, $2 \cdot B_{3,4}$, $2 \cdot B_{4,3}$, and the doubled bidirected path P_7 ([Remark A.41](#)).

The linear branch turns out to be richer than the double star first suggested. Every doubled bidirected spanning tree is extremal, confirmed exhaustively at $n = 4$ and 5 , and the saturated attachment lemma ([Lemma A.39](#)) explains why: attaching one further vertex to an already-saturated doubled tree can add at most $2(m-1)$ arcs, and equality holds exactly when the new vertex becomes a new bidirected leaf. So building up from a single doubled edge one such leaf at a time, in every possible order, generates exactly the doubled bidirected trees and nothing denser, which is what it means to say the doubled trees are the closure of a single doubled edge under maximal attachment. Several distinct extremisers tie at 24 arcs for $n = 7$, and sifting and classifying them is precisely what the enumeration must do.

It is worth naming what the closure removed, because the shape of the difficulty was misleading for a long time. The vertex-deletion route degraded as m grew, yielding the value but not uniqueness at $m \geq 4$ and stalling on the value itself at $m \geq 5$, which made those look like harder regimes of the problem. They are not. That was a limitation of one method, and the reachability skeleton simply does not have it, treating every m alike. The one question that genuinely survives is the uniqueness one above, stated in full among the open problems below: a finite, identified target rather than a mystery, and a natural goal for the next round of certified computation.

4.2 Two principal phenomena

Across all the variation, two phenomena stand out as the ones a first guess would not have predicted.

The first is the divergence at $m = 5$. The edge and vertex problems agree for $m \leq 4$, and it would be natural to expect them to agree forever. Instead they part company at the very next

value, with $k_5(n) = \lfloor 8n/3 \rfloor - 4$ pulling strictly above $\ell_5(n)$ from $n = 14$ onwards ([Theorem 1.6](#), and [Remark 1.7](#) for the counting convention that fixes the constant). The separation is genuinely a fact about m rather than about n : at $m = 5$ the two values still agree at every size up to $n = 13$ and only then come apart, and beyond $m = 5$ the vertex problem becomes genuinely hard and has stayed open since 1974.

The second is the quadratic bipartite phenomenon. In an undirected graph every edge contributes to the degree of both its endpoints, and Mader's theorem turns that symmetry into a hard ceiling: $\lfloor m(n-1)/2 \rfloor$ edges, linear in n . A digraph breaks the symmetry. Place all n vertices into two equal parts A and B and draw every arc from A to B , one direction only. The result has $\lfloor n^2/4 \rfloor$ arcs, quadratic in n , yet $\lambda^{\max} = 1$: every pair has at most one arc-disjoint route, because every route must cross the A -to- B wall exactly once and no arc crosses the other way to offer a second path. A graph whose arc count grows like n^2 can sit at connectivity one, something no undirected graph can do past a handful of vertices.

The initial conjecture for the directed case was linear, matching the undirected intuition. The one-directional bipartite construction refutes it outright, already at $m = 2$: for $n = 8$ the wall holds 16 arcs while the linear hub holds only 14. At $m = 3$ the refutation is the thirty-arc augmented bipartite graph of [Remark 1.12](#), which packs 30 arcs at $\lambda^{\max} = 2$ against the naive prediction of 27. The construction generalises to every m via [Construction 1.13](#), thickening the larger part B slightly so that each vertex of B absorbs $m - 2$ extra in-arcs from inside B : those extra arcs add exactly $m - 2$ short detours from A to each $b \in B$, bringing $\lambda^{\max}$ from 1 to $m - 1$ while the arc count climbs to $\lfloor (n + m - 2)^2/4 \rfloor$.

The quadratic term is what makes the directed cases the deepest in the thesis. The upper bound argument must explain why no configuration beats the bipartite wall, which requires producing the partition in the first place, ruling out backward arcs, and controlling the internal structure of the large part B . That structural task is exactly what [Conjecture A.21](#) has not yet resolved for $m \geq 3$. What [Proposition A.22](#) and [Theorem A.24](#) show is that the wall is at least the right shape, and the two say it in different currencies, which is worth keeping apart. In *size*, every feasible digraph is within $O_m(n)$ arcs of the wall, so the difficulty is a constant inside the second-order term rather than the phenomenon. In *shape*, every feasible digraph becomes a wall after deleting $O(\sqrt{m} n^{3/2})$ arcs, which is a weaker error term because it comes from the cruder of the two counts, the one that does not get to induct on a low-degree vertex. The undirected problem has no analogous obstacle, because Mader's degree-counting argument closes it in a few lines. The quadratic phenomenon is therefore not just a curiosity: it is the geometric reason the directed and undirected problems live on different levels of difficulty.

That same scarcity of extremal graphs is the thread through the remaining figures. [Figure 4.3](#) shows where the frontier runs: it plots every labelled graph at small n as a single point, its binding connectivity $\lambda^{\max}$ on the horizontal axis and its edge count on the vertical, so the whole population of graphs is visible at once. The points crowd into the low-connectivity, low-edge corner and thin out upward, where the blue extremal envelope, an integer staircase carrying one

dot per connectivity level, traces the densest graph available at each level. The empty region above that staircase is precisely the part the main theorems forbid a graph to occupy. The extremal problem is the question of where that staircase runs, and the figure shows it is a thin frontier with almost nothing pressed against it from below.

Figure 4.4 steps inside each graph to the level of individual pairs. For every graph in the full enumeration it records the connectivity of every vertex pair and pools all the observations, so the histogram counts not graphs but (graph, pair) observations. Each bar is split and stacked by the connectivity of the graph the pair came from: blue for pairs drawn from the lower-connectivity graphs, red for pairs from the higher-connectivity ones. The boundary between the two is set per variant at the midpoint of the connectivity range that variant's enumeration can reach, since the twelve variants span very different ranges and a single fixed boundary would leave some panels entirely one colour. A pair whose own connectivity exceeds the boundary can only come from a graph above it, so the bars to the right of the dashed line are wholly red, while to the left the two colours mix, since a high-connectivity graph still carries many low-connectivity pairs. The blue mass sits almost entirely at the lowest levels: high-connectivity pairs are rare even inside the graphs that have one.

Figure 4.5 then shows how the same graphs distribute by sheer edge count, with the same per-variant stacked colouring. At each edge count the blue part counts the lower-connectivity graphs and the red part the higher-connectivity ones, so the bar height is the total number of graphs there and the split shows how the two populations separate. The red mass sits to the right, since denser graphs tend to carry higher connectivity, and the dotted line marks the densest graph still on the low-connectivity side, landing far out in the sparse tail. The densest low-connectivity graphs, the ones the extremal problem is about, are exceptional in every variant.

Finally, Figure 4.6 lifts the whole study onto the (n, m) grid at once, drawing the best feasible edge count as a bar surface across every variant. The three proved variants rise as smooth linear or quadratic sheets that the formulas describe exactly, while the open variants show a visible step between the machine-proved blue bars and the search's purple lower bounds, so the surface makes the proved-versus-open divide of the trichotomy a single readable shape. Reading along the m axis recovers the linear scaling in the threshold, and reading along n recovers each variant's growth rate, which is the whole of Table 4.1 compressed into one object.

4.3 Summary of the twelve variants

Table 4.1 records the status of all twelve structural variants. Each entry is marked by how it is known: proved by hand here or in Appendix A, cited to the literature, conjectured with a proved lower bound, or, for the two rows where only the leading term is settled, proved asymptotically. No row now rests on a machine certificate alone. That category was live in earlier drafts, when the directed multigraph value was known only for the sizes the cut-counting optimisation could close, and Theorem A.30 retired it. The distinction is the honesty contract of Chapter 2 carried through to the summary. Figure 4.7 shows the same distinction as a picture before the table

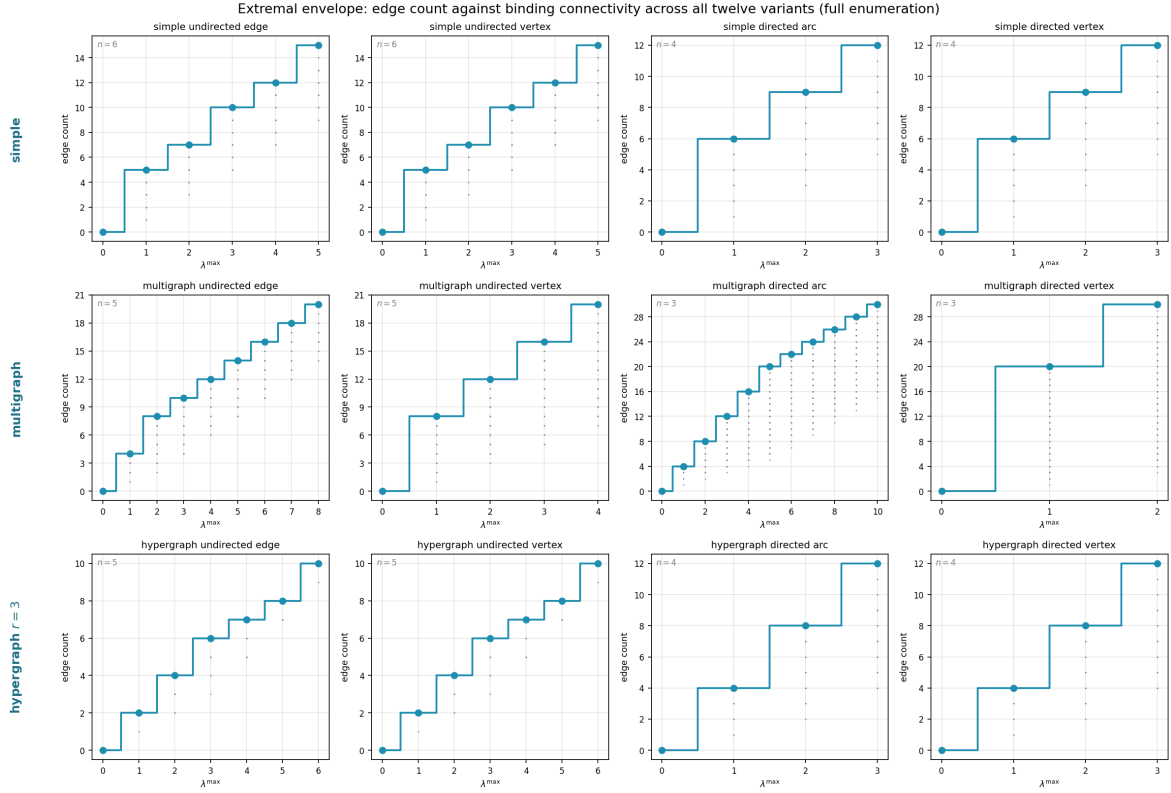


Figure 4.3. The complete landscape of all labelled graphs at small n (see panel annotations for each variant's n): every point is one graph, plotted at its binding connectivity $\lambda^{\max}$ horizontally and its edge count vertically. The blue staircase marking the densest graph at each connectivity level is the extremal envelope, the frontier the extremal problem asks about, with one dot per level so its exact value is legible. Graphs crowd the low-connectivity, low-edge region. The extremal graphs live at the top of the staircase. The emptiness above the envelope is exactly the region the main theorems forbid. The horizontal reach differs by panel because the connectivity that fits in a variant's enumeration n differs: arc-disjoint multigraph routes grow with edge multiplicity (the multigraph directed arc panel reaches $\lambda^{\max} = 10$ at $n = 3$), whereas vertex-disjoint routes are capped by the available internal vertices, so the multigraph directed *vertex* panel reaches only $\kappa^{\max} = 2$ at the same $n = 3$ (with a single internal vertex, a pair has at most the direct route plus one detour). This is a property of the separation, not of the search.

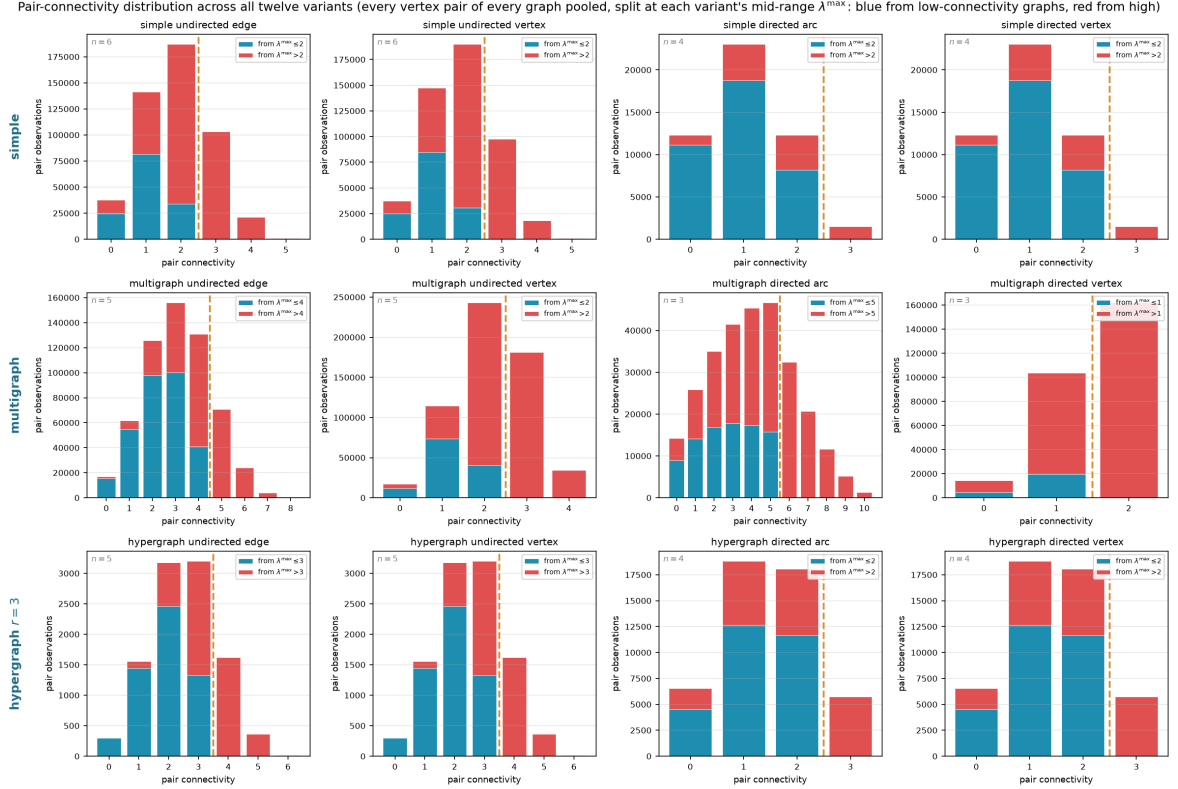


Figure 4.4. Pair-connectivity distribution across all twelve variants, pooling every vertex pair of every graph in the full enumeration. The horizontal axis is the connectivity of a single pair, and a bar's height is the number of (graph, pair) observations at that level. Each bar is split and stacked by the connectivity of the graph the pair was drawn from, blue from graphs at or below a per-panel boundary T and red from graphs above it, with the dashed line at T . The boundary is set per variant to the midpoint of the connectivity range that variant's enumeration reaches (each panel's legend gives its T), so the split stays informative in every panel instead of collapsing to one colour. Bars to the right of the dashed line are wholly red, since a pair of connectivity above T forces its graph above T . To the left the colours mix.

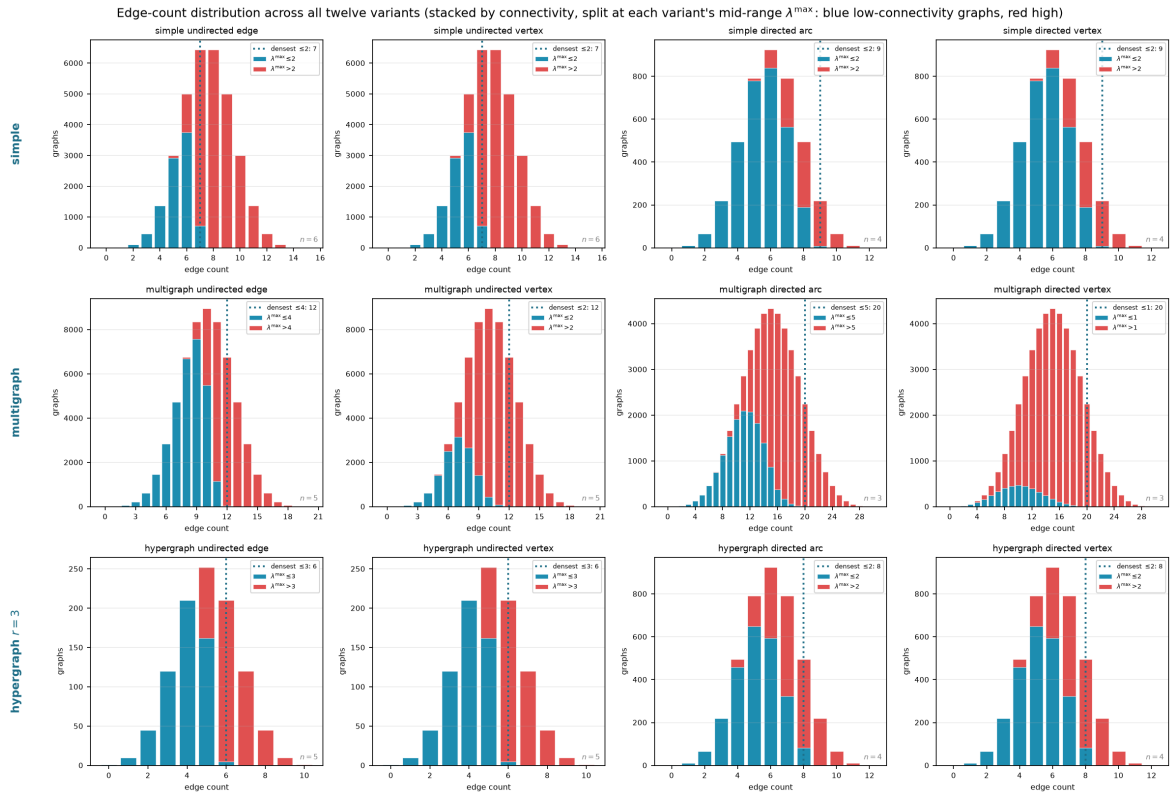


Figure 4.5. Edge-count distribution across all twelve variants, the same enumeration viewed by edge count rather than connectivity. Each bar is split and stacked by graph connectivity, blue for graphs at or below the per-panel boundary T and red above it (each panel's legend gives its T), so the full height is the total number of graphs at that edge count. The dotted vertical line marks the densest graph on the low-connectivity side. The reader can see directly how the two populations separate by edge count and how far low-connectivity graphs sit from the bulk.

Erdős 915: optimal bound over (n, m) , blue where proved, purple where only a search lower bound is known

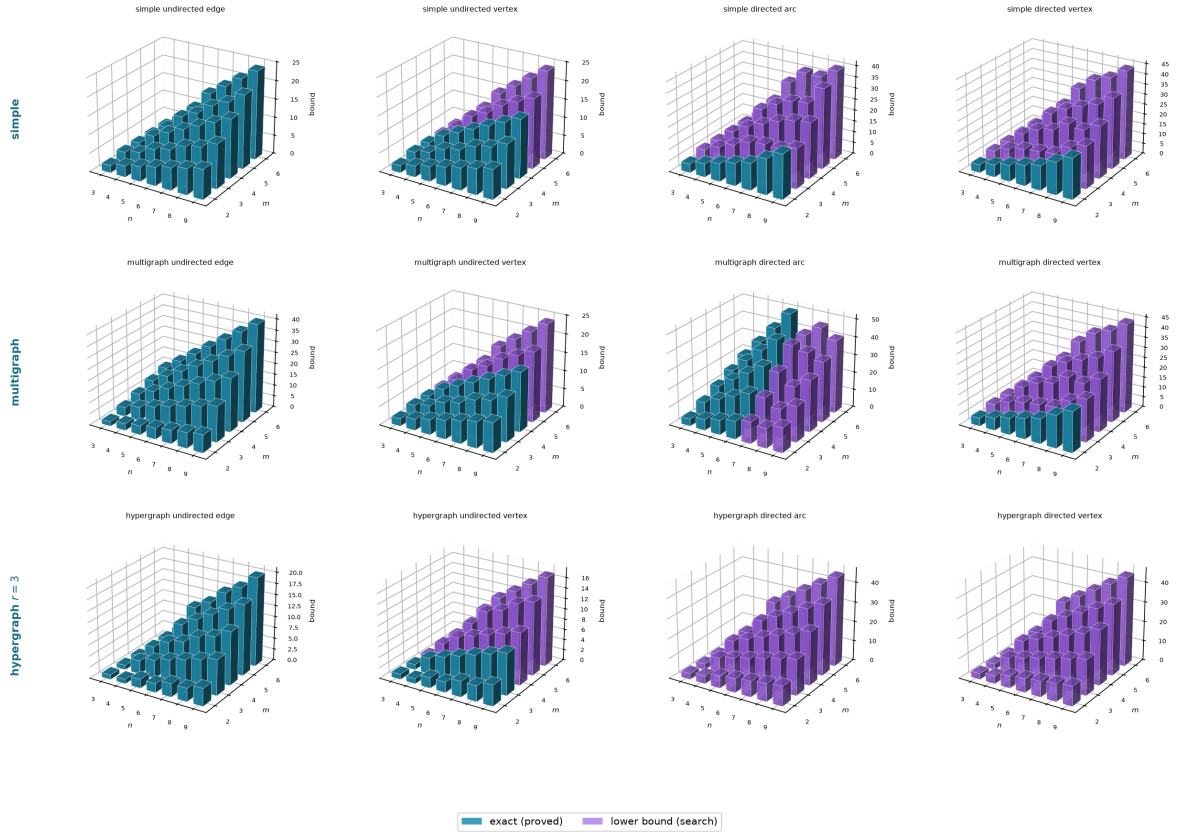


Figure 4.6. The optimal bound as a surface over the (n, m) grid for all twelve variants, every panel drawn from the same camera so the shapes compare directly. Each bar's height is the largest feasible edge count found by `solve` at that vertex count and forbidden threshold m . The single thing to read off is colour: a bar is *blue* exactly where the value is machine-proved and *purple* exactly where only a search lower bound is known, so each panel is a little map of what is settled. The three fully proved variants (Mader's undirected edge cases and the multigraph analogue) are solid blue sheets, rising linearly or quadratically as their formulas predict, while the open variants turn purple precisely where proof runs out. Reading along the m axis shows how the bound scales with the forbidden threshold, and reading along the n axis shows the growth rate for a fixed variant.

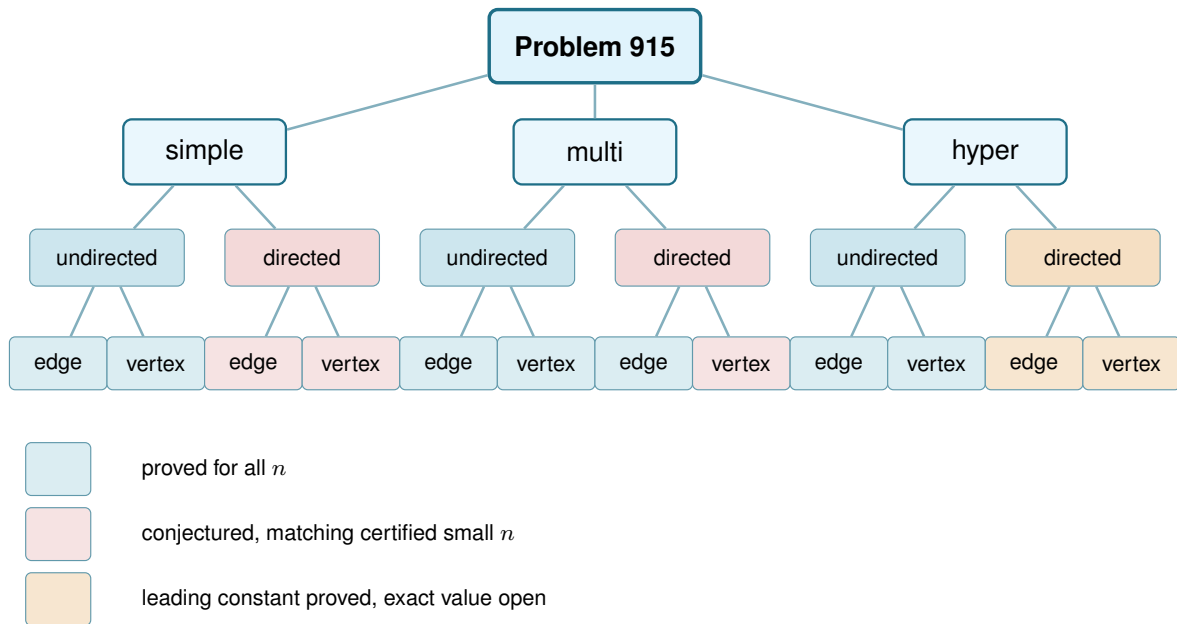


Figure 4.7. The same branching as Figure 1.7, now with every leaf resolved by the trichotomy of Section 3.6: blue is proved for all n , red is a conjectured formula backed by certified small cases and a matching search, and amber is a variant whose leading term is proved but whose exact value is unknown. Colour marks the status at general m , the harder regime this thesis mostly studies, and hides m -dependent exceptions and one convention that Table 4.1 states precisely. The one branch whose two leaves disagree is the directed multigraph: its edge leaf is blue by Theorem A.30, proved for every n and m , while its vertex leaf takes the red of the simple digraph it inherits, so that parent node is drawn in the weaker of the two colours. Every red leaf is a proved theorem at $m = 2$ and becomes conjectured only at $m \geq 3$, the directed edge leaf by Theorem 1.10 and the directed vertex leaves by the separate Theorem 1.11, which does not follow from it. Three blue leaves are proved only up to a finite m : the undirected vertex problem for simple graphs and for multigraphs ($m \leq 4$) and for hypergraphs ($m \leq 3$), open beyond. The two multigraph vertex leaves inherit whatever their simple counterparts have, because the counting convention of Remark 2.1 makes those two questions the same question.

spells out every case.

Table 4.2 answers a different question about the same twelve variants: not what is known, but which routine of the program knows it. Most variants share machinery rather than each needing its own, so the table is grouped by family, and every name in it has its own codecard box where it is built in Chapters 2 and 3.

4.3.1 Orientation models for the directed hypergraph

The directed rows of Table 4.1 read a directed Berge edge in one way, the *forward* one of a single tail fanning to $r - 1$ heads. It is not the only reading. In general a directed r -uniform hyperedge is a split of its vertices into a non-empty tail set T and head set H , a route entering at a tail and leaving at a head, and the gate checker of Chapter 2 measures any such split without change. Three models are worth naming, *forward* ($|T| = 1$), *backward* ($|H| = 1$), and *general* (any split, with genuinely mixed ones appearing once $r \geq 4$). The natural question, and one a reader of the directed figures tends to ask, is whether they multiply the table.

Separation	Model	Value	Status
<i>Undirected</i>			
edge	simple	$\lfloor m(n-1)/2 \rfloor$	proved (Mader)
edge	multi	$(m-1)(n-1)$	proved
edge	hyper	$\lfloor (m-1)(n-1)/(r-1) \rfloor$	proved (cited, Gomory-Hu)
vertex	simple	$\lfloor m(n-1)/2 \rfloor$ ($m \leq 4$), $\lfloor 8n/3 \rfloor - 4$ ($m = 5, n \geq 6, n \neq 7, 12$, see Remark 1.7)	cited, open $m \geq 6$
vertex	multi	adjacency count, so the simple value $\lfloor m(n-1)/2 \rfloor$ ($m \leq 4$)	convention (Remark 2.1)
vertex	hyper	$\lfloor \frac{(m-1)(n-1)}{r-1} \rfloor$ ($m \leq 3$), $\geq$ that for all m	proved $m \leq 3$ (Theorems A.50 and A.53), open $m \geq 4$
<i>Directed</i>			
arc	simple	$\max(2(n-1), \lfloor n^2/4 \rfloor)$ ($m = 2$), $n^2/4 + \Theta_m(n)$ (all m)	proved (Theorems 1.10 and A.24), conj. $m \geq 3$
arc	multi	$(m-1) \max(2(n-1), \lfloor n^2/4 \rfloor)$	proved, all n and m
arc	hyper	$(1 + o(1))(m-1)n^2/(4(r-1))$, exact value open	leading constant proved (Theorem A.59)
vertex	simple	$\max(2(n-1), \lfloor n^2/4 \rfloor)$ ($m = 2$), $n^2/4 + \Theta_m(n)$ (all m)	proved separately (Theorems 1.11 and A.28), conj. $m \geq 3$
vertex	multi	adjacency count, so the simple digraph value	convention (Remark 2.1)
vertex	hyper	$(1 + o(1))(m-1)n^2/(4(r-1))$, exact value open	leading constant proved (Theorem A.59)

Table 4.1. The twelve structural variants of Problem 915 and their status. Every simple-graph closed form here is stated for $n \geq m$, the hypothesis [Theorem 1.2](#), [Theorem 1.5](#) and [Conjecture 1.14](#) carry: below it the complete graph or digraph is itself feasible and the answer is the trivial maximum, which the formulas exceed. “conj.” means a conjecture with a proved matching lower bound. “Leading constant proved” means the n^2 term of the value is a theorem while the exact value is open, which is the strongest thing known about the two directed hypergraph rows. “Convention” marks the two rows whose value is fixed by the counting convention of [Remark 2.1](#) rather than by a theorem. The appearance threshold $p^* = m/n$ of [Theorem 1.16](#) is deliberately not a column here: it is proved for $G(n, p)$, extends to the vertex and directed analogues by [Remark A.70](#), and does *not* transfer to the hypergraph models, whose natural density scale is different ([Remark A.71](#)).

They do not, and two results say why. First, backward is not a new problem. Reversing every hyperarc turns a backward hypergraph into a forward one and reverses every route along the way, so the two models share every extremal number, edge and vertex, for all r , m and n ([Lemma A.56](#)). The backward column is the forward column read upside down. Second, the general model, though it admits more hyperedges on a single vertex set, obeys the same first-order bound. A hyperedge (T, H) occupies $|T||H| \geq r-1$ ordered tail-head pairs, each usable by at most $m-1$ disjoint one-step routes, so $(r-1)|E| \leq (m-1)n(n-1)$, exactly the forward cap of [Proposition A.55](#) ([Proposition A.57](#)). The forward construction is itself a set of general hyperedges, so general is squeezed between the same lower and upper bounds as forward, and its value too is quadratic in n . Those two bounds differ by a factor of four, so what this fixes is the order and not the leading constant, and it does not by itself force the two models to agree asymptotically. That last step is now taken. [Theorem A.60](#) closes the factor of four for the general model as well, so all three orientations share the leading term $(m-1)n^2/(4(r-1))$ under both separations, and the axis collapses asymptotically rather than merely by the evidence below. The obstacle it had to clear is the one the forward proof quietly avoided: with a single tail, entering and leaving hyperedges can never be confused, and with several tails they can, both because two midpoints may leave through one shared hyperedge and because a mixed hyperedge may be one target’s entrance and another’s exit. The fix is to take a *maximum* family of two-step routes rather than build a large one, and to read off why it stopped, since every target it leaves out must be blocked by a hyperedge already spent, and a hyperedge has only r vertices to block with.

Where the models part company is only at small sizes, and the program shows exactly where. [Table 4.3](#) lists the extremal hyperedge counts the branch-and-bound checker ([Table 4.2](#)) proves for the forward and general models. The method is the include-or-exclude search of [Chapter 2](#)

Variant family	Measure	Prove	Discover
Undirected, edge (all three models)	max_edge_connectivity max_hyperedge_connectivity	exhaustive enumeration (solve), Gomory–Hu cross-check	search_for_dense_graph
Undirected, vertex (all three models)	max_vertex_connectivity max_hyper_vertex_connectivity	exhaustive enumeration (solve)	search_for_dense_graph
Directed arc, simple	max_edge_connectivity	_exhaustive_directed: a pruned branch-and-bound	tabu_search_for_dense_graph
Directed arc, multigraph	max_edge_connectivity	the cut-counting optimisation (prove_directed_multigraph, prove_integral_arc_bound); the $n = 7$ classification (enumerate_extremal_directed_multigraphs_via_generation)	tabu_search_for_dense_graph
Directed vertex (simple and multi)	max_vertex_connectivity	_exhaustive_directed again, run under the vertex test (_vertex_flow_at_least), not inherited from the arc sweep	tabu_search_for_dense_graph
Directed hyper (arc and vertex)	hyper_connectivity	max_feasible_hyperedges: a pruned branch-and-bound	search_for_dense_graph

Table 4.2. The machinery of [Chapters 2 and 3](#) indexed by variant family rather than by function name. This is the cross-reference from a variant to the routine that handles it. The routine's own signature, inputs, and outputs are in its codecard box.

([Figure 2.9](#)) walked over the candidate hyperedges instead of the arc cells: a hyperedge is included only while the checker confirms the growing hypergraph stays feasible, a branch is dropped once even including everything that remains cannot beat the best count already found, and a sweep that finishes has considered every candidate set, so the value it reports is proved exact rather than searched. The general one is strictly denser precisely when vertices are scarce, $n \approx r$, where the richer set of orientations lets a single vertex-set carry more hyperedges. At $n = r = 4$ it reaches eight hyperarcs against forward's four, and the surplus grows with the connectivity budget. Once n exceeds r the computed gap closes, proved equal already at $r = 3$, $n = 4$, and larger searches find no separation thereafter. The reading that suggests itself is that once the hypergraph spreads past a single edge-set, the one-tail model already saturates the Menger budget and extra orientations buy nothing. Asymptotically that reading is now a theorem: [Theorem A.60](#) gives the two models the same leading term, so any gap between them is confined to lower-order terms and cannot grow with n^2 . At finite n it remains computational evidence, and the two statements should not be confused, since a difference of order n would be invisible to the leading constant and is exactly what [Table 4.3](#) displays at $n \approx r$. The orientation axis therefore collapses in the sense that matters for the shape of the answer, which is why Problem 915 stays a twelve-variant question and the general model enters as a refinement at the margin rather than a thirteenth column. What is left open is the finite comparison: whether the values agree exactly once n exceeds r , or there is a first size where they do not.

r	m	n	forward	general
3	3	3	3	4
3	3	4	8	8
4	3	4	4	6
4	4	4	4	8

Table 4.3. Extremal hyperedge counts the branch-and-bound checker proves exact for the forward and general directed r -uniform models, at the small sizes where they can differ. The general model is strictly denser only when $n \approx r$, doubling the forward count at $n = r = 4$, $m = 4$, and the gap closes once n exceeds r (proved equal at $r = 3$, $n = 4$). Edge- and vertex-disjoint counts agree throughout, and backward equals forward by [Lemma A.56](#).

4.4 Contributions and limitations of the program

The program turned a per-case craft into a uniform pipeline. Before it, each variant came with its own one-off computation and its own hand argument. After it, one representation holds every variant, one exact checker measures connectivity, one certifier proves small-case upper bounds, and one cooled search discovers extremisers and the lower bounds they witness. The rediscovery of [Chapter 3](#) is the evidence that the pipeline is faithful: pointed at solved cases, it returns the solved answers.

Its limits are as clear as its reach, and stating them is part of the honesty the thesis insists on. The search proves no upper bounds. It only ever exhibits graphs. The certifier proves upper bounds, but only for a fixed size, and the directed multigraph encoding closes unconditionally only up to $n = 6$. Past there the value is settled instead by the hand proof of [Appendix A](#), and the certifier contributed nothing at $n = 7$, where it exhausted its time limit without a verdict. The generation enumerator’s classification at $n = 7$ ([Remark A.41](#)) stands on its own as a machine-verified fact about which multigraphs attain the value. No amount of either replaces the genuinely structural gaps the thesis leaves open: the decomposition of [Conjecture A.21](#) that would turn [Conjecture 1.14](#) into a theorem, of which the backward-arc clause is the hardest part but not the whole, the directed vertex problem at $m \geq 3$, which Whitney’s inequality pointedly does not hand over with the arc case, and the undirected vertex-disjoint problem at $m \geq 6$, which has resisted for half a century and which the program can probe but not crack.

4.5 Open problems

- ★ The decomposition of [Conjecture A.21](#): prove that an extremal non-hub digraph admits a partition $A \cup B$ with every arc from A to B present, no arc inside A , and no arc from B back to A . That last clause is the backward-arc condition, and it is the one where the natural delete-and-compensate exchange fails to be monotone, but the other three are hypotheses too, and no argument here forces an arbitrary extremiser into that shape. The decomposition completes the upper bound in [Conjecture 1.14](#) for $m \geq 3$. Unconditionally, [Theorem A.24](#) already gives $\ell_m^{\text{dir}}(n) = n^2/4 + \Theta_m(n)$, so neither the leading constant nor the order of

the second-order term is at stake any more. What the decomposition is needed for is the *coefficient* of that linear term, which the conjecture puts at $(m-2)/2$ and the theorem bounds above by $4(m-1)$. Closing that factor of roughly eight is the smaller target, and [Remark A.25](#) narrows how it can be approached: the bound is already the exact maximum obtainable from the two inequalities the proof uses, so a sharper reading of them cannot help and a genuinely independent third inequality is required, one constraining the interference among the few expensive near-balanced vertices themselves.

- ★ The directed *vertex* problem at $m \geq 3$, which is a separate question and not a corollary of the arc one. Whitney's inequality makes the vertex-feasible family the larger of the two, so an arc upper bound does not restrict it, and [Conjecture A.21](#) would not settle it either. At $m = 2$ the two values agree ([Theorem 1.11](#)), proved by re-running the induction and its base cases under the stricter separation rather than by transfer. What is no longer open is the shape of the answer: [Theorem A.28](#) gives $k_m^{\text{dir}}(n) = n^2/4 + \Theta_m(n)$ unconditionally, so the two problems agree to first order and the vertex question, like the arc one, is now a question about the coefficient of a linear term. That bound does not come from the arc bound, which would be illegitimate, but from noticing that the route family the arc argument counts is internally vertex-disjoint as well as arc-disjoint, so it may be counted against κ instead. Whether the two values agree exactly for $m \geq 3$ is open. The exhaustive search run under both tests at $m = 3$ returns the same value at every n it reaches, tabulated in [Remark A.17](#), so there is no small counterexample, but that is evidence and not an argument. The useful next step is structural rather than numerical: find the first digraph in which some pair carries strictly more arc-disjoint routes than internally disjoint ones *and* that slack buys extra arcs, or show that it never can.

- ★ The multigraph vertex problem under the other counting convention, worked out in [Section A.9](#). Reading parallel copies as distinct routes makes this a genuinely new question, and the section settles $m = 2$ and exhibits several competing constructions. The natural guess from the $m \leq 4$ data, that a thickened tree becomes optimal once $n \geq m + 2$, is false for every $m \geq 5$: [Theorem A.63](#) shows that a bouquet of thickened cliques sharing one vertex beats the tree by an amount that grows linearly in n , and the true growth rate in m is $\Theta(m^2)$ rather than the $\Theta(m)$ the tree would give. That construction is a lower bound of the right order and not the answer: at $m = 5$, $n = 7$ it gives 27 where an unfinished exhaustive search had found 28, and the true value is 29. That the order is right is itself a theorem rather than an impression, since [Proposition A.64](#) supplies the matching upper bound $K_m^{\text{multi}}(n) \leq (m-1)k_m(n) < 2(m-1)^2n$, by observing that the underlying simple graph of a feasible multigraph is itself feasible for the simple vertex problem and then applying Mader's density theorem to it. So $K_m^{\text{multi}}(n) = \Theta(m^2n)$ and what is open is the constant.

Two things have since narrowed it. First, the value is exactly a *block* problem ([Theorem A.66](#)): writing the objective in closed form as $\sum_{uv \in E(G_0)} (m - \kappa_{G_0}(u, v))$ over the underlying simple graph makes it a sum of local terms, adjacent vertices lie in a common block and no route between them ever leaves it, so the total is additive over blocks and $K_m^{\text{multi}}(n)$ is a knapsack

over the best 2-connected block of each size. That turns an unbounded search into an enumeration of 2-connected graphs, and it settles the value at every $n \leq 8$ and $m \leq 8$ (Table A.2), with a clean split: thickened trees win for $m \leq 3$, the two agree at $m = 4$, and from $m = 5$ a single 2-connected block strictly beats every way of splitting the vertices, so the bouquet is a construction and not the shape of the answer. Second, the extremal blocks are visibly bipartite rather than complete, and thickening $K_{s,t}$ instead of K_r (Theorem A.67) raises the rate from $(\frac{1}{8} + o(1))m^2$ to $(3 - 2\sqrt{2} + o(1))m^2$, a factor of $8(3 - 2\sqrt{2}) \approx 1.373$, which brings the gap to the upper bound down from about 16 to $6 + 4\sqrt{2} \approx 11.66$. The optimal shape is lopsided, one side at the ceiling $m - 1$ and the other about $(\sqrt{2} - 1)m$, and the clique is the degenerate case that forces the two sides equal. What is open is the remaining constant, and the upper bound is now the side to attack, since it still charges every edge the full ceiling $m - 1$ and discards the correction $-\sum \pi$ that the whole section is really about.

- ★ Extremal uniqueness for directed multigraphs beyond the linear branch. Theorem A.30 settles the value $L_m^{\text{dir}}(n)$ for every n and m , and Lemma A.39 separately settles which multigraphs attain it on the linear branch, the doubled bidirected trees. On the quadratic branch the characterisation is now a theorem for a wide range of sizes rather than the single data point at $n = 7$, $m = 3$ that Remark A.41 supplies: for $n \geq \max(8, 2m)$ the extremiser is unique, the balanced one-directional complete bipartite digraph at multiplicity $m - 1$ (Theorem A.44). It had looked as though the reachability-skeleton argument could say nothing here, since it never needs to know which digraph it is peeling apart, and that is true of the proof read forwards. Read backwards from equality it is rigid: the skeleton count is maximised at exactly one value of the component number, which forces the extremiser to be acyclic, and the triangle-freeness that had only been used to make the skeleton small becomes, at equality, the equality case of Mantel's theorem and pins the skeleton to a balanced complete bipartite graph. What is left open is the range $n < 2m$, where the one step that needs room, converting a two-arc path in the skeleton into $\lfloor n/2 \rfloor$ arc-disjoint routes, no longer beats the budget $m - 1$. Losing that step removes a contradiction rather than supplying a counterexample, so the hypothesis reads as an artefact of the argument, but that is a guess and not a result.
- ★ The undirected vertex-disjoint problem for $m \geq 6$, open since 1974, where even the extremal structure is unknown.
- ★ The hypergraph vertex problem at $m \geq 4$. The problem is now settled at $m = 2$ and $m = 3$ for every uniformity and all n , with $k_m^{(r)}(n) = \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor = \ell_m^{(r)}(n)$ (Theorems A.50 and A.53). The incidence reduction stands for every m , but the rank bound that drives it (Lemma A.52) leans on the triconnected (Tutte/SPQR) decomposition exactly where $\kappa \leq 2$ lives. For $m = 4$ it would need a 4-connectivity analogue, where no comparably clean decomposition is available. Exhaustive computation finds no counterexample at $m = 4$ at any of eleven sizes, confirming both that $\lfloor 3(n-1)/(r-1) \rfloor$ is attained and that one more hyperedge is infeasible, so the agreement $k_m^{(r)} = \ell_m^{(r)}$ survives well past the theorem. That it must break eventually is not a guess: at $r = 2$ this problem is exactly the multigraph vertex problem of Section A.9, since a

2-uniform hyperedge is just an edge and every internally vertex-disjoint route of length two or more is automatically hyperedge-disjoint too, so the two disjointness requirements coincide and the two counting problems are the same problem under two names. There [Theorem A.63](#) breaks the formula for every $m \geq 5$. So the question is not whether the pattern fails but whether $r \geq 3$ postpones the failure past $m = 4$, which leaves the $m = 4$ row sitting on the boundary, and where it first breaks for $r \geq 3$ is unknown.

- ★ The directed hypergraph variants. The tail-and-heads model has first bounds and a quadratic lower-bound family ([Proposition A.55](#)): counting how many hyperedges may serve the same ordered pair of vertices caps the total at $(m-1)n(n-1)/(r-1)$, and a bipartite construction reaches $\approx (m-1)n^2/(4(r-1))$, so the value is quadratic in n . Those two bounds differ by a factor of four, which fixes the order and not the leading constant, and closing that factor was the first target. For the forward model it is now closed. [Theorem A.59](#) proves that the bipartite construction is asymptotically optimal, for the edge- and the vertex-disjoint separations at once, with the value $(1+o(1))(m-1)n^2/(4(r-1))$. The obstacle the appendix had recorded is real and the proof does not remove it: a hyperedge carries $r-1$ heads at once, so two-step routes through different midpoints need not be edge-disjoint. It absorbs it instead. Such a family can always be thinned to a genuine packing at a cost of a factor $r-1$, and that factor is affordable because of where it lands, inflating only the linear error term of the counting bound and never the $n^2/4$ that carries the constant, while a second and unrelated factor of $r-1$ is recovered when the auxiliary digraph is traded back for hyperedges. The orientation comparison is now closed to first order. Backward coincides with forward exactly, by reversal ([Lemma A.56](#)), and the general model, which the proof of [Theorem A.59](#) does not reach because it uses the single tail twice, is covered separately by [Theorem A.60](#): taking a maximum family of two-step routes instead of building a large one bounds the leftovers by the vertices of the hyperedges the family has already spent, which needs no independence between the entering and leaving sides and so survives several tails. All three orientations therefore share the leading term. What is still open here is the exact value at finite n , for every orientation, together with the finite form of the orientation comparison: the general model is strictly denser in the scarce-vertex regime $n \approx r$ ([Section 4.3.1](#)), and whether the two values agree exactly once n exceeds r , as the computed cells suggest, is not settled by an asymptotic statement.

The map is filled in where honest mathematics could fill it, and where it could not, the blanks are marked precisely, with a machine standing ready at each of them. That, finally, is the use of building the microscope: not that it answers every question, but that it shows exactly which questions are left, and hands the next reader a clean place to start.

Appendix A

Full Proofs and Computational Transcripts

This appendix holds the arguments deferred from the body: the ones that are either standard machinery or long case-checking, and whose length, not whose difficulty, kept them out of the narrative. The body carries the ideas. Here they are carried out in full. Because graphs are visual objects, the longer arguments here are accompanied by figures: each is drawn rather than merely described, and the prose points the reader to the picture at the moment it is needed.

How to read this appendix. The sections divide into standard machinery and the thesis's own proofs, and a reader chasing one thread of the body need not read all of them. The first two sections, on [Theorem A.1](#), [Theorem A.2](#), and Mader's theorem, are classical results included only so the body can cite them in the exact form it uses, and a reader who already trusts Menger, Gomory–Hu, and Mader may skip them. Six sections carry the load-bearing original arguments, and they fall into two groups. Three are about the directed models. The directed arc value at $m = 2$ supplies the finite base cases the body's induction rests on. The structural propositions on the directed frontier prove the unconditional quadratic bound, and then isolate the one counting fact behind it ([Lemma A.27](#)), which is what the last of the six reuses. The directed multigraph problem at large n is the longest argument here and the one behind the main directed contribution. Three more are about the other two models. The hypergraph vertex problem proves the incidence-rank lemma that settles the $m = 3$ hypergraph case. The multigraph vertex problem under the other convention withdraws a natural guess and replaces it with a construction and a matching order of growth. The directed hypergraph settles a leading constant, and is the section to read first for what [Lemma A.27](#) is good for beyond the case it was proved for. The remaining sections, on the hypergraph edge bounds and the random threshold, each support a single claim of the body and can be read on their own when that claim is reached. Finally, everything the program asserts is reproduced verbatim in the computational transcripts and the self-test section, so the machine proofs can be audited without running anything.

A.1 Menger, Gomory–Hu, and the degree bound

The two classical theorems below are the foundation of everything the thesis measures, and we state them at the level of generality at which the program actually applies them. A *capacitated graph*, or *weighted graph*, is a graph, directed or not, carrying a non-negative capacity $c(e)$ on

each edge or arc. A plain graph is read as the *unit-capacity* case $c \equiv 1$, and a multigraph as the *integer-capacity* case, one unit per parallel copy. A *flow* from u to v sends units along edges without exceeding any capacity, and the *capacity of a cut* is the total capacity of the edges crossing it. Both theorems below are theorems about capacitated graphs, and the thesis uses them mostly at unit and integer capacities, where the maximum flow is exactly the connectivity λ .

Theorem A.1 (Menger, as max-flow min-cut [2]). *In any capacitated digraph, and for distinct vertices u, v , the maximum flow from u to v equals the minimum capacity of a cut separating u from v . At unit capacities this is the combinatorial statement we cite throughout: the maximum number of pairwise edge-disjoint u – v paths equals the minimum number of edges whose removal separates them,*

$$\lambda_G(u, v) = \min\{|C| : C \subseteq E(G), G - C \text{ has no } u\text{--}v \text{ path}\}.$$

A directed capacitated graph is the most general object here, and every graph model the thesis uses is a special case of it. An undirected graph is the symmetric case, each edge a pair of opposite unit-capacity arcs. A multigraph is the integer-capacity case, with $\mu(u, v)$ units on the pair uv . The hypergraph version (Theorem A.45) is this very theorem applied to the helper-point network of Figure 2.5, a capacitated digraph. The internally vertex-disjoint version is obtained not by changing the theorem but by changing the graph it runs on, a device called *vertex splitting*: replace each internal vertex w by two copies joined by a single unit-capacity arc $w^- \rightarrow w^+$, send every arc that entered w into w^- and every arc that left w out of w^+ . A flow may now pass through w at most once, so edge-disjoint paths in the split graph are exactly internally vertex-disjoint paths in the original, and a minimum edge cut there is a minimum vertex cut here. This split graph is literally what the checker of Chapter 2 builds, which is why a single maximum-flow routine measures all four separations.

Theorem A.2 (Gomory–Hu [15]). *Every undirected capacitated graph G has a weighted tree T on the same vertex set, its Gomory–Hu tree, such that for all u, v the value $\lambda_G(u, v)$ equals the minimum edge weight on the u – v path in T , and each tree edge e corresponds to a minimum cut of G whose capacity is the weight of e . All $\binom{n}{2}$ pairwise values are thereby stored in the $n - 1$ weights of a single tree.*

Two features of the statement fix how far it reaches, and both are used below. Being a theorem about capacities, it applies verbatim to a simple graph (unit weights), to a multigraph (integer weights, the multiplicity μ as capacity), and, in the form of Theorem A.46, to the hyperedge-connectivity of a hypergraph. But it needs the cut function to be *symmetric*, which forces G to be undirected: a digraph has $\lambda(u, v) \neq \lambda(v, u)$ in general and admits no such tree, and the vertex-connectivity function fails symmetry for the same reason. So the Gomory–Hu tree is precisely the tool for the undirected edge problems, where it drives Mader’s theorem and its multigraph and hypergraph cousins (Theorems 1.2 and 1.3 and proposition 1.15) through one argument, and precisely the tool that vanishes once arcs turn one-way.

Existence rests on one property of cuts: writing $d(S)$ for the capacity of the cut $(S, V \setminus S)$, d is *submodular*, $d(A) + d(B) \geq d(A \cap B) + d(A \cup B)$ for any vertex sets A, B , which is exactly the inequality that lets a cheapest u – v cut crossing some other set S be traded for one that does not, without ever losing capacity. Building the tree is then $n - 1$ rounds of one minimum-cut computation each, each round splitting one bag of the current tree along a fresh minimum cut, with that trading property (the *uncrossing lemma*) guaranteeing no earlier round's cut is ever spoiled by a later one. The full construction, uncrossing induction included, is in Gomory and Hu's original paper [15].

The most basic constraint on local connectivity is local: a pair cannot have more independent routes than either endpoint has edges. The constructions below use it constantly.

Lemma A.3 (Degree bound). *For every pair of vertices of a graph G , $\lambda_G(u, v) \leq \min(\deg u, \deg v)$. In a digraph D , $\lambda_D(u, v) \leq \min(d^+(u), d^-(v))$.*

Proof. Every edge-disjoint u – v path uses a distinct edge at u , so there are at most $\deg u$ of them, and likewise at most $\deg v$. In the directed case each arc-disjoint path leaves u by a distinct out-arc and enters v by a distinct in-arc. $\square$

A.2 Mader's theorem in full

The upper bound of Theorem 1.2 rests on three short lemmas about a Gomory–Hu tree T of G . For an edge $e = \{u, v\}$ of G , write $\text{dist}_T(e)$ for the number of tree edges on the unique u – v path in T . Since T is a tree on the very same vertex set, $\text{dist}_T(e) \geq 1$ for every edge. Figure A.1 shows a small graph beside one of its Gomory–Hu trees and marks which edges have $\text{dist}_T(e) = 1$ and which have $\text{dist}_T(e) \geq 2$. It is worth keeping in view, since the three lemmas below are exactly statements about that picture. The first lemma is the one place the proof uses that G is *simple*, and it is exactly where the simple-graph bound improves on the multigraph bound.

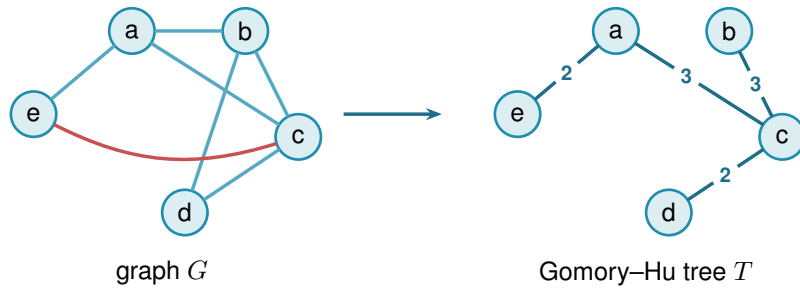


Figure A.1. A simple graph G (left) and a Gomory–Hu tree T of it (right). Both live on the same five vertices. The tree T has edges $ea(2)$, $ac(3)$, $cb(3)$, $cd(2)$, where each weight equals the local edge-connectivity of its endpoints in G . These four edges are also edges of G , so each has $\text{dist}_T(e) = 1$: these are the E_1 edges of Lemma A.4, and a *simple* graph admits at most $n - 1 = 4$ of them, one per tree edge. The remaining edges ab , bd , and the red chord ec all join tree-distance-2 pairs: the path in T from e to c is $e-a-c$, so $\text{dist}_T(ec) = 2$. Reading $\text{dist}_T(e)$ as the number of tree cuts an edge crosses makes $\sum_e \text{dist}_T(e) = \sum_t c(t)$ (Lemma A.6). Every edge crosses at least one cut, but only the $n - 1$ tree edges cross exactly one, which is the surplus that drives Lemma A.5 and ultimately the factor of two in Mader's bound.

Lemma A.4 (Distance-one edges). *At most $n - 1$ edges of a simple graph G satisfy $\text{dist}_T(e) = 1$.*

Proof. An edge $e \in E(G)$ has $\text{dist}_T(e) = 1$ if and only if its endpoints are adjacent in T . The tree T has exactly $n - 1$ edges, and since G is simple each pair of adjacent tree vertices supports at most one edge of G . $\square$

Lemma A.5 (Sum of tree distances). $\sum_{e \in E(G)} \text{dist}_T(e) \geq 2|E(G)| - (n - 1)$.

Proof. Partition $E(G)$ into $E_1 = \{e : \text{dist}_T(e) = 1\}$ and $E_{\geq 2} = \{e : \text{dist}_T(e) \geq 2\}$. Then

$$\sum_{e \in E(G)} \text{dist}_T(e) \geq |E_1| + 2|E_{\geq 2}| = 2|E(G)| - |E_1| \geq 2|E(G)| - (n - 1),$$

using [Lemma A.4](#) in the last step. $\square$

Lemma A.6 (Double-counting identity). *Let G carry unit capacities, so that each weight $c(t)$ of a Gomory–Hu tree T is the plain number of edges of G crossing the cut that t induces. Then*

$$\sum_{t \in E(T)} c(t) = \sum_{e \in E(G)} \text{dist}_T(e).$$

This is the form used for Mader’s theorem, where G is a simple graph and so unit-capacitated by default.

Proof. By [Theorem A.2](#) each tree edge t induces a cut of G of capacity $c(t)$, so $c(t) = \sum_{e \in E(G)} \mathbf{1}_{e \text{ crosses } t}$. Exchanging the order of summation,

$$\sum_{t \in E(T)} c(t) = \sum_{e \in E(G)} \sum_{t \in E(T)} \mathbf{1}_{e \text{ crosses } t} = \sum_{e \in E(G)} \text{dist}_T(e),$$

where the last equality holds because an edge $\{u, v\}$ crosses exactly those tree cuts induced by the tree edges on the unique u – v path in T . $\square$

Proof of [Theorem 1.2](#), upper bound. Let G be a simple graph on n vertices with $\lambda_G^{\max} \leq m - 1$ and let T be a Gomory–Hu tree of G . Each tree edge $t = \{x, y\}$ has weight $c(t) = \lambda_G(x, y) \leq m - 1$, so summing over the $n - 1$ tree edges, $\sum_t c(t) \leq (m - 1)(n - 1)$. Combining with [Lemmas A.5](#) and [A.6](#),

$$2|E(G)| - (n - 1) \leq \sum_{e \in E(G)} \text{dist}_T(e) = \sum_{t \in E(T)} c(t) \leq (m - 1)(n - 1).$$

Rearranging gives $2|E(G)| \leq m(n - 1)$, and since $|E(G)|$ is an integer, $|E(G)| \leq \left\lfloor \frac{m(n-1)}{2} \right\rfloor$. $\square$

Note where the factor of two came from: every edge crosses at least one tree cut, but a simple graph can afford at most $n - 1$ edges that cross only one, so on average each edge

is charged closer to twice. For multigraphs that refinement is unavailable, since parallel edges between tree-adjacent pairs each have $\text{dist}_T = 1$, and the same chain of inequalities stops at $|E| \leq \sum_e \text{dist}_T(e) \leq (m-1)(n-1)$, which is exactly the multigraph bound of [Theorem 1.3](#).

The lower bound is a pair of constructions, drawn side by side in [Figure A.2](#). The first is the shape the bound was guessed from, and works when $m-1$ divides $n-1$. The second generalises it to every $n \geq m \geq 2$.

Construction A.7 (Shared-clique graph). For $m \geq 3$ with $(m-1) \mid (n-1)$, take $k = (n-1)/(m-1)$ copies of the complete graph K_m and identify one vertex of each copy into a single shared hub vertex h . The result has $k(m-1) + 1 = n$ vertices and $k \binom{m}{2} = \frac{m(n-1)}{2}$ edges. Every non-hub vertex has degree $m-1$, and any pair of vertices contains a non-hub vertex, so $\lambda^{\max} \leq m-1$ by [Lemma A.3](#).

Construction A.8 (Hub-and-spokes graph). For $n \geq m \geq 2$, let $H_{m,n}$ have vertex set $\{h, v_1, \dots, v_{n-1}\}$ with (i) all $n-1$ hub edges $\{h, v_i\}$, and (ii) a spoke graph on $\{v_1, \dots, v_{n-1}\}$ with $\left\lfloor \frac{(m-2)(n-1)}{2} \right\rfloor$ edges in which every v_i has spoke degree at most $m-2$. Such a spoke graph exists by [Lemma A.9](#).

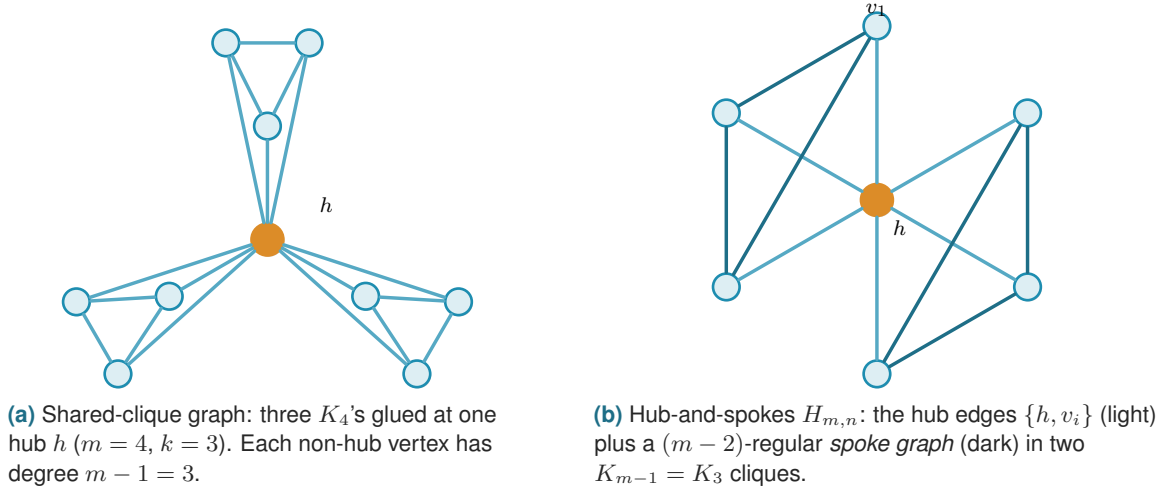


Figure A.2. The two extremal families. (a) The shared-clique graph of [Construction A.7](#) works when $(m-1) \mid (n-1)$: every pair contains a non-hub vertex of degree exactly $m-1$, so $\lambda^{\max} \leq m-1$ by the degree bound, while the edge count hits $\frac{m(n-1)}{2}$. (b) The hub-and-spokes graph $H_{m,n}$ of [Construction A.8](#) reaches the same bound for every $n \geq m$: each spoke vertex v_i has total degree $1 + (m-2) = m-1$, and choosing the spoke graph as disjoint cliques K_{m-1} recovers (a). The hub is drawn in orange in both.

Lemma A.9 (Near-regular graphs exist). For $N \geq 1$ and $0 \leq r \leq N-1$ there is a graph on N vertices with $\left\lfloor \frac{rN}{2} \right\rfloor$ edges and maximum degree at most r .

Proof. If rN is even we build an r -regular circulant graph on vertex set $\{0, \dots, N-1\}$: for even r connect each i to $i \pm j \pmod{N}$ for $j = 1, \dots, r/2$, and for odd r (then N is even) use $j = 1, \dots, (r-1)/2$ together with the diameter distance $N/2$, which contributes degree one. Both

are r -regular with $rN/2$ edges. If rN is odd, then r and N are both odd: take the $(r-1)$ -regular circulant with distances $1, \dots, (r-1)/2$ and add the near-perfect matching that pairs vertex i with $i + (N+1)/2 \pmod{N}$ for $i = 0, \dots, (N-3)/2$, which covers all vertices but one. The total is $\frac{(r-1)N}{2} + \frac{N-1}{2} = \frac{rN-1}{2} = \lfloor \frac{rN}{2} \rfloor$ edges, with one vertex of degree $r-1$ and all others of degree r . $\square$

Theorem A.10 (Extremal graphs exist). *For all $n \geq m \geq 2$ the graph $H_{m,n}$ has $\lfloor \frac{m(n-1)}{2} \rfloor$ edges and $\lambda^{\max} \leq m-1$.*

Proof. The edge count is $(n-1) + \lfloor \frac{(m-2)(n-1)}{2} \rfloor = \lfloor \frac{m(n-1)}{2} \rfloor$, since $(m-2)(n-1)$ and $m(n-1)$ have the same parity. For the connectivity, any two distinct vertices include at least one spoke vertex v_i , whose total degree is at most $1 + (m-2) = m-1$. [Lemma A.3](#) then gives $\lambda(u, v) \leq m-1$. When $(m-1) \mid (n-1)$, choosing the spoke graph as k disjoint cliques K_{m-1} recovers [Construction A.7](#), so the hub-and-spokes family genuinely generalises the shared-clique one. $\square$

Together the upper bound and [Theorem A.10](#) prove [Theorem 1.2](#). The proof also reveals which graphs are extremal: equality forces both inequalities in the chain to be tight.

Theorem A.11 (Characterisation of equality). *Let G be simple with $\lambda_G^{\max} \leq m-1$ and $|E(G)| = \lfloor \frac{m(n-1)}{2} \rfloor$, and let T be a Gomory–Hu tree of G . If $m(n-1)$ is even, then every tree edge has weight exactly $m-1$, every tree edge is also an edge of G , and every edge of G joins vertices at tree distance at most 2. If $m(n-1)$ is odd, there is exactly one unit of slack, located in exactly one of three places: one tree edge has weight $m-2$, or one tree edge is not an edge of G , or one edge of G joins vertices at tree distance 3. Everything else is tight.*

Proof. Write the chain of the upper-bound proof as three non-negative integer slacks, one for each inequality it uses:

$$S_1 = (m-1)(n-1) - \sum_t c(t), \quad S_2 = \sum_{e: \text{dist}_T(e) \geq 2} (\text{dist}_T(e) - 2), \quad S_3 = (n-1) - |E_1|.$$

In words, S_1 counts tree weights below $m-1$, S_2 counts edges beyond tree distance two, and S_3 counts tree edges that support no edge of G . Retracing the proof gives two expressions for the same quantity,

$$\sum_e \text{dist}_T(e) = 2|E| - |E_1| + S_2 \quad \text{and} \quad \sum_e \text{dist}_T(e) = \sum_t c(t) = (m-1)(n-1) - S_1.$$

Substituting $|E_1| = (n-1) - S_3$ into the first and equating it with the second,

$$\begin{aligned} 2|E| - (n-1) + S_2 + S_3 &= (m-1)(n-1) - S_1, \\ \text{so} \quad 2|E| &= m(n-1) - (S_1 + S_2 + S_3). \end{aligned}$$

The whole characterisation reads off that last identity. If $m(n-1)$ is even, equality $|E| = m(n-$

1)/2 forces $S_1 = S_2 = S_3 = 0$. If $m(n-1)$ is odd the floor absorbs exactly one unit, so $S_1 + S_2 + S_3 = 1$ and exactly one of the three slacks equals one. $\square$

For $m = 2$ the characterisation says the extremal graphs are exactly the spanning trees: every tree edge must carry weight one and lie in G , so G contains its own Gomory–Hu tree and has only $n - 1$ edges to spend. For $m = 4$ and $n \equiv 1 \pmod{3}$ it is consistent with the trees of K_4 blocks glued at shared cut vertices, the case of [Construction A.7](#) that the search of [Chapter 3](#) rediscovers.

A.3 The directed arc value at $m = 2$

Three short lemmas feed the induction. The first shows a hub forces a linear bound for every m , and is the proof behind the hub branch of [Conjecture 1.14](#). The three lemmas below and the proof of [Theorem 1.10](#) that they support are the author’s own, and its finite base cases $2 \leq n \leq 7$ were verified by the author’s program, through the exhaustive search of [Chapter 2](#) whose transcript is reproduced at the end of this appendix.

Lemma A.12 (Two-hop lemma). *Let D be a simple digraph with $\lambda_D^{\max} \leq m - 1$ containing a vertex r with $d^+(r) = n - 1$. Then every $v \neq r$ has at most $m - 2$ in-arcs from $V \setminus \{r, v\}$.*

Proof. If $u_1, \dots, u_{m-1}$ were distinct in-neighbours of v other than r , that is, distinct vertices each sending an arc to v , the paths $r \rightarrow v$ and $r \rightarrow u_i \rightarrow v$ for $i = 1, \dots, m-1$ would be m arc-disjoint routes, since the u_i are distinct and $d^+(r) = n - 1$ supplies every starting arc. That contradicts $\lambda_D(r, v) \leq m - 1$. $\square$

Theorem A.13 (Hub upper bound). *If a simple digraph D with $\lambda_D^{\max} \leq m - 1$ has a vertex r with $d^+(r) = n - 1$, then $|A(D)| \leq m(n - 1)$.*

Proof. Split the arcs into those leaving r (exactly $n - 1$), those entering r (at most $n - 1$), and the internal arcs avoiding r . By [Lemma A.12](#) each of the $n - 1$ other vertices receives at most $m - 2$ internal arcs, so there are at most $(m - 2)(n - 1)$ internal arcs, and the total is at most $m(n - 1)$. $\square$

Lemma A.14 (No transitive triangle). *A simple digraph with $\lambda^{\max} \leq 1$ contains no three vertices x, y, z with all of $(x, y), (y, z), (x, z)$ present, since $x \rightarrow z$ and $x \rightarrow y \rightarrow z$ would be two arc-disjoint routes.*

Lemma A.15 (Deletion monotonicity). *Write $D - v_0$ for the digraph obtained from D by deleting the vertex v_0 together with all arcs incident to it. For any digraph D and vertex v_0 ,*

$$\lambda^{\max}(D - v_0) \leq \lambda^{\max}(D) \quad \text{and} \quad \kappa^{\max}(D - v_0) \leq \kappa^{\max}(D),$$

since arc-disjoint paths in $D - v_0$ are still arc-disjoint paths in D , and internally vertex-disjoint paths in $D - v_0$ are still internally vertex-disjoint paths in D . The two statements are proved by

the same one-line observation because deleting a vertex only removes routes, never creates one, and it cannot make two surviving routes share anything they did not share before.

Proof of Theorem 1.10. Write $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$. A direct computation shows $2(n-1) \geq \lfloor n^2/4 \rfloor$ exactly for $n \leq 7$, with equality at $n = 7$, so $M(n) = 2(n-1)$ for $n \leq 7$ and $M(n) = \lfloor n^2/4 \rfloor$ for $n \geq 7$. The lower bound $\ell_2^{\text{dir}}(n) \geq M(n)$ is witnessed by the directed hub construction (Construction 1.9 at $m = 2$, the bidirected star) and the one-directional bipartite construction (Construction 1.8).

For the upper bound we show by strong induction on n that $\lambda_D^{\max} \leq 1$ forces $|A(D)| \leq M(n)$.

Base cases $2 \leq n \leq 7$. All six are settled uniformly by the pruned exhaustive search of Chapter 2, which walks every simple digraph with $\lambda^{\max} \leq 1$ on up to seven vertices and reports the maxima 2, 4, 6, 8, 10, 12, matching $M(n)$ at each size. The transcript is reproduced below, and the cut-counting certifier independently confirms the values within its reach through the $m = 2$ reduction of Corollary A.38. The two smallest are also immediate by hand, as a check on the machine rather than a separate argument: at $n = 2$ there are at most 2 arcs, and at $n = 3$ a fifth arc would leave at most one arc of the complete bidirected triangle missing, so three of the rest form a transitive triangle, which Lemma A.14 forbids. The point of the theorem is precisely that $n = 4, 5, 6, 7$ are *not* immediate by hand, which is what makes handing the case-check to the program the right move.

Inductive step $n \geq 8$. Let D be a simple digraph on n vertices with $\lambda_D^{\max} \leq 1$ and put $a := |A(D)|$. Choose v_0 minimising the total degree $d(v) = d^+(v) + d^-(v)$. Since $\sum_v d(v) = 2a$, averaging gives $d(v_0) \leq 2a/n$. By Lemma A.15 the digraph $D - v_0$ on $n-1 \geq 7$ vertices satisfies the induction hypothesis, so $a - d(v_0) \leq M(n-1) = \lfloor \frac{(n-1)^2}{4} \rfloor$. Combining,

$$a \leq \frac{2a}{n} + \left\lfloor \frac{(n-1)^2}{4} \right\rfloor \quad \implies \quad a \leq \frac{n}{n-2} \left\lfloor \frac{(n-1)^2}{4} \right\rfloor.$$

It remains to check that this right-hand side never exceeds $\lfloor n^2/4 \rfloor$, and the two parities behave differently enough to treat them in turn.

For even $n = 2k$ with $k \geq 4$, first simplify $\lfloor (n-1)^2/4 \rfloor$. Expanding $(2k-1)^2 = 4k^2 - 4k + 1$ and dividing by 4 gives $k^2 - k + \frac{1}{4}$, whose floor is $k(k-1)$. The bound then reads

$$a \leq \frac{2k}{2k-2} k(k-1) = \frac{k}{k-1} k(k-1) = k^2,$$

and since $\lfloor n^2/4 \rfloor = \lfloor (2k)^2/4 \rfloor = k^2$ as well, the bound lands exactly on the target with no slack left over.

For odd $n = 2k+1$ with $k \geq 4$, the same simplification gives $\lfloor (n-1)^2/4 \rfloor = \lfloor (2k)^2/4 \rfloor = k^2$, so the bound is $a \leq \frac{2k+1}{2k-1} k^2$. To compare this fraction with the target, split off its integer part by dividing $(2k+1)k^2$ by $2k-1$. The remainder is exactly k , because $(2k-1)(k^2+k) = 2k^3 + k^2 - k$

and $(2k+1)k^2 = 2k^3 + k^2$ differ by k , so

$$\frac{n}{n-2} \left\lfloor \frac{(n-1)^2}{4} \right\rfloor = \frac{(2k+1)k^2}{2k-1} = k(k+1) + \frac{k}{2k-1}.$$

The leftover fraction $\frac{k}{2k-1}$ lies strictly between 0 and 1 for every $k \geq 2$ (the induction here is already past the base cases, so $k \geq 4$), so the bound puts a at most an integer $k(k+1)$ plus a positive amount below one. Since a is itself an integer it cannot reach the next integer up, so $a \leq k(k+1)$. The target $\lfloor n^2/4 \rfloor = \lfloor (2k+1)^2/4 \rfloor = k^2 + k = k(k+1)$ agrees, so for odd n it is integrality, not the raw arithmetic, that closes the last fraction of a unit. In both parities $a \leq \lfloor n^2/4 \rfloor \leq M(n)$, completing the induction. $\square$

Two remarks on the shape of this proof. First, the minimum-degree deletion closes the step on its own, and no separate hub case is needed, because the averaging bound holds whether or not D contains a hub, although [Theorem A.13](#) shows directly that a hub digraph never exceeds the linear branch. Second, for even n the bound is exactly tight against the one-directional bipartite construction, which is $\frac{n}{2}$ -regular in total degree, so every step of the chain is achieved by the conjectured extremiser, and for odd n the slack $\frac{k}{2k-1} < 1$ is absorbed by integrality. The proof is long not because it is deep but because the two regimes of $M(n)$ must be carried through the arithmetic separately, and that bookkeeping is precisely what [Chapter 2](#) hands to the machine at the base cases.

Remark A.16 (Which way Whitney runs). It is worth pausing on the direction of Whitney's inequality before using it, because the tempting reading is the wrong one. From $\kappa_D(u, v) \leq \lambda_D(u, v)$ for every pair it follows that $\kappa_D^{\max} \leq \lambda_D^{\max}$, hence that

$$\{D : \lambda_D^{\max} \leq m-1\} \subseteq \{D : \kappa_D^{\max} \leq m-1\}.$$

Every digraph feasible for the *arc* problem is feasible for the *vertex* problem, and not the other way round. The vertex-feasible family is the larger of the two, so the inequality between the extremal numbers it hands over for free is

$$k_m^{\text{dir}}(n) \geq \ell_m^{\text{dir}}(n),$$

and an upper bound for the arc problem says nothing at all about the vertex problem. The gap is not vacuous: two routes that share an interior vertex but no arc are counted by λ and not by κ , so there really are digraphs in the difference of the two families. What Whitney gives is therefore exactly one thing, that an arc construction is an honest vertex *lower* bound, and the vertex upper bound has to be earned separately. The proof below earns it, and [Chapter 4](#) keeps the same discipline for $m \geq 3$, where the vertex problem stays open even though the arc conjecture is stated.

Proof of Theorem 1.11. Write $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$ as before.

Lower bound. The directed hub and the one-directional bipartite construction both have

$\lambda^{\max} = 1$, so by [Remark A.16](#) both have $\kappa^{\max} \leq 1$ and are feasible for the vertex problem. Hence $k_2^{\text{dir}}(n) \geq M(n)$.

Upper bound. We show by strong induction on n that $\kappa_D^{\max} \leq 1$ forces $|A(D)| \leq M(n)$. The induction is the one that proved [Theorem 1.10](#), and every step of it survives the change of separation, because the only two properties of the ceiling that argument uses are that it is inherited by vertex deletion and that it holds at the base sizes.

The base cases $2 \leq n \leq 7$ are settled by the same pruned exhaustive search of [Chapter 2](#), run with the vertex feasibility test in place of the arc one. It walks every simple digraph with $\kappa^{\max} \leq 1$ on up to seven vertices and reports the maxima 2, 4, 6, 8, 10, 12, matching $M(n)$ at each size. The transcript is reproduced in [Section A.11](#) beside the arc one.

For the inductive step $n \geq 8$, let D be a simple digraph on n vertices with $\kappa_D^{\max} \leq 1$ and put $a := |A(D)|$. Choose v_0 of minimum total degree, so $d(v_0) \leq 2a/n$ by averaging. By [Lemma A.15](#), which holds for κ exactly as it does for λ , the digraph $D - v_0$ on $n - 1 \geq 7$ vertices again satisfies $\kappa^{\max} \leq 1$, so the induction hypothesis gives $a - d(v_0) \leq M(n - 1) = \lfloor (n - 1)^2 / 4 \rfloor$. From here the two parity computations are verbatim those of [Theorem 1.10](#): they use only the two displayed inequalities and the integrality of a , no property of the separation. They yield $a \leq \lfloor n^2 / 4 \rfloor \leq M(n)$, completing the induction.

Hence $k_2^{\text{dir}}(n) = M(n) = \ell_2^{\text{dir}}(n)$. The two problems have the same answer at $m = 2$, but this is a computed coincidence of the two searches meeting at every base size, not a formal consequence of Whitney's inequality. $\square$

Remark A.17 (The directed vertex problem at $m \geq 3$). The proof above transfers the $m = 2$ induction but nothing more, and for $m \geq 3$ the vertex problem is genuinely separate: the arc conjecture, and the decomposition of [Conjecture A.21](#) that would prove it, both concern the smaller family and leave $k_m^{\text{dir}}(n)$ untouched. What can be said is computational. Running the exhaustive search of [Chapter 2](#) under both feasibility tests at $m = 3$ gives

n	2	3	4	5	6
$\ell_3^{\text{dir}}(n)$	2	6	9	12	15
$k_3^{\text{dir}}(n)$	2	6	9	12	15

with every entry proved complete, so the two separations agree at every size the exhaustion reaches, and from $n = 3$ onwards both match the conjectured $\max(m(n - 1), \lfloor (n + m - 2)^2 / 4 \rfloor)$. At $n = 2$ the conjecture's formula gives 3 where only 2 arcs exist at all, the usual degeneracy of the formula below $n = m$. Agreement over five sizes is evidence and not a proof, and the natural next question is structural rather than numerical: to separate the two problems one needs a digraph where some pair carries strictly more arc-disjoint routes than internally disjoint ones *and* where that slack pays for extra arcs. The two demands pull against each other, which is perhaps why no small example does both.

A.4 Structural propositions on the directed frontier

These are the proved ingredients of the reduction discussed in [Chapter 4](#). The section ends with what the reduction still assumes, stated as [Conjecture A.21](#), and with the one quadratic bound that assumes nothing, [Proposition A.22](#).

Proposition A.18 (Mutually unreachable out-neighbourhoods). *Let D be a simple digraph with $\lambda_D^{\max} \leq 1$. For any vertex u and distinct out-neighbours v, w of u , there is no directed path from v to w in $D - u$.*

Proof. If P were a directed v – w path avoiding u , then $u \rightarrow w$ and $u \rightarrow v \xrightarrow{P} w$ would be two arc-disjoint u – w routes: the first uses only the arc (u, w) , the second starts with (u, v) and continues along P , none of whose arcs touch u . This contradicts $\lambda_D(u, w) \leq 1$. The restriction to $D - u$ is not cosmetic, and [Figure A.3](#) draws both the argument and the reason for the restriction: in the digraph with arcs $(u, v), (v, u), (u, w)$ every local connectivity is 1, yet $v \rightarrow u \rightarrow w$ reaches w from v through u . $\square$



(a) If a directed $v \rightsquigarrow w$ path (red) existed in $D - u$, then $u \rightarrow w$ and $u \rightarrow v \rightsquigarrow w$ would be two arc-disjoint u – w routes, against $\lambda(u, w) \leq 1$.

(b) The counterexample $\{(u, v), (v, u), (u, w)\}$: $\lambda^{\max} = 1$, yet w is reachable from v , but only *through* u .

Figure A.3. Why [Proposition A.18](#) must exclude paths through u . (a) For out-neighbours v, w of u in a digraph with $\lambda^{\max} \leq 1$, no $v \rightsquigarrow w$ path can avoid u : such a path would combine with the lone arc $u \rightarrow w$ into two arc-disjoint routes. (b) The restriction to $D - u$ is essential. In the three-vertex digraph $\{(u, v), (v, u), (u, w)\}$ we have $\lambda^{\max} = 1$, yet v reaches w via $v \rightarrow u \rightarrow w$, a route that reuses the vertex u , which the proposition allows.

Proposition A.19 (Disjoint second out-neighbourhoods). *Let D be a simple digraph with $\lambda_D^{\max} \leq 1$, let $u \in V$, and let $v_1 \neq v_2$ be out-neighbours of u . Then $(N^+(v_1) \setminus \{u\}) \cap (N^+(v_2) \setminus \{u\}) = \emptyset$.*

Proof. A common out-neighbour $w \neq u$ would give the two arc-disjoint routes $u \rightarrow v_1 \rightarrow w$ and $u \rightarrow v_2 \rightarrow w$, contradicting $\lambda_D(u, w) \leq 1$. $\square$

Proposition A.20 (Two-hop budget on bipartite partitions). *Let D be a simple digraph with $\lambda_D^{\max} \leq m - 1$. If V admits a partition $A \cup B$ with $A \neq \emptyset$ and every arc from A to B present, then every $b \in B$ has in-degree at most $m - 2$ from inside B , and consequently the number of arcs inside B is at most $(m - 2)|B|$.*

Proof. Fix $a \in A$ and $b \in B$, and let $b'_1, \dots, b'_d$ be the in-neighbours of b inside B , where $d = \deg_B^-(b)$. The direct arc $a \rightarrow b$ together with the two-step detours $a \rightarrow b'_i \rightarrow b$ are $1 + d$ arc-disjoint a -to- b routes, all present because $A \rightarrow B$ is complete. Feasibility forces $1 + d \leq m - 1$, so $d \leq m - 2$, and summing over B bounds the internal arcs by $(m - 2)|B|$. $\square$

We now combine [Proposition A.20](#) with the best choice of partition. Suppose every arc of D either runs from A to B or stays inside B : the partition $A \rightarrow B$ is complete, no arc lies inside A , and crucially *no arc runs from B back to A* . Write $b = |B|$, so that $|A| = n - b$. The arcs then come in exactly two kinds. The arcs crossing the partition number $|A||B| = (n - b)b$, one for each ordered A – B pair. The arcs inside B number at most $(m - 2)b$: since no route between two vertices of B ever needs to leave B (there is no arc back to A to leave by), [Proposition A.20](#) applies to B on its own and caps the in-degree of each $b \in B$ from inside B at $m - 2$. There are no other arcs, so, adding the two kinds,

$$|A(D)| \leq (n - b)b + (m - 2)b = b(n + m - 2 - b).$$

The right-hand side is a downward parabola in b . Writing $s = n + m - 2$, it is $b(s - b) = -b^2 + sb$, maximised at $b = s/2$, so over integers the best partition takes $b = \lceil s/2 \rceil$ (equivalently $\lfloor s/2 \rfloor$), and the maximum value is

$$\max_b b(s - b) = \left\lfloor \frac{s^2}{4} \right\rfloor = \left\lfloor \frac{(n + m - 2)^2}{4} \right\rfloor.$$

This is the quadratic branch of [Conjecture 1.14](#), reached uniformly in m . It is worth being exact about how much was assumed, because the count is often summarised as resting on the absence of backward arcs alone, and that summary is too generous. Four things were granted at once: that a partition $V = A \cup B$ exists at all, that every arc from A to B is present, that no arc lies inside A , and that no arc runs from B back to A . Only the last of the four is the backward-arc condition. The first three describe the shape of the extremiser, and nothing proved so far forces an arbitrary extremal digraph to have that shape. The honest statement of the gap is therefore a structural one:

Conjecture A.21 (The missing decomposition). *Let n be large enough that the quadratic branch of [Conjecture 1.14](#) strictly exceeds the linear one, that is, $\lfloor (n + m - 2)^2/4 \rfloor > m(n - 1)$. Then every extremal digraph D on n vertices with $\lambda_D^{\max} \leq m - 1$ admits a partition $V(D) = A \cup B$ such that every arc from A to B is present, no arc lies inside A , and no arc runs from B to A .*

The hypothesis on n has to be there, and it is worth seeing why, because the obvious phrasing is too narrow. On the linear branch the extremisers are *not* just the hub. At $m = 2$, every bidirected spanning tree is feasible and has exactly $2(n - 1)$ arcs, since between two vertices the only route follows the unique tree path, so the whole extremal set on that branch is the set of bidirected trees. An exhaustive classification confirms it: at $n = 5$ and $n = 6$ there are exactly 3 and 6 extremal digraphs up to isomorphism, which are precisely the numbers of trees on 5 and

6 vertices, and none of them admits the decomposition. Excluding “the hub” would leave all the other trees behind, so the clean hypothesis is on the size rather than on the shape.

Given [Conjecture A.21](#) the count above closes [Conjecture 1.14](#) for $m \geq 3$. The backward-arc clause is the part where the natural delete-and-compensate exchange fails to be monotone, which is why it is the clause usually named, but the partition itself is not free either, and [Chapter 4](#) states the position that way. Testing the conjecture by exhaustion is harder than it looks: the crossover sits at $n = 8$ for $m = 2$ and at $n = 9$ for $m = 3$, so every size an exhaustive sweep can reach at $m = 3$ still lies on the linear branch, where the statement says nothing. That is the practical reason this conjecture has resisted a computational probe rather than a proof.

What can be proved without any of the four is the leading constant, and the argument is short enough to give in full. It also says something the conjecture does not: that every feasible digraph is within a lower-order number of arcs of being one-directional bipartite.

Proposition A.22 (Unconditional quadratic bound and stability). *Let D be a simple digraph on $n \geq 2$ vertices with $\lambda_D^{\max} \leq m - 1$. Then*

$$|A(D)| \leq \left\lfloor \frac{n^2}{4} \right\rfloor + \sqrt{m} n^{3/2}.$$

Moreover D can be made one-directional bipartite, meaning every remaining arc runs from one part to the other and none returns, by deleting at most $\sqrt{m} n^{3/2}$ arcs. Consequently, for each fixed m ,

$$\ell_m^{\text{dir}}(n) = \frac{n^2}{4} + O_m(n^{3/2}),$$

so the leading constant $\frac{1}{4}$ of [Conjecture 1.14](#) holds unconditionally, with no hypothesis on the structure of the extremiser.

Proof. Step 1, a budget on two-step routes. Fix an ordered pair (u, v) with $u \neq v$, and let $p_2(u, v)$ count the vertices x with both $u \rightarrow x$ and $x \rightarrow v$ present. Such an x is automatically distinct from u and from v , since D has no loops, so each one gives a genuine two-step route $u \rightarrow x \rightarrow v$. Two of these routes with different midpoints $x \neq x'$ share no arc, since the first uses (u, x) and (x, v) and the second uses (u, x') and (x', v) , and neither uses the arc (u, v) , which has no midpoint at all. So the direct arc, when present, together with all the two-step routes, is a family of pairwise arc-disjoint u -to- v routes, and feasibility caps its size:

$$\mathbf{1}_{(u,v) \in A(D)} + p_2(u, v) \leq \lambda_D(u, v) \leq m - 1.$$

Summing over all $n(n - 1)$ ordered pairs gives $|A(D)| + \sum_{u \neq v} p_2(u, v) \leq (m - 1)n(n - 1)$.

Step 2, from pairs to degrees. Write Δ for the number of digons of D , meaning the pairs of vertices carrying an arc in each direction. Three small facts turn Step 1 into a statement about degrees alone.

(2a) *Walks of length two, counted by their midpoint.*

$$\sum_{u,v \in V} p_2(u,v) = \sum_{x \in V} d^-(x) d^+(x),$$

where the left-hand sum runs over *all* ordered pairs, the diagonal $u = v$ included. Fixing a vertex x , a walk $u \rightarrow x \rightarrow v$ is nothing more than an independent choice of an in-neighbour u of x and an out-neighbour v of x , so x is the midpoint of exactly $d^-(x) d^+(x)$ of them.

(2b) *What the diagonal counts.* A diagonal term $p_2(u, u)$ counts the closed walks $u \rightarrow x \rightarrow u$, and each digon contributes two such walks, one based at each of its endpoints. Hence

$$\sum_{u \in V} p_2(u, u) = 2\Delta.$$

(2c) *Digons are few.* Distinct digons use disjoint pairs of arcs, so $2\Delta \leq |A(D)|$.

Splitting the all-pairs sum of (2a) along its diagonal and feeding in Step 1 now gives the inequality the rest of the argument runs on:

$$\begin{aligned} \sum_{x \in V} d^+(x) d^-(x) &= \sum_{u \neq v} p_2(u, v) + 2\Delta && \text{by (2a) and (2b),} \\ &\leq [(m-1)n(n-1) - |A(D)|] + 2\Delta && \text{by Step 1,} \\ &\leq [(m-1)n(n-1) - |A(D)|] + |A(D)| && \text{by (2c),} \\ &= (m-1)n(n-1). \end{aligned}$$

The point of the last two lines is that the digon term is absorbed exactly by the slack Step 1 left behind, so nothing is given away in passing from pairs to degrees. It is worth reading the conclusion in words: a vertex with a large in-degree *and* a large out-degree is expensive, because it manufactures two-step routes for many pairs at once, and the budget for those routes is only quadratic in n .

Step 3, the smaller half-degree is small on average. Since $\min(a, b) \leq \sqrt{ab}$ for non-negative a, b , and by the Cauchy-Schwarz inequality in the form $\sum_x \sqrt{a_x} \leq \sqrt{n \sum_x a_x}$,

$$\begin{aligned} \sum_{x \in V} \min(d^+(x), d^-(x)) &\leq \sum_{x \in V} \sqrt{d^+(x) d^-(x)} \leq \sqrt{n \sum_{x \in V} d^+(x) d^-(x)} \\ &\leq \sqrt{n \cdot (m-1)n(n-1)} = n\sqrt{(m-1)(n-1)} \leq \sqrt{m} n^{3/2}. \end{aligned}$$

Step 4, deleting the smaller half of every vertex. Call x *source-like* if $d^+(x) \geq d^-(x)$ and *sink-like* otherwise, and write S and T for the two classes, so $V = S \cup T$. Delete every in-arc of a source-like vertex and every out-arc of a sink-like vertex. A source-like x loses $d^-(x) = \min(d^+(x), d^-(x))$ arcs and a sink-like x loses $d^+(x) = \min(d^+(x), d^-(x))$ arcs, so the number of deleted arcs is at most $\sum_x \min(d^+(x), d^-(x))$, which Step 3 bounds by $\sqrt{m} n^{3/2}$. An arc (a, b)

survives only if a is not sink-like and b is not source-like, that is, only if $a \in S$ and $b \in T$. So what remains runs entirely from S to T , which is the one-directional bipartite shape, and it has at most $|S||T| \leq \lfloor n^2/4 \rfloor$ arcs. Adding the two counts gives the bound.

The asymptotic statement follows because the augmented bipartite construction ([Construction 1.13](#)) already gives $\ell_m^{\text{dir}}(n) \geq \lfloor (n+m-2)^2/4 \rfloor = n^2/4 + O_m(n)$, and $n^{3/2}$ dominates n . $\square$

The error term $n^{3/2}$ is larger than the $(m-2)n/2$ that [Conjecture 1.14](#) predicts, so the proposition as it stands does not separate the conjectured formula from its near neighbours. It can be sharpened to the right order, and the extra idea is small enough to be worth giving, because it explains where the $\sqrt{n}$ was being wasted.

Step 3 is lossy exactly when every vertex carries a middling degree. The Cauchy-Schwarz step is tight only if $d^+(x)d^-(x)$ is about $(m-1)n$ at every vertex, which forces every degree down to about $\sqrt{mn}$. But a digraph with $\Theta(n^2)$ arcs has average degree $\Theta(n)$, and at a vertex of large degree the constraint $d^+(x)d^-(x) \leq (m-1)n$ bites much harder: with $d^+ + d^-$ of order n , the smaller of the two must be of order m rather than $\sqrt{mn}$. The dense case and the tight case of Step 3 are therefore incompatible, and the way to exploit that is a dichotomy on the minimum degree, with vertex deletion to handle the sparse side.

Lemma A.23 (Small side of a high-degree vertex). *For any vertex x of a digraph with total degree $d(x) = d^+(x) + d^-(x) > 0$,*

$$\min(d^+(x), d^-(x)) \leq \frac{2 d^+(x) d^-(x)}{d(x)}.$$

Proof. Write $\mu = \min(d^+(x), d^-(x))$ and $M = \max(d^+(x), d^-(x))$, so $\mu + M = d(x)$ and $\mu M = d^+(x)d^-(x)$. Since $\mu \leq M$ we have $M \geq d(x)/2$, hence $d^+(x)d^-(x) = \mu M \geq \mu d(x)/2$, which rearranges to the claim. $\square$

Theorem A.24 (The quadratic bound with a linear error). *For all $m \geq 2$ and $n \geq 2$, every simple digraph D on n vertices with $\lambda_D^{\max} \leq m-1$ satisfies*

$$|A(D)| \leq \left\lfloor \frac{n^2}{4} \right\rfloor + 4(m-1)(n-1),$$

so for each fixed m ,

$$\ell_m^{\text{dir}}(n) = \frac{n^2}{4} + \Theta_m(n),$$

and both the leading constant $\frac{1}{4}$ and the order of the second-order term of [Conjecture 1.14](#) hold unconditionally.

Proof. Induction on n . For $n = 2$ a simple digraph has at most 2 arcs and the right-hand side is $1 + 4(m-1) \geq 5$, so the base holds. Let $n \geq 3$ and split on the minimum total degree.

Case 1: some vertex v has $d(v) < n/2$. Since $d(v)$ is an integer, $d(v) \leq \lfloor n/2 \rfloor$. By [Lemma A.15](#) the digraph $D - v$ on $n - 1$ vertices is still feasible, so the induction hypothesis applies to it and

$$|A(D)| \leq \left\lfloor \frac{(n-1)^2}{4} \right\rfloor + 4(m-1)(n-2) + \left\lfloor \frac{n}{2} \right\rfloor = \left\lfloor \frac{n^2}{4} \right\rfloor + 4(m-1)(n-2),$$

using [Lemma A.29](#) for the last step, and this is below the claimed bound.

Case 2: every vertex has $d(x) \geq n/2$. By [Lemma A.23](#) and Step 2 of [Proposition A.22](#),

$$\sum_{x \in V} \min(d^+(x), d^-(x)) \leq \sum_{x \in V} \frac{2d^+(x)d^-(x)}{d(x)} \leq \frac{4}{n} \sum_{x \in V} d^+(x)d^-(x) \leq \frac{4}{n} (m-1)n(n-1),$$

which is $4(m-1)(n-1)$. Step 4 of [Proposition A.22](#) deletes exactly that many arcs and leaves at most $\lfloor n^2/4 \rfloor$, so the bound holds in this case too.

For the two-sided asymptotic statement, [Construction 1.13](#) gives

$$\ell_m^{\text{dir}}(n) \geq \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor = \frac{n^2}{4} + \frac{(m-2)n}{2} + O_m(1),$$

so the second-order term is linear in n from both sides. □

What is left open is therefore a constant factor in one coefficient, not a shape. [Conjecture 1.14](#) says the second-order coefficient is exactly $(m-2)/2$, and the theorem pins it between that and $4(m-1)$, a factor of roughly eight. Closing that gap is where the structure of [Conjecture A.21](#) lives, and [Remark A.25](#) rules out the most obvious way to try.

Remark A.25 (Case 2 cannot be sharpened without a new inequality). One might hope to cut Case 2's constant using the observation that in the conjectured extremiser, one whole side has $\min(d^+, d^-) = 0$, so summing [Lemma A.23](#)'s worst case at every vertex looks wasteful. It is not: Case 2's bound is already the exact maximum obtainable from the two facts it uses, the budget $\sum_x d^+(x)d^-(x) \leq (m-1)n(n-1)$ of Step 2 in [Proposition A.22](#) and the floor $d(x) \geq n/2$, and no cleverer use of those two facts alone can do better.

Think of the budget as money and of $t_x = \min(d^+(x), d^-(x))$ as what each vertex buys with it. At a vertex with $d(x) = s \geq n/2$, buying t costs

$$d^+(x)d^-(x) = t(s-t) \geq t\left(\frac{n}{2} - t\right) =: c(t), \quad t \in \left[0, \frac{n}{4}\right],$$

the cheapest case being $s = n/2$, and the cap $t \leq n/4$ coming from $t \leq s/2$. Now c is concave with $c(0) = 0$, so its slope is decreasing and buying in bulk is cheaper per unit than spreading the purchase out. Concretely, given two vertices holding $0 < a \leq b$ strictly below the cap, shifting one unit from a to b leaves $a + b$ unchanged and lowers $c(a) + c(b)$. Repeating that exchange pushes every t_x to one of the two ends, so under a shared budget Q the true maximum of $\sum_x t_x$ puts as many vertices as the budget allows at the cap $t_x = n/4$ and the rest at $t_x = 0$. That is

exactly the qualitative picture the conjectured extremiser suggests, and since $c(n/4) = n^2/16$ it already gives

$$\sum_{x \in V} t_x \leq \underbrace{\frac{Q}{n^2/16}}_{\text{vertices at the cap}} \cdot \underbrace{\frac{n}{4}}_{\text{each buying}} = \frac{4Q}{n} = 4(m-1)(n-1)$$

once the number of vertices at the cap is below n , which is Case 2's constant unchanged.

The crude per-vertex bound therefore loses nothing, and the reason is that its two sources of slack close at that very point. [Lemma A.23](#) is an equality only when $d^+(x) = d^-(x)$, and the floor substitution $d(x) \rightarrow n/2$ is an equality only when $d(x) = n/2$. Both hold at once exactly when $d^+(x) = d^-(x) = n/4$, which is the vertex profile the true optimum concentrates on. Cutting the constant therefore needs a second and independent inequality, one that constrains interference among the handful of expensive near-balanced vertices themselves rather than a sharper reading of the one already in hand.

A.4.1 What that proof actually used

Neither the proposition nor the theorem above ever used feasibility twice. Each used it once, in Step 1, to cap a single explicit family of routes, and everything after that is arithmetic on in-degrees and out-degrees. Promoting that one cap from a deduction to a hypothesis costs nothing here and buys two further results below, one of them for a problem [Chapter 4](#) lists as untouched by the arc case.

Definition A.26 (Two-step budget). Let D be a simple digraph and let $C \geq 1$. Call D *C-budgeted* if every ordered pair (u, v) of distinct vertices satisfies

$$\mathbf{1}_{(u,v) \in A(D)} + p_2(u, v) \leq C,$$

where $p_2(u, v)$ counts, as in [Proposition A.22](#), the vertices $x \notin \{u, v\}$ with both (u, x) and (x, v) present.

Lemma A.27 (The counting core). *Every C-budgeted simple digraph D on $n \geq 2$ vertices satisfies*

$$|A(D)| \leq \left\lfloor \frac{n^2}{4} \right\rfloor + 4C(n-1).$$

Proof. Read the proofs of [Proposition A.22](#) and [Theorem A.24](#) again with C written wherever $m-1$ stood. Step 1 of the proposition is now the hypothesis rather than a deduction, and it was the only place feasibility ever entered, since Steps 2, 3 and 4 manipulate in-degrees and out-degrees and nothing else. The theorem's induction needs one further point, that the hypothesis is inherited by induced subdigraphs, which [Lemma A.15](#) supplied there by way of feasibility. Here it is immediate: deleting a vertex only deletes arcs and removes candidate midpoints, so neither term of the budget can rise. With those two observations the induction runs verbatim and delivers the stated bound. $\square$

The first application is the directed *vertex* problem. [Theorem 1.11](#) settles it at $m = 2$, and for $m \geq 3$ this thesis has been careful to record that it does not follow from the arc case: Whitney's $\kappa \leq \lambda$ makes the vertex-feasible family the larger of the two, so an arc upper bound restricts nothing there. That remains true of the *bound*. What does carry across is the route-counting, because the family Step 1 exhibits happens to be disjoint in the stronger sense as well, and once that is noticed the argument never needs the arc value at all.

Theorem A.28 (Directed vertex problem, leading constant and order). *For all $m \geq 2$ and $n \geq 2$, every simple digraph D on n vertices with $\kappa_D^{\max} \leq m - 1$ satisfies*

$$|A(D)| \leq \left\lfloor \frac{n^2}{4} \right\rfloor + 4(m-1)(n-1),$$

and consequently, for each fixed m ,

$$k_m^{\text{dir}}(n) = \frac{n^2}{4} + \Theta_m(n).$$

Proof. Fix an ordered pair (u, v) of distinct vertices. The direct arc (u, v) , when it is present, is a route with no interior vertex at all, and a two-step route $u \rightarrow x \rightarrow v$ has interior exactly $\{x\}$, one vertex, distinct from both u and v . Distinct midpoints give disjoint interiors, and the direct arc's interior is disjoint from all of them for want of any element, so these $\mathbf{1}_{(u,v) \in A(D)} + p_2(u, v)$ routes are pairwise internally vertex-disjoint and not merely arc-disjoint. Feasibility caps the family at $\kappa_D(u, v) \leq m - 1$, so D is $(m - 1)$ -budgeted and [Lemma A.27](#) applies.

For the lower bound, the augmented bipartite digraph of [Construction 1.13](#) has $\lambda^{\max} \leq m - 1$, hence $\kappa^{\max} \leq m - 1$ by Whitney, so for $n \geq m$ it witnesses $k_m^{\text{dir}}(n) \geq \lfloor (n + m - 2)^2 / 4 \rfloor$, which is $n^2/4 + \Theta_m(n)$. $\square$

The vertex problem at $m \geq 3$ is therefore open in the same narrow sense as the arc problem rather than in the wide one. Its leading constant is $\frac{1}{4}$ unconditionally, the term after it is linear in n , and since $\ell_m^{\text{dir}}(n) \leq k_m^{\text{dir}}(n)$ with both now inside $n^2/4 + \Theta_m(n)$, the two values agree to first order. What is left is the coefficient of the linear term, the same quantity [Conjecture A.21](#) is needed for on the arc side, together with the conjecture that the two values coincide exactly. It is worth being clear about what did and did not transfer. The upper bound of [Theorem A.24](#) did not: it is proved over the smaller family and says nothing here. The route family behind it did, and that is a statement about a construction rather than about a bound, which is why Whitney's inequality does not block it.

A.5 The directed multigraph problem, solved in full

The body proves $L_m^{\text{dir}}(n) = 2(n - 1)(m - 1)$ for $n \leq 6$ by an optimisation that scales m out of the problem entirely ([Theorem 2.2](#)), and the section just below, on the simple digraph at $m = 2$, closes every n by repeatedly deleting one vertex of least degree and averaging. It is natural

to hope the same trick settles the multigraph case at every m too, and for a long while it very nearly does: dividing every multiplicity by $m - 1$, as in the scaling step of [Chapter 2](#), turns a feasible multigraph into a weighting with values in $[0, 1]$ whose pairwise maximum flows are all at most one, and deleting a vertex of least weighted degree from that weighting drives the same averaging argument through cleanly on the linear branch, and on the quadratic branch whenever n is even. What breaks it is a single fractional remainder at odd n : a leftover of size $\frac{k}{2k-1}$ that an integer count could round away but a genuinely fractional weight cannot. Finding a specific vertex that is provably light enough to absorb that remainder turns out to resist every direct attempt.

This section abandons that attempt and proves the full value a different way. Instead of deleting one vertex at a time, it deletes one whole *sparse subgraph* at a time, chosen so that the deletion is guaranteed to lower every pair's connectivity by exactly one in a single stroke. That change removes the fractional step altogether, along with the restriction to a particular m or a particular parity of n .

Lemma A.29 (Arithmetic identity). *For all $n \geq 2$, $\left\lfloor \frac{(n-1)^2}{4} \right\rfloor + \lfloor n/2 \rfloor = \left\lfloor \frac{n^2}{4} \right\rfloor$.*

Proof. For $n = 2k$ the left side is $k(k-1) + k = k^2$, and for $n = 2k+1$ it is $k^2 + k = k(k+1)$. Both match the right side. $\square$

Theorem A.30 (Directed multigraph value, every n and m). *For all $n \geq 2$ and $m \geq 2$,*

$$L_m^{\text{dir}}(n) = (m-1) M(n), \quad M(n) := \max\left(2(n-1), \left\lfloor \frac{n^2}{4} \right\rfloor\right).$$

A.5.1 The lower bound: thickening

The lower bound is short, and it is worth isolating the one idea that makes it short: multiplying every arc's multiplicity by a fixed integer multiplies every cut's capacity, and hence every pairwise connectivity, by that same integer.

Lemma A.31 (Thickening). *Let D_0 be a directed multigraph and $k \geq 1$ an integer. Write $k \cdot D_0$ for the multigraph obtained by multiplying every multiplicity $\mu(u, v)$ by k . Then $\lambda_{k \cdot D_0}(u, v) = k \cdot \lambda_{D_0}(u, v)$ for every ordered pair $u \neq v$, and $|A(k \cdot D_0)| = k |A(D_0)|$.*

Proof. By [Theorem A.1](#), $\lambda_{D_0}(u, v)$ equals the minimum capacity, over all u - v cuts, of the total multiplicity crossing that cut. Multiplying every multiplicity by k multiplies the capacity of every cut by k at once, so it multiplies the minimum over all cuts by k as well, which is precisely $\lambda_{k \cdot D_0}(u, v) = k \cdot \lambda_{D_0}(u, v)$. The arc count scales by k because every one of its summands, each multiplicity, does. $\square$

Construction A.32 (Thickened tree and thickened bipartite digraph). For $m \geq 2$, take any spanning tree T of K_n and give every edge multiplicity $m - 1$ in each direction. Between two vertices, T offers only its one tree path, so this is $m - 1$ copies of the $m = 2$ bidirected tree of [Theorem 1.10](#), feasible at $\lambda^{\max} = 1$ there, and [Lemma A.31](#) with $k = m - 1$ gives $\lambda^{\max} = m - 1$ here

and an arc count of $2(n-1)(m-1)$. Alternatively, take the one-directional complete bipartite digraph of [Construction 1.8](#) (balanced parts A, B with $|A| = \lfloor n/2 \rfloor$, $|B| = \lceil n/2 \rceil$, feasible at $\lambda^{\max} = 1$) and give every arc multiplicity $m-1$: the same lemma gives $\lambda^{\max} = m-1$ and $(m-1) \lfloor n^2/4 \rfloor$ arcs.

Taking whichever of the two constructions is larger gives $L_m^{\text{dir}}(n) \geq (m-1)M(n)$ for every n and m , which is the lower bound half of [Theorem A.30](#). The rest of this section proves the matching upper bound.

A.5.2 The upper bound: a reachability skeleton

It helps to first ask what a single vertex deletion actually buys at $m=2$: removing a vertex can only remove routes passing through it, so no pair's connectivity ever rises, and the whole $m=2$ induction rests on that one guarantee together with an averaging bound on how much one well-chosen vertex carries. The multigraph problem at a general m wants the same two ingredients, a deletion that is guaranteed to lower every connectivity, paired with a bound on how much it removes, but there the natural analogue of “one vertex” breaks down: once multiplicities are free to sit anywhere between 0 and $m-1$, no single vertex is ever guaranteed to carry a controllable share of the arcs, which is exactly the fractional remainder above. The fix is to stop insisting on a vertex and delete a whole *subgraph* instead: one general enough to always exist, sparse enough to bound, and specific enough that removing it is guaranteed to lower every pairwise connectivity by exactly one.

Definition A.33 (Reachability skeleton). Let D be a directed multigraph. A spanning subdigraph $R \subseteq D$, using at most one copy of each arc it uses (no parallel copies, even if D has them), is a *reachability skeleton* of D if, for every pair of vertices $u \neq v$, u reaches v by some directed path in R exactly when u reaches v by some directed path in D .

A reachability skeleton keeps every yes-or-no fact about who can reach whom and discards everything else, in particular every parallel copy of every arc. It is a caricature of D : thin, but faithful on the one question connectivity actually depends on. The next two lemmas turn that faithfulness into the whole induction. The first says a reachability skeleton always exists and is never too big. The second says deleting it always lowers every connectivity by exactly one.

Lemma A.34 (A sparse reachability skeleton always exists). *Every loopless directed multigraph D on $n \geq 2$ vertices has a reachability skeleton R with $|A(R)| \leq M(n)$.*

Building R takes three steps: handle the traffic *inside* each strongly connected piece of D , handle the traffic *between* pieces, then count. A *strongly connected component* (an SCC) is a maximal set of vertices any two of which reach each other, and every directed multigraph partitions uniquely into its SCCs, since mutual reachability is an equivalence relation. [Figure A.4](#) carries one small running example through all three steps, and it is worth glancing at now, before the general argument, since every object named below appears in it concretely.

Step 1: inside each component. Fix one SCC of order $n_i \geq 1$ and pick any vertex r inside it as a root. Since r reaches every vertex of the component, a standard fact about digraphs gives a spanning *out-arborescence*: a tree of $n_i - 1$ arcs of D , every one pointing away from r , along which r alone reaches every other vertex of the component (grow it one hop at a time, always available since the component is strongly connected). Symmetrically, because every vertex of the component reaches r , there is a spanning *in-arborescence* of $n_i - 1$ arcs, every one pointing toward r , along which every vertex reaches r . Taking both trees together, any two vertices u, v of the component are joined by the route $u \rightarrow r \rightarrow v$, first the in-arborescence, then the out-arborescence, so the $2(n_i - 1)$ arcs of the two trees alone reconstruct *every* reachability relation inside this one component.

Step 2: between components. Contract each SCC to a single point and, between two contracted points, keep one arc whenever D has at least one arc from the first component to the second (dropping any further parallel arcs between the same two components at this stage). Call the result Q_0 , a digraph on q points, one per component. Two different SCCs can never reach each other in both directions, since that would make them one strongly connected component after all, so Q_0 has no directed cycle: it is a DAG (a directed acyclic graph). Reachability between two different components of D is, by this very construction, exactly reachability between the two corresponding points of Q_0 . Thin Q_0 down to an *inclusion-minimal* spanning subdigraph $Q \subseteq Q_0$ that still has the same reachability relation, deleting one arc at a time for as long as some arc is redundant, which must stop since Q_0 has only finitely many arcs.

The reason to thin Q_0 down to Q is that Q 's *underlying undirected graph*, meaning Q with the arrows erased, has no triangle, and this deserves seeing directly. Suppose three points x, y, z of Q were pairwise joined by some arc of Q . Since $Q \subseteq Q_0$ and Q_0 has no cycle, Q has none either, so between any two of x, y, z the arc runs in only one direction, never both. Three points pairwise joined with a consistent direction on each pair admit only one pattern, up to relabelling: a linear order, say $x \rightarrow y, y \rightarrow z$, and $x \rightarrow z$. But then $x \rightarrow z$ is redundant, since x already reaches z via y , so deleting it would leave the reachability relation of Q unchanged, contradicting that Q is inclusion-minimal. So no three points of Q can be pairwise joined at all: that is triangle-freeness of the underlying graph.

A triangle-free graph on q vertices has at most $\lfloor q^2/4 \rfloor$ edges, Mantel's theorem [16], the $r = 2$ case of the more general theorem of Turán [17]. Applied to Q ,

$$|A(Q)| \leq \left\lfloor \frac{q^2}{4} \right\rfloor.$$

Step 3: assembling R and counting. Build R from the two arborescences of Step 1 inside every SCC, together with one witness arc of D for every arc of Q (some arc of D realises it, by the very definition of Q_0). By Step 1, every pair inside one component reaches each other in R exactly as in D . By Step 2, and because within-component reachability is already secured, every pair in two different components reaches each other in R exactly as in D too: to travel from u to

v across the components Q 's path visits, reach each witness arc's tail inside its own component (Step 1), cross the witness arc, and repeat. Since $R \subseteq D$ throughout, R can never certify a reachability D does not already have, so R is a genuine reachability skeleton. Writing $n_1, \dots, n_q$ for the SCC orders, so $\sum_i n_i = n$, its arc count is

$$|A(R)| = \sum_{i=1}^q 2(n_i - 1) + |A(Q)| \leq 2(n - q) + \left\lfloor \frac{q^2}{4} \right\rfloor =: f(q).$$

It remains to bound $f(q)$ over the range $1 \leq q \leq n$ that the number of components can actually take. Its increments are

$$f(q+1) - f(q) = -2 + \left\lfloor \frac{q+1}{2} \right\rfloor,$$

using the identity $\lfloor (q+1)^2/4 \rfloor - \lfloor q^2/4 \rfloor = \lfloor (q+1)/2 \rfloor$, checked directly on the four residues of q modulo 4, and $\lfloor (q+1)/2 \rfloor$ never decreases as q grows. So the increments of f themselves never decrease: f may fall for a while and then rise, one single valley, but it can never fall, rise, and fall again. A function shaped like one valley always attains its largest value at one of the two ends of its domain, since any interior point is flanked by a point at least as low on its falling side and a point at least as high on its rising side, so no interior point can beat both ends at once. Hence

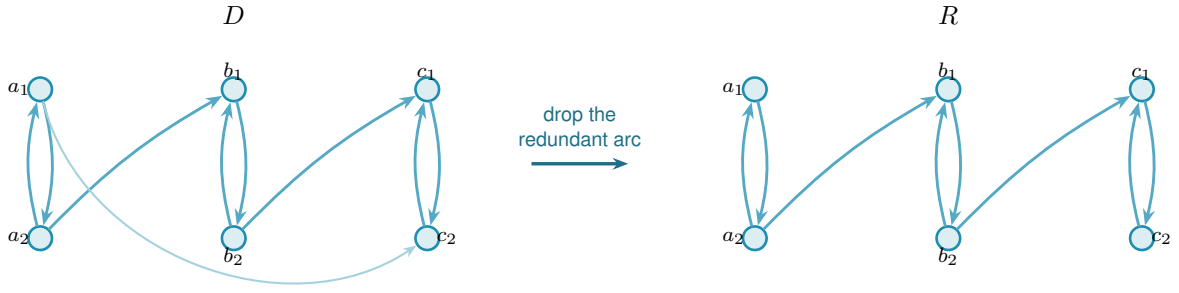
$$f(q) \leq \max(f(1), f(n)) \quad \text{for } 1 \leq q \leq n.$$

At $q = 1$ the whole digraph is one component, so $f(1) = 2(n-1) + 0$. At $q = n$ every component is a single vertex, so $f(n) = 0 + \lfloor n^2/4 \rfloor$. Hence $f(q) \leq \max(2(n-1), \lfloor n^2/4 \rfloor) = M(n)$ for every q in range, which is the lemma.

Lemma A.35 (Deleting a skeleton lowers every connectivity by one). *Let D be a directed multigraph with $\lambda^{\max}(D) \leq k$ for some $k \geq 1$, and let R be a reachability skeleton of D . Write $D - R$ for the multigraph obtained by deleting, from each multiplicity $\mu(u, v)$, the one copy (if any) that R uses on that pair. Then $\lambda^{\max}(D - R) \leq k - 1$.*

Proof. Fix an ordered pair $u \neq v$ and suppose $D - R$ carries $t \geq 1$ pairwise arc-disjoint $u-v$ paths. These use arcs of $D - R$ and hence of D , since $D - R \subseteq D$, so u reaches v in D . Because R is a reachability skeleton, u then also reaches v in R : some directed path P from u to v uses only arcs of R . Every arc of R has been subtracted out of $D - R$, so P shares no arc with any of the t paths already found inside $D - R$. Together, P and those t paths are $t + 1$ pairwise arc-disjoint $u-v$ paths in D , so $t + 1 \leq \lambda_D(u, v) \leq k$, that is, $t \leq k - 1$. Pairs with $t = 0$ trivially satisfy $t \leq k - 1$ once $k \geq 1$, so this holds for every pair, giving $\lambda^{\max}(D - R) \leq k - 1$. $\square$

Proof of Theorem A.30. The lower bound is [Construction A.32](#). For the upper bound, fix n and prove by induction on $k = m - 1 \geq 0$ that every directed multigraph D on n vertices with $\lambda^{\max}(D) \leq k$ has $|A(D)| \leq k M(n)$. At $k = 0$, every pair has $\lambda_D(u, v) \leq 0$, so no arc (u, v) can



redundant: a_1 already reaches c_2 via a_2, b_1, b_2, c_1

Figure A.4. A small instance of [Lemma A.34](#). In D (left) three bidirected pairs $A = \{a_1, a_2\}$, $B = \{b_1, b_2\}$, $C = \{c_1, c_2\}$ are the strongly connected components, joined by $a_2 \rightarrow b_1$, $b_2 \rightarrow c_1$, and $a_1 \rightarrow c_2$. Contracting each pair to a point gives a condensation Q_0 on three points, with all three arcs $A \rightarrow B$, $B \rightarrow C$, $A \rightarrow C$ present, exactly the transitively-closed triangle the proof rules out of the thinned Q : the arc $A \rightarrow C$ is redundant, since A already reaches C via B , so the transitive reduction Q keeps only $A \rightarrow B$ and $B \rightarrow C$, and the underlying graph of Q , a path on three points, is triangle-free as claimed. The reachability skeleton R (right) reflects exactly this: every SCC keeps both of its internal arcs (an in- and an out-arborescence on two vertices is just the pair itself), while of the two component-crossing arcs of D only the two witnesses of Q 's two arcs survive. The direct arc $a_1 \rightarrow c_2$ is dropped, yet a_1 still reaches c_2 in R : first $a_1 \rightarrow a_2$, then $a_2 \rightarrow b_1$, then $b_1 \rightarrow b_2$, then $b_2 \rightarrow c_1$, then $c_1 \rightarrow c_2$.

even be present, since a single arc is already one route, and $|A(D)| = 0 = 0 \cdot M(n)$.

For $k \geq 1$, take D with $\lambda^{\max}(D) \leq k$ and let R be a reachability skeleton, which exists with $|A(R)| \leq M(n)$ by [Lemma A.34](#). By [Lemma A.35](#), $D - R$ has $\lambda^{\max}(D - R) \leq k - 1$, and it is again a directed multigraph on the same n vertices, so the induction hypothesis gives $|A(D - R)| \leq (k - 1)M(n)$. Since R uses at most one copy of each arc it touches, the arcs of D split exactly into those of R and those left in $D - R$, so

$$|A(D)| = |A(R)| + |A(D - R)| \leq M(n) + (k - 1)M(n) = k M(n).$$

Taking $k = m - 1$ gives $L_m^{\text{dir}}(n) \leq (m - 1)M(n)$, matching the lower bound. $\square$

Nothing above depended on m or on the parity of n : the same three-step construction of R and the same one-line deletion bound run identically at every level. In particular, the induction closes exactly the finite computation the rest of this appendix once needed help to settle.

Corollary A.36. $L_3^{\text{dir}}(7) = 24$.

Proof. $M(7) = \max(12, 12) = 12$, so [Theorem A.30](#) at $m = 3$, $n = 7$ gives $L_3^{\text{dir}}(7) = 2 \cdot 12 = 24$. Concretely, the induction peels two reachability skeletons off any 7-vertex witness in turn, each of at most $M(7) = 12$ arcs by [Lemma A.34](#), first lowering $\lambda^{\max}$ from 2 to 1 and then from 1 to 0: $24 = 12 + 12$ spends the whole budget twice over, with nothing left for a third pass. This is a proof by hand of the one statement [Chapters 2 and 4](#) once recorded as an outstanding computation. That computation was the mixed-integer program `prove_integral_arc_bound(7, 3, 25)`, whose INFEASIBLE verdict would have settled the same fact, and which never returned one: it reached

its time limit on the bundled CBC solver, and a fourteen-hour uncapped enumeration was stopped without a verdict. It was never run on a stronger solver, and it no longer needs to be. The honest summary is that the machine route to this fact was abandoned unfinished and the hand proof above replaced it, rather than the two confirming each other. $\square$

Corollary A.37 (The fractional relaxation is integral). *For every $n \geq 2$, the m -free optimum $M^*(n)$ of [Chapter 2](#) equals $M(n)$, and it is attained by a $\{0, 1\}$ weight matrix. Consequently the scaling identity*

$$L_m^{\text{dir}}(n) = (m - 1) M^*(n)$$

holds for every n and every $m \geq 2$, with no hypothesis on the optimum.

Proof. For $M^*(n) \geq M(n)$, both constructions of [Construction A.32](#) at multiplicity one are $\{0, 1\}$ matrices with all pairwise maximum flows equal to one, of total weight $2(n - 1)$ and $\lfloor n^2/4 \rfloor$ respectively.

For the reverse, note first that the optimum is attained at a *rational* matrix. A weight matrix is feasible exactly when, for each ordered pair, some cut separating that pair has capacity at most one, so the feasible region is the union, over the finitely many ways of choosing one cut per pair, of the bounded rational polytopes cut out by those choices together with $0 \leq w \leq 1$. A linear objective over a bounded rational polytope attains its maximum at a rational vertex, and $M^*(n)$ is the largest of finitely many such maxima.

So let w be a feasible rational optimum and let q be a common denominator of its entries. Then qw has integer entries, and multiplying every weight by q multiplies every cut capacity by q and hence every maximum flow by q , exactly as in [Lemma A.31](#). So qw is a directed multigraph with $\lambda^{\max} \leq q$, that is, feasible at $m - 1 = q$, and [Theorem A.30](#) bounds its arc count by $q M(n)$. Dividing by q gives $M^*(n) = \sum_{u \neq v} w(u, v) \leq M(n)$.

The identity follows because [Theorem A.30](#) gives $L_m^{\text{dir}}(n) = (m - 1)M(n)$. $\square$

This settles the integrality question [Chapter 2](#) raises when it introduces $M^*(n)$, namely whether the heaviest fractional weight matrix is always a $\{0, 1\}$ one. It is, and the reason is worth noting because it is backwards from the usual direction of such arguments. Nothing here inspects the fractional problem at all. An integral theorem, proved for every m at once, is strong enough to be applied to the scaled-up matrix qw , and the fractional statement falls out of the integral one rather than the other way round. That is only possible because [Theorem A.30](#) holds at every m , however large, so no denominator is out of reach.

Corollary A.38 (The $m = 2$ case collapses to the simple digraph). *At $m = 2$ (so $k = 1$), [Theorem A.30](#) reads $L_2^{\text{dir}}(n) = M(n)$, matching [Theorem 1.10](#) exactly, and this is not a coincidence. A directed multigraph with $\lambda^{\max} \leq 1$ can carry no parallel arcs at all, since two parallel copies of one arc are already two arc-disjoint routes between their endpoints, so at $m = 2$ the multigraph problem and the simple digraph problem are literally the same problem. [Theorem A.30](#) recovers [Theorem 1.10](#) as its $k = 1$ case, rather than needing it as a separate input.*

Theorem A.30 settles the value at every n and m , but a value is not a full description of the extremal digraphs, and on the linear branch a companion fact about *which* multigraphs attain it is worth recording, because it comes almost for free and it is genuinely more than the value alone gives. Call a directed multigraph *everywhere-saturated* at level m when every ordered pair of distinct vertices has local connectivity exactly $m - 1$. The doubled bidirected spanning trees at multiplicity $m - 1$, the extremisers of the linear branch, are exactly such graphs: the $m - 1$ routes between two vertices run along the doubled tree path, and cutting one doubled edge of that path is a cut of $m - 1$ arcs.

Lemma A.39 (Saturated attachment). *Let $m \geq 2$ and let D_0 be a directed multigraph on at least two vertices with $\lambda_{D_0}(x, y) = m - 1$ for every ordered pair of distinct vertices. Let D be obtained from D_0 by adding one new vertex v together with any multiset of arcs incident to v . If $\lambda_D^{\max} \leq m - 1$, then $d(v) \leq 2(m - 1)$. If $d(v) = 2(m - 1)$, then every arc at v joins v to one single vertex u , with $\mu(v, u) = \mu(u, v) = m - 1$, and D is again everywhere-saturated.*

Proof. No mixed pair. Suppose v has an in-arc from u and an out-arc to w with $u \neq w$. By Menger and saturation D_0 carries $m - 1$ pairwise arc-disjoint u -to- w routes, and the route $u \rightarrow v \rightarrow w$ uses only arcs incident to v , so it is arc-disjoint from all of them and $\lambda_D(u, w) \geq m$, a contradiction. Hence v is a pure source, a pure sink, or every arc at v touches one single vertex u .

A pure source has $d(v) \leq m - 1$. Let $t = \sum_x \mu(v, x)$ and fix u with $\mu(v, u) \geq 1$. Any cut $(S, \bar{S})$ with $v \in S$ and $u \in \bar{S}$ has capacity $\sum_{x \in \bar{S}} \mu(v, x) + e_{D_0}(S \setminus \{v\}, \bar{S})$. If S contains a vertex x of D_0 , the second term is at least $\lambda_{D_0}(x, u) = m - 1$ and the first is at least $\mu(v, u)$. If $S = \{v\}$, the capacity is t . Every cut therefore has capacity at least $\mu(v, u) + \min(t - \mu(v, u), m - 1)$, and by max-flow min-cut so does $\lambda_D(v, u)$, which feasibility caps at $m - 1$. Were $t - \mu(v, u) \geq m - 1$, this would force $\mu(v, u) = 0$, a contradiction, so $t - \mu(v, u) < m - 1$ and the bound reads $t \leq m - 1$.

A pure sink has $d(v) \leq m - 1$. Reverse every arc: the reverse of an everywhere-saturated multigraph is everywhere-saturated, a pure sink becomes a pure source, and the previous step applies.

A single partner. If every arc at v touches one vertex u , then parallel arcs are already that many arc-disjoint one-step routes, so $\mu(v, u) \leq m - 1$ and $\mu(u, v) \leq m - 1$, giving $d(v) \leq 2(m - 1)$.

Equality. Since $2(m - 1) > m - 1$, a degree of $2(m - 1)$ rules out the pure cases, so v has a single partner at full multiplicity both ways. The result is feasible and again everywhere-saturated: a route through v must enter and leave through u , so it adds nothing to any flow between old vertices, and $\lambda_D(x, v) = \min(\lambda_{D_0}(x, u), \mu(u, v)) = m - 1$ for every old x , symmetrically for $\lambda_D(v, x)$. $\square$

Remark A.40 (The linear branch grows one leaf at a time). The equality case of [Lemma A.39](#) attaches a new leaf to an arbitrary vertex at full multiplicity, and that is the only attachment reaching $2(m - 1)$. Iterating from a single doubled edge, itself everywhere-saturated on two vertices, generates exactly the doubled bidirected spanning trees, so the extremal family of the linear branch is

closed under maximal attachment and grows one leaf at a time. The program confirms the lemma exhaustively at $m = 3$: over every doubled tree on five and six vertices and every one of the 3^{2n} possible attachment patterns, the densest feasible attachment has degree exactly $4 = 2(m - 1)$, and the degree-4 patterns are precisely the single-partner attachments at full multiplicity, one per choice of partner.

Remark A.41 (Which multigraphs attain 24 arcs at $n = 7$). [Lemma A.39](#) identifies the whole extremal family of the linear branch, but [Corollary A.36](#)'s value alone does not say which digraphs attain 24 arcs at $n = 7$, $m = 3$, the size where the linear and quadratic branches of $M(n)$ tie ($M(7) = \max(12, 12) = 12$). That finer question was settled separately, by machine, before the value itself had a proof by hand: the sound generation enumerator of [Chapter 2](#) (`enumerate_extremal_directed_multigraphs_via_generation(7, 3, 24, max_degree=8)`), which prunes only with proved bounds and is therefore complete at every n) ran for about seven hours across ten processor cores and returned exactly three isomorphism classes: the two bipartite shapes $2 \cdot B_{3,4}$ and $2 \cdot B_{4,3}$, and the doubled bidirected path P_7 , the one non-star tree whose maximum degree (2) is compatible with an 8-regular graph one vertex up. Each of the three was re-verified by the exact checker. This is a genuine complement to [Corollary A.36](#), not a step on the way to it: the reachability-skeleton induction proves the value 24 without ever needing to know which graphs attain it, and this classification answers the separate question of which ones do, at this one size.

A.5.3 Uniqueness on the quadratic branch

The classification above is one finite data point, and the linear branch is settled by [Lemma A.39](#), so what is missing is the quadratic branch at a general n . The reachability-skeleton proof looks unpromising for this, since it is designed never to need to know which multigraph it is taking apart. But that indifference is only in the direction of the *bound*. Run backwards, from the assumption that the bound is met exactly, the same construction is unexpectedly rigid: it pins down the number of strongly connected components, then the shape of the skeleton, and finally the multigraph.

Two consequences of equality do the work, and both come from reading the count of [Lemma A.34](#) as an equation rather than an inequality.

Lemma A.42 (What equality forces). *Let $n \geq 8$, so that $\lfloor n^2/4 \rfloor > 2(n - 1)$ and $M(n) = \lfloor n^2/4 \rfloor$, and let D be a directed multigraph on n vertices with $\lambda_D^{\max} \leq m - 1$ and $|A(D)| = (m - 1) \lfloor n^2/4 \rfloor$. Then*

- (1) D is acyclic, and
- (2) D has a reachability skeleton R whose underlying undirected graph is exactly the balanced complete bipartite graph $K_{\lfloor n/2 \rfloor, \lfloor n/2 \rfloor}$, so R is an acyclic orientation of it.

Proof. The proof of [Theorem A.30](#) peels reachability skeletons $R_1, \dots, R_{m-1}$, each deletion dropping every pairwise connectivity by exactly one, so that after $m - 1$ peels no arc sur-

vives and $|A(D)| = \sum_i |A(R_i)|$. Each $|A(R_i)| \leq M(n)$, so equality in the hypothesis forces $|A(R_i)| = M(n)$ for every i , and in particular for $R := R_1$.

Now read the count of [Lemma A.34](#) backwards. It gives $|A(R)| \leq f(q) := 2(n - q) + \lfloor q^2/4 \rfloor$, where q is the number of strongly connected components of D , so $f(q) \geq \lfloor n^2/4 \rfloor = f(n)$. The increments $f(q + 1) - f(q) = -2 + \lfloor (q + 1)/2 \rfloor$ are non-decreasing, so f is convex, and $f(n) - f(n - 1) = -2 + \lfloor n/2 \rfloor \geq 2$ for $n \geq 8$. A convex function that is strictly increasing at its right endpoint cannot equal its endpoint value anywhere earlier, since it would then have to be constant on the whole stretch in between. Hence $q = n$: every strongly connected component is a single vertex, which is (1).

With $q = n$ the two arborescences of Step 1 of that proof are empty, so R consists of the witness arcs of Q alone and $|A(Q)| = \lfloor n^2/4 \rfloor$. The underlying undirected graph of Q is triangle-free, proved there from the inclusion-minimality of Q . A triangle-free graph on n vertices with $\lfloor n^2/4 \rfloor$ edges meets Mantel's bound [16] with equality, and the extremal graph for Mantel's theorem is unique: it is $K_{\lfloor n/2 \rfloor, \lfloor n/2 \rfloor}$. That is (2), and R is acyclic because D is. $\square$

The next lemma is where the skeleton's minimality, which had only been used to make it small, is used to make it *shallow*. It costs nothing and it is what makes the theorem work.

Lemma A.43 (A minimal bipartite skeleton has no long path). *Let R be as in [Lemma A.42](#), with parts A and B . Then R contains no directed path with three arcs.*

Proof. Suppose $u \rightarrow v \rightarrow w \rightarrow x$ were such a path. A path in a bipartite graph alternates sides, so u and x lie on opposite sides, and since R 's underlying graph is the *complete* bipartite graph, R contains an arc between them. It cannot be $x \rightarrow u$, which would close a directed cycle against the path, and D is acyclic. So $u \rightarrow x$ lies in R . But then deleting $u \rightarrow x$ from R changes no reachability, since u still reaches x along the path, contradicting the inclusion-minimality of Q . $\square$

Theorem A.44 (Uniqueness on the quadratic branch). *Let $m \geq 2$ and $n \geq \max(8, 2m)$ with n on the quadratic branch. If D is a directed multigraph on n vertices with $\lambda_D^{\max} \leq m - 1$ and $|A(D)| = L_m^{\text{dir}}(n) = (m - 1) \lfloor n^2/4 \rfloor$, then*

$$D = (m - 1) \cdot B_{\lfloor n/2 \rfloor, \lfloor n/2 \rfloor},$$

the one-directional complete bipartite digraph on a balanced bipartition, every arc at multiplicity exactly $m - 1$. The extremiser is therefore unique up to relabelling and the choice of which side is the larger.

Proof. Take R , A and B from [Lemma A.42](#), and write $\alpha = |A| \geq \beta = |B| = \lfloor n/2 \rfloor$.

Step 1: R has no directed path with two arcs. Suppose $a \rightarrow b \rightarrow a'$ with $a, a' \in A$ and $b \in B$. If a had an incoming arc $b_0 \rightarrow a$ in R , then $b_0 \rightarrow a \rightarrow b \rightarrow a'$ would be a path with three arcs, which [Lemma A.43](#) forbids. So a has no incoming arc in R , and since R 's underlying graph is

complete bipartite, $a \rightarrow b''$ for every $b'' \in B$. Symmetrically a' has no outgoing arc, so $b'' \rightarrow a'$ for every $b'' \in B$. The β routes $a \rightarrow b'' \rightarrow a'$, one through each $b'' \in B$, use pairwise distinct arcs, and each of their arcs is present in D because $R \subseteq D$. Hence $\lambda_D(a, a') \geq \beta = \lfloor n/2 \rfloor \geq m$, against feasibility. The case of a path $b \rightarrow a \rightarrow b'$ with middle vertex in A is identical with the sides exchanged and gives $\lambda_D(b, b') \geq \alpha \geq \beta \geq m$, equally impossible. This is the one step that uses $n \geq 2m$.

Step 2: R is one-directional. By Step 1 no vertex of R has both an incoming and an outgoing arc, so every vertex is a source or a sink. Fix $a \in A$; it is adjacent in R to every vertex of B , so it is not isolated. If a is a source then every $b \in B$ has an incoming arc, hence is a sink, hence has no outgoing arc, so every arc at every b runs into it: all arcs of R run from A to B . If instead a is a sink the same argument gives all arcs from B to A . Relabelling the two sides, assume $R = B_{\alpha, \beta}$, all arcs from A to B .

Step 3: D has no arc outside R . R has the same reachability as D , and in a one-directional bipartite digraph the only ordered pairs joined by a directed path are the pairs $(a, b) \in A \times B$, each by the single arc $a \rightarrow b$. So if D had an arc (u, x) , then u reaches x in R , forcing $(u, x) \in A \times B$, and R already contains that arc. Hence D and R use exactly the same $\alpha\beta = \lfloor n^2/4 \rfloor$ ordered pairs.

Step 4: every multiplicity is $m - 1$. A pair joined by $\mu(a, b)$ parallel arcs has $\lambda_D(a, b) \geq \mu(a, b)$, so feasibility gives $\mu(a, b) \leq m - 1$ on each of the $\lfloor n^2/4 \rfloor$ pairs. Their total is $|A(D)| = (m - 1) \lfloor n^2/4 \rfloor$, so every one of them attains the ceiling. Hence $D = (m - 1) \cdot B_{\alpha, \beta}$, and $\alpha\beta = \lfloor n^2/4 \rfloor$ forces the bipartition to be balanced. $\square$

Three remarks. First, on where the hypothesis $n \geq 2m$ is used and what it costs. It enters once, in Step 1, where a two-arc path in the skeleton is converted into $\lfloor n/2 \rfloor$ arc-disjoint routes between one pair, and feasibility kills that only when $\lfloor n/2 \rfloor$ reaches m . For $n < 2m$ the theorem is not proved, and it is not known to fail either. What can be said is that nothing in the argument suggests a real boundary there: the step is a lower bound on one pair's connectivity, and losing it removes a contradiction rather than supplying a counterexample. Second, the regularity just mentioned is worth recording on its own, since it is what makes that enumeration finite in practice: deleting a vertex leaves at most $(m - 1)M(n - 1)$ arcs, so every vertex of an extremiser carries degree at least $(m - 1) \lfloor n/2 \rfloor$, and for even n the degree sum $2(m - 1) \lfloor n^2/4 \rfloor = n \cdot (m - 1)(n/2)$ leaves no slack, so every vertex has degree exactly $(m - 1)n/2$. Third, the shape of the argument is worth noticing. [Theorem A.30](#) is indifferent to which multigraph it dismantles, and [Chapter 4](#) records that indifference as the reason the value says nothing about uniqueness. That is true of the proof read forwards. Read backwards from equality, the same skeleton is a rigid object, and the two facts that had been mere bookkeeping, that f is maximised at one end and that a minimal skeleton is triangle-free, become a determination of q and an application of the equality case of Mantel's theorem.

A.6 The hypergraph bounds

The body's hypergraph claims rest on two theorems: a Menger theorem for Berge routes, which also certifies the helper-point checker of [Chapter 2](#), and a hypergraph Gomory–Hu theorem from the recent literature.

Theorem A.45 (Menger for hypergraphs). *For a hypergraph $\mathcal{H}$ and vertices u, v , the maximum number of hyperedge-disjoint Berge u – v paths equals the minimum size of a set C of hyperedges whose removal separates u from v .*

Proof. Build the helper-point network of [Figure 2.5](#): for each hyperedge e a helper arc $e^- \rightarrow e^+$ of capacity one, and arcs $x \rightarrow e^-$ and $e^+ \rightarrow x$ of unbounded capacity for each member $x \in e$. The proof matches the number of disjoint Berge paths to the size of a minimum cut by translating between Berge-path language and flow language in each direction: a family of hyperedge-disjoint Berge paths becomes a family of arc-disjoint flow paths, and conversely a flow becomes a family of Berge paths, as the next two paragraphs spell out in turn.

Berge paths become flow. A Berge path $u, e_1, v_1, \dots, e_k, v$ becomes the directed walk $u \rightarrow e_1^- \rightarrow e_1^+ \rightarrow v_1 \rightarrow \dots \rightarrow v$, crossing one helper arc per hyperedge it uses. The e_i along a single Berge path are distinct, and two hyperedge-disjoint Berge paths share no hyperedge at all, so they use disjoint sets of helper arcs and translate to *arc-disjoint* flow paths, paths that share no arc, each carrying one unit of flow.

Flow becomes Berge paths. Conversely, because every capacity is an integer, a maximum flow may be taken integral, and an integral flow of value k splits into k arc-disjoint unit paths. Reading a helper crossing $\dots x \rightarrow e^- \rightarrow e^+ \rightarrow y \dots$ back as the single hyperedge step x, e, y , that is, *suppressing* the two helper nodes, turns each unit path into a Berge path, and no two of them can share a hyperedge, since that would send two units across some capacity-one helper arc.

Cuts. Finally, the member arcs are uncapped, so any cut of finite capacity uses helper arcs only, and a set of helper arcs separates u from v in the network exactly when the corresponding hyperedges separate them in $\mathcal{H}$. The max-flow min-cut theorem equates the largest number of arc-disjoint flow paths with the smallest helper-arc cut, which is precisely the claim. $\square$

Theorem A.46 (Hypergraph Gomory–Hu). *Every r -uniform hypergraph $\mathcal{H}$ on vertex set V has a weighted tree T^* on V , its hypergraph Gomory–Hu tree, such that $\lambda_{\mathcal{H}}(u, v)$ equals the minimum weight on the u – v tree path, and each tree edge t induces a vertex partition $(S_t, V \setminus S_t)$ with $|\delta_{\mathcal{H}}(S_t)| = c^*(t)$, where $\delta_{\mathcal{H}}(S)$ is the set of hyperedges incident to both sides, that is, having at least one vertex in S and at least one outside it.*

This is the exact hypergraph echo of [Theorem A.2](#), the same tree-of-cuts summary of all pairwise connectivities, with the single change that the capacity of a cut now counts the hyperedges straddling it, $|\delta_{\mathcal{H}}(S)|$, in place of the edges. Gomory and Hu's construction [15] needs nothing

about the cut function beyond it being symmetric and submodular, and the hyperedge-cut function $|\delta_{\mathcal{H}}(\cdot)|$ has both properties exactly as the ordinary edge-cut function does, so the identical uncrossing construction builds this tree too. We use only the two properties in the statement, the path-minimum reading of $\lambda_{\mathcal{H}}$ and the identity $|\delta_{\mathcal{H}}(S_t)| = c^*(t)$.

Proof of Proposition 1.15. Upper bound. Let $\mathcal{H}$ be r -uniform with $\lambda_{\mathcal{H}}^{\max} \leq m - 1$, and let T^* be its hypergraph Gomory–Hu tree (Theorem A.46), a weighted tree on the same n vertices with weights c^* . The plan is to charge each hyperedge to tree edges and count the charges two ways, exactly as in the Mader proof, with the one change that a hyperedge spans r vertices rather than two.

Step 1: each hyperedge crosses at least $r - 1$ tree cuts. Fix a hyperedge e , a set of r vertices, and let $\text{ST}(e)$ be its *smallest connected subtree*, the minimal subtree of T^* that contains all r of them (its Steiner tree). A tree spanning at least r vertices has at least $r - 1$ edges, so $\text{ST}(e)$ has at least $r - 1$ of them. Each such edge t is a genuine cut for e : deleting t from T^* splits the vertices into the two sides $(S_t, V \setminus S_t)$, and because t lies on $\text{ST}(e)$ the hyperedge e has at least one vertex on each side. So e touches both sides, that is, $e \in \delta_{\mathcal{H}}(S_t)$. Therefore

$$|\{t \in E(T^*) : e \in \delta_{\mathcal{H}}(S_t)\}| \geq r - 1 \quad \text{for every hyperedge } e.$$

Step 2: count the incidences two ways. Sum that inequality over all hyperedges, then exchange the order of summation. This is legitimate because both of the middle sums below simply count the same pairs (e, t) with $e \in \delta_{\mathcal{H}}(S_t)$, once grouped by the hyperedge and once by the tree edge:

$$(r - 1) |E(\mathcal{H})| \leq \sum_e |\{t : e \in \delta_{\mathcal{H}}(S_t)\}| = \sum_t |\{e : e \in \delta_{\mathcal{H}}(S_t)\}| = \sum_t |\delta_{\mathcal{H}}(S_t)|.$$

Step 3: read off the bound. By the Gomory–Hu property each tree edge has $|\delta_{\mathcal{H}}(S_t)| = c^*(t)$. Moreover $c^*(t)$ is itself a local connectivity $\lambda_{\mathcal{H}}$ of t 's two endpoints: since those endpoints are adjacent in T^* , the u – v tree path between them is the single edge t , so the path-minimum reading of Theorem A.46 gives $\lambda_{\mathcal{H}}$ of that pair exactly $c^*(t)$, hence at most $m - 1$ by feasibility. Summing over the $n - 1$ tree edges, $\sum_t |\delta_{\mathcal{H}}(S_t)| = \sum_t c^*(t) \leq (m - 1)(n - 1)$. Chaining Steps 2 and 3,

$$(r - 1) |E(\mathcal{H})| \leq (m - 1)(n - 1),$$

and dividing by $r - 1$, then rounding down because $|E(\mathcal{H})|$ is an integer, gives $|E(\mathcal{H})| \leq \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$. *Lower bound.* When $(r - 1) \mid (n - 1)$, take a hub h and split the other $n - 1$ vertices into groups $G_1, G_2, \dots$ of size $r - 1$ each. Form the star hypertree whose hyperedges are $\{h\} \cup G_i$, one per group, and give each hyperedge multiplicity $m - 1$. Every non-hub vertex lies in exactly $m - 1$ hyperedges, which form a cut isolating it, so by Theorem A.45 every pair has Berge connectivity at most $m - 1$, while the count is $\frac{(m-1)(n-1)}{r-1}$. For general n under the hypothesis $m - 1 \leq \binom{n-2}{r-2}$, Theorem A.47 below attains the floor exactly, without even needing

repeated hyperedges. □

The multiplicities in the lower bound can in fact be avoided: for $r \geq 3$, *simple* r -uniform hypergraphs already attain the same bound, in sharp contrast to the graph case $r = 2$, where simplicity costs the factor that separates Mader's $\lfloor m(n-1)/2 \rfloor$ from the multigraph value $(m-1)(n-1)$.

Theorem A.47 (Simplicity is free for $r \geq 3$). *Let $r \geq 3$, $n \geq r$, and $2 \leq m$ with $m-1 \leq \binom{n-2}{r-2}$. Then the maximum number of hyperedges in a simple r -uniform hypergraph on n vertices with Berge edge-connectivity at most $m-1$ is exactly $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$.*

Proof. The upper bound is [Proposition 1.15](#), since a simple hypergraph is a multihypergraph. For the lower bound, fix a hub h and let $V' = V \setminus \{h\}$. By [Lemma A.49](#) applied with rank $r-1$ and degree ceiling $d = m-1$ on the $n-1$ vertices of V' , there is an $(r-1)$ -uniform simple hypergraph $\mathcal{G}^*$ on V' with maximum degree at most $m-1$ and exactly $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ hyperedges. Adding h to each hyperedge produces distinct r -sets, so the result $\mathcal{H}$ is simple, and every non-hub vertex w lies in $\deg_{\mathcal{G}^*}(w) \leq m-1$ hyperedges, which form a cut isolating w . By [Theorem A.45](#), $\lambda_{\mathcal{H}}^{\max} \leq m-1$. □

The sparse hypergraph the construction needs is a corollary of a classical partition theorem, which we state before using it.

Theorem A.48 (Baranyai [18]). *If r divides n , the $\binom{n}{r}$ distinct r -subsets of an n -element set can be partitioned into $\binom{n-1}{r-1}$ classes, each a perfect matching, meaning a family of n/r disjoint r -sets that together cover every vertex exactly once. More generally, for any prescribed class sizes $a_1 + \dots + a_t = \binom{n}{r}$, and with no divisibility assumed, the r -subsets can be partitioned into classes of exactly those sizes so that in the class of size a_i every vertex lies in either $\lfloor ra_i/n \rfloor$ or $\lceil ra_i/n \rceil$ of the sets, as evenly balanced as the size permits.*

The first statement is the memorable one. When r divides n the complete r -uniform hypergraph, the one holding every r -set, comes apart into $\binom{n-1}{r-1}$ perfect matchings, the exact hypergraph analogue of splitting a regular bipartite graph into perfect matchings by König's theorem. The proof is an induction that keeps the partial partition as balanced as possible at every step, an integer-flow argument in the spirit of Hall's marriage theorem rather than an explicit construction, and its full form is in Baranyai [18]. We need only the general equitable form, and only its conclusion about degrees.

Lemma A.49 (Sparse uniform hypergraphs exist). *Let $r \geq 2$, $n \geq r$, and $0 \leq d \leq \binom{n-1}{r-1}$. There exists an r -uniform simple hypergraph on n vertices with maximum degree at most d and exactly $\left\lfloor \frac{dn}{r} \right\rfloor$ hyperedges.*

Proof. Put $e := \left\lfloor \frac{dn}{r} \right\rfloor \leq \frac{n}{r} \binom{n-1}{r-1} = \binom{n}{r}$. Apply the equitable form of [Theorem A.48](#) with two classes, of sizes e and $\binom{n}{r} - e$. Within each class every vertex is covered either $\lfloor re/n \rfloor$ or $\lceil re/n \rceil$

times, so the class of size e is a simple r -uniform hypergraph with $e = \lfloor \frac{dn}{r} \rfloor$ hyperedges and maximum degree at most $\lceil re/n \rceil \leq \lceil d \rceil = d$, the last step because $re/n \leq r(dn/r)/n = d$ and d is an integer. $\square$

A.7 The hypergraph vertex problem

The whole vertex problem translates into a graph problem through one observation, the one the rest of this section rests on and which [Figure A.5](#) draws in full. Write $I(\mathcal{H})$ for the bipartite *incidence graph* of a hypergraph $\mathcal{H}$: one node per vertex, one node per hyperedge, an edge when the vertex lies in the hyperedge. A Berge path with distinct hyperedges is exactly a path of $I(\mathcal{H})$ between two vertex-class nodes. The interior of such a path consists of both kinds of node, the hyperedges it rides and the vertices where it changes hyperedge, so two paths of $I(\mathcal{H})$ are internally disjoint in the ordinary graph sense exactly when the two Berge routes share neither a hyperedge nor an intermediate vertex. That is precisely the separation fixed in [Chapter 1](#), and it is the reason both halves were required there. Hence

$$\kappa_{\mathcal{H}}(u, v) = \kappa_{I(\mathcal{H})}(u, v),$$

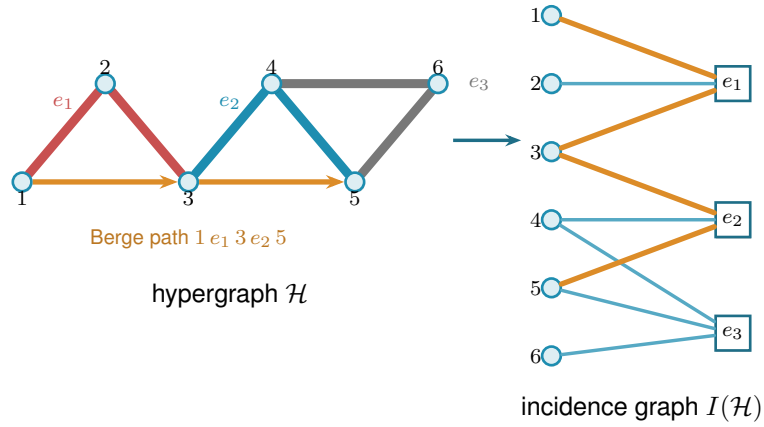
the maximum number of internally disjoint u - v paths in $I(\mathcal{H})$, and the hypergraph vertex problem asks for the maximum number of degree- r nodes on one side of a bipartite graph whose other side holds n nodes, subject to no two of those n nodes being joined by m internally disjoint paths. [Figure A.5a](#) traces a single Berge path through both pictures, and [Figure A.5b](#) shows the star hypertree of [Theorem A.50](#), whose incidence graph is a tree.

Theorem A.50 (Hypergraph vertex value at $m = 2$). *For $r \geq 2$ and $n \geq 1$, the maximum number of hyperedges of an r -uniform multihypergraph on n vertices with $\kappa^{\max} \leq 1$ is*

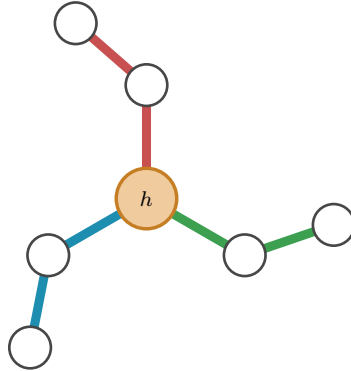
$$k_2^{(r)}(n) = \left\lfloor \frac{n-1}{r-1} \right\rfloor,$$

attained by the star hypertree. In particular $k_2^{(r)}(n) = \ell_2^{(r)}(n)$: the vertex and edge problems agree at $m = 2$ for every uniformity, and repeated hyperedges never help.

Proof. We claim $\kappa_{\mathcal{H}}^{\max} \leq 1$ holds if and only if $I(\mathcal{H})$ is a forest. If $I(\mathcal{H})$ contains a cycle $v_1, e_1, v_2, e_2, \dots, v_\ell, e_\ell, v_1$ (it alternates classes, so $\ell \geq 2$, all v_i and e_i distinct), then v_1, e_1, v_2 and $v_1, e_\ell, v_\ell, e_{\ell-1}, \dots, e_2, v_2$ are two Berge v_1 - v_2 paths with disjoint hyperedge sets and disjoint interior vertex sets, so $\kappa(v_1, v_2) \geq 2$. (At $\ell = 2$ this is the repeated pair: two hyperedges sharing two vertices.) Conversely, two such paths between u and v concatenate to a cycle of $I(\mathcal{H})$. This proves the claim. A forest on $n + q$ nodes has at most $n + q - 1$ edges, while $I(\mathcal{H})$ has exactly rq edges, so $rq \leq n + q - 1$, i.e. $q \leq (n-1)/(r-1)$, and q is an integer. The star hypertree has $\lfloor (n-1)/(r-1) \rfloor$ hyperedges and a tree as incidence graph (each block vertex lies in one hyperedge, every cycle would need two hyperedges sharing two vertices), so the floor is attained. $\square$



(a) A 3-uniform hypergraph $\mathcal{H}$ (left) and its bipartite incidence graph $I(\mathcal{H})$ (right): circles for vertices, hollow squares for hyperedges, an edge when the vertex lies in the hyperedge. The Berge path $1-e_1-3-e_2-5$ is traced in orange in both pictures. In $\mathcal{H}$ it rides the red line e_1 and the blue line e_2 , changing at the shared vertex 3, while the grey line e_3 lies off it. In $I(\mathcal{H})$ it is the ordinary path $1-e_1-3-e_2-5$ alternating sides.



(b) A star hypertree: every hyperedge $\{h\} \cup G_i$ shares only the hub h (orange), and the other $r-1=2$ vertices of each block sit in a single hyperedge. Its incidence graph is a tree, so $\kappa^{\max} \leq 1$.

Figure A.5. The translation the vertex problem rests on. Internally vertex-disjoint Berge paths in $\mathcal{H}$ correspond exactly to internally disjoint paths in $I(\mathcal{H})$, so $\kappa_{\mathcal{H}}(u, v) = \kappa_{I(\mathcal{H})}(u, v)$. **(top)** A hypergraph and its incidence graph with one Berge path drawn in both. **(bottom)** The star hypertree of [Theorem A.50](#), whose incidence graph is a tree.

Proposition A.51 (General lower bound). *For $r \geq 3$, $m \geq 2$ and $m - 1 \leq \binom{n-2}{r-2}$,*

$$k_m^{(r)}(n) \geq \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor,$$

attained by a simple hypergraph.

Proof. The simple construction of [Theorem A.47](#) has Berge edge-connectivity at most $m - 1$, and Whitney's inequality $\kappa \leq \lambda$ holds pairwise (an internally vertex-disjoint family is in particular hyperedge-disjoint under our definition), so the same hypergraph is feasible for the vertex problem. $\square$

The $m = 3$ case also closes completely. The engine is a rank bound for graphs in which one side of a partition must stay below local 3-connectivity. The hypergraph theorem then falls out of the incidence translation. Throughout, $\text{rank}(G) = |E(G)| - |V(G)| + 1$ denotes the cycle rank of a connected multigraph, that is, the number of independent cycles it carries. A tree has cycle rank 0, and each edge added beyond a spanning tree closes exactly one new cycle and raises the rank by one, so bounding the rank is the same as bounding how many independent cycles the graph can hold.

Primer: global connectivity and the pieces of a graph

The lemma below speaks of a graph being 2-connected or 3-connected and of taking it apart at small separators. These are the global cousins of the local $\kappa(u, v)$ from [Chapter 1](#), so we fix the words once. A graph on more than k vertices is *k-connected* if it stays connected after the removal of any $k - 1$ vertices. By Menger this is the same as asking that every pair of vertices be joined by at least k internally vertex-disjoint paths, so *k-connected* means $\kappa(u, v) \geq k$ for all pairs at once, the uniform version of the local measure. Thus 2-connected means no single vertex disconnects the graph, and 3-connected means no two vertices do. Three named obstructions go with this. A *cut vertex* is a vertex whose removal disconnects the graph, so a connected graph on at least three vertices is 2-connected exactly when it has no cut vertex. A *separating pair* is a set of two vertices whose removal disconnects the graph, the obstruction a 2-connected graph must lack in order to be 3-connected. Finally a *block* is a maximal piece with no cut vertex of its own, hence either a single bridge edge or a maximal 2-connected subgraph. Two blocks overlap in at most one vertex, always a cut vertex, and the blocks and cut vertices hang together in a tree, the *block-cut tree*, with one node per block and one per cut vertex. Step 1 of the proof is precisely an induction over that tree.

Lemma A.52 (Incidence rank lemma). *Let G be a connected multigraph with vertex set $X \cup Z$ such that (i) Z is independent, (ii) no edge incident to Z is parallel to another edge, (iii) every $z \in Z$ has degree at least 2, and (iv) $\kappa_G(x, x') \leq 2$ for all distinct $x, x' \in X$. Then $\text{rank}(G) \leq |X| - 1$.*

Proof. Induction on $|V| + |E|$. The proof peels G down to a core that cannot exist. Step 1 handles

a cut vertex by arguing block by block. Steps 2 and 3 remove a degree-two Z -vertex or X -vertex without changing the bound. What survives is 2-connected with minimum degree at least three, and Step 4 shows that such a graph already violates the connectivity ceiling (iv), so it never arises. *Base* $|V| \leq 2$. A single vertex has degree 0, too little for the degree-at-least-2 floor (iii) imposes on a Z -vertex, so it cannot lie in Z and must lie in X , giving $0 \leq |X| - 1$. On two vertices the same reasoning applies to either one: a vertex in Z could reach only the other vertex, and (ii) forbids that single edge from being parallel to another, so it would have degree at most 1, again short of (iii). Hence (i)–(iii) force both vertices into X . Parallel edges between them are internally disjoint paths (each has empty interior, so no two can share an interior vertex), so (iv) caps the multiplicity at 2 and $\text{rank} \leq 1 = |X| - 1$.

Step 1: a cut vertex. Suppose G has a cut vertex, so it is not yet 2-connected. Break it into its *blocks* $B_1, \dots, B_c$ with $c \geq 2$, a block being a maximal piece with no cut vertex of its own, that is, either a single bridge edge or a maximal 2-connected subgraph. Two blocks meet in at most one vertex, and any shared vertex is a cut vertex of G . No cycle can pass through two different blocks: since two blocks share at most one vertex, a cycle straddling both would have to visit that shared vertex twice, which a simple cycle cannot do. So the cycle rank adds up over blocks, $\text{rank}(G) = \sum_i \text{rank}(B_i)$, and it is enough to bound each $\text{rank}(B_i)$ by $|X \cap B_i| - 1$ and sum.

Each block inherits the hypotheses. A bridge block is one edge, so its rank is 0, and by (i) at least one endpoint lies in X , giving $0 \leq |X \cap B| - 1$. Every other block is 2-connected, hence has minimum degree at least 2, so (i) to (iv) hold inside it and the induction hypothesis gives $\text{rank}(B_i) \leq |X \cap B_i| - 1$.

Adding these requires care, because a cut vertex is shared by several blocks and would be counted several times. Write $b(v)$ for the number of blocks containing v , so $b(v) = 1$ except at cut vertices. Counting each X -vertex once for every block it lies in,

$$\sum_i |X \cap B_i| = |X| + \sum_{x \in X \text{ cut}} (b(x) - 1).$$

The number of blocks satisfies the block–cut tree identity $c = 1 + \sum_{v \text{ cut}} (b(v) - 1)$. This follows from counting the edges of the block-cut tree two ways. That tree has c block-nodes and one node per cut vertex, so being a tree its edge count is one less than its total number of nodes, $c + \#\{\text{cut vertices}\} - 1$. The same edge count also equals $\sum_{v \text{ cut}} b(v)$, since each cut vertex v is joined in the tree to exactly the $b(v)$ blocks that contain it. Equating the two expressions for the edge count and solving for c gives the identity. Subtracting this count of blocks from the previous display and separating the cut vertices into their X - and Z -parts leaves

$$\text{rank}(G) = \sum_i \text{rank}(B_i) \leq \sum_i (|X \cap B_i| - 1) = |X| - 1 - \sum_{z \in Z \text{ cut}} (b(z) - 1) \leq |X| - 1,$$

where the last inequality holds because each $b(z) - 1 \geq 0$, so the Z -cut sum only pushes the bound further below $|X| - 1$.

Step 2: a Z -vertex of degree two. From now on G is 2-connected, every cut vertex having been handled. Suppose some $z_0 \in Z$ has degree exactly 2. Both its neighbours lie in X by (i), and they are distinct, since two edges to the same vertex would be parallel at a Z -vertex, which (ii) forbids. *Suppress* z_0 by deleting it and joining its two neighbours x, x' with one new edge. This loses two edges and one vertex and gains one edge, so $|E|$ and $|V|$ each drop by one and $\text{rank} = |E| - |V| + 1$ is unchanged. The new edge runs X to X , so $|X|$ is unchanged and (i), (ii), (iii) still hold for the remaining Z -vertices. Replacing the path x, z_0, x' by a direct edge alters no route among the surviving vertices, so every $\kappa(x_1, x_2)$ is preserved and (iv) holds. The graph is strictly smaller, so the induction hypothesis applies and returns the bound for G .

Step 3: an X -vertex of degree two. Now G is 2-connected and, Step 2 being used up, every Z -vertex has degree at least 3. Suppose some $x_0 \in X$ has degree 2, and delete it. Deleting one vertex from a 2-connected graph leaves it connected. Stripping a degree-2 vertex removes two edges and one vertex, so rank drops by exactly one. Each neighbour of x_0 loses one edge, and a neighbour in Z only falls from degree at least 3 to degree at least 2, so (iii) survives, while deleting a vertex cannot raise a connectivity, so (iv) survives. Here $|X|$ drops by one, and $|X| \geq 2$ to begin with, for if $|X| \leq 1$ then by (i) and (ii) every z would send all its edges to one X -vertex without repeats, hence have degree at most 1, against (iii). The smaller graph satisfies the hypotheses, so induction gives $\text{rank}(G) - 1 \leq (|X| - 1) - 1$, that is $\text{rank}(G) \leq |X| - 1$.

Step 4: the remaining case is impossible. Now G is 2-connected with $|V| \geq 3$ and every degree at least 3. First, G is simple: a parallel class lies inside X by (ii), and if $\mu(x, x') \geq 2$ then a third internally disjoint route exists: walk from x to any third vertex w inside $G - x'$, then from w to x' inside $G - x$, both walks available because deleting a single vertex from a 2-connected graph leaves it connected, exactly as already used in Step 3. The concatenation visits x only first and x' only last, so it contains an $x-x'$ path with at least one interior vertex, giving $\kappa(x, x') \geq 3$, against (iv). If G is 3-connected, Menger puts every pair at $\kappa \geq 3$, so two vertices of X would violate the ceiling $\kappa(x, x') \leq 2$ that (iv) imposes on every pair drawn from X , forcing $|X| \leq 1$, and then every z has degree at most 1 by (i), absurd.

Otherwise G is 2-connected but not 3-connected, so it can be taken apart at its 2-vertex separators.

Primer: the triconnected (SPQR) decomposition

A 2-connected graph that is not 3-connected must have a separating pair, and cutting along every such pair sorts any 2-connected graph into a few standard shapes. This is a classical theorem of Tutte [19], made into a linear-time algorithm by Hopcroft and Tarjan [20].

Theorem (triconnected decomposition). Every 2-connected graph G on at least three vertices decomposes into a *tree of pieces*, unique up to relabelling, in which each piece is one of three kinds: a *cycle* of length at least three (an *S*-piece, for *series*), a *bond*, meaning two vertices joined by three or more parallel edges (a *P*-piece, for *parallel*), or a 3-connected simple graph on at least four vertices (an *R*-piece, for *rigid*). Two adjacent pieces share exactly

one *virtual edge* $\{a, b\}$, a placeholder that records a separating pair of G at which the two pieces were split apart. The three initials S , P , R , together with Q for a trivial single-edge piece, give the structure its usual name, the *SPQR tree*.

The tree is built by the recursion that also defines it. If G is already a cycle, a bond, or 3-connected, it is a single piece and the tree is one node. Otherwise choose a separating pair $\{a, b\}$, split G into the parts that this pair separates, add the virtual edge $\{a, b\}$ to each part so the split is remembered, and recurse on the parts. Every split makes the parts strictly smaller, so the recursion halts, and Tutte's theorem is that the resulting collection of pieces does not depend on the order in which the separating pairs are taken. The one fact the proof draws from all this is that a virtual edge $\{a, b\}$ always stands for a genuine a – b path through the rest of G , so an argument that must cross from a to b outside a given piece can always do so on the far side of that piece's virtual edge.

The decomposition just stated is sketched in [Figure A.6](#), which also marks the leaf at which the argument bites. Two properties are all the proof needs from it, and both are visible in the figure. First, a vertex of a piece that touches none of that piece's virtual edges belongs to that piece alone, so it keeps in G exactly the degree it has in the piece. Second, for any virtual edge $\{a, b\}$ the part of G on the far side of the cut contains a genuine a – b path whose interior avoids the piece, so the virtual edge can always be replaced by a real route.

Now G itself is none of the three piece types, since it is not a single cycle (its degrees are at least three), not a bond (it is simple), and not 3-connected (the case just handled), so the tree has at least two leaves. Let L be one, with unique virtual edge $\{a, b\}$. If L is a cycle, a vertex of L off $\{a, b\}$ has degree 2 in G , absurd. If L is a bond, it holds at least two real parallel edges, absurd in a simple graph. If L is 3-connected, then any two of its vertices are joined by three internally disjoint paths inside L , at most one through the virtual edge, which the far-side path replaces: so $\kappa_G(u, v) \geq 3$ for all $u, v \in V(L)$, forcing $|X \cap V(L)| \leq 1$ by (iv). But $|V(L)| \geq 4$ leaves a Z -vertex $z^* \notin \{a, b\}$. Its neighbours lie in $X \cap V(L)$ by (i), at most one vertex, reached by at most one edge since G is simple, giving degree at most 1, which is absurd. $\square$

Theorem A.53 (Hypergraph vertex value at $m = 3$). *For $r \geq 2$, every r -uniform multihypergraph on n vertices with $\kappa^{\max} \leq 2$ has at most $\left\lfloor \frac{2(n-1)}{r-1} \right\rfloor$ hyperedges. For $r \geq 3$ with $2 \leq \binom{n-2}{r-2}$ the bound is attained by a simple hypergraph, so*

$$k_3^{(r)}(n) = \left\lfloor \frac{2(n-1)}{r-1} \right\rfloor = \ell_3^{(r)}(n).$$

Proof. Apply [Lemma A.52](#) to each connected component of the incidence graph $I(\mathcal{H})$, with X the vertex class and Z the hyperedge class: Z is independent, incidences are simple even when hyperedges repeat, hyperedge-nodes have degree $r \geq 2$, components without an X -node are impossible, and $\kappa_I(u, v) = \kappa_{\mathcal{H}}(u, v) \leq 2$ for vertex pairs. Sum $\text{rank} \leq |X_c| - 1$ over the C components. The incidence graph $I(\mathcal{H})$ has rq edges (each of the q hyperedges contributes r

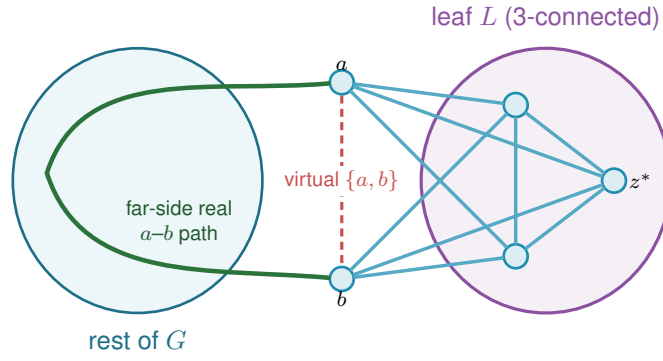


Figure A.6. The triconnected (Tutte/SPQR) decomposition in Step 4 of [Lemma A.52](#), schematic: G splits into a tree of pieces (blobs) glued along virtual edges, and the analysis works at a *leaf* L with its single virtual edge $\{a, b\}$ (dashed red). The far side of the tree supplies a *real* a - b path (green) that internally avoids L . If L is 3-connected, any two of its vertices are joined by three internally disjoint paths inside L , at most one using the virtual edge, and the far-side path replaces it, so $\kappa_G \geq 3$ for every pair in L . With $|V(L)| \geq 4$ that forces a hyperedge-node $z^* \notin \{a, b\}$ of degree at most one, which is the contradiction that closes the lemma.

incidences) and $n + q$ nodes, so $\sum_c \text{rank} = rq - (n + q) + C$, while $\sum_c (|X_c| - 1) = n - C$. The inequality reads $rq - (n + q) + C \leq n - C$, hence $q(r - 1) \leq 2(n - 1) + 2(1 - C) \leq 2(n - 1)$, the last step using $C \geq 1$. The lower bound is [Proposition A.51](#) at $m = 3$. $\square$

Remark A.54 (Edges as gates at $r = 2$, and the boundary of the method). At $r = 2$ the theorem speaks of multigraphs under the hyperedge-as-gate convention, in which parallel edges are distinct one-step routes, giving the multigraph value $2(n - 1)$ of [Theorem 1.3](#), complementing the body's convention for graphs, where parallel adjacencies count once. The proof of [Lemma A.52](#) leans on the triconnected decomposition exactly where $\kappa \leq 2$ lives. For $m = 4$, the analogous question asks for a rank bound relative to $\kappa \leq 3$, where no comparably clean decomposition exists. That the pattern $k_m^{(r)} = \ell_m^{(r)}$ must fail eventually is not merely an analogy with the graph case ($m = 5$, Sørensen-Thomassen), which is stated for simple graphs under the body's convention. It already fails inside this convention, at $r = 2$. The identification in the first sentence of this remark runs both ways: the $r = 2$ case of the present problem *is* the multigraph vertex problem of [Section A.9](#), since a route there is a family of edge-disjoint and internally disjoint paths, which is exactly [Lemma A.61](#). So [Theorem A.63](#) applies to it, and the value exceeds $\lfloor (m - 1)(n - 1)/(r - 1) \rfloor = (m - 1)(n - 1)$ for every $m \geq 5$ and every $n \geq 3$, by a margin that grows with n . The formula is therefore not a candidate for all m and r together, and the live question is only whether $r \geq 3$ postpones the failure past $m = 4$.

For $m = 4$ it does not fail anywhere within reach. An exhaustive search over r -uniform multihypergraphs confirms both halves of the value, that $\lfloor 3(n - 1)/(r - 1) \rfloor$ is attained and that one more hyperedge is infeasible, at $(n, r) = (4, 3), (5, 3), (5, 4), (6, 4), (6, 5), (7, 5), (7, 6), (8, 6), (8, 7), (9, 7)$ and $(9, 8)$. That is a wider net than the single cell $k_4^{(3)}(5) = 6$, and it is evidence rather than proof: the missing ingredient remains the 4-connectivity rank bound, and the $r = 2$ failure at $m = 5$ shows the surrounding claim has a genuine boundary rather than extending indefinitely.

A.8 The directed hypergraph

The remaining two of the twelve variants orient the Berge edge. They are grouped here rather than with the undirected hypergraph bounds above because the questions they raise are different ones: not which decomposition controls the rank of an incidence graph, but how much a single edge with one tail and many heads can carry, and whether the way it is oriented changes the answer. The first result is a pair of bounds a factor of four apart, the last closes that factor for the model the program measures.

Proposition A.55 (Directed hypergraphs: first bounds). *Model a directed r -uniform hyperedge as a tail vertex together with $r - 1$ head vertices, a Berge route entering at the tail and leaving at a head (the program's convention, drawn in Figure A.7). If every pair of vertices has at most $m - 1$ arc-disjoint directed Berge routes, then*

$$|E(\mathcal{H})| \leq \frac{(m - 1) n(n - 1)}{r - 1},$$

and for $m - 1 \leq \binom{n - \lceil n/2 \rceil - 1}{r - 2}$ there are feasible constructions with $\max_{1 \leq \alpha \leq n - 1} \alpha \left\lfloor \frac{(m - 1)(n - \alpha)}{r - 1} \right\rfloor \approx (m - 1)n^2 / (4(r - 1))$ hyperedges, so the value is quadratic in n .

Proof. Upper bound: a hyperedge with tail u serves the $r - 1$ ordered pairs (u, h) , h a head. If some ordered pair (u, v) were served by m distinct hyperedges, those would be m arc-disjoint one-step routes, so each ordered pair is served at most $m - 1$ times. Summing that cap over all $n(n - 1)$ ordered pairs counts every hyperedge once for each of its $r - 1$ tail-head pairs, so the sum is $(r - 1)|E|$ on one side and at most $(m - 1)n(n - 1)$ on the other, giving $(r - 1)|E| \leq (m - 1)n(n - 1)$. Lower bound: split V into tails A , $|A| = \alpha$, and heads B . By Lemma A.49 there is a simple $(r - 1)$ -uniform hypergraph $\mathcal{G}^*$ on B with maximum degree $m - 1$ and $\lfloor (m - 1)|B| / (r - 1) \rfloor$ edges. Give every tail $a \in A$ the hyperedges (a, H) , $H \in \mathcal{G}^*$. No vertex of B is a tail, so every route is a single step and $\lambda(a, b) = \deg_{\mathcal{G}^*}(b) \leq m - 1$, while pairs inside A or starting in B have no routes. $\square$

The forward model is one of three ways to orient a Berge edge. In general a directed r -uniform hyperedge is a split of its r vertices into a non-empty *tail set* T and a non-empty *head set* H , a route entering at a tail and leaving at a head. The *forward* model is $|T| = 1$, the *backward* model is $|H| = 1$, and the *general* model allows any split (genuinely mixed splits, both parts at least two, exist from $r \geq 4$). The next two results place all three. The first is that backward needs no separate study, and the second is that the general model, despite admitting more hyperedges on a single vertex set, obeys the very same first-order bound as forward.

Lemma A.56 (Backward is the reverse of forward). *Reversing every hyperarc, $(T, H) \mapsto (H, T)$, is an edge-count-preserving bijection between the forward directed r -uniform hypergraphs and the backward ones on the same vertices, and the reverse $\mathcal{H}^R$ satisfies $\lambda_{\mathcal{H}}(u, v) = \lambda_{\mathcal{H}^R}(v, u)$*



Figure A.7. How the thesis draws a single hyperedge on its r vertices, in the metro colours of Figure 1.3. *Left:* an undirected hyperedge is one metro line threading all its members, with no arrows. *Right:* a directed hyperedge, a tail with $r - 1$ heads, is drawn as a star, the tail sending one thick arc to each head, so a Berge route enters at the tail and leaves at any head. The hypergraph panels of Figure 1.4 use this same star. Where a directed hyperedge instead has to sit on a metro line it already occupies undirected, as in Figure 2.8, the line is oriented along its own length away from the tail rather than fanned out, which draws the same object with fewer crossings. Either way one rule recovers the tail from the picture: it is the single station of that line into which none of its own arrows points. The shape is a drawing convention, not a path. Each coloured line or star is a *single* edge on several vertices, not a sequence of ordinary edges, so the line on the left is one hyperedge and not three. One colour means one hyperedge, and where two of them meet they cross at a shared station rather than joining into a longer line. What might be mistaken for a hyperedge is a *route*, which does run edge by edge from one vertex to another, but a route is drawn as a thin thread laid over the lines, so the thick coloured hyperedge and the thin route never look alike.

and $\kappa_{\mathcal{H}}(u, v) = \kappa_{\mathcal{H}^R}(v, u)$. Hence $\lambda^{\max}$ and $\kappa^{\max}$ are invariant under reversal, and the backward edge and vertex extremal numbers equal the forward ones, for every r, m and n .

Proof. Reversal sends a backward hyperarc ($|T| = r - 1, |H| = 1$) to a forward one ($|T| = 1$) and is its own inverse, so it is an edge-count-preserving bijection of the two families. Let $u = x_0, e_1, x_1, \dots, e_k, x_k = v$ be a directed Berge route in $\mathcal{H}$, where x_{i-1} is a tail and x_i a head of e_i . The reversed edge $e_i^R = (H_i, T_i)$ has x_i as a tail and x_{i-1} as a head, so $v = x_k, e_k^R, x_{k-1}, \dots, e_1^R, x_0 = u$ is a directed Berge route from v to u in $\mathcal{H}^R$ on the same edges and the same internal vertices. This is a bijection between u - v routes in $\mathcal{H}$ and v - u routes in $\mathcal{H}^R$ that preserves which edges and which internal vertices any two routes share, so it carries an edge-disjoint (respectively internally vertex-disjoint) family to one of equal size. Therefore $\lambda_{\mathcal{H}}(u, v) = \lambda_{\mathcal{H}^R}(v, u)$ and $\kappa_{\mathcal{H}}(u, v) = \kappa_{\mathcal{H}^R}(v, u)$, and maximising over ordered pairs gives $\lambda^{\max}(\mathcal{H}) = \lambda^{\max}(\mathcal{H}^R)$ and the same for κ . Since reversal is a bijection preserving feasibility and edge count, the two models share every extremal number. $\square$

Proposition A.57 (The general model obeys the forward bound). *Let a directed r -uniform hyperedge be any split into a non-empty tail set T and head set H . If every ordered pair has at most $m - 1$ arc-disjoint, or at most $m - 1$ internally vertex-disjoint, directed Berge routes, then*

$$|E(\mathcal{H})| \leq \frac{(m - 1) n(n - 1)}{r - 1},$$

the same bound as the forward model of Proposition A.55. The forward construction of that proposition is itself a set of general hyperedges, so the general value obeys the same upper bound and inherits the same quadratic lower bound, and in particular is $\Theta(n^2)$.

Proof. Fix an ordered pair (u, v) . A hyperedge (T, H) with $u \in T$ and $v \in H$ carries the one-step route $u \rightarrow v$ that enters at the tail u and leaves at the head v . Distinct such hyperedges give routes that share no edge and no internal vertex, so if k hyperedges had u in their tail set and v in their head set then $\lambda(u, v) \geq k$ and $\kappa(u, v) \geq k$, so feasibility forces $k \leq m - 1$. Each hyperedge (T, H) is counted by exactly the $|T| |H|$ ordered pairs (u, v) with $u \in T$ and $v \in H$, so summing the per-pair bound over all pairs gives $\sum_{(T, H)} |T| |H| \leq (m - 1) n(n - 1)$. As $|T| + |H| = r$ with both parts at least one, $|T| |H| \geq 1 \cdot (r - 1) = r - 1$, whence $(r - 1)|E| \leq (m - 1)n(n - 1)$. The forward family of [Proposition A.55](#) consists of general hyperedges, so it witnesses the same $\approx (m - 1)n^2/(4(r - 1))$ lower bound. $\square$

The two models are therefore trapped between the same pair of bounds, which differ by a factor of 4. That is enough to fix the *order* of both values and not enough to fix either leading constant, so it does not follow that the forward and general models agree asymptotically, only that any disagreement between them lives inside that factor. The rest of this section closes that gap for the forward model, which leaves the comparison between the two models as the question that survives.

It is worth saying which end of that gap is the likely culprit, because the two bounds are not equally crude and the analogy with the simple digraph case is informative. The upper bound charges each hyperedge to the ordered tail-head pairs it occupies and then allows all $n(n - 1)$ ordered pairs to be used to capacity. The construction uses only the $\alpha(n - \alpha) \leq n^2/4$ pairs that run from the tail class to the head class, and leaves the pairs inside each class, and those running backwards, entirely idle. The factor of four is exactly that difference, n^2 against $n^2/4$, and it is the same difference that appears in the simple directed problem, where [Construction 1.13](#) also uses only a one-directional bipartite pattern. There, [Theorem A.24](#) now proves that the bipartite pattern is right to first order, that no arrangement recovers the idle pairs. If that phenomenon is a feature of directedness rather than of graphs specifically, then it is the upper bound here that should come down by a factor of four, not the construction that should climb, and that is what happens.

The obstacle to running the argument of [Theorem A.24](#) here is real and worth naming before it is disposed of, because the way around it is not to remove it. There, two-step routes through distinct midpoints were automatically arc-disjoint. Here a single hyperedge carries $r - 1$ heads at once, so a pair of two-step routes with different midpoints may enter through the very same hyperedge, and the family of two-step routes is not edge-disjoint at all. The resolution is to accept the loss. A packing can always be extracted from that family at a cost of a factor $r - 1$, and that factor turns out to be free, because of where it lands. The route count does not bound the hyperedges directly. It bounds the arcs of an auxiliary digraph, and there the factor $r - 1$ enters only the *linear* error term of [Lemma A.27](#), never the $n^2/4$ leading term. Meanwhile a second, independent factor of $r - 1$ is gained when the auxiliary digraph is exchanged back for hyperedges, and that one does divide the leading term. The two factors are not the same factor, and only one of them is paid where it matters.

Definition A.58 (One-step shadow). Let $\mathcal{H}$ be a forward directed r -uniform hypergraph on V . Its *one-step shadow* is the simple digraph $R(\mathcal{H})$ on V with an arc (u, v) exactly when some hyperedge of $\mathcal{H}$ has tail u and v among its heads. Write $\text{cod}(u, v)$ for the number of hyperedges realising that arc.

Theorem A.59 (The directed hypergraph constant). *Let $r \geq 2$ and $m \geq 2$. Every forward directed r -uniform multihypergraph $\mathcal{H}$ on $n \geq 2$ vertices with $\kappa_{\mathcal{H}}^{\max} \leq m - 1$ satisfies*

$$|E(\mathcal{H})| \leq \frac{m-1}{r-1} \left\lfloor \frac{n^2}{4} \right\rfloor + 4(m-1)^2(n-1).$$

The same bound therefore holds under $\lambda_{\mathcal{H}}^{\max} \leq m - 1$, which is the stronger hypothesis by $\kappa \leq \lambda$. Hence for fixed r and m both the edge- and the vertex-disjoint maxima are $(1 + o(1))(m - 1)n^2 / (4(r - 1))$, and the bipartite construction of [Proposition A.55](#) is asymptotically optimal for both.

Proof. Write $R = R(\mathcal{H})$ for the one-step shadow and note it has no loops, since a hyperedge's tail is not one of its heads.

Step 1: hyperedges against shadow arcs. Suppose $\text{cod}(u, v) = k$. Those k hyperedges each give a one-step Berge route from u to v , and the routes use one hyperedge apiece and different ones, so they are pairwise hyperedge-disjoint, while having no interior vertex at all they are internally vertex-disjoint as well. Hence $k \leq \kappa_{\mathcal{H}}(u, v) \leq m - 1$. Every hyperedge is counted by exactly its $r - 1$ tail-head pairs, so summing over the arcs of R ,

$$(r - 1) |E(\mathcal{H})| = \sum_{(u,v) \in A(R)} \text{cod}(u, v) \leq (m - 1) |A(R)|.$$

This is the count of [Proposition A.55](#) stopped one step earlier. That proof then bounded $|A(R)|$ by the trivial $n(n - 1)$, which is exactly where its factor of four was lost, and the rest of this argument replaces that step.

Step 2: the shadow is budgeted. Fix an ordered pair (u, v) of distinct vertices, and let $T \subseteq V$ collect v itself when $(u, v) \in A(R)$ together with every midpoint counted by $p_2(u, v)$, so $|T| = \mathbf{1}_{(u,v) \in A(R)} + p_2(u, v)$. Every $t \in T$ is a head of at least one hyperedge whose tail is u . Form the bipartite graph joining each $t \in T$ to each such hyperedge containing it, and let M be a maximum matching in it.

Then $|M| \geq |T| / (r - 1)$. To see it, take any $t \in T$ left unmatched by M . Every hyperedge containing t must already be matched, since otherwise M could be extended along that pair. So every element of T , matched or not, lies in some hyperedge used by M , and each hyperedge holds at most $r - 1$ elements of T , giving $|T| \leq |M| (r - 1)$.

Now read M as a family of routes. For a matched pair (v, e) take the one-step route $u \rightarrow v$ through e , and for a matched pair (x, e) with x a midpoint take $u \rightarrow x \rightarrow v$, entering through e and leaving through any hyperedge whose tail is x and which has v as a head, one of which exists

because $(x, v) \in A(R)$. These routes are pairwise hyperedge-disjoint: the entering hyperedges are distinct because M is a matching, the leaving hyperedges have pairwise distinct tails and so are distinct from each other, and no leaving hyperedge is an entering one, since the entering ones have tail u and the leaving ones have tail a midpoint, which is not u . They are also internally vertex-disjoint, since the one-step route has empty interior and each two-step route has for interior the single vertex it is matched to, and M matches distinct targets. Hence

$$\frac{\mathbf{1}_{(u,v) \in A(R)} + p_2(u, v)}{r-1} \leq |M| \leq \kappa_{\mathcal{H}}(u, v) \leq m-1,$$

so R is $(r-1)(m-1)$ -budgeted in the sense of [Definition A.26](#).

Step 3: assembling. [Lemma A.27](#) with $C = (r-1)(m-1)$ gives $|A(R)| \leq \lfloor n^2/4 \rfloor + 4(r-1)(m-1)(n-1)$, and substituting that into Step 1,

$$|E(\mathcal{H})| \leq \frac{m-1}{r-1} \left[\left\lfloor \frac{n^2}{4} \right\rfloor + 4(r-1)(m-1)(n-1) \right] = \frac{m-1}{r-1} \left\lfloor \frac{n^2}{4} \right\rfloor + 4(m-1)^2(n-1).$$

The lower bound is [Proposition A.55](#), whose bipartite family has

$$\max_{\alpha} \alpha \left\lfloor \frac{(m-1)(n-\alpha)}{r-1} \right\rfloor = \frac{(m-1)n^2}{4(r-1)} - O_{m,r}(n)$$

hyperedges, where the balanced split $\alpha = \lfloor n/2 \rfloor$ already attains the right-hand side on its own, and which is feasible for both separations at once, since it is λ -feasible and $\kappa \leq \lambda$. (The integer maximum is occasionally taken a little off balance, since the floor can favour an unbalanced split by a unit or two, but never by enough to move the n^2 term.) Upper and lower bound agree in their n^2 term, which is the asymptotic claim. $\square$

Two remarks on the statement. First, the two upper bounds are not comparable and both are worth keeping: the bound of [Proposition A.55](#) is the smaller one until n is around $16(m-1)(r-1)/3$, since the linear term here carries a constant of $4(m-1)^2$, and the theorem is an asymptotic improvement rather than a uniform one. Second, at $r=2$ the shadow is the digraph itself and cod is the multiplicity, so the theorem recovers the quadratic branch of [Theorem A.30](#) up to its linear error term, which is the consistency check one wants, though it does not reprove that theorem, which is exact.

A third remark used to say that the general orientation model was out of reach, and it is worth recording why, because the obstacle was real and the way past it is short. The proof above uses the forward model twice, once for the entering hyperedges and once for the leaving ones, in both cases to know that a hyperedge has a single tail. That is what makes Step 2's matching legitimate: the leaving hyperedges have pairwise distinct tails and so are automatically distinct from each other, and none of them can be an entering hyperedge, since the entering ones have tail u and the leaving ones do not. Allow several tails and both statements fail at once. Two midpoints may leave through one shared hyperedge, and from $r \geq 4$ a single hyperedge with two

tails and two heads may serve as one target's entrance and another's exit. So the conflicts run both ways and no matching on one side controls them.

The repair is to stop building a large family and instead take a maximum one and ask what stopped it. A maximum family cannot be extended, so every target it leaves unserved must be obstructed by a hyperedge the family has already spent, and a hyperedge has only r vertices in all to obstruct with. That counts the leftovers without ever needing the two sides to be independent, which is exactly what the general model denies.

Theorem A.60 (The general orientation model shares the constant). *Let $r \geq 2$ and $m \geq 2$, and let a directed r -uniform hyperedge be any split (T, H) into a non-empty tail set and a non-empty head set, as in [Proposition A.57](#). Every such multihypergraph $\mathcal{H}$ on $n \geq 2$ vertices with $\kappa_{\mathcal{H}}^{\max} \leq m - 1$ satisfies*

$$|E(\mathcal{H})| \leq \frac{m-1}{r-1} \left\lfloor \frac{n^2}{4} \right\rfloor + \frac{4(2r+1)(m-1)^2}{r-1} (n-1).$$

The same bound therefore holds under $\lambda_{\mathcal{H}}^{\max} \leq m - 1$. Hence for fixed r and m the general model has the same leading term $(1 + o(1))(m-1)n^2/(4(r-1))$ as the forward model, under both separations, and the bipartite construction of [Proposition A.55](#) is asymptotically optimal for it too.

Proof. Let R be the one-step shadow read in the general model, so $(u, v) \in A(R)$ exactly when some hyperedge has u among its tails and v among its heads, and $\text{cod}(u, v)$ counts the hyperedges realising it. A hyperedge's tail set and head set are disjoint, so R has no loops.

Step 1: hyperedges against shadow arcs. The $\text{cod}(u, v)$ hyperedges realising an arc give that many one-step routes, one hyperedge apiece and different ones, with no interior vertex, so they are pairwise hyperedge-disjoint and internally vertex-disjoint and $\text{cod}(u, v) \leq \kappa_{\mathcal{H}}(u, v) \leq m - 1$. A hyperedge (T, H) is counted by exactly the $|T| |H|$ ordered pairs it fills, and $|T| + |H| = r$ with both parts non-empty gives $|T| |H| \geq r - 1$. Hence

$$(r-1) |E(\mathcal{H})| \leq \sum_{(T,H) \in E(\mathcal{H})} |T| |H| = \sum_{(u,v) \in A(R)} \text{cod}(u, v) \leq (m-1) |A(R)|.$$

This is the count of [Proposition A.57](#) stopped one step early, exactly as Step 1 above stopped [Proposition A.55](#) one step early.

Step 2: the shadow is budgeted. Fix an ordered pair (u, v) of distinct vertices and let $\mathcal{T}$ collect v itself when $(u, v) \in A(R)$ together with every midpoint counted by $p_2(u, v)$, so $|\mathcal{T}| = \mathbf{1}_{(u,v) \in A(R)} + p_2(u, v)$. Call a hyperedge e an *entrance* for $t \in \mathcal{T}$ if u is a tail of e and t a head, and an *exit* for a midpoint t if t is a tail of e and v a head. Every $t \in \mathcal{T}$ has at least one entrance, and every midpoint at least one exit, by the definition of R . No hyperedge is both an entrance and an exit for the *same* t , since that would put t in its tail set and its head set at once.

Consider families of routes indexed by distinct targets: for the target v the one-step route

through an entrance, and for a midpoint t the two-step route $u \rightarrow t \rightarrow v$ through an entrance and an exit. Such a family is admissible when its hyperedges are pairwise distinct. Let $\mathcal{F}$ be one of maximum size M , serving the target set S , and let U be the hyperedges it uses, so $|U| \leq 2M$. The routes of $\mathcal{F}$ are pairwise hyperedge-disjoint by construction and internally vertex-disjoint because their interiors are the empty set and distinct singletons, so

$$M \leq \kappa_{\mathcal{H}}(u, v) \leq m - 1.$$

Now take $t \in \mathcal{T} \setminus S$. If some entrance $e \notin U$ existed and, for a midpoint, some exit $f \notin U$ as well, then $e \neq f$ by the paragraph above and the route through them could be added to $\mathcal{F}$, serving a target not yet served and using no hyperedge already spent. That contradicts maximality. So every entrance of t lies in U , or t is a midpoint all of whose exits lie in U . Either way t lies in the tail set or the head set of some member of U , and therefore

$$\mathcal{T} \setminus S \subseteq \bigcup_{e \in U} (T_e \cup H_e), \quad \text{so} \quad |\mathcal{T}| \leq M + \sum_{e \in U} (|T_e| + |H_e|) = M + r|U|.$$

The last equality is where the two sides are counted together rather than separately, and it is what keeps the constant at r rather than $2(r - 1)$. With $|U| \leq 2M$ this gives $|\mathcal{T}| \leq (2r + 1)M \leq (2r + 1)(m - 1)$, so R is $(2r + 1)(m - 1)$ -budgeted in the sense of [Definition A.26](#).

Step 3: assembling. [Lemma A.27](#) with $C = (2r + 1)(m - 1)$ gives $|A(R)| \leq \lfloor n^2/4 \rfloor + 4(2r + 1)(m - 1)(n - 1)$, and substituting into Step 1 gives the stated bound. The hypothesis $\lambda^{\max} \leq m - 1$ implies $\kappa^{\max} \leq m - 1$ by Whitney, so the bound holds there too. For the lower bound, the bipartite family of [Proposition A.55](#) consists of general hyperedges, so it witnesses $\approx (m - 1)n^2/(4(r - 1))$ here as well, and upper and lower bound agree in their n^2 term. $\square$

Two things about that proof are worth separating. The constant $2r + 1$ is not optimised and is not claimed to be sharp: it is what the crude estimate $|U| \leq 2M$ gives, and a search over deliberately adversarial instances, in which every target shares both its entrance and its exit with another, never drives $|\mathcal{T}|/\kappa(u, v)$ above $r - 1$, the forward model's own constant. Nothing rests on the difference, since the whole point of [Lemma A.27](#) is that this constant lands in the linear term and never touches the $n^2/4$ that carries the leading constant. The second point is what the theorem does to the orientation axis of [Section 4.3.1](#). Forward and backward were already known equal, exactly, by [Lemma A.56](#). The general model was known only to sit between the same two bounds a factor of four apart, which left room for it to be genuinely denser to first order. It is not. All three orientation models have the same leading term, and the computed differences between them at $n \approx r$ are a small-size effect that the asymptotics cannot see.

A.9 The multigraph vertex problem under the other convention

[Remark 2.1](#) collapses parallel copies in vertex mode, which makes the two multigraph vertex variants restatements of their simple counterparts and is why they carry no theorem of their own.

This section takes the other reading seriously, because under it the question is genuinely new, and records what can be proved and what the machine finds. Throughout, q parallel copies of the edge uv count as q internally vertex-disjoint u - v routes, since a single edge is a path with no interior. Write $K_m^{\text{multi}}(n)$ for the largest total multiplicity of a multigraph on n vertices with $\kappa^{\max} \leq m - 1$ under this reading.

The first thing to notice is that the two kinds of route never interfere, which splits the measure cleanly in two.

Lemma A.61 (Splitting the vertex measure). *Let G be a multigraph with multiplicity function μ and underlying simple graph G_0 . For any two vertices $u \neq v$, write $\pi(u, v)$ for the largest number of pairwise internally disjoint u - v paths of length at least two in G_0 . Then, under the convention above,*

$$\kappa_G(u, v) = \mu(u, v) + \pi(u, v).$$

Proof. A parallel copy of uv is a path with empty interior, so any two copies are internally disjoint from each other and from every longer route. A maximum family may therefore be assumed to contain all $\mu(u, v)$ copies, and what it can add to them is a family of pairwise internally disjoint routes of length at least two, whose interiors are disjoint sets of vertices. Since parallel copies are irrelevant to a route of length at least two, which is determined by its vertex sequence, the largest such family has size $\pi(u, v)$, computed in G_0 . $\square$

So feasibility says exactly that $\mu(u, v) \leq m - 1 - \pi(u, v)$ on every edge, and that $\pi(u, v) \leq m - 1$ on every pair. At $m = 2$ this is already decisive.

Theorem A.62 (The other convention at $m = 2$). $K_2^{\text{multi}}(n) = n - 1$, attained exactly by the spanning trees.

Proof. By Lemma A.61, $\kappa^{\max} \leq 1$ forces $\mu(u, v) + \pi(u, v) \leq 1$ on every edge. If G_0 contained a cycle, an edge uv of that cycle would have $\mu(u, v) \geq 1$ and $\pi(u, v) \geq 1$ from the rest of the cycle, a contradiction, so G_0 is a forest and every multiplicity is one. A forest has at most $n - 1$ edges, and a spanning tree attains it. $\square$

For $m \geq 3$ at least three constructions compete, and which one wins depends on how n compares with m , exactly as in every other variant of this thesis.

- ★ *The thickened tree.* Take a spanning tree and give every edge multiplicity $m - 1$. Tree edges have $\pi = 0$ and non-adjacent pairs have $\pi = 1$, so this is feasible for every $m \geq 2$, and it carries $(m - 1)(n - 1)$ edges, the multigraph edge value of Theorem 1.3.
- ★ *The thickened complete graph.* Give every edge of K_n the same multiplicity q . Here $\pi(u, v) = n - 2$, so feasibility asks $q \leq m + 1 - n$, and the count is $\binom{n}{2}(m + 1 - n)$, positive only when $n \leq m + 1$.

★ *The thickened theta.* Choose two *poles* and join each of the $n - 2$ remaining vertices to both of them at multiplicity $m - 2$, then join the poles to each other at multiplicity $m + 1 - n$ when that is positive. A pole and a middle vertex have $\pi = 1$, two middle vertices have $\pi = 2$, and the two poles have $\pi = n - 2$, so this is feasible exactly when $n \leq m + 1$, and it carries $2(n - 2)(m - 2) + \max(0, m + 1 - n)$ edges.

The theta family is the one that makes this problem genuinely different from the edge problem, and it was found by the search rather than by hand. Whenever it is feasible, which is to say for $n \leq m + 1$, it beats the thickened tree for every $m \geq 4$, and it is the extremiser in every such case the exhaustion has settled. Comparing the three counts, the tree leads once $n \geq m + 2$, where the other two die for lack of route budget.

Exhaustive search over all multigraphs with multiplicities in $\{0, \dots, m - 1\}$, run under this convention, gives the values in [Table A.1](#). The routine is `max_multigraph_vertex_standard(n, m)`, which is the same include-or-exclude branch and bound as everywhere else in [Chapter 2](#), with the checker's vertex mode switched to the second convention by one flag, `parallel_routes`, on `_split_capacity_matrix`. Every value below was computed twice, once by that routine and once by a separately written script sharing no code with it, and on the sixteen cells both have finished they agree exactly.

		$n = 2$	3	4	5	6	7
$m = 2$	value	1	2	3	4	5	6
	tree	1	2	3	4	5	6
$m = 3$	value	2	4	6	8	10	12
	tree	2	4	6	8	10	12
$m = 4$	value		6	9	12	15	
	tree		6	9	12	15	
	theta		6	9	12		
$m = 5$	value			14	19		
	tree			12	16		
	theta			14	19		
$m = 6$	value			19			
	tree			15			
	theta			19			

Table A.1. The multigraph vertex problem under the convention that parallel copies are distinct routes, computed exhaustively by `max_multigraph_vertex_standard`. Entries in bold are the cases where the value exceeds the multigraph edge value $(m - 1)(n - 1)$, and there the thickened theta is the extremiser. A blank cell is a size not computed, or one where the construction in that row does not apply. Cells with $n \leq m + 1$ are the regime in which the theta and complete constructions are feasible at all.

Two things stand out. First, the problem is *not* the edge problem in disguise: at $m = 5, n = 5$ the value is 19 against the edge value 16, so the separation axis genuinely buys something for multigraphs under this reading. Second, the small- n regime $n \leq m + 1$, where the complete

graph is feasible for the simple problems too, is tempting to read as the *only* place a cycle pays for itself, with the thickened tree optimal once $n \geq m + 2$. It is not: a single triangle grafted anywhere onto the thickened tree already beats it, for every $m \geq 5$ and every such n , which the next theorem makes precise.

By [Lemma A.61](#), a feasible multigraph on top of a fixed connected simple graph G_0 maximises its total multiplicity by setting $\mu(u, v) = m - 1 - \pi(u, v)$ on every edge, giving total $(m - 1)|E(G_0)| - \sum_{uv \in E(G_0)} \pi(u, v)$. Writing $|E(G_0)| = n - 1 + \text{rank}(G_0)$, the thickened tree is beaten by any G_0 with

$$\sum_{uv \in E(G_0)} \pi(u, v) < (m - 1) \text{rank}(G_0).$$

The obvious guess is that cycles must always pay for themselves against this bound, each independent cycle costing $m - 1$ units of route capacity. The next theorem shows a single triangle already fails to pay, for $m \geq 5$, and packing many of them compounds the failure into a gap that grows with n .

Theorem A.63 (A bouquet of thickened cliques, and how it beats the thickened tree). *Fix $3 \leq r \leq m$ and set $q = m + 1 - r \geq 1$, the multiplicity of a thickened K_r that saturates the vertex ceiling on r vertices (the “thickened complete graph” construction above, restricted to a block of size r rather than all of n). Let $1 \leq k \leq \lfloor (n - 1)/(r - 1) \rfloor$. Take k copies of K_r that all share one common vertex and are otherwise disjoint, thicken every edge of every copy to multiplicity q , and attach any leftover vertices as a pendant path at multiplicity $m - 1$. This multigraph is feasible, $\kappa^{\max} \leq m - 1$, and its total multiplicity is exactly*

$$(m - 1)(n - 1) + k \cdot \text{gain}(r, m), \quad \text{gain}(r, m) := \frac{(r - 1)(r - 2)(m - 1 - r)}{2}.$$

The count therefore exceeds the thickened tree’s $(m - 1)(n - 1)$ exactly when $\text{gain}(r, m) > 0$, that is when $r \leq m - 2$, which is possible for some r in range precisely when $m \geq 5$.

Proof. Write G_0 for the underlying simple graph.

Feasibility. The shared vertex is a cut vertex of G_0 , and so is every vertex along the pendant path, since the blocks meet only there. So a path of length ≥ 2 between two vertices of the same block cannot leave that block and come back: to do so it would have to pass through the shared vertex twice. Hence for an edge uv inside a block, every route counted by $\pi(u, v)$ stays inside its K_r , where deleting uv leaves u, v joined by exactly $r - 2$ pairwise internally-disjoint two-step routes, one through each of the other block vertices, and by Menger no more, since those same $r - 2$ vertices are a cut of that size. So $\pi(u, v) = r - 2$ and, by [Lemma A.61](#), $\kappa_G(u, v) = q + (r - 2) = m - 1$ exactly, never over. For u, v in different blocks, or with either endpoint on the pendant path, a single vertex separates them, the shared vertex in the first case and a path vertex in the second, so $\kappa_G(u, v) \leq 1$.

Count. Since $q \geq 1$, every block edge is genuinely present, so G_0 is connected and $|E(G_0)| = n - 1 + \text{rank}(G_0)$. This is where the hypothesis $r \leq m$ earns its keep: at $r = m + 1$ the multiplicity q would drop to zero, the blocks would carry no edges at all, and G_0 would fall apart into isolated vertices. Only the k blocks contribute to $\text{rank}(G_0)$ or to $\sum \pi$, since the pendant edges have $\pi = 0$ and lie in no cycle. Each K_r has rank $(r - 1)(r - 2)/2$ and contributes $\binom{r}{2}(r - 2)$ to the sum, since each of its $\binom{r}{2}$ edges has $\pi = r - 2$. Substituting into $(m - 1)(n - 1 + \text{rank}(G_0)) - \sum \pi$,

$$(m - 1) \left[n - 1 + k \frac{(r-1)(r-2)}{2} \right] - k \frac{r(r-1)(r-2)}{2} = (m - 1)(n - 1) + k \text{gain}(r, m),$$

$$\text{using } (m - 1) \frac{(r-1)(r-2)}{2} - \frac{r(r-1)(r-2)}{2} = \frac{(r-1)(r-2)[(m-1)-r]}{2} = \text{gain}(r, m). \quad \square$$

At $r = 3$, $\text{gain}(3, m) = m - 4$: zero at $m = 4$, matching the exhaustive finding that $m \leq 4$ has no counterexample, and strictly positive for every $m \geq 5$. So the guess that the thickened tree is optimal once $n \geq m + 2$ is false for every $m \geq 5$: it held only within the reach of the exhaustive table, $m \leq 4$. The smallest witness is small enough to check by hand. Take $m = 5$ and $n = 7$, the first size the guess covers. Put a triangle on $\{u_1, u_2, u_3\}$ at multiplicity 3 and hang a path of four further vertices off u_1 at multiplicity 4. Two triangle vertices have three parallel copies plus one route through the third vertex, so $\kappa = 4$, and every other pair is separated by a single cut vertex or joined only by its own parallel copies, so nothing exceeds $4 = m - 1$. The total multiplicity is $3 \cdot 3 + 4 \cdot 4 = 25$, one more than the $(m - 1)(n - 1) = 24$ the guess allows. Packing k blocks instead of one multiplies the excess by k , and k may be taken as large as $\lfloor (n - 1)/(r - 1) \rfloor$, so the gap over the thickened tree grows linearly in n .

Sharing a vertex is what buys that count. Disjoint blocks joined by bridges would fit only $\lfloor n/r \rfloor$ of them, since each bridge spends a vertex on nothing but the join, and the difference is real rather than cosmetic: at $m = 10$, $r = 6$ and $n = 11$ the shared-vertex form carries 150 against the bridged form's 120.

Optimising r widens the gap further. Treated as continuous, the gain per vertex $\text{gain}(r, m)/(r - 1)$ peaks near $r \sim (m + 1)/2$, where it is $\Theta(m^2)$, so the construction gives $K_m^{\text{multi}}(n) \geq ((m - 1) + \Theta(m^2))(n - 1)$ at fixed large m , a different order in m from the guess above.

Even this is not the truth, and by more than the first evidence suggested. At $m = 5$, $n = 7$ the construction above gives 27, and an exhaustive branch and bound, stopped by its time limit before it could finish, had already found a feasible multigraph with 28. The true value there is 29 (Table A.2), so neither the construction nor that partial search was near it. What is not open is the order, and to say that the bouquet is a lower bound of the right order there has to be a matching upper bound. There is one, and it comes from reading the problem through its own underlying simple graph rather than through cycle rank. After it, Section A.9.1 returns to the value itself and says what the right shape of the extremiser is, which is neither a tree nor a bouquet of cliques.

Proposition A.64 (An upper bound of the same order). *For all $m \geq 2$ and $n \geq 2$,*

$$K_m^{\text{multi}}(n) \leq (m-1)k_m(n) < 2(m-1)^2 n.$$

Proof. Let G be feasible, with underlying simple graph G_0 .

Every multiplicity is at most $m-1$. By [Lemma A.61](#), $\kappa_G(u, v) = \mu(u, v) + \pi(u, v) \geq \mu(u, v)$, so feasibility gives $\mu(u, v) \leq m-1$ on every pair and the total is at most $(m-1)|E(G_0)|$.

G_0 is itself feasible for the simple vertex problem. Take two vertices $u \neq v$. If they are adjacent in G_0 then $\mu(u, v) \geq 1$, and the routes of G_0 between them are the single edge together with the internally disjoint routes of length at least two, so $\kappa_{G_0}(u, v) = 1 + \pi(u, v) \leq \mu(u, v) + \pi(u, v) \leq m-1$. If they are not adjacent then $\kappa_{G_0}(u, v) = \pi(u, v) \leq m-1$ directly. So $\kappa_{G_0}^{\max} \leq m-1$ and $|E(G_0)| \leq k_m(n)$ by definition of k_m , which gives the first inequality.

The simple problem is linearly bounded. Suppose G_0 had average degree at least $4(m-1)$. By Mader's density theorem [\[21\]](#), a graph of average degree at least $4k$ contains a $(k+1)$ -connected subgraph, so with $k = m-1$ some subgraph $H \subseteq G_0$ is m -connected. An m -connected graph has more than m vertices and, by the global form of Menger's theorem, any two of them are joined by m internally disjoint paths inside H and hence inside G_0 . That contradicts $\kappa_{G_0}^{\max} \leq m-1$. So the average degree is below $4(m-1)$, that is $2|E(G_0)|/n < 4(m-1)$, giving $k_m(n) < 2(m-1)n$ and the second inequality. $\square$

Combining the two directions, at fixed m and $n \rightarrow \infty$,

$$\left(\frac{1}{8} + o_m(1)\right)m^2 n \leq K_m^{\text{multi}}(n) < 2m^2 n,$$

where the $o_m(1)$ vanishes as m grows, so $K_m^{\text{multi}}(n) = \Theta(m^2 n)$: the bouquet of [Theorem A.63](#) is a lower bound of the right order in n and in m both, and what separates the two sides is a constant factor tending to 16. [Theorem A.67](#) below improves the lower constant and brings that factor down to $6 + 4\sqrt{2} \approx 11.66$. The upper bound is deliberately crude, since it throws away the $-\sum \pi$ correction entirely and charges every edge the full $m-1$, and the first inequality $K_m^{\text{multi}}(n) \leq (m-1)k_m(n)$ is the more informative half: it says this variant can never outrun the simple vertex problem by more than the multiplicity ceiling, which ties the least understood of the twelve to the one with the longest history. It is tight at $m = 2$, where both sides read $n-1$.

A.9.1 The value is a block problem

The lower bounds above are constructions and the upper bound is a density count, and neither says what an extremiser looks like. This section says exactly what it looks like, in the sense that it reduces the whole question to one about 2-connected graphs, which is a genuine reduction and not a restatement: the number of vertices in play stops being n and becomes the size of a single block.

Two observations do it. The first is already in the section, one line further than it was taken.

Lemma A.65 (The objective in closed form). *For a simple graph G_0 on n vertices with $\kappa_{G_0}^{\max} \leq m - 1$, the largest total multiplicity of a feasible multigraph with underlying simple graph exactly G_0 is*

$$W_m(G_0) := \sum_{uv \in E(G_0)} (m - \kappa_{G_0}(u, v)),$$

and every simple graph arising as the underlying graph of a feasible multigraph satisfies $\kappa_{G_0}^{\max} \leq m - 1$. Hence

$$K_m^{\text{multi}}(n) = \max \{ W_m(G_0) : G_0 \text{ simple on } n \text{ vertices, } \kappa_{G_0}^{\max} \leq m - 1 \}.$$

Proof. By [Lemma A.61](#) feasibility reads $\mu(u, v) \leq m - 1 - \pi(u, v)$ on every edge, so the best choice is $\mu(u, v) = m - 1 - \pi(u, v)$, which is what the paragraph before [Theorem A.63](#) records. On an edge $\kappa_{G_0}(u, v) = 1 + \pi(u, v)$, since the routes of G_0 between adjacent u and v are the single edge together with the routes of length at least two. Substituting turns $m - 1 - \pi(u, v)$ into $m - \kappa_{G_0}(u, v)$ and the total into $W_m(G_0)$.

Two things have to be checked for that maximum to be attained by a multigraph over G_0 itself. Feasibility of G_0 is the middle step of [Proposition A.64](#). And the chosen multiplicities are all at least one, so the underlying graph really is G_0 and no edge silently disappears: $\kappa_{G_0}(u, v) \leq m - 1$ on an edge gives $\pi(u, v) \leq m - 2$, hence $\mu(u, v) = m - 1 - \pi(u, v) \geq 1$. Conversely any feasible multigraph is one of the multigraphs considered over its own underlying graph, so nothing is lost by maximising over G_0 first. $\square$

That the objective is a sum over edges of a *local* quantity is what makes the second observation bite. Recall that a *block* of a graph is a maximal connected subgraph with no cut vertex of its own, so every block is either 2-connected or a single edge, every edge lies in exactly one block, and two blocks meet in at most one vertex.

Theorem A.66 (The multigraph vertex problem is a knapsack over blocks). *For a 2-connected graph B , or a single edge, write*

$$g_m(b) := \max \{ W_m(B) : B \text{ 2-connected or an edge, } |V(B)| = b, \kappa_B^{\max} \leq m - 1 \},$$

with $g_m(b) := -\infty$ when no such B exists. Then for all $m \geq 2$ and $n \geq 2$,

$$K_m^{\text{multi}}(n) = \max \left\{ \sum_i g_m(b_i) : b_i \geq 2, \sum_i (b_i - 1) \leq n - 1 \right\},$$

the maximum over all finite multisets of block sizes. In particular the value is determined by the finitely many numbers $g_m(2), \dots, g_m(n)$.

Proof. The objective is additive over blocks. Let G_0 be feasible and let u, v be adjacent, lying in the block B . Every u - v path of G_0 stays inside B : a path leaving B must leave through a cut vertex, and to return it would have to pass through that same cut vertex a second time, which no

path does. So $\kappa_{G_0}(u, v) = \kappa_B(u, v)$, and since every edge lies in exactly one block,

$$W_m(G_0) = \sum_{\text{blocks } B} W_m(B).$$

Each block inherits feasibility, $\kappa_B^{\max} \leq \kappa_{G_0}^{\max} \leq m - 1$ by the same identity, so $W_m(B) \leq g_m(|V(B)|)$.

The sizes fit. If G_0 has c components and blocks $B_1, \dots, B_t$, then $\sum_i (|V(B_i)| - 1) = n - c - (\text{number of isolated vertices contributed}) \leq n - 1$: within one component, contracting the block tree gives $\sum_i (|V(B_i)| - 1)$ equal to the number of vertices of that component minus one, and summing over components loses one further vertex per extra component. With [Lemma A.65](#) this proves $\leq$.

Every multiset is realised. Given blocks $B_1, \dots, B_t$ with $\sum (b_i - 1) \leq n - 1$, take one common vertex and attach the B_i to it, pairwise disjoint otherwise, leaving any spare vertices isolated. The result uses $1 + \sum (b_i - 1) \leq n$ vertices. Its blocks are exactly the B_i , so by the first paragraph its κ agrees with κ_{B_i} on pairs inside a block and equals at most 1 on pairs split between two blocks, which the shared vertex separates. So the graph is feasible and scores $\sum_i W_m(B_i)$, and choosing each B_i optimal gives $\sum_i g_m(b_i)$. This proves $\geq$. $\square$

The reduction changes what has to be searched. Instead of all feasible multigraphs on n vertices, one enumerates 2-connected simple graphs on $b \leq n$ vertices, which `geng` does directly, evaluates W_m on each, and solves a small knapsack. Every value in [Table A.2](#) was produced that way, over all 2-connected graphs on at most eight vertices, and every cell the exhaustive routine had previously proved agrees with it.

	$n = 2$	3	4	5	6	7	8
$m = 2$	1	2	3	4	5	6	7
$m = 3$	2	4	6	8	10	12	14
$m = 4$	3	6	9	12	15	18	21
$m = 5$	4	9	14	19	24	29	34
$m = 6$	5	12	19	26	33	40	47
$m = 7$	6	15	24	33	42	52	62
$m = 8$	7	18	30	42	54	66	79

Table A.2. $K_m^{\text{multi}}(n)$, exact, from [Theorem A.66](#) together with an exhaustive sweep of the 2-connected graphs on at most eight vertices (1, 1, 3, 10, 56, 468 and 7123 of them). The knapsack shows a clean split. For $m \leq 3$ the optimum is a thickened tree, every block a single edge. At $m = 4$ the two agree, every split of the vertices scoring the same. From $m = 5$ on, a single 2-connected block strictly beats every split, so the extremiser is 2-connected and the bouquet is a construction rather than the answer. The bold cell is the one [Theorem A.63](#) reaches only 27 on and an unfinished search had found 28 on. As a check on the reduction itself, the exhaustive routine behind [Table A.1](#), which shares no reasoning with the block argument, was run against these values: it proves 23 of them within its time limit and agrees on every one, and the cells where it runs out of time return lower bounds that are consistent, the bold one among them.

What those blocks are is the useful part, and they are not cliques. At $m = 5$, $n = 6$ the

winning block is exactly $K_{2,4}$ and at $m = 6, n = 7$ exactly $K_{2,5}$. At $m = 7$ and $m = 8$ with $n = 7$ it is $K_{3,4}$ with two extra edges inside the smaller side, and the 29-witness at $m = 5, n = 7$ is $K_{2,4}$ with one further edge subdivided onto it. So the shape is bipartite, unbalanced, and sometimes decorated with a few extra edges, rather than complete. The clean member of that family already beats the bouquet of cliques outright.

Theorem A.67 (Thickened complete bipartite blocks). *Let $1 \leq s \leq t \leq m - 1$. Thicken every edge of $K_{s,t}$ to multiplicity $m - s$. The result is feasible, $\kappa^{\max} \leq m - 1$, on $s + t$ vertices with total multiplicity $st(m - s)$. Hanging k such blocks off one shared vertex gives, for $n = k(s + t - 1) + 1$,*

$$K_m^{\text{multi}}(n) \geq \frac{st(m - s)}{s + t - 1} (n - 1).$$

Optimised over s and t this rate is $(3 - 2\sqrt{2} + o_m(1))m^2$, attained at $t = m - 1$ and $s \sim (\sqrt{2} - 1)m$, against the $(\frac{1}{8} + o_m(1))m^2$ of [Theorem A.63](#). The improvement is by a factor $8(3 - 2\sqrt{2}) = 24 - 16\sqrt{2} \approx 1.373$.

Proof. Write S and T for the two sides, $|S| = s$ and $|T| = t$.

The connectivities of $K_{s,t}$. For $u \neq u'$ in S , the u - u' paths of length at least two are the t two-step routes through the vertices of T , and T is a cut of size t , so $\kappa(u, u') = \pi(u, u') = t$. Symmetrically $\kappa(v, v') = s$ for $v \neq v'$ in T . For adjacent $u \in S$ and $v \in T$, delete the edge uv : the sets $T \setminus \{v\}$ and $S \setminus \{u\}$ both separate u from v , so $\pi(u, v) \leq \min(s - 1, t - 1) = s - 1$, and the $s - 1$ routes u, v_i, u_i, v through distinct $u_i \in S \setminus \{u\}$ and distinct $v_i \in T \setminus \{v\}$ attain it, which needs $s - 1 \leq t - 1$. So $\pi(u, v) = s - 1$ and $\kappa(u, v) = s$.

Feasibility and count. By [Lemma A.61](#) the multigraph has $\kappa = (m - s) + (s - 1) = m - 1$ on an edge, and $\kappa = t \leq m - 1$ or $s \leq m - 1$ on a non-adjacent pair, so nothing exceeds the ceiling. The count is st edges at multiplicity $m - s$. Equivalently, by [Lemma A.65](#), $W_m(K_{s,t}) = st(m - s)$.

The bouquet. $K_{s,t}$ is 2-connected for $s \geq 2$ and is a single edge for $s = t = 1$, so [Theorem A.66](#) applies and k copies at one shared vertex are feasible with total $kst(m - s)$ on $n = k(s + t - 1) + 1$ vertices, which is the displayed rate.

The optimum. Fix $t = m - 1$, its largest permitted value, and put $s = \alpha m$. The rate becomes

$$\frac{s(m - 1)(m - s)}{s + m - 2} = \frac{\alpha(1 - \alpha)}{1 + \alpha} m^2 + O(m),$$

and $h(\alpha) = \alpha(1 - \alpha)/(1 + \alpha)$ has $h'(\alpha) = (1 - 2\alpha - \alpha^2)/(1 + \alpha)^2$, vanishing at $\alpha = \sqrt{2} - 1$, where $h = 3 - 2\sqrt{2}$. Since $3 - 2\sqrt{2} > \frac{1}{8}$, the family beats the clique bouquet, by the stated factor. $\square$

Two comments. First, the trade-off the optimum balances is visible in the formula: a bigger s buys more edges, st of them, and costs route capacity on each, $m - s$ rather than m , and the clique is the special case where the two sides are forced equal to one another and to the whole

block. Letting them differ is the entire gain, and the optimal shape is lopsided, one side of size about $0.41m$ against the other at the ceiling $m - 1$. Second, the bipartite family is not the answer either, since [Table A.2](#) already shows extra edges inside the small side helping at $m = 8$. What is now known is that the answer is a block problem, that its blocks are dense bipartite-like graphs rather than cliques, and that the constant lies between $3 - 2\sqrt{2}$ and 2.

A.10 The random threshold

The probabilistic ingredients are two Chernoff bounds, quoted in the form we use them.

Primer: what a Chernoff bound says

Both proofs below need to say that a sum of many independent yes-or-no trials almost never lands far from its average. That is the content of a *Chernoff bound*, and the intuition is worth stating even though we only quote the result. Let $X = X_1 + \dots + X_N$ count the successes among independent trials, with mean $\mu = \mathbb{E}[X]$. A Chernoff bound says the chance that X overshoots μ by a factor $1 + \delta$, or undershoots it by a factor $1 - \delta$, decays *exponentially* in μ , not merely polynomially as Chebyshev's inequality would give. The mechanism is a single trick. Rather than bound $\Pr[X \geq a]$ directly, bound $\Pr[e^{tX} \geq e^{ta}]$ by Markov's inequality for a parameter $t > 0$, use independence to factor $\mathbb{E}[e^{tX}] = \prod_i \mathbb{E}[e^{tX_i}]$ into a product over the trials, and then choose t to make the resulting bound as small as possible. The exponential in the answer is exactly the e^{tX} that was fed through Markov. We quote the two tails in the shape the threshold proof consumes.

Theorem A.68 (Chernoff bounds, see [22, Cor. 4.4, Thm. 4.4]). *Let X be a sum of independent $\{0, 1\}$ -valued random variables with mean μ . For $\delta > 0$, $\Pr[X \geq (1 + \delta)\mu] \leq \exp(-\frac{\min(\delta, \delta^2)\mu}{4})$, and for $\delta \in (0, 1)$, $\Pr[X \leq (1 - \delta)\mu] \leq \exp(-\frac{\delta^2\mu}{2})$.*

Corollary A.69. *Let $X \sim \text{Binomial}(n - 1, p)$ with $p = cm/n$ for a constant $c \in (0, 1)$. Then for all sufficiently large m , $\Pr[X \geq m] \leq e^{-\alpha m}$ with $\alpha = \alpha(c) > 0$.*

Proof. The mean is $\mu = (n - 1)p \leq cm$, so $m \geq (1 + \delta)\mu$ with $\delta \geq \frac{1-c}{c} > 0$, and [Theorem A.68](#) applies with $\mu \geq cm/2$ for $n \geq 2$, giving $\alpha = \min(\frac{1-c}{c}, (\frac{1-c}{c})^2) \frac{c}{8}$. $\square$

Proof of Theorem 1.16. Recall the hypotheses: $m = m(n) \geq 2$ with $m/\ln n \rightarrow \infty$ and $m = o(n)$, and $p = cm/n$ for a constant $c > 0$.

Case $c > 1$. The edge count $|E|$ of $G(n, p)$ is binomial with mean $\mu = \binom{n}{2}p = \frac{cm(n-1)}{2}$. With $\delta = 1 - \frac{1}{c}$ the lower-tail bound of [Theorem A.68](#) gives

$$\Pr\left[|E| \leq \frac{m(n-1)}{2}\right] \leq \exp\left(-\frac{(c-1)^2 m(n-1)}{4c}\right) \rightarrow 0,$$

so with high probability $|E| > \left\lfloor \frac{m(n-1)}{2} \right\rfloor$, and the contrapositive of Mader's theorem ([Theorem 1.2](#)) forces a pair with $\lambda \geq m$. Hence $\Pr[\lambda^{\max} \geq m] \rightarrow 1$.

Case $c < 1$. Each degree is $\text{Binomial}(n-1, p)$, so by [Corollary A.69](#) and a union bound over the n vertices,

$$\Pr[\Delta(G) \geq m] \leq n e^{-\alpha m} = e^{\ln n - \alpha m} \rightarrow 0,$$

since $m/\ln n \rightarrow \infty$. By the degree bound ([Lemma A.3](#)) $\lambda^{\max} \leq \Delta(G)$, so $\Pr[\lambda^{\max} \geq m] \rightarrow 0$. $\square$

Remark A.70 (The vertex and directed analogues). Below the threshold, $\kappa^{\max} \leq \lambda^{\max}$ gives the vertex statement for free, and in the random digraph $D(n, p)$, where each ordered pair's arc is tossed independently with probability p , the out-degree version of the same union bound applies. Above the threshold, the minimum degree exceeds m with high probability by the lower-tail bound. Mader's theorem alone does not carry this to the vertex problem, since $\kappa^{\max} \leq \lambda^{\max}$ runs the wrong way to deduce $\kappa^{\max} \geq m$ from the edge bound. What supplies it instead are the theorems of Bollobás [[23](#), Ch. 7] for $G(n, p)$, in hitting-time form due to Bollobás and Thomason [[24](#)], and its directed analogue [[25](#)], which state that in this regime the global connectivities equal the minimum degree (respectively the minimum semi-degree, the smaller of a vertex's in- and out-degree) with high probability, so $\kappa^{\max}$ and the directed $\lambda^{\max}$ also cross m at $p^* = m/n$. The growth hypothesis $m/\ln n \rightarrow \infty$ is what lets the union bound beat the factor n . For fixed m , the bound is vacuous and finer tools (Poisson approximation of low-degree vertex counts) would be needed.

Remark A.71 (How far the threshold reaches, and where it stops). [Theorem 1.16](#) is a statement about $G(n, p)$, and [Remark A.70](#) carries it to the vertex separation and to the directed model $D(n, p)$. It does not reach the hypergraph models, and the reason is a change of scale rather than a gap in the argument. The threshold sits where a vertex's expected degree reaches m , since below that the degree bound caps every local connectivity and above it the extremal bound forces a high pair. In $G(n, p)$ a vertex has expected degree $p(n-1)$, so the balance point is $p^* = m/n$ as stated. In the binomial r -uniform hypergraph, where each of the $\binom{n}{r}$ possible hyperedges is included independently with probability p , a vertex lies in an expected $p\binom{n-1}{r-1}$ hyperedges, so the balance point is

$$p^* = \frac{m}{\binom{n-1}{r-1}} = \Theta\left(\frac{m}{n^{r-1}}\right),$$

which for $r \geq 3$ is smaller than m/n by a factor of order n^{r-2} . Reading m/n as the hypergraph threshold would therefore place the transition far above where it happens: at $r = 3$ and $n = 8$, for instance, m/n is $3/8$ while $m/\binom{7}{2}$ is $3/21$, and by the former density the sampled hypergraph is long past the transition. [Figure B.8](#) accordingly sweeps each model in units of its own p^* rather than a shared p . A second limit of scope is worth repeating here: the theorem assumes $m/\ln n \rightarrow \infty$, so any picture at a fixed small m illustrates the shape of the transition without being an instance of the theorem.

A.11 Computational transcripts

The cut-counting optimisation formulation that certifies [Theorem 2.2](#) is given in [Chapter 2](#). Here are the solver transcripts that back it, together with the exhaustive-search output that settles the directed base cases. The complete commented source accompanies the document as a single self-contained file, `erdos915_unified.py` in the `program/` directory. Its core runs on `numpy` and `scipy` alone, the certifier adds `pulp`, and `matplotlib` and `networkx` are optional and used only for the figures. An optional compiled helper accelerates selected small hot paths but is not needed for correctness.

```

1 Cut-MILP certification of the directed multigraph problem
2  $M^*(n)$  is the scaled optimum;  $L_m^{\text{dir}}(n) = (m-1) M^*(n)$ .
3
4 n=3: status=OPTIMAL, M*=4.000, target=2(n-1)=4, proof=True, time=0.0s,  $L_3^{\text{dir}}(n)=8$ 
5 n=4: status=OPTIMAL, M*=6.000, target=2(n-1)=6, proof=True, time=0.3s,  $L_3^{\text{dir}}(n)=12$ 
6 n=5: status=OPTIMAL, M*=8.000, target=2(n-1)=8, proof=True, time=5.9s,  $L_3^{\text{dir}}(n)=16$ 
7 n=6: status=OPTIMAL, M*=10.000, target=2(n-1)=10, proof=True, time=1259.9s,  $L_3^{\text{dir}}(n)=20$ 

```

Listing A.1. Solver output: $M^*(n) = 2(n-1)$, optimal with zero gap, for $n = 3, 4, 5, 6$.

The cut-counting optimisation closes the gap up to $n = 5$ inside a short budget and reaches $n = 6$ in about twenty minutes. Past that it stalls, because a solver handles the yes-or-no cut labels by case-splitting, trying a label both ways and solving each half, and the number of cases it must open grows too fast at $n = 7$. The base cases of the directed induction ([Theorem 1.10](#)) run to $n = 7$, and those are settled instead by the pruned exhaustive search over all simple digraphs with $\lambda^{\max} \leq 1$. That search is exact and complete at these sizes, and it agrees with $\ell_2^{\text{dir}}(n) = M(n)$ throughout.

```

1 Exhaustive search over all simple digraphs with  $\lambda^{\max} \leq 1$  (directed,
  ↳  $m=2$ ).
2 Branch and bound; reported max arc count =  $\ell_2^{\text{dir}}(n) = \max(2(n-1), \text{floor}(n^2/4))$ .
3
4 n=2: max= 2, formula= 2, proved (complete), nodes=5, 0.0s
5 n=3: max= 4, formula= 4, proved (complete), nodes=22, 0.0s
6 n=4: max= 6, formula= 6, proved (complete), nodes=537, 0.0s
7 n=5: max= 8, formula= 8, proved (complete), nodes=17823, 0.1s
8 n=6: max=10, formula=10, proved (complete), nodes=788101, 9.1s
9 n=7: max=12, formula=12, proved (complete), nodes=45122129, 718.1s
10
11 The cut-MILP certifier of the multigraph problem closes  $n \leq 6$  ( $n = 6$  in
  ↳ about
12 21 minutes); the directed  $m = 2$  base cases through  $n = 7$  that the induction

```

13 requires are settled by this exhaustive search instead.

Listing A.2. Exhaustive-search output for the base cases $2 \leq n \leq 7$: the maximum arc count equals $M(n) = 2, 4, 6, 8, 10, 12$, proved complete at each n .

The vertex problem needs its own base cases, since [Remark A.16](#) shows they cannot be inherited from the arc ones, and the same driver supplies them with only the inner feasibility test swapped. The two separations agree at every size, which is what closes [Theorem 1.11](#).

```
1 Exhaustive search over all simple digraphs with kappa^max <= 1 (directed, m
  ↳ =2).
2 Branch and bound; reported max arc count = k_2^dir(n) = max(2(n-1), floor(n
  ↳ ^2/4)).
3
4 n=2: max= 2, formula= 2, proved (complete), nodes=5, 0.0s
5 n=3: max= 4, formula= 4, proved (complete), nodes=22, 0.0s
6 n=4: max= 6, formula= 6, proved (complete), nodes=537, 0.0s
7 n=5: max= 8, formula= 8, proved (complete), nodes=17823, 1.1s
8 n=6: max=10, formula=10, proved (complete), nodes=788101, 77.0s
9 n=7: max=12, formula=12, proved (complete), nodes=45122129, 3124.8s
10
11 Same driver, same prunes, same sizes as the arc run above. Only the
12 inner test changes, from arc-disjoint to internally vertex-disjoint.
13 The two separations agree at every base case, which is what the
14 vertex induction needs and what Whitney's inequality cannot supply.
```

Listing A.3. The same search run with the internally vertex-disjoint feasibility test, settling the base cases of [Theorem 1.11](#): the maximum arc count is again 2, 4, 6, 8, 10, 12, proved complete at each n .

A.12 How the program checks itself

The invariants are checked by the self-test at the foot of `erdos915_unified.py`, run simply with `python erdos915_unified.py`. Every check passes, grouped here by what they cover.

- ★ *Base values.* The checker against the proven $m = 2$ values 2, 4, 6, 8, the first three vertex base cases of [Theorem 1.11](#), and the fast vertex-flow test against the exact checker it stands in for.
- ★ *Sampled inequalities.* The counting step and both bounds of [Proposition A.22](#) and [Theorem A.24](#) on sampled feasible digraphs, including the high-degree case of the latter's proof, and Whitney's inequality on every sample.
- ★ *Constructions.* Two values of the alternative-convention multigraph vertex problem of [Section A.9](#), including the one that separates it from the edge problem, the named constructions against their predicted sizes and connectivities, and the Gomory–Hu shortcut against brute-force max-flow.

★ *The checker and driver.* The Berge connectivities of the hypergraph checker, the Monte Carlo appearance threshold, the cut-counting optima for small n , the search reaching the certified optimum, and the driver returning the proven value or a feasible witness in each mode.

A single companion script, `make_figures.py`, regenerates every figure in the thesis from the same file, with the mapping listed in `program/README.md`.

Appendix B

Supplementary Figures

This appendix collects the supplementary figures that the program produces but that the chapters reference rather than reproduce in full. Each one is named where it belongs in the body, and the file name printed in its caption matches the output of `make_figures.py` and the figure table in `program/README.md`. Figures already shown in the running text are not repeated here.

B.1 A gallery of extremal graphs

The extremal graphs are not abstractions. [Figure B.1](#) draws nine of the named families the body argues about, each at a deliberately large representative size rather than at the smallest case, so the structure is visible. These are the special, structured extremisers, not the trivial small trees. Four of the nine are directed hub constructions. The first is the bidirected star, the simple hub with arcs both ways to every leaf. The second is the directed double star, its multigraph analogue, with hub arcs at full multiplicity $m - 1$, which ties $2 \cdot B_{3,4}$ at the crossover. The third is the doubled bipartite wall $2 \cdot B_{3,4}$ itself, and the fourth is the dense bidirected complete graph. A fifth is the undirected counterpart of the double star, the multigraph star, with hub edges at full multiplicity $m - 1$. The remaining families are the star hypertrees, drawn as metro maps in which each hyperedge is one coloured line through its members. The program builds every one of these directly and, at the small sizes the enumeration reaches, also proves them optimal rather than merely feasible, so the gallery is the analysis made concrete at a size a reader can take in.

B.2 Search traces across the variants

[Figure 3.2](#) in [Chapter 3](#) follows one cooling run in detail. The five runs below repeat that experiment across the family. The first four cover every combination of the two matrix models, simple and multigraph, with the two directions, and the fifth changes the separation, so that the behaviour can be seen not to be special to one case. Each plots every visited graph by its binding connectivity (horizontal) against its size (vertical), colouring each point by the step at which it was visited, from the hot early exploration in warm tones to the converged tail in blue. In every one the walk strays past the feasibility boundary while hot and is driven back onto the densest feasible graph as the temperature falls, exactly as in the body figure. The seeds are fixed, so each picture reproduces verbatim.

The last of the five is the one worth comparing against the others. It runs the same directed multigraph problem under the *vertex* separation rather than the arc one, and the stricter test is

Extremal graphs of the named families, drawn large

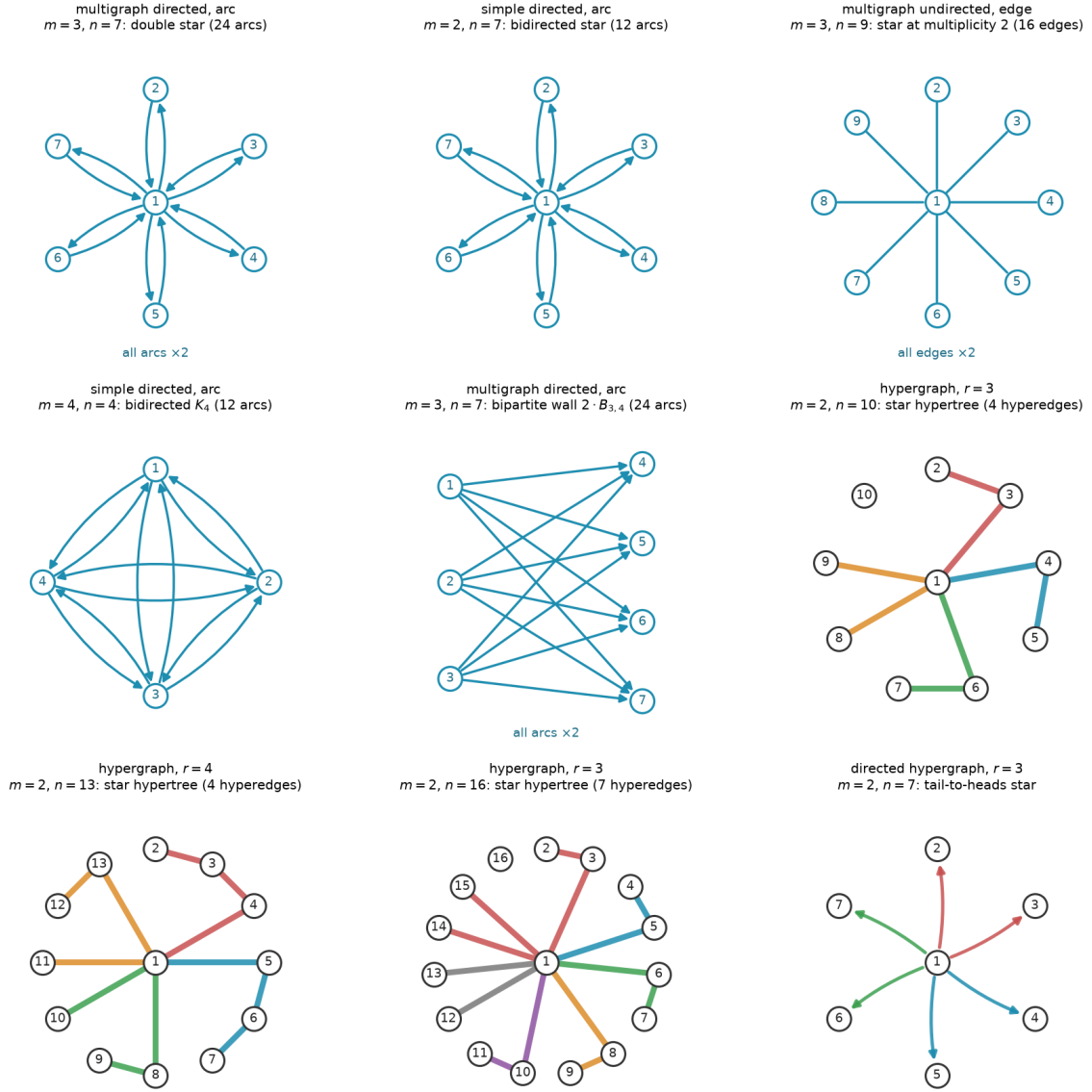


Figure B.1. Extremal graphs of the named families, drawn at a large representative size. *Top and middle rows:* matrix variants, with a central hub where the family has one. Every family here carries one uniform multiplicity, stated once per panel as “all arcs $\times k$ ” rather than repeated on each arc or drawn as parallel curves, and directed arcs carry arrowheads. *Hypergraph panels:* each hyperedge is a metro line through its member vertices, in its own colour, with the hub at the centre, and the directed hypergraph fans coloured arrows from each tail to its heads. Every graph shown is feasible at the stated m . The matrix and undirected-hypergraph families are extremal for their variant, the two directed stars drawn at the crossover $n = 7$, the largest size at which the linear star still ties the one-directional wall before the wall overtakes it. The bipartite wall $2 \cdot B_{3,4}$ is drawn at that same size to show the other side of the tie. It also carries 24 arcs, matching the double star’s count, and it is one of the three families the $n = 7$ classification of [Remark A.41](#) identifies, together with $2 \cdot B_{4,3}$ and the doubled bidirected path P_7 (the double star itself is not among them, since its hub degree exceeds that classification’s degree cap). The final panel is the directed-hypergraph analogue, a tail-to-heads star, shown for a variant whose exact value is still open even though its leading term is now settled ([Proposition A.55](#) and [theorem A.59](#)). File `extremal_graphs_gallery.png`.

visible in the trace: the walk spends a markedly larger share of its steps on the infeasible side of the boundary before the cooling pulls it back, which is what a harder constraint looks like from inside the search.

Search trace: simple undirected, $n = 7, m = 3$

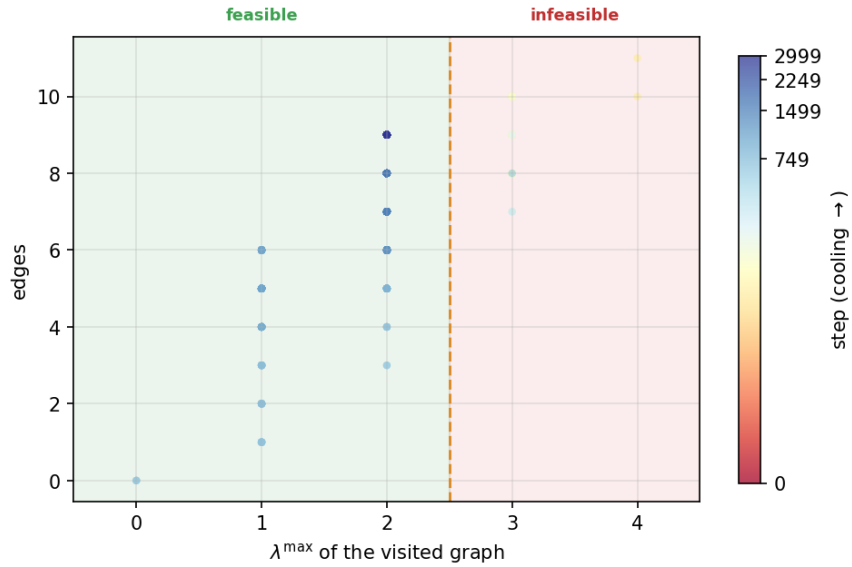


Figure B.2. Search trace for the simple undirected edge problem at $n = 7, m = 3$. File `trace_simple_undirected_n7_m3.png`.

B.3 The variant landscape at $m = 6$ and in three dimensions

This appendix shows the per-graph connectivity distribution at the fixed forbidden threshold $m = 6$, alongside a three-dimensional view of the appearance threshold (Figure B.8). They confirm that raising the forbidden threshold relocates the feasibility boundary without changing the qualitative picture. The pooled pair-connectivity and edge-count grids of Chapter 4 (Figures 4.4 and 4.5) already split each variant at its own mid-range connectivity, so they carry no separate $m = 6$ companion.

Search trace: multi directed, $n = 5, m = 3$

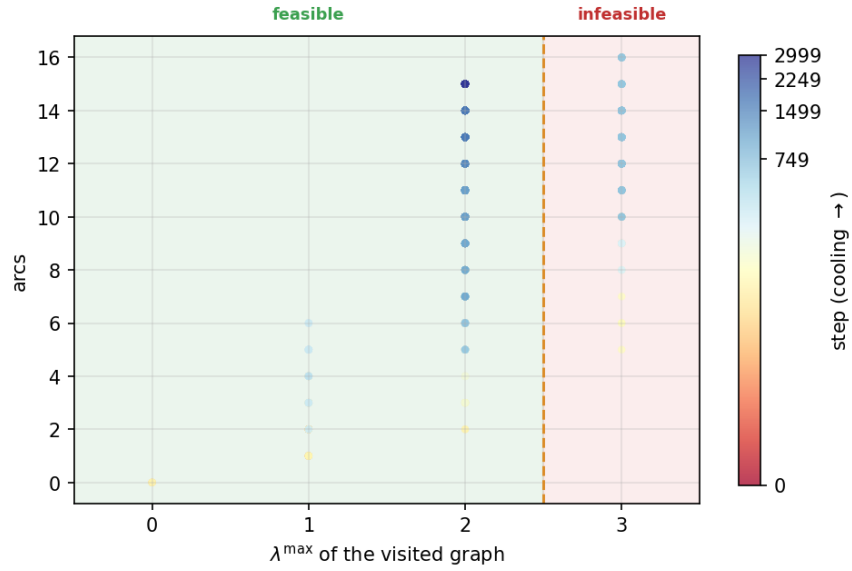


Figure B.3. Search trace for the multigraph directed arc problem at $n = 5, m = 3$. File `trace_multi_directed_n5_m3.png`.

Search trace: simple directed, $n = 5, m = 3$

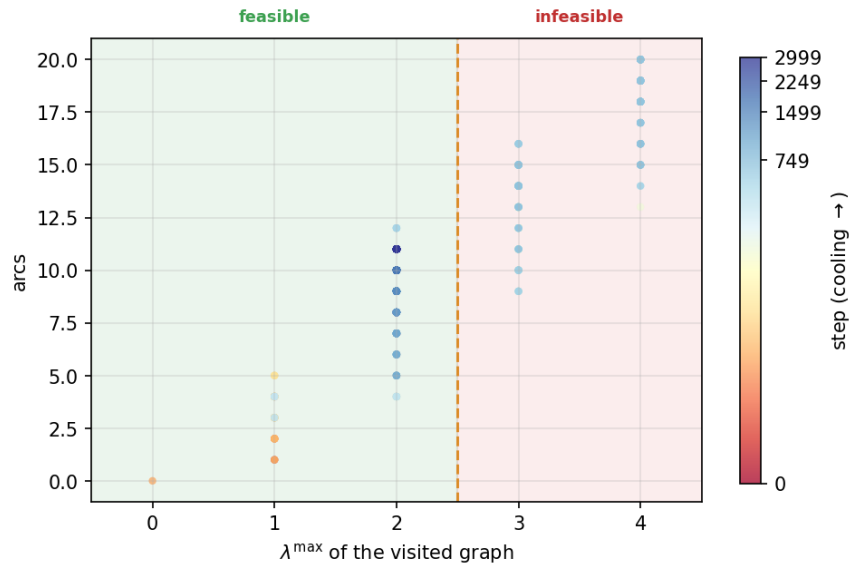


Figure B.4. Search trace for the simple directed arc problem at $n = 5, m = 3$. File `trace_simple_directed_n5_m3.png`.

Search trace: multi undirected, $n = 5, m = 3$

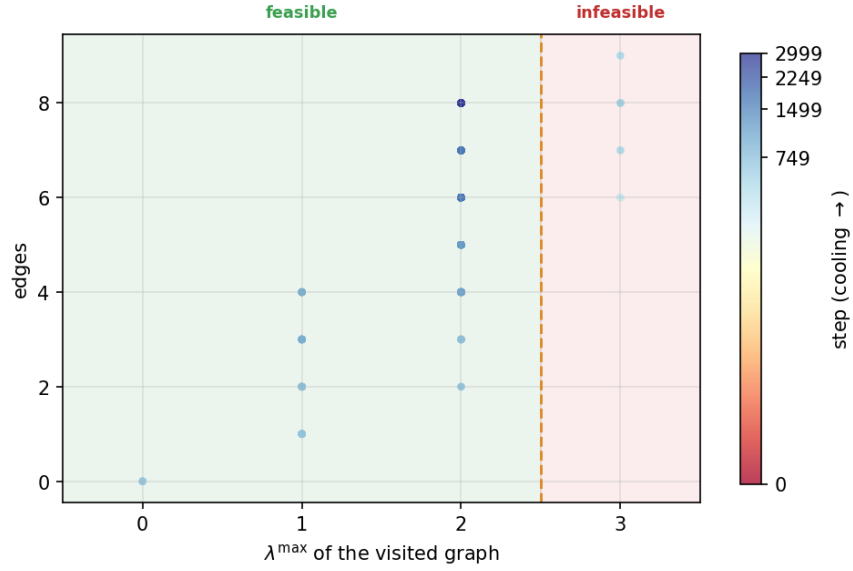


Figure B.5. Search trace for the multigraph undirected edge problem at $n = 5, m = 3$. File `trace_multi_undirected_n5_m3.png`.

Search trace: multi directed vertex-sep, $n = 4, m = 3$

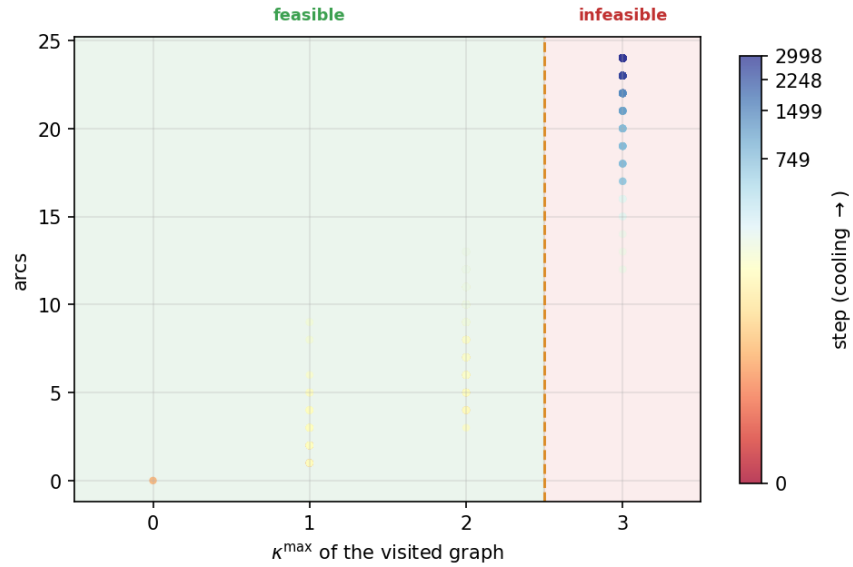


Figure B.6. Search trace for the multigraph directed problem at $n = 4, m = 3$ under the *vertex* separation, so the horizontal axis is $\kappa^{\max}$ rather than $\lambda^{\max}$. Compared with the arc runs above, the walk lingers much longer in the infeasible region before the cooling drives it back, the stricter separation seen from inside the search. File `trace_multi_directed_n4_m3_vertex.png`.

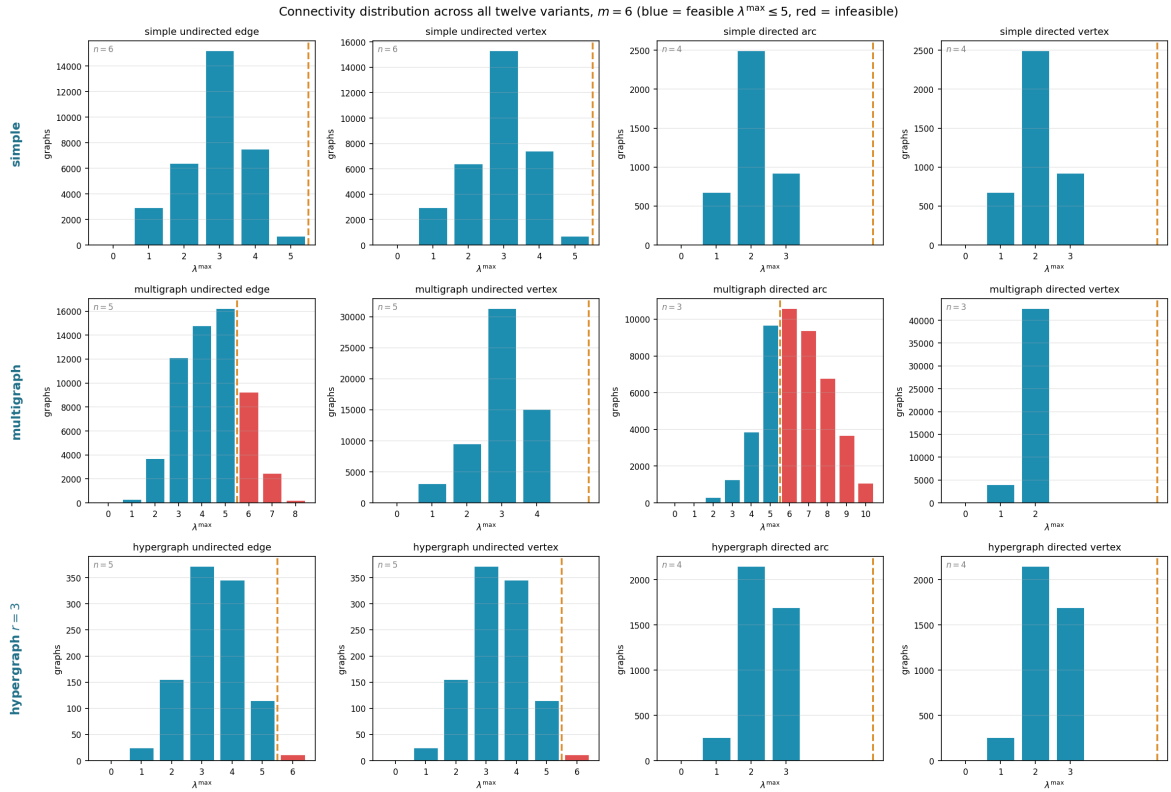


Figure B.7. How the full small- n enumeration distributes by binding connectivity $\lambda^{\max}$ across all twelve variants at the fixed forbidden threshold $m = 6$. Blue bars are feasible graphs ($\lambda^{\max} \leq 5$), red bars infeasible ones, and the dashed line marks the feasibility boundary. At this higher threshold most of the small graphs the enumeration reaches are feasible, so the blue mass dominates: raising m moves the boundary out past the bulk of the distribution. File `conn_dist_m6.png`.

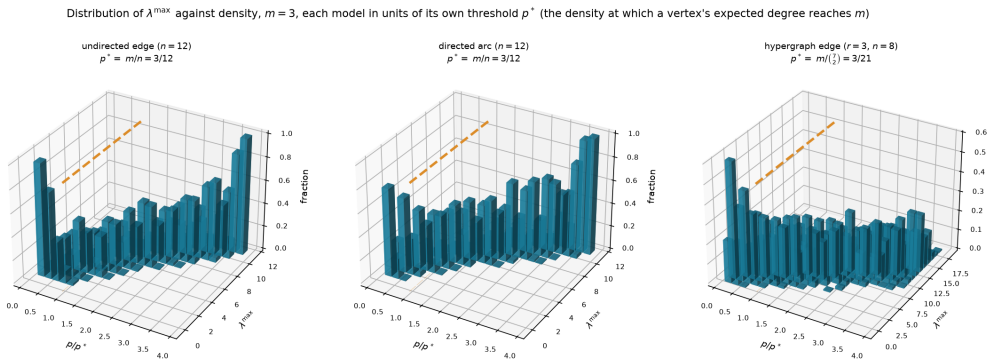


Figure B.8. The appearance threshold as a three-dimensional bar chart, for three representative variants (simple undirected edge and simple directed arc at $n = 12$, three-uniform hypergraph edge at $n = 8$), all at $m = 3$. Each panel plots the fraction of samples (height) reaching each binding connectivity $\lambda^{\max}$ (depth) against density (width), and the mass migrates from low to high connectivity as the density crosses the dashed line. The three models do *not* share a density scale, so each panel is swept in units of its *own* threshold p^* , the density at which a vertex's expected degree reaches m : that is m/n for the two graph models and $m/\binom{n-1}{r-1}$ for the hypergraph, smaller by a factor of order n^{r-2} (Remark A.71). Each panel title gives its own p^* . Only the two graph panels fall under Theorem 1.16, and even they sit outside its asymptotic regime $m/\ln n \rightarrow \infty$ at this fixed $m = 3$, so the figure illustrates the shape of the transition rather than instantiating the theorem. File `threshold_3d.png`.

Appendix C

The Complete Source Code

Every number in this thesis that was not proved by hand came out of the program printed here, and this appendix prints all of it, with nothing summarised away. The point is that a reader who doubts a computed value can find the exact lines that produced it, rather than taking a description of them on trust.

There is no need to read this front to back. It is laid out so that you can jump to whichever part you want to check. The listing is broken into the same four movements as the thesis itself, because the program is organised that way on purpose: it defines the objects, then measures them, then proves things about them, then searches for examples and draws the results. Each part below opens with a short account of what it does and which chapter it belongs to, and every listing carries line numbers so that a reference like “line 1412” has one unambiguous meaning. Where a line of code was too long for the page it is wrapped, and the continuation is marked with a $\hookrightarrow$ arrow rather than being silently cut.

Four files make up the whole of it. The main program, `erdos915_unified.py`, is the one the chapters discuss throughout and the only one that computes a reported result. `make_figures.py` redraws every picture in the thesis by calling into that program, so a figure can never drift away from the numbers behind it. `_erdos_fast.c` is a small optional accelerator: it speeds up two inner loops and changes no answer, and the program runs correctly without it. The `tests/` directory holds the suite that checks all of the above.

C.1 The main program

C.1.1 How the file opens

The program begins by saying what it is for, then imports what it needs and fixes a few constants. Two decisions made here shape everything after. The first is that the heavy scientific libraries are optional: only `numpy` and `scipy` are required, and `matplotlib`, `networkx` and `pulp` are each wrapped so that the file still loads without them, with the features that need them disabled rather than the whole program failing. The second is that every import is hoisted to the top, so there is one place to look to see what the program depends on.

```
1 """
2 erdos915_unified: the complete unified program for Erdos Problem 915, in one file.
3
4 This single module is the whole program, kept in one namespace so it can be read
```

```

5 top to bottom as a continuous story. It is meant to be read, not just run: every
6 non-trivial step carries an inline comment explaining *why* it is there, not
7 merely what it does. Running it ('python erdos915_unified.py') executes the
8 invariant self-check at the foot of the file; 'make_figures.py' next to it
9 regenerates the thesis figures.
10
11 -----
12 The problem, in one breath
13 -----
14 Erdos Problem 915 asks, in its many guises, for the densest graph on 'n'
15 vertices in which no pair of vertices has 'm' independent routes between them.
16 "Independent" can mean edge-disjoint or internally vertex-disjoint; the object
17 can be a simple graph, a multigraph, a digraph, or a hypergraph; and forbidding
18 'm' routes is exactly the constraint that the largest *local* connectivity
19 stays at or below 'm - 1'. Twelve concrete variants fall out of these
20 choices, and this program treats all of them with one representation and one
21 search.
22
23 -----
24 The epistemic spine: measure / prove / discover
25 -----
26 The whole architecture is organised around three different *kinds of claim* a
27 program can make about extremal values, and it keeps them scrupulously apart so
28 they can never silently masquerade as one another:
29
30 * MEASURE. The checker computes exact local connectivities by max-flow
31 (Menger's theorem). Whatever it reports is a fact about a concrete graph,
32 not an estimate. This is the trusted core; everything else is checked
33 against it.
34
35 * PROVE. The cut-counting method proves *upper* bounds for a fixed number of
36 vertices: it shows that no graph on 'n' vertices can be denser, with a
37 zero optimality gap, so an optimal solution is a genuine proof.
38
39 * DISCOVER. The random search finds concrete dense graphs, hence
40 *lower* bounds. A construction it returns is a witness that the extremal
41 value is at least so large; it never proves that no denser graph exists.
42
43 The closed-form 'bounds' library sits to the side as the theory's predictions
44 (proved, cited, or merely conjectured, each labelled honestly), so that "what
45 the program found" can always be held against "what the theory predicts" without
46 the two drifting apart. The Monte Carlo module sits in the DISCOVER/observe
47 corner: a sampled estimate is an estimate, used to make a proved threshold
48 *visible*, never to prove it.
49
50 -----
51 Chapter map (sections are grouped below by the thesis chapter they support)
52 -----
53 CHAPTER 1 The problem and its base cases
54     VARIANT, BOUNDS, HYPERGRAPH, CONSTRUCTIONS
55 CHAPTER 2 Proving bounds by machine
56     CHECKER, GOMORY-HU, PROVE (the brute-force and exhaustive provers
57     live inside the SOLVE driver, in Chapter 3)
58 CHAPTER 3 Discovering bounds by search
59     SENSITIVITY, SEARCH, SOLVE (the one driver: prove or discover)
60 CHAPTER 4 Synthesis and results
61     MONTE CARLO, FIGURES, __main__ (the invariant self-check suite)
62

```

```

63 A bold CHAPTER divider precedes each group, so as you scroll you always know
64 which chapter's machinery you are reading. Each section keeps its own banner.
65 """
66
67 # =====
68 # Hoisted imports
69 # -----
70 # All external dependencies, gathered into a single block.  “from __future__
71 # import annotations” must be (and is) the very first statement so that
72 # string-form type hints work uniformly. Everything in this program lives in one
73 # namespace, so there are no intra-module imports to follow.
74 # =====
75 from __future__ import annotations
76
77 import copy
78 import json
79 import math
80 import pickle
81 import random
82 import shutil
83 import statistics
84 import subprocess
85 import time
86 import warnings
87 from collections import Counter, deque
88 from collections.abc import Iterable
89 from dataclasses import dataclass, field
90 from itertools import combinations, combinations_with_replacement, permutations,
    ↪ product
91 from pathlib import Path
92 from typing import Callable, Iterator
93
94 import concurrent.futures
95
96 import numpy as np
97 from scipy.sparse import csr_matrix as _csr
98 from scipy.sparse.csgraph import maximum_flow as _csgraph_maxflow
99
100 # networkx is OPTIONAL. Every connectivity measure the thesis reports is computed
101 # through scipy's integer max-flow on a capacity matrix, so the core checker,
102 # search, provers, and enumeration run on numpy + scipy alone. networkx is needed
103 # only for the Gomory-Hu tree (one figure/analysis helper), which raises a clear
104 # message if called without it installed.
105 try:
106     import networkx as nx
107     NETWORKX_AVAILABLE = True
108 except ImportError:
109     nx = None
110     NETWORKX_AVAILABLE = False
111
112
113 def _require_networkx(feature: str):
114     """Raise a clear error when an optional networkx-only feature is used headless
    ↪ """
115     if not NETWORKX_AVAILABLE:
116         raise ImportError(
117             f"{feature} needs networkx (pip install networkx). The core
    ↪ connectivity "

```



```

118         f"measures do not: they run on numpy + scipy alone."
119     )
120
121     # pulp is OPTIONAL. It backs the MILP certifier (prove_directed_multigraph and
122     # prove_integral_arc_bound): one solver-agnostic cut-counting model that CBC (pulp
123     ↪ 's
124     # bundled solver) runs by default and Gurobi runs by a one-line switch. The model
125     ↪ ,
126     # checker, search, enumeration, and sampling do not need it.
127     try:
128         import pulp
129         # pulp 3.x emits migration notices for its 4.0 API (LpVariable construction,
130         # the PULP_CBC_CMD name). We use the stable 3.x API on purpose and pin it in
131         # requirements.txt, so silence that forward-looking noise.
132         import warnings as _warnings
133         _warnings.filterwarnings("ignore", message=r".*PuLP 4\.0.*",
134                                 category=DeprecationWarning)
135         PULP_AVAILABLE = True
136     except ImportError:
137         pulp = None
138         PULP_AVAILABLE = False
139
140     def _require_pulp():
141         """Raise a clear error when the MILP certifier is used without pulp installed.
142         ↪ """
143         if not PULP_AVAILABLE:
144             raise ImportError(
145                 "the MILP certifier needs pulp (pip install pulp). The checker, search
146                 ↪ , "
147                 "and enumeration do not: they run on numpy + scipy alone."
148             )
149
150     import ctypes as _ct
151     _C = None
152     _so_path = Path(__file__).with_name("_erdos_fast.so")
153     if _so_path.exists():
154         _C = _ct.CDLL(str(_so_path))
155         _C.tiny_maxflow.restype = _ct.c_int
156         _C.tiny_maxflow.argtypes = [
157             _ct.POINTER(_ct.c_int), _ct.c_int, _ct.c_int, _ct.c_int, _ct.c_int,
158         ]
159         _C.max_connectivity_exceeds.restype = _ct.c_int
160         _C.max_connectivity_exceeds.argtypes = [
161             _ct.POINTER(_ct.c_int), _ct.c_int, _ct.c_int, _ct.c_int,
162         ]
163         _C.canonical_form_min.restype = None
164         _C.canonical_form_min.argtypes = [
165             _ct.POINTER(_ct.c_int), _ct.c_int, _ct.POINTER(_ct.c_int),
166         ]
167     C_EXTENSION_LOADED = _C is not None
168
169     # matplotlib is OPTIONAL: it is needed only to render the figures (the plot_*
170     # functions in the FIGURES section). It needs its backend chosen before pyplot is
171     # imported, so that the figure code runs head-less (writes PNG files, never opens
172     ↪ a
173     # window). When it is absent the model, checker, provers, search, and enumeration
174     # all still run; only figure generation is unavailable.

```

```

171 try:
172     import matplotlib
173     matplotlib.use("Agg")
174     import matplotlib.pyplot as plt # noqa: E402 (after backend selection)
175     from matplotlib.ticker import MaxNLocator # noqa: E402 (integer-only axis
        ↳ ticks)
176     from matplotlib.colors import PowerNorm # noqa: E402
177     from matplotlib.gridspec import GridSpec # noqa: E402
178     from matplotlib.patches import FancyArrowPatch, Patch # noqa: E402
179     from matplotlib.transforms import blended_transform_factory # noqa: E402
180     from mpl_toolkits.mplot3d import Axes3D # noqa: F401 (registers the '3d'
        ↳ projection)
181     MATPLOTLIB_AVAILABLE = True
182 except ImportError:
183     plt = None
184     MaxNLocator = None
185     PowerNorm = None
186     GridSpec = None
187     FancyArrowPatch = None
188     Patch = None
189     blended_transform_factory = None
190     MATPLOTLIB_AVAILABLE = False

```

C.1.2 The objects: graphs, hypergraphs, and the values we already know

This part corresponds to [Chapter 1](#). It defines what a graph is for the purposes of this thesis, which is a single square table of numbers: entry (u, v) records how many parallel connections run from u to v . That one table covers all eight of the matrix variants at once, because a Variant record says whether the graph is directed and whether it is simple, and those two switches are the only difference between them. Hypergraphs need a different shape, so they are stored as a list of the sets of vertices each hyperedge joins.

After the objects come the closed-form values the thesis proves or conjectures, each written as a small function whose docstring records whether it is proved, conjectured, or a bound only. Then come the named constructions, the specific graphs that achieve those values, so that a claimed bound can always be met with a concrete example rather than an existence argument.

```

193 #####
194 ##
195 ## CHAPTER 1 -- THE PROBLEM AND ITS BASE CASES
196 ## Objects, variants, and the closed-form values we already know.
197 ##
198 #####
199
200 # --- VARIANT: mu[u,v] = multiplicity from u to v; simple = {0,1}; undirected =
    ↳ symmetric ---
201
202
203 @dataclass(frozen=True)
204 class Variant:
205     """Which kind of graph object we are working with.
206
207     Attributes:
208         directed: 'True' for digraphs (arcs), 'False' for graphs (edges).

```

```

209         simple: ‘‘True‘‘ caps every multiplicity at one. ‘‘False‘‘ allows
210             arbitrarily many parallel edges or arcs (a multigraph).
211         name: a short human-readable label, used in figures and reports.
212     """
213
214     directed: bool
215     simple: bool
216     #loops: bool
217     name: str = ""
218
219     def describe(self) -> str:
220         """Return a readable description such as ‘‘simple directed“‘.
221         # Compose the label from the two boolean axes; used in plot titles.
222         words = ["simple" if self.simple else "multi",
223                 "directed" if self.directed else "undirected"]
224         return " ".join(words)
225
226
227     # The four matrix-representable variants, named for convenience. Random graphs
228     # reuse these (a sampled simple undirected graph is SIMPLE_UNDIRECTED).
229     SIMPLE_UNDIRECTED = Variant(directed=False, simple=True, name="simple undirected")
230     MULTI_UNDIRECTED = Variant(directed=False, simple=False, name="multi undirected")
231     SIMPLE_DIRECTED = Variant(directed=True, simple=True, name="simple directed")
232     MULTI_DIRECTED = Variant(directed=True, simple=False, name="multi directed")
233
234
235     class Graph:
236         """A graph or digraph stored as an integer multiplicity matrix.
237
238         The single source of truth is ‘‘self.mu‘‘, an ‘‘n by n‘‘ array of
239         non-negative integers with a zero diagonal (no self-loops). For an
240         undirected variant the matrix is kept symmetric at all times, so the
241         invariant ‘‘mu[u, v] == mu[v, u]‘‘ may be relied upon everywhere.
242
243         All mutating methods respect the variant: they keep an undirected graph
244         symmetric and never let a simple graph exceed multiplicity one.
245         """
246
247         def __init__(self, num_vertices: int, variant: Variant):
248             if num_vertices < 0:
249                 raise ValueError("number of vertices must be non-negative")
250             self.variant = variant
251             # The whole graph is this one matrix; integer dtype keeps it exact.
252             self.mu = np.zeros((num_vertices, num_vertices), dtype=int)
253
254             # -- basic queries -----
255
256             @property
257             def num_vertices(self) -> int:
258                 """The number of vertices ‘‘n‘‘.
259                 return self.mu.shape[0]
260
261             def vertices(self) -> range:
262                 """The vertex set ‘‘{0, 1, ..., n - 1}‘‘ as a range.
263                 return range(self.num_vertices)
264
265             def multiplicity(self, u: int, v: int) -> int:
266                 """Number of parallel edges or arcs from ‘‘u‘‘ to ‘‘v‘‘.

```

```

267         return int(self.mu[u, v])
268
269     def has_edge(self, u: int, v: int) -> bool:
270         """Whether at least one edge or arc joins 'u' to 'v'."""
271         return self.mu[u, v] > 0
272
273     def edge_count(self) -> int:
274         """Total number of edges or arcs, counted with multiplicity.
275
276         For an undirected graph each edge is stored twice in the matrix, so we
277         count the strict upper triangle. For a digraph every ordered pair
278         counts once.
279         """
280         if self.variant.directed:
281             return int(self.mu.sum())
282         # Strict upper triangle (k=1) avoids double-counting the symmetric pair.
283         return int(np.triu(self.mu, k=1).sum())
284
285     def out_degree(self, v: int) -> int:
286         """Number of edges or arcs leaving 'v' (with multiplicity)."""
287         return int(self.mu[v, :].sum())
288
289     def in_degree(self, v: int) -> int:
290         """Number of edges or arcs entering 'v' (with multiplicity)."""
291         return int(self.mu[:, v].sum())
292
293     def degree(self, v: int) -> int:
294         """Total degree of 'v'."""
295
296         For a digraph this is the sum of in- and out-degree. For an undirected
297         graph in- and out-degree coincide, so we report the out-degree.
298         """
299         if self.variant.directed:
300             return self.out_degree(v) + self.in_degree(v)
301         return self.out_degree(v)
302
303     def edges(self) -> Iterator[tuple[int, int, int]]:
304         """Yield '(u, v, multiplicity)' for every present edge or arc.
305
306         For an undirected graph each edge is yielded once, with 'u < v'.
307         """
308         n = self.num_vertices
309         for u in range(n):
310             # Undirected: only scan v > u so each edge is emitted once.
311             # Directed: scan all v so both arc directions are emitted.
312             start = u + 1 if not self.variant.directed else 0
313             for v in range(start, n):
314                 if u != v and self.mu[u, v] > 0:
315                     yield u, v, int(self.mu[u, v])
316
317     # -- mutation -----
318
319     def add_edge(self, u: int, v: int, count: int = 1) -> None:
320         """Add 'count' parallel edges or arcs from 'u' to 'v'."""
321
322         A simple graph silently saturates at multiplicity one. An undirected
323         graph updates both matrix entries to stay symmetric.
324         """

```

```

325     self._require_distinct(u, v)
326     new_value = self.mu[u, v] + count
327     if self.variant.simple:
328         new_value = min(new_value, 1) # enforce the 0/1 simple-graph cap
329     self._assign(u, v, new_value)
330
331 def remove_edge(self, u: int, v: int, count: int = 1) -> None:
332     """Remove ``count`` parallel edges or arcs (never below zero)."""
333     self._require_distinct(u, v)
334     new_value = max(0, self.mu[u, v] - count)
335     self._assign(u, v, new_value)
336
337 def set_multiplicity(self, u: int, v: int, value: int) -> None:
338     """Set the multiplicity from ``u`` to ``v`` directly."""
339     self._require_distinct(u, v)
340     if value < 0:
341         raise ValueError("multiplicity must be non-negative")
342     if self.variant.simple and value > 1:
343         raise ValueError("a simple graph cannot have multiplicity above one")
344     self._assign(u, v, value)
345
346 def copy(self) -> "Graph":
347     """Return an independent copy with the same variant and edges."""
348     clone = Graph(self.num_vertices, self.variant)
349     clone.mu = self.mu.copy() # deep-copy the matrix so edits don't alias
350     return clone
351
352 # -- internal helpers -----
353
354 def _assign(self, u: int, v: int, value: int) -> None:
355     """Write ``value`` into the matrix, mirroring it when undirected."""
356     self.mu[u, v] = value
357     # Maintain the symmetry invariant for undirected graphs in one place.
358     if not self.variant.directed:
359         self.mu[v, u] = value
360
361 def _require_distinct(self, u: int, v: int) -> None:
362     if u == v:
363         raise ValueError("self-loops are not allowed in this model")
364
365 def __repr__(self) -> str:
366     return (f"Graph(n={self.num_vertices}, variant={self.variant.describe()!r}
367             ↪ }, "
368             f"edges={self.edge_count()}))"
369
370 # --- BOUNDS: closed-form extremal values; docstrings record proved/conjectured
371     ↪ status ---
372
373 def simple_undirected_edge(n: int, m: int) -> int:
374     """``ell_m(n) = floor(m (n-1) / 2)``. Proved (Mader, 1973)."""
375     return (m * (n - 1)) // 2
376
377
378 def multigraph_undirected_edge(n: int, m: int) -> int:
379     """``L_m(n) = (m-1)(n-1)``. Proved (multi-tree construction and bound)."""
380     return (m - 1) * (n - 1)

```

```

381
382
383 def simple_undirected_vertex_le4(n: int, m: int) -> int:
384     """'k_m(n) = floor(m (n-1) / 2)' for 'm <= 4'. Cited (Leonard, 1972).'''
385     if m > 4:
386         raise ValueError("this equality is only known for m <= 4")
387     # For m <= 4 the vertex problem coincides with the edge problem.
388     return simple_undirected_edge(n, m)
389
390
391 def simple_undirected_vertex_m5(n: int) -> int:
392     """'k_5(n) = floor(8n/3) - 4' for 'n >= 6', 'n != 7', 'n != 12'.
393
394     Cited (Sorensen-Thomassen, Theorem 4). Their paper counts the least edge
395     total that FORCES a 5-rail and prints 'floor(8n/3) - 3'; this thesis
396     counts the largest that avoids one, which is exactly one less. The two
397     sizes the closed form misses are given there separately, and below n = 6
398     the formula is not claimed at all.
399     """
400     if n < 6:
401         raise ValueError("k_5(n) is only determined for n >= 6")
402     if n == 7:
403         return 15
404     if n == 12:
405         return 27
406     return (8 * n) // 3 - 4
407
408
409 def directed_arc_m2(n: int) -> int:
410     """'ell_2^dir(n) = max(2(n-1), floor(n^2/4))'. Proved (induction on n)."""
411     # The max picks whichever of the two competing constructions wins at this n:
412     # the linear hub (double star) versus the quadratic one-directional wall.
413     return max(2 * (n - 1), (n * n) // 4)
414
415
416 def directed_arc_lower_bound(n: int, m: int) -> int:
417     """Lower bound and conjectured value for 'ell_m^dir(n)', 'm >= 2'.
418
419     'max(m(n-1), floor((n+m-2)^2/4))'. The two branches are the hub
420     construction ('const:directed-hub') and the shifted-partition augmented
421     bipartite construction ('const:augmented-bipartite'). Proved as a lower
422     bound for all 'm'; conjectured to be tight for 'm >= 3'
423     ('conj:dir-arc'), proved tight for 'm = 2'.
424     """
425     hub_branch = m * (n - 1)
426     bipartite_branch = (n + m - 2) ** 2 // 4
427     return max(hub_branch, bipartite_branch)
428
429
430 def hypergraph_edge(n: int, m: int, r: int) -> int:
431     """'floor((m-1)(n-1)/(r-1))' hyperedges for the 'r'-uniform edge problem.
432
433     Proved modulo the hypergraph Gomory-Hu theorem; the star hypertree attains
434     the 'm = 2' case and a sparse hypergraph attains the general bound.
435     """
436     return ((m - 1) * (n - 1)) // (r - 1)
437
438

```

```

439 def directed_multigraph_arc(n: int, m: int) -> int:
440     """ $L_m^{\text{dir}}(n) = (m-1) \max(2(n-1), \lfloor n^2/4 \rfloor)$  Proved, all  $n$  and  $m$ .
441
442     The double star dominates the one-directional bipartite branch up to the
443     crossover at ' $n = 7$ ', where the two tie; past it the quadratic branch
444     takes over. Proved outright by ' $\text{thm:dir-multi-full}$ ' (peel one sparse
445     reachability-preserving subgraph per unit of ' $m-1$ '), which superseded the
446     earlier route that was certified only for ' $n \leq 6$ ' by counting cuts.
447     """
448     double_star_branch = 2 * (n - 1) * (m - 1)
449     bipartite_branch = (m - 1) * ((n * n) // 4)
450     return max(double_star_branch, bipartite_branch)
451
452
453 # --- HYPERGRAPH: stored as incidence list; Berge-path connectivity via helper-
454     ↪ node max-flow ---
455
456 class Hypergraph:
457     """A hypergraph stored by its incidence (a list of hyperedges).
458
459     An undirected hyperedge is a set of vertices, crossed in either direction. A
460     directed hyperedge (directed=True) splits an ' $r$ '-set into a non-empty
461     tail set ' $T$ ' and a non-empty head set ' $H$ ': a Berge route may enter it only
462     at a tail and leave only at a head. Two storage forms are accepted, and the
463     checker treats them identically:
464
465     * the legacy forward form ' $(\text{tail}, \text{frozenset}(\text{heads}))$ ' with a single integer
466       tail and ' $r - 1$ ' heads ( $|T| = 1$ ), the one-tail / many-head model the
467       rest of the thesis uses; and
468     * the general form ' $(\text{frozenset}(\text{tails}), \text{frozenset}(\text{heads}))$ ' for any split,
469       which subsumes forward ( $|T| = 1$ ), backward ( $|H| = 1$ ) and, from
470       ' $r \geq 4$ ', genuinely mixed ( $|T|, |H| \geq 2$ ) hyperarcs.
471
472     For ' $r = 2$ ' every form collapses to a single arc, so a directed graph is the
473     two-vertex special case.
474     """
475
476     def __init__(self, num_vertices: int, hyperedges: Iterable = (),
477                 *, directed: bool = False):
478         self.num_vertices = num_vertices
479         self.directed = directed
480         # Each entry is a frozenset, or a ' $(\text{tail}, \text{frozenset}(\text{heads}))$ ' pair.
481         self.hyperedges: list = []
482         for edge in hyperedges:
483             self.add_hyperedge(edge)
484
485     def add_hyperedge(self, edge) -> None:
486         """Add an undirected vertex set, or a directed ' $(\text{tail}, \text{heads})$ ' pair.
487
488         A directed edge may be the forward form ' $(\text{tail: int}, \text{heads})$ ' or the
489         general form ' $(\text{tails: frozenset}, \text{heads: frozenset})$ '; the general form is
490         kept verbatim, the forward form is kept verbatim too (no normalisation),
491         so existing forward results are unchanged byte-for-byte.
492         """
493         if self.directed:
494             first, raw_heads = edge
495             if isinstance(first, (set, frozenset)): # general (tails, heads)

```

```

496         tails, heads = frozenset(first), frozenset(raw_heads)
497         members = tails | heads
498         if (tails & heads) or not tails or not heads or len(members) < 2:
499             raise ValueError("a directed hyperedge needs disjoint, non-
↳ empty tail and head sets")
500         if any(v < 0 or v >= self.num_vertices for v in members):
501             raise ValueError("hyperedge refers to a vertex outside the
↳ graph")
502         self.hyperedges.append((tails, heads))
503         return
504     tail, heads = first, frozenset(raw_heads)    # legacy forward (tail,
↳ heads)
505     members = heads | {tail}
506     if tail in heads or len(members) < 2:
507         raise ValueError("a directed hyperedge needs a tail and a distinct
↳ head")
508     if any(v < 0 or v >= self.num_vertices for v in members):
509         raise ValueError("hyperedge refers to a vertex outside the graph")
510     self.hyperedges.append((tail, heads))
511 else:
512     vertices = frozenset(edge)    # a hyperedge is a set of distinct
↳ vertices
513     if len(vertices) < 2:
514         raise ValueError("a hyperedge must contain at least two vertices")
515     if any(v < 0 or v >= self.num_vertices for v in vertices):
516         raise ValueError("hyperedge refers to a vertex outside the graph")
517     self.hyperedges.append(vertices)
518
519 def members(self, edge) -> frozenset:
520     """The vertex set of a stored hyperedge, directed or not."""
521     if self.directed:
522         tails, heads = _dir_tails_heads(edge)
523         return tails | heads
524     return edge
525
526 def vertices(self) -> range:
527     return range(self.num_vertices)
528
529 def edge_count(self) -> int:
530     """Number of hyperedges."""
531     return len(self.hyperedges)
532
533 def incident_hyperedges(self, v: int) -> list[int]:
534     """Indices of the hyperedges containing vertex 'v'."""
535     return [i for i, edge in enumerate(self.hyperedges) if v in self.members(
↳ edge)]
536
537
538 def _dir_tails_heads(edge) -> tuple[frozenset, frozenset]:
539     """Tail and head sets of a stored directed hyperedge, as frozensets.
540
541     Accepts the legacy forward form '(tail:int, heads)' and the general form
542     '(tails:frozenset, heads:frozenset)', so every consumer of a directed
543     hyperedge can read one '(tails, heads)' shape regardless of how it was
544     built.
545     """
546     first, heads = edge
547     if isinstance(first, (set, frozenset)):

```



```

548         return frozenset(first), frozenset(heads)
549     return frozenset((first,)), frozenset(heads)
550
551
552 def _hyper_capacity_matrix(hypergraph: Hypergraph, *, vertex_split: bool = False):
553     """Integer capacity matrix of the hypergraph flow network (one per variant).
554
555     Each hyperedge becomes an in/out gate joined by a capacity-1 arc, so at most
556     one disjoint Berge route may traverse it: this is the hyperedge-as-node trick.
557     With 'vertex_split' each ORIGINAL vertex is also split into an in/out pair
558     of capacity one, so the flow counts internally vertex-disjoint Berge routes
559     instead of edge-disjoint ones. For a directed hypergraph the gate is entered
560     only from a tail and left only toward a head (one tail for the forward model,
561     several for the general one); for an undirected one every member links to the
562     gate both ways. The hypergraph variants are thus the same construction with a
563     boolean or two flipped, not separate measures.
564
565     Vertices index '0..base-1' ('base = 2n' split, else 'n'); hyperedge gate
566     'i' indexes 'base+2i' (in) and 'base+2i+1' (out). 'leave(v)'/enter(
        ↪ v)'
567
568     return the index a route leaves/enters vertex 'v' through, so the same scipy
569     max-flow that serves plain graphs serves hypergraphs too.
570
571     """
572     n = hypergraph.num_vertices
573     base = 2 * n if vertex_split else n
574     size = base + 2 * len(hypergraph.hyperedges)
575     cap = np.zeros((size, size), dtype=int)
576     leave = (lambda v: 2 * v + 1) if vertex_split else (lambda v: v)
577     enter = (lambda v: 2 * v) if vertex_split else (lambda v: v)
578     if vertex_split:
579         for v in range(n):
580             cap[2 * v, 2 * v + 1] = 1 # one route through each vertex
581     for index, edge in enumerate(hypergraph.hyperedges):
582         gate_in, gate_out = base + 2 * index, base + 2 * index + 1
583         cap[gate_in, gate_out] = 1 # one route through each
        ↪ hyperedge
584     if hypergraph.directed:
585         tails, heads = _dir_tails_heads(edge)
586         for tail in tails: # enter the gate from any tail
587             cap[leave(tail), gate_in] = _UNBOUNDED
588         for head in heads: # leave the gate toward any head
589             cap[gate_out, enter(head)] = _UNBOUNDED
590     else:
591         for vertex in edge:
592             cap[leave(vertex), gate_in] = _UNBOUNDED
593             cap[gate_out, enter(vertex)] = _UNBOUNDED
594     return cap, size, leave, enter
595
596
597 def hyper_connectivity(hypergraph: Hypergraph, source: int, target: int,
598     *, vertex_split: bool = False) -> int:
599     """Edge- or vertex-disjoint Berge routes from 'source' to 'target'.
600
601     A single scipy integer max-flow on the matrix above. An isolated endpoint
602     (in no hyperedge) has no incident arcs, so the flow is then zero.
603
604     """
605     cap, _, leave, enter = _hyper_capacity_matrix(hypergraph, vertex_split=
        ↪ vertex_split)

```

```

603     if vertex_split:
604         # Endpoints must not be the bottleneck: only interior vertices are capped.
605         cap[2 * source, 2 * source + 1] = _UNBOUNDED
606         cap[2 * target, 2 * target + 1] = _UNBOUNDED
607     return int(_csgraph_maxflow(_csr(cap, dtype=int), leave(source), enter(target)
        ↪ ).flow_value)
608
609
610 def max_hyper_connectivity(hypergraph: Hypergraph, *, vertex_split: bool = False)
        ↪ -> int:
611     """ $\lambda^{\max}$  (edge) or  $\kappa^{\max}$  (vertex) over all pairs."""
612     return max((hyper_connectivity(hypergraph, source, target, vertex_split=
        ↪ vertex_split)
        for source, target in _pairs(hypergraph)), default=0)
613
614
615
616 def hyperedge_connectivity(hypergraph: Hypergraph, source: int, target: int) ->
        ↪ int:
617     """Maximum number of edge-disjoint Berge  $s$ - $t$  paths."""
618     return hyper_connectivity(hypergraph, source, target, vertex_split=False)
619
620
621 def max_hyperedge_connectivity(hypergraph: Hypergraph) -> int:
622     """ $\lambda^{\max}$  for the undirected edge hypergraph problem."""
623     return max_hyper_connectivity(hypergraph, vertex_split=False)
624
625
626 # --- CONSTRUCTIONS: named extremal graphs; each states its edge count and  $\lambda^{\max}$ 
        ↪ max ---
627
628
629 def double_star(n: int, m: int, directed: bool = True) -> Graph:
630     """The double star: one hub joined to every leaf with multiplicity  $m-1$ .
631
632     For a digraph the hub carries  $m-1$  arcs in both directions to each
633     leaf, attaining  $2(n-1)(m-1)$  arcs with  $\lambda^{\max} = m-1$ . For an
634     undirected multigraph it attains  $(m-1)(n-1)$  edges. This is the
635     small- $n$  extremiser of the multigraph problems.
636     """
637     variant = MULTI_DIRECTED if directed else MULTI_UNDIRECTED
638     graph = Graph(n, variant)
639     hub = 0 # vertex 0 is the central hub; everything else is a leaf
640     for leaf in range(1, n):
641         graph.set_multiplicity(hub, leaf, m - 1)
642         if directed:
643             graph.set_multiplicity(leaf, hub, m - 1) # arcs in both directions
644     return graph
645
646
647 def one_directional_bipartite(n: int) -> Graph:
648     """All arcs from one part to the other:  $\lfloor n^2/4 \rfloor$  arcs,  $\lambda^{\max} =
        ↪ 1$ .
649
650     The vertices split into a smaller part  $A$  and a larger part  $B$  and
651     every arc runs  $A \rightarrow B$ . No two vertices have two arc-disjoint routes, so
652     this single-direction wall of arcs is feasible for  $m = 2$  while holding
653     quadratically many arcs: the phenomenon that has no undirected analogue.
654     """

```

```

655 graph = Graph(n, SIMPLE_DIRECTED)
656 size_a = n // 2 # smaller part; the  $n^{2/4}$  maximum is at the balanced split
657 part_a = range(size_a)
658 part_b = range(size_a, n)
659 for a in part_a:
660     for b in part_b:
661         graph.add_edge(a, b) # every arc runs A -> B only (one direction)
662 return graph
663
664
665 def augmented_bipartite(n: int, m: int) -> Graph:
666     """The conjectured directed extremiser for ' $m \geq 2$ '.|B| = \text{ceil}((n+m-2)/2)' and ' $|A| = n - |B|$ '. Fill every arc
669     ' $A \rightarrow B$ ' (the one-directional wall), then give each ' $B$ -vertex
670     ' $m - 2$ ' extra in-arcs from its circulant predecessors inside ' $B$ '.
671     Because no arc leaves ' $B$ ' toward ' $A$ ', a route between two ' $B$ '
672     vertices never escapes ' $B$ ', and a route from ' $A$ ' to ' $b$ ' uses the
673     direct arc plus one short detour through each of ' $b$ 's ' $m - 2$ '
674     in-arcs, giving ' $\lambda^{\max} = m - 1$ '. The arc count is
675     ' $\text{floor}((n+m-2)^{2/4})$ ' ('const:augmented-bipartite' in the thesis).
676
677     Requires ' $n \geq m$ ' so that ' $|A| \geq 1$ ' and the circulant offsets stay
678     distinct. At ' $m = 2$ ' or ' $m = 3$ ' the partition equals the balanced
679     ' $(\text{floor}(n/2), \text{ceil}(n/2))$ ' split and the formula reduces to
680     ' $\text{floor}(n^{2/4}) + (m-2)\text{ceil}(n/2)$ '. At ' $m \geq 4$ ' the shifted partition
681     gives strictly more arcs. For ' $m = 3, n = 10$ ' this is the 30-arc
682     counterexample.
683     """
684     size_b = (n + m - 2 + 1) // 2 #  $\text{ceil}((n+m-2)/2)$ 
685     size_a = n - size_b
686     if size_a < 1:
687         raise ValueError("augmented_bipartite needs  $n \geq m$ ")
688
689     graph = Graph(n, SIMPLE_DIRECTED)
690     part_a = range(size_a)
691     part_b = list(range(size_a, n))
692
693     # Complete wall of arcs from the small part to the large part.
694     for a in part_a:
695         for b in part_b:
696             graph.add_edge(a, b)
697
698     # Thicken B by a circulant giving every B-vertex in-degree exactly  $m - 2$ .
699     # Each offset adds a full cycle of arcs within B;  $m - 2$  offsets give each
700     # B-vertex  $m - 2$  extra in-arcs from other B-vertices (the short detours).
701     for offset in range(1, m - 1):
702         for i in range(size_b):
703             source = part_b[i]
704             target = part_b[(i + offset) % size_b]
705             graph.add_edge(source, target)
706     return graph
707
708
709 def clique_tree(clique_size: int, blocks_added: int) -> Graph:
710     """A tree of cliques (a ' $K$ '-tree): the chordal scaffold of the vertex case.
711
712     Begin with a complete graph ' $K_r$ ' on ' $r = \text{clique\_size}$ ' vertices, then

```

```

713 repeatedly attach a new vertex to the previous ‘r - 1’ vertices, which
714 always form a clique. The result is a maximal chordal graph of treewidth
715 ‘r - 1’ and *global* vertex connectivity ‘r - 1’.
716
717 A caution that the thesis makes explicit: the controlled quantity here is the
718 global connectivity, not ‘kappa^max’. Attaching a vertex to a triangle
719 raises the local connectivity of that triangle’s vertices above ‘r - 1’, so
720 a ‘k’-tree is not itself feasible for the ‘kappa^max’ problem. It appears
721 in the thesis to illustrate the chordal structure underlying the
722 Sorensen-Thomassen analysis of the ‘m = 5’ divergence, not as a feasible
723 extremiser.
724 """
725 if clique_size < 2:
726     raise ValueError("clique_size must be at least two")
727 n = clique_size + blocks_added
728 graph = Graph(n, SIMPLE_UNDIRECTED)
729
730 # The seed clique on the first r vertices.
731 for u in range(clique_size):
732     for v in range(u + 1, clique_size):
733         graph.add_edge(u, v)
734
735 # Each new vertex joins the r - 1 most recently added vertices.
736 # Those r - 1 always form a clique, which keeps the graph chordal (a k-tree).
737 for new_vertex in range(clique_size, n):
738     attach_to = range(new_vertex - (clique_size - 1), new_vertex)
739     for old_vertex in attach_to:
740         graph.add_edge(new_vertex, old_vertex)
741 return graph
742
743
744 def complete_uniform_hypergraph(n: int, r: int) -> Hypergraph:
745     """The complete ‘r’-uniform hypergraph: every ‘r’-subset is a hyperedge.
746
747     It is dense and highly connected; for example  $K_4^{\{3\}}$  has Berge
748     edge-connectivity three between every pair, exceeding the two-hyperedge count
749     of the pairs because longer Berge routes also contribute.
750     """
751     # One hyperedge per r-subset of the vertex set.
752     return Hypergraph(n, (set(subset) for subset in combinations(range(n), r)))
753
754
755 def star_hypertree(n: int, r: int) -> Hypergraph:
756     """A hub joined to disjoint blocks: the feasible extremiser at ‘m = 2’.
757
758     Vertex ‘0’ is the hub, and the remaining ‘n - 1’ vertices are split into
759     blocks of size ‘r - 1’; each block together with the hub forms one
760     hyperedge. No two vertices share more than one hyperedge and the only routes
761     between blocks pass through the single hub, so the Berge edge-connectivity is
762     one everywhere, and the construction holds  $\lfloor (n-1)/(r-1) \rfloor$ 
763     hyperedges, the ‘m = 2’ case of the hypergraph edge bound.
764     """
765     if r < 2:
766         raise ValueError("hyperedge size r must be at least two")
767     hub = 0
768     others = list(range(1, n))
769     hypergraph = Hypergraph(n)
770     # Walk the non-hub vertices in blocks of r - 1; each block plus the hub is

```

```

771     # one hyperedge (a "star" of r-sets all sharing the single hub).
772     for start in range(0, len(others) - (r - 2), r - 1):
773         block = others[start:start + (r - 1)]
774         if len(block) == r - 1: # drop a trailing partial block
775             hypergraph.add_hyperedge([hub, *block])
776     return hypergraph

```

C.1.3 Measuring, and proving

This part corresponds to [Chapter 2](#). Measuring comes first. The question “how many independent routes join these two vertices” is answered exactly, by turning it into a maximum-flow problem and solving that, which is Menger’s theorem used as an algorithm. The same routine answers all four versions of the question, because whether routes must avoid shared edges or shared intermediate vertices is controlled by one flag that decides how the flow network is built, and hypergraphs are handled by giving each hyperedge a gate of its own.

Proving comes second and is a different kind of work. To show that no graph of a given size can beat a bound, it is not enough to fail to find one. The program instead writes the whole question as a single optimisation problem whose variables describe an arbitrary graph and whose constraints say that every pair of vertices admits a small cut. If that problem has no solution, no such graph exists, and that is a proof rather than an absence of evidence.

```

779 #####
780 ##
781 ## CHAPTER 2 -- PROVING BOUNDS BY MACHINE
782 ## Measure connectivity exactly, then prove small-case upper bounds.
783 ##
784 #####
785
786 # --- CHECKER: exact max-flow connectivity; vertex_split=True gives kappa^max ---
787
788 # A capacity large enough to never be the bottleneck of any cut we build.
789 _UNBOUNDED = 10 ** 9
790
791
792 # -----
793 # One flow network and one measure, parameterised by edge vs vertex
794 # -----
795 # Every measure here is a single scipy integer max-flow on a capacity matrix
796 # (Menger). EDGE mode runs directly on the multiplicity matrix mu: each unit of
797 # capacity is one parallel edge, so a max-flow of value f is f edge-disjoint
798 # routes. VERTEX mode runs on the 2n x 2n split matrix (see
799     ↪ _split_capacity_matrix)
800 # where each vertex becomes a capacity-one in/out gate, so a route uses a vertex
801     ↪ at
802 # most once. Edge vs vertex is one boolean, not a second code path.
803
804 def local_connectivity(graph: Graph, source: int, target: int,
805     *, vertex_split: bool = False) -> int:
806     """Disjoint ‘source’-‘target’ routes by a single max-flow (Menger).
807
808     Edge-disjoint with ‘vertex_split=False’; internally vertex-disjoint with
809     ‘vertex_split=True’. An isolated endpoint yields zero (no augmenting path).

```

```

808     """
809     if not vertex_split:
810         # scipy.sparse.csgraph.maximum_flow is a C implementation; faster than
811         # NetworkX for the small integer capacity matrices we use.
812         return int(_csgraph_maxflow(_csr(graph.mu, dtype=int), source, target).
813             ↪ flow_value)
814
815     # Vertex mode: the same scipy max-flow on the split matrix. Uncapping the two
816     # endpoints' own in->out gates makes Menger count internally vertex-disjoint
817     # routes (only interior vertices stay capacity-capped at one).
818     cap, _ = _split_capacity_matrix(graph)
819     cap[2 * source, 2 * source + 1] = _UNBOUNDED
820     cap[2 * target, 2 * target + 1] = _UNBOUNDED
821     return int(_csgraph_maxflow(_csr(cap, dtype=int), 2 * source + 1, 2 * target).
822         ↪ flow_value)
823
824 def max_connectivity(graph: Graph, *, vertex_split: bool = False) -> int:
825     """The largest local connectivity over all pairs: ' $\lambda^{\max}$ ' (edge) or
826     ' $\kappa^{\max}$ ' (vertex). For a digraph every ordered pair is examined, for an
827     undirected graph each unordered pair once.
828     """
829     if not vertex_split:
830         # Build the CSR matrix once and reuse it for every pair.
831         csr = _csr(graph.mu, dtype=int)
832         return max((int(_csgraph_maxflow(csr, s, t).flow_value)
833             for s, t in _pairs(graph)), default=0)
834     return max((local_connectivity(graph, s, t, vertex_split=True)
835         for s, t in _pairs(graph)), default=0)
836
837 # -----
838 # Named measures: thin views on the single path above (so the rest of the
839 # program can pass ' $\max\_edge\_connectivity$ ' / ' $\max\_vertex\_connectivity$ ' around
840 # by name -- just as ' $hyperedge\_connectivity$ ' names one branch of
841 # ' $hyper\_connectivity$ ' in the hypergraph section).
842 # -----
843
844 def local_edge_connectivity(graph: Graph, source: int, target: int) -> int:
845     """Maximum number of edge-disjoint ' $source$ '-' $target$ ' paths."""
846     return local_connectivity(graph, source, target, vertex_split=False)
847
848 def max_edge_connectivity(graph: Graph) -> int:
849     """' $\lambda^{\max}(G)$ ': the largest local edge-connectivity over all pairs."""
850     return max_connectivity(graph, vertex_split=False)
851
852
853 def local_vertex_connectivity(graph: Graph, source: int, target: int) -> int:
854     """Maximum number of internally vertex-disjoint ' $source$ '-' $target$ ' paths."
855     ↪
856     return local_connectivity(graph, source, target, vertex_split=True)
857
858 def max_vertex_connectivity(graph: Graph) -> int:
859     """' $\kappa^{\max}(G)$ ': the largest local vertex-connectivity over all pairs."""
860     return max_connectivity(graph, vertex_split=True)
861
862

```

```

863 def min_vertex_connectivity(graph: Graph) -> int:
864     """The global vertex connectivity  $\kappa(G)$ : the smallest over all pairs.
865
866     This is the classical connectivity of a graph (the least number of vertices
867     whose removal disconnects it). It is distinct from  $\kappa^{\max}$ : a  $\kappa$ -
        ↳ tree,
868     for instance, is globally  $\kappa$ -connected yet has pairs with higher local
869     connectivity.
870     """
871     pairs = list(_pairs(graph))
872     if not pairs:
873         return 0
874     # Global connectivity is the MINIMUM local connectivity over all pairs.
875     return min(local_vertex_connectivity(graph, source, target)
876                for source, target in pairs)
877
878
879 # -----
880 # Shared helpers
881 # -----
882
883 def _pairs(obj):
884     """Yield the vertex pairs to test: ordered if directed, unordered if not.
885
886     Works for both a Graph (directedness on variant) and a Hypergraph
887     (directedness on the object), so one pair sweep serves every variant.
888     """
889     directed = obj.directed if hasattr(obj, "directed") else obj.variant.directed
890     n = obj.num_vertices
891     for source in range(n):
892         # Directed objects need both (s, t) and (t, s); undirected need each
893         # unordered pair once, so we start the inner loop above the source.
894         start = 0 if directed else source + 1
895         for target in range(start, n):
896             if source != target:
897                 yield source, target
898
899
900 # -----
901 # Capped feasibility predicate (used inside the search, Section 7)
902 # -----
903 # The exact measures above answer "what is  $\lambda^{\max} / \kappa^{\max}$ ". The search
904 # only ever needs the cheaper YES/NO question "is it greater than k?", because
905 # its energy depends on the value solely through the excess  $\max(0, \text{value} - (m-1))$ .
906 # A capped flow answers exactly that and stops the moment the answer is known, so
907 # this predicate is equivalent to max_connectivity(graph, ...) > k but usually
908 # far cheaper: each pair's flow halts after  $k+1$  augmenting paths, and the pair
909 # loop halts at the first violating pair. The exact value is recomputed only on
910 # the rare step where the search genuinely needs it (an infeasible proposal).
911 # DEPENDENCIES: _pairs (above), _tiny_maxflow (Section ENUMERATE), _UNBOUNDED.
912
913 def _split_capacity_matrix(graph: Graph,
914                            parallel_routes: bool = False) -> tuple[np.ndarray, int]:
915     ↳ ]:
916     """The  $2n \times 2n$  capacity matrix of the vertex-split network.
917
918     The vertex-mode network as a plain matrix, consumed by both the exact measure
919     (:func:`local_connectivity` via scipy max-flow) and the capped search

```

```

    ↪ predicate
919 (:func:'exceeds_bound' via :func:'_tiny_maxflow'). Vertex 'v' becomes an
920 in-copy '2v' and an out-copy '2v+1' joined by an internal arc of capacity
921 one (so any route uses 'v' at most once), and each adjacency 'u -> v'
922 becomes an arc '(2u+1) -> (2v)'. The caller uncaps the two endpoints' own
923 in->out gates so Menger counts internally disjoint routes (see
924 :func:'exceeds_bound').
925
926 'parallel_routes' selects the counting convention for parallel copies, the
927 choice discussed at length in the thesis (sec:parallel-convention). The
928 default False caps every adjacency at one, so a bundle of parallel edges is a
929 single direct route and a multigraph measures as its underlying simple graph:
930 that is the convention the twelve variants use. Setting it True gives the
931 adjacency capacity 'mu(u,v)', so 'q' parallel copies count as 'q'
932 internally disjoint routes, which is the alternative convention explored in
933 sec:multi-vertex-standard.
934 """
935 n = graph.num_vertices
936 size = 2 * n
937 cap = np.zeros((size, size), dtype=int)
938 for v in range(n):
939     cap[2 * v, 2 * v + 1] = 1 # internal gate: one route through v
940 for u in range(n):
941     for v in range(n):
942         if u != v and graph.mu[u, v] > 0:
943             cap[2 * u + 1, 2 * v] = (int(graph.mu[u, v]) if parallel_routes
944                                     else 1)
945 return cap, size
946
947
948 def max_multigraph_vertex_standard(n: int, m: int,
949                                   deadline: float | None = None) -> tuple[int,
950                                   ↪ Graph | None, bool]:
951     """Exhaustive maximum for the multigraph vertex problem, OTHER convention.
952
953     Maximises the total multiplicity of an undirected multigraph on 'n'
954     vertices subject to ' $\kappa^{\max} \leq m - 1$ ' when parallel copies are counted
955     as distinct internally vertex-disjoint routes. Every multiplicity is capped
956     at 'm - 1' because 'm' parallel copies already exceed the ceiling on
957     their own. Branch and bound over the pairs in a fixed order, trying the
958     largest multiplicity first, pruning on feasibility (which is monotone) and
959     on the incumbent. Returns '(best_total, witness, completed)'.
960     """
961     pairs = list(combinations(range(n), 2))
962     cap = m - 1
963     graph = Graph(n, MULTI_UNDIRECTED)
964     best = [0, None]
965     timed_out = [False]
966
967     def feasible() -> bool:
968         return not exceeds_bound(graph, m - 1, separation="vertex",
969                                 parallel_routes=True)
970
971     def recurse(i: int, total: int) -> None:
972         if deadline is not None and time.time() > deadline:
973             timed_out[0] = True
974             return
975         if total + cap * (len(pairs) - i) <= best[0]:

```



```

975         return
976     if i == len(pairs):
977         if total > best[0]:
978             best[0] = total
979             best[1] = graph.copy()
980         return
981     u, v = pairs[i]
982     for q in range(cap, -1, -1):
983         graph.set_multiplicity(u, v, q)
984         if q == 0 or feasible():
985             recurse(i + 1, total + q)
986         if timed_out[0]:
987             break
988     graph.set_multiplicity(u, v, 0)
989
990 recurse(0, 0)
991 return best[0], best[1], not timed_out[0]
992
993
994 def exceeds_bound(graph: Graph, k: int, *, separation: str = "edge",
995                  parallel_routes: bool = False) -> bool:
996     """'True' iff ' $\lambda^{\max}(G) > k$ ' (edge) or ' $\kappa^{\max}(G) > k$ ' (vertex).
997
998     Equivalent to ' $\text{max\_connectivity}(\text{graph}, \text{vertex\_split}=(\text{separation}=='\text{vertex}')) >$ 
999      $\hookrightarrow k$ '
1000
1001     but with two early exits: each pair's capped flow stops after ' $k+1$ '
1002      $\hookrightarrow$  augmenting
1003
1004     paths, and the pair loop stops at the first violating pair. On an infeasible
1005     graph this typically returns after a single pair. The pair set is exactly the
1006     one :func:'max_connectivity' iterates, so the predicate matches it pair for
1007      $\hookrightarrow$  pair.
1008
1009     ' $\text{parallel\_routes}$ ' is passed through to :func:'_split_capacity_matrix' and
1010     only affects vertex mode: it selects the convention in which parallel copies
1011     are distinct routes, used by :func:'max_multigraph_vertex_standard'.
1012     """
1013     n = graph.num_vertices
1014     if separation == "edge":
1015         mu = graph.mu
1016         if _C is not None and n <= 16:
1017             flat, ptr = _c_flat(mu)
1018             directed = int(graph.variant.directed)
1019             return bool(_C.max_connectivity_exceeds(ptr, n, k, directed))
1020         for s, t in _pairs(graph):
1021             if _tiny_maxflow(mu, n, s, t, k):
1022                 return True
1023         return False
1024     if separation == "vertex":
1025         cap, size = _split_capacity_matrix(graph, parallel_routes)
1026         for s, t in _pairs(graph):
1027             # Uncap the endpoints' own in->out gates so Menger counts INTERNAL
1028             # routes (mirrors local_connectivity's vertex mode); flow leaves s's
1029             # out-copy (2s+1) and arrives at t's in-copy (2t).
1030             cap[2 * s, 2 * s + 1] = _UNBOUNDED
1031             cap[2 * t, 2 * t + 1] = _UNBOUNDED
1032             hit = _tiny_maxflow(cap, size, 2 * s + 1, 2 * t, k)
1033             cap[2 * s, 2 * s + 1] = 1 # restore the gates for the next
1034              $\hookrightarrow$  pair

```

```

1029         cap[2 * t, 2 * t + 1] = 1
1030         if hit:
1031             return True
1032         return False
1033     raise ValueError("separation must be 'edge' or 'vertex'")
1034
1035
1036 # --- GOMORY-HU: encode all pairwise edge-connectivities in n-1 flows (undirected
    ↪ only) ---
1037
1038
1039 def _undirected_capacity_graph(graph: Graph) -> nx.Graph:
1040     """Build the undirected capacitated graph that Gomory-Hu consumes."""
1041     if graph.variant.directed:
1042         raise ValueError("Gomory-Hu trees are defined for undirected graphs only")
1043     capacity_graph = nx.Graph()
1044     capacity_graph.add_nodes_from(graph.vertices())
1045     # One undirected edge per adjacency, capacity = multiplicity (its edge count).
1046     for u, v, multiplicity in graph.edges():
1047         capacity_graph.add_edge(u, v, capacity=float(multiplicity))
1048     return capacity_graph
1049
1050
1051 def gomory_hu_tree(graph: Graph) -> nx.Graph:
1052     """Return a Gomory-Hu tree of ‘graph’ as a weighted ‘networkx’ tree.
1053
1054     Each tree edge carries a ‘weight’ equal to the capacity of the minimum cut
1055     it represents. This is the one place that genuinely needs networkx’s
1056     Gomory-Hu routine; it backs a figure, not any reported bound.
1057     """
1058     _require_networkx("gomory_hu_tree")
1059     return nx.gomory_hu_tree(_undirected_capacity_graph(graph), capacity="capacity
    ↪ ")
1060
1061
1062 def max_edge_connectivity_via_tree(graph: Graph) -> int:
1063     """‘lambda^max(G)’ read off the Gomory-Hu tree as its heaviest edge.
1064
1065     Returns ‘0’ for graphs with fewer than two vertices or no edges.
1066     """
1067     if graph.num_vertices < 2:
1068         return 0
1069     tree = gomory_hu_tree(graph)
1070     # READ-OFF: lambda(u, v) = lightest tree edge on the u-v path, so the LARGEST
1071     # such bottleneck over all pairs is simply the heaviest edge in the tree.
1072     weights = [data["weight"] for _, _, data in tree.edges(data=True)]
1073     if not weights:
1074         return 0
1075     return int(round(max(weights)))
1076
1077
1078 # --- PROVE: MILP for M*(n) via cut-counting (scipy/HiGHS or Gurobi backend) ---
1079 # L_m^dir(n) = (m-1)*M*(n) where M*(n) = max sum(w) s.t. maxflow(s,t;w)<=1 all
    ↪ pairs.
1080 # Flow constraint encoded exactly: choose a cut x in {0,1}^n per pair, cap
    ↪ crossing
1081 # weight via p; zero MIP gap = proof. Shared helpers build constraints for both
    ↪ backends.

```

```

1082
1083
1084 @dataclass
1085 class ProofResult:
1086     """The outcome of a proof run."""
1087
1088     n: int
1089     status: str          # OPTIMAL, LIMIT, INFEASIBLE, UNBOUNDED, ...
1090     scaled_optimum: float #  $M*(n)$  = max total weight, exact if status OPTIMAL
1091     relative_gap_zero: bool # whether the solver closed the gap to zero
1092     solve_seconds: float
1093     weight_matrix: np.ndarray | None # a witnessing matrix  $w$ , if one was found
1094
1095     def value_for(self, m: int) -> int:
1096         """The proved directed multigraph value ' $L_m^{dir}(n) = (m-1) M*(n)$ '."""
1097         # Undo the  $(m-1)$  scaling: one proved  $M*(n)$  yields every  $m$ .
1098         return (m - 1) * int(round(self.scaled_optimum))
1099
1100     def is_proof(self) -> bool:
1101         """Whether this run is a genuine upper-bound proof (optimal, gap zero)."""
1102         # Only an OPTIMAL solve with a closed gap counts as a proof; a LIMIT
1103         # (timed out) result is merely a feasible lower bound, never a proof.
1104         return self.status == "OPTIMAL" and self.relative_gap_zero
1105
1106
1107 def _ordered_pairs(n: int) -> list[tuple[int, int]]:
1108     return [(u, v) for u in range(n) for v in range(n) if u != v]
1109
1110
1111 def _two_hop_triples(n: int) -> list[tuple[int, int, int]]:
1112     # All (source, middle, target) with three distinct vertices: the  $s \rightarrow x \rightarrow t$ 
1113     # detours that the two-hop inequality reasons about.
1114     return [(s, x, t)
1115             for s in range(n) for t in range(n) if s != t
1116             for x in range(n) if x != s and x != t]
1117
1118
1119 # Proved values of  $M*(k)$  for the deletion cuts below (proved:  $M*(k) =$ 
1120 #  $2(k-1)$  for  $k \leq 6$ , proved by this very optimisation with zero gap).
1121 _PROVEN_MSTAR = {2: 2, 3: 4, 4: 6, 5: 8, 6: 10}
1122
1123
1124 # -----
1125 # The cut-counting MILP: one PuLP model for both provers, any solver
1126 # -----
1127 # The two provers below are the SAME cut-counting optimisation over two weight
1128 # domains: the fractional one (prove_directed_multigraph, weights  $w$  in  $[0, 1]$ , cut
1129 # cap 1) maximises the total weight  $M*(n)$ ; the integral one (prove_integral_arc_
1130 # bound, multiplicities  $\mu$  in  $\{0..m-1\}$ , cut cap  $m-1$ ) decides feasibility at a
1131     ↪ fixed
1132 # arc target. One builder serves both, and any MILP solver runs it: CBC (bundled
1133 # with pulp) by default, Gurobi by a one-line switch. Because the cut formulation
1134     ↪ proof.
1135
1136 _PULP_STATUS = {"Optimal": "OPTIMAL", "Infeasible": "INFEASIBLE",
1137                 "Unbounded": "UNBOUNDED"}
1138
1139

```

```

1138
1139 def _pick_solver(time_limit: float, show_log: bool, use_gurobi: bool | None):
1140     """Choose the MILP solver.  'gapRel=0' demands a closed gap, which is what
1141     makes an OPTIMAL verdict a proof rather than a heuristic.
1142
1143     'use_gurobi': 'True' forces Gurobi (error if it is not usable), 'False'
1144     forces CBC, 'None' uses Gurobi when its licence is active and CBC otherwise.
1145     Switching to Gurobi is the whole change needed to attack the open n=7 facts.
1146     """
1147     if use_gurobi is not False:
1148         gurobi = pulp.GUROBI_CMD(msg=show_log, timeLimit=time_limit,
1149                                 options=[("MIPGap", 0)])
1150         if gurobi.available():
1151             return gurobi
1152         if use_gurobi is True:
1153             raise RuntimeError(
1154                 "use_gurobi=True but no usable Gurobi was found. This path runs "
1155                 "Gurobi through PuLP's GUROBI_CMD, which calls the gurobi_cl "
1156                 "command line, so gurobipy is not required: check that "
1157                 "'gurobi_cl --version' runs and that the licence is active."
1158             )
1159     return pulp.PULP_CBC_CMD(msg=show_log, timeLimit=time_limit, gapRel=0.0)
1160
1161
1162 def _cut_counting_model(n: int, *, cap: float, integer: bool, two_hop: bool,
1163                        symmetry: bool, deletion: bool, degree_pair: bool):
1164     """Build the shared cut-counting MILP as a PuLP model (no objective set yet).
1165
1166     For every ordered pair (s, t) the model CHOOSES one cut: the side indicators
1167     fix s on the source side (constant 1) and t on the sink side (constant 0), and
1168     every other vertex takes a binary side. The helper p picks up each crossing
1169     arc's weight through 'p >= w + cap*(x_u - x_v) - cap', which forces 'p >= w
1170     ↪ exactly on a crossing arc (x_u = 1, x_v = 0) and leaves p free at 0 otherwise.
1171     Capping the crossing total at 'cap' says maxflow(s, t) <= cap, and letting
1172     ↪ the
1173     optimisation pick the cut means it picks the MINIMUM cut, so this is exactly
1174     maxflow(s, t) <= cap, the true feasible region. 'cap=1' with continuous
1175     weights is the scaled prover; 'cap=m-1' with integer weights the arc one.
1176
1177     The optional families are all proved-valid inequalities that tighten the
1178     relaxation without ever moving the optimum: 'two_hop' (a two-arc-disjoint
1179     detour bound via z = min of the two hops), 'degree_pair' (a flow lower bound
1180     on each pair), 'deletion' (an induced-subgraph bound from proved smaller
1181     M*(k)), and 'symmetry' (a degree ordering that prunes relabelled duplicates)
1182     ↪ .
1183
1184     Returns '(prob, w)' with 'w' the weight variables keyed by ordered pair.
1185     """
1186     pairs = _ordered_pairs(n)
1187     prob = pulp.LpProblem("erdos915", pulp.LpMaximize)
1188     category = "Integer" if integer else "Continuous"
1189     w = {(u, v): pulp.LpVariable(f"w_{u}_{v}", 0, cap, cat=category)
1190          for (u, v) in pairs}
1191
1192     for (s, t) in pairs:
1193         # s is always on the source side, t always on the sink side: constants,
1194         ↪ not
1195         # variables (which also avoids pulp resetting a Binary's bounds to [0, 1])

```

```

1192     ↪ .
1193     side = {u: 1 if u == s else 0 if u == t
1194             else pulp.LpVariable(f"x_{s}_{t}_{u}", cat="Binary")
1195             for u in range(n)}
1196     crossing = []
1197     for (u, v) in pairs:
1198         p_uv = pulp.LpVariable(f"p_{s}_{t}_{u}_{v}", 0, cap)
1199         prob += p_uv >= w[(u, v)] + cap * (side[u] - side[v]) - cap
1200         crossing.append(p_uv)
1201         prob += pulp.lpSum(crossing) <= cap          # crossing weight <= cap
1202
1203     if two_hop:
1204         z = {}
1205         for (s, x, t) in _two_hop_triples(n):
1206             z[(s, x, t)] = pulp.LpVariable(f"z_{s}_{x}_{t}", 0, cap)
1207             selector = pulp.LpVariable(f"b_{s}_{x}_{t}", cat="Binary")
1208             # z = min(w[s,x], w[x,t]): two upper bounds, and the selector forces z
1209             # up to whichever argument is the minimum (big-M with M = cap).
1210             prob += z[(s, x, t)] <= w[(s, x)]
1211             prob += z[(s, x, t)] <= w[(x, t)]
1212             prob += z[(s, x, t)] >= w[(s, x)] - cap * (1 - selector)
1213             prob += z[(s, x, t)] >= w[(x, t)] - cap * selector
1214         for (s, t) in pairs:
1215             prob += w[(s, t)] + pulp.lpSum(
1216                 z[(s, x, t)] for x in range(n) if x != s and x != t) <= cap
1217
1218     if degree_pair:
1219         # d+(s) + d-(t) - w[s,t] <= (n-1)*cap. This is the two-hop bound
1220         # aggregated: w[s,t] + sum_x min(w[s,x], w[x,t]) <= cap, plus each middle
1221         # max(w[s,x], w[x,t]) <= cap, sums to (n-1)*cap. It is a genuine cut, NOT
1222         # trivially satisfied: the box bound alone allows the LHS up to (2n-3)*cap
1223         ↪ ,
1224         # and a two-route witness reaches 2(n-1)*cap (e.g. n=4: s->x1->t, s->x2->t
1225         # gives LHS 4 > 3). Dominated by two_hop when that family is on, but
1226         # standalone-useful when it is off; valid in both weight domains.
1227         for (s, t) in pairs:
1228             prob += (pulp.lpSum(w[(s, x)] for x in range(n) if x != s)
1229                     + pulp.lpSum(w[(x, t)] for x in range(n) if x != t)
1230                     - w[(s, t)]) <= (n - 1) * cap
1231
1232     if deletion:
1233         for k, holes in ((n - 1, 1), (n - 2, 2)):
1234             if k not in _PROVEN_MSTAR:
1235                 continue
1236             bound = cap * _PROVEN_MSTAR[k]          # induced subgraph on k vertices
1237             for gone in combinations(range(n), holes):
1238                 prob += pulp.lpSum(w[(a, b)] for (a, b) in pairs
1239                                     if a not in gone and b not in gone) <= bound
1240
1241     if symmetry:
1242         degree = [pulp.lpSum(w[(u, v)] for (u, v) in pairs if v0 in (u, v))
1243                   for v0 in range(n)]
1244         for v0 in range(n - 1):
1245             prob += degree[v0] <= degree[v0 + 1]
1246
1247     return prob, w

```

```

1248 def prove_directed_multigraph(
1249     n: int,
1250     *,
1251     time_limit: float = 1500.0,
1252     use_gurobi: bool | None = None,
1253     use_two_hop: bool = True,
1254     use_symmetry_breaking: bool = True,
1255     use_deletion_cuts: bool = False,
1256     show_solver_log: bool = False,
1257 ) -> ProofResult:
1258     """Prove  $M \times (n)$  for the directed multigraph arc problem.
1259
1260     Returns a :class:`ProofResult`. When its :meth:`is_proof` is true the value
1261     is a proven upper bound, and  $\text{value\_for}(m)$  gives  $L_m^{\text{dir}}(n)$  for every
1262      $m$ .  $\text{use\_gurobi}$  picks the solver (see :func:`_pick_solver`); the
1263     ↪ default
1264     uses Gurobi when its licence is active and CBC otherwise.
1265     """
1266     _require_pulp()
1267     prob, w = _cut_counting_model(
1268         n, cap=1.0, integer=False, two_hop=use_two_hop,
1269         symmetry=use_symmetry_breaking, deletion=use_deletion_cuts,
1270         degree_pair=use_deletion_cuts,
1271     )
1272     prob += pulp.lpSum(w.values()) # maximise total weight =  $M \times (n)$ 
1273     ↪ )
1274
1275     start = time.time()
1276     prob.solve(_pick_solver(time_limit, show_solver_log, use_gurobi))
1277     elapsed = time.time() - start
1278     status = _PULP_STATUS.get(pulp.LpStatus[prob.status], "LIMIT")
1279
1280     weight_matrix = None
1281     if status == "OPTIMAL":
1282         weight_matrix = np.zeros((n, n))
1283         for (u, v), variable in w.items():
1284             weight_matrix[u, v] = variable.value() or 0.0
1285
1286     return ProofResult(
1287         n=n, status=status,
1288         scaled_optimum=(pulp.value(prob.objective) if status == "OPTIMAL"
1289                        else float("nan")),
1290         relative_gap_zero=(status == "OPTIMAL"),
1291         solve_seconds=elapsed, weight_matrix=weight_matrix,
1292     )

```

C.1.4 Searching

This part corresponds to [Chapter 3](#). Where proving is out of reach, the program looks for good examples instead. It starts from a graph, repeatedly makes a small random change, and keeps the change if it helps. Keeping only improvements gets stuck in the first decent shape it finds, so early on it also accepts some changes that make things worse, and it does this less and less as it goes, in the manner of hot metal cooling into an ordered state. A second engine, tabu search, walks the same landscape with a different rule: it always takes the best available move

but refuses to revisit recent positions, which is what stops it circling.

Two ideas make this fast enough to be useful. The search is told which edges are load-bearing, so that it changes those rather than wasting moves on edges that carry nothing. And it almost never measures connectivity fully: adding one connection can raise the count by at most one and removing one can never raise it at all, so most steps can be settled by arithmetic on a bound the search already carries, with an exact measurement only when the answer is genuinely in doubt. This part ends with `solve`, the single entry point that either proves or searches depending on one argument.

```
1293 #####
1294 ##
1295 ## CHAPTER 3 -- DISCOVERING BOUNDS BY SEARCH
1296 ## The temperature search, and the one driver that proves or discovers.
1297 ##
1298 #####
1299
1300 # --- SENSITIVITY:  $\sigma(e) = \lambda_{\max}(G) - \lambda_{\max}(G-e)$ ; dial for the
    ↳ cooling schedule ---
1301
1302 # A connectivity measure maps a graph to its  $\lambda_{\max}$  or  $\kappa_{\max}$ .
1303 ConnectivityMeasure = Callable[[Graph], int]
1304
1305
1306 def edge_sensitivity(
1307     graph: Graph,
1308     u: int,
1309     v: int,
1310     connectivity: ConnectivityMeasure = max_edge_connectivity,
1311 ) -> int:
1312     """Return ' $\sigma(e)$ ' for the edge or arc ' $e = (u, v)$ '.\lambda_{\max} dropped: positive means e was load-bearing.
1325     return before - after
1326
1327
1328 def sensitivity_map(
1329     graph: Graph,
1330     connectivity: ConnectivityMeasure = max_edge_connectivity,
1331 ) -> dict[tuple[int, int], int]:
1332     """Return ' $\{(u, v): \sigma((u, v))\}$ ' over all present edges or arcs.
1333
1334     Useful for visualising an extremiser: high-sensitivity edges are the load
1335     bearers, low-sensitivity edges are slack.
1336
1337     The baseline connectivity ' $\text{before} = \text{connectivity}(\text{graph})$ ' is the SAME for
```

```

1338     every edge, so it is measured once here rather than re-measured inside each
1339     :func:'edge_sensitivity' call. This changes no value (the result equals
1340     '{e: edge_sensitivity(graph, *e) for e in graph.edges()}'); it only avoids
1341     recomputing the shared baseline once per edge.
1342     """
1343     before = connectivity(graph)
1344     result: dict[tuple[int, int], int] = {}
1345     for u, v, _ in graph.edges():
1346         reduced = graph.copy()
1347         reduced.set_multiplicity(u, v, 0) # delete the WHOLE adjacency (all
            ↳ copies)
1348         result[(u, v)] = before - connectivity(reduced)
1349     return result
1350
1351
1352 # --- SEARCH: simulated annealing;  $E = -|E(G)| + \text{penalty} \cdot \max(0, \lambda^{\max - (m-1)})$ 
            ↳ ---
1353 # Metropolis rule with geometric cooling; removal proposals biased by edge
            ↳ sensitivity.
1354
1355
1356 @dataclass
1357 class SearchStep:
1358     """One step of the search, recorded for plotting the temperature trace."""
1359
1360     step: int
1361     temperature: float
1362     energy: float
1363     edge_count: int
1364     connectivity: int
1365     accepted: bool
1366     seconds: float = 0.0 # wall-clock since the search began (for timed
            ↳ traces)
1367     best_feasible: int = 0 # densest feasible graph seen so far (for
            ↳ convergence)
1368
1369
1370 @dataclass
1371 class SearchResult:
1372     """The outcome of a search: the best feasible graph and the full trace."""
1373
1374     best_graph: Graph
1375     best_edge_count: int
1376     feasible_found: bool
1377     history: list[SearchStep] = field(default_factory=list)
1378
1379     def acceptance_rate(self) -> float:
1380         """Fraction of proposed moves that were accepted."""
1381         if not self.history:
1382             return 0.0
1383         return sum(record.accepted for record in self.history) / len(self.history)
1384
1385
1386 def _connectivity_measure(separation: str):
1387     """Return the connectivity function named by 'separation'."""
1388     if separation == "edge":
1389         return max_edge_connectivity
1390     if separation == "vertex":

```



```

1391         return max_vertex_connectivity
1392     raise ValueError("separation must be 'edge' or 'vertex'")
1393
1394
1395 def _energy(graph: Graph, m: int, measure, penalty: float) -> float:
1396     """The energy  $-|E| + \text{penalty} * \text{excess connectivity}$ """
1397     # Excess is how far connectivity exceeds the allowed  $m-1$ ; zero when feasible.
1398     excess = max(0, measure(graph) - (m - 1))
1399     # Lower energy is better: more edges (negative term) and no excess.
1400     return -graph.edge_count() + penalty * excess
1401
1402
1403 def _proposal_energy(
1404     proposal: Graph,
1405     *,
1406     is_add: bool,
1407     feasible_current: bool,
1408     lam_ub: int,
1409     m: int,
1410     penalty: float,
1411     measure_full,
1412     separation: str,
1413 ) -> tuple[float, bool]:
1414     """Return  $(\_energy(proposal), \text{proposal\_is\_feasible})$  doing the least work.
1415
1416     The energy  $-|E| + \text{penalty} * \max(0, \text{connectivity} - (m-1))$  depends on the
1417     connectivity only through the excess, and by monotonicity (Facts M1/M2 in the
1418     thesis) that excess is provably zero on most steps, so no flow is needed:
1419
1420     - a removal from a feasible graph stays feasible (M1), excess 0;
1421     - an addition while  $\text{lam\_ub} \leq m-2$  stays feasible (M2), excess 0.
1422
1423     Only when neither shortcut applies do we run one capped predicate, and only
1424      $\hookrightarrow$  when
1425     that reports an infeasible proposal do we spend the exact  $\text{measure\_full}$  to
1426      $\hookrightarrow$  get
1427     the penalty term right. The float returned is byte-for-byte the one  $\text{\_energy}$ 
1428      $\hookrightarrow$ 
1429     would return (the same arithmetic on the same excess), so the search
1430      $\hookrightarrow$  trajectory
1431     is unchanged. The boolean is the EXACT feasibility of  $\text{proposal}$ .
1432     """
1433     edges = proposal.edge_count()
1434     # Decide excess = max(0, measure(proposal) - (m-1)) with the least work.
1435     if (is_add and lam_ub <= m - 2) or (not is_add and feasible_current):
1436         excess = 0 # M2 (add) or M1 (remove): provably
1437          $\hookrightarrow$  feasible
1438     elif not exceeds_bound(proposal, m - 1, separation=separation):
1439         excess = 0 # at the boundary but the capped flow
1440          $\hookrightarrow$  clears it
1441     else:
1442         excess = measure_full(proposal) - (m - 1) # infeasible: exact penalty
1443          $\hookrightarrow$  term
1444     # Identical arithmetic to _energy, so the returned float matches it exactly.
1445     return -edges + penalty * excess, excess == 0
1446
1447 def _candidate_pairs(graph: Graph) -> list[tuple[int, int]]:

```

```

1442     """All ordered pairs (directed) or unordered pairs (undirected)."""
1443     n = graph.num_vertices
1444     pairs = []
1445     for u in range(n):
1446         start = 0 if graph.variant.directed else u + 1
1447         for v in range(start, n):
1448             if u != v:
1449                 pairs.append((u, v))
1450     return pairs
1451
1452
1453 def _propose_removal(
1454     graph: Graph,
1455     temperature: float,
1456     rng: random.Random,
1457     cached_sensitivity: dict[tuple[int, int], int] | None,
1458 ) -> tuple[int, int]:
1459     """Choose a present edge to thin out, biased toward free (low sigma) edges.
1460
1461     Without a sensitivity cache the choice is uniform. With one, edge ‘e’ is
1462     drawn with weight ‘exp(-sigma(e) / T)’: at high ‘T’ the weights flatten
1463     (uniform exploration), at low ‘T’ they concentrate on ‘sigma = 0’ edges.
1464     """
1465     present = [(u, v) for u, v, _ in graph.edges()]
1466     if cached_sensitivity is None:
1467         return rng.choice(present) # no guidance: uniform removal
1468     # exp(-sigma/T): big sigma (load-bearing) is suppressed, and the suppression
1469     # sharpens as T -> 0. max(T, eps) guards against division by zero.
1470     weights = [
1471         math.exp(-cached_sensitivity.get((u, v), 0) / max(temperature, 1e-9))
1472         for (u, v) in present
1473     ]
1474     return rng.choices(present, weights=weights, k=1)[0]
1475
1476
1477 def search_for_dense_graph(
1478     variant: Variant,
1479     n: int,
1480     m: int,
1481     *,
1482     separation: str = "edge",
1483     steps: int = 4000,
1484     initial_temperature: float = 3.0,
1485     cooling: float = 0.9985,
1486     penalty: float = 6.0,
1487     max_multiplicity: int | None = None,
1488     sensitivity_guided: bool = True,
1489     bias_refresh: int = 200,
1490     seed: int = 0,
1491     deadline: float | None = None,
1492     record_exact_connectivity: bool = False,
1493     reference_mode: bool = False,
1494 ) -> SearchResult:
1495     """Find the densest feasible graph by a guided random search (simulated
1496         ↪ annealing).
1497
1498     Args:
1499         variant: which graph model to search in.

```

```

1499     n: number of vertices.
1500     m: forbidden value. Feasibility means ‘connectivity <= m - 1’.
1501     separation: ‘“edge”’ for arc/edge-disjoint, ‘“vertex”’ for
1502         internally vertex-disjoint paths.
1503     steps: number of proposed moves.
1504     initial_temperature: starting temperature ‘T_0’.
1505     cooling: factor (< 1) the temperature is multiplied by each step.
1506     penalty: weight of the connectivity violation in the energy.
1507     max_multiplicity: cap on parallel edges (defaults to ‘m - 1’, a simple
1508         variant is always capped at one regardless).
1509     sensitivity_guided: bias removals toward free edges (the temperature
1510         dial described in the module docstring).
1511     bias_refresh: recompute the sensitivity cache every this many steps.
1512     seed: random seed, fixed so figures are reproducible.
1513     record_exact_connectivity: log the EXACT ‘lambda^max’ per step in
1514         ‘SearchStep.connectivity’ (the figure-trace needs it). When False,
1515         the cheaper maintained upper bound ‘lam_ub’ is logged instead, which
1516         does not affect the search at all (the energy never reads this field).
1517     reference_mode: INTERNAL, test only. Force the slow, exact per-step
1518         energy (call :func:‘_energy’ and :func:‘max_connectivity’ directly)
1519         instead of the capped/monotone fast path. Used by the equivalence
1520         test to prove the fast path reproduces this reference step for step.
1521
1522     Returns:
1523         An :class:‘SearchResult’ with the best feasible graph found and the
1524         full step-by-step history.
1525
1526     The fast path is *trajectory-preserving*: it computes the identical per-step
1527     energy as ‘reference_mode’ (proven by ‘test_search.
1528         ↳ test_fast_path_matches_exact’),
1529     so every accept/reject decision, the random-number draw order, and the
1530         ↳ returned
1531     witness are unchanged. It only avoids flows whose result the energy cannot
1532     depend on (Facts M1/M2: a removal from a feasible graph stays feasible, an
1533     addition while ‘lam_ub <= m-2’ stays feasible), and resyncs the upper bound
1534     ‘lam_ub’ to the exact value at the ‘bias_refresh’ cadence.
1535     """
1536     rng = random.Random(seed)
1537     measure = _connectivity_measure(separation)
1538     # Multiplicity cap: simple graphs are always 0/1; multigraphs default to m-1.
1539     cap = 1 if variant.simple else (max_multiplicity if max_multiplicity else m -
1540         ↳ 1)
1541     pairs = None # built lazily once we know we need it
1542
1543     current = Graph(n, variant) # start from the empty graph: feasible, energy 0
1544     current_energy = _energy(current, m, measure, penalty)
1545
1546     best_graph = current.copy()
1547     best_edges = 0
1548     feasible_found = True # the empty graph is feasible
1549
1550     # State carried across steps so the energy can avoid recomputing connectivity
1551     # from scratch (see the trajectory-preserving note in the docstring):
1552     # feasible -- is ‘current’ feasible (lambda^max <= m-1)? Exact at all times
1553         ↳ .
1554     # lam_ub -- a SOUND upper bound on lambda^max(current). Facts M1/M2 keep
1555         ↳ it
1556     #
1557         ↳ sound: +1 on an accepted addition, unchanged on a removal.

```

```

1552     feasible = True                # the empty graph is feasible
1553     lam_ub = 0                     #  $\lambda^{\max}(\text{empty graph}) = 0$ 
1554
1555     temperature = initial_temperature
1556     cached_sensitivity = None
1557     history: list[SearchStep] = []
1558     clock_start = time.perf_counter() # for the timed convergence trace only
1559
1560     for step in range(steps):
1561         # Periodically recompute the sensitivity map that guides removals. It
1562         # is expensive, so we cache it and refresh only every bias_refresh steps.
1563         if sensitivity_guided and step % bias_refresh == 0 and current.edge_count
1564             ↪ () > 0:
1565             cached_sensitivity = sensitivity_map(current, measure)
1566         # Resync the upper bound to the truth at the same cadence (the bound can
1567         # only drift upward over a run of removals, which costs extra capped
1568         ↪ checks
1569         # but never correctness). This step already pays for flows above, and the
1570         # resync touches neither the energy nor the RNG, so the trajectory is safe
1571         ↪ .
1572         if step % bias_refresh == 0:
1573             lam_ub = measure(current)
1574             feasible = lam_ub <= m - 1
1575
1576     proposal = current.copy()
1577     # Flip a coin: remove an edge (only if some exist) or add one.
1578     removing = current.edge_count() > 0 and rng.random() < 0.5
1579     if removing:
1580         # Sensitivity-guided removal: prefers free edges, especially cold.
1581         u, v = _propose_removal(current, temperature, rng, cached_sensitivity)
1582         proposal.remove_edge(u, v)
1583     else:
1584         if pairs is None:
1585             pairs = _candidate_pairs(current)
1586         # Only pairs that have room under the multiplicity cap can grow.
1587         addable = [(u, v) for (u, v) in pairs if proposal.multiplicity(u, v) <
1588             ↪ cap]
1589         if not addable:
1590             continue # nothing to add; skip this step
1591         u, v = rng.choice(addable)
1592         proposal.add_edge(u, v)
1593     is_add = not removing
1594
1595     # The proposal's energy and exact feasibility. reference_mode is the slow
1596     ↪ ,
1597     # exact path the equivalence test compares against; the fast path returns
1598     # the IDENTICAL energy by Facts M1/M2 (see _proposal_energy).
1599     if reference_mode:
1600         proposal_energy = _energy(proposal, m, measure, penalty)
1601         proposal_feasible = measure(proposal) <= m - 1
1602     else:
1603         proposal_energy, proposal_feasible = _proposal_energy(
1604             proposal, is_add=is_add, feasible_current=feasible, lam_ub=lam_ub,
1605             m=m, penalty=penalty, measure_full=measure, separation=separation)
1606     change = proposal_energy - current_energy
1607     # ACCEPT OR NOT: always take an improvement (change <= 0); take a
1608     ↪ worsening
1609     # move only with probability  $\exp(-\text{change} / T)$ . As T falls this

```

```

1604         ↪ probability
1605     # shrinks, so the walk increasingly refuses to go uphill.
1606     accept = change <= 0 or rng.random() < math.exp(-change / max(temperature,
1607         ↪ 1e-9))
1608
1609     if accept:
1610         current, current_energy = proposal, proposal_energy
1611         # Carry the exact feasibility and update the bound by M2 (an accepted
1612         # addition can raise  $\lambda^{\max}$  by at most 1; a removal cannot raise
1613         ↪ it).
1614         feasible = proposal_feasible
1615         if is_add:
1616             lam_ub += 1
1617         # Track the best FEASIBLE graph found (within bound and denser than
1618         ↪ any
1619         # seen so far): this is the lower-bound witness. 'feasible' is exact
1620         # here, so the witness is certified with no extra flow.
1621         if feasible and current.edge_count() > best_edges:
1622             best_graph, best_edges = current.copy(), current.edge_count()
1623             feasible_found = True
1624
1625     # The connectivity field feeds fig:trace only. Log the exact value when
1626     ↪ the
1627     # figure asks for it (or in reference_mode), else the maintained upper
1628     ↪ bound.
1629
1630     if record_exact_connectivity or reference_mode:
1631         recorded_connectivity = measure(current)
1632         # Cheap audit in the exact paths: the maintained bound must never
1633         # underestimate, and the feasibility flag must match the truth.
1634         assert lam_ub >= recorded_connectivity, (lam_ub, recorded_connectivity
1635             ↪ )
1636         assert feasible == (recorded_connectivity <= m - 1)
1637     else:
1638         recorded_connectivity = lam_ub
1639     history.append(SearchStep(
1640         step=step,
1641         temperature=temperature,
1642         energy=current_energy,
1643         edge_count=current.edge_count(),
1644         connectivity=recorded_connectivity,
1645         accepted=accept,
1646         seconds=time.perf_counter() - clock_start,
1647         best_feasible=best_edges,
1648     ))
1649     temperature *= cooling # cool down:  $T \leftarrow \text{cooling} * T$  each step
1650
1651     # Respect the deadline when supplied; check every 100 steps to amortise
1652     # the time.time() call.
1653     if deadline is not None and step % 100 == 99 and time.time() > deadline:
1654         break
1655
1656     return SearchResult(best_graph, best_edges, feasible_found, history)
1657
1658 def _neighbour_moves(graph: Graph, cap: int) -> list[tuple[int, int, int]]:
1659     """Every single add ('+1') or remove ('-1') of one edge/arc under the cap.
1660
1661     The shared neighbourhood of both discoverers: one ordered pair for a digraph,

```

```

1655     one unordered pair for an undirected graph, addable while below the
1656     multiplicity ‘‘cap’’ and removable while above zero.
1657     """
1658     n = graph.num_vertices
1659     directed = graph.variant.directed
1660     moves: list[tuple[int, int, int]] = []
1661     for u in range(n):
1662         for v in range(n):
1663             if u == v or (not directed and v < u):
1664                 continue
1665             mult = graph.multiplicity(u, v)
1666             if mult < cap:
1667                 moves.append((u, v, +1))
1668             if mult > 0:
1669                 moves.append((u, v, -1))
1670     return moves
1671
1672
1673 def tabu_search_for_dense_graph(
1674     variant: Variant,
1675     n: int,
1676     m: int,
1677     *,
1678     separation: str = "edge",
1679     steps: int = 4000,
1680     penalty: float = 6.0,
1681     tenure: int = 7,
1682     stall_limit: int = 40,
1683     max_multiplicity: int | None = None,
1684     seed: int = 0,
1685     deadline: float | None = None,
1686     record_exact_connectivity: bool = True,
1687 ) -> SearchResult:
1688     """Find the densest feasible graph by tabu search, the deterministic twin of
1689     :func:‘search_for_dense_graph‘.
1690
1691     Same energy ‘‘-|E| + penalty * max(0, connectivity - (m-1))’’ and the same
1692     neighbourhood (:func:‘_neighbour_moves‘) as the annealer, but a different
1693     engine. Each step scans the whole neighbourhood and takes the best move whose
1694     pair is not on the tabu list, forbids that pair for ‘‘tenure’’ steps, and
1695     perturbs from the incumbent once it has stalled for ‘‘stall_limit’’ steps.
1696     Aspiration overrides the tabu flag for any move that beats the global best.
1697     There is no temperature and no accept-worse coin: the only randomness is the
1698     seed driving the perturbation kicks, so two runs from one seed are identical.
1699
1700     Returns the same :class:‘SearchResult‘ as the annealer (the best feasible
1701         ↳ graph
1702     and the full timed history), so the two engines are interchangeable through
1703         ↳ the
1704     ‘‘method’’ argument of :func:‘best_of_searches‘ and :func:‘solve‘.
1705     """
1706     rng = random.Random(seed)
1707     measure = _connectivity_measure(separation)
1708     cap = 1 if variant.simple else (max_multiplicity if max_multiplicity else m -
1709         ↳ 1)
1710
1711     current = Graph(n, variant) # start empty: feasible, energy 0
1712     best_graph, best_edges = current.copy(), 0

```

```

1710     tabu: dict[tuple[int, int], int] = {}
1711     stall = 0
1712     history: list[SearchStep] = []
1713     clock_start = time.perf_counter()
1714
1715     def trial_energy_and_feasibility(graph: Graph) -> tuple[float, bool]:
1716         """The annealer's energy and exact feasibility, by the same capped trick.
1717
1718         'exceeds_bound' decides feasibility with an early-exit capped flow, and
1719         the full 'measure' runs only on the rare infeasible trial that needs its
1720         penalty term, so most moves cost one cheap predicate instead of a full
1721         all-pairs flow. The returned float equals '_energy' exactly.
1722         """
1723         edges = graph.edge_count()
1724         if not exceeds_bound(graph, m - 1, separation=separation):
1725             return float(-edges), True # feasible: excess is zero
1726         return -edges + penalty * (measure(graph) - (m - 1)), False
1727
1728     for step in range(steps):
1729         if deadline is not None and time.time() > deadline:
1730             break
1731         best_move: tuple[int, int, int] | None = None
1732         best_move_energy: float | None = None
1733         best_move_global = False
1734         for (u, v, d) in _neighbour_moves(current, cap):
1735             trial = current.copy()
1736             trial.add_edge(u, v) if d > 0 else trial.remove_edge(u, v)
1737             trial_energy, trial_feasible = trial_energy_and_feasibility(trial)
1738             improves_global = trial_feasible and trial.edge_count() > best_edges
1739             if tabu.get((u, v), 0) > step and not improves_global:
1740                 continue # tabu, with no aspiration to override
1741             if (best_move_energy is None or trial_energy < best_move_energy
1742                 or (improves_global and not best_move_global)):
1743                 best_move, best_move_energy = (u, v, d), trial_energy
1744                 best_move_global = improves_global
1745         if best_move is None: # every move tabu: clear the list,
1746             ↪ retry
1747             tabu.clear()
1748             continue
1749
1750         u, v, d = best_move
1751         current.add_edge(u, v) if d > 0 else current.remove_edge(u, v)
1752         tabu[(u, v)] = step + tenure
1753
1754         connectivity = measure(current) # computed anyway for best-tracking
1755         if connectivity <= m - 1 and current.edge_count() > best_edges:
1756             best_graph, best_edges = current.copy(), current.edge_count()
1757             stall = 0
1758         else:
1759             stall += 1
1760         history.append(SearchStep(
1761             step=step, temperature=0.0, energy=best_move_energy,
1762             edge_count=current.edge_count(),
1763             connectivity=connectivity if record_exact_connectivity else -1,
1764             accepted=True, # tabu always takes its best legal move
1765             seconds=time.perf_counter() - clock_start,
1766             best_feasible=best_edges,
1767         ))

```

```

1767
1768         if stall >= stall_limit:                # perturb from the incumbent and reset
1769             current = best_graph.copy()
1770             for _ in range(rng.randint(2, 4)):
1771                 kicks = _neighbour_moves(current, cap)
1772                 if kicks:
1773                     a, b, dd = rng.choice(kicks)
1774                     current.add_edge(a, b) if dd > 0 else current.remove_edge(a, b
1775                                     ↪ )
1776
1777             tabu.clear()
1778             stall = 0
1779
1780
1781     return SearchResult(best_graph, best_edges, feasible_found=True, history=
1782         ↪ history)
1783
1784
1785 def _run_one_search(args: tuple) -> SearchResult:
1786     """Module-level worker for parallel restarts (must be picklable for
1787         ↪ ProcessPoolExecutor)."""
1788     variant, n, m, seed, method, search_options = args
1789     engine = tabu_search_for_dense_graph if method == "tabu" else
1790         ↪ search_for_dense_graph
1791     return engine(variant, n, m, seed=seed, **search_options)
1792
1793
1794 def best_of_searches(
1795     variant: Variant,
1796     n: int,
1797     m: int,
1798     *,
1799     restarts: int = 6,
1800     seed: int = 0,
1801     parallel: bool = True,
1802     method: str = "sa",
1803     **search_options,
1804 ) -> SearchResult:
1805     """Run several independent discovery schedules and keep the densest result.
1806
1807     A single search can settle into a local optimum, so for reliable
1808     rediscovery we restart from a fresh random seed a handful of times and return
1809     the run that found the most edges. Any keyword understood by the chosen
1810     engine ('steps', 'penalty', 'separation', and so on) is forwarded
1811     unchanged.
1812
1813     Args:
1814         parallel: when 'True' (default), run the restarts concurrently using
1815             :class:`~concurrent.futures.ProcessPoolExecutor`. Each restart has
1816             a distinct seed so the runs are genuinely independent. Falls back
1817             to sequential execution if multiprocessing is unavailable.
1818         method: '"sa"' (default, simulated annealing,
1819             :func:`search_for_dense_graph`) or '"tabu"'
1820             (:func:`tabu_search_for_dense_graph`). Both optimise the same energy
1821             over the same neighbourhood and return the same result type.
1822
1823     """
1824     if method not in ("sa", "tabu"):
1825         raise ValueError("method must be 'sa' or 'tabu'")
1826     task_args = [(variant, n, m, seed + r, method, search_options) for r in range(
1827         ↪ restarts)]

```



```

1820
1821     if parallel and restarts > 1:
1822         try:
1823             with concurrent.futures.ProcessPoolExecutor() as executor:
1824                 results = list(executor.map(_run_one_search, task_args))
1825             except (OSError, concurrent.futures.BrokenExecutor) as exc:
1826                 # Multiprocessing is genuinely unavailable here (a sandbox with no
1827                 # fork/spawn, or a worker that could not start). Fall back to a
1828                 # sequential run, but say so rather than masking the loss of cores.
1829                 # A bug inside the worker is NOT caught here, so it fails loudly.
1830                 warnings.warn(
1831                     f"parallel restarts unavailable ({exc!r}); running sequentially",
1832                     RuntimeWarning, stacklevel=2,
1833                 )
1834                 results = [_run_one_search(a) for a in task_args]
1835             else:
1836                 results = [_run_one_search(a) for a in task_args]
1837
1838         return max(results, key=lambda r: r.best_edge_count)
1839
1840
1841 # --- SOLVE: unified driver; exhaustive=True proves, exhaustive=False discovers
1842 ↪ ---
1843
1844 @dataclass
1845 class SolveResult:
1846     """The outcome of a solve, carrying an honest bound label."""
1847
1848     n: int
1849     m: int
1850     variant: str          # human label, e.g. "simple directed"
1851     separation: str       # "edge" or "vertex"
1852     value: int            # the extremal count we report
1853     bound: str            # "exact" / "lower" / "upper"
1854     method: str           # how the value was obtained
1855     seconds: float
1856     complete: bool        # an exhaustive run that actually finished
1857     witness: object | None # a Graph/Hypergraph attaining "value", if any
1858     note: str = ""
1859
1860     @property
1861     def proven(self) -> bool:
1862         """Whether "value" is a proved exact extremal value."""
1863         return self.bound == "exact"
1864
1865     def describe(self) -> str:
1866         """One readable line summarising the result."""
1867         relation = {"exact": "=", "lower": ">=", "upper": "<="}[self.bound]
1868         line = (f"{self.variant} ({self.separation}), n={self.n}, m={self.m}: "
1869               f"value {relation}{self.value} "
1870               f"[{self.bound}, {self.method}, {self.seconds:.1f}s]")
1871         return line if not self.note else f"{line}\n    note: {self.note}"
1872
1873
1874 def _variant_for(directed: bool, simple: bool) -> Variant:
1875     """Pick the matching named Variant constant from the two booleans."""
1876     if directed:

```

```

1877         return SIMPLE_DIRECTED if simple else MULTI_DIRECTED
1878     return SIMPLE_UNDIRECTED if simple else MULTI_UNDIRECTED
1879
1880
1881 def _matrix_cells(n: int, directed: bool) -> list[tuple[int, int]]:
1882     """The off-diagonal matrix cells a graph of this kind may fill.
1883
1884     Directed graphs vary every ordered pair; undirected graphs vary only the
1885     upper triangle, since ‘‘set_multiplicity’’ mirrors the lower half for us.
1886     """
1887     if directed:
1888         return [(u, v) for u in range(n) for v in range(n) if u != v]
1889     return [(u, v) for u in range(n) for v in range(n) if u < v]
1890
1891
1892 def _directed_witness(n: int, m: int, simple: bool) -> Graph | None:
1893     """The densest named directed construction that is feasible for ‘‘(n, m)’’.
1894
1895     Used to exhibit a concrete graph attaining the proved value: the better
1896     of the double star and the (augmented) one-directional bipartite, or
1897     ‘‘None’’ if neither applies to this case.
1898     """
1899     candidates: list[Graph] = []
1900     if not (simple and m > 2):
1901         # a simple star needs m == 2
1902         candidates.append(double_star(n, m, directed=True)) # 2(n-1)(m-1) arcs
1903     if m == 2:
1904         candidates.append(one_directional_bipartite(n)) # floor(n^2/4) arcs
1905     elif (m - 2) < (n - n // 2):
1906         # augmented needs m-2 < ceil(n/2)
1907         candidates.append(augmented_bipartite(n, m))
1908     feasible = [g for g in candidates if max_edge_connectivity(g) <= m - 1]
1909     return max(feasible, key=lambda g: g.edge_count()) if feasible else None
1910
1911 def _brute_force_matrix(
1912     variant: Variant, n: int, m: int, separation: str, deadline: float,
1913 ) -> tuple[int, Graph | None, bool]:
1914     """Exhaustively maximise edges over every graph of ‘‘variant’’ on ‘‘n’’.
1915
1916     Returns ‘‘(best_edge_count, witness, completed)’’. ‘‘completed’’ is False if
1917     the budget ran out first, in which case the value is only a lower bound.
1918     A simple cell ranges over ‘‘{0, 1}’’; a multigraph cell over ‘‘{0, ..., m-1}’’
1919     because multiplicity ‘‘m’’ already gives ‘‘m’’ parallel disjoint routes.
1920     """
1921     measure = _connectivity_measure(separation)
1922     cells = _matrix_cells(n, variant.directed)
1923     span = 2 if variant.simple else m # cell value lives in range(span)
1924     best_count, best_graph, completed = 0, None, True
1925     base = Graph(n, variant)
1926     for tick, values in enumerate(product(range(span), repeat=len(cells))):
1927         if tick % 256 == 0 and time.time() > deadline:
1928             completed = False # ran out before exhausting the space
1929             break
1930         candidate = base.copy()
1931         for (u, v), value in zip(cells, values):
1932             if value:
1933                 candidate.set_multiplicity(u, v, value)
1934         if measure(candidate) <= m - 1 and candidate.edge_count() > best_count:
1935             best_count, best_graph = candidate.edge_count(), candidate.copy()

```

```

1935     return best_count, best_graph, completed
1936
1937
1938 def _search_within_budget(
1939     variant: Variant, n: int, m: int, separation: str,
1940     deadline: float, seed: int, method: str = "sa",
1941 ) -> SearchResult:
1942     """Repeated search restarts until the deadline; keep the densest run."""
1943     engine = tabu_search_for_dense_graph if method == "tabu" else
        ↪ search_for_dense_graph
1944     best: SearchResult | None = None
1945     restart = 0
1946     while True:
1947         # Pass the deadline through so a single long restart can be cut short.
1948         outcome = engine(variant, n, m, separation=separation,
1949                         steps=4000, seed=seed + restart, deadline=deadline)
1950         if best is None or outcome.best_edge_count > best.best_edge_count:
1951             best = outcome
1952             restart += 1
1953         if time.time() > deadline or restart >= 200:
1954             break
1955     assert best is not None
1956     return best
1957
1958
1959 def _hyperedge_candidates(n: int, r: int, directed: bool, *, kind: str = "forward"
        ↪ ) -> list:
1960     """Every possible 'r'-uniform hyperedge.
1961
1962     Undirected: every 'r'-subset. Directed: the 'kind' selects the
1963     orientation model.
1964
1965     * "forward" (default): one tail, 'r - 1' heads, stored in the legacy
1966       '(tail:int, heads)' form so existing forward results are untouched.
1967     * "backward": 'r - 1' tails, one head (the arc-reversal dual of
1968       forward), stored in the general '(tails, heads)' form.
1969     * "general": every split of an 'r'-subset into a non-empty tail set and
1970       a non-empty head set, i.e. forward, backward, and (from 'r >= 4') mixed.
1971     """
1972     if not directed:
1973         return [frozenset(s) for s in combinations(range(n), r)]
1974     if kind == "forward":
1975         # One tail and r-1 heads chosen from the rest (legacy form).
1976         return [(tail, frozenset(heads))
1977                 for tail in range(n)
1978                 for heads in combinations([v for v in range(n) if v != tail], r -
        ↪ 1)]
1979     if kind == "backward":
1980         # r-1 tails and one head: the reverse of every forward arc.
1981         return [(frozenset(tails), frozenset((head,)))
1982                 for head in range(n)
1983                 for tails in combinations([v for v in range(n) if v != head], r -
        ↪ 1)]
1984     if kind == "general":
1985         # Every ordered (tails, heads) split of each r-subset, both sides non-
        ↪ empty.
1986         out: list = []
1987         for subset in combinations(range(n), r):

```

```

1988         for mask in range(1, (1 << r) - 1):          # exclude all-heads and all-
               ↪ tails
1989             tails = frozenset(subset[i] for i in range(r) if mask & (1 << i))
1990             heads = frozenset(subset[i] for i in range(r) if not (mask & (1 <<
               ↪ i)))
1991             out.append((tails, heads))
1992         return out
1993     raise ValueError(f"unknown directed hyperedge kind {kind!r}")
1994
1995
1996 def _brute_force_hypergraph(
1997     n: int, r: int, m: int, deadline: float,
1998     *, directed: bool = False, vertex_split: bool = False, kind: str = "forward",
1999 ) -> tuple[int, Hypergraph | None, bool]:
2000     """Exhaustively maximise hyperedges over all ‘r’-uniform hypergraphs.
2001
2002     Each possible hyperedge is present or absent, so the search visits
2003     ‘2 ** (#candidates)’ hypergraphs; only tiny ‘(n, r)’ finish. The
2004     ‘directed’ and ‘vertex_split’ flags select which of the four hypergraph
2005     measures decides feasibility, exactly as ‘solve’ passes them through, and
2006     ‘kind’ selects the directed orientation model (forward/backward/general).
2007     """
2008     candidates = _hyperedge_candidates(n, r, directed, kind=kind)
2009     best_count, best_h, completed = 0, None, True
2010     for tick, mask in enumerate(product((0, 1), repeat=len(candidates))):
2011         if tick % 256 == 0 and time.time() > deadline:
2012             completed = False
2013             break
2014         chosen = [candidates[i] for i, on in enumerate(mask) if on]
2015         hypergraph = Hypergraph(n, chosen, directed=directed)
2016         if (max_hyper_connectivity(hypergraph, vertex_split=vertex_split) <= m - 1
2017             and hypergraph.edge_count() > best_count):
2018             best_count, best_h = hypergraph.edge_count(), hypergraph
2019     return best_count, best_h, completed
2020
2021
2022 def _random_hypergraph_search(
2023     n: int, r: int, m: int, deadline: float, seed: int,
2024     *, directed: bool = False, vertex_split: bool = False, kind: str = "forward",
2025 ) -> tuple[int, Hypergraph | None]:
2026     """Greedy randomised growth: add random hyperedges while feasible, restart.
2027
2028     A discovery heuristic for the hypergraph model (search is matrix-only):
2029     each pass shuffles the candidate hyperedges and adds each one that keeps the
2030     Berge connectivity within ‘m - 1’; the densest pass within budget wins. The
2031     feasible hypergraph it returns is the easy construction behind a lower bound.
2032     ‘kind’ selects the directed orientation model.
2033     """
2034     rng = random.Random(seed)
2035     candidates = _hyperedge_candidates(n, r, directed, kind=kind)
2036     best_count, best_h = 0, None
2037     while time.time() < deadline:
2038         order = candidates[:]
2039         rng.shuffle(order)
2040         hypergraph = Hypergraph(n, directed=directed)
2041         for edge in order:
2042             hypergraph.add_hyperedge(edge)
2043             if max_hyper_connectivity(hypergraph, vertex_split=vertex_split) > m -

```

```

2044         ↪ 1:
2045         hypergraph.hyperedges.pop()    # this one broke feasibility; undo
2046     if hypergraph.edge_count() > best_count:
2047         best_count, best_h = hypergraph.edge_count(), hypergraph
2048     return best_count, (best_h if best_h is not None else Hypergraph(n, directed=
2049         ↪ directed))
2050
2051 def _flow_at_least(cap: dict[tuple[int, int], int], num_nodes: int,
2052     source: int, sink: int, k: int) -> bool:
2053     """True if 'k' rounds of Ford--Fulkerson find an augmenting path each time.
2054
2055     The shared engine behind :func: '_arc_flow_at_least' and
2056     :func: '_vertex_flow_at_least', which differ only in how they build 'cap'
2057     (the plain arc network versus the vertex-split network) and in what
2058     'source'/'sink'/'num_nodes' mean for that network. Each round finds
2059     any augmenting path in the residual (forward arcs and the reverse arcs
2060     left by earlier augmentations) by DFS and pushes one unit of flow along
2061     it; the first round with no path means fewer than 'k' disjoint routes
2062     exist. Mutates 'cap' in place as the residual network, so callers pass
2063     a freshly built dict.
2064
2065     """
2066     for _ in range(k):
2067         prev = {source: source}
2068         stack = [source]                    # DFS for any residual augmenting path
2069         while stack:
2070             x = stack.pop()
2071             if x == sink:
2072                 break
2073             for y in range(num_nodes):
2074                 if y not in prev and cap.get((x, y), 0) > 0:
2075                     prev[y] = x
2076                     stack.append(y)
2077             if sink not in prev:
2078                 return False                # no further path: fewer than k exist
2079         node = sink                        # walk back, pushing one unit of flow
2080         while node != source:
2081             p = prev[node]
2082             cap[(p, node)] -= 1
2083             cap[(node, p)] = cap.get((node, p), 0) + 1
2084             node = p
2085     return True
2086
2087 def _arc_flow_at_least(out_adj: list[set[int]], n: int, s: int, t: int,
2088     k: int) -> bool:
2089     """True if there are at least 'k' arc-disjoint 's'-'t' paths.
2090
2091     Builds the plain unit-capacity arc network and hands it to
2092     :func: '_flow_at_least'. Used as the inner feasibility test of the
2093     exhaustive digraph search, where the question is only whether the
2094     local connectivity has reached the forbidden value 'm' (so 'k = m').
2095
2096     """
2097     cap: dict[tuple[int, int], int] = {}
2098     for a in range(n):
2099         for b in out_adj[a]:
2100             cap[(a, b)] = 1
2101     return _flow_at_least(cap, n, s, t, k)

```

```

2100
2101
2102 def _vertex_flow_at_least(out_adj: list[set[int]], n: int, s: int, t: int,
2103                             k: int) -> bool:
2104     """True if there are at least 'k' internally vertex-disjoint 's'-'t'
        ↪ paths.
2105
2106     The vertex twin of :func: '_arc_flow_at_least': builds the split network
2107     instead of the plain one, every vertex 'x' becoming an entry copy
2108     '2x' and an exit copy '2x+1' joined by a single unit-capacity arc so
2109     a path may pass through 'x' at most once, with the two endpoints
2110     keeping an uncapped gate since a route is allowed to start at 's' and
2111     finish at 't' (a direct arc 's -> t' therefore counts as one route
2112     with no interior, exactly as the thesis's separation axis intends), and
2113     hands that network to the same :func: '_flow_at_least' engine.
2114     """
2115     cap: dict[tuple[int, int], int] = {}
2116     for x in range(n):
2117         cap[(2 * x, 2 * x + 1)] = k if (x == s or x == t) else 1
2118     for a in range(n):
2119         for b in out_adj[a]:
2120             cap[(2 * a + 1, 2 * b)] = 1
2121     return _flow_at_least(cap, 2 * n, 2 * s + 1, 2 * t, k)
2122
2123
2124 def _exhaustive_directed(
2125     n: int, m: int, separation: str, deadline: float,
2126     stats: dict[str, int] | None = None,
2127 ) -> tuple[int, Graph | None, bool]:
2128     """Prove the simple directed maximum by a pruned exhaustive search.
2129
2130     This is the honest version of the thesis's base-case search: branch and bound
2131     over every simple digraph on 'n' vertices, keeping the densest one whose
2132     connectivity stays within 'm - 1'. Two prunes make it finish where naive
2133     enumeration cannot. Feasibility is monotone, so the include branch adds an
2134     arc only when the digraph is still feasible (an infeasible prefix can never be
2135     rescued by adding more arcs), and the bound prune drops a subtree once even
2136     taking every remaining arc could not beat the best found. Both separations
2137     use the same incremental feasibility test: a new arc can only lift the
2138     connectivity of a pair whose source reaches its tail and whose sink is
2139     reachable from its head, so only those pairs are re-measured, with
2140     :func: '_arc_flow_at_least' or :func: '_vertex_flow_at_least' according to the
2141     separation. Returns '(max_count, witness, completed)'; 'completed' is
2142     False if the time budget ran out first, leaving the value a lower bound.
2143     Pass 'stats' to receive the number of search-tree nodes visited.
2144     """
2145     pairs = [(u, v) for u in range(n) for v in range(n) if u != v]
2146     total = len(pairs)
2147     out: list[set[int]] = [set() for _ in range(n)]
2148     inc: list[set[int]] = [set() for _ in range(n)]
2149     # Seed the best at one below a known construction so the bound prune bites
2150     # immediately, and the search still records an actual witness when it ties it.
2151     # The same seed is valid in both separations: by Whitney's inequality
2152     #  $\kappa \leq \lambda$ , a digraph feasible for the arc problem is feasible for the
2153     # vertex problem too, so an arc construction is an honest vertex lower bound.
2154     seed = _directed_witness(n, m, simple=True)
2155     seed_value = seed.edge_count() if seed is not None else 0
2156     best_count = [max(0, seed_value - 1)]

```

```

2157 best_arcs: list[tuple[int, int]] = []
2158 timed_out = [False]
2159
2160 def reaches_to(target: int) -> set[int]:
2161     seen, stack = {target}, [target]    # reverse reachability (can reach t)
2162     while stack:
2163         x = stack.pop()
2164         for y in inc[x]:
2165             if y not in seen:
2166                 seen.add(y); stack.append(y)
2167     return seen
2168
2169 def reaches_from(source: int) -> set[int]:
2170     seen, stack = {source}, [source]    # forward reachability (s can reach)
2171     while stack:
2172         x = stack.pop()
2173         for y in out[x]:
2174             if y not in seen:
2175                 seen.add(y); stack.append(y)
2176     return seen
2177
2178 flow_at_least = (_arc_flow_at_least if separation == "edge"
2179                 else _vertex_flow_at_least)
2180
2181 def feasible_after(u: int, v: int) -> bool:
2182     for s in reaches_to(u):
2183         for t in reaches_from(v):
2184             if s != t and flow_at_least(out, n, s, t, m):
2185                 return False    # this pair reached m disjoint paths
2186     return True
2187
2188 def recurse(idx: int, count: int) -> None:
2189     if stats is not None:
2190         stats["nodes"] = stats.get("nodes", 0) + 1
2191     if time.time() > deadline:
2192         timed_out[0] = True
2193         return
2194     if count + (total - idx) <= best_count[0]:
2195         return    # cannot beat the incumbent: prune
2196     if idx == total:
2197         if count > best_count[0]:
2198             best_count[0] = count
2199             best_arcs[:] = [(a, b) for a in range(n) for b in out[a]]
2200         return
2201     u, v = pairs[idx]
2202     out[u].add(v); inc[v].add(u)    # include the arc, if still feasible
2203     if feasible_after(u, v):
2204         recurse(idx + 1, count + 1)
2205     out[u].discard(v); inc[v].discard(u)
2206     if timed_out[0]:
2207         return
2208     recurse(idx + 1, count)    # exclude the arc
2209
2210 recurse(0, 0)
2211 witness = None
2212 if best_arcs:
2213     witness = Graph(n, SIMPLE_DIRECTED)
2214     for (a, b) in best_arcs:

```

```

2215         witness.mu[a, b] = 1
2216     elif seed is not None:
2217         witness = seed # max equalled the seed construction
2218         # A completed search always reaches at least the seed's count, so the max
2219         # below only matters on a timeout before any recorded leaf: there it lifts
2220         # the reported lower bound from seed_value - 1 (the pruning incumbent) to
2221         # the seed witness's own arc count, keeping value and witness consistent.
2222         return max(best_count[0], seed_value), witness, not timed_out[0]
2223
2224
2225 def solve(
2226     n: int, m: int, *,
2227     directed: bool = False, simple: bool = True,
2228     hypergraph: bool = False, r: int = 3,
2229     exhaustive: bool = False, separation: str = "edge",
2230     max_seconds: float = 60.0, seed: int = 0, method: str = "tabu",
2231 ) -> SolveResult:
2232     """The single driver: prove the exact value, or discover a dense example.
2233
2234     Discovery defaults to "method="tabu": tabu search is the stronger engine on
2235     the harder directed problems and the one used to solve the problems in this
2236     work. Simulated annealing ("method="sa") is kept for the self-check and
2237     verification, where the lighter engine is enough.
2238
2239     Args:
2240         n, m: the vertices and the forbidden number of independent routes.
2241         directed, simple: the two matrix axes (ignored when "hypergraph").
2242         hypergraph, r: switch to the "r"-uniform hypergraph model instead.
2243         exhaustive: "True" to PROVE the optimum, "False" to DISCOVER one.
2244         separation: "edge" or "vertex" disjointness (matrix models).
2245         max_seconds: wall-clock budget; always respected.
2246         seed: random seed for the discovery searches.
2247
2248     Returns:
2249         A :class:'SolveResult' whose "bound" is "exact" (proved),
2250         "lower" (a witness), or "upper" (proved, no matching witness).
2251     """
2252     start = time.time()
2253     deadline = start + max_seconds
2254
2255     # ----- the hypergraph model -----
2256     # Same driver, a different internal measure: the edge/vertex separation and
2257     # the direction pick which of the four Berge measures decides feasibility.
2258     if hypergraph:
2259         vertex_split = (separation == "vertex")
2260         label = f"{r}-uniform {'directed ' if directed else ''}hypergraph"
2261         if exhaustive:
2262             value, witness, done = _brute_force_hypergraph(
2263                 n, r, m, deadline, directed=directed, vertex_split=vertex_split)
2264             bound = "exact" if done else "lower"
2265             method = "brute-force enumeration"
2266             note = "" if done else "budget ran out; value is only a lower bound"
2267         else:
2268             value, witness = _random_hypergraph_search(
2269                 n, r, m, deadline, seed, directed=directed, vertex_split=
2270                     ↪ vertex_split)
2271             bound, done, method = "lower", True, "randomised greedy search"
2272             note = "discovery only ever yields a lower bound"

```



```

2272         return SolveResult(n, m, label, separation, value, bound, method,
2273                             time.time() - start, done, witness, note)
2274
2275     # ----- the matrix models -----
2276     variant = _variant_for(directed, simple)
2277     label = variant.describe()
2278
2279     # DISCOVER: search within the budget; the witness found is a lower bound.
2280     if not exhaustive:
2281         if method not in ("sa", "tabu"):
2282             raise ValueError("method must be 'sa' or 'tabu'")
2283         result = _search_within_budget(variant, n, m, separation, deadline, seed,
2284                                       ↪ method)
2285         method_label = "tabu search" if method == "tabu" else "random search"
2286         return SolveResult(
2287             n, m, label, separation, result.best_edge_count, "lower",
2288             method_label, time.time() - start, True,
2289             result.best_graph, "discovery only ever yields a lower bound")
2290
2291     # EXHAUSTIVE, simple directed: a pruned exhaustive digraph search is exact.
2292     # This is the prover for the m=2 base cases of the directed theorem, and it
2293     # reaches n = 6, 7 where the cut-counting cannot close the gap in any sane
2294     ↪ time.
2295     if directed and simple:
2296         value, witness, done = _exhaustive_directed(n, m, separation, deadline)
2297         bound = "exact" if done else "lower"
2298         method = "exhaustive digraph search (branch and bound)"
2299         note = "" if done else "budget ran out; value is only a lower bound"
2300         return SolveResult(n, m, label, separation, value, bound, method,
2301                             time.time() - start, done, witness, note)
2302
2303     # EXHAUSTIVE, directed MULTIGRAPH arc problem: the cut-counting is the prover.
2304     if directed and separation == "edge":
2305         budget = max(1.0, deadline - time.time())
2306         proof = prove_directed_multigraph(n, time_limit=budget)
2307         if proof.is_proof():
2308             # We only reach here for a multigraph, where (m-1) M*(n) IS the value.
2309             value = proof.value_for(m) # (m-1) * M*(n)
2310             witness = _directed_witness(n, m, simple)
2311             return SolveResult(n, m, label, separation, value, "exact",
2312                               "cut-counting (gap zero)", time.time() - start,
2313                               True, witness, "")
2314         # Timed out: fall back to the best construction as a lower bound.
2315         witness = _directed_witness(n, m, simple)
2316         low = witness.edge_count() if witness else 0
2317         return SolveResult(
2318             n, m, label, separation, low, "lower",
2319             "cut-counting hit the time limit; reporting a construction",
2320             time.time() - start, False, witness,
2321             "raise max_seconds to let the cut-counting close the gap")
2322
2323     # EXHAUSTIVE otherwise (undirected, or vertex separation): brute force.
2324     value, witness, done = _brute_force_matrix(
2325         variant, n, m, separation, deadline)
2326     bound = "exact" if done else "lower"
2327     method = "brute-force enumeration"
2328     note = (" if done else "budget ran out; value is only a lower bound "
2329            "(no cut-counting exists for this case)")

```

```

2328     return SolveResult(n, m, label, separation, value, bound, method,
2329                        time.time() - start, done, witness, note)

```

C.1.5 The random model

Here begins the material behind [Chapter 4](#). This first piece builds random graphs and measures them, which is how the thesis looks at the typical case rather than the extreme one. Each of the six models gets its own sampler, and the multigraph samplers use a two-stage rule: decide whether a pair is connected at all, then decide how many parallel copies it gets, which recovers the ordinary random graph when the second stage is switched off.

```

2332 #####
2333 ##
2334 ## CHAPTER 4 -- SYNTHESIS AND RESULTS
2335 ## Sampling the random model, the thesis figures, and the self-check.
2336 ##
2337 #####
2338
2339 # --- MONTE CARLO: sample G(n,p) graphs, measure with exact checker, average ---
2340
2341
2342 def sample_random_graph(n: int, p: float, directed: bool, rng: random.Random) ->
↳ Graph:
2343     """Draw one sample of G(n, p) (or the random digraph D(n, p)).
2344
2345     Each possible edge is included independently with probability 'p': unordered
2346     pairs for an undirected graph, ordered pairs for a digraph.
2347     """
2348     variant = SIMPLE_DIRECTED if directed else SIMPLE_UNDIRECTED
2349     graph = Graph(n, variant)
2350     for u in range(n):
2351         # Undirected: scan v > u (each pair once). Directed: scan all v.
2352         start = 0 if directed else u + 1
2353         for v in range(start, n):
2354             if u != v and rng.random() < p: # include this edge with prob p
2355                 graph.add_edge(u, v)
2356     return graph
2357
2358
2359 def estimate_appearance_probability(
2360     n: int, p: float, m: int, *, trials: int = 200,
2361     separation: str = "edge", directed: bool = False, seed: int = 0,
2362 ) -> float:
2363     """Estimate the probability that a sample has  $\lambda^{\max} \geq m$ ."""
2364     rng = random.Random(seed)
2365     measure = _connectivity_measure(separation)
2366     # Count the fraction of samples that already exhibit m independent routes
2367     # somewhere ( $\lambda^{\max} \geq m$ ): the empirical appearance probability.
2368     appearances = sum(
2369         measure(sample_random_graph(n, p, directed, rng)) >= m
2370         for _ in range(trials)
2371     )
2372     return appearances / trials
2373
2374

```

```

2375 def connectivity_distribution(
2376     n: int, p: float, *, trials: int = 200,
2377     separation: str = "edge", directed: bool = False, seed: int = 0,
2378 ) -> list[int]:
2379     """Return the list of  $\lambda^{\max}$  (or  $\kappa^{\max}$ ) over samples.
2380
2381     Same arguments as the estimators above. The returned list has one entry per
2382     sample, so its histogram shows how the binding connectivity is distributed
2383     for that kind of graph. Calling this with the same ‘seed’ and ‘(n, p)’
2384     but different ‘separation’ reuses the very same random graphs, which is why
2385     the edge and vertex distributions can be compared sample by sample.
2386     """
2387     rng = random.Random(seed)
2388     measure = _connectivity_measure(separation)
2389     # Same seed + same (n, p) reproduces the identical sequence of graphs, so an
2390     # edge run and a vertex run can be compared sample by sample (Whitney check).
2391     return [measure(sample_random_graph(n, p, directed, rng)) for _ in range(
        ↪ trials)]
2392
2393
2394 @dataclass
2395 class ThresholdCurve:
2396     """The appearance probability sampled across a range of densities."""
2397
2398     n: int
2399     m: int
2400     probabilities: list[tuple[float, float]] # (p, estimated probability)
2401
2402     @property
2403     def predicted_threshold(self) -> float:
2404         """The proved location  $p^* = m/n$ ."""
2405         return self.m / self.n
2406
2407
2408 def threshold_curve(
2409     n: int, m: int, p_values: list[float], *, trials: int = 200,
2410     separation: str = "edge", directed: bool = False, seed: int = 0,
2411 ) -> ThresholdCurve:
2412     """Sweep ‘p’ and return the appearance probability at each value."""
2413     points = [
2414         (p, estimate_appearance_probability(
2415             n, p, m, trials=trials, separation=separation, directed=directed, seed
2416             ↪ =seed))
2417         for p in p_values
2418     ]
2419     return ThresholdCurve(n=n, m=m, probabilities=points)
2420
2421 # --- edge against vertex disjointness in the random model -----
2422 #
2423 # Whitney's inequality  $\kappa^{\max} \leq \lambda^{\max}$  leaves open whether the two
2424 # binding connectivities of a graph actually differ. For the EXTREMAL problem
2425 # they first part company at  $m = 5$  (Sorensen-Thomassen). The function below lets
2426 # the same split be watched in the random model: it draws samples of  $G(n, p)$  and
2427 # measures both maxima on each one. On small graphs, where the binding
2428 # connectivity stays at or below four, the two coincide on every sample; on
2429 # larger, denser graphs, where it climbs past five, a pair can carry more
2430 # edge-disjoint routes than internally vertex-disjoint ones, and  $\kappa^{\max}$  drops

```

```

2431 # below  $\lambda^{\max}$  on a growing fraction of samples.
2432
2433
2434 def edge_vertex_distribution(
2435     n: int, p: float, *, trials: int = 300, seed: int = 0,
2436 ) -> tuple[list[int], list[int]]:
2437     """Return paired ( $\lambda^{\max}$ ,  $\kappa^{\max}$ ) over the same samples of  $G(n, p)$ .
2438
2439     Both lists are measured on the identical sequence of random graphs, because
2440     connectivity_distribution reuses the seed, so the two can be compared sample
2441     by sample: the fraction of entries with  $\kappa^{\max} < \lambda^{\max}$  is exactly how
2442     often vertex disjointness is strictly more demanding than edge disjointness
2443     at that size and density.
2444     """
2445     lambda_max = connectivity_distribution(
2446         n, p, trials=trials, separation="edge", seed=seed)
2447     kappa_max = connectivity_distribution(
2448         n, p, trials=trials, separation="vertex", seed=seed)
2449     return lambda_max, kappa_max
2450
2451
2452 # --- SAMPLING ALL VARIANTS: Bernoulli for simple/hypergraph, hurdle-geometric for
2453     ↪ multigraph ---
2454
2455 def sample_random_multigraph(
2456     n: int, p: float, alpha: float, directed: bool, rng: random.Random,
2457     *, cap: int | None = None,
2458 ) -> Graph:
2459     """Random multigraph with hurdle-geometric edge multiplicities.
2460
2461     Each cell is empty with probability ' $1 - p$ '; otherwise it carries one copy
2462     plus a Geometric(' $\alpha$ ') number of further copies, so the multiplicity
2463     decays exponentially with rate ' $\alpha$ ' and ' $\alpha = 0$ ' recovers simple
2464     ' $G(n, p)$ '. With ' $\text{cap}$ ' set (use ' $m - 1$ ') a single fat edge cannot
2465     trivially break feasibility, so the panel stays a structural question.
2466     """
2467     variant = MULTI_DIRECTED if directed else MULTI_UNDIRECTED
2468     graph = Graph(n, variant)
2469     for u in range(n):
2470         start = 0 if directed else u + 1
2471         for v in range(start, n):
2472             if u == v or rng.random() >= p:
2473                 continue
2474             k = 1
2475             while rng.random() < alpha:
2476                 k += 1
2477             graph.set_multiplicity(u, v, k if cap is None else min(k, cap))
2478     return graph
2479
2480
2481 def sample_random_hypergraph(
2482     n: int, p: float, r: int, directed: bool, rng: random.Random,
2483 ) -> Hypergraph:
2484     """Random ' $r$ '-uniform hypergraph: each candidate hyperedge included w.p. ' $p$ '
2485     ↪ '.,'"""
2486     chosen = [c for c in _hyperedge_candidates(n, r, directed) if rng.random() < p
2487     ↪ ]

```

```

2486     return Hypergraph(n, chosen, directed=directed)
2487
2488
2489 def _sample_variant(n, p, *, directed, simple, hypergraph, r, alpha, cap, rng):
2490     """Draw one sample of whichever generative model the flags select."""
2491     if hypergraph:
2492         return sample_random_hypergraph(n, p, r, directed, rng)
2493     if simple:
2494         return sample_random_graph(n, p, directed, rng)
2495     return sample_random_multigraph(n, p, alpha, directed, rng, cap=cap)
2496
2497
2498 def _measure_variant(obj, *, separation, hypergraph):
2499     """Binding connectivity of a sampled object under the chosen separation."""
2500     if hypergraph:
2501         return max_hyper_connectivity(obj, vertex_split=(separation == "vertex"))
2502     return _connectivity_measure(separation)(obj)
2503
2504
2505 def _mean_binding_degree(obj, *, directed, hypergraph):
2506     """The degree that the degree bound caps connectivity by, averaged over
        ↪ vertices.
2507
2508     Undirected: total incident multiplicity. Directed: ‘min(d+, d-)’ (the
2509     quantity bounding an arc-disjoint count). Hypergraph: the Berge degree, the
2510     number of hyperedges through a vertex.
2511     """
2512     n = obj.num_vertices
2513     if hypergraph:
2514         deg = [0] * n
2515         for edge in obj.hyperedges:
2516             for v in obj.members(edge):
2517                 deg[v] += 1
2518         return sum(deg) / n
2519     if directed:
2520         return sum(min(obj.out_degree(v), obj.in_degree(v)) for v in range(n)) / n
2521     return sum(obj.degree(v) for v in range(n)) / n
2522
2523
2524 def _p_for_target_degree(target_deg, n, *, directed, simple, hypergraph, r, alpha)
2525     ↪ :
2526     """Presence probability ‘p’ that puts the expected degree near ‘target_deg
2527     ↪ ‘.
2528
2529     Only an aiming heuristic -- the figures plot the *measured* mean degree -- so
2530     the exact value of the cap or of the geometric tail does not need inverting.
2531     """
2532     if hypergraph:
2533         per_vertex = math.comb(n - 1, r - 1)
2534         if directed:
2535             per_vertex += (n - 1) * math.comb(n - 2, r - 2)
2536         return min(1.0, target_deg / max(per_vertex, 1))
2537     copies = 1.0 if simple else 1.0 / (1.0 - alpha)
2538     return min(1.0, target_deg / ((n - 1) * copies))
2539
2540
2541 # Twelve sampled panels, mirroring _VARIANT_ENUM_CONFIGS but at sampling sizes
2542 # well past the enumeration wall (n=6). ‘sample_n’ is chosen per variant so

```

```

2541 # the binding-connectivity sweep stays a few minutes overall.
2542 _VARIANT_SAMPLE_CONFIGS: list[dict] = [
2543     dict(key="simple_undirected_edge", title="simple undirected edge",
2544         directed=False, simple=True, hypergraph=False, r=3, separation="edge",
2545         ↪ sample_n=26),
2546     dict(key="simple_undirected_vertex", title="simple undirected vertex",
2547         directed=False, simple=True, hypergraph=False, r=3, separation="vertex",
2548         ↪ sample_n=18),
2549     dict(key="simple_directed_edge", title="simple directed arc",
2550         directed=True, simple=True, hypergraph=False, r=3, separation="edge",
2551         ↪ sample_n=14),
2552     dict(key="simple_directed_vertex", title="simple directed vertex",
2553         directed=True, simple=True, hypergraph=False, r=3, separation="vertex",
2554         ↪ sample_n=14),
2555     dict(key="multi_undirected_edge", title="multigraph undirected edge",
2556         directed=False, simple=False, hypergraph=False, r=3, separation="edge",
2557         ↪ sample_n=22),
2558     dict(key="multi_undirected_vertex", title="multigraph undirected vertex",
2559         directed=False, simple=False, hypergraph=False, r=3, separation="vertex",
2560         ↪ sample_n=16),
2561     dict(key="multi_directed_edge", title="multigraph directed arc",
2562         directed=True, simple=False, hypergraph=False, r=3, separation="edge",
2563         ↪ sample_n=12),
2564     dict(key="multi_directed_vertex", title="multigraph directed vertex",
2565         directed=True, simple=False, hypergraph=False, r=3, separation="vertex",
2566         ↪ sample_n=12),
2567     dict(key="hyper_undirected_edge", title="hypergraph undirected edge",
2568         directed=False, simple=True, hypergraph=True, r=3, separation="edge",
2569         ↪ sample_n=12),
2570     dict(key="hyper_undirected_vertex", title="hypergraph undirected vertex",
2571         directed=False, simple=True, hypergraph=True, r=3, separation="vertex",
2572         ↪ sample_n=12),
2573     dict(key="hyper_directed_edge", title="hypergraph directed arc",
2574         directed=True, simple=True, hypergraph=True, r=3, separation="edge",
2575         ↪ sample_n=9),
2576     dict(key="hyper_directed_vertex", title="hypergraph directed vertex",
2577         directed=True, simple=True, hypergraph=True, r=3, separation="vertex",
2578         ↪ sample_n=9),
2579 ]

```

C.1.6 Drawing the figures

Every picture in the thesis is produced by the routines below, from the same data structures the rest of the program uses. They are collected here rather than in the figure script next door so that a plot and the computation behind it stay in one file.

```

2570 # --- FIGURES: headless matplotlib PNGs; each function takes a path and writes a
2571     ↪ file ---
2572
2573 _KUL_BLUE = "#1D8DB0" # feasible / proved
2574 _KUL_DARK = "#1E6E87" # row labels, known-max lines
2575 _KUL_LIGHT = "#52BDEC" # named branches
2576 _WARM = "#DC8C28" # feasibility boundary, certain interval
2577 _RED = "#E05050" # infeasible bars
2578 _GREEN = "#3CA050" # machine-checked exact markers
2579 _VIOLET = "#9B5DE5" # search lower-bound markers

```

```

2579 _GUESS = "#E6B800"          # "guess" interpolation: a clear yellow/gold, kept well
2580                                # apart from the red conjecture curve (proved=blue,
2581                                # conjectured=red, guess=yellow)
2582
2583 # Four-level sensitivity palette: cool (sigma=0) -> hot (sigma=max).
2584 _SIGMA_PALETTE = ["#52BDEC", "#F9C74F", "#F4914B", "#DC8C28"] # 0,1,2,3+
2585
2586
2587 def plot_directed_crossover(m: int, max_n: int, path: str | Path) -> None:
2588     """Plot the two competing directed branches and their maximum versus 'n'.
2589
2590     Shows how the linear hub branch 'm(n-1)' is overtaken by the quadratic
2591     augmented-bipartite branch near 'n ~ 2m': the heart of the directed story.
2592     """
2593     ns = list(range(2, max_n + 1))
2594     hub = [m * (n - 1) for n in ns] # linear hub branch
2595     # Quadratic branch: the shifted-partition augmented bipartite count
2596     # floor((n+m-2)^2/4) of const: augmented-bipartite (at m <= 3 it equals the
2597     # balanced-partition count floor(n^2/4) + (m-2)ceil(n/2)).
2598     bipartite = [(n + m - 2) ** 2 // 4 for n in ns]
2599     envelope = [directed_arc_lower_bound(n, m) for n in ns] # their pointwise max
2600
2601     plt.figure(figsize=(7, 4.3))
2602     plt.plot(ns, hub, "--", color=_KUL_DARK, label=r"hub branch $m(n-1)$")
2603     plt.plot(ns, bipartite, "--", color=_WARM,
2604              label=r"bipartite branch $\lfloor (n+m-2)^2/4 \rfloor$")
2605     plt.plot(ns, envelope, "-", color=_KUL_BLUE, linewidth=2.4, label="their
2606             ↪ maximum")
2607
2608     # Mark the crossover: the first n at which the quadratic branch overtakes the
2609     # linear one, the order (linear in m) where the directed story changes hands.
2610     cross_n = next((n for n, h, b in zip(ns, hub, bipartite) if b > h), None)
2611     if cross_n is not None:
2612         y_cross = directed_arc_lower_bound(cross_n, m)
2613         plt.axvline(cross_n, color="gray", linestyle=":", linewidth=1.3, alpha
2614                    ↪ =0.8)
2615         plt.scatter([cross_n], [y_cross], color=_KUL_BLUE, zorder=5,
2616                    s=45, edgecolor="white", linewidth=0.8)
2617         plt.annotate(f"crossover at $n = {cross_n}$",
2618                    xy=(cross_n, y_cross),
2619                    xytext=(cross_n - 2.8, max(envelope) * 0.48),
2620                    fontsize=9.5, color="black", ha="center",
2621                    arrowprops=dict(arrowstyle="->", color="gray", lw=1.1))
2622
2623     plt.xlabel("number of vertices $n$")
2624     plt.ylabel("arcs")
2625     plt.title(f"Directed lower bound at $m = {m}$")
2626     plt.legend(loc="upper left")
2627     plt.grid(True, alpha=0.3)
2628     _save(path)
2629
2630 def plot_edge_vertex_divergence(max_n: int, path: str | Path) -> None:
2631     """Show agreement through m<=4 and divergence at m=5, all starting at the same
2632     ↪ n.
2633
2634     All curves begin at n_start so the reader can compare directly.
2635     For m=2,3,4 the two problems coincide (one line each, blue shades).

```

```

2634     At m=5 they part: edge below, vertex above, with the gap shaded.
2635     """
2636     n_start = 4 # common start for all curves
2637     ns = list(range(n_start, max_n + 1))
2638
2639     plt.figure(figsize=(8, 5))
2640
2641     # m=2,3,4: edge == vertex -- one curve per m
2642     agree_palette = ["#AED9EE", "#5CB4D9", _KUL_DARK]
2643     for m_val, color in zip([2, 3, 4], agree_palette):
2644         vals = [simple_undirected_edge(n, m_val) for n in ns]
2645         plt.plot(ns, vals, "--", color=color, linewidth=1.8, alpha=0.9,
2646                 label=f"$m={m_val}$: edge $=$ vertex")
2647
2648     # m=5: edge and vertex. Sorensen-Thomassen only determine  $k_5$  from  $n = 6$ ,
2649     # so the vertex curve starts there rather than at  $n_{\text{start}}$ . The two agree
2650     # for every  $n \leq 13$  and first come apart at  $n = 14$ , which is what the
2651     # shading marks.
2652     ns5 = [n for n in ns if n >= 6]
2653     edge5 = [simple_undirected_edge(n, 5) for n in ns5]
2654     vert5 = [simple_undirected_vertex_m5(n) for n in ns5]
2655     plt.plot(ns5, edge5, "--", color=_KUL_BLUE, linewidth=2.4,
2656             label=r"$m=5$: edge $\ell_5(n)=\lfloor 5(n-1)/2 \rfloor$")
2657     plt.plot(ns5, vert5, "-", color=_WARM, linewidth=2.4,
2658             label=r"$m=5$: vertex $k_5(n)=\lfloor 8n/3 \rfloor - 4$")
2659     # shade only where the gap is positive
2660     plt.fill_between(ns5, edge5, vert5, where=[v > e for v, e in zip(vert5, edge5)
2661         ↪ ],
2662                     alpha=0.15, color=_WARM)
2663
2664     plt.xlabel("number of vertices $n$")
2665     plt.ylabel("maximum edges")
2666     plt.title("Agreement through $m \leq 4$, first divergence at $m = 5$")
2667     plt.legend(fontsize=9, loc="upper left")
2668     plt.grid(True, alpha=0.3)
2669     _save(path)
2670
2671     # NOTE: plot_degree_threshold generated the 2-D random-sampling threshold figure
2672     # (the old Figure 4.2) removed from the thesis on 2026-06-20. Kept here so the
2673     # threshold phenomenon can be restored; see
2674     # research_notes/removed_threshold_phenomenon.md for the recipe. (Its 3-D
2675     # companion plot_conn_threshold_3d is still used, for the appendix.)
2676     def plot_degree_threshold(
2677         path: str | Path, *, n: int = 12, m: int = 4, alpha: float = 0.5,
2678         trials: int = 70, seed: int = 7,
2679         targets=(0.3, 0.5, 0.65, 0.8, 0.95, 1.1, 1.3, 1.6, 2.0),
2680     ) -> None:
2681         """Appearance probability against mean binding-degree, across every model.
2682
2683         Six random models -- simple, multigraph and 3-uniform hypergraph, each
2684         undirected and directed -- are swept in density and plotted by their
2685         *measured* mean binding degree (in units of ‘m’) against the probability
2686         that a sample already contains a pair of (Berge) edge-connectivity at least
2687         ‘m’. The vertical line at degree ‘= m’ is the asymptotic threshold the
2688         degree argument of thm:gnp-threshold predicts; at the fixed small ‘m’ a
2689         computation can reach the curves are ORDERED rather than coincident -- a
2690         directed multigraph forces the pair at the least degree (heavy parallel edges

```



```

2691     concentrate connectivity), a directed hypergraph at the most (a hyperedge
2692     serves only one Berge route), with the simple cases between. Universality is
2693     the ‘‘m/ln n -> infinity’’ limit; the spread here is the finite-m gap.
2694     """
2695     models = [
2696         ("simple, undirected", dict(directed=False, simple=True,
2697             ↪ hypergraph=False, r=3), _KUL_LIGHT, "o", "-"),
2698         ("simple, directed", dict(directed=True, simple=True,
2699             ↪ hypergraph=False, r=3), _KUL_BLUE, "s", "-"),
2700         ("multigraph, undirected", dict(directed=False, simple=False,
2701             ↪ hypergraph=False, r=3), _WARM, "o", "--"),
2702         ("multigraph, directed", dict(directed=True, simple=False,
2703             ↪ hypergraph=False, r=3), "#C0392B", "s", "--"),
2704         ("$$-uniform hyper, undir.", dict(directed=False, simple=True,
2705             ↪ hypergraph=True, r=3), _VIOLET, "^", ":"),
2706         ("$$-uniform hyper, dir.", dict(directed=True, simple=True,
2707             ↪ hypergraph=True, r=3), _GREEN, "^", ":"),
2708     ]
2709     plt.figure(figsize=(7.8, 5.0))
2710     for label, spec, colour, mk, ls in models:
2711         xs, ys = [], []
2712         for t in targets:
2713             p = _p_for_target_degree(t * m, n, alpha=alpha, **spec)
2714             rng = random.Random(seed)
2715             degs, hits = [], 0
2716             for _ in range(trials):
2717                 obj = _sample_variant(n, p, alpha=alpha, cap=m - 1, rng=rng, **
2718                     ↪ spec)
2719                 degs.append(_mean_binding_degree(
2720                     obj, directed=spec["directed"], hypergraph=spec["hypergraph"])
2721                     ↪ )
2722                 if _measure_variant(obj, separation="edge",
2723                     ↪ hypergraph=spec["hypergraph"]) >= m:
2724                     hits += 1
2725             xs.append(sum(degs) / len(degs) / m)
2726             ys.append(hits / trials)
2727         plt.plot(xs, ys, ls, marker=mk, color=colour, markersize=4.5,
2728             ↪ linewidth=1.8, label=label)
2729     plt.axvline(1.0, color="black", linestyle="--", linewidth=1.6,
2730         ↪ label=r"asymptotic threshold (degree $= m$)")
2731     plt.axhline(0.5, color="gray", linestyle=":", linewidth=1.0, alpha=0.6)
2732     plt.xlabel(r"mean binding degree, in units of $m$")
2733     plt.ylabel(r"$\hat{P}\backslash, \backslash, \text{binding connectivity} \geq m \backslash, $")
2734     plt.title(f"Appearance vs expected degree, by model "
2735         ↪ f"$n = {n}$, $m = {m}$, multiplicities capped at $m-1$")
2736     plt.ylim(-0.03, 1.03)
2737     plt.legend(fontsize=8.5, loc="lower right")
2738     plt.grid(True, alpha=0.3)
2739     _save(path)
2740
2741 def plot_connectivity_distribution(series: dict, n: int, p: float, path: str |
2742     ↪ Path) -> None:
2743     """Plot overlaid histograms of the binding connectivity for several models.
2744
2745     ‘‘series’’ maps a label to a list of connectivity values (one per random
2746     sample), as produced by :func:‘montecarlo.connectivity_distribution’. The
2747     integer-valued histograms are drawn on shared integer bins so the shapes can

```

```

2740     be compared directly across the kinds of graph.
2741     """
2742     # Shared integer bins across every series so the shapes line up exactly.
2743     all_values = [value for values in series.values() for value in values]
2744     low, high = min(all_values), max(all_values)
2745     levels = list(range(low, high + 1))
2746     colours = [_KUL_BLUE, _WARM, _KUL_DARK, _KUL_LIGHT]
2747     group_count = len(series)
2748     bar_width = 0.8 / group_count # split each integer slot among the series
2749
2750     plt.figure(figsize=(7, 4.3))
2751     for index, ((label, values), colour) in enumerate(zip(series.items(), colours)
2752     ↪ ):
2753         counts = [values.count(level) for level in levels]
2754         # Offset each series' bars so grouped bars sit side by side per level.
2755         offset = (index - (group_count - 1) / 2) * bar_width
2756         positions = [level + offset for level in levels]
2757         plt.bar(positions, counts, width=bar_width, color=colour, label=label,
2758                 edgecolor="white", linewidth=0.4)
2759     plt.xticks(levels)
2760     plt.xlabel("binding connectivity")
2761     plt.ylabel("number of samples")
2762     plt.title(f"Connectivity over random graphs, $n = {n}$, $p = {p}$")
2763     plt.legend()
2764     plt.grid(True, axis="y", alpha=0.3)
2765     _save(path)
2766
2767 _FRACTION_LABEL = {0.25: "1/4", 0.5: "1/2", 0.75: "3/4"}
2768
2769
2770 def plot_edge_vertex_histograms(
2771     data: dict[float, tuple[list[int], list[int]]],
2772     n: int,
2773     p_values: list[float],
2774     path: str | Path,
2775 ) -> None:
2776     """Six histograms of the binding connectivity in  $G(n, p)$ , at one size  $n$ .
2777
2778     ‘data’ maps each density ‘p’ to a pair ‘(lambda_max_values,
2779     kappa_max_values)’ measured on the same samples, from
2780     :func:‘montecarlo.edge_vertex_distribution’. The grid is two rows by three
2781     columns: the top row is the edge-connectivity ‘lambda^max’ and the bottom
2782     row the vertex-connectivity ‘kappa^max’, one column per density. Reading a
2783     column top to bottom compares the two distributions at a fixed density, so
2784     Whitney’s inequality shows up as the vertex histogram sitting at or to the
2785     left of the edge one. The fraction of samples with ‘kappa^max <
2786     lambda^max’ is printed on each vertex panel. All six share one axis range.
2787
2788     """
2789     rows = [(r"edge  $\lambda^{\max}$ ", 0, _KUL_BLUE),
2790             (r"vertex  $\kappa^{\max}$ ", 1, _WARM)]
2791     all_values = [v for p in p_values for series in data[p] for v in series]
2792     lo, hi = min(all_values), max(all_values)
2793     bins = range(lo, hi + 2) # one integer-wide bin per connectivity value
2794     trials = len(data[p_values[0]][0])
2795
2796     fig, axes = plt.subplots(2, len(p_values), figsize=(11, 6),
2797                             sharex=True, sharey=True)

```

```

2797 for col, p in enumerate(p_values):
2798     lam, kap = data[p]
2799     gap = 100 * sum(k < 1 for k, l in zip(kap, lam)) / len(lam)
2800     plabel = _FRACTION_LABEL.get(p, f"{p:.2f}")
2801     for label, ridx, colour in rows:
2802         axis = axes[ridx][col]
2803         values = lam if ridx == 0 else kap
2804         axis.hist(values, bins=bins, align="left", color=colour,
2805                 edgecolor="white", linewidth=0.4, rwidth=0.9)
2806         axis.axvline(statistics.mean(values), color="black", linestyle="--",
2807                 linewidth=1.2)
2808         if ridx == 0:
2809             axis.set_title(f"$p = {plabel}$")
2810         else:
2811             axis.set_xlabel("binding connectivity")
2812             axis.text(0.03, 0.92, fr"$\kappa\!<\!\lambda$: {gap:.0f}%",
2813                     transform=axis.transAxes, ha="left", va="top",
2814                     fontsize=9, color=_WARM)
2815         if col == 0:
2816             axis.set_ylabel(f"{label}\nsamples")
2817         axis.grid(True, axis="y", alpha=0.3)
2818
2819 fig.suptitle(fr"Distribution of the binding connectivity in ${n},p$ "
2820             fr"over {trials} samples", fontsize=12)
2821 _save(path)
2822
2823
2824 def plot_search_trace(result: SearchResult, path: str | Path,
2825                     optimum: int | None = None,
2826                     ceiling: int | None = None,
2827                     *,
2828                     show_cooling: bool = True,
2829                     show_histogram: bool = True,
2830                     connectivity_label: str = r"$\lambda^{\max}$",
2831                     edge_label: str = "arcs",
2832                     title: str = "Cooling toward an extremal graph") -> None:
2833     """Plot the search trajectory; optionally include the cooling schedule.
2834
2835     Layout: [cooling schedule (optional)] | [scatter] | [visit histogram].
2836
2837     The scatter colours points by step number with a heavily nonlinear
2838     PowerNorm (gamma=0.2, RdYlBu colourmap) so that only the brief hot
2839     exploration phase (early steps) stands out in warm reds/yellows, while
2840     the long converged tail is uniformly blue. When ‘ceiling’ is given the
2841     feasibility boundary is marked and "feasible"/"infeasible" labels are
2842     placed just above the axes box using a blended data/axes transform, so
2843     they sit centred over their respective shaded regions regardless of zoom.
2844
2845     The rightmost panel is an optional horizontal density histogram showing how
2846     many search steps landed at each edge-count level; it shares the y-axis with
2847     the scatter so the bars align with the dots.
2848
2849     New keyword arguments (all optional, all have sensible defaults):
2850     - ‘show_cooling’: set False to omit the left cooling-schedule panel.
2851     - ‘show_histogram’: set False to omit the right visit-density panel.
2852     - ‘connectivity_label’: x-axis label for the scatter (default  $\lambda^{\max}$ ).
2853     - ‘edge_label’: y-axis label (default "arcs"; use "edges" for undirected).
2854     - ‘title’: figure supitle.

```

```

2855     """
2856
2857     steps = [record.step for record in result.history]
2858     temperature = [record.temperature for record in result.history]
2859     edges = [record.edge_count for record in result.history]
2860     connectivity = [record.connectivity for record in result.history]
2861
2862     # Build layout: cooling (optional) / scatter / histogram (optional).
2863     # Histogram axes use no sharey -- that avoids a tight_layout warning, so the
2864     # y-limits are synced to the scatter manually after it is fully drawn.
2865     if show_cooling and show_histogram:
2866         fig = plt.figure(figsize=(13.5, 4.8))
2867         gs = GridSpec(1, 3, figure=fig, width_ratios=[1.0, 1.6, 0.28],
2868                     wspace=0.40)
2869         ax_cool = fig.add_subplot(gs[0])
2870         ax_scatter = fig.add_subplot(gs[1])
2871         ax_hist = fig.add_subplot(gs[2])
2872     elif show_cooling and not show_histogram:
2873         fig = plt.figure(figsize=(11.5, 4.8))
2874         gs = GridSpec(1, 2, figure=fig, width_ratios=[1.0, 1.6], wspace=0.40)
2875         ax_cool = fig.add_subplot(gs[0])
2876         ax_scatter = fig.add_subplot(gs[1])
2877         ax_hist = None
2878     elif (not show_cooling) and show_histogram:
2879         fig = plt.figure(figsize=(8.5, 4.8))
2880         gs = GridSpec(1, 2, figure=fig, width_ratios=[1.6, 0.28], wspace=0.12)
2881         ax_cool = None
2882         ax_scatter = fig.add_subplot(gs[0])
2883         ax_hist = fig.add_subplot(gs[1])
2884     else:
2885         fig = plt.figure(figsize=(6.6, 4.8))
2886         ax_cool = None
2887         ax_scatter = fig.add_subplot(1, 1, 1)
2888         ax_hist = None
2889
2890     # Left: cooling schedule.
2891     if ax_cool is not None:
2892         ax_cool.plot(steps, temperature, color=_WARM, linewidth=1.8)
2893         ax_cool.set_xlabel("step")
2894         ax_cool.set_ylabel("temperature")
2895         ax_cool.set_title("Cooling schedule")
2896         ax_cool.grid(True, alpha=0.3)
2897
2898     # Middle: scatter of (connectivity, edge_count), coloured by step.
2899     ax = ax_scatter
2900     lo = min(connectivity) - 0.5
2901     hi = max(connectivity) + 0.5
2902     if ceiling is not None:
2903         boundary = ceiling + 0.5
2904         ax.axvspan(lo, boundary, color=_GREEN, alpha=0.10)
2905         ax.axvspan(boundary, hi, color=_RED, alpha=0.10)
2906         ax.axvline(boundary, color=_WARM, linestyle="--", linewidth=1.4)
2907         ax.set_xlim(lo, hi)
2908         # Labels centred over each shaded region, placed just above the axes box.
2909         # blended_transform mixes data-x (tracks the region) with axes-fraction-y
2910         # (1.03 is always just above the top of the box regardless of zoom level).
2911         xt = blended_transform_factory(ax.transData, ax.transAxes)
2912         ax.text((lo + boundary) / 2, 1.03, "feasible",

```

```

2913         transform=xt, ha="center", va="bottom",
2914         fontsize=8.5, color=_GREEN, fontweight="bold", clip_on=False)
2915     ax.text((boundary + hi) / 2, 1.03, "infeasible",
2916           transform=xt, ha="center", va="bottom",
2917           fontsize=8.5, color="#C03030", fontweight="bold", clip_on=False)
2918
2919     # Nonlinear colour norm: gamma < 1 squashes large step values toward 1,
2920     # mapping the long converged tail (most points) to the BLUE end of RdYlBu
2921     # while only the first few hot steps appear in warm reds and yellows.
2922     max_step = max(steps) if steps else 1
2923     norm = PowerNorm(gamma=0.2, vmin=0, vmax=max_step)
2924     sc = ax.scatter(connectivity, edges, c=steps, cmap="RdYlBu",
2925                   norm=norm, s=16, alpha=0.75, linewidths=0)
2926
2927     if optimum is not None:
2928         ax.axhline(optimum, color=_KUL_DARK, linestyle="--", linewidth=1.4,
2929                   label=f"certified optimum = {optimum}")
2930     if optimum is not None and ceiling is not None:
2931         ax.plot([ceiling], [optimum], "*", color="#D11A2A", markersize=20,
2932               markeredgewidth=0.9, zorder=5,
2933               label="densest feasible graph")
2934     ax.set_xlabel(connectivity_label + " of the visited graph")
2935     ax.set_ylabel(edge_label)
2936     # No panel title: the supitle and the feasible/infeasible labels describe
2937     # the panel; a panel title would overlap with the labels above the axes box.
2938     ax.xaxis.set_major_locator(MaxNLocator(integer=True))
2939     ax.grid(True, alpha=0.3)
2940     handles, labels = ax.get_legend_handles_labels()
2941     if labels:
2942         ax.legend(loc="upper left", fontsize=8, framealpha=0.9)
2943     cbar = fig.colorbar(sc, ax=ax, shrink=0.92)
2944     cbar.set_label(r"step (cooling $\rightarrow$)")
2945     cbar.set_ticks([0, max_step // 4, max_step // 2,
2946                   3 * max_step // 4, max_step])
2947
2948     # Right: horizontal histogram -- visit density per edge-count level.
2949     # Sync y-limits with the scatter manually (avoids tight_layout warning
2950     # that sharey axes cause inside _save's plt.tight_layout() call).
2951     if ax_hist is not None:
2952         visit_counts = Counter(edges)
2953         arc_vals = sorted(visit_counts)
2954         ax_hist.barh(arc_vals, [visit_counts[a] for a in arc_vals],
2955                   height=0.72, color=_KUL_BLUE, alpha=0.75, linewidth=0)
2956         ax_hist.set_ylim(ax.get_ylim())
2957         ax_hist.set_xlabel("visits", fontsize=8)
2958         ax_hist.set_title("Distribution", fontsize=9)
2959         ax_hist.tick_params(axis="y", left=False, labelleft=False)
2960         ax_hist.xaxis.set_major_locator(MaxNLocator(integer=True, nbins=3))
2961         ax_hist.tick_params(labelsize=7)
2962         ax_hist.grid(True, axis="x", alpha=0.3)
2963
2964     fig.suptitle(title, fontsize=12, y=1.04)
2965     _save(path)
2966
2967
2968 def draw_graph_with_sensitivity(
2969     graph: Graph,
2970     path: str | Path,

```

```

2971     layout: dict | None = None,
2972     show_sigma: bool = False,
2973     node_labels: dict | None = None,
2974 ) -> None:
2975     """Draw a directed multigraph with every parallel arc drawn, coloured by sigma
        ↳ .

2976
2977     Each parallel copy of an adjacency is rendered as its own curved arc, so the
2978     multiplicity ‘‘mu(u, v)’’ is read by *counting* arcs rather than from a label,
2979     and the whole adjacency is coloured by its sensitivity ‘‘sigma(e)’’ -- how far
2980     ‘‘lambda^max’’ falls when the adjacency is deleted -- on a four-level
2981     cool-to-hot palette. An over-provisioned adjacency (several parallel copies
2982     but a downstream bottleneck) therefore shows as a bundle of arcs in a cool
2983     colour: the visual point that multiplicity and load-bearing role differ.

2984
2985     Pass a ‘‘layout’’ dict mapping vertex id to (x, y) and a ‘‘node_labels’’ dict
2986     to override the numeric labels. ‘‘show_sigma’’ is accepted for backward
2987     compatibility and ignored (sigma is always shown).
2988     """

2989
2990     sensitivities = sensitivity_map(graph)
2991     if layout is None:
2992         # Evenly spaced on the unit circle (the same picture nx.circular_layout
2993         # draws), computed directly so drawing needs no networkx.
2994         verts = list(graph.vertices())
2995         layout = {v: (math.cos(2 * math.pi * i / len(verts)),
2996                     math.sin(2 * math.pi * i / len(verts)))
2997                 for i, v in enumerate(verts)}
2998     labels = (node_labels if node_labels is not None
2999              else {v: str(v) for v in graph.vertices()})
3000
3001     fig, ax = plt.subplots(figsize=(8.6, 6.2))
3002
3003     for (u, v), sigma in sensitivities.items():
3004         mu = graph.multiplicity(u, v)
3005         colour = _SIGMA_PALETTE[min(sigma, len(_SIGMA_PALETTE) - 1)]
3006         # Symmetric fan of curvatures, one per parallel copy.
3007         if mu <= 1:
3008             rads = [0.0]
3009         else:
3010             spread = 0.13 * (mu - 1)
3011             rads = [-spread + 2 * spread * k / (mu - 1) for k in range(mu)]
3012         for rad in rads:
3013             ax.add_patch(FancyArrowPatch(
3014                 layout[u], layout[v], connectionstyle=f"arc3,rad={rad}",
3015                 arrowstyle="-|>", mutation_scale=13, lw=2.3, color=colour,
3016                 shrinkA=13, shrinkB=13, zorder=1))
3017             # One sigma label per adjacency, at the chord midpoint.
3018             (x0, y0), (x1, y1) = layout[u], layout[v]
3019             ax.text((x0 + x1) / 2, (y0 + y1) / 2, f"$\\sigma={sigma}$",
3020                    fontsize=9, ha="center", va="center", zorder=3,
3021                    bbox={"boxstyle": "round,pad=0.18", "fc": "white",
3022                        "ec": colour, "alpha": 0.9})
3023
3024     xs = [p[0] for p in layout.values()]
3025     ys = [p[1] for p in layout.values()]
3026     ax.scatter(xs, ys, s=620, c=_KUL_DARK, edgecolors="white", linewidths=1.2,
3027               zorder=2)

```

```

3028     for v, (x, y) in layout.items():
3029         ax.text(x, y, str(labels.get(v, v)), color="white", fontweight="bold",
3030                ha="center", va="center", zorder=4)
3031
3032     mx = (max(xs) - min(xs)) * 0.12 + 0.3
3033     my = (max(ys) - min(ys)) * 0.12 + 0.3
3034     ax.set_xlim(min(xs) - mx, max(xs) + mx)
3035     ax.set_ylim(min(ys) - my, max(ys) + my)
3036     ax.set_aspect("equal")
3037     ax.axis("off")
3038     ax.set_title("Parallel arcs drawn explicitly, coloured by sensitivity "
3039                 "$\\sigma$: ncool ($\\sigma=0$, free) $\\to$ warm (load-bearing)"
3040                 "\u2192",
3041                 fontsize=12)
3042     _save(path)
3043
3044 def _running_best_trace(result: SearchResult) -> tuple[list[float], list[int]]:
3045     """Best feasible edge count so far against wall-clock seconds, from a history.
3046
3047     Reads the timed :class:`SearchStep` log (its ``best_feasible`` field already
3048     carries the densest feasible graph seen up to each step) and returns the timed
3049     convergence curve both engines record in their native fast modes.
3050     """
3051     xs = [record.seconds for record in result.history]
3052     ys = [record.best_feasible for record in result.history]
3053     return xs, ys
3054
3055 def _settle_time(xs: list[float], ys: list[int]) -> float:
3056     """Wall-clock second at which a trace first reached its own final best value.
3057
3058     The convergence curves plateau long before the time budget runs out, so this
3059     is what the x-axis should track: cropping to the budget would leave a long
3060     flat tail and the cropped view would barely change when the budget does.
3061     """
3062     if not xs:
3063         return 0.0
3064     final = ys[-1]
3065     return next((x for x, y in zip(xs, ys) if y == final), xs[-1])
3066
3067
3068 def plot_sa_vs_tabu_convergence(
3069     path: str | Path, *,
3070     cases: tuple[tuple[int, int], ...] = ((5, 3), (7, 3)),
3071     budget: float = 8.0,
3072     seed: int = 0,
3073     xmaxs: tuple[float, ...] | None = None,
3074     settle_margin: float = 1.3,
3075 ) -> None:
3076     """Best-feasible-value against wall-clock for annealing vs tabu search.
3077
3078     For each ``(n, m)`` it runs one simulated-annealing schedule
3079     (:func:`search_for_dense_graph`) and one tabu schedule
3080     (:func:`tabu_search_for_dense_graph`) on the directed multigraph at an equal
3081     wall-clock ``budget``, and overlays their timed best-so-far traces against the
3082     conjectured optimum. The chosen cases are where the two engines diverge: tabu
3083     assembles the bipartite extremiser and reaches the optimum while annealing

```

```

3085     plateaus a few arcs short. The time axis is wall-clock, so this one figure is
3086     a representative timed run on the reference machine rather than a seed-exact
3087     reproduction like the others.
3088
3089     By default each panel's x-axis stops shortly after the slower engine settles
3090     ('settle_margin' times the later of the two plateau times), so the view
3091     tracks the actual run instead of a fixed window: cropping to a hard-coded
3092     second count leaves a long flat tail because both engines converge well
3093     inside one second. Pass 'xmaxes' to force fixed per-panel limits instead.
3094     """
3095     if not MATPLOTLIB_AVAILABLE:
3096         raise RuntimeError("matplotlib is required for figures")
3097     fig, axes = plt.subplots(1, len(cases), figsize=(11, 4.3))
3098     if len(cases) == 1:
3099         axes = [axes]
3100     for i, (ax, (n, m)) in enumerate(zip(axes, cases)):
3101         sa = search_for_dense_graph(MULTI_DIRECTED, n, m, seed=seed, steps=10**7,
3102                                   deadline=time.time() + budget)
3103         tb = tabu_search_for_dense_graph(MULTI_DIRECTED, n, m, seed=seed,
3104                                         steps=10**7, deadline=time.time() +
3105                                         ↪ budget)
3106
3107         settle = 0.0
3108         for res, label, colour in ((sa, "simulated annealing", _KUL_BLUE),
3109                                   (tb, "tabu search", _WARM)):
3110             xs, ys = _running_best_trace(res)
3111             ax.step(xs, ys, where="post", label=label, color=colour, linewidth
3112                   ↪ =2.0)
3113             settle = max(settle, _settle_time(xs, ys))
3114         opt = directed_multigraph_arc(n, m)
3115         ax.axhline(opt, linestyle=":", color=_KUL_DARK, linewidth=1.6,
3116                   label=f"optimum  $L_3^{\{\mathrm{dir}\}}(\{n\}) = \{opt\}$")
3117         ax.set_title(f"directed multigraph, $n = \{n\}$, $m = \{m\}$", fontsize=11)
3118         ax.set_xlabel("wall-clock seconds", fontsize=9.5)
3119         ax.set_ylabel("densest feasible arc count", fontsize=9.5)
3120         # Crop the long flat tail: stop just after the slower engine settles, so
3121         # the rise fills the panel instead of being squeezed against the y-axis.
3122         # A fixed second count cannot do this, since the settle time shifts run to
3123         # run; an explicit xmaxes overrides for a reproducible fixed window.
3124         if xmaxes is not None and i < len(xmaxes):
3125             ax.set_xlim(0, xmaxes[i])
3126         else:
3127             ax.set_xlim(0, max(settle * settle_margin, 0.5))
3128         ax.grid(True, alpha=0.3)
3129         ax.legend(fontsize=8.5, loc="lower right")
3130         # Fewer x-ticks declutters the time axis: the action concentrates early
3131         # and dense ticks add noise.
3132         if MaxNLocator is not None:
3133             ax.xaxis.set_major_locator(MaxNLocator(nbins=5))
3134     _save(path)
3135
3136     def _save(path: str | Path, *, tight: bool = True) -> None:
3137         """Tighten the layout, write the file, and close the figure.
3138
3139         'tight=False' skips 'tight_layout' for the 3-D figures, where it
3140         misbehaves with the projected axes (they manage their own spacing).
3141         """
3142         Path(path).parent.mkdir(parents=True, exist_ok=True)$ 
```



```

3141     if tight:
3142         # tight_layout can warn when figures contain colorbars or shared axes;
3143         # bbox_inches="tight" in savefig handles the actual bounding box so the
3144         # layout warning is cosmetic and safe to suppress.
3145         with warnings.catch_warnings():
3146             warnings.simplefilter("ignore", UserWarning)
3147             plt.tight_layout()
3148     plt.savefig(path, dpi=150, bbox_inches="tight")
3149     plt.close() # release the figure so head-less runs don't leak memory
3150
3151
3152 def plot_complexity_growth(path: str | Path) -> None:
3153     """The combinatorial explosion: number of graphs to enumerate, by direction.
3154
3155     Two panels, directed (left) and undirected (right), sharing one axis range.
3156     Each plots, on a log scale, how many graphs blind enumeration must visit
3157     against the vertex count 'n', for the simple model, two multigraph
3158     connectivity caps, and (dotted) the 3-uniform hypergraph, with a dashed line
3159     at a generous budget of 109 candidates. The count does not depend on the
3160     edge-versus-vertex separation, because both search the very same set of
3161     graphs, so one pair of panels covers all of those variants at once.
3162     Direction doubles the number of cells to fill, which is the whole difference
3163     between the two panels.
3164     """
3165     ns = list(range(2, 13))
3166
3167     def log10_pow(base: float, exponent: float) -> float:
3168         return exponent * math.log10(base)
3169
3170     def curves(directed: bool):
3171         # Each model fills a number of off-diagonal cells; the candidate count is
3172         # (values per cell) ^ (number of cells). Labels stay plain -- the precise
3173         # exponential forms live in the caption, not the legend. Direction
3174         # multiplies the cell count: by 2 for graphs (ordered vs unordered pairs,
3175         #  $n(n-1)$  vs  $C(n,2)$ ), but by 3 for 3-uniform hypergraphs, since a directed
3176         # hyperedge is a tail plus 2 heads ( $n \cdot C(n-1,2) = 3 \cdot C(n,3)$  candidates).
3177         cells = (lambda n: n * (n - 1)) if directed else (lambda n: n * (n - 1) /
3178             ↪ 2)
3179         hyper_cells = ((lambda n: n * (n - 1) * (n - 2) / 2) if directed
3180             else (lambda n: n * (n - 1) * (n - 2) / 6))
3181         return [
3182             ("simple / random", _KUL_BLUE, "-",
3183              [log10_pow(2, cells(n)) for n in ns]),
3184             ("multigraph $m=3$", _KUL_DARK, "-",
3185              [log10_pow(3, cells(n)) for n in ns]),
3186             ("multigraph $m=4$", _WARM, "-",
3187              [log10_pow(4, cells(n)) for n in ns]),
3188             ("$$$-uniform hypergraph", _VIOLET, ":",
3189              [log10_pow(2, hyper_cells(n)) for n in ns]),
3190         ]
3191
3192     fig, axes = plt.subplots(1, 2, figsize=(11, 4.6), sharey=True)
3193     for ax, directed, title in [(axes[0], True, "directed graphs"),
3194                                (axes[1], False, "undirected graphs")]:
3195         for label, colour, style, vals in curves(directed):
3196             ax.plot(ns, vals, style, color=colour, marker="o", markersize=3.5,
3197                     label=label)
3198     ax.axhline(9, color="gray", linestyle="--", linewidth=1,

```

```

3198         label=r"$10^{\{9\}}$ budget")
3199     ax.set_title(title)
3200     ax.set_xlabel("vertices $n$")
3201     ax.grid(True, alpha=0.3)
3202     # Cap the shared vertical axis at 100 so the slowest curve (the directed
3203     # 3-uniform hypergraph) runs off the top instead of compressing every other
3204     # curve into a thin band near the bottom. sharey=True propagates the limit.
3205     axes[0].set_ylim(0, 100)
3206     axes[0].annotate(r"$3$-uniform directed" + "\n" + r"continues off the top",
3207                     xy=(9.55, 99.3), xytext=(7.5, 92), fontsize=8, color=_VIOLET,
3208                     ha="center", va="center",
3209                     arrowprops=dict(arrowstyle="->", color=_VIOLET, lw=1.0))
3210     axes[0].set_ylabel(r"$\log_{10}$ (number of graphs)")
3211     axes[0].legend(fontsize=8.5, loc="upper left", title="model")
3212     fig.suptitle("Number of graphs blind enumeration must visit",
3213                 fontsize=12)
3214     _save(path)
3215
3216
3217 def _circle_layout(n, hub=None):
3218     """Vertex positions for a panel: all on a unit circle, or the ‘hub’ vertex
3219     at the centre and the rest on the circle, which draws stars and hub graphs
3220     far more clearly than a plain circle does."""
3221     if hub is None:
3222         angs = [math.pi / 2 - 2 * math.pi * k / n for k in range(n)]
3223         return [(math.cos(a), math.sin(a)) for a in angs]
3224     pos = [(0.0, 0.0) for _ in range(n)]
3225     others = [k for k in range(n) if k != hub]
3226     for idx, k in enumerate(others):
3227         a = math.pi / 2 - 2 * math.pi * idx / len(others)
3228         pos[k] = (math.cos(a), math.sin(a))
3229     return pos
3230
3231
3232 def _draw_extremal_panel(ax, matrix, *, directed: bool, hub=None) -> None:
3233     """Draw one extremal multiplicity matrix on ‘ax’ as a graph.
3234
3235     Vertices sit on a circle (or, when ‘hub’ is given, that vertex sits at the
3236     centre and the rest on the circle). An adjacency of multiplicity one is a
3237     single line (an arrowed line when ‘directed’), and a higher multiplicity is
3238     shown by a small count label rather than parallel curves, so the picture
3239     stays legible. Opposite arcs of a bidirected pair are bent apart.
3240
3241     Every named family the gallery draws is *uniform*: all its adjacencies carry
3242     one and the same multiplicity. Repeating that count on every arc buries the
3243     structure under labels (worst on the bidirected ‘K4’, twelve labels over
3244     six crossing arcs), so a uniform multiplicity above one is stated once in a
3245     corner note, and the per-arc labels are kept only for the genuinely mixed
3246     case. ‘hub’ may also be an explicit list of ‘(x, y)’ positions, used
3247     for the bipartite wall panel, where two columns read far better than a circle.
3248     """
3249     n = len(matrix)
3250     pos = hub if isinstance(hub, list) else _circle_layout(n, hub)
3251     mults = {int(matrix[i][j]) for i in range(n) for j in range(n)
3252              if i != j and matrix[i][j] > 0}
3253     uniform = mults.pop() if len(mults) == 1 else None
3254     per_arc_labels = uniform is None
3255

```

```

3256 def edge(i, j, rad, arrow):
3257     ax.add_patch(FancyArrowPatch(
3258         pos[i], pos[j], connectionstyle=f"arc3,rad={rad}",
3259         arrowstyle="->" if arrow else "-"), mutation_scale=12,
3260         lw=1.7, color=_KUL_BLUE, shrinkA=10, shrinkB=10, zorder=1))
3261
3262 def mlabel(i, j, rad, mu):
3263     if mu <= 1:
3264         return
3265     mx, my = (pos[i][0] + pos[j][0]) / 2, (pos[i][1] + pos[j][1]) / 2
3266     dx, dy = pos[j][0] - pos[i][0], pos[j][1] - pos[i][1]
3267     length = math.hypot(dx, dy) or 1.0
3268     mx += (rad * 0.9 + 0.06) * (-dy) / length
3269     my += (rad * 0.9 + 0.06) * (dx) / length
3270     ax.text(mx, my, f"$\\times\\mu$", fontsize=8, ha="center", va="center",
3271             color=_KUL_DARK, zorder=3,
3272             bbox=dict(boxstyle="round,pad=0.08", fc="white", ec="none"))
3273
3274 if directed:
3275     for i in range(n):
3276         for j in range(n):
3277             if i == j or matrix[i][j] == 0:
3278                 continue
3279             rad = 0.16 if matrix[j][i] else 0.0
3280             edge(i, j, rad, True)
3281             if per_arc_labels:
3282                 mlabel(i, j, rad, matrix[i][j])
3283 else:
3284     for i in range(n):
3285         for j in range(i + 1, n):
3286             if matrix[i][j] == 0:
3287                 continue
3288             edge(i, j, 0.0, False)
3289             if per_arc_labels:
3290                 mlabel(i, j, 0.0, matrix[i][j])
3291 for k, (x, y) in enumerate(pos):
3292     ax.scatter([x], [y], s=300, c="white", edgecolors=_KUL_BLUE,
3293               linewidths=1.6, zorder=2)
3294     ax.text(x, y, str(k + 1), ha="center", va="center", fontsize=9,
3295            zorder=3, color=_KUL_DARK)
3296 if uniform is not None and uniform > 1:
3297     word = "arcs" if directed else "edges"
3298     ax.text(0.0, -1.34, f"all {word} $\\times\\{uniform}$", ha="center",
3299            va="center", fontsize=9, color=_KUL_DARK)
3300 ax.set_xlim(-1.45, 1.45)
3301 ax.set_ylim(-1.45, 1.45)
3302
3303
3304 # metro-line palette for hypergraph panels: red, blue, green, orange, purple, grey
3305 _GALLERY_METRO = ["#C85050", "#1D8DB0", "#3CA050", "#DC8C28", "#8C50A0", "#787878"
3306                  ↪ ]
3307
3308 def _draw_hyper_panel(ax, hyperedges, n, *, directed: bool, hub=None) -> None:
3309     """Draw a hypergraph extremiser as a metro map: vertices on a circle (or a
3310     hub at the centre), each hyperedge a thick coloured line through its members
3311     when undirected, or a fan of coloured arrows from its tail when directed."""
3312     pos = _circle_layout(n, hub)

```

```

3313     for idx, he in enumerate(hyperedges):
3314         col = _GALLERY_METRO[idx % len(_GALLERY_METRO)]
3315         if directed:
3316             tail, heads = he
3317             tx, ty = pos[tail]
3318             for h in heads:
3319                 hx, hy = pos[h]
3320                 ax.add_patch(FancyArrowPatch(
3321                     (tx, ty), (hx, hy), arrowstyle="-|>", mutation_scale=11,
3322                     lw=2.4, color=col, alpha=0.85, shrinkA=11, shrinkB=11,
3323                     zorder=1, connectionstyle="arc3,rad=0.10"))
3324         else:
3325             pts = [pos[v] for v in sorted(he)]
3326             cx = sum(p[0] for p in pts) / len(pts)
3327             cy = sum(p[1] for p in pts) / len(pts)
3328             pts.sort(key=lambda p: math.atan2(p[1] - cy, p[0] - cx))
3329             ax.plot([p[0] for p in pts], [p[1] for p in pts], "-", color=col,
3330                     lw=4.2, solid_capstyle="round", solid_joinstyle="round",
3331                     alpha=0.85, zorder=1)
3332     for k, (x, y) in enumerate(pos):
3333         ax.scatter([x], [y], s=300, c="white", edgecolors="#333333",
3334                   linewidths=1.5, zorder=2)
3335         ax.text(x, y, str(k + 1), ha="center", va="center", fontsize=9,
3336                zorder=3, color="#222222")
3337 ax.set_xlim(-1.45, 1.45)
3338 ax.set_ylim(-1.45, 1.45)
3339
3340
3341 def plot_extremal_gallery(path: str | Path, *,
3342                          gallery_json: str | Path | None = None) -> None:
3343     """A gallery of the named extremal graphs, drawn at a large representative
3344         ↪ size.
3345
3346     These are the structured extremisers the analysis identifies, not the
3347     trivial small trees: the directed double and bidirected stars, the doubled
3348     bipartite wall, the dense bidirected complete graph, the multigraph star at
3349     full multiplicity, and the star hypertrees (drawn as metro maps). The program
3350     ↪ both builds these and,
3351     at small 'n', proves them optimal by the enumeration of
3352     :func:'gallery_extremal_graphs'. The 'gallery_json' argument is accepted for
3353     backward compatibility and ignored.
3354     """
3355     if not MATPLOTLIB_AVAILABLE:
3356         raise RuntimeError("matplotlib is required for figures")
3357     # dense bidirected complete graphs, built directly
3358     bidir_k4 = Graph(4, SIMPLE_DIRECTED)
3359     for i in range(4):
3360         for j in range(4):
3361             if i != j:
3362                 bidir_k4.set_multiplicity(i, j, 1)
3363     # 2.B_{3,4}: the doubled one-directional bipartite wall, one of the three
3364     # proved 24-arc extremisers at n = 7, m = 3 (rem:odd-step-roadmap, fact (b)).
3365     # It ties the double star at the crossover, so the gallery shows both branches
3366     ↪ .
3367     bip_2b34 = Graph(7, MULTI_DIRECTED)
3368     for a in range(3):
3369         for b in range(3, 7):
3370             bip_2b34.set_multiplicity(a, b, 2)

```

```

3368 bip_pos = [(-0.85, 0.85), (-0.85, 0.0), (-0.85, -0.85),
3369             (0.85, 1.05), (0.85, 0.35), (0.85, -0.35), (0.85, -1.05)]
3370 multi_star = Graph(9, MULTI_UNDIRECTED) # undirected star, mult 2
3371 for i in range(1, 9):
3372     multi_star.set_multiplicity(0, i, 2)
3373 dir_hypertree = [(0, frozenset({1, 2})), (0, frozenset({3, 4})),
3374                 (0, frozenset({5, 6}))] # directed star hypertree on 7
3375                                     ↪ vertices
3376 # ("g"/"h"/"hd", object, directed, hub, title)
3377 panels = [
3378     ("g", double_star(7, 3, directed=True), True, 0,
3379      "multigraph directed, arc\n$m=3$, $n=7$: double star ($24$ arcs)"),
3380     ("g", double_star(7, 2, directed=True), True, 0,
3381      "simple directed, arc\n$m=2$, $n=7$: bidirected star ($12$ arcs)"),
3382     ("g", multi_star, False, 0,
3383      "multigraph undirected, edge\n$m=3$, $n=9$: star at multiplicity $2$ (
3384       ↪ $16$ edges)"),
3385     ("g", bidir_k4, True, None,
3386      "simple directed, arc\n$m=4$, $n=4$: bidirected $K_4$ ($12$ arcs)"),
3387     ("g", bip_2b34, True, bip_pos,
3388      "multigraph directed, arc\n$m=3$, $n=7$: bipartite wall $2$ \cdot $B_{3,4}$
3389       ↪ $ ($24$ arcs)"),
3390     ("h", star_hypertree(10, 3), False, 0,
3391      "hypergraph, $r=3$\n$m=2$, $n=10$: star hypertree ($4$ hyperedges)"),
3392     ("h", star_hypertree(13, 4), False, 0,
3393      "hypergraph, $r=4$\n$m=2$, $n=13$: star hypertree ($4$ hyperedges)"),
3394     ("h", star_hypertree(16, 3), False, 0,
3395      "hypergraph, $r=3$\n$m=2$, $n=16$: star hypertree ($7$ hyperedges)"),
3396     ("hd", (dir_hypertree, 7), True, 0,
3397      "directed hypergraph, $r=3$\n$m=2$, $n=7$: tail-to-heads star"),
3398 ]
3399 fig, axes = plt.subplots(3, 3, figsize=(12, 12))
3400 for ax, (kind, obj, directed, hub, title) in zip(axes.flat, panels):
3401     if kind == "g":
3402         _draw_extremal_panel(ax, obj.mu.tolist(), directed=directed, hub=hub)
3403     elif kind == "h":
3404         _draw_hyper_panel(ax, list(obj.hyperedges), obj.num_vertices,
3405                           directed=False, hub=hub)
3406     else: # "hd": (hyperedges, n)
3407         hes, nn = obj
3408         _draw_hyper_panel(ax, hes, nn, directed=True, hub=hub)
3409     ax.set_title(title, fontsize=9.5)
3410     ax.set_aspect("equal")
3411     ax.axis("off")
3412 fig.suptitle("Extremal graphs of the named families, drawn large", fontsize
3413             ↪ =14)
3414 _save(path)
3415
3416 _GUESS_STYLE = (0, (1, 1)) # dense dots: "we are very much guessing"
3417
3418 def plot_variant_grid(panels: list[dict], path: str | Path,
3419                      m: int | None = None) -> None:
3420     """All twelve variants on one grid, in the proved/conjectured/guessed language
3421     ↪ .

```

```

3421     ‘‘panels‘‘ is a row-major list of twelve dictionaries (rows = model simple,
3422     multi, hyper; columns = undirected edge, undirected vertex, directed arc,
3423     directed vertex). Each panel may carry any of:
3424
3425     ‘‘proved‘‘ ‘‘(xs, ys)‘‘ solid blue line, a theorem holding for all ‘‘n‘‘;
3426     ‘‘conj‘‘ ‘‘(xs, ys)‘‘ solid red line, a conjecture formula;
3427     ‘‘guess‘‘ ‘‘(xs, ys)‘‘ solid yellow line, a bare extrapolation of the
3428     machine points where no formula is known;
3429     ‘‘band‘‘ ‘‘(xs, lo, hi)‘‘ the certain interval, an easy construction below
3430     and the trivial maximum edge count above;
3431     ‘‘exact‘‘ ‘‘(xs, ys)‘‘ filled squares, sizes the machine proved;
3432     ‘‘search‘‘ ‘‘(xs, ys)‘‘ open circles, the search lower bounds.
3433
3434     The point is one honest picture, distinguished by colour rather than dash
3435     pattern: a blue line is settled, a red line is conjectured, a yellow line is
3436     a guess, and the shaded band is the interval we are certain the truth lies in.
3437     """
3438     def draw_panel(ax, panel):
3439         band = panel.get("band")
3440         if band is not None:
3441             xs, lo, hi = band
3442             ax.fill_between(xs, lo, hi, color=_WARM, alpha=0.12,
3443                             label="certain interval")
3444             ax.plot(xs, lo, "-", color=_WARM, linewidth=0.8, alpha=0.6)
3445             ax.plot(xs, hi, "-", color=_WARM, linewidth=0.8, alpha=0.6)
3446         for branch in panel.get("branches", []):
3447             bxs, bys, blabel = branch
3448             ax.plot(bxs, bys, ":", color=_KUL_LIGHT, linewidth=1.3, label=blabel)
3449         if panel.get("proved") is not None:
3450             xs, ys = panel["proved"]
3451             ax.plot(xs, ys, "-", color=_KUL_BLUE, linewidth=2.3, label="proved")
3452         if panel.get("conj") is not None:
3453             xs, ys = panel["conj"]
3454             ax.plot(xs, ys, "-", color=_RED, linewidth=2.0, label="conjectured")
3455         if panel.get("guess") is not None:
3456             xs, ys = panel["guess"]
3457             ax.plot(xs, ys, "-", color=_GUESS, linewidth=2.0,
3458                     label="guess (interpolated)")
3459         if panel.get("exact") is not None:
3460             xs, ys = panel["exact"]
3461             if len(xs):
3462                 ax.plot(xs, ys, "s", color=_GREEN, markersize=7,
3463                         label="machine-checked (exact)")
3464         if panel.get("search") is not None:
3465             xs, ys = panel["search"]
3466             if len(xs):
3467                 ax.plot(xs, ys, "o", mfc="none", mec=_VIOLET, mew=1.8,
3468                         markersize=8, label="search (lower bound)")
3469         ax.set_title(panel["title"], fontsize=9.5)
3470         ax.set_xlabel("vertices $n$", fontsize=8.5)
3471         ax.set_ylabel(panel.get("ylabel", "edges"), fontsize=8.5)
3472         ax.tick_params(labelsize=8)
3473         ax.grid(True, alpha=0.3)
3474         ax.legend(fontsize=6.8, loc="upper left", framealpha=0.85)
3475
3476     # The line styles (solid / dashed / dotted) are named in the per-panel
3477     # legends and spelled out in the caption, so the supitle stays short and
3478     # just records the variant family and the threshold m.

```

```

3479     subtitle = "Erdős 915 across the twelve variants"
3480     if m is not None:
3481         subtitle += fr",  $m = {m}$"
3482     _variant_panel_grid(draw_panel, configs=panels, subtitle=subtitle, path=path,
3483                         subtitle_fontsize=13)

```

C.1.7 Walking the whole space

At small sizes it is possible to visit every graph there is and record what it looks like, which is what these routines do. The result is the distribution behind the landscape figures, and, more importantly, a way to confirm a claimed maximum by exhaustion rather than by search. The three-dimensional surfaces are built on the same idea, run across a grid of sizes and route limits and cached so that the sweep is paid for once.

```

3486 # --- ENUMERATION LANDSCAPES: visit every labeled graph, collect (edges, lambda^
    ↪ max) ---
3487
3488 # Row-major table of all twelve variants, matching gather_variant_grid.
3489 # "enum_n" is the vertex count used for full enumeration.
3490 # "max_mult" caps per-cell multiplicity for multigraph variants.
3491 _VARIANT_ENUM_CONFIGS: list[dict] = [
3492     # row 1 -- simple
3493     dict(key="simple_undirected_edge", title="simple undirected edge",
3494           directed=False, simple=True, hypergraph=False, r=3, separation="edge",
3495           enum_n=6, max_mult=1),
3496     dict(key="simple_undirected_vertex", title="simple undirected vertex",
3497           directed=False, simple=True, hypergraph=False, r=3, separation="vertex",
3498           enum_n=6, max_mult=1),
3499     dict(key="simple_directed_edge", title="simple directed arc",
3500           directed=True, simple=True, hypergraph=False, r=3, separation="edge",
3501           enum_n=4, max_mult=1),
3502     dict(key="simple_directed_vertex", title="simple directed vertex",
3503           directed=True, simple=True, hypergraph=False, r=3, separation="vertex",
3504           enum_n=4, max_mult=1),
3505     # row 2 -- multigraph
3506     dict(key="multi_undirected_edge", title="multigraph undirected edge",
3507           directed=False, simple=False, hypergraph=False, r=3, separation="edge",
3508           enum_n=5, max_mult=2),
3509     dict(key="multi_undirected_vertex", title="multigraph undirected vertex",
3510           directed=False, simple=False, hypergraph=False, r=3, separation="vertex",
3511           enum_n=5, max_mult=2),
3512     dict(key="multi_directed_edge", title="multigraph directed arc",
3513           directed=True, simple=False, hypergraph=False, r=3, separation="edge",
3514           enum_n=3, max_mult=5),
3515     dict(key="multi_directed_vertex", title="multigraph directed vertex",
3516           directed=True, simple=False, hypergraph=False, r=3, separation="vertex",
3517           enum_n=3, max_mult=5),
3518     # row 3 -- hypergraph r=3
3519     dict(key="hyper_undirected_edge", title="hypergraph undirected edge",
3520           directed=False, simple=True, hypergraph=True, r=3, separation="edge",
3521           enum_n=5, max_mult=1),
3522     dict(key="hyper_undirected_vertex", title="hypergraph undirected vertex",
3523           directed=False, simple=True, hypergraph=True, r=3, separation="vertex",
3524           enum_n=5, max_mult=1),
3525     dict(key="hyper_directed_edge", title="hypergraph directed arc",

```

```

3526         directed=True, simple=True, hypergraph=True, r=3, separation="edge",
3527         enum_n=4, max_mult=1),
3528     dict(key="hyper_directed_vertex", title="hypergraph directed vertex",
3529         directed=True, simple=True, hypergraph=True, r=3, separation="vertex",
3530         enum_n=4, max_mult=1),
3531 ]
3532
3533
3534 def enumerate_all_graphs(
3535     n: int, *,
3536     directed: bool,
3537     simple: bool,
3538     hypergraph: bool = False,
3539     r: int = 3,
3540     max_mult: int = 3,
3541     separation: str = "edge",
3542 ) -> list[tuple[int, int]]:
3543     """Return '(edge_count, lambda_max)' for every labeled graph of this type on
3544         ↪ 'n' vertices.
3545
3546     For simple graphs each cell is 0/1; for multigraphs each cell ranges over
3547     '{0,...,max_mult}'. For hypergraphs each candidate hyperedge is present or
3548     absent. The list covers the full labeled graph space (isomorphic copies
3549     included), which is what we want for the distribution over the search space.
3550
3551     """
3552     results: list[tuple[int, int]] = []
3553
3554     if hypergraph:
3555         vertex_split = (separation == "vertex")
3556         candidates = _hyperedge_candidates(n, r, directed)
3557         for mask in product((0, 1), repeat=len(candidates)):
3558             chosen = [candidates[i] for i, flag in enumerate(mask) if flag]
3559             h = Hypergraph(n, chosen, directed=directed)
3560             conn = max_hyper_connectivity(h, vertex_split=vertex_split)
3561             results.append((h.edge_count(), conn))
3562         return results
3563
3564     variant = _variant_for(directed, simple)
3565     measure = _connectivity_measure(separation)
3566     cells = _matrix_cells(n, directed)
3567     span = 2 if simple else (max_mult + 1)
3568     base = Graph(n, variant)
3569     for values in product(range(span), repeat=len(cells)):
3570         candidate = base.copy()
3571         for (u, v), value in zip(cells, values):
3572             if value:
3573                 candidate.set_multiplicity(u, v, value)
3574         conn = measure(candidate)
3575         results.append((candidate.edge_count(), conn))
3576     return results
3577
3578 def _pair_connectivities(obj, *, separation: str, hypergraph: bool) -> list[int]:
3579     """Local connectivity of every vertex pair of one graph or hypergraph.
3580
3581     The same pair sweep 'max_connectivity' runs, but keeping each value rather
3582     than only the maximum. For the edge case the capacity matrix is built once
3583     and reused across pairs (as 'max_connectivity' does); vertex and hypergraph

```



```

3583     cases reuse their named per-pair routines.
3584     """
3585     vsplit = (separation == "vertex")
3586     if hypergraph:
3587         return [hyper_connectivity(obj, s, t, vertex_split=vsplit)
3588                 for s, t in _pairs(obj)]
3589     if not vsplit:
3590         csr = _csr(obj.mu, dtype=int)
3591         return [int(_csgraph_maxflow(csr, s, t).flow_value) for s, t in _pairs(obj)
3592                 ↪ ]
3593     return [local_connectivity(obj, s, t, vertex_split=True) for s, t in _pairs(
3594             ↪ obj)]
3595
3596 def enumerate_pair_connectivities(
3597     n: int, *,
3598     directed: bool,
3599     simple: bool,
3600     hypergraph: bool = False,
3601     r: int = 3,
3602     max_mult: int = 3,
3603     separation: str = "edge",
3604 ) -> dict[tuple[int, int], int]:
3605     """Pooled 2-D histogram '{(lambda_max, pair_conn): count}' over all graphs.
3606
3607     For every labeled graph on 'n' vertices and every vertex pair, record the
3608     pair's local connectivity tagged with the graph's own 'lambda^max'. The
3609     result is a small threshold-independent table: at plot time a feasibility
3610     threshold 'T' splits each pair-connectivity column into observations from
3611     graphs that stay at 'lambda^max <= T' and those that exceed it. Much
3612     smaller than the raw observation list (a few dozen entries per variant), so
3613     it pickles compactly even though the sweep itself enumerates every graph.
3614     """
3615     table: Counter = Counter()
3616
3617     def record(obj):
3618         pcs = _pair_connectivities(obj, separation=separation, hypergraph=
3619             ↪ hypergraph)
3620         lmax = max(pcs, default=0)
3621         for c in pcs:
3622             table[(lmax, c)] += 1
3623
3624     if hypergraph:
3625         candidates = _hyperedge_candidates(n, r, directed)
3626         for mask in product((0, 1), repeat=len(candidates)):
3627             chosen = [candidates[i] for i, flag in enumerate(mask) if flag]
3628             record(Hypergraph(n, chosen, directed=directed))
3629         return dict(table)
3630
3631     variant = _variant_for(directed, simple)
3632     cells = _matrix_cells(n, directed)
3633     span = 2 if simple else (max_mult + 1)
3634     base = Graph(n, variant)
3635     for values in product(range(span), repeat=len(cells)):
3636         candidate = base.copy()
3637         for (u, v), value in zip(cells, values):
3638             if value:
3639                 candidate.set_multiplicity(u, v, value)

```

```

3638         record(candidate)
3639     return dict(table)
3640
3641
3642 def compute_pair_enumeration_cache(
3643     cache_path: str | Path = "figures/pair_enumeration_cache.pkl",
3644     configs: list[dict] | None = None,
3645 ) -> dict[str, dict[tuple[int, int], int]]:
3646     """Per-variant pooled pair-connectivity tables, loading from cache if present.
3647
3648     The pooled analogue of :func:`compute_enumeration_cache`: one
3649     '{(lambda_max, pair_conn): count}' table per variant. Slow on first run
3650     (it re-sweeps the enumeration measuring every pair), cached thereafter.
3651     """
3652     cache_path = Path(cache_path)
3653     if configs is None:
3654         configs = _VARIANT_ENUM_CONFIGS
3655
3656     if cache_path.exists():
3657         with open(cache_path, "rb") as f:
3658             cached = pickle.load(f)
3659     else:
3660         cached = {}
3661
3662     changed = False
3663     for cfg in configs:
3664         key = cfg["key"]
3665         if key in cached:
3666             continue
3667         print(f"pooling pairs for {cfg['title']} (n={cfg['enum_n']})...", flush=
3668             ↪ True)
3669         cached[key] = enumerate_pair_connectivities(
3670             cfg["enum_n"],
3671             directed=cfg["directed"], simple=cfg["simple"],
3672             hypergraph=cfg["hypergraph"], r=cfg["r"],
3673             max_mult=cfg["max_mult"], separation=cfg["separation"])
3674         changed = True
3675
3676     if changed:
3677         cache_path.parent.mkdir(parents=True, exist_ok=True)
3678         with open(cache_path, "wb") as f:
3679             pickle.dump(cached, f)
3680
3681     return cached
3682
3683 def compute_enumeration_cache(
3684     cache_path: str | Path = "figures/enumeration_cache.pkl",
3685     configs: list[dict] | None = None,
3686 ) -> dict[str, list[tuple[int, int]]]:
3687     """Return the full enumeration for every variant, loading from cache if
3688     ↪ available.
3689
3690     On first run this takes several minutes; subsequent runs load the pickle
3691     instantly. The cache lives at 'cache_path' relative to the caller's
3692     working directory.
3693     """
3694     cache_path = Path(cache_path)

```

```

3694     if configs is None:
3695         configs = _VARIANT_ENUM_CONFIGS
3696
3697     if cache_path.exists():
3698         with open(cache_path, "rb") as f:
3699             cached = pickle.load(f)
3700     else:
3701         cached = {}
3702
3703     changed = False
3704     for cfg in configs:
3705         key = cfg["key"]
3706         if key in cached:
3707             continue
3708         print(f"    enumerating {cfg['title']} (n={cfg['enum_n']})...", flush=True)
3709         data = enumerate_all_graphs(
3710             cfg["enum_n"],
3711             directed=cfg["directed"],
3712             simple=cfg["simple"],
3713             hypergraph=cfg["hypergraph"],
3714             r=cfg["r"],
3715             max_mult=cfg["max_mult"],
3716             separation=cfg["separation"],
3717         )
3718         cached[key] = data
3719         changed = True
3720
3721     if changed:
3722         cache_path.parent.mkdir(parents=True, exist_ok=True)
3723         with open(cache_path, "wb") as f:
3724             pickle.dump(cached, f)
3725
3726     return cached
3727
3728
3729 def plot_scatter_lambda_edges(
3730     enum_data: dict[str, list[tuple[int, int]]],
3731     path: str | Path,
3732 ) -> None:
3733     """Extremal envelope of edge count against binding connectivity, all twelve
3734         ↪ variants.
3735
3736     For each variant the full enumeration is drawn as a faint grey cloud (one
3737     point per labeled graph) with the binding connectivity ' $\lambda^{\max}$ ' on the
3738     horizontal axis and the edge count on the vertical, matching the
3739     edges-against-parameter layout of the all-variant grid. The extremal
3740     envelope -- the densest feasible graph available at each connectivity level
3741     -- is overlaid as an integer staircase with one dot per level, so every
3742     connectivity value carries an exact extremal edge count and the frontier is
3743     read off without any interpolation between levels.
3744
3745     """
3746     def draw_panel(ax, cfg):
3747         key = cfg["key"]
3748         data = enum_data.get(key, [])
3749         if not data:
3750             ax.set_title(cfg["title"], fontsize=9.5)
3751             return

```

```

3751     edges_arr = [e for e, _ in data]
3752     conn_arr = [c for _, c in data]
3753
3754     # Faint cloud of every graph: connectivity (x) against edge count (y).
3755     ax.scatter(conn_arr, edges_arr, s=1.5, c="grey", alpha=0.25,
3756               linewidths=0, rasterized=True)
3757
3758     # Extremal envelope e(lambda) = densest feasible graph with lambda^max
3759     # at most this level: a non-decreasing integer staircase. Reading it at
3760     # lambda = m-1 gives the extremal value of Problem 915 at this n.
3761     levels = sorted(set(conn_arr))
3762     running, env_edges = 0, []
3763     for lv in levels:
3764         running = max(running, max(e for e, c in data if c == lv))
3765         env_edges.append(running)
3766     ax.step(levels, env_edges, where="mid", color=_KUL_BLUE, linewidth=1.8,
3767            zorder=3)
3768     ax.plot(levels, env_edges, "o", color=_KUL_BLUE, markersize=6,
3769            zorder=4, label="extremal envelope")
3770
3771     ax.set_title(cfg["title"], fontsize=9.5)
3772     ax.set_xlabel(r"$\lambda^{\max}$", fontsize=8.5)
3773     ax.set_ylabel("edge count", fontsize=8.5)
3774     ax.tick_params(labelsize=8)
3775     # Both axes count discrete objects (independent routes, edges).
3776     ax.xaxis.set_major_locator(MaxNLocator(integer=True))
3777     ax.yaxis.set_major_locator(MaxNLocator(integer=True))
3778     ax.grid(True, alpha=0.3)
3779     ax.text(0.02, 0.98, f"$n={cfg['enum_n']}$", transform=ax.transAxes,
3780            ha="left", va="top", fontsize=8, color="grey")
3781
3782     _variant_panel_grid(
3783         draw_panel, path=path,
3784         suptitle=("Extremal envelope: edge count against binding connectivity "
3785                  "across all twelve variants (full enumeration)"),
3786         suptitle_fontsize=13)
3787
3788
3789 def _variant_panel_grid(draw_panel, *, suptitle: str, path: str | Path,
3790                        configs=None, suptitle_fontsize: float = 12.0,
3791                        row_label_fontsize: float = 12.0) -> None:
3792     """Shared scaffold for every twelve-panel variant grid (three model rows by
3793     four columns): the distribution grids, the proved/conjectured bound grid, the
3794     sampled grid, and the extremal-envelope scatter. Builds the axes, calls
3795     ‘draw_panel(ax, cfg)’ on each panel config in turn, adds the per-row model
3796     labels and the suptitle, and saves. Only the per-panel drawing and the panel
3797     configs differ between the grids, so the caller supplies both; everything else
3798     (layout, row labels, save) lives here once. ‘configs’ defaults to the
3799     twelve enumeration variants but may be any 12-item list (e.g. the sampled
3800     configs or a precomputed ‘panels’ list).
3801     """
3802     if configs is None:
3803         configs = _VARIANT_ENUM_CONFIGS
3804     fig, axes = plt.subplots(3, 4, figsize=(16, 11))
3805     for cfg, ax in zip(configs, axes.flat):
3806         draw_panel(ax, cfg)
3807     for row, name in enumerate(("simple", "multigraph", "hypergraph $r=3$")):
3808         axes[row, 0].annotate(name, xy=(-0.32, 0.5), xycoords="axes fraction",

```

```

3809             rotation=90, ha="center", va="center",
3810             fontsize=row_label_fontsize, fontweight="bold",
3811             color=_KUL_DARK)
3812 fig.suptitle(suptitle, fontsize=suptitle_fontsize)
3813 fig.tight_layout(rect=(0.02, 0.0, 1.0, 0.97))
3814 _save(path)
3815
3816
3817 def plot_conn_dist_grid(
3818     enum_data: dict[str, list[tuple[int, int]]],
3819     m: int,
3820     path: str | Path,
3821     known_maxima: dict[str, int] | None = None,
3822 ) -> None:
3823     """12-panel histogram of lambda_max distribution, all variants, fixed m.
3824
3825     Bars to the left of the feasibility boundary m-1 are coloured blue (the
3826     graphs we study); bars at or above m are coloured red (infeasible for this
3827     m). A vertical dashed line marks the boundary, and a dotted line marks
3828     the known or conjectured maximum if supplied in 'known_maxima'.
3829     """
3830     threshold = m - 1 # feasible means lambda_max <= threshold
3831
3832     def draw_panel(ax, cfg):
3833         key = cfg["key"]
3834         data = enum_data.get(key, [])
3835         if not data:
3836             ax.set_title(cfg["title"], fontsize=9)
3837             return
3838
3839         conn_vals = [c for _, c in data]
3840         lo, hi = min(conn_vals), max(conn_vals)
3841         levels = list(range(lo, hi + 1))
3842         counts = [conn_vals.count(lv) for lv in levels]
3843         bar_colours = [_KUL_BLUE if lv <= threshold else _RED for lv in levels]
3844         ax.bar(levels, counts, color=bar_colours, edgecolor="white", linewidth
3845             ↳ =0.4)
3846
3847         ax.axvline(threshold + 0.5, color=_WARM, linestyle="--", linewidth=1.8)
3848         if known_maxima and key in known_maxima:
3849             ax.axvline(known_maxima[key], color=_KUL_DARK, linestyle=":",
3850                 linewidth=1.6, label=f"known max = {known_maxima[key]}")
3851
3852         ax.set_title(cfg["title"], fontsize=9.5)
3853         ax.set_xlabel(r"$\lambda^{\max}$", fontsize=8.5)
3854         ax.set_ylabel("graphs", fontsize=8.5)
3855         ax.tick_params(labelsize=8)
3856         # Connectivity is integer-valued: force integer ticks so the directed
3857         # panels stop showing spurious half-integer labels (0.5, 1.5, ...).
3858         ax.set_xticks(levels)
3859         ax.grid(True, axis="y", alpha=0.3)
3860         # The feasibility boundary is identical in all twelve panels, so it is
3861         # explained once in the title and caption rather than repeated as a
3862         # legend in every panel. A legend is drawn only when a per-panel known
3863         # maximum line is present (a value that genuinely differs by panel).
3864         if known_maxima and key in known_maxima:
3865             ax.legend(fontsize=6.8, loc="upper right", framealpha=0.85)
3866         ax.text(0.02, 0.98, f"$n={cfg['enum_n']}$", transform=ax.transAxes,

```

```

3866         ha="left", va="top", fontsize=8, color="grey")
3867
3868     suprtile = (fr"Connectivity distribution across all twelve variants, $m = {m}$
3869         ↳ "
3870         fr"(blue = feasible $\lambda^{\{\max\}} \leq \{\text{threshold}\}$, red =
3871             ↳ infeasible)")
3872     _variant_panel_grid(draw_panel, suprtile=suprtile, path=path,
3873         row_label_fontsize=12)
3874
3875 def _midrange_lambda_threshold(hi: int) -> int:
3876     """A per-variant connectivity boundary at the midpoint of the achievable range
3877         ↳ .
3878
3879     The distribution figures split graphs by ‘‘lambda^max’’ into a blue (low) and
3880     a red (high) population. A single global boundary collapses some panels to
3881     ↳ one
3882     colour (every graph below it, or almost none), because the twelve variants
3883     reach very different connectivity ranges at the sizes enumeration allows.
3884     Splitting instead at ‘‘round(hi/2)’’ -- the middle of what each enumeration
3885     ↳ can
3886     reach -- keeps both populations visible in every panel, scaled to what is
3887     possible there. Floored at one so the boundary is never zero, which would
3888     push every graph above it and leave the blue (low) side empty.
3889     """
3890     return max(1, round(hi / 2))
3891
3892 def plot_pair_conn_dist_grid(
3893     pair_data: dict[str, dict[tuple[int, int], int]],
3894     path: str | Path,
3895 ) -> None:
3896     """12-panel histogram of per-PAIR connectivity, pooled over the full
3897         ↳ enumeration.
3898
3899     Where :func:‘plot_conn_dist_grid’ plots one ‘‘lambda^max’’ per graph, this
3900     pools every vertex pair of every graph: one observation per (graph, pair).
3901     Each bar is split and STACKED by the connectivity of the graph the pair came
3902     from -- blue (bottom) for pairs from graphs whose ‘‘lambda^max’’ stays at or
3903     below the per-panel boundary ‘‘T’’, red (top) for pairs from graphs above it
3904     -- so the bar’s full height is the true number of observations at that level.
3905
3906     ‘‘T’’ is set per variant by :func:‘_midrange_lambda_threshold’, not by a
3907     ↳ single
3908     global ‘‘m’’: at a fixed boundary some panels are entirely one colour, while
3909     ↳ the
3910     mid-range split keeps both populations visible everywhere. Pairs whose own
3911     connectivity exceeds ‘‘T’’ can only come from graphs above ‘‘T’’, so those
3912     ↳ bars
3913     are wholly red; below ‘‘T’’ the colours mix, since a high-connectivity graph
3914     still has many low-connectivity pairs.
3915     """
3916     def draw_panel(ax, cfg):
3917         table = pair_data.get(cfg["key"], {})
3918         if not table:
3919             ax.set_title(cfg["title"], fontsize=9)
3920             return

```

```

3915     hi = max(lmax for (lmax, _pc) in table)
3916     T = _midrange_lambda_threshold(hi)
3917     levels = sorted({pc for (_lmax, pc) in table})
3918     blue = [sum(c for (lmax, pc), c in table.items()
3919                 if pc == lv and lmax <= T) for lv in levels]
3920     red = [sum(c for (lmax, pc), c in table.items()
3921                if pc == lv and lmax > T) for lv in levels]
3922
3923     ax.bar(levels, blue, color=_KUL_BLUE, edgecolor="white", linewidth=0.4,
3924            label=fr"from $\lambda^{\{\max\}}\!\!\leq\!\{T\}$")
3925     ax.bar(levels, red, bottom=blue, color=_RED, edgecolor="white",
3926            linewidth=0.4, label=fr"from $\lambda^{\{\max\}}\!\!>\!\{T\}$")
3927     ax.axvline(T + 0.5, color=_WARM, linestyle="--", linewidth=1.8)
3928
3929     ax.set_title(cfg["title"], fontsize=9.5)
3930     ax.set_xlabel("pair connectivity", fontsize=8.5)
3931     ax.set_ylabel("pair observations", fontsize=8.5)
3932     ax.tick_params(labels=8)
3933     ax.set_xticks(levels)
3934     ax.grid(True, axis="y", alpha=0.3)
3935     ax.legend(fontsize=6.8, loc="upper right", framealpha=0.85)
3936     ax.text(0.02, 0.98, f"$n={cfg['enum_n']}$", transform=ax.transAxes,
3937            ha="left", va="top", fontsize=8, color="grey")
3938
3939     supitle = ("Pair-connectivity distribution across all twelve variants "
3940               "(every vertex pair of every graph pooled, split at each variant's
3941                ↪ ")
3942     r"mid-range $\lambda^{\max}$: blue from low-connectivity graphs,
3943        ↪ red from high)")
3944     _variant_panel_grid(draw_panel, supitle=supitle, path=path,
3945                          row_label_fontsize=12)
3946
3947 def plot_edge_dist_grid(
3948     enum_data: dict[str, list[tuple[int, int]]],
3949     path: str | Path,
3950 ) -> None:
3951     """12-panel histogram of edge count distribution, all variants.
3952
3953     Each bar is split and STACKED by graph connectivity: blue (bottom) for graphs
3954     whose ‘‘ $\lambda^{\max}$ ’’ stays at or below the per-panel boundary ‘‘ $T$ ’’, red (top)
3955     for graphs above it, so the full bar height is the true number of graphs at
3956     that edge count. ‘‘ $T$ ’’ is the midpoint of each variant’s achievable
3957     connectivity range (:func:‘_midrange_lambda_threshold’); a single global
3958     boundary leaves some panels all one colour, the mid-range split keeps both
3959     populations visible. A dotted line marks the densest graph at or below ‘‘ $T$ ’’.
3960
3961     """
3962     def draw_panel(ax, cfg):
3963         key = cfg["key"]
3964         data = enum_data.get(key, [])
3965         if not data:
3966             ax.set_title(cfg["title"], fontsize=9)
3967             return
3968
3969         hi = max(c for _e, c in data)
3970         T = _midrange_lambda_threshold(hi)
3971         max_e = max(e for e, _ in data)
3972         levels = list(range(0, max_e + 1))

```

```

3971     blue = [0] * (max_e + 1)
3972     red = [0] * (max_e + 1)
3973     for e, c in data:
3974         (blue if c <= T else red)[e] += 1
3975
3976     # Stacked, not overlaid: low-connectivity graphs (blue) on the bottom,
3977     # high-connectivity (red) on top, so the full height is the true count.
3978     ax.bar(levels, blue, color=_KUL_BLUE, edgecolor="white", linewidth=0.3,
3979           label=fr"$\lambda^{\{\max\}}\!\leq\!\{T\}$")
3980     ax.bar(levels, red, bottom=blue, color=_RED, edgecolor="white",
3981           linewidth=0.3, label=fr"$\lambda^{\{\max\}}\!>\!\{T\}$")
3982
3983     # The densest graph in the lower-connectivity population: read straight
3984     # off the data, so it lands on the right edge of the blue mass.
3985     if any(blue):
3986         kmax = max(e for e in levels if blue[e])
3987         ax.axvline(kmax, color=_KUL_DARK, linestyle=":", linewidth=1.8,
3988               label=fr"densest $\leq\!\{T\}$: {kmax}")
3989
3990     ax.set_title(cfg["title"], fontsize=8.5)
3991     ax.set_xlabel("edge count", fontsize=7.5)
3992     ax.set_ylabel("graphs", fontsize=7.5)
3993     ax.tick_params(labels=7)
3994     # Edge counts are integers: keep the tick labels integer-valued.
3995     ax.xaxis.set_major_locator(MaxNLocator(integer=True))
3996     ax.grid(True, axis="y", alpha=0.25)
3997     ax.legend(fontsize=6.2, loc="upper right", framealpha=0.8)
3998     ax.text(0.98, 0.02, f"$n={cfg['enum_n']}$", transform=ax.transAxes,
3999           ha="right", va="bottom", fontsize=7, color="grey")
4000
4001     suptitle = ("Edge-count distribution across all twelve variants "
4002               r"(stacked by connectivity, split at each variant's mid-range "
4003               r"$\lambda^{\{\max\}}$: blue low-connectivity graphs, red high)")
4004     _variant_panel_grid(draw_panel, suptitle=suptitle, path=path,
4005                       row_label_fontsize=11)
4006
4007
4008 # --- 3-D BOUND SURFACE: cache solve() over (variant, n, m) grid;
4009     ↪ plot_variant_3d_surfaces draws it ---
4010
4010 # The two multigraph vertex problems reduce exactly to the simple ones: vertex
4011 # mode is blind to parallel copies (see _split_capacity_matrix), so a multigraph
4012 # and its underlying simple graph have the same vertex connectivity. We mirror
4013 # the simple-vertex surface into these keys rather than searching multigraphs,
4014 # whose degenerate free-parallel fills would report a lower bound far above the
4015 # true (simple) value and spike the plot.
4016 _SURFACE_ALIASED_VERTEX = {
4017     "multi_undirected_vertex": "simple_undirected_vertex",
4018     "multi_directed_vertex":   "simple_directed_vertex",
4019 }
4020
4021 # 12 variant configs for the surface computation.
4022 _SURFACE_VARIANT_CONFIGS: list[dict] = [
4023     dict(key="simple_undirected_edge", title="simple undirected edge",
4024           directed=False, simple=True, hypergraph=False, r=3, separation="edge"),
4025     dict(key="simple_undirected_vertex", title="simple undirected vertex",
4026           directed=False, simple=True, hypergraph=False, r=3, separation="vertex")
4027     ↪ ,

```



```

4027     dict(key="simple_directed_edge",      title="simple directed arc",
4028           directed=True, simple=True, hypergraph=False, r=3, separation="edge"),
4029     dict(key="simple_directed_vertex",    title="simple directed vertex",
4030           directed=True, simple=True, hypergraph=False, r=3, separation="vertex")
4031     ↪ ,
4032     dict(key="multi_undirected_edge",    title="multigraph undirected edge",
4033           directed=False, simple=False, hypergraph=False, r=3, separation="edge"),
4034     dict(key="multi_undirected_vertex",  title="multigraph undirected vertex",
4035           directed=False, simple=False, hypergraph=False, r=3, separation="vertex")
4036     ↪ ,
4037     dict(key="multi_directed_edge",      title="multigraph directed arc",
4038           directed=True, simple=False, hypergraph=False, r=3, separation="edge"),
4039     dict(key="multi_directed_vertex",    title="multigraph directed vertex",
4040           directed=True, simple=False, hypergraph=False, r=3, separation="vertex")
4041     ↪ ,
4042     dict(key="hyper_undirected_edge",    title="hypergraph undirected edge",
4043           directed=False, simple=True, hypergraph=True, r=3, separation="edge"),
4044     dict(key="hyper_undirected_vertex",  title="hypergraph undirected vertex",
4045           directed=False, simple=True, hypergraph=True, r=3, separation="vertex")
4046     ↪ ,
4047     dict(key="hyper_directed_edge",      title="hypergraph directed arc",
4048           directed=True, simple=True, hypergraph=True, r=3, separation="edge"),
4049     dict(key="hyper_directed_vertex",    title="hypergraph directed vertex",
4050           directed=True, simple=True, hypergraph=True, r=3, separation="vertex")
4051     ↪ ,
4052 ]
4053
4054 def _surface_known_value(vkey: str, n: int, m: int) -> int | None:
4055     """The proved exact value at '(n, m)' for 'vkey', or 'None' if open here
4056     ↪ .
4057
4058     Returns the closed-form extremal value precisely in the regime where the
4059     thesis proves it, so the surface can colour those bars as exact (blue)
4060     rather than leaving every bar a search lower bound. Outside the proved
4061     regime it returns 'None' and the caller falls back to a discovery search
4062     (a purple lower bound). Each value is capped at the trivial maximum for
4063     its model, since the closed form overshoots in the small-'n' large-'m'
4064     corner where no graph that dense exists.
4065     """
4066     tri_simple = n * (n - 1) // 2
4067     tri_dir = n * (n - 1)
4068     tri_hyp = math.comb(n, 3)
4069     if vkey == "simple_undirected_edge":      # Mader, all m
4070         return min(simple_undirected_edge(n, m), tri_simple)
4071     if vkey == "simple_undirected_vertex":    # Leonard, m<=4
4072         return min(simple_undirected_edge(n, m), tri_simple) if m <= 4 else None
4073     if vkey == "simple_directed_edge":       # exact only at m=2
4074         return min(directed_arc_lower_bound(n, 2), tri_dir) if m == 2 else None
4075     if vkey == "simple_directed_vertex":     # exact only at m=2
4076         return min(directed_arc_lower_bound(n, 2), tri_dir) if m == 2 else None
4077     if vkey == "multi_undirected_edge":     # multi-tree bound, all m
4078         return min(multigraph_undirected_edge(n, m), (m - 1) * tri_simple)
4079     if vkey == "multi_undirected_vertex":   # = simple vertex, m<=4
4080         return min(simple_undirected_edge(n, m), tri_simple) if m <= 4 else None
4081     if vkey == "multi_directed_edge":       # cut-counting proves n<=6
4082         return min(directed_multigraph_arc(n, m), (m - 1) * tri_dir) if n <= 6
4083     ↪ else None

```

```

4078     if vkey == "multi_directed_vertex":           # = simple digraph, exact m=2
4079         return min(directed_arc_lower_bound(n, 2), tri_dir) if m == 2 else None
4080     if vkey == "hyper_undirected_edge":           # incidence-rank, all m
4081         return min(hypergraph_edge(n, m, 3), tri_hyp)
4082     if vkey == "hyper_undirected_vertex":         # incidence-rank lemma, m<=3
4083         return min(hypergraph_edge(n, m, 3), tri_hyp) if m <= 3 else None
4084     # hyper_directed_edge, hyper_directed_vertex: open (new model), no formula.
4085     return None
4086
4087
4088 def compute_surface_cache(
4089     n_range: tuple[int, int] = (3, 9),
4090     m_range: tuple[int, int] = (2, 6),
4091     max_seconds: float = 20.0,
4092     cache_path: str | Path = "figures/surface_cache.json",
4093 ) -> dict:
4094     """Compute the optimal bound at every (variant, n, m) triple, save to a JSON
4095         ↳ cache.
4096
4097     Returns the cache dict: ‘{variant_key: {str(n): {str(m): {value, bound}}}}’.
4098     Where the value is proved (see :func:‘_surface_known_value’) the bound is
4099     recorded as ‘“exact”‘ using the closed form; otherwise a discovery search
4100     supplies a ‘“lower”‘ bound. Missing entries are skipped on later runs.
4101     """
4102     cache_path = Path(cache_path)
4103     if cache_path.exists():
4104         with open(cache_path) as f:
4105             cache = json.load(f)
4106     else:
4107         cache = {}
4108
4109     changed = False
4110     ns = list(range(n_range[0], n_range[1] + 1))
4111     ms = list(range(m_range[0], m_range[1] + 1))
4112
4113     for cfg in _SURFACE_VARIANT_CONFIGS:
4114         vkey = cfg["key"]
4115         if vkey in _SURFACE_ALIASED_VERTEX:
4116             continue # filled by mirroring its simple counterpart, below
4117         if vkey not in cache:
4118             cache[vkey] = {}
4119         for n in ns:
4120             sn = str(n)
4121             if sn not in cache[vkey]:
4122                 cache[vkey][sn] = {}
4123             for m in ms:
4124                 sm = str(m)
4125                 known = _surface_known_value(vkey, n, m)
4126                 if known is not None:
4127                     # Proved regime: use the closed form, mark it exact (blue).
4128                     # Always (re)write, so an older cache whose every cell was
4129                     # tagged "lower" is corrected here without recomputation.
4130                     entry = {"value": known, "bound": "exact"}
4131                     if cache[vkey][sn].get(sm) != entry:
4132                         cache[vkey][sn][sm] = entry
4133                         changed = True
4134                     continue
4135             if sm in cache[vkey][sn]:

```

```

4135         continue # open cell already searched; keep the lower bound
4136     print(f"    surface: {cfg['title']}    n={n}    m={m}", flush=True)
4137     res = solve(
4138         n, m,
4139         directed=cfg["directed"],
4140         simple=cfg["simple"],
4141         hypergraph=cfg["hypergraph"],
4142         r=cfg["r"],
4143         exhaustive=False,
4144         separation=cfg["separation"],
4145         max_seconds=max_seconds,
4146         seed=0,
4147     )
4148     cache[vkey][sn][sm] = {"value": res.value, "bound": res.bound}
4149     changed = True
4150
4151     # Mirror the simple-vertex surfaces into their multigraph aliases, so the two
4152     # provably-equal problems show identical bars instead of two independent
4153     # (and possibly degenerate) searches.
4154     for multi_key, simple_key in _SURFACE_ALIASED_VERTEX.items():
4155         if simple_key in cache and cache.get(multi_key) != cache[simple_key]:
4156             cache[multi_key] = copy.deepcopy(cache[simple_key])
4157             changed = True
4158
4159     if changed:
4160         cache_path.parent.mkdir(parents=True, exist_ok=True)
4161         with open(cache_path, "w") as f:
4162             json.dump(cache, f)
4163
4164     return cache
4165
4166
4167 def plot_variant_3d_surfaces(
4168     cache_path: str | Path,
4169     path: str | Path,
4170 ) -> None:
4171     """3-D bar chart of the optimal bound over the (n, m) grid, all twelve
4172         ↳ variants.
4173
4174     Each bar's height is the bound value; exact points are blue, lower-bound
4175     points are purple. The 3x4 layout matches the flat grid figure.
4176     """
4177     with open(cache_path) as f:
4178         cache = json.load(f)
4179
4180     fig = plt.figure(figsize=(20, 13))
4181     row_names = ("simple", "multigraph", "hypergraph $r=3$")
4182
4183     for idx, cfg in enumerate(_SURFACE_VARIANT_CONFIGS):
4184         vkey = cfg["key"]
4185         ax = fig.add_subplot(3, 4, idx + 1, projection="3d")
4186
4187         variant_data = cache.get(vkey, {})
4188         ns_exact, ms_exact, vs_exact = [], [], []
4189         ns_lower, ms_lower, vs_lower = [], [], []
4190
4191         for sn, m_dict in sorted(variant_data.items(), key=lambda x: int(x[0])):
4192             n = int(sn)

```

```

4192         for sm, entry in sorted(m_dict.items(), key=lambda x: int(x[0])):
4193             m_val = int(sm)
4194             v = entry["value"]
4195             if entry["bound"] == "exact":
4196                 ns_exact.append(n)
4197                 ms_exact.append(m_val)
4198                 vs_exact.append(v)
4199             else:
4200                 ns_lower.append(n)
4201                 ms_lower.append(m_val)
4202                 vs_lower.append(v)
4203
4204     dx = dy = 0.6
4205     if ns_lower:
4206         ax.bar3d(
4207             [x - dx / 2 for x in ns_lower],
4208             [y - dy / 2 for y in ms_lower],
4209             [0] * len(vs_lower),
4210             dx, dy, vs_lower,
4211             color=_VIOLET, alpha=0.7, shade=True,
4212             edgecolor="white", linewidth=0.25,
4213         )
4214     if ns_exact:
4215         ax.bar3d(
4216             [x - dx / 2 for x in ns_exact],
4217             [y - dy / 2 for y in ms_exact],
4218             [0] * len(vs_exact),
4219             dx, dy, vs_exact,
4220             color=_KUL_BLUE, alpha=0.9, shade=True,
4221             edgecolor="white", linewidth=0.25,
4222         )
4223
4224     # Same camera for every panel so the twelve are read side by side, and a
4225     # z-axis anchored at 0 so bar heights are comparable within a panel.
4226     ax.view_init(elev=24, azim=-58)
4227     peak = max(vs_exact + vs_lower, default=1)
4228     ax.set_zlim(0, peak * 1.05)
4229     for pane in (ax.xaxis, ax.yaxis, ax.zaxis):
4230         pane.pane.set_alpha(0.04)
4231
4232     ax.set_title(cfg["title"], fontsize=7.5, pad=2)
4233     ax.set_xlabel("$n$", fontsize=7, labelpad=2)
4234     ax.set_ylabel("$m$", fontsize=7, labelpad=2)
4235     ax.set_zlabel("bound", fontsize=7, labelpad=2)
4236     ax.tick_params(labelsize=6)
4237
4238     # Row labels
4239     for row, name in enumerate(row_names):
4240         row_ax = fig.axes[row * 4]
4241         row_ax.text2D(-0.28, 0.5, name, transform=row_ax.transAxes,
4242                      rotation=90, ha="center", va="center",
4243                      fontsize=11, fontweight="bold", color=_KUL_DARK)
4244
4245     # Manual legend patches
4246     legend_elems = [
4247         Patch(facecolor=_KUL_BLUE, alpha=0.85, label="exact (proved)"),
4248         Patch(facecolor=_VIOLET, alpha=0.65, label="lower bound (search)"),
4249     ]

```

```

4250     fig.legend(handles=legend_elems, loc="lower center", ncol=2,
4251                fontsize=10, framealpha=0.9)
4252
4253     fig.suptitle("Erdős 915: optimal bound over  $(n, m)$ ",
4254                "blue where proved, purple where only a search lower bound is
4255                ↪ known",
4256                fontsize=13, y=0.99)
4257     _save(path, tight=False)
4258
4259 # --- 3-D THRESHOLD HISTOGRAM:  $\lambda^{\max}$  distribution vs  $p$  for three variants ---
4260
4261
4262 def plot_conn_threshold_3d(
4263     path: str | Path,
4264     n: int = 30,
4265     m: int = 3,
4266     p_values: list[float] | None = None,
4267     samples: int = 400,
4268     seed: int = 7,
4269 ) -> None:
4270     """3-D bar chart of the  $\lambda^{\max}$  distribution over density, for three
4271     ↪ variants.
4272
4273      $x$  = density in units of that model's OWN threshold,  $y$  =  $\lambda^{\max}$  value,
4274      $z$  = fraction of samples. Each panel is swept in units of its own  $p^*$ , and the
4275     threshold line sits at 1 in all three, because the models do NOT share a
4276     density scale. The threshold is where a vertex's expected degree reaches  $m$ ,
4277     which for a graph is  $p^*(n-1) = m$ , giving  $p^* = m/n$ , but for an  $r$ -uniform
4278     binomial hypergraph is  $p^*C(n-1, r-1) = m$ , giving  $p^* = m/C(n-1, r-1) =$ 
4279      $\Theta(m/n^{(r-1)})$ . Plotting the hypergraph panel against  $m/n$  would put the
4280     line in the wrong place by a factor of order  $n^{(r-2)}$ . Only the graph panels
4281     are covered by thm:gnp-threshold; the hypergraph panel is an observation.
4282
4283     Three panels: undirected edge (uses ' $n$ '), directed arc (uses ' $n$ '),
4284     hypergraph edge (uses  $\min(n, 8)$  to keep flow computation fast).
4285
4286     # Sweep in units of the model's own threshold, so all three panels are
4287     # directly comparable and each transition is visible where it actually is.
4288     ratios = ([i / 5.0 for i in range(1, 20)] if p_values is None else None)
4289
4290     rng = random.Random(seed)
4291     # Hypergraph flow computation scales poorly: cap  $n$  at 8 for that panel.
4292     n_hyper = min(n, 8)
4293
4294     # The hypergraph threshold is  $m / C(n-1, r-1)$  with  $r = 3$ . Write the binomial
4295     # with its arguments already evaluated: matplotlib's mathtext lays out
4296     # " $\binom{8-1}{2}$ " so that the "-1" reads as a superscript on the 8, which
4297     # renders as a garbled binomial rather than "7 choose 2".
4298     hyper_top = n_hyper - 1
4299     hyper_pstar = math.comb(hyper_top, 2)
4300     variants_3d = [
4301         dict(label=f"undirected edge ( $n=\{n\}$ )", directed=False, separation="edge"
4302             ↪ ,
4303             hypergraph=False, panel_n=n, p_star=m / n,
4304             star_text=f" $m/n = \{m\}/\{n\}$ "),
4305         dict(label=f"directed arc ( $n=\{n\}$ )", directed=True, separation="edge"
4306             ↪ ,

```

```

4304         hypergraph=False, panel_n=n, p_star=m / n,
4305         star_text=f"$m/n = {m}/{n}$"),
4306     dict(label=f"hypergraph edge ($r=3$, $n={n\_hyper}$)", directed=False,
4307         separation="edge", hypergraph=True, panel_n=n_hyper,
4308         p_star=m / hyper_pstar,
4309         star_text=(fr"$m/\binom{{{hyper\_top}}}{{2}} = $"
4310                   fr"{m}/{hyper\_pstar}$")),
4311 ]
4312
4313 fig = plt.figure(figsize=(18, 6))
4314
4315 for panel_idx, vdesc in enumerate(variants_3d):
4316     ax = fig.add_subplot(1, 3, panel_idx + 1, projection="3d")
4317     panel_n = vdesc["panel_n"]
4318     p_star = vdesc["p_star"]
4319     panel_ps = (p_values if p_values is not None
4320                else [min(1.0, r * p_star) for r in ratios])
4321
4322     all_conn_vals: list[int] = []
4323     dist_by_p: list[tuple[float, list[int]]] = []
4324
4325     for p in panel_ps:
4326         conn_list: list[int] = []
4327         for _ in range(samples):
4328             if vdesc["hypergraph"]:
4329                 cand = _hyperedge_candidates(panel_n, 3, vdesc["directed"])
4330                 chosen = [e for e in cand if rng.random() < p]
4331                 h = Hypergraph(panel_n, chosen, directed=vdesc["directed"])
4332                 c = max_hyper_connectivity(h, vertex_split=False)
4333             else:
4334                 g = sample_random_graph(panel_n, p, vdesc["directed"], rng)
4335                 if vdesc["separation"] == "edge":
4336                     c = max_edge_connectivity(g)
4337                 else:
4338                     c = max_vertex_connectivity(g)
4339             conn_list.append(c)
4340         dist_by_p.append((p, conn_list))
4341         all_conn_vals.extend(conn_list)
4342
4343     if not all_conn_vals:
4344         continue
4345
4346     lo, hi = 0, max(all_conn_vals)
4347     levels = list(range(lo, hi + 1))
4348
4349     # x is the density in units of this panel's own threshold.
4350     xs = [p / p_star for p, _ in dist_by_p]
4351     width = 0.9 * min((b - a) for a, b in zip(xs, xs[1:])) if len(xs) > 1 else
4352         ↪ 0.1
4353     for x, (_, conn_list) in zip(xs, dist_by_p):
4354         total = len(conn_list)
4355         for lv in levels:
4356             frac = conn_list.count(lv) / total
4357             if frac > 0:
4358                 ax.bar3d(x - width / 2, lv - 0.4, 0, width, 0.8, frac,
4359                         color=_KUL_BLUE, alpha=0.7, shade=True)
4360
4361     # The threshold sits at 1 in these units, by construction.

```

```

4361         for lv in levels:
4362             ax.plot([1.0, 1.0], [lv - 0.4, lv + 0.4], [0, 0],
4363                     color=_WARM, linewidth=0.5, alpha=0.4)
4364         zmax = max(conn_list.count(lv) / len(conn_list)
4365                     for _, conn_list in dist_by_p
4366                     for lv in levels
4367                     if conn_list.count(lv) > 0)
4368         ax.plot([1.0, 1.0], [lo, hi], [zmax * 0.9, zmax * 0.9],
4369                 color=_WARM, linewidth=2.5, linestyle="--", label="$p / p^* = 1$")
4370
4371         ax.set_title(f"{vdesc['label']}\n$p^* = $ {vdesc['star_text']}", fontsize
4372                     ↪ =9)
4373         ax.set_xlabel("$p / p^*$", fontsize=9)
4374         ax.set_ylabel(r"$\lambda^{\{\max\}}$", fontsize=9)
4375         ax.set_zlabel("fraction", fontsize=9)
4376         ax.tick_params(labelsize=7)
4377
4378     fig.suptitle(
4379         fr"Distribution of $\lambda^{\{\max\}}$ against density, $m = {m}$, each
4380         ↪ ↪ model "
4381         fr"in units of its own threshold $p^*$ (the density at which a vertex's "
4382         fr"expected degree reaches $m$)",
4383         fontsize=11,
4384     )
4385     _save(path, tight=False)

```

C.1.8 The open variants

These routines exist to attack the questions the thesis leaves open. They measure the hypergraph vertex problem, work with fractional weightings where the integer problem is out of reach, and provide the exhaustive maximisers used to test conjectures at the sizes where testing is still possible. Several results in [Appendix A](#) were found, or in two cases refuted, by running exactly these functions.

```

4386 # --- OPEN-VARIANT EXPLORATION: hypergraph vertex connectivity, fractional search
4387 ↪ ↪ tools ---
4388
4389 def max_hyper_vertex_connectivity(hypergraph: Hypergraph) -> int:
4390     """'kappa^max' for the hypergraph vertex problem (readability wrapper)."""
4391     return max_hyper_connectivity(hypergraph, vertex_split=True)
4392
4393
4394 def _hyper_codegree_ok(edges, bound: int) -> bool:
4395     """Necessary feasibility filter: two vertices in > 'bound' common
4396     hyperedges already carry that many one-step disjoint Berge routes."""
4397
4398     codegree: Counter = Counter()
4399     for edge in edges:
4400         for pair in combinations(sorted(edge), 2):
4401             codegree[pair] += 1
4402             if codegree[pair] > bound:
4403                 return False
4404     return True
4405

```

```

4406
4407 def hyper_vertex_feasible_exists(n: int, r: int, m: int, q: int,
4408                                 multi: bool = True) -> bool:
4409     """Exhaustively decide whether some ‘r’-uniform (multi-)hypergraph on
4410     ‘n’ vertices with ‘q’ hyperedges has ‘kappa^max <= m - 1’.

```



```

4463     two prunings. Feasibility is monotone (adding a hyperedge never lowers any
4464     Berge connectivity), so once the active set is infeasible no superset can be
4465     feasible and the include branch is dropped; and a branch is cut once
4466     ‘‘active + remaining’’ cannot beat the best feasible count already seen. A
4467     short randomised warm start raises that incumbent first, so the bound prune
4468     bites early. ‘‘seed_lb’’ is an externally known feasible lower bound (e.g.
         ⇨ the
4469     forward value when maximising over the general model, which contains every
4470     forward hypergraph) used to start the incumbent; it never changes the proven
4471     optimum, only the pruning.
4472     """
4473     deadline = time.time() + time_limit
4474     # Warm start: a short greedy randomised lower bound sharpens the bound prune;
4475     # kept brief so the branch and bound, which does the actual proving, keeps
4476     # the budget.
4477     warm_share = min(0.3 * time_limit, 1.5)
4478     lb, _ = _random_hypergraph_search(
4479         n, r, m, time.time() + warm_share, seed=0,
4480         directed=directed, vertex_split=vertex_split, kind=kind)
4481     lb = max(lb, seed_lb)
4482
4483     candidates = _hyperedge_candidates(n, r, directed, kind=kind)
4484     total = len(candidates)
4485     best = [lb]
4486     timed_out = [False]
4487     active = Hypergraph(n, directed=directed)
4488
4489     def dfs(pos: int, count: int) -> None:
4490         if timed_out[0]:
4491             return
4492         if count + (total - pos) <= best[0]:
4493             return # cannot beat the incumbent
4494         if time.time() > deadline:
4495             timed_out[0] = True
4496             return
4497         if pos == total:
4498             if count > best[0]:
4499                 best[0] = count
4500             return
4501         # Include candidates[pos], but only if the set stays feasible.
4502         active.add_hyperedge(candidates[pos])
4503         if max_hyper_connectivity(active, vertex_split=vertex_split) <= m - 1:
4504             dfs(pos + 1, count + 1)
4505         active.hyperedges.pop()
4506         # Exclude candidates[pos].
4507         dfs(pos + 1, count)
4508
4509     dfs(0, 0)
4510     return best[0], (not timed_out[0])
4511
4512
4513 def _c_flat(mu: np.ndarray) -> tuple[np.ndarray, "_ct.POINTER[_ct.c_int]"]:
4514     """Return a contiguous int32 copy of ‘‘mu’’ and a ctypes c_int pointer into it
         ⇨ ."""
4515     flat = np.ascontiguousarray(mu, dtype=np.int32)
4516     ptr = flat.ctypes.data_as(_ct.POINTER(_ct.c_int))
4517     return flat, ptr # caller must keep flat alive while ptr is in use
4518

```

```

4519
4520 def _tiny_maxflow(mu: np.ndarray, n: int, s: int, t: int, cap: int) -> bool:
4521     """Return True iff the integer max-flow from s to t under capacities 'mu'
4522     exceeds 'cap'. Plain DFS augmentation (Ford-Fulkerson with a stack, not
4523     BFS/Edmonds-Karp); built for n <= 7, where it beats both the networkx call
4524     and scipy's C 'maximum_flow' by orders of magnitude inside the hot
4525     enumeration/search loops (measured ~16x faster than scipy at n=7: the
4526     library calls' per-call sparse-matrix construction and dispatch overhead
4527     dominates at this size, while the 'cap' lets this routine stop after
4528     cap+1 augmenting paths). Correct for integer capacities: each augmentation
4529     increases flow by at least 1, so the loop terminates in at most
4530     max-flow-value iterations."""
4531     if _C is not None and n <= 7:
4532         flat, ptr = _c_flat(mu)
4533         return bool(_C.tiny_maxflow(ptr, n, s, t, cap))
4534     residual = mu.astype(int)          # astype already returns a fresh, mutable
4535     ↪ copy
4536     flow = 0
4537     while flow <= cap:
4538         parent = [-1] * n
4539         parent[s] = s
4540         queue = [s]
4541         while queue and parent[t] == -1:
4542             u = queue.pop()
4543             for v in range(n):
4544                 if parent[v] == -1 and residual[u, v] > 0:
4545                     parent[v] = u
4546                     queue.append(v)
4547             if parent[t] == -1:
4548                 return False # flow is maximal and <= cap
4549             bottleneck = 10 ** 9
4550             v = t
4551             while v != s:
4552                 bottleneck = min(bottleneck, residual[parent[v], v])
4553                 v = parent[v]
4554             v = t
4555             while v != s:
4556                 residual[parent[v], v] -= bottleneck
4557                 residual[v, parent[v]] += bottleneck
4558                 v = parent[v]
4559             flow += bottleneck
4560     return True
4561
4562 def _canonical_form(mu: np.ndarray) -> bytes:
4563     """A canonical key for the isomorphism class of a directed multigraph.
4564
4565     The key is the lexicographically smallest flattened multiplicity matrix over
4566     all vertex relabellings: two multiplicity matrices have the same key iff one
4567     is a vertex permutation of the other. This is the SOURCE OF TRUTH for
4568     isomorphism here, with no dependency on any external library, so the
4569     enumeration's correctness never rests on an outside binary. For 'n=7' it is
4570     5040 permutations per graph, which is cheap on CPU; storing one key per class
4571     is what bounds the enumeration's memory to the number of classes rather than
4572     the number of labelled copies.
4573     """
4574     n = mu.shape[0]
4575     if _C is not None and n <= 7:

```

```

4576         flat, ptr = _c_flat(mu)
4577         out = np.empty(n * n, dtype=np.int32)
4578         out_ptr = out.ctypes.data_as(_ct.POINTER(_ct.c_int))
4579         _C.canonical_form_min(ptr, n, out_ptr)
4580         return out.tobytes()
4581     best: bytes | None = None
4582     for perm in permutations(range(n)):
4583         p = list(perm)
4584         cand = mu[np.ix_(p, p)].tobytes()
4585         if best is None or cand < best:
4586             best = cand
4587     assert best is not None # n >= 1, so at least the identity permutation exists
4588     return best
4589
4590
4591 def enumerate_extremal_directed_multigraphs(
4592     n: int, m: int, target_arcs: int, max_degree: int | None = None,
4593     up_to_iso: bool = True,
4594 ) -> list[np.ndarray]:
4595     """Exhaustively list the directed multigraphs on 'n' vertices with
4596     multiplicities in {0..m-1}, ' $\lambda^{\max} \leq m-1$ ', exactly 'target_arcs'
4597     arcs and (optionally) maximum total degree at most 'max_degree'."""
4598
4599     This is the finite-base tool for the m = 3 chain (see claude.md and the
4600     commented rem:odd-step-roadmap): the characterisations at n = 4, 5 feed
4601     the recursion behind statement (b) at n = 7. DFS in vertex-block order
4602     with three prunings: multiplicity capacity, the induced-subgraph arc bound
4603     on every completed prefix, and exact flow checks on completed prefixes
4604     (induced flows only grow, so a violation is final).
4605
4606     SOUNDNESS: the induced-subgraph bound is a PROVED upper bound for j <= 6
4607     (from the cut-counting MILP; see 'known' dict below). For j >= 7 it
4608     falls back to the CONJECTURED bipartite bound  $\lfloor j^2/4 \rfloor$ , which is not
4609     yet proved. Therefore this function is a sound complete search only for
4610     n <= 6. For n >= 7 it is a heuristic lower-bound search: it may miss
4611     extremal graphs whose j=7 prefix exceeds the bipartite bound (if the
4612     conjecture is false) or whose search branch was pruned by it.
4613     Returns multiplicity matrices, deduplicated up to vertex permutation when
4614     'up_to_iso' is set. Deduplication is STREAMED through a canonical form
4615     (:func: '_canonical_form') as graphs are found, so peak memory is bounded by
4616     the number of isomorphism classes rather than the number of labelled copies
4617     (the previous buffer-everything approach exhausted RAM on n=7). The returned
4618     list, and its order, are unchanged by this.
4619     """
4620     # Ordered pairs grouped so that all pairs inside {0..j} precede vertex j+1.
4621     order: list[tuple[int, int]] = []
4622     for j in range(1, n):
4623         for i in range(j):
4624             order.append((i, j))
4625             order.append((j, i))
4626     block_end = {j: 2 * ((j + 1) * j // 2) for j in range(1, n)} # prefix length
4627     # SOUNDNESS NOTE: for j <= 6 the prefix cap (m-1)*M*(j) is a PROVED upper
4628     # bound, so pruning at that cap is safe. For j >= 7, _PROVEN_MSTAR.get(j, 0)
4629     #   ↪ == 0
4630     # and we fall back to (m-1)*floor(j^2/4), which equals the conjectured
4631     # bipartite bound and is NOT yet proved. Any call with n >= 7 therefore
4632     # uses an unverified pruning at the j=7 boundary: the enumeration is sound
4633     # as a lower-bound search but cannot certify completeness for n >= 7.

```

```

4633     # To produce a proof for n=7 the j=7 pruning must be disabled or replaced
4634     # by a proved bound (e.g. from the MILP once M*(7) is confirmed).
4635     prefix_cap = {j: (m - 1) * max(_PROVEN_MSTAR.get(j, 0), (j * j) // 4)
4636                  for j in range(2, n + 1)}
4637
4638     mu = np.zeros((n, n), dtype=int)
4639     total_pairs = len(order)
4640     # Stream results straight into the deduplicated output instead of buffering
4641     # every labelled feasible matrix first. Memory is then bounded by the number
4642     # of ISOMORPHISM CLASSES (one canonical key in 'seen', one representative in
4643     # the list), not by the number of labelled copies, which is what previously
4644     # blew up to tens of gigabytes on n=7. The set of matrices returned, and
4645     # their order, are identical to the old buffer-then-dedup code.
4646     seen: set[bytes] = set()
4647     representatives: list[np.ndarray] = []
4648
4649     def _record(matrix: np.ndarray) -> None:
4650         # The DFS only checked prefixes {0..j} for j < n, so confirm the full
4651         # n-vertex graph is feasible before keeping it (cheap, and done first so a
4652         # rejected graph never pays for the canonical form).
4653         if any(_tiny_maxflow(matrix, n, s, t, m - 1)
4654              for s in range(n) for t in range(n) if s != t):
4655             return
4656         if up_to_iso:
4657             canon = _canonical_form(matrix)
4658             if canon in seen:
4659                 return
4660             seen.add(canon)
4661             representatives.append(matrix.copy())
4662
4663     def dfs(pos: int, arcs: int) -> None:
4664         if arcs + (m - 1) * (total_pairs - pos) < target_arcs or arcs >
4665             ↪ target_arcs:
4666             return
4667         if pos == total_pairs:
4668             if arcs == target_arcs:
4669                 _record(mu)
4670                 return
4671         completed = next((j for j in range(1, n) if block_end[j] == pos), None)
4672         if completed is not None:
4673             j = completed + 1 # prefix {0..completed} has j vertices
4674             if arcs > prefix_cap.get(j, 10 ** 9):
4675                 return
4676             for s in range(j):
4677                 for t in range(j):
4678                     if s != t and _tiny_maxflow(mu, n, s, t, m - 1):
4679                         return
4680         u, v = order[pos]
4681         for value in range(m):
4682             mu[u, v] = value
4683             if max_degree is None or (mu[u, :].sum() + mu[:, u].sum() <=
4684                 ↪ max_degree
4685                 and mu[v, :].sum() + mu[:, v].sum() <=
4686                 ↪ max_degree):
4687                 dfs(pos + 1, arcs + value)
4688             mu[u, v] = 0
4689     dfs(0, 0)

```

```

4688     return representatives
4689
4690
4691 def _geng_support_graphs(
4692     n: int, min_edges: int, max_edges: int, geng_path: str = "geng",
4693 ) -> Iterator[list[tuple[int, int]]]:
4694     """Yield one edge list per isomorphism class of simple graph on 'n'
4695     vertices with 'min_edges' to 'max_edges' edges, via nauty's 'geng'.
4696
4697     Each edge is an ordered tuple '(u, v)' with 'u < v'; isolated vertices
4698     are kept ('geng' always emits all 'n' vertices). 'geng' writes the
4699     non-isomorphic graphs in graph6 to stdout and we parse each with _graph6_edges
4700     ↳ .
4701
4702     Raises 'RuntimeError' if 'geng' is not on PATH.
4703     """
4704     exe = shutil.which(geng_path)
4705     if exe is None:
4706         raise RuntimeError(
4707             f'nauty's '{geng_path}' was not found on PATH. Install nauty "
4708             "(e.g. the 'nauty' package) or pass geng_path=...; the "
4709             "generation enumerator needs it."
4710         )
4711     if max_edges < min_edges:
4712         return
4713     proc = subprocess.run(
4714         [exe, str(n), f"{min_edges}:{max_edges}"],
4715         capture_output=True, check=True,
4716     )
4717     for line in proc.stdout.splitlines():
4718         line = line.strip()
4719         if not line:
4720             continue
4721         yield _graph6_edges(line)
4722
4723 def _graph6_edges(data: bytes) -> list[tuple[int, int]]:
4724     """Decode one graph6 line (geng's output format) into its '(i, j)' edges, '
4725     ↳ i < j'.
4726
4727     graph6 stores 'n' in the first byte (offset 63), then the upper-triangle
4728     adjacency bits column by column (for 'j' from 1, all 'i < j'), six bits
4729     ↳ per
4730     subsequent byte, high bit first. We only ever generate 'n <= 12' supports,
4731     ↳ so
4732     the single-byte header case is all that is needed and the format needs no
4733     networkx to read.
4734     """
4735     values = [byte - 63 for byte in data]
4736     n = values[0]
4737     bits = [(value >> shift) & 1 for value in values[1:] for shift in range(5, -1,
4738     ↳ -1)]
4739     index = 0
4740     edges: list[tuple[int, int]] = []
4741     for j in range(1, n):
4742         for i in range(j):
4743             if bits[index]:
4744                 edges.append((i, j))
4745             index += 1

```

```

4741     return edges
4742
4743
4744 def _decorate_support_worker(task: tuple) -> list[np.ndarray]:
4745     """Every feasible extremal decoration of ONE undirected support graph.
4746
4747     Module-level and picklable, so :func:`
4748         ↪ enumerate_extremal_directed_multigraphs_via_generation`
4749     can fan the supports out across processes (each support is independent, and
4750     two non-isomorphic supports can never yield the same multigraph). Returns one
4751     multiplicity matrix per decoration, deduplicated within this support when
4752     ‘‘up_to_iso‘‘. The decoration logic and its pruning are identical to the
4753     sequential generator; only the ownership of the scratch arrays moves here.
4754     """
4755     n, m, target_arcs, max_degree, up_to_iso, edges = task
4756     per_edge_max = 2 * (m - 1)
4757     # Per-edge states (a, b) = (mu[u,v], mu[v,u]) excluding (0,0): a support edge
4758     # carries at least one arc in some direction.
4759     states = [(a, b) for a in range(m) for b in range(m) if (a, b) != (0, 0)]
4760     mu = np.zeros((n, n), dtype=int)
4761     deg = np.zeros(n, dtype=int)
4762     out_deg = np.zeros(n, dtype=int)
4763     in_deg = np.zeros(n, dtype=int)
4764     seen: set[bytes] = set()
4765     reps: list[np.ndarray] = []
4766
4767     def feasible_prefix(j: int) -> bool:
4768         cap = m - 1
4769         for s in range(j):
4770             if out_deg[s] <= cap:
4771                 continue
4772             for t in range(j):
4773                 if s != t and in_deg[t] > cap and _tiny_maxflow(mu, n, s, t, cap):
4774                     return False
4775         return True
4776
4777     # Decorate in vertex-block order: pairs sorted by larger endpoint, then
4778     # smaller, so after the last pair whose larger endpoint is w the induced
4779     # subgraph on {0..w} is complete and the prefix bound for size w+1 applies.
4780     sps = sorted(edges, key=lambda e: (e[1], e[0]))
4781     bound: list[int | None] = [None] * len(sps)
4782     for i, (_, w) in enumerate(sps):
4783         if i + 1 == len(sps) or sps[i + 1][1] > w:
4784             bound[i] = w + 1
4785
4786     def decorate(idx: int, arcs: int) -> None:
4787         remaining = len(sps) - idx
4788         if arcs + remaining > target_arcs:
4789             return
4790         if arcs + remaining * per_edge_max < target_arcs:
4791             return
4792         if idx == len(sps):
4793             # every support edge decorated
4794             if arcs == target_arcs and feasible_prefix(n):
4795                 if up_to_iso:
4796                     canon = _canonical_form(mu)
4797                     if canon in seen:
4798                         return
4799                     seen.add(canon)

```

```

4798         reps.append(mu.copy())
4799     return
4800     u, v = sps[idx]
4801     j = bound[idx]          # prefix size to verify after this edge, or None
4802     for a, b in states:
4803         s = a + b
4804         if arcs + s > target_arcs:
4805             continue
4806         if max_degree is not None and (deg[u] + s > max_degree
4807                                         or deg[v] + s > max_degree):
4808             continue
4809         mu[u, v] = a
4810         mu[v, u] = b
4811         deg[u] += s
4812         deg[v] += s
4813         out_deg[u] += a; in_deg[v] += a
4814         out_deg[v] += b; in_deg[u] += b
4815         ok = True
4816         if j is not None:      # block boundary: prefix {0..j-1} is complete
4817             cap = _PROVEN_MSTAR.get(j)      # j == prefix size
4818             if cap is not None and arcs + s > (m - 1) * cap:
4819                 ok = False                  # PROVED induced-arc bound
4820             elif not feasible_prefix(j):
4821                 ok = False
4822         if ok:
4823             decorate(idx + 1, arcs + s)
4824             deg[u] -= s
4825             deg[v] -= s
4826             out_deg[u] -= a; in_deg[v] -= a
4827             out_deg[v] -= b; in_deg[u] -= b
4828             mu[u, v] = 0
4829             mu[v, u] = 0
4830
4831     decorate(0, 0)
4832     return reps
4833
4834
4835 def enumerate_extremal_directed_multigraphs_via_generation(
4836     n: int, m: int, target_arcs: int, max_degree: int | None = None,
4837     up_to_iso: bool = True, geng_path: str = "geng", parallel: bool = True,
4838 ) -> list[np.ndarray]:
4839     """Sound, geng-seeded twin of :func:'enumerate_extremal_directed_multigraphs'.
4840
4841     Lists every directed multigraph on 'n' vertices with multiplicities in
4842     {0..m-1}, ' $\lambda^{\max} \leq m-1$ ', exactly 'target_arcs' arcs and
4843     (optionally) maximum total degree ' $\leq \max\_degree$ ', deduplicated up to
4844     isomorphism. Same object and same return format (multiplicity matrices) as
4845     the DFS enumerator.
4846
4847     Following J. Goedgebeur's suggestion, the underlying undirected SUPPORT
4848     graph (the pairs carrying at least one arc) is generated once per
4849     isomorphism class by nauty's 'geng'; each support is then decorated, in
4850     vertex-block order, with a directed multiplicity pair '(mu[u,v], mu[v,u])'
4851     in {0..m-1}2 minus (0,0) per support edge. Pruning is the running arc
4852     total, the degree cap, and -- at each completed prefix on 'j' vertices --
4853     the PROVED induced-arc bound 'arcs(prefix) <= (m-1) * M*(j)' together with a
4854     prefix feasibility check (induced max-flows only grow, so a prefix that
4855     already exceeds the cap is dead).

```

```

4856
4857 Soundness for every ‘n’. Two directed multigraphs with non-isomorphic
4858 supports are non-isomorphic, and every isomorphism class with a given
4859 support class has a labelling whose support equals geng’s representative, so
4860 decorating every representative and canonical-deduplicating returns exactly
4861 one matrix per isomorphism class. Crucially the induced-arc bound is used
4862 ONLY for ‘j <= 6’, where ‘M*(j)’ is proved; for ‘j >= 7’ no arc bound
4863 is applied (only the global ‘target_arcs’ and feasibility). This is the
4864 difference from :func:‘enumerate_extremal_directed_multigraphs’, which falls
4865 back to the CONJECTURED ‘floor(j^2/4)’ at ‘j >= 7’ and is therefore a
4866 complete search only for ‘n <= 6’. This generator is complete for all
4867 ‘n’ (it never assumes an unproved value), so it can certify the finite
4868 ‘n = 7’ classification once it finishes.
4869
4870 Requires nauty’s ‘geng’ on PATH. For ‘n <= 6’ it returns the same set
4871 of isomorphism classes as :func:‘enumerate_extremal_directed_multigraphs’;
4872 ‘tests/test_solve’ checks that equality.
4873
4874 With ‘parallel’ (default) the supports are decorated concurrently across
4875 processor cores via a :class:‘concurrent.futures.ProcessPoolExecutor’, since
4876 each support is independent. This is where the ‘n = 7’ classification
4877 becomes practical on a multi-core machine: the work scales with the number of
4878 supports, which is exactly what fans out. It falls back to a sequential run
4879 (with a warning) where multiprocessing is unavailable, and the result is
4880 identical either way.
4881 """
4882 if m < 1:
4883     raise ValueError("m must be >= 1")
4884 per_edge_max = 2 * (m - 1)          # most arcs one undirected pair can hold
4885 if per_edge_max == 0:              # m == 1: only the empty graph is
4886     ↪ feasible
4887     return [np.zeros((n, n), dtype=int)] if target_arcs == 0 else []
4888 min_edges = (target_arcs + per_edge_max - 1) // per_edge_max
4889 max_edges = min(target_arcs, n * (n - 1) // 2)
4890
4891 # geng emits each undirected support iso-class once, and decorating one
4892 # support is independent of the others (two non-isomorphic supports can never
4893 # yield the same multigraph), so the supports fan out across processes, each
4894 # with its own scratch arrays inside _decorate_support_worker. The PROVED
4895 # M*(j) bound is applied there only for j <= 6; j >= 7 gets no arc bound, so
4896 # the search stays sound at every n.
4897 tasks = [(n, m, target_arcs, max_degree, up_to_iso, edges)
4898         for edges in _geng_support_graphs(n, min_edges, max_edges, geng_path)
4899         ↪ ]
4900
4901 representatives: list[np.ndarray] = []
4902 if parallel and len(tasks) > 1:
4903     try:
4904         with concurrent.futures.ProcessPoolExecutor() as executor:
4905             # chunksize=1: supports vary wildly in cost, so hand them out one
4906             # at a time to keep every core busy to the end.
4907             for reps in executor.map(_decorate_support_worker, tasks,
4908                                     chunksize=1):
4909                 representatives.extend(reps)
4910     except (OSError, concurrent.futures.BrokenExecutor) as exc:
4911         warnings.warn(
4912             f"parallel support enumeration unavailable ({exc!r}); running "
4913             "sequentially", RuntimeWarning, stacklevel=2,

```



```

4912         )
4913         for task in tasks:
4914             representatives.extend(_decorate_support_worker(task))
4915     else:
4916         for task in tasks:
4917             representatives.extend(_decorate_support_worker(task))
4918
4919     if up_to_iso:
4920         # Distinct geng supports give non-isomorphic multigraphs, so this final
4921         # canonical pass is only a safety net over the process-local per-support
4922         # dedup; it keeps exactly one representative per isomorphism class.
4923         seen: set[bytes] = set()
4924         unique: list[np.ndarray] = []
4925         for matrix in representatives:
4926             key = _canonical_form(matrix)
4927             if key not in seen:
4928                 seen.add(key)
4929                 unique.append(matrix)
4930         representatives = unique
4931     return representatives

```

C.1.9 The gallery

Given a size and a route limit, these routines find every extremal graph there is, up to relabelling, and count the symmetries of each. Deciding when two graphs are the same graph drawn differently is done by trying all relabellings and keeping the smallest resulting table, which is slow in principle but perfectly quick at the sizes involved.

```

4934 # --- GALLERY: all non-isomorphic extremal graphs for small (n, m); entry point
4935     ↪ gallery_extremal_graphs ---
4936
4937 def _graph_from_mu(mu: np.ndarray, variant: Variant) -> Graph:
4938     """Wrap a multiplicity matrix in a fresh Graph (zero-copy)."""
4939     g = Graph(mu.shape[0], variant)
4940     g.mu[:] = mu
4941     return g
4942
4943
4944 def _aut_count_matrix(mu: np.ndarray) -> int:
4945     """Count vertex permutations that preserve ‘mu’ (equals |Aut(G)|)."""
4946     n = mu.shape[0]
4947     canon = _canonical_form(mu)
4948     return sum(1 for perm in permutations(range(n))
4949               if mu[np.ix_(list(perm), list(perm))].tobytes() == canon)
4950
4951
4952 def _dir_relabel_key(edge, p: list) -> tuple:
4953     """Permutation-relabelled key of one directed hyperedge.
4954
4955     Keeps the legacy forward shape ‘(int, tuple_of_heads)’ for forward edges
4956     (so their canonical keys and dedup counts are unchanged) and uses
4957     ‘(tuple_of_tails, tuple_of_heads)’ for general edges.
4958     """
4959     first, heads = edge
4960     if isinstance(first, (set, frozenset)):

```

```

4961         return (tuple(sorted(p[t] for t in first)), tuple(sorted(p[h] for h in
4962             ↪ heads)))
4963     return (p[first], tuple(sorted(p[h] for h in heads)))
4964
4965 def _hyper_canonical(hyperedges: list, n: int, directed: bool) -> tuple:
4966     """Canonical key for a hyperedge collection under vertex permutation.
4967
4968     Each edge is a frozenset (undirected) or a directed hyperedge in either the
4969     forward ‘(tail, heads)’ or general ‘(tails, heads)’ form. Returns the
4970     lex-minimum over all n! relabellings.
4971     """
4972     best: tuple | None = None
4973     for perm in permutations(range(n)):
4974         p = list(perm)
4975         if directed:
4976             relabeled: tuple = tuple(sorted(_dir_relabel_key(edge, p) for edge in
4977                 ↪ hyperedges))
4978         else:
4979             relabeled = tuple(sorted(
4980                 tuple(sorted(p[v] for v in edge)) for edge in hyperedges
4981             ))
4982             if best is None or relabeled < best:
4983                 best = relabeled
4984     return best # type: ignore[return-value]
4985
4986 def _aut_count_hyper(hyperedges: list, n: int, directed: bool) -> int:
4987     """Count vertex permutations that preserve a hyperedge collection."""
4988     canon = _hyper_canonical(hyperedges, n, directed)
4989
4990     def _relabel(p: list) -> tuple:
4991         if directed:
4992             return tuple(sorted(_dir_relabel_key(edge, p) for edge in hyperedges))
4993         return tuple(sorted(
4994             tuple(sorted(p[v] for v in edge)) for edge in hyperedges
4995         ))
4996
4997     return sum(1 for perm in permutations(range(n))
4998         if _relabel(list(perm)) == canon)
4999
5000
5001 def _hyper_to_lists(hyperedges: list, directed: bool) -> list:
5002     """Convert hyperedges to JSON-serialisable sorted integer lists.
5003
5004     Forward edges keep their ‘[tail, [heads]]’ shape; general edges use
5005     ‘[[tails], [heads]]’.
5006     """
5007     if directed:
5008         out = []
5009         for edge in hyperedges:
5010             first, heads = edge
5011             if isinstance(first, (set, frozenset)):
5012                 out.append([sorted(first), sorted(heads)])
5013             else:
5014                 out.append([first, sorted(heads)])
5015         return out
5016     return [sorted(edge) for edge in hyperedges]

```

```

5017
5018
5019 def _enum_matrix_extremals(
5020     variant: Variant,
5021     n: int,
5022     m: int,
5023     separation: str,
5024     target: int,
5025     deadline: float,
5026 ) -> tuple[list[np.ndarray], bool]:
5027     """DFS over all (variant, n, m) multiplicity matrices achieving exactly ‘
        ↳ target‘ edges.

5028
5029     Generalises :func:‘enumerate_extremal_directed_multigraphs‘ to all four
5030     matrix variants (simple/multi × directed/undirected) and to vertex as well
5031     as edge separation. Returns ‘(reps, timed_out)’ where ‘reps’ are
5032     deduplicated representatives (one per isomorphism class).
5033
5034     Pruning strategy: arc-count reachability at every step; prefix edge-
5035     connectivity check at each completed vertex block (edge separation only;
5036     vertex separation checks only the final graph because the per-node cost is
5037     too high for DFS).
5038     """
5039     max_mult = 1 if variant.simple else (m - 1)
5040     is_directed = variant.directed
5041
5042     # Vertex-block order: all pairs involving vertex j before j+1 appears.
5043     order: list[tuple[int, int]] = []
5044     for j in range(1, n):
5045         for i in range(j):
5046             order.append((i, j))
5047             if is_directed:
5048                 order.append((j, i))
5049     total = len(order)
5050
5051     # Position at which the subgraph on {0..j} is fully decided.
5052     # Directed: j*(j+1) positions; undirected: j*(j+1)//2 positions.
5053     step_per_vertex = 2 if is_directed else 1
5054     block_end: dict[int, int] = {
5055         j: step_per_vertex * j * (j + 1) // 2 for j in range(1, n)
5056     }
5057
5058     mu = np.zeros((n, n), dtype=int)
5059     seen: set[bytes] = set()
5060     reps: list[np.ndarray] = []
5061     timed_out = [False]
5062
5063     def _feasible_full() -> bool:
5064         g = _graph_from_mu(mu, variant)
5065         if separation == "edge":
5066             return max_edge_connectivity(g) <= m - 1
5067         return max_vertex_connectivity(g) <= m - 1
5068
5069     def dfs(pos: int, edges: int) -> None:
5070         if timed_out[0]:
5071             return
5072         if time.time() > deadline:
5073             timed_out[0] = True

```

```

5074         return
5075     if edges + max_mult * (total - pos) < target or edges > target:
5076         return
5077     if pos == total:
5078         if edges == target and _feasible_full():
5079             canon = _canonical_form(mu)
5080             if canon not in seen:
5081                 seen.add(canon)
5082                 reps.append(mu.copy())
5083         return
5084         # At each block boundary: prune if the prefix subgraph is already
5085         # infeasible (edge connectivity only -- cheap with _tiny_maxflow).
5086     completed_j = next(
5087         (j for j in range(1, n) if block_end.get(j) == pos), None
5088     )
5089     if completed_j is not None and separation == "edge":
5090         sub_n = completed_j + 1
5091         if any(_tiny_maxflow(mu, n, s, t, m - 1)
5092              for s in range(sub_n) for t in range(sub_n) if s != t):
5093             return
5094     u, v = order[pos]
5095     for val in range(max_mult + 1):
5096         mu[u, v] = val
5097         if not is_directed:
5098             mu[v, u] = val
5099         dfs(pos + 1, edges + val)
5100     mu[u, v] = 0
5101     if not is_directed:
5102         mu[v, u] = 0
5103
5104     dfs(0, 0)
5105     return reps, timed_out[0]
5106
5107
5108 def _enum_hyper_extremals(
5109     directed: bool,
5110     vertex_split: bool,
5111     r: int,
5112     n: int,
5113     m: int,
5114     target: int,
5115     deadline: float,
5116     *,
5117     kind: str = "forward",
5118 ) -> tuple[list[list], bool]:
5119     """DFS over all r-uniform hypergraphs on n vertices with exactly ‘target’
5120     ↪ edges.
5121
5122     Iterates candidate hyperedges in order (include / exclude) and prunes
5123     immediately when adding a hyperedge pushes the connectivity above m-1
5124     (monotonicity: adding edges never decreases connectivity, so no superset
5125     can be feasible at that branch either). ‘kind’ selects the directed
5126     orientation model. Returns ‘(reps, timed_out)’ where each representative
5127     in ‘reps’ is a list of JSON-serialisable sorted vertex lists.
5128     """
5129     candidates = _hyperedge_candidates(n, r, directed, kind=kind)
5130     total = len(candidates)
5131     active: list = []

```

```

5131     seen: set[tuple] = set()
5132     reps: list[list] = []
5133     timed_out = [False]
5134
5135     def dfs(pos: int, count: int) -> None:
5136         if timed_out[0]:
5137             return
5138         if time.time() > deadline:
5139             timed_out[0] = True
5140             return
5141         if count + (total - pos) < target or count > target:
5142             return
5143         if pos == total:
5144             if count == target:
5145                 canon = _hyper_canonical(active, n, directed)
5146                 if canon not in seen:
5147                     seen.add(canon)
5148                     reps.append(_hyper_to_lists(active, directed))
5149             return
5150         # Include candidates[pos] and recurse only if still feasible.
5151         active.append(candidates[pos])
5152         h = Hypergraph(n, active, directed=directed)
5153         if max_hyper_connectivity(h, vertex_split=vertex_split) <= m - 1:
5154             dfs(pos + 1, count + 1)
5155         active.pop()
5156         # Exclude candidates[pos].
5157         dfs(pos + 1, count)
5158
5159     dfs(0, 0)
5160     return reps, timed_out[0]
5161
5162
5163 def gallery_extremal_graphs(
5164     max_n: int = 7,
5165     max_m: int = 4,
5166     r: int = 3,
5167     time_per_case: float = 3.0,
5168 ) -> dict:
5169     """Collect every non-isomorphic extremal graph for all 12 variants at small n,
5170         ↪ m.
5171
5172     For each (variant, n, m) triple where the enumeration finishes within
5173         'time_per_case' seconds, the function:
5174
5175     1. runs a short repeated search to discover the extremal edge count;
5176     2. exhaustively enumerates every graph achieving that count;
5177     3. deduplicates by isomorphism class and records 'n! / |Aut(G)|' (the
5178         number of labelled copies) per class.
5179
5180     Returns a JSON-serialisable dict with structure::
5181
5182         result[variant_key]["n={n}_m={m}"] = {
5183             "extremal_value": int,
5184             "classes": [{"repr": <matrix or edge list>, "labelled_count": int}],
5185             "total_labelled": int,      # sum of labelled_count over all classes
5186             "complete": bool,          # False when the deadline was hit
5187         }

```

```

5188 For matrix variants ‘repr’ is the multiplicity matrix (list of lists of
5189 int). For hypergraph variants it is a list of hyperedges: each edge is a
5190 sorted list of vertex indices (undirected) or ‘[tail, [head, ...]]’
5191 (directed).
5192
5193 The twelve variant keys are:
5194 ‘simple_undirected_{edge,vertex}’,
5195 ‘simple_directed_{edge,vertex}’,
5196 ‘multi_undirected_{edge,vertex}’,
5197 ‘multi_directed_{edge,vertex}’,
5198 ‘hyper_undirected_r{r}_{edge,vertex}’,
5199 ‘hyper_directed_r{r}_{edge,vertex}’.
5200
5201 The two ‘multi*_vertex’ keys report the reduced SIMPLE problem, because
5202 the thesis’s vertex measure is blind to parallel copies and the variant
5203 collapses onto the simple one (see the config comment below).
5204 """
5205 # The two multi-vertex rows run on the SIMPLE variant: the checker’s vertex
5206 # mode gives every adjacency capacity one, so parallel copies never change
5207 # kappa and a raw multigraph search would just fill every cell to m-1 for
5208 # free (it once reported a "36-arc extremal  $K_4$  at multiplicity 3"). The
5209 # thesis reduces these variants to the simple ones (tab:summary), and the
5210 # gallery must report the reduced problem, exactly as the variant grids do.
5211 matrix_configs: list[tuple[str, Variant, str]] = [
5212     ("simple_undirected_edge", SIMPLE_UNDIRECTED, "edge"),
5213     ("simple_undirected_vertex", SIMPLE_UNDIRECTED, "vertex"),
5214     ("simple_directed_edge", SIMPLE_DIRECTED, "edge"),
5215     ("simple_directed_vertex", SIMPLE_DIRECTED, "vertex"),
5216     ("multi_undirected_edge", MULTI_UNDIRECTED, "edge"),
5217     ("multi_undirected_vertex", SIMPLE_UNDIRECTED, "vertex"),
5218     ("multi_directed_edge", MULTI_DIRECTED, "edge"),
5219     ("multi_directed_vertex", SIMPLE_DIRECTED, "vertex"),
5220 ]
5221 hyper_configs: list[tuple[str, bool, bool]] = [
5222     ("hyper_undirected_r{r}_edge", False, False),
5223     ("hyper_undirected_r{r}_vertex", False, True),
5224     ("hyper_directed_r{r}_edge", True, False),
5225     ("hyper_directed_r{r}_vertex", True, True),
5226 ]
5227
5228 result: dict = {}
5229
5230 for key, variant, separation in matrix_configs:
5231     result[key] = {}
5232     for n in range(2, max_n + 1):
5233         for m in range(2, max_m + 1):
5234             case_deadline = time.time() + time_per_case
5235             # Step 1: discover the extremal value via repeated search.
5236             # 2000 steps/restart is enough to converge for n <= 6;
5237             # we run at least 3 seeds before the deadline to be robust
5238             # against unlucky single-seed runs.
5239             best_val = 0
5240             seed = 0
5241             while time.time() < case_deadline - 2.0 or seed < 3:
5242                 if time.time() > case_deadline - 0.5:
5243                     break
5244                 res = search_for_dense_graph(
5245                     variant, n, m, separation=separation,

```

```

5246         steps=2000, seed=seed)
5247         best_val = max(best_val, res.best_edge_count)
5248         seed += 1
5249         # Step 2: enumerate all graphs at that value.
5250         enum_deadline = min(case_deadline, time.time() + 2.0)
5251         reps, timed_out = _enum_matrix_extremals(
5252             variant, n, m, separation, best_val, enum_deadline)
5253         # Step 3: count automorphisms and labelled copies.
5254         fn = math.factorial(n)
5255         classes = [
5256             {"repr": mu.tolist(),
5257              "labelled_count": fn // _aut_count_matrix(mu)}
5258             for mu in reps
5259         ]
5260         result[key][f"n={n}_m={m}"] = {
5261             "extremal_value": best_val,
5262             "classes": classes,
5263             "total_labelled": sum(c["labelled_count"] for c in classes),
5264             "complete": not timed_out,
5265         }
5266
5267     for key, directed, vertex_split in hyper_configs:
5268         result[key] = {}
5269         for n in range(r, max_n + 1):
5270             for m in range(2, max_m + 1):
5271                 case_deadline = time.time() + time_per_case
5272                 # Brute-force search (small n): finds the extremal value.
5273                 best_val, _, completed = _brute_force_hypergraph(
5274                     n, r, m, case_deadline,
5275                     directed=directed, vertex_split=vertex_split)
5276                 if not completed:
5277                     result[key][f"n={n}_m={m}"] = {
5278                         "extremal_value": best_val, "classes": [],
5279                         "total_labelled": 0, "complete": False,
5280                     }
5281                     continue
5282                 # Enumerate all achieving best_val.
5283                 enum_deadline = min(case_deadline, time.time() + 2.0)
5284                 hyper_reps, timed_out = _enum_hyper_extremals(
5285                     directed, vertex_split, r, n, m, best_val, enum_deadline)
5286                 fn = math.factorial(n)
5287                 classes = []
5288                 for edge_lists in hyper_reps:
5289                     raw = [(row[0], frozenset(row[1])) for row in edge_lists]
5290                     if directed else [frozenset(e) for e in edge_lists]
5291                     classes.append({
5292                         "repr": edge_lists,
5293                         "labelled_count": fn // _aut_count_hyper(raw, n, directed)
5294                         ↪ ,
5295                     })
5296                 result[key][f"n={n}_m={m}"] = {
5297                     "extremal_value": best_val,
5298                     "classes": classes,
5299                     "total_labelled": sum(c["labelled_count"] for c in classes),
5300                     "complete": not timed_out,
5301                 }
5302
5303     return result

```

```

5303
5304
5305 def save_gallery_json(gallery: dict, path: str | Path) -> None:
5306     """Write the gallery dict to a JSON file (pretty-printed)."""
5307     with open(path, "w") as f:
5308         json.dump(gallery, f, indent=2)
5309
5310
5311 def prove_integral_arc_bound(n: int, m: int, target: int, *,
5312                             time_limit: float = 3000.0,
5313                             use_gurobi: bool | None = None,
5314                             show_solver_log: bool = False) -> str:
5315     """Decide by MILP whether some directed multigraph on ‘n’ vertices with
5316     multiplicities in {0..m-1} and ‘lambda^max <= m-1’ has >= ‘target’ arcs.
5317
5318     Returns "INFEASIBLE" (proving  $L_m^{\text{dir}}(n) < \text{target}$ ), "FEASIBLE", or
5319     "LIMIT". This is the integral companion of the fractional prover: the
5320     m = 3 chain (rem:odd-step-roadmap) needs only  $L_3^{\text{dir}}(7) = 24$ , i.e.
5321     INFEASIBLE at target 25, which is a far friendlier MILP than  $M^*(7)$  because the
5322     weights themselves are integers. Same exact :func:‘_cut_counting_model’ with
5323     cap = m-1 and integer multiplicities, all its proved-valid families on, plus
5324     ↪ the
5325     one constraint that the multiplicities total at least ‘target’.
5326     """
5327     _require_pulp()
5328     prob, w = _cut_counting_model(
5329         n, cap=float(m - 1), integer=True, two_hop=True,
5330         symmetry=True, deletion=True, degree_pair=True,
5331     )
5332     prob += pulp.lpSum(w.values()) >= target          # the arc target
5333     prob += 0                                         # feasibility, no objective
5334
5335     prob.solve(_pick_solver(time_limit, show_solver_log, use_gurobi))
5336     status = pulp.LpStatus[prob.status]
5337     if status == "Infeasible":
5338         return "INFEASIBLE"
5339     if status == "Optimal":
5340         return "FEASIBLE"
5341     return "LIMIT"
5342
5343 def _float_maxflow_value(capacity: np.ndarray, source: int, target: int,
5344                         eps: float = 1e-12) -> float:
5345     """Max-flow value for FLOAT capacities (Edmonds-Karp, shortest augmenting path
5346     ↪ ).
5347
5348     scipy’s csgraph max-flow is integer-only, so the fractional check below needs
5349     its own tiny routine. BFS augmenting paths give a polynomial bound
5350     ↪ independent
5351     of the capacity magnitudes, so the loop terminates even on irrational-looking
5352     floats; ‘eps’ treats a near-zero residual as saturated.
5353     """
5354     n = capacity.shape[0]
5355     residual = capacity.astype(float).copy()
5356     flow = 0.0
5357     while True:
5358         parent = [-1] * n
5359         parent[source] = source

```



```

5358     queue = deque([source])
5359     while queue and parent[target] == -1:
5360         u = queue.popleft()           # FIFO: shortest augmenting
           ↪ path
5361         for v in range(n):
5362             if parent[v] == -1 and residual[u, v] > eps:
5363                 parent[v] = u
5364                 queue.append(v)
5365         if parent[target] == -1:
5366             return flow
5367         bottleneck = float("inf")
5368         v = target
5369         while v != source:
5370             bottleneck = min(bottleneck, residual[parent[v], v])
5371             v = parent[v]
5372         v = target
5373         while v != source:
5374             residual[parent[v], v] -= bottleneck
5375             residual[v, parent[v]] += bottleneck
5376             v = parent[v]
5377         flow += bottleneck
5378
5379
5380 def fractional_flows_feasible(weights: np.ndarray, tol: float = 1e-9) -> bool:
5381     """Check the scaled multigraph constraint: all pairwise max-flows <= 1.
5382
5383     ‘‘weights‘‘ is an ‘‘n x n‘‘ matrix with entries in [0,1] (zero diagonal),
5384     the scaled multiplicity matrix mu/(m-1) of lem:scaling-reduction.
5385     """
5386     n = weights.shape[0]
5387     capacity = np.where(weights > 1e-12, weights, 0.0).astype(float)
5388     np.fill_diagonal(capacity, 0.0)
5389     for s in range(n):
5390         for t in range(n):
5391             if t != s and _float_maxflow_value(capacity, s, t) > 1 + tol:
5392                 return False
5393     return True
5394
5395
5396 def fractional_search(n: int, objective: str = "min_degree", steps: int = 6000,
5397                     seed: int = 0, start: str = "bipartite") -> tuple[np.ndarray
           ↪ , float]:
5398     """Hunt for a fractional counterexample to the odd-n directed multigraph
5399     statements (conj:min-degree and the total-weight bound).
5400
5401     ‘‘objective‘‘: "min_degree" maximises the minimum weighted total degree
5402     (conjecture: cannot exceed  $k = (n-1)/2$  for odd n); "total" maximises the
5403     total weight (conjecture: cannot exceed  $\max(2(n-1), \lfloor n^2/4 \rfloor)$ ).
5404     ‘‘start‘‘: "bipartite" (the conjectured extremiser), "zero", or "random".
5405     Returns the best weighting found and its objective value. A value
5406     exceeding the conjectured ceiling would refute the conjecture for every
5407     m - 1 divisible by the weight denominators; none was found at n = 7, 9.
5408     """
5409     rng = random.Random(seed)
5410     weights = np.zeros((n, n))
5411     if start == "bipartite":
5412         for a in range(n // 2):
5413             for b in range(n // 2, n):

```

```

5414         weights[a, b] = 1.0
5415     elif start == "random":
5416         weights = np.array([[rng.random() if i != j else 0.0 for j in range(n)]
5417                             for i in range(n)])
5418         while not fractional_flows_feasible(weights):
5419             weights *= 0.85
5420
5421     def score(w: np.ndarray) -> float:
5422         if objective == "total":
5423             return float(w.sum())
5424         degrees = w.sum(axis=0) + w.sum(axis=1)
5425         return float(degrees.min()) + 1e-3 * float(w.sum())
5426
5427     current = score(weights)
5428     best, best_w = current, weights.copy()
5429     for it in range(steps):
5430         temperature = 0.08 * (1 - it / steps) + 1e-4
5431         u, v = rng.randrange(n), rng.randrange(n)
5432         if u == v:
5433             continue
5434         old = weights[u, v]
5435         weights[u, v] = min(1.0, max(0.0, old + (rng.random() * 2 - 1)
5436                                     * max(0.02, temperature)))
5437         if weights[u, v] == old:
5438             continue
5439         value = score(weights)
5440         accept = value > current or rng.random() < math.exp((value - current)
5441                                                             / temperature)
5442         if accept and fractional_flows_feasible(weights):
5443             current = value
5444             if value > best:
5445                 best, best_w = value, weights.copy()
5446         else:
5447             weights[u, v] = old
5448     return best_w, best
5449
5450
5451 # --- SELF-CHECK: python erdos915_unified.py runs every invariant; exits 0 on all-
5452 ↪ PASS ---
5453
5454 _failures = 0
5455
5456 def check(label: str, condition: bool) -> None:
5457     """Print a PASS/FAIL line and tally failures."""
5458     global _failures
5459     print(f" {'PASS'} if condition else 'FAIL' } {label}")
5460     if not condition:
5461         _failures += 1
5462
5463
5464 def section(title: str) -> None:
5465     """Print a section header."""
5466     print(f"\n{title}")

```

C.1.10 The self-check

The file ends with a suite of checks that runs when the program is executed directly. It re-derives known values, tests each construction against its predicted size and connectivity, confirms the fast routines agree with the slow ones they stand in for, and verifies the bounds proved in [Appendix A](#) on sampled graphs. It prints a line per check and fails loudly rather than quietly if any of them stops holding, which is what makes it useful as a regression test after an edit.

```
5469 def _run_checks() -> int:
5470     """Run every invariant check; return 1 on any failure, else 0.
5471
5472     This is the program checking itself end to end, using the bare top-level
5473     names defined above in this single file.
5474     """
5475     section("Checker: the directed m=2 values ell_2^dir(n) = 2, 4, 6, 8")
5476     for n, expected in [(2, 2), (3, 4), (4, 6), (5, 8)]:
5477         graph = double_star(n, m=2, directed=True)
5478         check(f"double star n={n}: {graph.edge_count()} arcs, lambda^max={
5479             ↪ max_edge_connectivity(graph)}",
5480               graph.edge_count() == expected and max_edge_connectivity(graph) ==
5481               ↪ 1)
5482
5483     section("Checker: one-directional bipartite is quadratic and feasible")
5484     for n in range(2, 8):
5485         graph = one_directional_bipartite(n)
5486         check(f"bipartite n={n}: {graph.edge_count()} arcs (floor(n^2/4)={ (n*n)
5487             ↪ //4 })",
5488               graph.edge_count() == (n * n) // 4 and max_edge_connectivity(graph)
5489               ↪ == 1)
5490
5491     section("Checker: the undirected cycle is the infeasible witness for m=2")
5492     cycle = Graph(6, SIMPLE_UNDIRECTED)
5493     for v in range(6):
5494         cycle.add_edge(v, (v + 1) % 6)
5495     check(f"C_6 has lambda^max={max_edge_connectivity(cycle)}",
5496           ↪ max_edge_connectivity(cycle) == 2)
5497
5498     if NETWORKX_AVAILABLE:
5499         section("Gomory-Hu shortcut agrees with brute-force lambda^max")
5500         tree = clique_tree(3, 4)
5501         check(f"GH={max_edge_connectivity_via_tree(tree)} vs direct={
5502             ↪ max_edge_connectivity(tree)}",
5503               max_edge_connectivity_via_tree(tree) == max_edge_connectivity(tree))
5504     else:
5505         section("Gomory-Hu shortcut skipped (optional networkx not installed)")
5506
5507     section("Vertex connectivity: K_4-trees are globally 3-connected")
5508     for blocks in range(0, 5):
5509         ktree = clique_tree(4, blocks)
5510         check(f"K_4-tree on {ktree.num_vertices} vertices: kappa={
5511             ↪ min_vertex_connectivity(ktree)}",
5512               min_vertex_connectivity(ktree) == 3)
5513
5514     section("Constructions: the augmented bipartite conjecture values")
5515     counterexample = augmented_bipartite(10, 3)
```

```

5509     check(f"m=3,n=10: {counterexample.edge_count()} arcs, lambda^max={
5510         ↪ max_edge_connectivity(counterexample)}",
5511         counterexample.edge_count() == 30 and max_edge_connectivity(
5512             ↪ counterexample) == 2)
5511     for n, m in [(8, 3), (10, 4), (12, 4), (10, 5)]:
5512         graph = augmented_bipartite(n, m)
5513         expected = (n + m - 2) ** 2 // 4
5514         check(f"n={n},m={m}: {graph.edge_count()} arcs (expected {expected}),
5515             ↪ lambda^max={max_edge_connectivity(graph)}",
5516             graph.edge_count() == expected and max_edge_connectivity(graph) == m
5517             ↪ - 1)
5516
5517     section("Hypergraphs: Berge connectivity, exact and generalising the graph
5518         ↪ case")
5518     k43 = complete_uniform_hypergraph(4, 3)
5519     check(f"complete 3-uniform K_4^(3): Berge lambda^max = {
5520         ↪ max_hyperedge_connectivity(k43)} (expect 3)",
5521         max_hyperedge_connectivity(k43) == 3)
5520
5521     for n in [3, 4, 5]:
5522         complete = complete_uniform_hypergraph(n, 2)
5523         check(f"2-uniform K_{n} reduces to edge-connectivity {
5524             ↪ max_hyperedge_connectivity(complete)} (expect {n-1})",
5525             max_hyperedge_connectivity(complete) == n - 1)
5524
5525     for n, r in [(7, 3), (10, 4), (13, 4)]:
5526         star = star_hypertree(n, r)
5527         check(f"star hypertree n={n}, r={r}: {star.edge_count()} edges (expect {(n
5528             ↪ -1)/(r-1))}, "
5529             f"lambda^max={max_hyperedge_connectivity(star)}",
5530             star.edge_count() == (n - 1) // (r - 1) and
5531             ↪ max_hyperedge_connectivity(star) == 1)
5530
5531     section("Hypergraphs: directed orientation models (forward / backward /
5532         ↪ general)")
5532     # The general gate admits a mixed arc {0,1} -> {2,3}: a route enters at a tail
5533     # and leaves at a head, so 0->2 carries one route and 2->0 carries none.
5534     mixed = Hypergraph(4, [(frozenset({0, 1}), frozenset({2, 3}))], directed=True)
5535     check("mixed arc {0,1}->{2,3}: lambda(0,2)=1 and lambda(2,0)=0",
5536         hyper_connectivity(mixed, 0, 2) == 1 and hyper_connectivity(mixed, 2, 0)
5537         ↪ == 0)
5537
5538     # Backward is the arc-reversal dual of forward, so their extremal numbers
5539     ↪ agree.
5538     fwd, _ = max_feasible_hyperedges(4, 3, 3, directed=True, kind="forward",
5539         ↪ time_limit=1.5)
5539     bwd, _ = max_feasible_hyperedges(4, 3, 3, directed=True, kind="backward",
5540         ↪ time_limit=1.5)
5540
5541     check(f"forward == backward at n=4, r=3, m=3 ({fwd} == {bwd})", fwd == bwd)
5541     # General contains forward and strictly beats it where vertices are scarce
5542     # (n = r); both values here are proved exact by the branch and bound.
5543     g3, e3 = max_feasible_hyperedges(3, 3, 3, directed=True, kind="general",
5544         ↪ time_limit=1.5)
5544     f3, _ = max_feasible_hyperedges(3, 3, 3, directed=True, kind="forward",
5545         ↪ time_limit=1.5)
5545
5546     check(f"general 4 > forward 3 at n=3, r=3, m=3 (got {g3} vs {f3}, exact={e3})"
5547         ↪ ,
5548         g3 == 4 and f3 == 3 and e3)
5546
5547     g4, e4 = max_feasible_hyperedges(4, 4, 4, directed=True, kind="general",
5548         ↪ time_limit=10.0)

```

```

5548     f4, _ = max_feasible_hyperedges(4, 4, 4, directed=True, kind="forward",
5549         ↪ time_limit=3.0)
5549     check(f"general 8 doubles forward 4 at n = r = 4, m = 4 ({g4} vs {f4}, exact={
5550         ↪ e4})",
5551         g4 == 8 and f4 == 4 and e4)
5551
5552     section("Monte Carlo: the appearance probability crosses the m/n scale")
5553     n, m = 30, 3
5554     below = estimate_appearance_probability(n, 0.3 * m / n, m, trials=120, seed=1)
5555     above = estimate_appearance_probability(n, 1.8 * m / n, m, trials=120, seed=1)
5556     check(f"P[appears] rises from {below:.2f} (well below m/n) to {above:.2f} (
5557         ↪ well above)",
5558         below < 0.5 < above)
5558
5559     section("Monte Carlo: vertex-connectivity never exceeds edge-connectivity (
5560         ↪ Whitney)")
5560     edge_dist = connectivity_distribution(20, 0.25, trials=40, separation="edge",
5561         ↪ seed=3)
5561     vertex_dist = connectivity_distribution(20, 0.25, trials=40, separation="
5562         ↪ vertex", seed=3)
5562     check("kappa^max <= lambda^max on every one of the 40 identical samples",
5563         all(kappa <= lam for kappa, lam in zip(vertex_dist, edge_dist)))
5564
5565     section("Edge vs vertex in G(n,p): they agree at small m, part company at
5566         ↪ large m")
5566     lam_small, kap_small = edge_vertex_distribution(5, 0.5, trials=120, seed=7)
5567     small_gap = sum(k < 1 for k, l in zip(kap_small, lam_small))
5568     check("n=5: kappa^max = lambda^max on every sample (no gap)",
5569         small_gap == 0 and all(k <= 1 for k, l in zip(kap_small, lam_small)))
5570     lam_large, kap_large = edge_vertex_distribution(16, 0.25, trials=200, seed=7)
5571     large_gap = sum(k < 1 for k, l in zip(kap_large, lam_large))
5572     check(f"n=16: Whitney holds and kappa^max < lambda^max on {large_gap} of 200
5573         ↪ samples",
5574         large_gap > 0 and all(k <= 1 for k, l in zip(kap_large, lam_large)))
5574
5575     if not PULP_AVAILABLE:
5576         section("Prover sections skipped (optional pulp not installed)")
5577     else:
5578         section("Prover: cut-counting proves M*(n) = 2(n-1) optimal for small n")
5579         # The thesis (ch2) claims  $L_m^{dir}(n) = 2(n-1)(m-1)$  is proved for ALL  $n \leq$ 
5580         ↪ 6
5580         # via the cut-counting MILP (thm:dir-multi-small). n=3,4,5 all finish in
5581         # well under 60 s and are verified here. n=6 takes ~1315 s (~22 min) on a
5582         # typical laptop and is NOT included in this fast loop -- it must be
5583         # verified separately:
5584         #
5585         # result = prove_directed_multigraph(6, time_limit=2000.0)
5586         # assert result.is_proof() and round(result.scaled_optimum) == 10
5587         #
5588         # The confirmed run (2026-06-13) is logged in program/logs/selftest_check.
5589         ↪ log
5589         # (search "n=6 OPTIMAL M*(6)=10 in 1315s"). Omitting n=6 from this loop
5590         ↪ is
5590         # not an error in the proof -- the proof ran -- but it means this self-
5591         ↪ test
5591         # alone does not reproduce the full  $n \leq 6$  certification.
5592         for n in [3, 4, 5]:
5593             result = prove_directed_multigraph(n, time_limit=300.0)

```

```

5594         check(f"n={n}: status={result.status}, M*={result.scaled_optimum:.1f},
        ↪ proof={result.is_proof()}"),
5595         result.is_proof() and round(result.scaled_optimum) == 2 * (n -
        ↪ 1))
5596
5597     section("Prover: the valid inequalities sharpen but never move the optimum
        ↪ ")
5598     # Turning the two-hop and symmetry-breaking rows off leaves the BARE exact
5599     # cut formulation, which must still prove the same  $M^*(3) = 4$ . If any row
        ↪ we
5600     # call a "valid inequality" were in fact invalid, it would cut a feasible
5601     # point and shift this optimum -- so this is the regression tripwire that
5602     # guards the prover's soundness, not just a re-run of the value.
5603     bare = prove_directed_multigraph(3, time_limit=120.0,
5604                                     use_two_hop=False, use_symmetry_breaking=
        ↪ False)
5605     check(f" $M^*(3)$  with the tighteners off = {bare.scaled_optimum:.1f} (expect
        ↪ 4), proof={bare.is_proof()}"),
5606         bare.is_proof() and round(bare.scaled_optimum) == 4)
5607
5608     section("Prover (integral): an INFEASIBLE verdict is a genuine upper-bound
        ↪ proof")
5609     #  $L_3^{\text{dir}}(4) = 12$ : a 12-arc multigraph exists, no 13-arc one does. This
5610     # exercises the exact mechanism behind the thesis's  $L_3(7) = 24$  result
5611     # (INFEASIBLE one above the optimum) on a value small enough to re-run
        ↪ here,
5612     # so a regression in the integral encoding cannot pass unnoticed.
5613     feasible_at_12 = prove_integral_arc_bound(4, 3, 12, time_limit=120.0)
5614     infeasible_at_13 = prove_integral_arc_bound(4, 3, 13, time_limit=120.0)
5615     check(f"target 12 -> {feasible_at_12} (expect FEASIBLE), "
5616           f"target 13 -> {infeasible_at_13} (expect INFEASIBLE)",
5617           feasible_at_12 == "FEASIBLE" and infeasible_at_13 == "INFEASIBLE")
5618
5619     section("Search: rediscovering  $\text{ell}_2^{\text{dir}}(4) = 6$  by random search")
5620     result = search_for_dense_graph(SIMPLE_DIRECTED, n=4, m=2, steps=6000, seed=0)
5621     check(f"best found = {result.best_edge_count} arcs, feasible  $\lambda^{\text{max}}=\{$ 
        ↪  $\text{max\_edge\_connectivity}(\text{result.best\_graph})\}$ ",
5622           result.best_edge_count == 6 and  $\text{max\_edge\_connectivity}(\text{result.best\_graph})$ 
        ↪ <= 1)
5623
5624     section("Driver: one solve() call handles each kind of question")
5625     # Exhaustive directed: the cut-counting proves  $\text{ell}_2^{\text{dir}}(4) = 6$  exactly.
5626     proved = solve(4, 2, directed=True, simple=True, exhaustive=True,
5627                  max_seconds=120.0)
5628     check(f"solve exhaustive simple-directed n=4,m=2: {proved.value} ({proved.
        ↪ bound})",
5629           proved.proven and proved.value == 6)
5630     # Discovery finds the same 6 arcs as a lower bound within a short budget.
5631     # The checks use sa (lighter); tabu is the default for actually solving.
5632     found = solve(4, 2, directed=True, simple=True, exhaustive=False,
5633                  max_seconds=3.0, method="sa")
5634     check(f"solve discovery simple-directed n=4,m=2: {found.value} ({found.bound})
        ↪ ",
5635           found.bound == "lower" and found.value == 6)
5636     # The VERTEX separation is a separate question, not a corollary of the arc
5637     # one: Whitney's  $\kappa \leq \lambda$  makes the vertex-feasible family the LARGER
5638     # of the two, so an arc upper bound does not restrict it. thm:dir-vertex-m2
5639     # needs its own base cases, and these are the first three of them.

```

```

5640 for base_n, base_value in ((3, 4), (4, 6), (5, 8)):
5641     v_value, _, v_done = _exhaustive_directed(base_n, 2, "vertex",
5642                                             time.time() + 120.0)
5643     check(f"exhaustive simple-directed VERTEX n={base_n},m=2: {v_value}",
5644           v_done and v_value == base_value)
5645     # The vertex flow helper must agree with the exact checker it stands in for.
5646     _vg = Graph(4, SIMPLE_DIRECTED)
5647     for _a, _b in ((0, 1), (0, 2), (1, 3), (2, 3)):
5648         _vg.mu[_a, _b] = 1
5649     _vout = [{_b for _b in range(4) if _vg.mu[_a, _b]} for _a in range(4)]
5650     check("vertex flow helper agrees with the exact checker on a theta digraph",
5651           _vertex_flow_at_least(_vout, 4, 0, 3, 2)
5652           and not _vertex_flow_at_least(_vout, 4, 0, 3, 3)
5653           and local_connectivity(_vg, 0, 3, vertex_split=True) == 2)
5654     # prop:dir-arc-stability, the unconditional quadratic bound. Both the
5655     # load-bearing counting step (sum of d+ d- capped by m n(n-1)) and the bound
5656     # it yields are checked on sampled feasible digraphs.
5657     _rng = random.Random(3)
5658     _worst_slack = None
5659     _count_ok = True
5660     for _ in range(120):
5661         _sn, _sm = _rng.randint(3, 7), _rng.randint(2, 4)
5662         _sg = Graph(_sn, SIMPLE_DIRECTED)
5663         for _a in range(_sn):
5664             for _b in range(_sn):
5665                 if _a != _b and _rng.random() < 0.6:
5666                     _sg.mu[_a, _b] = 1
5667         while max_edge_connectivity(_sg) > _sm - 1: # prune down to feasible
5668             _present = [(a, b) for a in range(_sn) for b in range(_sn) if _sg.mu[a
5669                                     ↪ , b]]
5670             _a, _b = _rng.choice(_present)
5671             _sg.mu[_a, _b] = 0
5672             _dout = [int(_sg.mu[x].sum()) for x in range(_sn)]
5673             _din = [int(_sg.mu[:, x].sum()) for x in range(_sn)]
5674             # The sharp form of the counting step: (m-1) n(n-1), not m n(n-1).
5675             if sum(_dout[x] * _din[x] for x in range(_sn)) > (_sm - 1) * _sn * (_sn -
5676                                     ↪ 1):
5677                 _count_ok = False
5678                 _slack = ((_sn * _sn) // 4 + math.sqrt(_sm) * _sn ** 1.5
5679                           - int(_sg.mu.sum()))
5680                 _worst_slack = _slack if _worst_slack is None else min(_worst_slack,
5681                                     ↪ _slack)
5682             # thm:dir-arc-linear-error, the O_m(n) bound.
5683             if int(_sg.mu.sum()) > (_sn * _sn) // 4 + 4 * (_sm - 1) * (_sn - 1):
5684                 _count_ok = False
5685             # Case 2 of its proof: min total degree >= n/2 forces the linear bound
5686             # on the sum of the smaller half-degrees.
5687             if min(_dout[x] + _din[x] for x in range(_sn)) >= _sn / 2:
5688                 if (sum(min(_dout[x], _din[x]) for x in range(_sn))
5689                     > 4 * (_sm - 1) * (_sn - 1)):
5690                     _count_ok = False
5691     check("prop:dir-arc-stability and thm:dir-arc-linear-error hold on 120 "
5692           f"samled feasible digraphs (tightest slack {_worst_slack:.1f})",
5693           _count_ok and _worst_slack >= 0)
5694     # sec:multi-vertex-standard: the OTHER counting convention is a different
5695     # problem, and the theta construction beats the thickened tree at m=5, n=4.
5696     _mv4, _, _mv4_done = max_multigraph_vertex_standard(4, 3)
5697     _mv5, _, _mv5_done = max_multigraph_vertex_standard(4, 5)

```

```

5695     check(f"multigraph vertex, other convention:  $K_3(4) = \{\_mv4\} = (m-1)(n-1)$ ",
5696           \_mv4_done and \_mv4 == 6)
5697     check(f"multigraph vertex, other convention:  $K_5(4) = \{\_mv5\} > 12$ , the "
5698           "multigraph edge value, so the two problems differ",
5699           \_mv5_done and \_mv5 == 14)
5700     # Exhaustive undirected by brute force: ell_2(5) = n-1 = 4 (a spanning tree).
5701     tree = solve(5, 2, directed=False, simple=True, exhaustive=True,
5702                 max_seconds=30.0)
5703     check(f"solve exhaustive simple-undirected n=5,m=2: {tree.value} ({tree.bound}
5704            $\hookrightarrow$  })",
5705           tree.proven and tree.value == 4)
5706     # Hypergraph discovery returns a feasible lower bound for the r=3, m=2 case.
5707     hyper = solve(7, 2, hypergraph=True, r=3, exhaustive=False, max_seconds=3.0,
5708                  method="sa")
5709     check(f"solve discovery 3-uniform hypergraph n=7,m=2: {hyper.value} ({hyper.
5710            $\hookrightarrow$  bound})",
5711           hyper.bound == "lower" and hyper.value == 3)
5712
5713     # --- open-variant exploration coverage -----
5714     # These three functions are the program's only handle on the OPEN parts of
5715     # the problem (hypergraph vertex value, the fractional relaxation behind the
5716     # odd-n conjectures). The first author-reviewed draft of this block ran
5717     # exhaustive n>=7 sweeps and 6000-step searches, which added minutes; the
5718     # checks below cover the same code paths at n<=5 so they finish in ~2s.
5719     section("Open variants: hypergraph vertex value ( $\kappa^{\max}$ ) at small n")
5720     # m=2 uses the incidence-rank/forest witness, m=3 the brute-force
5721     # feasibility sweep; both confirm value = floor((m-1)(n-1)/(r-1)).
5722     for n, r, m in [(4, 3, 2), (5, 3, 2), (4, 3, 3), (5, 3, 3)]:
5723         target = ((m - 1) * (n - 1)) // (r - 1)
5724         check(f"hypergraph vertex value at n={n}, r={r}, m={m} = {target}",
5725               verify_hyper_vertex_value(n, r, m))
5726
5727     section("Open variants: the fractional [0,1]-weight checker is exact")
5728     # A directed path keeps every max-flow at 1 (feasible); two internally
5729     # disjoint s->t routes push the s->t flow to 2 (infeasible). This is the
5730     # relaxation that fractional_search optimises over.
5731     path_w = np.zeros((4, 4))
5732     path_w[0, 1] = path_w[1, 2] = path_w[2, 3] = 1.0
5733     two_route = np.zeros((4, 4))
5734     two_route[0, 1] = two_route[0, 2] = two_route[1, 3] = two_route[2, 3] = 1.0
5735     check("path weighting feasible, two-route weighting infeasible",
5736           fractional_flows_feasible(path_w)
5737           and not fractional_flows_feasible(two_route))
5738
5739     section("Open variants: fractional search respects the odd-n ceilings")
5740     # Short n=5 runs (the conjectured bipartite point is rigid). The
5741     # min_degree score carries a 1e-3 * total-weight tie-breaker, so its
5742     # comparison allows that slack; a genuine refutation would clear the
5743     # ceiling by a full unit, far above it.
5744     _, total_best = fractional_search(5, "total", steps=1500, seed=1)
5745     check(f"fractional total weight at n=5 stays <= 8: {total_best:.3f}",
5746           total_best <= 8 + 1e-6)
5747     _, deg_best = fractional_search(5, "min_degree", steps=1500, seed=1)
5748     check(f"fractional min degree at n=5 stays <= k=2 (+tie-break slack): {
5749            $\hookrightarrow$  deg_best:.3f}",
5750           deg_best <= 2 + 1e-3 * 5 * 5)

```



```

5749     print(f"\n{'ALL CHECKS PASSED' if _failures == 0 else f'{'_failures'} CHECK(S)
        ↳ FAILED'}")
5750     return 1 if _failures else 0
5751
5752
5753 if __name__ == "__main__":
5754     print(f"C extension: {'LOADED (_erdos_fast.so)' if C_EXTENSION_LOADED else '
        ↳ not found (pure Python)'}")
5755     raise SystemExit(_run_checks())

```

C.2 The figure script

This is the only script that writes image files. It imports the main program and calls its plotting routines, so it holds no mathematics of its own. Keeping it separate means the figures can be regenerated in one command without running anything else, and every random choice it makes is seeded, so running it twice produces identical files.

```

1  """
2  Regenerate every figure the thesis uses, from the one program next door.
3
4  This is the only figure script. It imports the single program
5  ‘erdos915_unified.py’ and calls its plotting routines, so the figures can never
6  drift from the code that produces the numbers. Run it from this ‘program/’
7  directory:
8
9      python make_figures.py
10
11 Each figure is written into ‘../figures/’. Every randomised search uses a
12 fixed seed, so the output is reproducible.
13 """
14
15 from __future__ import annotations
16
17 import math
18 import sys
19 from pathlib import Path
20
21 import numpy as np
22
23 sys.path.insert(0, str(Path(__file__).resolve().parent))
24
25 from erdos915_unified import ( # noqa: E402
26     MULTI_DIRECTED,
27     MULTI_UNDIRECTED,
28     SIMPLE_DIRECTED,
29     SIMPLE_UNDIRECTED,
30     Graph,
31     _VARIANT_ENUM_CONFIGS,
32     search_for_dense_graph,
33     compute_enumeration_cache,
34     compute_pair_enumeration_cache,
35     compute_surface_cache,
36     connectivity_distribution,
37     gallery_extremal_graphs,
38     directed_arc_lower_bound,

```

```

39     directed_multigraph_arc ,
40     draw_graph_with_sensitivity ,
41     edge_vertex_distribution ,
42     hypergraph_edge ,
43     multigraph_undirected_edge ,
44     plot_complexity_growth ,
45     plot_extremal_gallery ,
46     plot_conn_dist_grid ,
47     plot_conn_threshold_3d ,
48     plot_connectivity_distribution ,
49     plot_pair_conn_dist_grid ,
50     plot_directed_crossover ,
51     plot_edge_vertex_divergence ,
52     plot_edge_vertex_histograms ,
53     plot_edge_dist_grid ,
54     plot_scatter_lambda_edges ,
55     plot_search_trace ,
56     plot_sa_vs_tabu_convergence ,
57     plot_variant_3d_surfaces ,
58     plot_variant_grid ,
59     save_gallery_json ,
60     simple_undirected_edge ,
61     solve ,
62 )
63
64 FIGURES = Path(__file__).resolve().parent.parent / "figures"
65
66
67 # -----
68 # The all-variant grid: proved / conjectured / guessed, with machine points.
69 # Every number below comes from one driver, “solve”: exhaustive for an exact
70 # point, discovery for a search lower bound. The formula curves come from the
71 # closed forms; the guess is a fit; the band is [construction LB, trivial max].
72 # -----
73
74 def _exact_points(ns, m, budget, **kw):
75     """Machine-proved sizes: increasing n until the exhaustion no longer finishes.
76     ↳ """
77     xs, ys = [], []
78     for n in ns:
79         res = solve(n, m, exhaustive=True, max_seconds=budget, **kw)
80         if res.bound == "exact":
81             xs.append(n)
82             ys.append(res.value)
83         else:
84             break # once it cannot finish, larger n cannot either
85     return xs, ys
86
87 def _search_points(ns, m, budget, **kw):
88     """Discovery lower bounds: a concrete feasible graph at each n (the LB
89     ↳ construction)."""
90     xs, ys = [], []
91     for n in ns:
92         res = solve(n, m, exhaustive=False, max_seconds=budget, seed=0, **kw)
93         xs.append(n)
94         ys.append(res.value)
95     return xs, ys

```

```

95
96
97 def _band(search, trivial_max, ns):
98     """The certain interval: an easy construction below, the trivial maximum above
99         ↳ .
100
101     The lower edge is the densest feasible graph the search exhibited (a real
102     lower bound); the upper edge is the most edges any graph on n vertices can
103     hold at all (a loose but certain upper bound). The truth lies in between.
104     """
105     found = dict(zip(search[0], search[1]))
106     lower = [min(found.get(n, 0), trivial_max(n)) for n in ns]
107     upper = [trivial_max(n) for n in ns]
108     return list(ns), lower, upper
109
110 def _extend_lower_bounds(search, lb_fn, ns):
111     """Raise the search lower bounds to a known verified construction where it
112         ↳ wins.
113
114     Past the size the quick search reaches, an explicit construction the checker
115     confirms is a better (still honest) lower bound, so we report the best of the
116     two at each n, exactly as the rediscovery section does by hand.
117     """
118     found = dict(zip(search[0], search[1]))
119     for n in ns:
120         found[n] = max(found.get(n, 0), lb_fn(n))
121     xs = sorted(found)
122     return xs, [found[n] for n in xs]
123
124 def _reconcile_panel(panel: dict) -> dict:
125     """Make a panel internally consistent before plotting.
126
127     Three honest invariants, enforced here once for every panel rather than
128     scattered through the construction code:
129
130     1. **Nothing exceeds a proved optimum.** A machine-checked exact value is
131     the true maximum at that 'n'; no proved/conjectured curve, no
132     named-construction lower bound, and no search circle may sit above it.
133     (This is what produced the green square *below* its own curve: a closed
134     form -- e.g. the hypergraph bound capped at the complete hypergraph -- can
135     overshoot what is actually achievable at small 'n'.)
136
137     2. **Search lower bounds rise with 'n'.

```

```

151         if curve is None:
152             return None
153         xs, ys = curve
154         return xs, [min(y, exact[x]) if x in exact else y for x, y in zip(xs, ys)]
155
156     # invariant 1 applied to every formula line and to the named branches
157     for key in ("proved", "conj"):
158         if panel.get(key) is not None:
159             panel[key] = cap_to_exact(panel[key])
160     if panel.get("branches"):
161         panel["branches"] = [(cap_to_exact((bx, by)), lbl)
162                               for bx, by, lbl in panel["branches"]]
163
164     # invariants 1 then 2 applied to the search circles
165     if panel.get("search") is not None:
166         sx, sy = panel["search"]
167         sy = [min(y, exact[x]) if x in exact else y for x, y in zip(sx, sy)]
168         running, mono = -1, []
169         for y in sy:
170             running = max(running, y)
171             mono.append(running)
172         panel["search"] = (sx, mono)
173
174     # invariant 3b: the open-panel guess is the interpolation through those
175     # (monotone, exact-capped) points, so it can never ride above them.
176     if panel.get("guess") == "search":
177         panel["guess"] = panel.get("search")
178
179     return panel
180
181
182 def gather_variant_grid(m=3, exact_budget=4.0, search_budget=0.4,
183                        ↪ open_search_budget=4.0):
184     """Build the twelve panels of the all-variant grid (row-major, model x col).
185     ‘m’ is the forbidden connectivity value shown in every panel.
186
187     Two uniformity rules make the twelve panels directly comparable:
188
189     1. Every matrix panel plots its search lower bounds over the *same* vertex
190        range ‘matrix_ns’ and every hypergraph panel over ‘hyper_ns’, so each
191        panel carries the same number of data points -- no panel trails off early.
192     2. For every panel where a construction is known (proved or conjectured) the
193        search lower bound is raised to that named construction with
194        :func:‘_extend_lower_bounds’, exactly as the caption of the figure and
195        the rediscovery section state: the reported circle at each ‘n’ is the
196        better of the cooled search and the verified construction. This is the
197        honest "lands on the curve" rule, applied to *all* panels rather than
198        only the two directed ones it used to cover, so the proved curves pass
199        through their circles instead of floating above a jagged search tail.
200
201     Open panels also get a verified construction planted, not only the proved
202     ones: the open undirected vertex panels get the proved EDGE value (Whitney,
203      $\kappa \leq \lambda$ , so the edge extremiser is vertex-feasible) and the two
204     directed-hypergraph panels get the proved bipartite family of
205     prop:dir-hyper-first. This matters because at ‘search_budget=0.4’ the
206     vertex-mode search on a large matrix graph barely completes one move, so
207     the RAW result actively *degrades* with growing ‘n’ (measured:

```

```

208     n=6/10/14/16 gave 15/12/5/4 at m=6), and the monotone running-max in
209     :func:‘_reconcile_panel‘ then freezes the plotted curve at its early peak,
210     a flat line that looks like a finding but is really budget starvation.
211     ‘‘open_search_budget‘‘ (default 4s, ~10x ‘‘search_budget‘‘) still gives
212     those panels a stronger walk, since the search can genuinely beat the
213     planted construction where the open problem is richer (it does at m=6,
214     n=9 on the undirected vertex panel).
215     """
216     panels = []
217     # Uniform x-ranges for the search / construction curves. The machine-PROVED
218     # (exact) points are genuinely limited by enumeration cost and stop early; the
219     # search lower bounds and the formula curves are cheap, so they run well past
220     # that, to show each variant's trend and let the search discover dense
221     # ("maybe extremal") graphs at sizes exhaustion cannot reach.
222     matrix_ns = list(range(2, 17)) # all matrix panels, n = 2..16
223     hyper_ns = list(range(2, 13)) # all hypergraph panels, n = 2..12
224
225     def tri_undirected(n):
226         return n * (n - 1) // 2
227
228     def tri_hyper(n, r=3):
229         return math.comb(n, r)
230
231     def tri_dir_hyper(n, r=3):
232         return n * math.comb(n - 1, r - 1)
233
234     # Construction lower bounds (verified, proved-or-conjectured values), each
235     # capped at the trivial maximum for its model. The cap matters at small n:
236     # a closed form like Mader's floor(m(n-1)/2) is the value only for n >= m;
237     # for n < m the complete graph K_n already has lambda^max = n-1 <= m-1, so it
238     # is feasible and denser, and the true value is comb(n,2). Without the cap
239     # the plotted curve and its lower-bound circles float ABOVE the exact squares,
240     # which is impossible (a lower bound can never exceed the proved optimum).
241     def lb_simple_edge(n):
242         return min(simple_undirected_edge(n, m), n * (n - 1) // 2)
243
244     def lb_multi_edge(n):
245         return min(multigraph_undirected_edge(n, m), (m - 1) * (n * (n - 1) // 2))
246
247     def lb_dir(n):
248         return min(directed_arc_lower_bound(n, m), n * (n - 1))
249
250     def lb_multi_dir(n):
251         return min(directed_multigraph_arc(n, m), (m - 1) * n * (n - 1))
252
253     def lb_hyper_edge(n):
254         return min(hypergraph_edge(n, m, 3), math.comb(n, 3))
255
256     def lb_dir_hyper(n):
257         # The PROVED bipartite family of prop:dir-hyper-first at r = 3:
258         # alpha tails, and on the other n - alpha vertices a shared near-regular
259         # head graph of max degree m - 1 (realisable iff m - 1 <= n - alpha - 1),
260         # giving alpha * floor((m-1)(n-alpha)/2) single-step hyperarcs with
261         # lambda^max = kappa^max = m - 1. Sound for BOTH separations, since
262         # every route in it is a single tail -> head step.
263         best = 0
264         for alpha in range(1, n - m + 1):
265             best = max(best, alpha * ((m - 1) * (n - alpha) // 2))

```

```

266         return min(best, n * math.comb(n - 1, 2))
267
268     # Undirected vertex is proved for  $m \leq 4$  (Leonard/Whitney), open for  $m > 5$ .
269     vert_proved = (m <= 4)
270     # Hypergraph vertex is proved for  $m \leq 3$  (incidence-rank lemma, thm:hyper-vertex
271     #  $\hookrightarrow -m3$ ).
272     hyper_vert_proved = (m <= 3)
273
274     def searched(ns, lb_fn, **solve_kw):
275         """Search over the uniform range, then raise to the known construction."""
276         se = _search_points(ns, m, search_budget, **solve_kw)
277         if lb_fn is not None:
278             se = _extend_lower_bounds(se, lb_fn, ns)
279         return se
280
281     # ----- row 1: simple -----
282     # (1) simple undirected edge -- proved (Mader) for all  $m$ .
283     ex = _exact_points(range(2, 9), m, exact_budget,
284                        directed=False, simple=True, separation="edge")
285     se = searched(matrix_ns, lb_simple_edge,
286                  directed=False, simple=True, separation="edge")
287     panels.append(dict(
288         title=f"undirected edge,  $m=\{m\}$  (proved)", ylabel="edges",
289         proved=(matrix_ns, [lb_simple_edge(n) for n in matrix_ns]),
290         exact=ex, search=se))
291
292     # (2) simple undirected vertex -- proved for  $m \leq 4$ , open for  $m > 5$ .
293     ex2 = _exact_points(range(2, 8), m, exact_budget,
294                        directed=False, simple=True, separation="vertex")
295     if vert_proved:
296         se2 = searched(matrix_ns, lb_simple_edge,
297                       directed=False, simple=True, separation="vertex")
298         panels.append(dict(
299             title=f"undirected vertex,  $m=\{m\}$  (proved)", ylabel="edges",
300             proved=(matrix_ns, [lb_simple_edge(n) for n in matrix_ns]),
301             exact=ex2, search=se2))
302     else:
303         se2 = _search_points(matrix_ns, m, open_search_budget,
304                              directed=False, simple=True, separation="vertex")
305         # Whitney:  $\kappa \leq \lambda$ , so Mader's edge extremiser is vertex-feasible
306         # and the proved edge value is an honest lower bound on the open vertex
307         # panel too. Without it the starved vertex-mode search freezes into a
308         # flat plateau at large  $n$  (the "cut off" look); the search may still beat
309         # the edge value where the vertex problem is genuinely richer.
310         se2 = _extend_lower_bounds(se2, lb_simple_edge, matrix_ns)
311         band2 = _band(se2, tri_undirected, matrix_ns)
312         panels.append(dict(
313             title=f"undirected vertex,  $m=\{m\}$  (open)", ylabel="edges",
314             guess="search",
315             band=band2, exact=ex2, search=se2))
316
317     # (3) simple directed arc -- conjectured.
318     ex3 = _exact_points(range(2, 8), m, exact_budget,
319                        directed=True, simple=True, separation="edge")
320     se3 = searched(matrix_ns, lb_dir,
321                   directed=True, simple=True, separation="edge")
322     panels.append(dict(
323         title=f"directed arc,  $m=\{m\}$  (conjectured)", ylabel="arcs",

```

```

323     conj=(matrix_ns, [lb_dir(n) for n in matrix_ns]),
324     branches=[(matrix_ns, [min(m * (n - 1), n * (n - 1)) for n in matrix_ns],
325         "hub $m(n{-}1)$"),
326         # Shifted-partition augmented bipartite floor((n+m-2)^2/4)
327         # (the corrected conj:dir-arc branch; the old balanced count
328         # floor(n^2/4)+(m-2)ceil(n/2) agrees at m<=3 but is smaller
329         # at m>=4 and would sit below the conjecture curve at m=6).
330         (matrix_ns,
331             [min((n + m - 2) ** 2 // 4, n * (n - 1))
332             for n in matrix_ns], "bipartite")],
333     exact=ex3, search=se3))
334
335 # (4) simple directed vertex -- conjectured (= arc).
336 ex4 = _exact_points(range(2, 8), m, exact_budget,
337     directed=True, simple=True, separation="vertex")
338 se4 = searched(matrix_ns, lb_dir,
339     directed=True, simple=True, separation="vertex")
340 panels.append(dict(
341     title=f"directed vertex, $m={m}$ (conjectured)", ylabel="arcs",
342     conj=(matrix_ns, [lb_dir(n) for n in matrix_ns]),
343     exact=ex4, search=se4))
344
345 # ----- row 2: multigraph -----
346 # (5) multigraph undirected edge -- proved for all m.
347 ex5 = _exact_points(range(2, 7), m, exact_budget,
348     directed=False, simple=False, separation="edge")
349 se5 = searched(matrix_ns, lb_multi_edge,
350     directed=False, simple=False, separation="edge")
351 panels.append(dict(
352     title=f"undirected edge, $m={m}$ (proved)", ylabel="edges",
353     proved=(matrix_ns, [lb_multi_edge(n) for n in matrix_ns]),
354     exact=ex5, search=se5))
355
356 # (6) multigraph undirected vertex -- equals simple (parallel edges irrelevant
357     ↳ for vertex cuts).
358 multi_vert_label = ("proved, $=$ simple" if vert_proved else "open, $=$ simple
359     ↳ ")
360 if vert_proved:
361     panels.append(dict(
362         title=f"undirected vertex, $m={m}$ ({multi_vert_label})", ylabel="
363             ↳ edges",
364         proved=(matrix_ns, [lb_simple_edge(n) for n in matrix_ns]),
365         exact=ex2, search=se2))
366 else:
367     panels.append(dict(
368         title=f"undirected vertex, $m={m}$ ({multi_vert_label})", ylabel="
369             ↳ edges",
370         guess="search",
371         band=_band(se2, tri_undirected, matrix_ns), exact=ex2, search=se2))
372
373 # (7) multigraph directed arc -- proved for all n and m (thm:dir-multi-full).
374 ex7 = _exact_points(range(3, 6), m, 10.0,
375     directed=True, simple=False, separation="edge")
376 se7 = searched(matrix_ns, lb_multi_dir,
377     directed=True, simple=False, separation="edge")
378 panels.append(dict(
379     title=f"directed arc, $m={m}$ (multigraph, proved)", ylabel="arcs",
380     proved=(matrix_ns, [lb_multi_dir(n) for n in matrix_ns]),

```

```

377         exact=ex7, search=se7))
378
379     # (8) multigraph directed vertex -- conjectured (reduces to simple digraph).
380     panels.append(dict(
381         title=f"directed vertex, $m={m}$ (conjectured, $=$ simple)", ylabel="arcs
382         ↔ ",
383         conj=(matrix_ns, [lb_dir(n) for n in matrix_ns]),
384         exact=ex4, search=se4))
385
386     # ----- row 3: hypergraph (r=3) -----
387     # (9) hypergraph undirected edge -- proved for all m.
388     ex9 = _exact_points(range(2, 8), m, exact_budget,
389         hypergraph=True, r=3, directed=False, separation="edge")
389     se9 = searched(hyper_ns, lb_hyper_edge,
390         hypergraph=True, r=3, directed=False, separation="edge")
391     panels.append(dict(
392         title=f"undirected edge, $m={m}$ (proved)", ylabel="hyperedges",
393         proved=(hyper_ns, [lb_hyper_edge(n) for n in hyper_ns]),
394         exact=ex9, search=se9))
395
396     # (10) hypergraph undirected vertex -- PROVED for m<=3 (incidence-rank lemma),
397     #       open for m>=4.
398     ex10 = _exact_points(range(2, 7), m, exact_budget,
399         hypergraph=True, r=3, directed=False, separation="vertex"
400         ↔ )
401
402     if hyper_vert_proved:
403         se10 = searched(hyper_ns, lb_hyper_edge,
404             hypergraph=True, r=3, directed=False, separation="vertex")
405         panels.append(dict(
406             title=f"undirected vertex, $m={m}$ (proved)", ylabel="hyperedges",
407             proved=(hyper_ns, [lb_hyper_edge(n) for n in hyper_ns]),
408             exact=ex10, search=se10))
409     else:
410         se10 = _search_points(hyper_ns, m, open_search_budget,
411             hypergraph=True, r=3, directed=False, separation="
412             ↔ vertex")
413
414         # Whitney again: the proved hyperedge value is a lower bound for the
415         # vertex separation (its extremiser is vertex-feasible). The exact cap
416         # in _reconcile_panel corrects the small-n corner where the simple
417         # attainment condition m-1 <= n-2 has not kicked in yet.
418         se10 = _extend_lower_bounds(se10, lb_hyper_edge, hyper_ns)
419         panels.append(dict(
420             title=f"undirected vertex, $m={m}$ (open)", ylabel="hyperedges",
421             guess="search",
422             band=_band(se10, tri_hyper, hyper_ns), exact=ex10, search=se10))
423
424     # (11) hypergraph directed arc -- OPEN (new directed Berge model).
425     ex11 = _exact_points(range(2, 6), m, exact_budget,
426         hypergraph=True, r=3, directed=True, separation="edge")
427     se11 = _search_points(hyper_ns, m, open_search_budget,
428         hypergraph=True, r=3, directed=True, separation="edge")
429     # The proved bipartite construction of prop:dir-hyper-first is the named
430     # lower bound here (the search alone slips below the quadratic at larger n).
431     se11 = _extend_lower_bounds(se11, lb_dir_hyper, hyper_ns)
432     panels.append(dict(
433         title=f"directed arc, $m={m}$ (open, new model)", ylabel="hyperarcs",
434         guess="search",
435         band=_band(se11, tri_dir_hyper, hyper_ns), exact=ex11, search=se11))

```



```

432
433 # (12) hypergraph directed vertex -- OPEN (new directed Berge model).
434 ex12 = _exact_points(range(2, 6), m, exact_budget,
435                     hypergraph=True, r=3, directed=True, separation="vertex")
436 se12 = _search_points(hyper_ns, m, open_search_budget,
437                     hypergraph=True, r=3, directed=True, separation="vertex"
438                     ↪ )
439 # Same construction: all its routes are single tail -> head steps, so it is
440 # feasible for the vertex separation at the same value.
441 se12 = _extend_lower_bounds(se12, lb_dir_hyper, hyper_ns)
442 panels.append(dict(
443     title=f"directed vertex, $m={m}$ (open, new model)", ylabel="hyperarcs",
444     guess="search",
445     band=_band(se12, tri_dir_hyper, hyper_ns), exact=ex12, search=se12))
446
447 return [_reconcile_panel(p) for p in panels]
448
449 def main() -> None:
450     FIGURES.mkdir(parents=True, exist_ok=True)
451
452     plot_directed_crossover(m=3, max_n=24, path=FIGURES / "directed_crossover.png"
453     ↪ )
454     print("wrote directed_crossover.png")
455
456     plot_edge_vertex_divergence(max_n=35, path=FIGURES / "edge_vertex_divergence.
457     ↪ png")
458     print("wrote edge_vertex_divergence.png")
459
460     # Figure 3.2: cooling trace for the directed multigraph n=4, m=3 case.
461     # record_exact_connectivity logs the true lambda^max per step (more expensive
462     # but needed for an exact scatter; the search itself uses the cheaper upper
463     ↪ bound).
464     run = search_for_dense_graph(MULTI_DIRECTED, n=4, m=3, steps=3000, seed=1,
465     record_exact_connectivity=True)
466     plot_search_trace(run, path=FIGURES / "temperature_trace.png",
467     optimum=12, ceiling=2, show_histogram=False)
468     print(f"wrote temperature_trace.png (search reached {run.best_edge_count} arcs
469     ↪ of 12)")
470
471     # Additional scatter-only traces for other variants (no cooling panel needed
472     # since the cooling schedule is the same across all cases).
473     trace_cases = [
474         # (variant, n, m, filename_stem, ceiling, connectivity_label, edge_label,
475         ↪ title)
476         (SIMPLE_UNDIRECTED, 7, 3, "trace_simple_undirected_n7_m3", 2,
477         r"$\lambda^{\max}$", "edges",
478         r"Search trace: simple undirected, $n=7$, $m=3$"),
479         (SIMPLE_DIRECTED, 5, 3, "trace_simple_directed_n5_m3", 2,
480         r"$\lambda^{\max}$", "arcs",
481         r"Search trace: simple directed, $n=5$, $m=3$"),
482         (MULTI_UNDIRECTED, 5, 3, "trace_multi_undirected_n5_m3", 2,
483         r"$\lambda^{\max}$", "edges",
484         r"Search trace: multi undirected, $n=5$, $m=3$"),
485         (MULTI_DIRECTED, 5, 3, "trace_multi_directed_n5_m3", 2,
486         r"$\lambda^{\max}$", "arcs",
487         r"Search trace: multi directed, $n=5$, $m=3$"),
488         (MULTI_DIRECTED, 4, 3, "trace_multi_directed_n4_m3_vertex", 2,

```

```

484         r"$\kappa^{\{\max\}}$", "arcs",
485         r"Search trace: multi directed vertex-sep, $n=4$, $m=3$"),
486     ]
487 for variant, n, m, stem, ceil_val, clabel, elabel, ttitle in trace_cases:
488     sep = "vertex" if "vertex" in stem else "edge"
489     tr = search_for_dense_graph(variant, n=n, m=m, separation=sep,
490                                steps=3000, seed=1,
491                                record_exact_connectivity=True)
492     plot_search_trace(tr, path=FIGURES / f"{stem}.png",
493                      ceiling=ceil_val, show_cooling=False,
494                      show_histogram=False,
495                      connectivity_label=clabel, edge_label=elabel,
496                      title=ttitle)
497     print(f"wrote {stem}.png (best={tr.best_edge_count})")
498
499     # A larger directed multigraph: four s->t routes of differing capacity plus a
500     # dead end. Parallel arcs are drawn explicitly, so multiplicity is read by
501     # counting. The s->a route is over-provisioned (mu=3 into a but a single arc
502     # a->t out), so it appears as three arcs in a COOL colour (sigma=1): the
503     # visual point that multiplicity and load-bearing role are not the same.
504     sens = Graph(8, MULTI_DIRECTED) # 0=s 1=t 2=a 3=b 4=c 5=d 6=e 7=f
505     sens.set_multiplicity(0, 2, 3); sens.set_multiplicity(2, 1, 1) # s->a->t,
506         ↳ over-provisioned in
507     sens.set_multiplicity(0, 3, 3); sens.set_multiplicity(3, 1, 3) # s->b->t, cap
508         ↳ 3 (load-bearing)
509     sens.set_multiplicity(0, 4, 2); sens.set_multiplicity(4, 1, 2) # s->c->t, cap
510         ↳ 2
511     sens.set_multiplicity(0, 6, 1); sens.set_multiplicity(6, 1, 1) # s->e->t, cap
512         ↳ 1
513     sens.set_multiplicity(0, 7, 2); sens.set_multiplicity(7, 1, 2) # s->f->t, cap
514         ↳ 2
515     sens.set_multiplicity(0, 5, 2) # s->d, dead
516         ↳ end
517
518     sens_layout = {0: (-3.2, 0.0), 1: (3.2, 0.0),
519                   2: (0.0, 2.6), 3: (0.0, 1.3), 4: (0.0, 0.0),
520                   6: (0.0, -1.3), 7: (0.0, -2.6), 5: (-3.2, -2.4)}
521     sens_labels = {0: "s", 1: "t", 2: "a", 3: "b", 4: "c", 5: "d", 6: "e", 7: "f"}
522     draw_graph_with_sensitivity(
523         sens, path=FIGURES / "sensitivity_mixed.png",
524         layout=sens_layout, node_labels=sens_labels)
525     print("wrote sensitivity_mixed.png")
526
527     # Annealing vs tabu convergence (Ch.3): wall-clock timed, so a representative
528     # run rather than a seed-exact one (see the figure caption).
529     plot_sa_vs_tabu_convergence(FIGURES / "sa_vs_tabu_convergence.pdf",
530                                cases=((5, 3), (7, 3)), budget=8.0, seed=0)
531     print("wrote sa_vs_tabu_convergence.pdf")
532
533     plot_complexity_growth(path=FIGURES / "complexity_growth.png")
534     print("wrote complexity_growth.png")
535
536     # Gallery of machine-found extremal graphs (reads extremal_gallery.json).
537     plot_extremal_gallery(FIGURES / "extremal_graphs_gallery.png")
538     print("wrote extremal_graphs_gallery.png")
539
540     # Bounds grids: one for m=3 and one for m=6 (more purple dots).
541     for m_val in (3, 6):
542         print(f"gathering all-variant grid m={m_val} (runs solve many times)...")

```

```

536     panels = gather_variant_grid(m=m_val)
537     out = FIGURES / f"variant_bounds_m{m_val}.png"
538     plot_variant_grid(pannels, path=out, m=m_val)
539     print(f"wrote {out.name}")
540
541     # -----
542     # Full-enumeration scatter and distribution figures.
543     # The enumeration cache is written once to figures/enumeration_cache.pkl.
544     # -----
545     print("building enumeration cache (slow on first run, cached thereafter)...")
546     enum_cache_path = FIGURES / "enumeration_cache.pkl"
547     enum_data = compute_enumeration_cache(cache_path=enum_cache_path)
548     print("enumeration cache ready")
549
550     plot_scatter_lambda_edges(enum_data, path=FIGURES / "scatter_lambda_edges.png"
551                               ↪ )
552     print("wrote scatter_lambda_edges.png")
553
554     # Pooled per-pair connectivity: every vertex pair of every enumerated graph,
555     # tagged with its graph's lambda_max. Slow first run, cached thereafter.
556     print("building pair-connectivity cache (slow on first run, cached thereafter)
557           ↪ ...)")
558     pair_cache_path = FIGURES / "pair_enumeration_cache.pkl"
559     pair_data = compute_pair_enumeration_cache(cache_path=pair_cache_path)
560     print("pair-connectivity cache ready")
561
562     # Pooled per-pair connectivity histograms (Ch.4): each panel stacks blue/red
563     # at its own mid-range connectivity boundary, so the split is meaningful in
564     # every variant rather than collapsing to one colour at a fixed m.
565     plot_pair_conn_dist_grid(pair_data, path=FIGURES / "pair_conn_dist.png")
566     print("wrote pair_conn_dist.png")
567
568     # Edge-count histograms (Ch.4): same mid-range stacked colouring.
569     plot_edge_dist_grid(enum_data, path=FIGURES / "edges_dist.png")
570     print("wrote edges_dist.png")
571
572     # Per-graph connectivity distribution at the fixed forbidden threshold m=6
573     # (App. B), showing the m=6 feasibility split; m=3 body version removed
574     ↪ 2026-06-20.
575     plot_conn_dist_grid(enum_data, 6, path=FIGURES / "conn_dist_m6.png")
576     print("wrote conn_dist_m6.png")
577
578     # -----
579     # 3-D bound surface over the (n, m) grid.
580     # Uses a JSON cache (slow first run, instant thereafter).
581     # -----
582     surface_cache_path = FIGURES / "surface_cache.json"
583     print("building surface cache (slow on first run, cached thereafter)...")
584     compute_surface_cache(
585         n_range=(3, 9), m_range=(2, 6),
586         max_seconds=20, cache_path=surface_cache_path)
587     plot_variant_3d_surfaces(surface_cache_path, path=FIGURES / "
588                               ↪ variant_surface_3d.png")
589     print("wrote variant_surface_3d.png")
590
591     # -----
592     # 3-D threshold histogram (appendix): how the lambda_max distribution shifts
593     # with density p, for three representative variants. Kept as a standalone

```

```

590     # appendix figure (fig:threshold-3d) after the Figure 4.2 threshold story was
591     # removed from the body on 2026-06-20.
592     # -----
593     # No p_values: each panel sweeps in units of its own threshold, since the
594     # graph and hypergraph models do not share a density scale.
595     plot_conn_threshold_3d(
596         path=FIGURES / "threshold_3d.png",
597         n=12, m=3, samples=150, seed=7)
598     print("wrote threshold_3d.png")
599
600     # NOTE: the 2-D random-sampling threshold figures (degree_threshold,
601     # sampled_variant_grid) were removed from the thesis on 2026-06-20. Their
602     # generators are kept in erdos915_unified.py and the restore recipe is in
603     # research_notes/removed_threshold_phenomenon.md.
604
605     # Edge vs vertex disjointness in  $G(16, p)$  at three densities (Chapter 1).
606     p_values = [0.25, 0.5, 0.75]
607     data = {pp: edge_vertex_distribution(16, pp, trials=400, seed=7)
608             for pp in p_values}
609     plot_edge_vertex_histograms(data, 16, p_values,
610                                FIGURES / "edge_vertex_sampling.png")
611     print("wrote edge_vertex_sampling.png")
612
613     # Gallery of extremal graphs: all 12 variants  $\times$  small  $(n, m)$ , ~3s per case.
614     print("building extremal graph gallery (this takes a few minutes)...")
615     gallery_path = FIGURES / "extremal_gallery.json"
616     gal = gallery_extremal_graphs(max_n=7, max_m=4, r=3, time_per_case=3.0)
617     save_gallery_json(gal, gallery_path)
618     total_classes = sum(
619         len(cases.get("classes", []))
620         for variant_data in gal.values()
621         for cases in variant_data.values()
622     )
623     print(f"wrote {gallery_path.name} ({total_classes} iso-classes across all
624         ↳ variants)")
625
626     if __name__ == "__main__":
627         main()

```

C.3 The C accelerator

Two operations dominate the running time at the sizes the searches reach: deciding whether a pair of vertices already has too many independent routes, and reducing a graph to a standard form so that two drawings of the same graph can be recognised as equal. Both are rewritten here in C, which is compiled separately and loaded if it is present.

Nothing depends on it. Every function below has a pure Python counterpart in the main program, the two are tested against each other, and if the compiled library is missing the program simply uses the Python version. The size limits in the code are deliberate: the fixed-size buffers are safe only up to the stated number of vertices, and above it the program falls back rather than risking a buffer it cannot hold.

```

1  /*
2  * _erdos_fast.c - hot-path C helpers for erdos915_unified.py
3  *
4  * Compile: gcc -O3 -march=native -shared -fPIC -o _erdos_fast.so _erdos_fast.c
5  *
6  * All functions take a flat row-major int32 array 'mu' of size n*n.
7  * Access: mu[u*n + v] is the capacity (multiplicity) of arc u->v.
8  */
9
10 #include <stdint.h>
11 #include <string.h>
12 #include <stdlib.h>
13
14 /* ----- */
15 /* tiny_maxflow */
16 /* DFS Ford-Fulkerson; returns 1 iff max-flow(s,t) > cap. */
17 /* Stops as soon as it finds cap+1 augmenting paths (early exit). */
18 /* n <= 7 in practice; residual lives on the stack. */
19 /* ----- */
20 int tiny_maxflow(const int *mu, int n, int s, int t, int cap)
21 {
22     /* local mutable residual; sized for n up to 16 (vertex-split at n=8) */
23     int res[256];
24     memcpy(res, mu, (size_t)(n * n) * sizeof(int));
25
26     int flow = 0;
27     while (flow <= cap) {
28         /* DFS to find an augmenting path */
29         int parent[16];
30         for (int i = 0; i < n; i++) parent[i] = -1;
31         parent[s] = s;
32
33         /* stack-based DFS (max depth = n vertices) */
34         int stack[16];
35         int sp = 0;
36         stack[sp++] = s;
37
38         while (sp > 0 && parent[t] == -1) {
39             int u = stack[--sp];
40             for (int v = 0; v < n; v++) {
41                 if (parent[v] == -1 && res[u * n + v] > 0) {
42                     parent[v] = u;
43                     stack[sp++] = v;
44                 }
45             }
46         }
47
48         if (parent[t] == -1)
49             return 0; /* max-flow <= cap */
50
51         /* find bottleneck */
52         int bottleneck = 1 << 30;
53         for (int v = t; v != s; ) {
54             int u = parent[v];
55             int cap_uv = res[u * n + v];
56             if (cap_uv < bottleneck) bottleneck = cap_uv;
57             v = u;
58         }

```

```

59
60     /* augment */
61     for (int v = t; v != s; ) {
62         int u = parent[v];
63         res[u * n + v] -= bottleneck;
64         res[v * n + u] += bottleneck;
65         v = u;
66     }
67
68     flow += bottleneck;
69 }
70 return 1; /* max-flow > cap */
71 }
72
73 /* ----- */
74 /* max_connectivity_exceeds */
75 /* Sweeps all pairs; returns 1 iff any pair has flow > k. */
76 /* directed=1: all ordered pairs; directed=0: unordered pairs. */
77 /* ----- */
78 int max_connectivity_exceeds(const int *mu, int n, int k, int directed)
79 {
80     for (int s = 0; s < n; s++) {
81         int start = directed ? 0 : s + 1;
82         for (int t = start; t < n; t++) {
83             if (s == t) continue;
84             if (tiny_maxflow(mu, n, s, t, k))
85                 return 1;
86         }
87     }
88     return 0;
89 }
90
91 /* ----- */
92 /* canonical_form_min */
93 /* Writes the lex-min row-major permutation of the n×n matrix into */
94 /* out (n×n int32 values). Uses Heap's algorithm to enumerate n! */
95 /* permutations. At n=7: 5040 iterations × 49 ints each. */
96 /* ----- */
97 void canonical_form_min(const int *mu, int n, int *out)
98 {
99     int perm[7], best[49], cand[49];
100     int c[7]; /* Heap's algorithm counters */
101     int nn = n * n;
102
103     for (int i = 0; i < n; i++) { perm[i] = i; c[i] = 0; }
104
105     /* identity permutation is the first candidate */
106     for (int r = 0; r < n; r++)
107         for (int col = 0; col < n; col++)
108             best[r * n + col] = mu[perm[r] * n + perm[col]];
109
110     /* Heap's algorithm */
111     int i = 1;
112     while (i < n) {
113         if (c[i] < i) {
114             /* swap */
115             if (i % 2 == 0) {
116                 int tmp = perm[0]; perm[0] = perm[i]; perm[i] = tmp;

```

```

117         } else {
118             int tmp = perm[c[i]]; perm[c[i]] = perm[i]; perm[i] = tmp;
119         }
120
121         /* build candidate permuted matrix */
122         for (int r = 0; r < n; r++)
123             for (int col = 0; col < n; col++)
124                 cand[r * n + col] = mu[perm[r] * n + perm[col]];
125
126         /* update best if cand is lex-smaller */
127         if (memcmp(cand, best, (size_t)nn * sizeof(int)) < 0)
128             memcpy(best, cand, (size_t)nn * sizeof(int));
129
130         c[i]++;
131         i = 1;
132     } else {
133         c[i] = 0;
134         i++;
135     }
136 }
137
138 memcpy(out, best, (size_t)nn * sizeof(int));
139 }

```

C.4 The test suite

The suite checks the program against facts established independently of it: closed forms proved by hand, constructions whose size and connectivity are known in advance, and small cases where an exhaustive walk gives the answer outright. Several tests compare two routines that are supposed to agree, such as a fast path against the slow one it replaces, which is what catches an optimisation that changes an answer. Tests needing an optional library skip themselves when it is absent, so the suite passes on a minimal installation rather than reporting failures for tools it was never given.

C.4.1 Shared setup

```

1  """Test suite for the unified Erdos-915 program.
2
3  The tests import 'erdos915_unified' from the parent directory, so we put that
4  directory on the path here, once, for every test module. The whole suite runs
5  with the standard library alone:
6
7      cd program
8      python -m unittest discover -s tests
9
10 (no pytest required, though pytest will also discover and run these if present).
11 """
12
13 import sys
14 from pathlib import Path
15
16 sys.path.insert(0, str(Path(__file__).resolve().parent.parent))

```

C.4.2 The graph model

```
1 """The graph model: the multiplicity matrix and every operation on it."""
2
3 import unittest
4
5 from erdos915_unified import (
6     Graph,
7     MULTI_DIRECTED,
8     MULTI_UNDIRECTED,
9     SIMPLE_DIRECTED,
10    SIMPLE_UNDIRECTED,
11 )
12
13
14 class VariantBooleans(unittest.TestCase):
15     def test_the_two_booleans(self):
16         self.assertEqual((SIMPLE_UNDIRECTED.directed, SIMPLE_UNDIRECTED.simple), (
17             ↪ False, True))
18         self.assertEqual((MULTI_UNDIRECTED.directed, MULTI_UNDIRECTED.simple), (
19             ↪ False, False))
20         self.assertEqual((SIMPLE_DIRECTED.directed, SIMPLE_DIRECTED.simple), (True
21             ↪ , True))
22         self.assertEqual((MULTI_DIRECTED.directed, MULTI_DIRECTED.simple), (True,
23             ↪ False))
24
25     def test_describe_is_nonempty(self):
26         for variant in (SIMPLE_UNDIRECTED, MULTI_UNDIRECTED, SIMPLE_DIRECTED,
27             ↪ MULTI_DIRECTED):
28             self.assertTrue(variant.describe())
29
30
31 class EmptyGraph(unittest.TestCase):
32     def test_fresh_graph_is_empty(self):
33         g = Graph(5, SIMPLE_UNDIRECTED)
34         self.assertEqual(g.num_vertices, 5)
35         self.assertEqual(list(g.vertices()), list(range(5)))
36         self.assertEqual(g.edge_count(), 0)
37         self.assertEqual(list(g.edges()), [])
38         self.assertFalse(g.has_edge(0, 1))
39
40
41 class UndirectedModel(unittest.TestCase):
42     def test_add_edge_is_symmetric(self):
43         g = Graph(3, SIMPLE_UNDIRECTED)
44         g.add_edge(0, 1)
45         self.assertTrue(g.has_edge(0, 1))
46         self.assertTrue(g.has_edge(1, 0))
47         self.assertEqual(g.multiplicity(0, 1), 1)
48         self.assertEqual(g.multiplicity(1, 0), 1)
49         self.assertEqual(g.edge_count(), 1)
50         self.assertEqual(list(g.edges()), [(0, 1, 1)])
51
52     def test_simple_graph_saturates_at_one(self):
53         g = Graph(3, SIMPLE_UNDIRECTED)
54         g.add_edge(0, 1)
55         g.add_edge(0, 1)
56         self.assertEqual(g.multiplicity(0, 1), 1)
```



```

52         self.assertEqual(g.edge_count(), 1)
53
54     def test_multigraph_keeps_parallel_edges(self):
55         g = Graph(3, MULTI_UNDIRECTED)
56         g.add_edge(0, 1, 3)
57         self.assertEqual(g.multiplicity(0, 1), 3)
58         self.assertEqual(g.multiplicity(1, 0), 3)
59         self.assertEqual(g.edge_count(), 3)
60         self.assertEqual(g.degree(0), 3)
61         self.assertEqual(g.degree(1), 3)
62
63     def test_undirected_degree_counts_incident_multiplicity(self):
64         g = Graph(3, MULTI_UNDIRECTED)
65         g.add_edge(0, 1, 2)
66         g.add_edge(0, 2, 1)
67         self.assertEqual(g.degree(0), 3)
68         self.assertEqual(g.edge_count(), 3)
69
70
71 class DirectedModel(unittest.TestCase):
72     def test_arc_is_one_directional(self):
73         g = Graph(3, SIMPLE_DIRECTED)
74         g.add_edge(0, 1)
75         self.assertTrue(g.has_edge(0, 1))
76         self.assertFalse(g.has_edge(1, 0))
77         self.assertEqual(g.edge_count(), 1)
78         self.assertEqual(g.out_degree(0), 1)
79         self.assertEqual(g.in_degree(1), 1)
80         self.assertEqual(g.in_degree(0), 0)
81         self.assertEqual(g.degree(0), 1)
82         self.assertEqual(g.degree(1), 1)
83
84     def test_directed_multiplicities_are_independent(self):
85         g = Graph(3, MULTI_DIRECTED)
86         g.add_edge(0, 1, 2)
87         g.add_edge(1, 0, 1)
88         self.assertEqual(g.multiplicity(0, 1), 2)
89         self.assertEqual(g.multiplicity(1, 0), 1)
90         self.assertEqual(g.edge_count(), 3)
91         self.assertEqual(g.out_degree(0), 2)
92         self.assertEqual(g.in_degree(0), 1)
93         self.assertEqual(g.degree(0), 3)
94
95     def test_edges_yields_both_arc_directions(self):
96         g = Graph(3, SIMPLE_DIRECTED)
97         g.add_edge(0, 1)
98         g.add_edge(1, 0)
99         self.assertEqual(sorted(g.edges()), [(0, 1, 1), (1, 0, 1)])
100
101
102 class Mutation(unittest.TestCase):
103     def test_remove_never_goes_below_zero(self):
104         g = Graph(3, MULTI_UNDIRECTED)
105         g.add_edge(0, 1, 3)
106         g.remove_edge(0, 1, 5)
107         self.assertEqual(g.multiplicity(0, 1), 0)
108         self.assertFalse(g.has_edge(0, 1))
109

```

```

110     def test_set_multiplicity_directly(self):
111         g = Graph(3, MULTI_DIRECTED)
112         g.set_multiplicity(0, 1, 4)
113         self.assertEqual(g.multiplicity(0, 1), 4)
114         self.assertEqual(g.multiplicity(1, 0), 0)
115
116     def test_simple_rejects_multiplicity_above_one(self):
117         g = Graph(3, SIMPLE_UNDIRECTED)
118         with self.assertRaises(ValueError):
119             g.set_multiplicity(0, 1, 2)
120
121     def test_negative_multiplicity_rejected(self):
122         g = Graph(3, MULTI_UNDIRECTED)
123         with self.assertRaises(ValueError):
124             g.set_multiplicity(0, 1, -1)
125
126     def test_self_loops_rejected(self):
127         g = Graph(3, MULTI_UNDIRECTED)
128         with self.assertRaises(ValueError):
129             g.add_edge(1, 1)
130
131     def test_copy_is_independent(self):
132         g = Graph(3, MULTI_UNDIRECTED)
133         g.add_edge(0, 1, 2)
134         clone = g.copy()
135         clone.add_edge(0, 2, 1)
136         self.assertEqual(g.edge_count(), 2)
137         self.assertEqual(clone.edge_count(), 3)
138
139
140 if __name__ == "__main__":
141     unittest.main()

```

C.4.3 The closed-form bounds

```

1  """The closed-form bounds library: every formula, on a range of inputs."""
2
3  import unittest
4
5  from erdos915_unified import (
6      directed_arc_lower_bound,
7      directed_arc_m2,
8      directed_multigraph_arc,
9      hypergraph_edge,
10     multigraph_undirected_edge,
11     simple_undirected_edge,
12     simple_undirected_vertex_le4,
13     simple_undirected_vertex_m5,
14 )
15
16
17 class Bounds(unittest.TestCase):
18     def test_simple_undirected_edge(self):
19         self.assertEqual(simple_undirected_edge(5, 3), 6)
20         for n in range(2, 12):
21             for m in range(2, 6):
22                 self.assertEqual(simple_undirected_edge(n, m), (m * (n - 1)) // 2)

```

```

23
24 def test_multigraph_undirected_edge(self):
25     for n in range(2, 12):
26         for m in range(2, 6):
27             self.assertEqual(multigraph_undirected_edge(n, m), (m - 1) * (n -
                ↪ 1))
28
29 def test_vertex_le4_matches_edge_and_caps_at_four(self):
30     for m in range(2, 5):
31         self.assertEqual(simple_undirected_vertex_le4(10, m),
                ↪ simple_undirected_edge(10, m))
32     with self.assertRaises(ValueError):
33         simple_undirected_vertex_le4(10, 5)
34
35 def test_vertex_m5(self):
36     # Sorensen-Thomassen Theorem 4 covers n >= 6 apart from n = 7 and 12.
37     for n in range(6, 20):
38         if n in (7, 12):
39             continue
40         self.assertEqual(simple_undirected_vertex_m5(n), (8 * n) // 3 - 4)
41
42 def test_vertex_m5_matches_the_papers_own_values(self):
43     # Every value Sorensen-Thomassen state explicitly, converted. Their
44     # f_5 counts the least edge total that FORCES a 5-rail, one more than
45     # k_5 here, so each entry below is their printed figure minus one.
46     for n, f5 in {7: 16, 12: 28, 13: 31, 14: 34, 15: 37}.items():
47         self.assertEqual(simple_undirected_vertex_m5(n), f5 - 1, f"n={n}")
48
49 def test_vertex_m5_undetermined_below_six(self):
50     for n in (2, 3, 4, 5):
51         with self.assertRaises(ValueError):
52             simple_undirected_vertex_m5(n)
53
54 def test_vertex_m5_diverges_from_the_edge_value_at_fourteen(self):
55     # The paper proves f_5(n) = floor(5(n-1)/2) + 1 throughout 6 <= n <= 13,
56     # so the two problems agree there and first come apart at n = 14.
57     for n in range(6, 14):
58         self.assertEqual(simple_undirected_vertex_m5(n),
59             simple_undirected_edge(n, 5), f"n={n}")
60     for n in range(14, 40):
61         self.assertGreater(simple_undirected_vertex_m5(n),
62             simple_undirected_edge(n, 5), f"n={n}")
63
64 def test_directed_arc_m2_branches_and_crossover(self):
65     self.assertEqual(directed_arc_m2(4), 6) # linear hub branch wins
66     self.assertEqual(directed_arc_m2(7), 12) # the two branches meet
67     self.assertEqual(directed_arc_m2(8), 16) # quadratic branch wins
68     for n in range(2, 20):
69         self.assertEqual(directed_arc_m2(n), max(2 * (n - 1), (n * n) // 4))
70
71 def test_directed_arc_lower_bound(self):
72     self.assertEqual(directed_arc_lower_bound(10, 3), 30) # the 30-arc graph
73     self.assertEqual(directed_arc_lower_bound(4, 2), 6)
74     for n in range(3, 14):
75         for m in range(2, 5):
76             expected = max(m * (n - 1), (n + m - 2) ** 2 // 4)
77             self.assertEqual(directed_arc_lower_bound(n, m), expected)
78

```

```

79     def test_hypergraph_edge(self):
80         self.assertEqual(hypergraph_edge(7, 2, 3), 3)
81         self.assertEqual(hypergraph_edge(13, 2, 4), 4)
82         for n in range(3, 14):
83             for m in range(2, 5):
84                 for r in range(2, 5):
85                     self.assertEqual(hypergraph_edge(n, m, r), ((m - 1) * (n - 1))
86                                     ↪ // (r - 1))
87
88     def test_directed_multigraph_arc(self):
89         self.assertEqual(directed_multigraph_arc(4, 3), 12)
90         self.assertEqual(directed_multigraph_arc(6, 2), 10)
91         for n in range(2, 12):
92             for m in range(2, 6):
93                 expected = max(2 * (n - 1) * (m - 1), (m - 1) * ((n * n) // 4))
94                 self.assertEqual(directed_multigraph_arc(n, m), expected)
95
96 if __name__ == "__main__":
97     unittest.main()

```

C.4.4 The named constructions

```

1  """The named extremal constructions: their exact sizes and connectivities."""
2
3  import unittest
4
5  from erdos915_unified import (
6      augmented_bipartite,
7      clique_tree,
8      double_star,
9      max_edge_connectivity,
10     min_vertex_connectivity,
11     one_directional_bipartite,
12 )
13
14
15 class Constructions(unittest.TestCase):
16     def test_double_star_directed(self):
17         for n in range(2, 7):
18             for m in range(2, 5):
19                 g = double_star(n, m, directed=True)
20                 self.assertEqual(g.edge_count(), 2 * (n - 1) * (m - 1))
21                 self.assertEqual(max_edge_connectivity(g), m - 1)
22
23     def test_double_star_undirected(self):
24         for n in range(2, 7):
25             for m in range(2, 5):
26                 g = double_star(n, m, directed=False)
27                 self.assertEqual(g.edge_count(), (m - 1) * (n - 1))
28                 self.assertEqual(max_edge_connectivity(g), m - 1)
29
30     def test_one_directional_bipartite_is_quadratic_and_feasible(self):
31         for n in range(2, 8):
32             g = one_directional_bipartite(n)
33             self.assertEqual(g.edge_count(), (n * n) // 4)
34             self.assertEqual(max_edge_connectivity(g), 1)

```

```

35
36     def test_augmented_bipartite_conjecture_values(self):
37         self.assertEqual(augmented_bipartite(10, 3).edge_count(), 30)
38         for n, m in [(8, 3), (10, 4), (12, 4), (10, 5)]:
39             g = augmented_bipartite(n, m)
40             self.assertEqual(g.edge_count(), (n + m - 2) ** 2 // 4)
41             self.assertEqual(max_edge_connectivity(g), m - 1)
42
43     def test_augmented_bipartite_guards_its_precondition(self):
44         with self.assertRaises(ValueError):
45             augmented_bipartite(3, 5) # needs n >= m
46
47     def test_clique_tree_is_globally_connected(self):
48         for blocks in range(0, 5):
49             self.assertEqual(min_vertex_connectivity(clique_tree(4, blocks)), 3)
50
51
52 if __name__ == "__main__":
53     unittest.main()

```

C.4.5 The connectivity checker

```

1  """The connectivity checker (edge and vertex) and the Gomory-Hu shortcut."""
2
3  import random
4  import unittest
5
6  from erdos915_unified import (
7      Graph,
8      MULTI_DIRECTED,
9      MULTI_UNDIRECTED,
10     NETWORKX_AVAILABLE,
11     SIMPLE_DIRECTED,
12     SIMPLE_UNDIRECTED,
13     _vertex_flow_at_least,
14     exceeds_bound,
15     local_edge_connectivity,
16     local_vertex_connectivity,
17     max_connectivity,
18     max_edge_connectivity,
19     max_edge_connectivity_via_tree,
20     max_vertex_connectivity,
21     sample_random_graph,
22 )
23
24
25 def random_multigraph(variant, n, rng, max_mult=2):
26     """A random graph of ‘variant’ on ‘n’ vertices for predicate testing.
27
28     Simple variants take multiplicities in ‘{0, 1}’; multigraph variants in
29     ‘{0, ..., max_mult}’. Directed variants fill every ordered pair, undirected
30     variants the upper triangle (‘set_multiplicity’ mirrors the lower half).
31     """
32     g = Graph(n, variant)
33     top = 1 if variant.simple else max_mult
34     for u in range(n):
35         start = 0 if variant.directed else u + 1

```

```

36         for v in range(start, n):
37             if u != v:
38                 g.set_multiplicity(u, v, rng.randint(0, top))
39     return g
40
41
42 def complete_graph(n, variant=SIMPLE_UNDIRECTED):
43     g = Graph(n, variant)
44     for u in range(n):
45         for v in range(u + 1, n):
46             g.add_edge(u, v)
47     return g
48
49
50 def star(n):
51     g = Graph(n, SIMPLE_UNDIRECTED)
52     for leaf in range(1, n):
53         g.add_edge(0, leaf)
54     return g
55
56
57 def cycle(n):
58     g = Graph(n, SIMPLE_UNDIRECTED)
59     for v in range(n):
60         g.add_edge(v, (v + 1) % n)
61     return g
62
63
64 class EdgeConnectivity(unittest.TestCase):
65     def test_tree_is_one(self):
66         self.assertEqual(max_edge_connectivity(star(6)), 1)
67
68     def test_cycle_is_two(self):
69         self.assertEqual(max_edge_connectivity(cycle(6)), 2)
70
71     def test_complete_graph(self):
72         for n in (3, 4, 5):
73             self.assertEqual(max_edge_connectivity(complete_graph(n)), n - 1)
74             self.assertEqual(local_edge_connectivity(complete_graph(n), 0, 1), n -
75                 ↪ 1)
76
77     def test_parallel_edges_count_as_routes(self):
78         g = Graph(3, MULTI_UNDIRECTED)
79         g.add_edge(0, 1, 4)
80         self.assertEqual(local_edge_connectivity(g, 0, 1), 4)
81         self.assertEqual(max_edge_connectivity(g), 4)
82
83     def test_directed_arc_disjoint_routes(self):
84         g = Graph(3, SIMPLE_DIRECTED)
85         g.add_edge(0, 1) # direct route
86         g.add_edge(0, 2) # detour 0 -> 2 -> 1
87         g.add_edge(2, 1)
88         self.assertEqual(local_edge_connectivity(g, 0, 1), 2)
89
90     def test_removing_an_edge_never_raises_lambda(self):
91         rng = random.Random(11)
92         for _ in range(20):
93             g = sample_random_graph(8, 0.4, False, rng)

```

```

93         before = max_edge_connectivity(g)
94         for u, v, _count in list(g.edges()):
95             reduced = g.copy()
96             reduced.set_multiplicity(u, v, 0)
97             self.assertEqual(max_edge_connectivity(reduced), before)
98
99
100 class VertexConnectivity(unittest.TestCase):
101     def test_complete_graph(self):
102         for n in (3, 4, 5):
103             self.assertEqual(max_vertex_connectivity(complete_graph(n)), n - 1)
104             self.assertEqual(local_vertex_connectivity(complete_graph(n), 0, 1), n
105                 ↪ - 1)
106
107     def test_whitney_inequality_on_random_graphs(self):
108         rng = random.Random(0)
109         for _ in range(40):
110             g = sample_random_graph(12, 0.3, False, rng)
111             self.assertEqual(max_vertex_connectivity(g), max_edge_connectivity
112                 ↪ (g))
113
114 class VertexFlowPredicate(unittest.TestCase):
115     """‘_vertex_flow_at_least’ is the exhaustive search’s inner vertex test.
116
117     It must agree with the exact checker it replaces on every (pair, k), or the
118     base cases of the directed vertex theorem would rest on a faster wrong test.
119     """
120     def test_matches_the_exact_checker(self):
121         rng = random.Random(17)
122         for _ in range(150):
123             n = rng.randint(2, 6)
124             g = Graph(n, SIMPLE_DIRECTED)
125             out = [set() for _ in range(n)]
126             for a in range(n):
127                 for b in range(n):
128                     if a != b and rng.random() < 0.35:
129                         g.mu[a, b] = 1
130                         out[a].add(b)
131             for s in range(n):
132                 for t in range(n):
133                     if s == t:
134                         continue
135                     exact = local_vertex_connectivity(g, s, t)
136                     for k in range(1, 5):
137                         self.assertEqual(_vertex_flow_at_least(out, n, s, t, k),
138                             exact >= k)
139
140     def test_a_direct_arc_is_one_route_with_no_interior(self):
141         # s -> t plus the detour s -> x -> t: two internally disjoint routes.
142         out = [{1, 2}, set(), {1}, set()]
143         self.assertTrue(_vertex_flow_at_least(out, 4, 0, 1, 2))
144         self.assertFalse(_vertex_flow_at_least(out, 4, 0, 1, 3))
145
146
147 @unittest.skipUnless(NETWORKX_AVAILABLE, "Gomory-Hu trees need the optional
148     ↪ networkx")

```

```

148 class GomoryHu(unittest.TestCase):
149     def test_tree_matches_direct_sweep(self):
150         for g in (cycle(6), complete_graph(5), star(7)):
151             self.assertEqual(max_edge_connectivity_via_tree(g),
152                             ↪ max_edge_connectivity(g))
153
154     def test_multigraph_matches(self):
155         g = Graph(4, MULTI_UNDIRECTED)
156         g.add_edge(0, 1, 3)
157         g.add_edge(1, 2, 2)
158         g.add_edge(2, 3, 1)
159         g.add_edge(0, 3, 2)
160         self.assertEqual(max_edge_connectivity_via_tree(g), max_edge_connectivity(
161             ↪ g))
162
163 class CappedPredicate(unittest.TestCase):
164     """'exceeds_bound' must agree with the exact checker, pair set for pair set.
165
166     This pins the cheap capped predicate the search relies on to the trusted
167     exact 'max_connectivity' over every variant, separation and threshold.
168 """
169     def test_predicate_matches_exact_value(self):
170         variants = (SIMPLE_UNDIRECTED, MULTI_UNDIRECTED,
171                     SIMPLE_DIRECTED, MULTI_DIRECTED)
172         rng = random.Random(2024)
173         for variant in variants:
174             for n in (3, 4, 5):
175                 for _ in range(8):
176                     g = random_multigraph(variant, n, rng)
177                     for sep in ("edge", "vertex"):
178                         split = sep == "vertex"
179                         exact = max_connectivity(g, vertex_split=split)
180                         for k in range(-1, exact + 2):
181                             self.assertEqual(
182                                 exceeds_bound(g, k, separation=sep),
183                                 exact > k,
184                                 msg=f"{variant.name} n={n} sep={sep} k={k} "
185                                     f"exact={exact}",
186                             )
187
188     def test_predicate_does_not_mutate_the_graph(self):
189         rng = random.Random(7)
190         g = random_multigraph(MULTI_DIRECTED, 5, rng)
191         before = g.mu.copy()
192         exceeds_bound(g, 1, separation="vertex")
193         exceeds_bound(g, 1, separation="edge")
194         self.assertTrue((g.mu == before).all())
195
196
197 if __name__ == "__main__":
198     unittest.main()

```

C.4.6 Hypergraphs

```

1 """The hypergraph model and Berge connectivity by the helper-point network."""

```



```

2
3 import unittest
4
5 from erdos915_unified import (
6     Hypergraph,
7     complete_uniform_hypergraph,
8     hyper_connectivity,
9     hyperedge_connectivity,
10    max_feasible_hyperedges,
11    max_hyperedge_connectivity,
12    star_hypertree,
13    _hyperedge_candidates,
14 )
15
16
17 class BergeConnectivity(unittest.TestCase):
18     def test_single_hyperedge_carries_one_route(self):
19         h = Hypergraph(3, [[0, 1, 2]])
20         self.assertEqual(h.edge_count(), 1)
21         self.assertEqual(hyperedge_connectivity(h, 0, 1), 1)
22         self.assertEqual(hyperedge_connectivity(h, 0, 2), 1)
23         self.assertEqual(hyperedge_connectivity(h, 1, 2), 1)
24
25     def test_two_hyperedges_give_two_routes(self):
26         h = Hypergraph(4, [[0, 1, 2], [0, 1, 3]])
27         self.assertEqual(h.edge_count(), 2)
28         self.assertEqual(hyperedge_connectivity(h, 0, 1), 2)
29         self.assertEqual(len(h.incident_hyperedges(0)), 2)
30         self.assertEqual(len(h.incident_hyperedges(2)), 1)
31
32     def test_complete_three_uniform_k4(self):
33         self.assertEqual(max_hyperedge_connectivity(complete_uniform_hypergraph(4,
34             ↪ 3)), 3)
35
36     def test_two_uniform_reduces_to_edge_connectivity(self):
37         for n in (3, 4, 5):
38             self.assertEqual(max_hyperedge_connectivity(
39                 ↪ complete_uniform_hypergraph(n, 2)), n - 1)
40
41     def test_star_hypertree(self):
42         for n, r in [(7, 3), (10, 4), (13, 4)]:
43             h = star_hypertree(n, r)
44             self.assertEqual(h.edge_count(), (n - 1) // (r - 1))
45             self.assertEqual(max_hyperedge_connectivity(h), 1)
46
47     def test_bad_hyperedge_size_rejected(self):
48         with self.assertRaises(ValueError):
49             star_hypertree(7, 1)
50
51 class DirectedOrientationModels(unittest.TestCase):
52     """Forward / backward / general directed hyperedge models."""
53
54     def test_general_gate_is_one_way(self):
55         # A mixed arc {0,1} -> {2,3}: a route enters at a tail, leaves at a head.
56         h = Hypergraph(4, [(frozenset({0, 1}), frozenset({2, 3}))], directed=True)
57         self.assertEqual(sorted(h.members(h.hyperedges[0])), [0, 1, 2, 3])
58         self.assertEqual(hyper_connectivity(h, 0, 2), 1) # tail -> head

```

```

58         self.assertEqual(hyper_connectivity(h, 2, 0), 0)    # head -> tail: none
59
60     def test_forward_legacy_equals_general_form(self):
61         legacy = Hypergraph(3, [(0, frozenset({1, 2}))], directed=True)
62         general = Hypergraph(3, [(frozenset({0}), frozenset({1, 2}))], directed=
        ↪ True)
63         for s, t in [(0, 1), (0, 2), (1, 2)]:
64             self.assertEqual(hyper_connectivity(legacy, s, t),
65                             hyper_connectivity(general, s, t))
66
67     def test_candidate_counts_split_at_r3(self):
68         # At r = 3 every split has a singleton side, so general = forward +
        ↪ backward.
69         fwd = _hyperedge_candidates(4, 3, True, kind="forward")
70         bwd = _hyperedge_candidates(4, 3, True, kind="backward")
71         gen = _hyperedge_candidates(4, 3, True, kind="general")
72         self.assertEqual(len(fwd), 12)
73         self.assertEqual(len(bwd), 12)
74         self.assertEqual(len(gen), 24)
75
76     def test_forward_backward_duality(self):
77         # Arc reversal makes backward the dual of forward: same extremal numbers.
78         for n, m in [(3, 3), (4, 3), (4, 2)]:
79             f, _ = max_feasible_hyperedges(n, 3, m, directed=True, kind="forward",
        ↪ time_limit=2.0)
80             b, _ = max_feasible_hyperedges(n, 3, m, directed=True, kind="backward"
        ↪ , time_limit=2.0)
81             self.assertEqual(f, b)
82
83     def test_general_beats_forward_when_vertices_scarce(self):
84         # n = r is the regime where the richer orientation set packs more arcs.
85         g3, e3 = max_feasible_hyperedges(3, 3, 3, directed=True, kind="general",
        ↪ time_limit=2.0)
86         f3, _ = max_feasible_hyperedges(3, 3, 3, directed=True, kind="forward",
        ↪ time_limit=2.0)
87         self.assertTrue(e3)
88         self.assertEqual((g3, f3), (4, 3))
89
90     def test_general_contains_forward(self):
91         # General can never be worse than forward (forward hypergraphs are general
        ↪ ).
92         for n, m in [(4, 3), (5, 2)]:
93             f, _ = max_feasible_hyperedges(n, 3, m, directed=True, kind="forward",
        ↪ time_limit=2.0)
94             g, _ = max_feasible_hyperedges(n, 3, m, directed=True, kind="general",
95                                         time_limit=3.0, seed_lb=f)
96             self.assertGreaterEqual(g, f)
97
98
99 if __name__ == "__main__":
100     unittest.main()

```

C.4.7 Edge sensitivity

```

1  """Edge sensitivity: the load-bearing measure that guides the search."""
2
3  import random

```

```

4 import unittest
5
6 from erdos915_unified import (
7     Graph,
8     MULTI_DIRECTED,
9     MULTI_UNDIRECTED,
10    SIMPLE_DIRECTED,
11    SIMPLE_UNDIRECTED,
12    double_star,
13    edge_sensitivity,
14    max_edge_connectivity,
15    max_vertex_connectivity,
16    one_directional_bipartite,
17    sensitivity_map,
18 )
19
20
21 def _random_graph(variant, n, rng, top):
22     g = Graph(n, variant)
23     cap = 1 if variant.simple else top
24     for u in range(n):
25         start = 0 if variant.directed else u + 1
26         for v in range(start, n):
27             if u != v:
28                 g.set_multiplicity(u, v, rng.randint(0, cap))
29     return g
30
31
32 class Sensitivity(unittest.TestCase):
33     def test_absent_edge_has_zero_sensitivity(self):
34         g = Graph(3, SIMPLE_UNDIRECTED)
35         self.assertEqual(edge_sensitivity(g, 0, 1), 0)
36
37     def test_sensitivity_matches_its_definition(self):
38         g = one_directional_bipartite(6)
39         before = max_edge_connectivity(g)
40         for (u, v), sigma in sensitivity_map(g).items():
41             reduced = g.copy()
42             reduced.set_multiplicity(u, v, 0)
43             self.assertEqual(sigma, before - max_edge_connectivity(reduced))
44             self.assertGreaterEqual(sigma, 0) # removing an edge never raises
45             ↪ lambda
46
47     def test_map_covers_exactly_the_present_edges(self):
48         g = double_star(5, 3, directed=True)
49         self.assertEqual(len(sensitivity_map(g)), sum(1 for _ in g.edges()))
50
51     def test_map_equals_per_edge_sensitivity(self):
52         # The before-once refactor must return the IDENTICAL dict as the old
53         # per-edge definition, across variants and both connectivity measures.
54         variants = (SIMPLE_UNDIRECTED, MULTI_UNDIRECTED,
55                     SIMPLE_DIRECTED, MULTI_DIRECTED)
56         rng = random.Random(99)
57         for variant in variants:
58             for measure in (max_edge_connectivity, max_vertex_connectivity):
59                 for _ in range(6):
60                     g = _random_graph(variant, 5, rng, top=2)
61                     expected = {(u, v): edge_sensitivity(g, u, v, measure)}

```

```

61         for u, v, _ in g.edges():
62             self.assertEqual(sensitivity_map(g, measure), expected)
63
64
65 if __name__ == "__main__":
66     unittest.main()

```

C.4.8 The search

```

1  """The temperature search: it rediscovers the known small extremal values.
2
3  The search is seeded, so these outcomes are deterministic and reproducible.
4  """
5
6  import unittest
7
8  from erdos915_unified import (
9      MULTI_DIRECTED,
10     MULTI_UNDIRECTED,
11     SIMPLE_DIRECTED,
12     SIMPLE_UNDIRECTED,
13     best_of_searches,
14     max_connectivity,
15     max_edge_connectivity,
16     search_for_dense_graph,
17     tabu_search_for_dense_graph,
18 )
19
20
21 class Search(unittest.TestCase):
22     def test_search_rediscovers_directed_m2(self):
23         result = search_for_dense_graph(SIMPLE_DIRECTED, n=4, m=2, steps=6000,
24             ↳ seed=0)
25         self.assertEqual(result.best_edge_count, 6) #  $ell_2^{dir}(4) = 6$ 
26         self.assertTrue(result.feasible_found)
27         self.assertLessEqual(max_edge_connectivity(result.best_graph), 1)
28
29     def test_acceptance_rate_is_a_fraction(self):
30         result = search_for_dense_graph(SIMPLE_DIRECTED, n=4, m=2, steps=6000,
31             ↳ seed=0)
32         self.assertTrue(0.0 <= result.acceptance_rate() <= 1.0)
33         self.assertEqual(len(result.history), 6000)
34
35     def test_search_rediscovers_undirected_m3(self):
36         #  $ell_3(n) = \lfloor 3(n-1)/2 \rfloor$ :  $n=4 \rightarrow 4$ ,  $n=5 \rightarrow 6$ .
37         for n, expected in [(4, 4), (5, 6)]:
38             result = best_of_searches(SIMPLE_UNDIRECTED, n, 3, restarts=6, steps
39                 ↳ =3000, seed=0)
40             self.assertEqual(result.best_edge_count, expected)
41             self.assertLessEqual(max_edge_connectivity(result.best_graph), 2)
42
43 class TabuSearch(unittest.TestCase):
44     """Tabu search is the deterministic twin of the annealer: same energy and
45     neighbourhood, so it rediscovers the same small extremal values.
46     """
47
48

```

```

46     def test_tabu_rediscovered_directed_m2(self):
47         result = tabu_search_for_dense_graph(SIMPLE_DIRECTED, n=4, m=2,
48                                             steps=150, seed=0)
49         self.assertEqual(result.best_edge_count, 6) # ell_2^dir(4) = 6
50         self.assertTrue(result.feasible_found)
51         self.assertLessEqual(max_edge_connectivity(result.best_graph), 1)
52
53     def test_tabu_rediscovered_undirected_m3(self):
54         result = tabu_search_for_dense_graph(SIMPLE_UNDIRECTED, n=4, m=3,
55                                             steps=150, seed=0)
56         self.assertEqual(result.best_edge_count, 4) # ell_3(4) = 4
57         self.assertLessEqual(max_edge_connectivity(result.best_graph), 2)
58
59     def test_best_of_searches_accepts_tabu_method(self):
60         result = best_of_searches(SIMPLE_DIRECTED, 4, 2, restarts=2,
61                                 method="tabu", parallel=False, steps=120)
62         self.assertEqual(result.best_edge_count, 6)
63
64     def test_unknown_method_rejected(self):
65         with self.assertRaises(ValueError):
66             best_of_searches(SIMPLE_UNDIRECTED, 4, 3, method="genetic")
67
68
69 class FastPathEquivalence(unittest.TestCase):
70     """The fast (capped + monotone) search must reproduce the slow exact search
71     step for step. This is the "nothing under the rug" guarantee: the speedup
72     changes no reported value because it changes no single step of the walk.
73     """
74
75     def test_fast_path_matches_exact(self):
76         variants = (SIMPLE_UNDIRECTED, MULTI_UNDIRECTED,
77                   SIMPLE_DIRECTED, MULTI_DIRECTED)
78         for variant in variants:
79             for n in (3, 4, 5):
80                 for m in (2, 3):
81                     for sep in ("edge", "vertex"):
82                         for seed in range(2):
83                             opts = dict(separation=sep, steps=300, seed=seed)
84                             ref = search_for_dense_graph(
85                                 variant, n, m, reference_mode=True, **opts)
86                             fast = search_for_dense_graph(variant, n, m, **opts)
87                             tag = (f"{variant.name} n={n} m={m} sep={sep} "
88                                   f"seed={seed}")
89                             # Bit-for-bit: same energies, same accept/reject
90                             # decisions, same density at every step.
91                             self.assertEqual(
92                                 [s.energy for s in fast.history],
93                                 [s.energy for s in ref.history], tag)
94                             self.assertEqual(
95                                 [s.accepted for s in fast.history],
96                                 [s.accepted for s in ref.history], tag)
97                             self.assertEqual(
98                                 [s.edge_count for s in fast.history],
99                                 [s.edge_count for s in ref.history], tag)
100                             self.assertEqual(
101                                 fast.best_edge_count, ref.best_edge_count, tag)
102                             self.assertTrue(
103                                 (fast.best_graph.mu == ref.best_graph.mu).all(),

```

```

104         tag)
105         # The returned witness is genuinely feasible by the
106         # exact checker (certified, not just flagged).
107         split = sep == "vertex"
108         self.assertEqual(
109             max_connectivity(fast.best_graph,
110                             vertex_split=split),
111             m - 1, tag)
112
113
114 if __name__ == "__main__":
115     unittest.main()

```

C.4.9 The prover

```

1  """The cut-counting prover: a zero-gap solve is a genuine upper-bound proof."""
2
3  import os
4  import unittest
5
6  import numpy as np
7
8  from erdos915_unified import (
9      MULTI_DIRECTED,
10     PULP_AVAILABLE,
11     Graph,
12     _canonical_form,
13     enumerate_extremal_directed_multigraphs,
14     max_edge_connectivity,
15     prove_directed_multigraph,
16 )
17
18
19 @unittest.skipUnless(PULP_AVAILABLE, "the MILP certifier needs the optional pulp")
20 class Prover(unittest.TestCase):
21     def test_small_optima_are_proved(self):
22         # M*(n) = 2(n-1), optimal with zero gap, for the reachable sizes.
23         for n in (3, 4, 5):
24             result = prove_directed_multigraph(n, time_limit=120.0)
25             self.assertTrue(result.is_proof())
26             self.assertEqual(result.status, "OPTIMAL")
27             self.assertEqual(round(result.scaled_optimum), 2 * (n - 1))
28
29     def test_one_solve_settles_every_m(self):
30         result = prove_directed_multigraph(4, time_limit=120.0)
31         self.assertTrue(result.is_proof())
32         for m in (2, 3, 4, 5):
33             # L_m^dir(4) = (m-1) * M*(4) = (m-1) * 6.
34             self.assertEqual(result.value_for(m), (m - 1) * 2 * (4 - 1))
35
36
37 class EnumerationDedup(unittest.TestCase):
38     """The streamed canonical-form dedup keeps one representative per iso-class
39     and returns exactly the known small extremal sets.
40
41     The active path stays at n in {3, 4} (sub-second). The n=5 enumeration is
42     minutes long (the DFS, not the dedup), so its known count is checked only when

```

```

43     'RUN_SLOW_ENUM' is set; the dedup logic it exercises is identical to n=4.
44     """
45
46     # m=3 directed multigraph, target = 4(n-1) arcs: the doubled bidirected
47     # spanning trees. Known counts (claude.md): n=3 -> 1 (path), n=4 -> 2
48     # (star, path), n=5 -> 3 (star, broom, path).
49     FAST_COUNTS = {3: 1, 4: 2}
50
51     def test_known_iso_class_counts(self):
52         for n, expected in self.FAST_COUNTS.items():
53             reps = enumerate_extremal_directed_multigraphs(n, 3, 4 * (n - 1))
54             self.assertEqual(len(reps), expected, f"n={n}")
55
56     def test_representatives_are_distinct_and_feasible(self):
57         for n in self.FAST_COUNTS:
58             reps = enumerate_extremal_directed_multigraphs(n, 3, 4 * (n - 1))
59             keys = [_canonical_form(mu) for mu in reps]
60             # One representative per class: all canonical keys distinct.
61             self.assertEqual(len(set(keys)), len(keys), f"n={n}")
62             for mu in reps:
63                 g = Graph(n, MULTI_DIRECTED)
64                 g.mu = mu.copy()
65                 self.assertEqual(g.edge_count(), 4 * (n - 1))
66                 self.assertLessEqual(max_edge_connectivity(g), 2, f"n={n}")
67
68     def test_dedup_preserves_the_iso_classes(self):
69         # The deduped run must cover exactly the iso-classes of the full labelled
70         # run: no class invented, none dropped.
71         for n in self.FAST_COUNTS:
72             labelled = enumerate_extremal_directed_multigraphs(
73                 n, 3, 4 * (n - 1), up_to_iso=False)
74             deduped = enumerate_extremal_directed_multigraphs(n, 3, 4 * (n - 1))
75             self.assertEqual(
76                 {_canonical_form(mu) for mu in labelled},
77                 {_canonical_form(mu) for mu in deduped},
78                 f"n={n}")
79             # Every labelled matrix is itself one of the kept iso-classes.
80             kept = {_canonical_form(mu) for mu in deduped}
81             for mu in labelled:
82                 self.assertIn(_canonical_form(mu), kept, f"n={n}")
83
84     def test_canonical_form_is_permutation_invariant(self):
85         rng = np.random.default_rng(0)
86         for n in (3, 4, 5, 6):
87             mu = rng.integers(0, 3, size=(n, n))
88             np.fill_diagonal(mu, 0)
89             key = _canonical_form(mu)
90             for _ in range(5):
91                 p = rng.permutation(n)
92                 self.assertEqual(_canonical_form(mu[np.ix_(p, p)]), key, f"n={n}")
93
94     @unittest.skipUnless(os.environ.get("RUN_SLOW_ENUM"),
95                          "n=5 enumeration is minutes long; set RUN_SLOW_ENUM=1")
96     def test_n5_known_count(self):
97         reps = enumerate_extremal_directed_multigraphs(5, 3, 16)
98         self.assertEqual(len(reps), 3) # star, broom, path
99
100

```

```

101 if __name__ == "__main__":
102     unittest.main()

```

C.4.10 The random model

```

1  """The random model: the appearance threshold and the Whitney relation.
2
3  All sampling is seeded, so these outcomes are deterministic.
4  """
5
6  import unittest
7
8  from erdos915_unified import (
9      connectivity_distribution,
10     edge_vertex_distribution,
11     estimate_appearance_probability,
12     threshold_curve,
13 )
14
15
16 class MonteCarlo(unittest.TestCase):
17     def test_appearance_probability_crosses_m_over_n(self):
18         n, m = 30, 3
19         below = estimate_appearance_probability(n, 0.3 * m / n, m, trials=120,
20             ↳ seed=1)
21         above = estimate_appearance_probability(n, 1.8 * m / n, m, trials=120,
22             ↳ seed=1)
23         self.assertLess(below, 0.5)
24         self.assertGreater(above, 0.5)
25
26     def test_whitney_holds_on_every_sample(self):
27         edge = connectivity_distribution(20, 0.25, trials=40, separation="edge",
28             ↳ seed=3)
29         vertex = connectivity_distribution(20, 0.25, trials=40, separation="vertex",
30             ↳ seed=3)
31         self.assertEqual(len(edge), 40)
32         for kappa, lam in zip(vertex, edge):
33             self.assertLessEqual(kappa, lam)
34
35     def test_edge_and_vertex_agree_at_small_n(self):
36         lam, kap = edge_vertex_distribution(5, 0.5, trials=120, seed=7)
37         self.assertEqual(sum(k < 1 for k, l in zip(kap, lam)), 0) # no gap at n =
38             ↳ 5
39         self.assertTrue(all(k <= 1 for k, l in zip(kap, lam))) # Whitney
40
41     def test_edge_and_vertex_part_company_at_larger_n(self):
42         lam, kap = edge_vertex_distribution(16, 0.25, trials=200, seed=7)
43         gap = sum(k < 1 for k, l in zip(kap, lam))
44         self.assertGreater(gap, 0)
45         self.assertTrue(all(k <= 1 for k, l in zip(kap, lam)))
46
47     def test_threshold_curve_reports_the_predicted_threshold(self):
48         curve = threshold_curve(20, 2, [0.05, 0.2], trials=20, seed=0)
49         self.assertAlmostEqual(curve.predicted_threshold, 2 / 20)
50         self.assertEqual(len(curve.probabilities), 2)
51
52

```



```

48 if __name__ == "__main__":
49     unittest.main()

```

C.4.11 The solver

```

1  """The single solver: one solve() call handles every kind of question."""
2
3  import shutil
4  import unittest
5
6  import itertools
7
8  import numpy as np
9
10 from erdos915_unified import (
11     solve,
12     enumerate_extremal_directed_multigraphs as _dfs_enum,
13     enumerate_extremal_directed_multigraphs_via_generation as _gen_enum,
14     _decorate_support_worker,
15     _canonical_form,
16     PULP_AVAILABLE,
17 )
18
19
20 class Solve(unittest.TestCase):
21     def test_exhaustive_simple_directed_is_exact(self):
22         r = solve(4, 2, directed=True, simple=True, exhaustive=True, max_seconds
23             ↳ =120.0)
24         self.assertTrue(r.proven)
25         self.assertEqual(r.bound, "exact")
26         self.assertEqual(r.value, 6)
27
28     def test_discovery_returns_a_lower_bound(self):
29         # verification stays on sa; tabu is the default for solving the problems
30         r = solve(4, 2, directed=True, simple=True, exhaustive=False,
31             max_seconds=3.0, method="sa")
32         self.assertEqual(r.bound, "lower")
33         self.assertEqual(r.value, 6)
34
35     def test_exhaustive_undirected_finds_the_spanning_tree(self):
36         r = solve(5, 2, directed=False, simple=True, exhaustive=True, max_seconds
37             ↳ =30.0)
38         self.assertTrue(r.proven)
39         self.assertEqual(r.value, 4) # a spanning tree on 5 vertices
40
41     @unittest.skipUnless(PULP_AVAILABLE,
42         "solve() proves this one through the MILP certifier, "
43         "which needs the optional pulp")
44     def test_directed_multigraph_is_proved(self):
45         r = solve(4, 3, directed=True, simple=False, exhaustive=True, max_seconds
46             ↳ =120.0)
47         self.assertTrue(r.proven)
48         self.assertEqual(r.value, 12) #  $L_3^{dir}(4) = 2(n-1)(m-1) = 12$ 
49
50     def test_vertex_separation_undirected(self):
51         #  $k_2(n) = n - 1$ : no cycle is allowed, so a spanning tree is extremal.
52         r = solve(5, 2, directed=False, simple=True, separation="vertex",

```

```

50         exhaustive=True, max_seconds=30.0)
51     self.assertTrue(r.proven)
52     self.assertEqual(r.value, 4)
53
54     def test_hypergraph_discovery(self):
55         r = solve(7, 2, hypergraph=True, r=3, exhaustive=False, max_seconds=3.0,
56                 method="sa")
57         self.assertEqual(r.bound, "lower")
58         self.assertEqual(r.value, 3)
59
60     def test_result_describe_is_readable(self):
61         r = solve(4, 2, directed=True, simple=True, exhaustive=True, max_seconds
62                 ↪ =120.0)
63         self.assertIn("value", r.describe())
64
65 @unittest.skipIf(shutil.which("geng") is None, "nauty's geng is not installed")
66 class GengGeneration(unittest.TestCase):
67     """The geng-seeded enumerator must return exactly the same isomorphism
68     classes as the DFS enumerator wherever the DFS one is a sound complete
69     search ( $n \leq 6$ ). This pins down the new generation pipeline against the
70     method whose values are already in the thesis."""
71
72     @staticmethod
73     def _classes(reps):
74         return {_canonical_form(mu) for mu in reps}
75
76     def test_matches_dfs_at_n4(self):
77         #  $n = 4$ ,  $m = 3$ : full target range and a degree cap. Fast, and the DFS
78         # enumerator is provably complete here, so equality is the gold standard.
79         for target, cap in [(12, None), (10, None), (8, None), (8, 6), (6, None)]:
80             with self.subTest(target=target, cap=cap):
81                 a = self._classes(_dfs_enum(4, 3, target, max_degree=cap))
82                 b = self._classes(_gen_enum(4, 3, target, max_degree=cap))
83                 self.assertEqual(a, b)
84
85     def test_extremal_is_the_doubled_spanning_trees(self):
86         # At  $n \leq 6$  the extremal directed multigraphs are exactly the doubled
87         # bidirected spanning trees, one per unlabelled tree: 2 on 4 vertices.
88         reps = _gen_enum(4, 3, 12) #  $L_3^{\text{dir}}(4) = 2(n-1)(m-1) = 12$ 
89         self.assertEqual(len(self._classes(reps)), 2)
90
91
92 class SupportWorker(unittest.TestCase):
93     """The per-support decorator is the unit the generation enumerator fans out
94     across processes. Over every support it must reproduce the DFS classes, so
95     the parallel refactor is covered here even without geng on PATH."""
96
97     @staticmethod
98     def _all_supports(n, min_e, max_e):
99         # Every non-isomorphic simple graph on  $n$  vertices in the edge range, the
100         # output geng would feed the enumerator, by brute force + canonical dedup.
101         pairs = [(i, j) for i in range(n) for j in range(i + 1, n)]
102         seen, out = set(), []
103         for mask in itertools.product((0, 1), repeat=len(pairs)):
104             if not (min_e <= sum(mask) <= max_e):
105                 continue
106             mu = np.zeros((n, n), dtype=int)

```

```

107         edges = []
108         for (i, j), bit in zip(pairs, mask):
109             if bit:
110                 mu[i, j] = mu[j, i] = 1
111                 edges.append((i, j))
112             key = _canonical_form(mu)
113             if key not in seen:
114                 seen.add(key)
115                 out.append(edges)
116         return out
117
118     def test_worker_union_matches_dfs_at_n4(self):
119         n, m, target = 4, 3, 12
120         supports = self._all_supports(n, (target + 3) // 4, min(target, n * (n -
121             ↪ 1) // 2))
122         reps = []
123         for edges in supports:
124             reps.extend(_decorate_support_worker((n, m, target, None, True, edges)
125             ↪ ))
126         worker = {_canonical_form(mu) for mu in reps}
127         dfs = {_canonical_form(mu) for mu in _dfs_enum(n, m, target)}
128         self.assertEqual(worker, dfs)
129
130     if __name__ == "__main__":
131         unittest.main()

```

Bibliography

- [1] Béla Bollobás and Pál Erdős. Extremal problems in graph theory. *Matematikai Lapok*, 13:143–152, 1962. Original statement of Problem 915.
- [2] Karl Menger. Zur allgemeinen Kurventheorie. *Fundamenta Mathematicae*, 10:96–115, 1927. The original statement of Menger’s theorem.
- [3] Pál Erdős. Erdős problems. <https://www.erdosproblems.com/915>. Problem 915.
- [4] Wolfgang Mader. Ein Extremalproblem des Zusammenhangs von Graphen. *Mathematische Zeitschrift*, 131:223–231, 1973. Solves Erdős Problem 915 for edge-disjoint paths.
- [5] Hassler Whitney. Congruent graphs and the connectivity of graphs. *American Journal of Mathematics*, 54(1):150–168, 1932.
- [6] John L. Leonard. On graphs with at most four line-disjoint paths connecting any two vertices. *Journal of Combinatorial Theory, Series B*, 13:242–250, 1972.
- [7] B. A. Sørensen and Carsten Thomassen. On k -rails in graphs. *Journal of Combinatorial Theory, Series B*, 17(2):143–159, 1974. Theorem 4 determines $f_5(n) = \lfloor 8n/3 \rfloor - 3$ for $n \geq 6$, $n \neq 7$, $n \neq 12$. Their $f_k(n)$ is the least edge count that forces a k -rail, one more than this thesis’s $k_m(n)$, so the value here is $\lfloor 8n/3 \rfloor - 4$.
- [8] John L. Leonard. On a conjecture of Bollobás and Erdős. *Periodica Mathematica Hungarica*, 3(3–4):281–284, 1973. Proves $\ell_m(n) = k_m(n)$ for $m \leq 4$.
- [9] Brendan D. McKay and Adolfo Piperno. Practical graph isomorphism, II. *Journal of Symbolic Computation*, 60:94–112, 2014. Describes the nauty/Traces canonical-labelling tools underlying geng.
- [10] Brendan D. McKay and Stanisław P. Radziszowski. $R(4, 5) = 25$. *Journal of Graph Theory*, 19(3):309–322, 1995. Exhaustive-generation determination of a Ramsey number.
- [11] Marijn J. H. Heule, Oliver Kullmann, and Victor W. Marek. Solving and verifying the Boolean Pythagorean triples problem via cube-and-conquer. In *Theory and Applications of Satisfiability Testing (SAT 2016)*, volume 9710 of *Lecture Notes in Computer Science*, pages 228–245. Springer, 2016. The certified SAT proof is roughly 200 terabytes in DRAT format.
- [12] Marijn J. H. Heule. Schur number five. In *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence (AAAI-18)*, pages 6598–6606. AAAI Press, 2018.
- [13] Alexander A. Razborov. Flag algebras. *The Journal of Symbolic Logic*, 72(4):1239–1282, 2007.

- [14] L. R. Ford and D. R. Fulkerson. Maximal flow through a network. *Canadian Journal of Mathematics*, 8:399–404, 1956.
- [15] Ralph E. Gomory and Tien Chung Hu. Multi-terminal network flows. *Journal of the Society for Industrial and Applied Mathematics*, 9(4):551–570, 1961. Introduces the Gomory-Hu tree structure.
- [16] W. Mantel. Problem 28. *Wiskundige Opgaven*, 10:60–61, 1907. The triangle-free extremal bound, the $r = 2$ case of Turán's theorem.
- [17] Pál Turán. On an extremal problem in graph theory. *Matematikai és Fizikai Lapok*, 48:436–452, 1941.
- [18] Zsolt Baranyai. On the factorization of the complete uniform hypergraph. In *Infinite and Finite Sets (Colloq., Keszthely, 1973), Vol. I*, volume 10 of *Colloq. Math. Soc. Janos Bolyai*, pages 91–108. North-Holland, 1975.
- [19] W. T. Tutte. *Connectivity in Graphs*. University of Toronto Press, Toronto, 1966.
- [20] John E. Hopcroft and Robert E. Tarjan. Dividing a graph into triconnected components. *SIAM Journal on Computing*, 2(3):135–158, 1973.
- [21] Wolfgang Mader. Existenz n -fach zusammenhängender teilgraphen in Graphen genügend großer Kantendichte. *Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg*, 37:86–97, 1972. Average degree at least $4k$ forces a $(k+1)$ -connected subgraph.
- [22] Wolfgang Mulzer. Five proofs of Chernoff's bound with applications. *Bulletin of the EATCS*, 124, 2018. Accessible survey of Chernoff bound variants.
- [23] Béla Bollobás. *Random Graphs*, volume 73 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, 2 edition, 2001. Chapter 7 gives $\kappa(G(n, p)) = \delta(G(n, p))$ with high probability in this regime.
- [24] Béla Bollobás and Andrew Thomason. Random graphs of small order. In *Random Graphs '83*, volume 28 of *Annals of Discrete Mathematics*, pages 47–97. North-Holland, 1985. Hitting-time form of the connectivity result.
- [25] Alan Frieze and Michał Karoński. *Introduction to Random Graphs*. Cambridge University Press, 2016. Chapter 7 covers connectivity in $G(n, p)$ and $D(n, p)$; the directed analogue of Bollobás's $\kappa = \delta$ result appears in Section 7.3.

